Critical Jumps for Rough Polynomial Carleson Operators with Angular Hardy Data and Finite Radial Variation

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Abstract

We prove critical-jump estimates for sharply truncated rough singular integrals with polynomial phases. For mean-zero angular data in $H^1(\mathbb{S}^{n-1})$, every $1 < p < \infty$, and every fixed radial amplitude of finite ρ-variation, $1 \le \rho < \infty$, the polynomial modulation supremum and the jump-amplitude supremum are both controlled pointwise before taking the L^p norm. The bound is linear in the angular Hardy norm and the radial variation norm. For each fixed polynomial phase, mean-zero $L\log L$ angular data also give a coefficient-uniform weak type $(1,1)$ estimate for the pointwise critical-jump functional.

The strong estimate combines a pointwise dyadic martingale inequality, rough oscillatory correlations, and polynomial tile geometry. A uniform conical estimate retains the elliptic cutoff and radial amplitude created by affine normalization of spherical atoms. The weak endpoint uses phase-adapted cancellation and packet estimates that remain uniform under concentration of the bad data.

The estimates concern complete truncation trajectories and imply maximal truncation, r-variation for every $r > 2$, and principal-value convergence. In dimensions at least two, the mean-zero angular $L\log L$ class is strictly contained in $H^1(\mathbb{S}^{n-1})$. Counterexamples exclude unrestricted bounded radial amplitudes and true quadratic output variation.

AI-use disclosure. The author selected the research objectives through interaction with AI systems; the mathematical arguments and derivations were produced by those systems.

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1 Introduction and main results

A maximal-truncation estimate controls only the largest excursion of an operator trajectory. An r-variation estimate with $r > 2$ also quantifies convergence, but the endpoint $r = 2$ is generally inaccessible. The critical replacement is the pointwise 2-jump functional

$$\mathbf{J}(F)(x) := \sup_{\lambda>0} \lambda N_\lambda(F(x))^{1/2}. \tag{1.1}$$

The spatial norm is taken only after this pointwise supremum; in particular, the endpoint quantity is

$$\|\mathbf{J}(F)\|_{L^{1,\infty}} = \left\| \sup_{\lambda>0} \lambda N_\lambda(F)^{1/2} \right\|_{L^{1,\infty}}.$$

The functional $\mathbf{J}(F)$ measures disjoint oscillations at every amplitude and controls r-variation for every $r > 2$. Its weak spatial norm is stronger than the conventional mixed quasi-seminorm $\sup_{\lambda>0} \|\lambda N_\lambda(F)^{1/2}\|_{L^{1,\infty}}$; Proposition 27.1 proves strictness using finite jump paths. We prove two complementary forms of this estimate: a strong L^p theorem with a pointwise supremum over all bounded-degree polynomial phases, and a coefficient-uniform weak $(1,1)$ theorem for each fixed phase. Both concern the complete continuous sharp-truncation trajectory. The strong modulation theorem and the fixed-phase weak theorem both admit every fixed finite radial variation exponent. No nondegeneracy, smallness, or genericity condition is imposed on the polynomial phase. In particular, the results include convolution phases, genuinely nonconvolution phases, and mixed phases that vanish on the diagonal.

The strong modulation theorem assumes mean-zero angular data in $H^1(\mathbb{S}^{n-1})$. The fixed-phase weak endpoint assumes mean-zero angular $L\log L$ data. The pointwise phase supremum includes linear phases and general even angular components. Its proof combines polynomial tile geometry, selector-dependent rough estimates, and a conical atomic reduction. The complementary weak endpoint at $P = 0$ strengthens the formulation posed by Jones–Seeger–Wright [26]; phase-adapted cancellation extends it to every fixed polynomial phase.

1.1 Notation and truncation trajectories

Let $d\sigma$ be the usual surface measure on $\mathbb{S}^{n-1}$. Put

$$K_\Omega(z) = \frac{\Omega(z/|z|)}{|z|^n}, \qquad z \neq 0.$$

For $A = \|\Omega\|_1 > 0$ define the homogeneous gauge

$$\mathfrak{L}(\Omega) = A + \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| \log^+\left(\frac{|\Omega(\theta)|}{A}\right) d\sigma(\theta), \tag{1.2}$$

and set $\mathfrak{L}(0) = 0$. This is equivalent, with dimensional constants, to the homogeneous Luxemburg $L\log L$ norm. Its homogeneity and the boundedness of the odd and even projections

$$\Omega_o(\theta) = \frac{\Omega(\theta) - \Omega(-\theta)}{2}, \qquad \Omega_e(\theta) = \frac{\Omega(\theta) + \Omega(-\theta)}{2}$$

give

$$\mathfrak{L}(\Omega_o) + \mathfrak{L}(\Omega_e) \le C\mathfrak{L}(\Omega). \tag{1.3}$$

The elementary normalization details are recorded in Lemma 15.1.

On bounded compactly supported inputs define

$$T^P_{\Omega,\varepsilon} f(x) = \int_{|x-y|>\varepsilon} e^{iP(x,y)} K_\Omega(x-y) f(y)\, dy.$$

For L^p data with $1 < p < \infty$, the simultaneous construction is given in Sections 12–13. The L^1 endpoint construction appears in Section 25. Both constructions are characterized by the finite-annulus identity and convergence to zero as $\varepsilon \to \infty$.

Let E be a linearly ordered set and let $F = (F_t)_{t\in E}$ be a scalar path. For $\lambda > 0$, its paired strict jump number is

$$N_{\lambda,E}(F) := \sup\big(\{0\} \cup \mathscr{W}_{\lambda,E}(F)\big) \in \mathbb{N}_0 \cup \{\infty\},$$

where $\mathscr{W}_{\lambda,E}(F)$ is the set of all $N \in \mathbb{N}$ for which there are $s_i, t_i \in E$, $1 \le i \le N$, satisfying

$$s_1 < t_1 \le s_2 < t_2 \le \cdots \le s_N < t_N, \qquad |F_{t_i} - F_{s_i}| > \lambda \quad (1 \le i \le N).$$

Thus a path with no strict λ-jump has jump number zero, while a path with arbitrarily large finite witnesses has jump number infinity. For $r > 0$, its homogeneous r-variation is

$$V_E^r(F) := \sup\left(\{0\} \cup \left\{ \left(\sum_{i=1}^{N} |F_{t_i} - F_{t_{i-1}}|^r \right)^{1/r} : N \in \mathbb{N}, \quad t_0 < \cdots < t_N, \quad t_i \in E \right\}\right).$$

When the index set is displayed or unambiguous, we omit the subscript E; thus the main positive-radius paths use $N_\lambda(F)$ and $V^r(F_t : t > 0)$. For a restriction to an ordered subset $D \subset E$, put

$$\mathbf{J}_D(F) := \sup_{\lambda>0} \lambda N_{\lambda,D}(F|_D)^{1/2},$$

and omit D when it is the full parameter set. For a measurable function H, set $\|H\|_{L^{1,\infty}} = \sup_{\tau>0} \tau|\{x : |H(x)| > \tau\}|$. Call a path coordinatewise measurable if every F_t is measurable in the spatial variable. For such a path define $\mathbf{J}(F)$ pointwise by (1.1), initially as an extended-valued pointwise functional without a separate measurability assertion. We call the path jump-separable if there are a countable parameter set $D \subset E$ and one common full-measure spatial set on which, for every $\lambda > 0$, its full and D-restricted jump numbers agree. Whenever $\mathbf{J}(F)$ is measurable—in particular, for every jump-separable coordinatewise measurable path—and $1 < p < \infty$, set

$$\mathscr{J}_{2,p}(F) = \|\mathbf{J}(F)\|_p = \left\| \sup_{\lambda>0} \lambda N_\lambda(F)^{1/2} \right\|_p, \qquad \mathscr{J}_{2,1,\infty}(F) = \|\mathbf{J}(F)\|_{L^{1,\infty}}. \tag{1.4}$$

Because all jumps are strict, the supremum in λ may be restricted to $\mathbb{Q}_+$ without changing $\mathbf{J}(F)$. More precisely, if $D \subset E$ is countable and the coordinates F_t are measurable for $t \in D$, then, for every $N \ge 1$,

$$\{N_\lambda(F_t : t \in D) \ge N\} = \bigcup_{(s_i,t_i)_{i=1}^N \in \mathscr{O}_N(D)} \bigcap_{i=1}^{N} \{|F_{t_i} - F_{s_i}| > \lambda\},$$

where $\mathscr{O}_N(D)$ is the countable set of ordered paired $2N$-tuples in D. Thus the jump count on D is measurable, and every jump-separable coordinatewise measurable path has a measurable pointwise jump functional. If the path is locally continuous in the truncation parameter, its jump counts on a dense D agree with those on all positive parameters by Lemma 15.6. These observations and the rational λ-supremum prove measurability of $\mathbf{J}(F)$.

We use the following common-representative convention throughout. Whenever one of the continuum-parameter operator trajectories constructed below is asserted pointwise on the full parameter interval, its construction is carried out on one spatial set X_0 of full measure, independent of the parameter, on which the parameter path is locally continuous (or locally absolutely continuous when that stronger property is used) and all stated finite-annulus identities hold simultaneously. All coordinates are set equal to zero on X_0^c. Thus a countable dense parameter set determines all jumps and variations by Lemma 15.6, and the preceding countable-union argument proves their measurability. This convention does not impose continuity on the abstract scalar paths used in the jump lemmas. We do not pass from coordinatewise almost-everywhere identities at uncountably many parameters to a common

operator representative without an explicit finite-witness or anchored uniform-Cauchy argument. Joint measurability in the parameter is stated separately whenever Fubini or a parameter integral requires it.

Throughout, C and c may change from line to line. Every constant in a global theorem or corollary depends only on the parameters displayed there; in particular, it is independent of the support and location of the input and of every approximation used to construct the operator. Constants inside proofs may depend on fixed smooth cutoffs and normalized coordinate charts chosen once and for all. Only an explicitly localized statement may depend on support or chart data named in that statement. No constant depends on the coefficients of P, on truncation or dyadic scales, or on auxiliary finite cutoffs. Every coefficient norm is a fixed Euclidean norm on the relevant finite-dimensional polynomial space; changing that norm changes only the permitted structural constants.

The angular Hardy space. For $n \geq 2$, $H^1(\mathbb{S}^{n-1})$ denotes the real atomic Hardy space on the sphere, with complex-valued functions permitted. Write $m_n = \sigma(\mathbb{S}^{n-1})$. Its atoms are the constant atom $a_* = m_n^{-1}$ and the cancellative atoms a satisfying

$$\operatorname{supp} a \subset B, \qquad \int a \, d\sigma = 0, \qquad \|a\|_\infty \leq \sigma(B)^{-1},$$

where B is a geodesic ball or the whole sphere. The norm is

$$\|\Omega\|_{H^1(\mathbb{S}^{n-1})} = \inf\left\{ \sum_\nu |\lambda_\nu| : \Omega = \sum_\nu \lambda_\nu a_\nu \text{ in } L^1 \right\}.$$

The series are absolutely summable in L^1. This is the usual compact real Hardy space; the atomic formulation and the small-cap formulation used below are related in Section 10. We write $H^1_0 = \{\Omega \in H^1 : \int \Omega \, d\sigma = 0\}$. For $n = 1$, the sphere consists of two points; we use its finite-dimensional L^1 norm as the Hardy norm and impose the same mean-zero condition. The variable in this Hardy space is the angular variable. Spatial inputs in the main strong theorem belong to $L^p(\mathbb{R}^n)$.

For $1 \leq \rho < \infty$ and a pointwise bounded radial function $b : (0,\infty) \to \mathbb{C}$, put

$$\|b\|_{\mathscr{V}^\rho} := \|b\|_\infty + V^\rho(b;(0,\infty)). \tag{1.5}$$

A function of finite ρ-variation is regulated. In (1.5), $\|b\|_\infty$ denotes the pointwise supremum of the chosen representative, consistently with the pointwise definition of V^ρ. Whenever point values enter a Stieltjes formula, we use the representative

$$b^+(t) := \lim_{s \downarrow t} b(s), \qquad t > 0.$$

Indeed, simultaneous approximation of the points of any finite partition from the right gives $V^\rho(b^+;(0,\infty)) \leq V^\rho(b;(0,\infty))$, and the same right-limit argument gives $\|b^+\|_\infty \leq \|b\|_\infty$. To quantify the exceptional point values, fix $\delta > 0$ and distinct $t_1 < \cdots < t_N$ such that $|b(t_i) - b^+(t_i)| > \delta$. Choose, successively and sufficiently close to the right of each t_i, points $s_i \in (t_i, t_{i+1})$ (with the evident convention for $i = N$) such that $|b(s_i) - b^+(t_i)| < \delta/2$. The ordered partition containing every t_i, s_i then gives

$$N(\delta/2)^\rho \leq V^\rho(b;(0,\infty))^\rho.$$

Thus there are only finitely many such points for each δ, and $b = b^+$ off a countable set. Polar coordinates then show that $b(|z|) = b^+(|z|)$ for almost every $z \in \mathbb{R}^n$, so replacing b by b^+ changes none of the Lebesgue integral operators below and does not increase $\|b\|_{\mathscr{V}^\rho}$. Initially for bounded compactly supported inputs define

$$T^{P,b}_{\Omega,\varepsilon} f(x) = \int_{|x-y|>\varepsilon} e^{iP(x,y)} b(|x-y|) K_\Omega(x-y) f(y) \, dy.$$

1.2 Principal results

Theorem 1.1 (Angular Hardy data and maximally modulated finite radial variation)**.** *Let $n \geq 1$, $d \geq 0$, $1 < p < \infty$, and $1 \leq \rho < \infty$. Let $\Omega \in H^1_0(\mathbb{S}^{n-1})$, and fix a radial function $b : (0,\infty) \to \mathbb{C}$ with $\|b\|_{\wedge\rho} < \infty$. For every $f \in L^p(\mathbb{R}^n)$,*

$$\left\| \sup_{\deg P \leq d} \sup_{\lambda > 0} \lambda N_\lambda (T^{P,b}_{\Omega,t} f : t > 0)^{1/2} \right\|_p \leq C_{n,d,p,\rho} \|\Omega\|_{H^1(\mathbb{S}^{n-1})} \|b\|_{\wedge\rho} \|f\|_p. \tag{1.6}$$

Here P ranges over all real polynomials on $\mathbb{R}^n \times \mathbb{R}^n$ of total degree at most d. For each selected trajectory, the same polynomial is used at every truncation parameter. Both suprema in (1.6) *are pointwise suprema inside the spatial norm. The amplitude b is fixed; the constant is uniform over b subject to its displayed norm.*

There is one full-measure set, independent of P and $t > 0$, on which all positive truncations are absolutely convergent and depend jointly continuously on t and the polynomial coefficients. Their absolute tails tend to zero as $t \to \infty$, uniformly over P.

Theorem 1.1 uses the standard angular Hardy space defined above. The corresponding strong estimate for mean-zero $L\log L$ angular data is a direct special case of Theorem 1.1, by the continuous inclusion $L\log L(\mathbb{S}^{n-1}) \subset H^1(\mathbb{S}^{n-1})$, proved below. For $n \geq 2$ this inclusion is strict, including on the mean-zero subspaces. The phase supremum is taken pointwise before the spatial norm. It includes the maximally truncated polynomial Carleson operator: every finite-annulus integral is a single increment of the trajectory. The resulting maximal-truncation, $r > 2$ variation, and simultaneous principal-value conclusions are collected in Section 14.

The next theorem gives the complementary spatial endpoint. Its strong estimate follows from Theorem 1.1 with $b = 1$ and the $L\log L$ inclusion; Part II also provides an independent fixed-phase proof, whose localized estimates enter the weak-type and radial-amplitude arguments.

Theorem 1.2 (Fixed-phase critical jumps and the weak endpoint)**.** *Let $n \in \mathbb{N}$ and $d \in \mathbb{N}_0$. Let P be a real polynomial on $\mathbb{R}^n \times \mathbb{R}^n$ of total degree at most d, and let $\Omega \in L\log L(\mathbb{S}^{n-1})$ satisfy $\int \Omega \, d\sigma = 0$. Then the pointwise critical-jump functional of the sharp-truncation trajectory satisfies the coefficient-uniform estimates*

$$\mathscr{J}_{2,p}(T^P_{\Omega,\varepsilon} f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{L}(\Omega) \|f\|_p, \qquad 1 < p < \infty, \tag{1.7}$$

$$\mathscr{J}_{2,1,\infty}(T^P_{\Omega,\varepsilon} f : \varepsilon > 0) \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1. \tag{1.8}$$

Estimate (1.7) *holds for every $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$; estimate* (1.8) *holds for every $f \in L^1(\mathbb{R}^n)$. The constants are independent of every coefficient of P.*

For each fixed Ω as above and each fixed f in the data class of the asserted estimate, Proposition 17.6 in the strong case and Section 25 in the weak case construct the entire trajectory on one full-measure set, on which all of its finite-annulus identities hold and

$$T^P_{\Omega,\varepsilon} f(x) \longrightarrow 0 \qquad \text{as } \varepsilon \to \infty \tag{1.9}$$

holds. These two properties, rather than the estimates alone, determine the trajectory uniquely: the finite-annulus identities make the difference of two such representatives independent of ε on that common full-measure set, and the zero anchor makes this difference vanish. When $1 < p < \infty$, the convergence in (1.9) *also holds in $L^p(\mathbb{R}^n)$.*

Put

$$\kappa_r = \left(\sum_{k=1}^{\infty} k^{-r/2} \right)^{1/r} \leq \left(\frac{r}{r-2} \right)^{1/r}, \qquad r > 2. \tag{1.10}$$

Corollary 1.3 (Variation and convergence consequences)**.** *Under the hypotheses of Theorem 1.2, for every $r > 2$,*

$$\|V^r(T^P_{\Omega,\varepsilon} f : \varepsilon > 0)\|_p \le C_{n,d,p}\kappa_r \mathfrak{L}(\Omega)\|f\|_p, \qquad 1 < p < \infty, \tag{1.11}$$

$$\|V^r(T^P_{\Omega,\varepsilon} f : \varepsilon > 0)\|_{L^{1,\infty}} \le C_{n,d}\kappa_r \mathfrak{L}(\Omega)\|f\|_1. \tag{1.12}$$

Here (1.11) *holds for every $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$, and* (1.12) *holds for every $f \in L^1(\mathbb{R}^n)$. The factor κ_r is needed only for the V^r conclusions. The zero anchor and* (15.7) *give directly*

$$\left\| \sup_{\varepsilon>0} |T^P_{\Omega,\varepsilon} f| \right\|_p \le C_{n,d,p}\mathfrak{L}(\Omega)\|f\|_p, \qquad 1 < p < \infty,$$

$$\left\| \sup_{\varepsilon>0} |T^P_{\Omega,\varepsilon} f| \right\|_{L^{1,\infty}} \le C_{n,d}\mathfrak{L}(\Omega)\|f\|_1,$$

with no κ_r loss. The finiteness of V^r implies that $\lim_{\varepsilon\downarrow 0} T^P_{\Omega,\varepsilon} f$ exists almost everywhere in the respective ranges. Indeed, if a scalar path F_ε with finite V^r were not Cauchy as $\varepsilon \downarrow 0$, there would be $\delta > 0$ and, recursively, pairs

$$0 < s_{k+1} < t_{k+1} < s_k < t_k, \qquad |F_{t_k} - F_{s_k}| > \delta.$$

For every N, inserting the endpoints of the first N pairs in increasing order into one variation partition would give $V^r(F)^r \ge N\delta^r$, a contradiction. Completeness of $\mathbb{C}$ therefore gives the asserted endpoint limit. This argument uses only finite variation; the logically separate zero anchor at infinity is what gives the maximal estimate above.

Consequently, the principal-value limit $T^P_\Omega f := \lim_{\varepsilon\downarrow 0} T^P_{\Omega,\varepsilon} f$ defines an operator from L^1 to $L^{1,\infty}$ with bound $C_{n,d}\mathfrak{L}(\Omega)$. For general P, this answers, in the mean-zero $L\log L$ class, the rough-kernel weak type $(1,1)$ problem posed by Ding and Lai [21, §6.1]; the maximal-truncation and critical-jump bounds are stronger.

Theorem 1.4 (Coefficient-uniform bounds for every finite radial variation exponent)**.** *Let the hypotheses on n, d, P, and Ω be those of Theorem 1.2. Fix $1 \le \rho < \infty$, and let $b : (0,\infty) \to \mathbb{C}$ be pointwise bounded with $\|b\|_{\mathscr{V}^\rho} < \infty$. Then*

$$\|\mathbf{J}(T^{P,b}_{\Omega,\varepsilon} f : \varepsilon > 0)\|_p \le C_{n,d,p,\rho}\|b\|_{\mathscr{V}^\rho}\mathfrak{L}(\Omega)\|f\|_p, \qquad 1 < p < \infty, \tag{1.13}$$

$$\|\mathbf{J}(T^{P,b}_{\Omega,\varepsilon} f : \varepsilon > 0)\|_{L^{1,\infty}} \le C_{n,d,\rho}\|b\|_{\mathscr{V}^\rho}\mathfrak{L}(\Omega)\|f\|_1. \tag{1.14}$$

Estimate (1.13) *holds for every $f \in L^p(\mathbb{R}^n)$, and* (1.14) *holds for every $f \in L^1(\mathbb{R}^n)$. The constants are independent of every coefficient of P and of the location of the radial variation of b. The trajectory is uniquely determined by its finite-annulus increments and the zero-at-infinity anchor. For every output exponent $r > 2$, the corresponding strong and weak r-variation conclusions hold with the additional factor κ_r. The maximal-truncation estimates follow instead from the zero anchor and* (15.7)*, with the same right-hand sides as* (1.13)–(1.14) *and no κ_r loss. Almost-everywhere convergence as $\varepsilon \downarrow 0$ follows from the finiteness of V^r.*

The dependence on the fixed finite exponent ρ is not asserted to be uniform. Proposition 27.3 shows that the conclusion is false at $\rho = \infty$. The range $1 \le \rho < 2$ also admits the direct pathwise proof of Proposition 26.1; the general proof, including the critical and supercritical radial ranges, is given in Section 26.

For a bounded function b of finite total variation, write

$$\|b\|_{\mathrm{BV}} = \|b\|_\infty + \mathrm{Var}_{(0,\infty)}(b).$$

The case $\rho = 1$ of Theorem 1.4 is complemented by the following direct Stieltjes transfer, whose constant is obtained without the operator-specific radial argument.

Corollary 1.5 (Radial BV transfer)**.** *Under the hypotheses of Theorem 1.2, let $b:(0,\infty)\to\mathbb{C}$ be bounded and of finite total variation. All conclusions of Theorem 1.2 and Corollary 1.3 remain valid with $T^{P}_{\Omega,\varepsilon}$ replaced by $T^{P,b}_{\Omega,\varepsilon}$ and with each right-hand side multiplied by*

$$\|b\|_{\mathrm{BV}} = \|b\|_{\infty} + \mathrm{Var}_{(0,\infty)}(b).$$

The constants are independent of where the variation of b is concentrated.

The principal estimates have the scopes summarized in Table 1. The strong theorem combines angular H^1 data, every fixed finite radial variation exponent, and the pointwise polynomial phase supremum. The weak endpoint remains a coefficient-uniform fixed-phase result under angular $L\log L$. No supremum over radial amplitudes is included in these statements. Proposition 14.6 gives a one-dimensional obstruction to retaining weak type $(1,1)$ under the phase supremum when $d \geq 2$.

Result	Angular class	Polynomial phases	Radial amplitude	Spatial estimate
Theorem 1.1	H^1_0	Pointwise supremum	Fixed $b \in \mathscr{V}^{\rho}$, $\rho < \infty$	Strong L^p, $1 < p < \infty$
Theorem 1.2, weak endpoint	Mean-zero $L\log L$	Each fixed phase	$b \equiv 1$	Weak type $(1,1)$
Theorem 1.4, weak endpoint	Mean-zero $L\log L$	Each fixed phase	Fixed $b \in \mathscr{V}^{\rho}$, $\rho < \infty$	Weak type $(1,1)$

Table 1: Scope of the principal estimates. Constants are independent of the polynomial coefficients; every radial exponent satisfies $1 \leq \rho < \infty$.

1.3 Relation to earlier work

Static rough polynomial theory. For $P = 0$, the strong and weak theory of rough homogeneous singular integrals at the $L\log L$ angular scale was developed in [12, 23, 18, 36]. For arbitrary $P(x,y)$, Jiang–Lu [25] studied the static principal-value operator with $\Omega \in L\log L$ and a radial factor in BV, including an L^p boundedness criterion and weighted estimates. Ojanen [34] proved weighted strong estimates for the same polynomial-phase, angular-$L\log L$, radial-BV class under the geometric hypotheses formulated there. Al-Salman and Grafakos [3] more recently obtained static L^p bounds for arbitrary $P(x,y)$, angular $L\log L$ data, and a rough radial factor satisfying an L^{γ}-type condition. The corresponding rough-kernel weak type $(1,1)$ endpoint for the principal-value operator was explicitly listed as an open problem by Ding and Lai [21, §6.1]. Theorem 1.2 and Corollary 1.3 answer it in the mean-zero $L\log L$ class, with the stronger critical-jump control of the complete sharp-truncation trajectory.

Smooth coefficient-uniform weak theory. Ricci and Stein [35] established coefficient-uniform strong L^p estimates for smooth kernels. Chanillo and Christ [16] proved coefficient-uniform weak type $(1,1)$ for arbitrary polynomial phases and smooth Calderón–Zygmund convolution kernels. Their argument removes pure phases, normalizes a leading coefficient by dilation, and uses a phase-adapted double induction; their final remarks also yield L^p bounds, $1 < p < \infty$, for the maximal truncation. Rough $L\log L$ kernels fall outside that theory. The global phase flag introduced below is designed to retain coefficient uniformity while accommodating rough annular decay and phases that degenerate on the diagonal.

Maximal rough polynomial theory. Representative rough polynomial estimates include [32, 2]; polynomial maximal-truncation estimates for smooth Calderón–Zygmund kernels include [28].

Ding and Liu [22, Theorem 1.1] proved L^p bounds, $1 < p < \infty$, for the pointwise coefficient supremum of principal-value operators with mean-zero angular $H^1(\mathbb{S}^{n-1})$ kernels, $n \geq 2$, and radial phases

$Q_\lambda(|z|) = \sum_{k=2}^{d} \lambda_k |z|^k$. On the common subclass of even-power radial phases, Theorem 1.1 strengthens this principal-value estimate to critical-jump control of the complete sharp-truncation trajectory, with every fixed finite radial variation exponent. Their odd radial powers lie outside the polynomial phase class considered here.

Choudhary, Shrivastava, and Shuin [17] state coefficient-uniform sparse and strong maximal estimates for arbitrary P when $\Omega \in L^\infty$. For $P = 0$, Lai [29, Theorem 1.1] proves the weak $(1,1)$ maximal estimate at the $L\log L$ angular endpoint. In the present paper, maximal truncation is obtained as a consequence of the stronger critical-jump theorem.

Chen and Ji [7, Theorems 1.1–1.2] establish coefficient-uniform weak $(1,1)$ bounds for the principal-value operator and its full maximal truncation in the same mean-zero $L\log L$ angular class. For $b \equiv 1$, our fixed-phase critical-jump estimate strictly strengthens both conclusions. Indeed, the zero-at-infinity anchor gives

$$T_P^* f(x) \le \mathbf{J}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0),$$

and finite jumps also yield the principal-value limit. The additional fluctuation control is strictly stronger than maximal control: a finite path alternating between zero and one can have arbitrarily large jump functional while its maximum remains one. The related weak endpoint was publicly stated by Tsengmath on August 31, 2026 [38].

Fluctuation theory. For sharply truncated Hilbert transforms, Campbell, Jones, Reinhold, and Wierdl [13, Theorems 1.2–1.4 and Conjecture 7.2] established strong and weak r-variation estimates for $r > 2$, the associated noncritical jump bounds, and short 2-variation estimates, and isolated the critical jump exponent. Their higher-dimensional work [14] developed singular-integral variation theory at the rough $L\log L$ scale. Jones–Seeger–Wright [26, Theorem 1.6] proved the conventional strong critical-jump inequality for the unphased rough trajectory under $\Omega \in L^a$, $a > 1$, and posed the corresponding weak-type $(1,1)$ estimate for rough homogeneous kernels [26]. Their formulation places the supremum in the jump amplitude outside the spatial norm. The $P = 0$ case of (1.8) places that supremum pointwise inside $L^{1,\infty}$, and hence strictly strengthens the problem they posed.

Chen, Ding, Hong, and Liu [15] state weighted jump estimates with the amplitude supremum inside the spatial norm. The proof of their smooth-averaging proposition invokes a martingale estimate from [10]; the stated estimate in that source places the amplitude supremum outside the spatial norm. Section 3 proves the required unweighted pointwise martingale inequality directly; Lemma 4.5 gives its smooth-averaging form, and Lemma 15.4 derives the directional singular-integral and one-sided-average estimates. Ding, Hong, and Liu [20, Theorem 1.2(ii)] state, with the angular size tacitly normalized, the unphased strong critical-jump theorem at the $L\log L$ endpoint. Bhojak and Shrivastava [6, Theorem 1.1] subsequently state the unphased weak critical-jump and weak r-variation endpoints. Remark 28.1 explains why neither of the latter two arguments is used here.

Polynomial modulation and rough kernels. Becker, van Doorn, Jamneshan, Srivastava, and Thiele [4] construct a polynomial tile geometry and prove Carleson estimates for one-sided Hölder kernels under a non-tangential maximal L^2 hypothesis. We use the spatial and phase-cell geometry, the combinatorial decomposition, and the geometric stopping lemmas from that work. The analytic estimates for the rough kernels considered here are established in Part I; they are not consequences of the Hölder-kernel theorem alone.

Becker, Jamneshan, and Thiele [5, Theorem 1.3 and Corollary 1.4] prove strong L^p r-variation estimates, $1 < p < \infty$ and $r > 2$, for sharp truncations of Hölder Calderón–Zygmund kernels satisfying exact two-sided annular cancellation. Their estimate takes the supremum over bounded-degree polynomial input modulations pointwise inside the L^p norm and holds more generally on homogeneous Lie groups. In Euclidean space, this formulation already includes two-variable polynomial phases: for each fixed output point x, write

$$Q_x(y) = P(x,y) - P(x,0), \qquad T_{\Omega,t}^P f(x) = e^{iP(x,0)} T_{\Omega,t}^{Q_x} f(x).$$

The factor $e^{iP(x,0)}$ is independent of the truncation parameter. Consequently, the smooth-kernel result is not confined to phases that are independent of the output variable.

Combining their one-dimensional theorem with the method of rotations also gives the fixed-phase, coefficient-uniform strong $r > 2$ estimates for odd angular kernels in L^1, including the entire mean-zero angular class in dimension one. That argument does not treat the general even rough component at the $L\log L$ scale, the critical 2-jump estimate, or the spatial weak endpoint. Theorem 1.1 treats the mean-zero angular H^1 class at the critical jump scale, with a pointwise phase supremum and every fixed finite radial variation exponent. Corollary 14.4 records the corresponding L^1 angular improvement for odd kernels.

Contributions and complementary endpoints. Theorem 1.1 gives the strong phase-supremum estimate under angular H^1 and finite radial variation. Its bounded-angular stage is uniform for conically supported elliptic truncations; cancellative atomic summation then gives the Hardy norm directly. The radial factor and the sharp cutoff undergo the same affine transformation. Finite-ρ oscillatory decay and the actual paired-witness structure give the estimates needed for phase selection, including $\rho \geq 2$. Theorem 1.2 and Theorem 1.4 establish the complementary fixed-phase weak endpoint under angular $L\log L$. This weak conclusion is not inferred from the strong Hardy theorem.

1.4 Sharpness and limiting cases

The strong modulation theorem reaches the critical jump exponent 2; the fixed-phase theorem additionally reaches spatial weak type $(1,1)$. For $n \geq 2$, the inherited obstruction at the $L\log L$ threshold concerns the Orlicz scale, rather than optimality among all angular function spaces. Al-Qassem, Cheng, and Pan [1, Theorem 1.3(b)] construct mean-zero angular functions in $L(\log L)^{1-\varepsilon}$, for every $\varepsilon > 0$, whose associated scalar rough oscillatory integrals fail to be uniformly bounded over polynomial phases. This does not conflict with the strict inclusion $L\log L \subset H^1$: the Orlicz classes $L(\log L)^{1-\varepsilon}$ are not thereby asserted to lie in H^1. No optimality statement for the angular Hardy class is made. Proposition 14.7 also shows why the angular statement cannot be read as a spatial $H^1(\mathbb{R}^n) \to L^1$ theorem.

The radial and fluctuation endpoints are settled separately in Section 27. Theorem 1.4 holds for each fixed $1 \leq \rho < \infty$, but a one-dimensional Hilbert multiplier example excludes $\rho = \infty$. An abstract scalar-path example shows that the direct Young–Stieltjes transfer necessarily stops before $\rho = 2$, even though operator-specific orthogonality recovers every finite ρ. Finally, an exact Mellin representation reduces Poisson 2-variation to sharp Hilbert truncations and, together with the finite Gaussian obstruction proved below, proves that the pointwise critical jump cannot be strengthened to true output V^2. The precise optimal growth of the $r > 2$ constants as $r \downarrow 2$ is not addressed.

The radial theory also contains a distinct shellwise refinement for the Hilbert transform. If

$$R_2(b) = \left(\sum_{j \in \mathbb{Z}} V^2(b; (2^j, 2^{j+1}))^2 \right)^{1/2} < \infty,$$

then Proposition 26.31 gives

$$\|\mathbf{J}(H^b_\varepsilon f : \varepsilon > 0)\|_2 \leq C\big(\|b\|_\infty + R_2(b)\big)\|f\|_2.$$

The condition is genuinely shellwise: a bounded function that is constant on each open dyadic shell has $R_2(b) = 0$, even when its values oscillate between successive shells and its global finite-exponent variation is infinite. This refinement is independent of the global finite-variation hypothesis in Theorem 1.4.

1.5 Proof strategy and organization

Angular concentration and sharp truncation geometry. A small mean-zero cap atom can be stretched to a uniformly bounded angular function in a fixed cone. The transformation changes

Euclidean sharp truncations into elliptic ones. The estimate in Theorem 2.2 is uniform as the transverse singular values tend to zero, provided the angular support stays in the fixed nondegenerate cone. The exact cutoff is retained throughout. This permits atomic summation before any conjugate-exponent extrapolation over density levels.

Collective phase selection. A normalized jump witness has equal-amplitude interval coefficients. Single-shell Fourier decay therefore controls the high-frequency part after a phase-local smoothing. Near phase blocks are estimated by this decay; far blocks are estimated by the rough equal-shell correlation. The resulting collective remainder is summed before absolute values are taken over the large tiles. This order avoids a spatial-overlap loss in the antichain estimate. For forests, the separated-tree estimate retains norms on the intersection of the two top cubes, with the kernel regularity constant separated from the fixed tile geometry.

The weak endpoint requires a different organization of the oscillatory analysis. Three features of the fixed-phase argument are particularly important.

Phase degeneracy. Neither a displayed monomial coefficient nor the diagonal restriction of the phase provides a stable normalization. Simultaneous translation mixes individual coefficients, and the alternating bilinear phase $x_1y_2 - x_2y_1$ vanishes on the diagonal. Section 16 constructs a finite global flag on polynomial phases modulo pure input and output phases. The transverse rows are placed in reduced row-echelon form, and the final nonterminal rows consist of convolution phases and alternating bilinear forms. Dilation normalizes the first nonzero flag coordinate, while simultaneous translation preserves that coordinate and creates only later ones. This triangular structure yields coefficient-uniform annular decay and closes the local analysis by a finite backward induction over the flag.

Modulation-sensitive cancellation. The input modulations generated by a nonconvolution phase change the cancellation condition on each bad cube. The phase-adapted Calderón–Zygmund decomposition supplies

$$\int_Q e^{iP(c_Q,y)} b_Q(y)\,dy = 0,$$

and anchors the operator phase at the same parent center. Low phase coordinates are Fourier-expanded with coefficients fixed before the output point is evaluated; large coordinates give scale-covariant oscillatory decay. Proposition 19.2 presents this interaction in a bounded-density model. The concentration-stable packet estimates in Section 20 implement it under the cancellation and average mass bounds of the actual bad functions.

Output-dependent selectors. A jump linearization chooses parameter intervals as functions of the output point. The endpoint argument first imposes finite cube, shell, and parameter cutoffs and assigns every relevant increment a fixed rank. A variational Rademacher–Menshov estimate then controls every interval sum through that ranking, without introducing an output-dependent sign choice. The phase-adapted bad functions enter directly through cancellation and the average density bound $\|h_Q\|_1 \lesssim \alpha|Q|$. On each shell, a descending homogeneous-degree packet induction combines Fourier expansion of low blocks with a concentration-stable high-block estimate. Across high shells, an exact oriented two-ray calculation retains a coefficient of size R^m after all projective cancellations and lower-degree perturbations. A one-sided polynomial incidence estimate then sums the small-cube mesh errors geometrically while the larger atom remains fixed. The resulting fixed-sign all-shell estimate is precisely the input needed by the rank argument, and it requires no L^∞ decomposition of the bad data.

The pointwise martingale, smooth-averaging, and one-dimensional jump inputs are proved in this paper. The martingale proof uses a greedy-exit potential and two good-lambda inequalities on finite binary trees; the square-function and maximal estimates are proved there as well. A separate Brownian argument gives a stopped continuous-path estimate. The smooth square-function comparisons use classical Littlewood–Paley estimates and the Fefferman–Stein vector-valued maximal inequality. The

remaining specialized inputs include the long–short mechanisms of Jones–Seeger–Wright, the Hilbert short-variation theorem of Campbell–Jones–Reinhold–Wierdl, and Seeger's separated-packet estimates. The precise geometric input from Becker–van Doorn–Jamneshan–Srivastava–Thiele is stated in Section 5. The sharpness argument uses a finite Gaussian obstruction and de Leeuw's restriction principle; Jones–Wang's Poisson counterexample [27] provides background only. Path normability is proved directly in Lemma 2.1; no interchange of amplitude supremum and spatial norm is used.

Organization and dependencies. Part I proves Theorem 1.1; Part II develops the independent fixed-phase endpoint, radial-amplitude, and sharpness theory. A section-by-section guide opens each part. The correlation inputs for Part I are Lemma 20.5, Lemma 26.9, and Proposition 26.11. Their proofs in Part II do not use Theorem 1.1. The explicit path norm in Lemma 2.1 is an independent algebraic statement used in both parts.

Part I
Maximal polynomial modulation

Section 2 reduces the phase-supremum problem to a finite-path estimate for conically supported elliptic truncations. Sections 4–5 establish the annular analysis and specify the polynomial tile geometry. Section 6 proves the collective antichain estimate; Sections 7–9 develop the localized tree estimates and sum the forests in the full spatial range. The spherical atomic decomposition, affine normalization, and trajectory completion occupy Sections 10–13. Section 14 collects the maximal, variation, convergence, and odd-kernel consequences, together with the obstruction to a phase-supremum weak endpoint.

2 Maximal modulation and the finite-path reduction

The proof of Theorem 1.1 begins with a finite-path linearization. At each output point, one polynomial is used for every increment in the selected jump family; the polynomial may vary between output points. Both selections are made before the spatial norm is taken.

The principal analytic reduction is a uniform estimate for conically supported elliptic truncations. After this estimate has been established, a cancellative spherical atomic decomposition yields the angular endpoint. The polynomial geometry is taken from [4]; the required rough-kernel operator estimates are proved in the following sections.

2.1 A norm for paired jump paths

For an ordered parameter set D put

$$\mathscr{N}_D(F) = \sup_{N\ge 1} \sup_{\substack{s_1<t_1\le\cdots\le s_N<t_N\\ s_i,t_i\in D}} N^{-1/2} \sum_{i=1}^{N} |F_{t_i} - F_{s_i}|. \tag{2.1}$$

The empty supremum is zero.

Lemma 2.1. *For every scalar path,*

$$\mathbf{J}_D(F) \le \mathscr{N}_D(F) \le 2\mathbf{J}_D(F). \tag{2.2}$$

For finite D, $\mathscr{N}_D$ is a norm modulo constant paths. If F^z is coordinatewise integrable against the total variation of a complex measure μ, then

$$\mathscr{N}_D\left(\int F^z\, d\mu(z)\right) \le \int \mathscr{N}_D(F^z)\, d|\mu|(z). \tag{2.3}$$

Proof. A strict λ-jump witness with N pairs contributes more than $\lambda\sqrt{N}$ to (2.1). Conversely, fix a paired family and arrange the moduli of its increments in decreasing order c_k^*. The largest k increments, restored to temporal order, remain admissible. Thus $c_k^*\sqrt{k} \le \mathbf{J}_D(F)$. Consequently $N^{-1/2}\sum_{k\le N} c_k^* \le 2\mathbf{J}_D(F)$. This proves (2.2). Subadditivity follows by applying the ordinary triangle inequality to each paired family. A one-pair test detects every coordinate difference. Finally apply the integral triangle inequality to each finite paired family and take the supremum. □

The explicit norm $\mathscr{N}_D$ supplies the finite-path normability used in Lemma 15.3. The proof above concerns the pointwise functional itself. The mixed-norm interpolation formulation in [33] is a separate statement and is not needed for this step.

Let $\mathscr{P}_d$ denote the real polynomials on $\mathbb{R}^n$ of degree at most d, modulo constants. For fixed x, $Q_{P,x}(y) = P(x,y) - P(x,0)$ belongs to this space, and

$$T^{P,b}_{\Omega,t} f(x) = e^{iP(x,0)} T^{Q_{P,x},b}_{\Omega,t} f(x).$$

Conversely, every $Q(y)$ occurs as $P(x,y) = Q(y)$. The two pointwise phase suprema are therefore equal. It suffices to consider input phases Q.

For finite phase and truncation sets, all selections below can be made measurably, with a deterministic rule for ties. Linearizing $\max_Q \mathscr{N}_D(T^{Q,b} f)$ gives a measurable Q_x of finite range and

$$h_x(r) = N(x)^{-1/2} \sum_{\nu=1}^{N(x)} \zeta_\nu(x) \mathbf{1}_{(a_\nu(x), b_\nu(x)]}(r), \quad |\zeta_\nu| \le 1, \tag{2.4}$$

where $a_1 < b_1 \le \cdots \le a_N < b_N$. The zero operator is used if there is no nonempty witness. We have

$$\|h_x\|_\infty \le N(x)^{-1/2}, \qquad V^2(h_x;\mathbb{R}) \le C. \tag{2.5}$$

Indeed, after zero extension the step function has at most $2N+2$ changes, each of modulus at most $2N^{-1/2}$; deletion of repeated coordinates gives the same bound for every variation partition. An intersection with one radial interval retains the form (2.4), up to a common coefficient of modulus at most one. This special form is retained whenever a pathwise estimate is used. An arbitrary bounded-V^2 radial multiplier is not substituted for it.

Fix $1 \le \rho < \infty$ and the radial amplitude b from Theorem 1.1. Set $q_* := \max\{2, \rho\}$. The product variation inequality gives

$$\|bh_x\|_\infty + V^{q_*}(bh_x) \le C\|b\|_{\mathscr{V}^\rho}. \tag{2.6}$$

Indeed $V^{q_*}(b) \le V^\rho(b)$ and $V^{q_*}(h_x) \le V^2(h_x)$, and $V^{q_*}(uv) \le \|u\|_\infty V^{q_*}(v) + \|v\|_\infty V^{q_*}(u)$. Multiplication by an interval indicator adds at most two endpoint jumps, so the same estimate holds on each shell after zero extension. This finite-variation fact is used only in oscillatory correlation estimates. In unphased path estimates, b remains part of the fixed kernel and h_x retains its actual witness form (2.4).

2.2 The conical analytic theorem

Fix $c_0 \in (0,1)$. Let A be a real matrix with $\|A\| \le 1$ and put $\rho_A(z) = |Az|$. Suppose

$$\int \Theta \, d\sigma = 0, \qquad \Theta \in L^\infty(\mathbb{S}^{n-1}), \qquad \operatorname{supp}\Theta \subset \{\theta : |A\theta| \ge c_0\}. \tag{2.7}$$

No lower bound on the least singular value of A is imposed. Choose a measurable representative of Θ that vanishes everywhere on $\{\theta : |A\theta| < c_0\}$. The full weighted kernel is defined to be zero wherever $\rho_A(z) = 0$ or $\Theta(z') = 0$; the value $b(0)$ is never evaluated. In particular, $A = 0$ gives the zero operator. Define

$$S^{Q,A,b}_{\Theta,t} f(x) = \int_{\rho_A(x-y)>t} e^{iQ(y)} K_\Theta(x-y) b(\rho_A(x-y)) f(y)\, dy.$$

Theorem 2.2 (Uniform conical estimate)**.** *For all data in* (2.7)*, every fixed $b \in \mathscr{V}^\rho$ with $1 \le \rho < \infty$, all finite $\mathscr{F} \subset \mathscr{P}_d$ and $D_0 \subset (0,\infty)$, and every bounded compactly supported f,*

$$\left\| \max_{Q\in\mathscr{F}} \mathscr{N}_{D_0}(S^{Q,A,b}_{\Theta,t} f) \right\|_p \le C_{n,d,p,c_0,\rho} \|\Theta\|_\infty \|b\|_{\mathscr{V}^\rho} \|f\|_p, \qquad 1 < p < \infty. \tag{2.8}$$

The constant is independent of A, $\mathscr{F}$, and D_0.

By homogeneity it suffices to assume $\|\Theta\|_\infty \le 1$ and $\|b\|_{\mathscr{V}^\rho} \le 1$. The geometry is Euclidean. The function ρ_A enters both the radial amplitude and the cutoff; the spatial grid and the polynomial metric are Euclidean. All subsequent analytic constants may depend on the fixed finite exponent ρ.

3 A pointwise jump inequality for dyadic martingales

We prove the unweighted martingale estimate with the jump-amplitude supremum *inside* the spatial norm. A bound obtained by taking that supremum outside L^p is not used in this argument.

For a continuous scalar path F on an interval I, let

$$\mathscr{N}_I(F) = \sup_{N\ge 1} \sup_{\substack{s_1<t_1\le\cdots\le s_N<t_N \\ s_i,t_i\in I}} N^{-1/2} \sum_{i=1}^{N} |F(t_i) - F(s_i)|.$$

The same definition is used for a finite ordered sequence. It is unchanged by addition of a constant and is homogeneous. Its relation to the paired strict jump functional is

$$\mathbf{J}_I(F) \le \mathscr{N}_I(F) \le 2\mathbf{J}_I(F). \tag{3.1}$$

Indeed, a strict jump witness gives the first inequality. For the second, arrange a witness's increment magnitudes as c_k^* in decreasing order. Its largest k increments still form an admissible paired family, so $c_k^*\sqrt{k} \le \mathbf{J}_I(F)$. Now use $\sum_{k=1}^N k^{-1/2} \le 2\sqrt{N}$.

Lemma 3.1 (Concatenation)**.** *If $a \le s \le t$, then*

$$\mathscr{N}_{[a,t]}(F) \le \mathscr{N}_{[a,s]}(F) + \mathscr{N}_{[s,t]}(F). \tag{3.2}$$

Proof. Fix a witness with N pairs in $[a,t]$. At most one pair crosses s. Split that increment at s and apply the triangle inequality to it. After zero intervals are removed, each side is an admissible paired family containing at most N pairs. If their cardinalities are N_1, N_2, their contributions divided by $\sqrt{N}$ are at most $\sqrt{N_1/N}\mathscr{N}_{[a,s]}(F)$ and $\sqrt{N_2/N}\mathscr{N}_{[s,t]}(F)$. Each coefficient is at most one. Take the supremum over the original witness. □

3.1 The binary martingale estimate

Theorem 3.2 (Pointwise critical jumps for dyadic martingales)**.** *Let $1 < p < \infty$ and let $(M_k)_{k=0}^N$ be a finite martingale on a finite binary probability tree, each branch having conditional probability $1/2$. Then*

$$\|\mathscr{N}(M_k : 0 \le k \le N)\|_{L^p} \le C_p \|M_N\|_{L^p}. \tag{3.3}$$

Consequently, if E_j denotes expectation on the dyadic cubes of sidelength 2^j in $\mathbb{R}^n$, then

$$\left\| \sup_{\lambda>0} \lambda N_\lambda(E_j f : j \in \mathbb{Z})^{1/2} \right\|_{L^p(\mathbb{R}^n)} \le \|\mathscr{N}(E_j f : j \in \mathbb{Z})\|_{L^p(\mathbb{R}^n)} \le C_p \|f\|_{L^p(\mathbb{R}^n)}. \tag{3.4}$$

The constants are independent of the length of the martingale. For $1 < p \le 2$ they may be chosen at most $Cp/(p-1)$.

Proof. We give a direct finite-tree proof. First take one root and real values. Label the nodes at depth k by $0 \le r < 2^k$, with children $2r, 2r+1$, and give each terminal leaf mass 2^{-N}. The martingale identity says that the two increments from any parent are d and $-d$, with no restriction that $d \neq 0$. Expectations are taken with respect to this finite probability law. Along a branch put

$$Q_k = \sum_{i=1}^{k} (M_i - M_{i-1})^2, \quad S = \sqrt{Q_N}, \quad X = \max_{0 \le k \le N} |M_k|, \quad A_k = \mathscr{N}(M_i : 0 \le i \le k), \quad A = A_N.$$

Write $J(F) = \sup_{\lambda > 0} \lambda N_\lambda(F)^{1/2}$. We use the deterministic comparison $J(F) \le \mathscr{N}(F) \le 2J(F)$ from (3.1). To recall why the upper bound has an absolute constant, arrange the lengths of any m paired increments in decreasing order $a_1 \ge \cdots \ge a_m$. Their subfamilies show $a_i\sqrt{i} \le J(F)$, by taking thresholds increasing to a_i. Hence $\sum_{i=1}^m a_i \le J(F)\sum_{i=1}^m i^{-1/2} \le 2J(F)\sqrt{m}$. The lower bound follows by using a jump witness in the definition of $\mathscr{N}$. Splitting every paired increment at a common time also gives

$$A_k \le A_{k-1} + |M_k - M_{k-1}|, \qquad A_N \le A_k + \mathscr{N}(M_i : k \le i \le N).$$

Indeed, each of the two resulting families has at most the original number of pairs; divide their sums by that original square root. Empty or zero-length pieces contribute zero. In particular A_k is nondecreasing, starts at zero, and all quantities above are finite.

A tail estimate with a quadratic budget. Fix $a > 0$. Starting with base M_0, accept a new greedy exit whenever the distance from the current base first exceeds a, and reset the base to that new value. Let G_k be the number of completed exits and b_k the current base at time k. For $s \ge 0$ consider the nonnegative potential

$$Z_k = \cosh(sa)^{G_k} \cosh\big(s(M_k - b_k)\big) \exp(-s^2 Q_k/2).$$

It starts at 1. Before resetting a child, its unfinished factor is $\cosh(s(M_{k+1} - b_k))$. If an exit is accepted, replacing this factor by $\cosh(sa)$ can only decrease it. At a parent whose increment size is d, the average of the two unreset factors is

$$\frac{\cosh(s(M_k - b_k + d)) + \cosh(s(M_k - b_k - d))}{2} = \cosh\big(s(M_k - b_k)\big)\cosh(sd).$$

Both children add d^2 to Q_k, and $\cosh(sd) \le \exp(s^2 d^2/2)$. The latter inequality follows, for example, by integrating $\tanh t \le t$ for $t \ge 0$ and using evenness. Thus the average of the child potentials is at most the parent potential, and induction on the finite levels gives $\mathbb{E}Z_N \le 1$. Since $\cosh t \ge e^t/2$, for $m \ge 1$ and $v > 0$ this implies

$$\mathbb{P}(G_N \ge m,\ Q_N \le v) \le 2^m \exp(s^2 v/2 - msa).$$

Taking $s = ma/v$ proves

$$\mathbb{P}(G_N \ge m,\ Q_N \le v) \le 2^m \exp\left(-\frac{m^2 a^2}{2v}\right).$$

This controls the amplitude supremum *inside* the probability. If a pair has length greater than $2a$, there must be a greedy exit in its half-open time interval: otherwise both endpoints are within a of the same base. Ordered disjoint pairs therefore require distinct exits. If $J(M) > u > 0$, choose a witness with m pairs of length greater than λ, where $\lambda\sqrt{m} > u$, and choose $j \ge 0$ so that $4^j \le m < 4^{j+1}$. Retaining 4^j pairs gives lengths greater than $u/(2 \cdot 2^j)$. The greedy count at threshold $a_j = u/(4 \cdot 2^j)$ is consequently at least 4^j. A countable union bound, followed by the preceding estimate, yields

$$\mathbb{P}(J(M) > u,\ Q_N \le v) \le \sum_{j=0}^{\infty} 2^{4^j} \exp\left(-\frac{4^j u^2}{32v}\right).$$

If $64v\log 2 \le u^2$, put $h = u^2/(32v) \ge 2\log 2$. Since $4^j \ge j+1$, each summand is at most $\exp(-(j+1)h/2)$, and $e^{-h/2} \le 1/2$. Summing this geometric majorant gives the depth-independent bound

$$\mathbb{P}(J(M) > u,\ Q_N \le v) \le 2\exp\left(-\frac{u^2}{64v}\right).$$

The argument applies verbatim on the descendants of every node, with their normalized fair probability law.

The path-norm good-lambda inequality. Take $0 < \delta \le 1$ with $64\delta^2 \log 2 \le 1$, and set $\varepsilon(\delta) = 2\exp(-1/(64\delta^2))$. On $\{A > 4u,\ S \le \delta u\}$, let σ be the first level where $A_\sigma > u$. Each increment has size at most $S \le u$, so the preceding one-step estimate gives $A_\sigma \le 2u$. The concatenation estimate then forces the suffix path norm to exceed $2u$, hence its J to exceed u. Its quadratic variation is at most $(\delta u)^2$. The first-crossing nodes have disjoint descendant cylinders, whose union is exactly $\{A > u\}$. On each cylinder the preceding suffix estimate has conditional probability at most $\varepsilon(\delta)$. Multiplying by the cylinder masses and summing gives

$$\mathbb{P}(A > 4u,\ S \le \delta u) \le \varepsilon(\delta)\mathbb{P}(A > u), \qquad u > 0.$$

Consequently, for any $p > 0$, layer cake on this finite probability space gives

$$\big(4^{-p} - \varepsilon(\delta)\big)\mathbb{E}A^p \le \delta^{-p}\mathbb{E}S^p.$$

Indeed, first bound $\mathbb{P}(A > 4u)$ by the right-hand side of the good-lambda inequality plus $\mathbb{P}(S > \delta u)$, and integrate against $pu^{p-1}\,du$. All moments are finite here, so absorption is legitimate. The choice

$$\delta_p = \big(64(2p+3)\log 2\big)^{-1/2}$$

satisfies the preceding restrictions and $\varepsilon(\delta_p)4^p = 1/4$. Thus

$$\|A\|_p \le D_p\|S\|_p, \qquad D_p = \delta_p^{-1}\big(4^{-p} - \varepsilon(\delta_p)\big)^{-1/p} < \infty,$$

with no dependence on the depth or the martingale values.

Square function, maximal function, and terminal value. We next prove the needed square-function estimate, rather than assuming it. For a real binary martingale Y and a number $R \ge 0$, freeze both descendants of a node before a step unless the current value has absolute value at most R and both child values have absolute value at most $3R$; once frozen, stay frozen. This is a predictable rule on the tree, so the resulting process $\widetilde{Y}$ is still a martingale. If $|Y_0| \le R$, it is bounded by $3R$. On any branch on which $\max|Y_k| \le R$, no step is frozen: the unused sibling is twice the parent minus the chosen child, so its absolute value is at most $3R$. Therefore $Q_N(\widetilde{Y}) = Q_N(Y)$ on those branches. Averaging the identity

$$\frac{(y+d)^2 + (y-d)^2}{2} = y^2 + d^2$$

over the successive levels gives $\mathbb{E}Q_N(\widetilde{Y}) = \mathbb{E}\widetilde{Y}_N^2 - Y_0^2 \le 9R^2$. For $v > 0$ we obtain

$$\mathbb{P}\big(Q_N(Y) \ge v,\ \max_k |Y_k| \le R\big) \le \frac{9R^2}{v}.$$

If $|Y_0| > R$, the event is empty, so the same bound remains true.

Return to M, and take $u > 0$, $0 \le R \le u/2$. On $\{Q_N > 4u^2,\ X \le R\}$ stop at the first node with $Q_k > u^2$. The last increment has absolute value at most $2R$, so the quadratic variation at that node is at most $u^2 + 4R^2 \le 2u^2$. The suffix therefore has quadratic variation at least $2u^2$ and all its values have absolute value at most R. Apply the preceding bound, with $v = 2u^2$, on each first-crossing cylinder. Their disjoint union is $\{Q_N > u^2\}$, and hence

$$\mathbb{P}(S > 2u,\ X \le \eta u) \le \tfrac{9}{2}\eta^2\mathbb{P}(S > u), \qquad 0 < \eta \le \tfrac{1}{2}.$$

Another layer-cake integration shows

$$\|S\|_p \le B_p\|X\|_p, \qquad B_p = \eta^{-1}\big(2^{-p} - \tfrac{9}{2}\eta^2\big)^{-1/p},$$

whenever the quantity in parentheses is positive. For every $p > 0$ one may take $\eta = 1/(4(2^p+1))$, since $(9/2)\eta^2 2^p = 9\cdot 2^p/(32(2^p+1)^2) < 1$.

For completeness, the finite-tree maximal estimate has exactly the factor $p/(p-1)$ when $p > 1$. On every first-crossing cylinder of $\{|M_k| > t\}$, the node value is the average of its terminal values, so its absolute value is at most the average of $|M_N|$. Multiplying by the cylinder masses and summing gives

$$t\,\mathbb{P}(X > t) \le \mathbb{E}\big(|M_N|\mathbf{1}_{\{X>t\}}\big).$$

Integrating against $pt^{p-2}\,dt$ and then applying Hölder gives

$$\mathbb{E}X^p \le \frac{p}{p-1}\mathbb{E}(|M_N|X^{p-1}) \le \frac{p}{p-1}\|M_N\|_p\|X\|_p^{p-1}.$$

Division when $\|X\|_p > 0$, with the zero case immediate, proves $\|X\|_p \le p(p-1)^{-1}\|M_N\|_p$. Combining the three estimates proves the real assertion with $K_p = D_p B_p\, p/(p-1)$ for every $1 < p < \infty$.

For $1 < p \le 2$, fix $\delta = 1/32$ and $\eta = 1/16$ in the preceding good-lambda estimates. Then $4^{-p} - \varepsilon(\delta) \ge 1/32$ and $2^{-p} - (9/2)\eta^2 \ge 1/8$, so D_p and B_p are uniformly bounded in this range. Consequently $K_p \le Cp/(p-1)$.

Complex values and arbitrary initial roots. The real and imaginary parts satisfy the same midpoint identities, and

$$\mathscr{N}(M) \le \mathscr{N}(\mathrm{Re}\,M) + \mathscr{N}(\mathrm{Im}\,M).$$

Minkowski's inequality and $|\mathrm{Re}\,M_N|, |\mathrm{Im}\,M_N| \le |M_N|$ therefore give $\|J(M)\|_p \le 2K_p\|M_N\|_p$ and $\|\mathscr{N}(M)\|_p \le 4K_p\|M_N\|_p$.

If the initial sigma field has finitely many atoms, index them by r and write their masses as $w_r \ge 0$. A leaf below root r has mass $w_r 2^{-N}$, since each of the N conditional branch probabilities is $1/2$. Labeling such leaves identifies all integrals of ancestral-path functionals with $\sum_r w_r\mathbb{E}_r$, where $\mathbb{E}_r$ is the normalized binary average on that root. Zero-mass roots contribute zero. If H_r is either of the two path functionals and the one-root constant is C_p, raise that estimate to the pth power before summing:

$$\sum_r w_r\mathbb{E}_r H_r^p \le C_p^p \sum_r w_r\mathbb{E}_r|M_N|^p.$$

Taking the pth root proves exactly the same estimate on the whole forest, with no loss depending on the number or masses of the initial atoms. This argument needs neither equal root masses nor $\sum_r w_r = 1$; in the probability case the latter sum is 1. The same identity holds for nonnegative extended weights and nonnegative integrals, with the usual convention $0\cdot\infty = 0$. This proves (3.3), including nontrivial finite initial sigma fields and zero martingale increments.

Euclidean dyadic averages at all integer scales. Fix integers $a \le b$ and a half-open dyadic root cube Q of side 2^b. Normalize Lebesgue measure on Q by $|Q|^{-1}$. Successively bisect in the n coordinate directions, and repeat this for $b-a$ generations. This is a fair binary tree of depth $n(b-a)$: its leaves are exactly the subcubes $q \subset Q$ of side 2^a, each of normalized mass $2^{-n(b-a)}$. Give a terminal leaf the value

$$f_q = \frac{1}{|q|}\int_q f.$$

For $f \in L^p(\mathbb{R}^n)$ and $p > 1$ these integrals exist by Hölder on finite-volume cubes; global L^1 is not required. At any intermediate node use the average of its terminal leaf values. Splitting its equal-mass descendants into two equal halves proves the midpoint identity. At binary levels nt, additivity of the integral and equality of the leaf volumes identify that node value *exactly* with the average over the containing cube of side 2^{b-t}. Thus the sampled branch values at those levels are exactly $E_{b-t}f(x)$ for $x \in Q$.

Restricting a path to these levels cannot increase J. The complex binary estimate and Jensen on each terminal cube give

$$\frac{1}{|Q|}\int_Q J(E_{b-t}f : 0 \le t \le b-a)^p \le (2K_p)^p \sum_{q\subset Q} \frac{|q|}{|Q|}|f_q|^p \le \frac{(2K_p)^p}{|Q|}\int_Q |f|^p,$$

where the sum is over terminal cubes. The Jensen step follows from $|f_q| \le |q|^{-1}\int_q |f|$ and Hölder on q. Multiplying by $|Q|$ and summing over the countable disjoint cubes at scale b yields

$$\|J(E_j f : a \le j \le b)\|_{L^p(\mathbb{R}^n)} \le 2K_p \|f\|_{L^p(\mathbb{R}^n)}.$$

Here reversal of the finite order preserves every paired functional: reverse the list of pairs and interchange their endpoints. Neither the binary insertion, the normalization, Jensen, nor root-cube summation has introduced a dimension or length factor. If $n = 0$, all scale averages coincide and the conclusion is immediate.

Finally take $[a,b] = [-L,L]$ and let L increase. Every finite paired witness on $\mathbb{Z}$ is contained in one of these windows, so both their J's and their $\mathscr{N}$'s increase to the corresponding full-integer quantities. The cube averages are measurable step functions; the path norm is a supremum over the countably many finite integer pair families. For J, strict jump inequalities permit restriction of λ to positive rational numbers, so it too is measurable. Thus monotone convergence gives the full-integer J estimate with constant $2K_p$, with the amplitude supremum still inside L^p. Applying the pointwise comparison $J \le \mathscr{N} \le 2J$ gives (3.4) with $C_p = 4K_p$. All constants were chosen before the tree, its depth, the dimension, the root scales, and the input; the displayed numerical bounds give exactly the asserted range $1 < p \le 2$. □

3.2 A continuous-path counterpart

The finite-tree estimate has a continuous-path counterpart for Brownian motion stopped at an arbitrary stopping time. Let B be a real standard Brownian motion starting at zero, with continuous paths and its usual augmented filtration. We use the strong Markov property and Brownian scaling; see [11]. For deterministic $T > 0$, write $A_T = \mathscr{N}_{[0,T]}(B)$.

Lemma 3.3 (A deterministic-time Brownian tail)**.** *There are absolute constants $C, c, u_0 > 0$ such that*

$$\mathbb{P}(A_T > u) \le C\exp(-cu^2/T) \qquad (u \ge u_0\sqrt{T}). \tag{3.5}$$

In particular, A_T has moments of every finite positive order.

Proof. We first work on a finite deterministic time set $\mathscr{T} = \{0 = t_0 \le t_1 \le \cdots \le t_N \le T\}$; repeated times are harmless. For $a > 0$, start with greedy base $b_0 = B_{t_0}$ and count $G_0 = 0$. At step i, replace the base by B_{t_i} and increase the count by one precisely when $|B_{t_i} - b_{i-1}| > a$; otherwise keep both values. Write $G_a(\mathscr{T}) = G_N$. Every strict paired $2a$-jump on this time set contains a distinct greedy reset, because without a reset both endpoint values are within a of the same base. Thus

$$N_{2a,\mathscr{T}}(B) \le G_a(\mathscr{T}). \tag{3.6}$$

For $s \ge 0$, put $W_i = \cosh(sa)^{G_i}\cosh(s(B_{t_i} - b_i))$. The reset rule and monotonicity of cosh on $[0,\infty)$ give

$$W_i \le \cosh(sa)^{G_{i-1}}\cosh(s(B_{t_i} - b_{i-1})).$$

If Z is a centered Gaussian of variance v, its exponential moment formula gives $\mathbb{E}\cosh(s(u+Z)) = \cosh(su)e^{vs^2/2}$. The increment $B_{t_i} - B_{t_{i-1}}$ is independent of the entire finite past, including its measurable greedy state. Induction therefore gives $\mathbb{E}W_N \le e^{s^2T/2}$, with all quantities integrable at each finite step.

On $\{G_a(\mathscr{T}) \geq m\}$ one has $W_N \geq \cosh(sa)^m$. Markov's inequality and $\cosh(sa) \geq e^{sa}/2$, followed by $s = ma/T$, yield

$$\mathbb{P}(G_a(\mathscr{T}) \geq m) \leq 2^m \exp\left(-\frac{m^2a^2}{2T}\right). \tag{3.7}$$

This argument uses only deterministic-time independent Gaussian increments, not a stopping-time or strong-Markov assertion.

If $\mathbf{J}_{\mathscr{T}}(B) > u > 0$, choose a strict λ-jump witness with M pairs and $\lambda\sqrt{M} > u$. For $j = \lfloor \log_4 M \rfloor$, there are at least 4^j pairs and $\lambda > u/(2\cdot 2^j)$. With $a_j = u/(4\cdot 2^j)$, (3.6) gives $G_{a_j}(\mathscr{T}) \geq 4^j$. Consequently,

$$\mathbb{P}(\mathbf{J}_{\mathscr{T}}(B) > u) \leq \sum_{j\geq 0} 2^{4^j} \exp\left(-\frac{4^j u^2}{32T}\right) \leq 2\exp\left(-\frac{u^2}{64T}\right) \quad \text{if } u^2 \geq 64T\log 2.$$

For the last inequality put $v = u^2/(32T) \geq 2\log 2$. Each summand is at most $e^{-4^j v/2}$, hence at most $e^{-v/2}2^{-j}$ because $4^j \geq j+1$. The bound is uniform in the finite time set. Exhaust a countable dense subset of $[0,T]$ by increasing finite sets and use continuity from below for the superlevel events. Continuity of the Brownian paths and the finite strict-witness inequalities identify the resulting jump functional with the one on the whole interval. This also supplies its measurability. By (3.1), $A_T \leq 2\mathbf{J}_{[0,T]}(B)$, so

$$\mathbb{P}(A_T > u) \leq 2e^{-u^2/(256T)} \qquad (u \geq 16\sqrt{T}).$$

The finite-pair norm itself is measurable by the same dense-time approximation of finite families. Its tail implies almost-sure finiteness and, for every $u > 0$, the convenient bound $\mathbb{P}(A_T > u) \leq 2e\,e^{-u^2/(256T)}$. Layer cake and $u = \sqrt{T}\,v$ now give, for every $p > 0$,

$$\mathbb{E}A_T^p \leq \left(2ep\int_0^\infty v^{p-1}e^{-v^2/256}\,dv\right) T^{p/2} < \infty.$$

The integral is finite at zero since $p > 0$ and at infinity by Gaussian decay. This proves both claims with absolute tail constants. □

Lemma 3.4 (Continuity of the prefix norm)**.** *Outside one null set, the process $A_t = \mathscr{N}_{[0,t]}(B)$, $t \geq 0$, is finite, increasing, and continuous. It is adapted.*

Proof. Continuity of B permits rational endpoints in every finite witness on a deterministic compact interval, with its right endpoint included when necessary. This proves measurability and adaptedness of A_t. It is increasing by inclusion of parameter intervals.

Fix an integer $M \geq 1$, let $h_k = 2^{-k}$, and cover all intervals $[s,t] \subset [0,M]$ of length at most h_k by the deterministic intervals $[jh_k, (j+2)h_k]$, $0 \leq j \leq \lceil M/h_k \rceil$. Brownian stationary increments and Lemma 3.3 give, for sufficiently large fixed L,

$$\sum_{k\geq 1} \mathbb{P}\left(\max_{0\leq j\leq \lceil M/h_k\rceil} \mathscr{N}_{[jh_k,(j+2)h_k]}(B) > L\sqrt{kh_k}\right) < \infty.$$

Indeed the number of intervals is $O_M(2^k)$, whereas each probability is at most $C\exp(-cL^2k/2)$. The Borel–Cantelli lemma implies that the norm of every subinterval of $[0,M]$ of length at most h_k is at most $L\sqrt{kh_k}$ for all sufficiently large k. By (3.2),

$$0 \leq A_t - A_s \leq \mathscr{N}_{[s,t]}(B) \qquad (0 \leq s \leq t \leq M).$$

It follows that A is uniformly continuous on $[0,M]$. Its finiteness there follows from finiteness of A_M. Intersect over integer M. Modify A to be zero on the resulting null exceptional set. The usual augmented filtration contains that null set at time zero, so the modification is still adapted and has continuous paths. □

Lemma 3.5 (Brownian jump control up to a stopping time)**.** *For every $p > 0$ there is $C_p < \infty$ such that, for every stopping time τ,*

$$\mathbb{E}\big[\mathscr{N}_{[0,\tau]}(B)^p\big] \le C_p \mathbb{E}\left[\sup_{0\le t\le\tau} |B_t|^p\right]. \tag{3.8}$$

Both sides are allowed to be infinite. On $\{\tau = \infty\}$ the left side is the supremum over all finite-time witnesses.

Proof. We use the continuous adapted version of A_t from Lemma 3.4; it agrees with the original finite-witness norm simultaneously at all finite times outside one null set. Write $M_\tau = \sup_{0\le t\le\tau}|B_t|$. For a continuous path W, define the integer exit envelope

$$E_n(W) = \{|W_j| \le 1 \text{ for every integer } 0 \le j \le n\}, \qquad H(W) = \sup_{n\ge 0} \mathbf{1}_{E_n(W)} \mathscr{N}_{[0,n+1]}(W).$$

If $|W_t| \le 1$ for $0 \le t \le T < \infty$, then $\mathscr{N}_{[0,T]}(W) \le H(W)$: take $n = \lfloor T \rfloor$. The envelope is measurable by the countable dense-time representation of each finite-interval norm.

Let Z be a standard normal variable and set $q_0 = \mathbb{P}(|Z| \le 2) < 1$. On $E_n(B)$ all n successive unit increments have modulus at most two. Independence of the Brownian increments therefore gives $\mathbb{P}(E_n(B)) \le q_0^n$. If $E_n(B)$ fails, every nonzero term in $H(B)$ has index less than n, so $H(B) \le A_{n+1}$. The explicit bound used to prove (3.5) consequently gives

$$\mathbb{P}(H(B) > u) \le q_0^n + 2\exp\left(-\frac{u^2}{256(n+1)}\right) \qquad \big(u \ge 16\sqrt{n+1}\big).$$

In particular, with the numerical choices

$$R_n = 32(n+1), \qquad b_n = q_0^n + 2e^{-4(n+1)}, \qquad \delta_n = \frac{1}{64(n+1)},$$

we have $\mathbb{P}(H(B) > R_n) \le b_n$ and $b_n \to 0$. Thus $H(B)$ is finite almost surely. At these scales write $q(\delta_n) = b_n$ for the small-probability factor. Fix $p > 0$, and choose the least $n = n(p)$ with $b_n < 2^{-p}$, independently of the process and stopping time. Put $R = R_n$, $b = b_n$, and $\delta = \delta_n$.

First suppose $\tau \le m$ for some deterministic $m \ge 1$. For $\lambda > 0$ let $\sigma = \inf\{t \ge 0 : A_t \ge \lambda\}$ and $\widehat{\sigma} = \sigma \wedge m$. These are stopping times, and $E = \{\sigma \le \tau\} = \{A_\tau \ge \lambda\}$ belongs to the sigma field at $\widehat{\sigma}$, since $\tau \le m$. Set $a = 2\delta\lambda = \lambda/R$. The strong Markov property at the finite time $\widehat{\sigma}$ and Brownian scaling show that

$$W_t = a^{-1}\big(B_{\widehat{\sigma}+a^2 t} - B_{\widehat{\sigma}}\big)$$

is a standard Brownian motion independent of that sigma field. On $\{A_\tau > 2\lambda,\ M_\tau \le \delta\lambda\}$ we have $\widehat{\sigma} = \sigma \le \tau$ and $A_\sigma = \lambda$, with no overshoot. Concatenation and the maximal bound give

$$\mathscr{N}_{[\sigma,\tau]}(B) > \lambda, \qquad |B_t - B_\sigma| \le 2\delta\lambda = a \quad (\sigma \le t \le \tau).$$

With $T = (\tau - \sigma)/a^2$, positive linear time change and homogeneity therefore imply

$$H(W) \ge \mathscr{N}_{[0,T]}(W) = a^{-1}\mathscr{N}_{[\sigma,\tau]}(B) > R.$$

The event under consideration is thus contained, up to a null set, in $E \cap \{H(W) > R\}$. The envelope tail and independence yield

$$\mathbb{P}(A_\tau > 2\lambda,\ M_\tau \le \delta\lambda) \le b\,\mathbb{P}(A_\tau \ge \lambda),$$

and hence

$$\mathbb{P}(A_\tau > 2\lambda) \le \mathbb{P}(M_\tau > \delta\lambda) + b\,\mathbb{P}(A_\tau \ge \lambda). \tag{3.9}$$

Since $A_\tau \le A_m$, its pth moment is finite by Lemma 3.3. Multiply (3.9) by $p\lambda^{p-1}$ and integrate. The strict and non-strict layer-cake formulas give the same moment, including when the distribution has atoms, so

$$2^{-p}\mathbb{E}A_\tau^p \le \delta^{-p}\mathbb{E}M_\tau^p + b\,\mathbb{E}A_\tau^p.$$

Absorbing the finite last term proves the bounded-stopping estimate with the positive finite, process-independent constant

$$C_p = \frac{\delta_{n(p)}^{-p}}{2^{-p} - b_{n(p)}}.$$

No finiteness assumption on $\mathbb{E}M_\tau^p$ is needed: if it is infinite the inequality is automatic.

For an arbitrary extended stopping time, set $\tau_m = \tau \wedge m$. Every finite witness in $[0,\tau]$ has all its endpoints in $[0,\tau_m]$ for some integer m, even on $\{\tau = \infty\}$. Consequently the original quantities satisfy exactly

$$\mathscr{N}_{[0,\tau]}(B) = \sup_m \mathscr{N}_{[0,\tau_m]}(B), \qquad M_\tau = \sup_m M_{\tau_m},$$

with both sequences increasing. The stopped quantities at each finite horizon are measurable on the completed space, by adapted continuous paths and countable dense-time approximation. They, and then their countable suprema above, are measurable up to null sets for the original measure; their nonnegative expectations are unchanged by completion. Apply the bounded estimate to τ_m and use monotone convergence. This gives (3.8) for every $p > 0$, with both expectations permitted to be infinite and with the original finite-witness interpretation at infinity. □

4 Uniform annular estimates and rough correlations

Throughout this section the hypotheses of (2.7) hold and $\|\Theta\|_\infty \le 1$, $\|b\|_{\mathscr{V}^\rho} \le 1$. All compact radial support intervals and all Euclidean enlargements are fixed. Their constants may depend on c_0.

Fix $1 \le \rho < \infty$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$ have integral zero, let $\|A\| \le 1$, and suppose

$$\operatorname{supp}\Theta \subset \{\theta : |A\theta| \ge c_0\}, \qquad 0 < c_0 < 1.$$

Write $g_A(\theta) = |A\theta|$. Every integrand containing $g_A(\theta)^{-1}$ is restricted to $\{g_A \ge c_0\}$ and is defined to be zero elsewhere. Let b be a fixed bounded measurable function on $(0,\infty)$ of finite ρ-variation. The radial function is fixed throughout every phase supremum. By homogeneity, the proofs below assume

$$\|\Theta\|_\infty \le 1, \qquad \|b\|_\infty + V^\rho(b;(0,\infty)) \le 1.$$

The constants may depend on n, c_0, ρ, and subsequently on the polynomial degree and the spatial exponent, but not on A, the radial location of the variation, any phase coefficient, or finite cutoffs. A fixed right-continuous representative of b may be used. All the integrals are Lebesgue integrals; no Stieltjes measure associated with b is introduced.

4.1 A one-dimensional oscillatory estimate

Lemma 4.1. *Let $I \subset I_0$, where I_0 is a fixed bounded interval. If a is bounded and $V^\rho(a;I) < \infty$, then*

$$\left| \int_I e^{i\lambda s} a(s)\, ds \right| \le C_{I_0,\rho}(\|a\|_\infty + V^\rho(a;I)) \min\{1, |\lambda|^{-1/\rho}\}.$$

The same conclusion holds with $e^{i\lambda s}$ replaced by $e^{i\phi(s)}$ when ϕ' is monotone and $|\phi'| \ge L \ge 1$, with $L^{-1/\rho}$ on the right.

Proof. Only $L \ge 1$ needs proof. Partition I into consecutive half-open intervals I_m, with the final endpoint included in the last interval, of length at most L^{-1} and at least $(2L)^{-1}$ except for at most one final interval. There are at most $C_{I_0}L$ such intervals. Choose one sample s_m from each interval, in increasing order. Since the intervals have disjoint interiors, concatenation of finite partitions and passage to the supremum give

$$\sum_m V^\rho(a;I_m)^\rho \le V^\rho(a;I)^\rho.$$

Endpoint choices can equivalently be treated by taking interior limits; the integral does not depend on their values. Consequently the error in replacing a by $a(s_m)$ is at most

$$\sum_m |I_m| \operatorname{osc}_{I_m} a \le L^{-1}(C_{I_0}L)^{1-1/\rho} V^\rho(a;I) \le CL^{-1/\rho} V^\rho(a;I).$$

Let $w_m = \int_{I_m} e^{i\phi(s)}\,ds$. The first derivative van der Corput estimate gives $|\sum_{m=u}^{v} w_m| \le CL^{-1}$ for every consecutive block. This estimate follows directly by integrating $e^{i\phi} = (i\phi')^{-1}(e^{i\phi})'$ and using monotonicity of $1/\phi'$; thus its constant is independent of the upper size of ϕ'. Abel summation now gives

$$\left|\sum_m a(s_m) w_m\right| \le CL^{-1}\left(\|a\|_\infty + \sum_m |a(s_{m+1}) - a(s_m)|\right) \le CL^{-1/\rho}(\|a\|_\infty + V^\rho(a;I)).$$

Combining the two estimates proves the assertion. □

4.2 Weighted partial shells

For $j \in \mathbb{Z}$ and $1 \le u \le 2$, set

$$\nu_{j,u}^{A,b}(z) = K_\Theta(z) b(|Az|) \mathbf{1}_{\{2^j u < |Az| \le 2^{j+1}\}}.$$

Write $\nu_j = \nu_{j,1}^{A,b}$. Polar coordinates and the substitution $v = r g_A(\theta)$ give the exact Fourier identity

$$\widehat{\nu}_{j,u}^{A,b}(\xi) = \int_{\mathbb{S}^{n-1}} \Theta(\theta) \int_u^2 b(2^j v) e^{-i2^j v \theta\cdot\xi/g_A(\theta)} \frac{dv}{v}\, d\sigma(\theta). \tag{4.1}$$

In particular, every partial shell has zero mass. The cancellation is valid for arbitrary b because the inner integral at $\xi = 0$ is independent of θ.

Lemma 4.2. *Put $s = 2^j|\xi|$ and $\alpha = 1/(2\rho)$. Then*

$$\sup_{1\le u\le 2} |\widehat{\nu}_{j,u}^{A,b}(\xi)| \le C_\rho \min\{s, s^{-\alpha}\}, \tag{4.2}$$

$$\operatorname*{ess\,sup}_{1\le u\le 2} |\partial_u \widehat{\nu}_{j,u}^{A,b}(\xi)| \le C \min\{s, 1\}. \tag{4.3}$$

The derivative is the derivative of an absolutely continuous function of u. The assertions are uniform in j,A, and the location of the variation of b.

Proof. The low-frequency estimate follows by subtracting the zero-frequency value in (4.1), using $g_A \ge c_0$ on the angular support and $|e^{it} - 1| \le |t|$. The product rule for variation and monotone composition show that

$$\|v \mapsto b(2^j v)/v\|_{L^\infty([1,2])} + V^\rho(v \mapsto b(2^j v)/v; [1,2]) \le C_\rho.$$

Lemma 4.1 therefore bounds the inner integral in (4.1) by

$$C_\rho \min\{1, (s|e\cdot\theta|)^{-1/\rho}\}, \qquad e = \xi/|\xi|.$$

For $n \ge 2$ the spherical measure of $\{\theta : |e\cdot\theta| < s^{-1/2}\}$ is at most $C_n s^{-1/2}$ when $s \ge 4$. On its complement the preceding bound is at most $C_\rho s^{-1/(2\rho)}$. This proves the asserted high-frequency estimate. For $n = 1$ there is no small-projection set and the stronger bound $C_\rho s^{-1/\rho}$ applies. The remaining bounded range of s is absorbed in the constant.

Differentiation of the lower endpoint in the Lebesgue integral yields for almost every u

$$\partial_u \widehat{\nu}_{j,u}^{A,b}(\xi) = -\frac{b(2^j u)}{u} \int \Theta(\theta) e^{-i2^j u\theta\cdot\xi/g_A(\theta)}\, d\sigma(\theta).$$

The trivial estimate gives the upper bound by a constant. Subtraction of 1 from the exponential, followed by angular cancellation, gives the upper bound Cs. □

The same conclusions hold for any fixed ratio of shell endpoints and for a fixed lacunary sequence $R_j = D^j$, after splitting each shell into a fixed number of dyadic shells. Constants may depend on D; the positive exponent $1/(2\rho)$ remains available.

4.3 The static unphased conical operator

Lemma 4.3 (Static conical bound)**.** *Under the normalized hypotheses of this section, the unphased static operator $S_\Theta^{0,A,b}$ is bounded on $L^p(\mathbb{R}^n)$ for every $1<p<\infty$, with norm at most $C_{n,p,c_0,\rho}$. On smooth compactly supported inputs it equals the principal value of the conical integral. The constants are independent of the least singular value of A.*

Proof. Choose a smooth radial function χ equal to one on $|\xi|\le 1$ and zero on $|\xi|\ge 2$, and put

$$\psi(\xi)=\chi(\xi)-\chi(2\xi), \qquad \widetilde{\psi}(\xi)=\psi(\xi/2)+\psi(\xi)+\psi(2\xi)=\chi(\xi/2)-\chi(4\xi). \tag{4.4}$$

Then $\operatorname{supp}\psi\subset\{1/2\le|\xi|\le 2\}$, $\operatorname{supp}\widetilde{\psi}\subset\{1/4\le|\xi|\le 4\}$, and $\widetilde{\psi}=1$ on $\operatorname{supp}\psi$. Telescoping gives $\sum_{m\in\mathbb{Z}}\psi(2^{-m}\xi)=1$ for $\xi\neq 0$; this is a dual-cutoff resolution, not a square-partition assertion. Let $\Delta_{m,j}$ and $\widetilde{\Delta}_{m,j}$ have multipliers $\psi(2^{j-m}\xi)$ and $\widetilde{\psi}(2^{j-m}\xi)$, respectively. Thus

$$\widetilde{\Delta}_{m,j}\Delta_{m,j}=\Delta_{m,j}, \qquad f=\sum_m \widetilde{\Delta}_{m,j}\Delta_{m,j}f \quad \text{in } L^p, \quad 1<p<\infty.$$

Indeed, finite sums telescope to differences of smooth low-pass operators; their limits at small and large physical scales are f and zero in L^p, respectively. Inverse Fourier transformation gives Schwartz kernels for both cutoffs, and the multiplier identity gives exact convolution reproduction. With the convention $\widehat{f}_{\mathrm{ang}}(\xi)=\int e^{-ix\cdot\xi}f(x)\,dx$, the precise conversion is $\widehat{f}_{\mathrm{ang}}(\xi)=\widehat{f}_{\mathrm{cyc}}(\xi/(2\pi))$; no normalization factor changes the reproduction identity. For finite shell sums, define

$$S_m f=\sum_j \nu_j * \widetilde{\Delta}_{m,j}\Delta_{m,j}f.$$

Every summand has Fourier support in the annulus $|\xi|\simeq 2^{m-j}$; these annuli have bounded overlap as j varies. Plancherel and Lemma 4.2 imply

$$\|S_m f\|_2\le Ca_m\|f\|_2, \qquad a_m=\begin{cases}2^m, & m\le 0,\\ 2^{-\alpha m}, & m>0.\end{cases} \tag{4.5}$$

The support and size of ν_j give $|\nu_j * g|\le CMg$, where M is the Hardy–Littlewood maximal operator. For every $1<r<\infty$, Littlewood–Paley almost orthogonal summation and the Fefferman–Stein vector maximal inequality give

$$\begin{aligned}\|S_m f\|_r &\le C_r\left\|\left(\sum_j|\nu_j*\widetilde{\Delta}_{m,j}\Delta_{m,j}f|^2\right)^{1/2}\right\|_r\\ &\le C_r\left\|\left(\sum_j|M(\widetilde{\Delta}_{m,j}\Delta_{m,j}f)|^2\right)^{1/2}\right\|_r\le C_r\|f\|_r.\end{aligned}$$

All these estimates are uniform over finite shell sets. Interpolating with (4.5), choosing r on the other side of a fixed p from 2, gives

$$\|S_m\|_{p\to p}\le C_{p,\rho}a_m^{\theta_p}, \qquad \theta_p>0.$$

Thus $\sum_m S_m$ converges in the operator norm on L^p. The individual j sums converge by the Littlewood–Paley square-function estimate and the vector maximal inequality; their tails tend to zero by dominated convergence in the input Littlewood–Paley square function. These facts construct a bounded operator $S_\Theta^{0,A,b}$ with

$$\|S_\Theta^{0,A,b}f\|_p\le C_{n,p,c_0,\rho}\|f\|_p. \tag{4.6}$$

No static theorem for a homogeneous radial kernel has been applied to the nonhomogeneous factor b.

For smooth compactly supported inputs this operator is the usual principal value, in the following precise sense. At the origin, subtract $f(x)$ on each partial shell; its integral against the kernel is zero, and the remaining integrand is bounded in absolute integral by $C\|\nabla f\|_\infty dr$. At infinity there is no issue on compactly supported inputs for each fixed x. Finite symmetric shell sums converge in L^p by the construction above, and the preceding cancellation argument gives their pointwise principal-value limit on smooth compactly supported inputs. A subsequence of the L^p convergence identifies these two limits almost everywhere. For $p > 1$ every positive truncation is absolutely convergent, since the kernel restricted to $|Az| > t$ and to the angular support belongs to $L^{p'}$ with norm at most $Ct^{-n/p}$. The construction above also shows that its dyadic tails converge in L^p and vanish in L^p at infinity: first fix m and use the input square-function tail, then sum in m using $a_m^{\theta_p}$. For each positive dyadic cutoff, absolute convergence of the integral and a subsequence of the L^p shell convergence identify the tail sum with that integral. Taking the intersection over countably many dyadic cutoffs gives their common representative. □

4.4 Comparison of smooth averages with a dyadic martingale

Let E_j denote conditional expectation on the standard dyadic cubes of sidelength 2^j in $\mathbb{R}^n$. The scale index is decreasing in the filtration; reversal of a finite ordered set does not change the paired-increment norm. Let Φ be any Schwartz function with $\int \Phi = 1$, and let $\Phi_t = t^{-n}\Phi(t^{-1}\cdot)$ for $t > 0$.

Lemma 4.4. *For every $1 < p < \infty$,*

$$\left\| \left(\sum_{j\in\mathbb{Z}} |\Phi_{2^j} * f - E_j f|^2 \right)^{1/2} \right\|_p \le C_{n,p,\Phi} \|f\|_p. \tag{4.7}$$

Proof. Set $D_m = E_{m-1} - E_m$. Then $D_m f$ is constant on dyadic cubes of sidelength 2^{m-1}, and its integral over each dyadic cube of sidelength 2^m is zero. Also $f = \sum_m D_m f$ in L^p. Indeed $E_j f \to f$ in L^p as $j \to -\infty$, by approximation with smooth functions and the contraction property of E_j, and $E_j f \to 0$ in L^p as $j \to \infty$, first for bounded compactly supported functions and then by density. The orthogonality identity in L^2 is $\sum_m \|D_m f\|_2^2 = \|f\|_2^2$.

We first establish

$$\|(\Phi_{2^{m+k}} * -E_{m+k}) D_m f\|_2 \le C 2^{-|k|/2} \|D_m f\|_2. \tag{4.8}$$

For $k \ge 0$, $E_{m+k} D_m f = 0$. Write $u = D_m f$, put $s = 2^{m+k}$, and use its zero integral on each dyadic cube Q of sidelength 2^m:

$$\Phi_{2^{m+k}} * u(x) = \sum_Q \int_Q \{\Phi_{2^{m+k}}(x-y) - \Phi_{2^{m+k}}(x-c_Q)\} u(y)\, dy.$$

The mean value theorem and the Schwartz estimates, using $|y - c_Q| \le C 2^m \le Cs$, give

$$|\Phi_{2^{m+k}}(x-y) - \Phi_{2^{m+k}}(x-c_Q)| \le C_N 2^{-k} s^{-n} (1 + |x-y|/s)^{-N}.$$

For $N > n$, convolution with the positive decreasing envelope is bounded by $CMu(x)$. Hence $\|\Phi_{2^{m+k}} * u\|_2 \le C 2^{-k} \|u\|_2$.

For $k \le -1$, one has $E_{m+k} u = u$. Put $L = 2^{m-1}$ and $s = 2^{m+k} \le L$. A function constant on cubes in the grid of sidelength L obeys the translation bound

$$\|u(\cdot - y) - u\|_2 \le C_n \min\{1, (|y|/L)^{1/2}\} \|u\|_2. \tag{4.9}$$

To verify it for $|y| \le L$, the integrand vanishes unless x and $x - y$ belong to distinct grid cubes. Inside each cube, this exceptional set is contained in boundary strips of total volume at most $C_n(|y|/L)|Q|$. Since u is constant on each cube, integration of $2|u(x)|^2 + 2|u(x-y)|^2$ over the exceptional set proves the square

of (4.9). The change of variable $x-y=z$ handles the second term with the same bound. For $|y|>L$, translation invariance of the L^2 norm gives the stated trivial bound. Since $\int\Phi=1$, Minkowski and (4.9) give

$$\begin{aligned}\|\Phi_{2^{m+k}}*u-u\|_2 &\le \int |\Phi_{2^{m+k}}(y)|\,\|u(\cdot-y)-u\|_2\,dy\\ &\le C(s/L)^{1/2}\|u\|_2\int|\Phi(z)||z|^{1/2}\,dz \le C2^{k/2}\|u\|_2.\end{aligned}$$

This proves (4.8) in both ranges.

For fixed k, define the linear vector-valued map

$$\mathscr{G}_k f=\{(\Phi_{2^{m+k}}*-E_{m+k})D_m f\}_{m\in\mathbb{Z}}.$$

By (4.8) and orthogonality,

$$\|\mathscr{G}_k f\|_{L^2(\ell^2)}\le C2^{-|k|/2}\|f\|_2.$$

For every $1<r<\infty$, the bound $|\Phi_{2^j}*g|\le CMg$, the identities $E_{m+k}D_m f\in\{0,D_m f\}$, the Fefferman–Stein vector maximal inequality, and the classical martingale square-function inequality give

$$\begin{aligned}\|\mathscr{G}_k f\|_{L^r(\ell^2)} &\le C\left\|\left(\sum_m|M(D_m f)|^2\right)^{1/2}\right\|_r+\left\|\left(\sum_m|D_m f|^2\right)^{1/2}\right\|_r\\ &\le C_r\left\|\left(\sum_m|D_m f|^2\right)^{1/2}\right\|_r\le C_r\|f\|_r.\end{aligned}$$

All inequalities first hold for finite m-sets with uniform constants. Interpolating the bounds $L^2\to L^2(\ell^2)$ and $L^r\to L^r(\ell^2)$ for this linear map, taking r beyond a fixed p from 2, gives

$$\|\mathscr{G}_k f\|_{L^p(\ell^2)}\le C_p2^{-\varepsilon_p|k|}\|f\|_p,\qquad \varepsilon_p>0.$$

Finally, the exact coordinate identity

$$\Phi_{2^j}*f-E_j f=\sum_{k\in\mathbb{Z}}(\Phi_{2^j}*-E_j)D_{j-k}f$$

holds in L^p. The summands, regarded as an $\ell^2(j)$-valued function, have the norms just estimated after relabeling $j=m+k$. Minkowski and summation over k prove (4.7). Convergence in $L^p(\ell^2)$ also justifies the coordinate identity for the resulting sequence. □

Lemma 4.5 (Smooth pointwise jumps)**.** *For every $1<p<\infty$,*

$$\|\mathscr{N}(\Phi_{2^j}*f:j\in\mathbb{Z})\|_p\le C_{n,p,\Phi}\|f\|_p.$$

The same estimate holds for any fixed geometric subsequence $\Phi_{D^j}*f$ when D is a positive integral power of 2.

Proof. The paired norm triangle inequality and the elementary sequence bound give

$$\mathscr{N}(\Phi_{2^j}*f:j)\le\mathscr{N}(E_j f:j)+2\left(\sum_j|\Phi_{2^j}*f-E_j f|^2\right)^{1/2}.$$

The first term is bounded by Theorem 3.2, using the equivalence of the paired norm with the pointwise jump functional. The second is bounded by Lemma 4.4. Restriction of an ordered path to a subsequence can only decrease its paired norm. □

4.5 The unphased critical-jump estimate

Proposition 4.6 (Unphased conical jumps)**.** *Under the normalized hypotheses of this section, for every $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathscr{N}(S^{0,A,b}_{\Theta,t} f : t > 0)\|_p \le C_{n,p,c_0,\rho} \|f\|_p. \tag{4.10}$$

The pointwise trajectory is the absolutely convergent positive-radius integral and tends to zero at infinity. The estimate is uniform in A and in the location of the variation of b.

Proof. Let $F_j = \sum_{k \ge j} \nu_k * f$ and take a smooth low-pass multiplier $\widehat{\Phi}$ equal to 1 near zero and equal to 0 outside a fixed ball. Put $S = S^{0,A,b}_{\Theta}$. Then

$$\mathscr{E}_j := F_j - \Phi_{2^j} * Sf = (I - \Phi_{2^j}*) \sum_{r \ge 0} \nu_{j+r} * f - \Phi_{2^j} * \sum_{r<0} \nu_{j+r} * f.$$

Use the exact resolution $f = \sum_m \widetilde{\Delta}_{m,j} \Delta_{m,j} f$ from (4.4). For fixed m, r, Plancherel, the shell envelope, and interpolation with the uniform vector maximal bound yield

$$\Big\| \{ \nu_{j+r} * \widetilde{\Delta}_{m,j} \Delta_{m,j} f \}_j \Big\|_{L^p(\ell^2)} \le C_{p,\rho} 2^{-c_{p,\rho}|m+r|} \|f\|_p.$$

Insertion of $I - \Phi_{2^j}*$ or $\Phi_{2^j}*$ preserves the uniform vector bound. Moreover the first factor vanishes on the support of $\Delta_{m,j}$ for all sufficiently negative m, and the second vanishes there for all sufficiently positive m. Consequently both double sums

$$\sum_{r \ge 0} \sum_{m \ge -C} 2^{-c|m+r|}, \qquad \sum_{r<0} \sum_{m \le C} 2^{-c|m+r|}$$

converge. Minkowski therefore gives

$$\|\{\mathscr{E}_j\}_j\|_{L^p(\ell^2)} \le C_{p,\rho} \|f\|_p. \tag{4.11}$$

For every scalar sequence v_j, the paired-increment norm satisfies $\mathscr{N}(v_j) \le 2(\sum_j |v_j|^2)^{1/2}$. Lemma 4.5 gives

$$\|\mathscr{N}(\Phi_{2^j} * g : j \in \mathbb{Z})\|_p \le C_p \|g\|_p.$$

Its proof uses the pointwise martingale theorem (Theorem 3.2) and a square-function comparison proved above. No exchange of a jump-amplitude supremum with an L^p norm is made. Combining this estimate with (4.6) and (4.11) controls the long jumps.

For the short jumps, let $G_{j,u} = \nu^{A,b}_{j,u} * g$. This path is absolutely continuous in u, and $G_{j,2} = 0$. On a fixed parameter interval, the elementary inequalities

$$V^2(G)^2 \le 2 \sup|G| \int |G'|, \qquad \sup|G|^2 \le 2 \int |G||G'|$$

imply

$$V^2(G) \le C \|G\|^{1/4}_{L^2_u} \|G'\|^{3/4}_{L^2_u}. \tag{4.12}$$

Applying this pointwise and then Hölder and Plancherel gives, for an input with Fourier support $|\xi| \simeq 2^{m-j}$,

$$\|V^2(\nu^{A,b}_{j,u} * g : u \in [1,2])\|_2 \le C_\rho b_m \|g\|_2, \qquad b_m = \begin{cases} 2^m, & m \le 0, \\ 2^{-m/(8\rho)}, & m > 0. \end{cases} \tag{4.13}$$

The uniform vector L^r bound follows from the pointwise estimate

$$V^2(\nu^{A,b}_{j,u} * g : u) \le V^1(\nu^{A,b}_{j,u} * g : u) \le \int_{2^j < |Az| \le 2^{j+1}} |K_\Theta(z)| \, |g(x-z)| \, dz \le CMg(x).$$

It is the integral over the full shell that is dominated by the maximal operator; no pointwise domination of a single spherical endpoint integral is needed. Summing (4.13) in j in $L^2(\ell^2)$, interpolating with the vector L^r bound, and summing in m proves

$$\left\| \left(\sum_j V^2(\nu_{j,u}^{A,b} * f : u)^2 \right)^{1/2} \right\|_p \le C_{p,\rho} \|f\|_p.$$

The deterministic long–short jump inequality in Lemma 15.2 now yields the asserted inequality (4.10). Every estimate may first be applied to finite phase-independent paths; the paired norm is the supremum of finite witnesses. Thus passage to all positive radii uses monotone convergence over rational witnesses and local absolute continuity of the integral trajectory. □

4.6 Coefficient estimates used for radial oscillation

Lemma 4.7 (A polynomial sublevel estimate)**.** *On a fixed box in $\mathbb{R}^k$, nonzero polynomials of degree at most D satisfy*

$$|\{x : |p(x)| \le t\|p\|_{\mathrm{coeff}}\}| \le C_{k,D} t^{\alpha_{k,D}}, \qquad 0 < t < 1,$$

for some $\alpha_{k,D} > 0$. The same statement holds for a finite polynomial coefficient vector that is not identically zero, with its Euclidean norm in place of $|p|$.

Proof. For $D = 0$ the assertion is immediate. In one variable with $D \ge 1$, if a subset has measure μ, it contains $D+1$ points with consecutive separation at least $c_D\mu$. Lagrange interpolation then bounds the coefficient norm by $C_D\mu^{-D}$ times the maximum of $|p|$ at these points. This proves the assertion with exponent $1/D$. For several variables, view p as a polynomial in its last variable. One coefficient polynomial in the remaining variables has coefficient norm at least a fixed multiple of $\|p\|_{\mathrm{coeff}}$. Delete where that coefficient is less than t^a times its norm, apply the induction hypothesis there, and apply the one-dimensional result on all remaining sections. Choose a fixed $a \in (0,1)$ to balance the two powers. For a vector polynomial $P(x) = \sum_\beta a_\beta x^\beta$, let $M = M(k,D)$ be the number of allowed monomials and choose a_{β_0} of largest Euclidean norm. Then $\|a_{\beta_0}\| \ge M^{-1/2}\|P\|_{\mathrm{coeff}}$. Projecting onto $u = a_{\beta_0}/\|a_{\beta_0}\|$ gives the scalar polynomial $q = \langle u, P\rangle$, whose β_0 coefficient equals $\|a_{\beta_0}\|$, while $|q(x)| \le \|P(x)\|$. Thus the vector sublevel set is contained in the scalar sublevel set at threshold $t\sqrt{M}$. Apply the scalar estimate when $t\sqrt{M} < 1$ and the trivial box bound otherwise. After increasing the constant, this proves the claim independently of the number of vector components. □

Lemma 4.8 (Polynomial radial oscillation with V^{q_*} profiles)**.** *Let $d \ge 1$ and let S be a real polynomial on a fixed annulus, of degree at most d, with nonconstant coefficient norm Λ. Let v have bounded $L^\infty + V^{q_*}$ norm and fixed compact radial support. If A_0 is Lipschitz on the annulus, then*

$$\left| \int e^{iS(z)} K_\Theta(z) v(|Az|) A_0(z)\, dz \right| \le C(1+\Lambda)^{-\kappa}(\|A_0\|_\infty + \mathrm{Lip}(A_0)), \tag{4.14}$$

for some $\kappa > 0$. Mean zero of Θ is not needed in this lemma.

Proof. For a one-variable phase of bounded degree and coefficient size G, the BV oscillatory bound follows by applying Lemma 4.7 to its derivative, followed by integration by parts on the boundedly many derivative-monotonicity intervals. It has the form $C(1+G)^{-c}(\|v\|_\infty + \mathrm{Var}\, v)$. For $0 < u < 1$, a normalized finite-q_*-variation amplitude admits a step approximation v_u with uniform error at most $2u$, at most Cu^{-q_*} intervals, and

$$\|v_u\|_\infty + \mathrm{Var}(v_u) \le C(1 + u^{1-q_*}).$$

One obtains it by recording consecutive values whenever their distance from the last recorded value reaches u; one-sided limits are used at discontinuities. Every retained nonzero increment has modulus at least u, their q_*-powers sum to at most a fixed constant, and therefore their moduli sum to at most

u^{1-q_*} times that constant. There are only finitely many retained increments; interval endpoints are irrelevant for the Lebesgue integral. The bounded-variation estimate and the approximation error give $C[(1+G)^{-c}u^{1-q_*}+u]$. Taking $u=(1+G)^{-c/q_*}$ gives a positive power depending only on the degree and q_*. A detailed equivalent finite-variation argument is given in Lemma 26.6 and Corollary 26.8.

Write $S(r\theta)=\sum_{k=0}^{d} r^k S_k(\theta)$. At least one $k\geq 1$ has $\|S_k\|_{\mathrm{coeff}}\geq c\Lambda$. On fixed rational sphere charts, the restriction of a nonzero homogeneous polynomial is injective in coefficients after clearing the fixed denominator. Lemma 4.7 therefore bounds the directions where its size is smaller than Λu by Cu^{α}. Elsewhere the radial polynomial has coefficient size at least $c\Lambda u$. Along each ray $|Az|=rg_A(\theta)$, and monotone composition preserves V^{q_*}. The multiplier $r^{-1}A_0(r\theta)$ has the required uniform variation on the fixed radial interval. Apply the one-variable estimate, bound the bad directions absolutely, and optimize u. This proves (4.14). □

4.7 Equal-shell correlations for conical profiles

Proposition 4.9 (Conical equal-shell correlation)**.** *Let $d\geq 1$. For two profiles a,c of uniformly bounded $L^\infty+V^{q_*}$ norm, supported in a fixed compact subset of $(0,\infty)$, put*

$$k_a^A(z)=K_\Theta(z)a(|Az|),\qquad k_c^A(z)=K_\Theta(z)c(|Az|).$$

There are $C,\gamma_*>0$, depending on n,d,c_0,q_* and the fixed support annuli but independent of the matrix degeneration, such that

$$\left|\int e^{iS(z)}\overline{k_a^A(z)}k_c^A(z-h)\,dz\right|\leq C\min\{1,(\Lambda|h|^d)^{-\gamma_*}\} \tag{4.15}$$

when $n\geq 2$. Here Λ is the nonconstant coefficient norm of S, and the right side is interpreted as C if $h=0$ or $\Lambda=0$. For $n=1$ it may be replaced by $C\min\{1,\Lambda^{-\gamma_*}\}$.

Proof. We give the extension of Lemma 20.5 and Proposition 26.11; their rational phase and coefficient estimates do not use the shape of the radial profile. By (2.7), both Euclidean radii lie in a fixed compact annulus. Insert a smooth Euclidean radial cutoff equal to one there. This is an exact insertion and fixes all Euclidean Jacobian support bounds independently of A.

For $h\neq 0$, use the two-ray coordinates in the proof of Lemma 20.5. On a normalized transverse block the profile arguments become

$$\gamma_1(X,Y)=\frac{|A\mathscr{O}(Ue_1+\alpha\omega)|}{|q(X,Y)|},\qquad \gamma_2(X,Y)=\frac{|A\mathscr{O}(Ve_1+\alpha\omega)|}{|q(X,Y)|},$$

where $U=U_0+HX$, $V=V_0+HY$, $q=\Delta+X-Y$, and q has fixed sign and is bounded away from zero on the enlarged block. The square of either numerator is a nonnegative quadratic polynomial, say $p_A(U)$. A horizontal derivative can vanish only where

$$\tfrac{H}{2}p_A'(U)q-p_A(U)=0.$$

This equation has bounded degree. The zeros of p_A add at most two break points; identically zero sections are constant. The other sections are treated identically. Thus both arguments have uniformly bounded numbers of monotonicity intervals. On each such interval, $V^{q_*}(a\circ\gamma_i)\leq V^{q_*}(a)$, with zero extension accounting for support crossings. The product variation inequality and the fixed smooth Euclidean cutoffs now give uniformly bounded sectional V^{q_*} norms for the complete amplitude. This verifies the amplitude and Schur hypotheses of Lemma 26.9, used with exponent q_*.

For clarity, the phase lower bound is unchanged. With $\varphi=\sum_{m=0}^{d}\varphi_m$ in homogeneous pieces and $f_m(X)=\varphi_m(U_0+HX,\alpha)$, its pullback satisfies

$$q^{d+2}\partial_X\partial_Y\left(\sum_{m=0}^{d}f_m(X)q^{-m}\right)=\sum_{m=1}^{d}q^{d-m}\{mqf_m'-m(m+1)f_m\}.$$

The coefficient of q^{d-m} is $-m(m+1)f_m+(m+1)f'_{m+1}$, with $f_{d+1}=0$. Backward recovery of the f_m gives a uniform inverse on the fixed coefficient spaces. Affine substitution then gives the factor $(\alpha H)^d$ in Lemma 20.6. The matrix A does not occur in this calculation.

The one-variable rational estimate and its TT^* proof in Lemma 26.9 now apply. The rough angular factors remain respectively functions of the two ray variables; they are used as bounded L^2 input and output functions and are not differentiated. The transverse-band bound is

$$C\alpha^{n-1}\int_{\mathbb{S}^{n-2}}\min\{1,[\Gamma(\omega)(\alpha H)^d]^{-c_1}\}\,d\omega.$$

The chart coefficient estimate and polynomial sublevel bound used in Lemma 20.5 give a positive power in $\Lambda(\alpha H)^d$. Summing dyadic α gives (4.15), after decreasing the exponent. The cases $h=0$ and constant S use the size bound. In dimension one, a bounded partition into signed overlap intervals reduces directly to the one-variable V^{q_*} estimate. □

5 A fixed polynomial tile geometry

For $d=0$, Theorem 2.2 is Proposition 4.6, since constant phases do not affect the path norm. In this section and the ensuing polynomial tile analysis assume $d\ge 1$.

This section specifies the external geometric input. The analytic estimates for its operators are not imported from a smooth-kernel Carleson theorem.

Fix first the analytic exponent $\beta=1/(16\rho)$, furnished by the shell estimates and the single-tree proof below independently of the grid base. Choose a fixed integer K with $K\beta\ge 2$. Then choose one sufficiently large structural integer $a\ge 4$ for the Euclidean space and the polynomial space $\mathscr{P}_d$, with ball metric given by the oscillation of the difference of two polynomials. The finite-dimensional norm comparisons and Lemma 4.7 verify the compatibility and polynomial cancellation conditions of [4]. Set $D=2^{100a^2}$. Increasing a once, before any finite cutoffs are imposed, is permitted. It will never depend on a smoothing parameter, an angular atom, or a density generation.

At scale j use the disjoint radial shell

$$c_0D^j/(4D)<\rho_A(z)\le c_0D^j/4. \tag{5.1}$$

These shells partition the positive radial axis. On the angular support,

$$c_0D^j/(4D)<|z|\le D^j/4,\qquad |k_{j,x}(z)|\le CD^{-jn},$$

where $k_{j,x}=K_\Theta b(\rho_A)h_x(\rho_A)\mathbf{1}_{\text{shell }j}$. The wider fixed shell ratio only changes structural constants in Section 4.

Fix bounded sets F,G and a finite-range measurable phase choice Q_x. The finite spatial grid and nested polynomial cells are those of [4, Proposition 2.1]. A tile p consists of a spatial cube I_p, a scale $s(p)$, a phase cell Ω_p, and its center polynomial Q_p. Write $r_p=D^{s(p)}$ and $c_p=c(I_p)$. In particular,

$$B(c_p,r_p/4)\subset I_p\subset B(c_p,4r_p).$$

The polynomial metric d_p is the oscillation metric on $B(c_p,r_p)$. Changing this ball by any one fixed factor changes constants only. Phase cells at a fixed spatial cube partition the finite phase range; nested spatial cubes have nested or disjoint corresponding phase cells. The order is $p\le p'$ if $I_p\subset I_{p'}$ and $\Omega_{p'}\subset\Omega_p$.

Same-family nets and full ambient phase cells. The finite-net selection in the proof of the geometric input must be made in the full ambient phase metric, not merely after intersecting balls with the given finite range $\mathscr{Q}$ of the phase choice. For any such finite family and any one ball metric d, choose a maximal $3/5$-separated subfamily $Z\subset\mathscr{Q}$. Then every member of $\mathscr{Q}$ has distance strictly less than $3/5$

from Z, and the full ambient open $3/10$ balls about distinct centers in Z are disjoint. Both assertions follow directly from maximality and the triangle inequality. The centers remain in the original family. In contrast, disjointness only after intersecting with $\mathscr{Q}$ is insufficient: for $\mathscr{Q} = \{0, 2/5\} \subset \mathbb{R}$ the two ambient radius-$3/10$ balls meet at $1/5$, although their intersections with $\mathscr{Q}$ are disjoint. This corrects the intermediate selection step in [4, Lemma 3.2 and (3.1)], without changing the geometric statement being used.

Enumerate $Z = \{z_i\}$. Assign each point at distance less than $7/10$ from Z to its nearest center, resolving ties by the least index, and call the resulting ambient cells V_i. They are disjoint, cover $\mathscr{Q}$, and satisfy

$$B_d(z_i, 3/10) \subset V_i \subset B_d(z_i, 7/10). \tag{5.2}$$

For the first inclusion, the triangle inequality and separation show that every other center is strictly farther away. The cells are measurable whenever the distance coordinates are measurable, since their definition uses finitely many strict or non-strict comparisons.

We construct the nested phase cells recursively along each ball chain. Number levels from coarse to fine, write d_k for their polynomial oscillation metrics, and suppose

$$d_{k+1}(P,Q) \le \tfrac{1}{10} d_k(P,Q) \quad \text{for every } P, Q. \tag{5.3}$$

At each level choose the preceding separated net from the same $\mathscr{Q}$, with centers $z_{k,i}$ and base cells $V_{k,i}$. Start with $C_{0,i} = V_{0,i}$ and define

$$C_{k+1,i} = B_{d_{k+1}}(z_{k+1,i}, 1/5) \cup \bigcup_{j : z_{k,j} \in V_{k+1,i}} C_{k,j}. \tag{5.4}$$

Induction gives pairwise disjoint cells at each level, coverage of $\mathscr{Q}$, and

$$B_{d_k}(z_{k,i}, 1/5) \subset C_{k,i} \subset B_{d_k}(z_{k,i}, 1).$$

Indeed, each inherited cell has radius less than $1/10$ in d_{k+1} about its old center. An intersection with a new $1/5$ core forces that old center into the corresponding $3/10$ core and hence into the same base cell. This proves disjointness; the triangle inequality gives outer radius less than $1/10 + 7/10 < 1$. Each old center belongs to $\mathscr{Q}$, hence is assigned, proving coverage and containment of each old cell in a new one. Iteration gives nesting or disjointness at arbitrary depth. Measurability propagates by finite unions.

For polynomial oscillation metrics, the contraction is uniform along ball chains with bounded center displacement. More precisely, for fixed degree, dimension, and $A \ge 0$, one can choose an integer J such that

$$c_{k+1} = c_k + r_k a_k, \qquad r_{k+1} = r_k / 2^J, \qquad |(a_k)_i| \le A$$

implies (5.3). Finite-dimensional polynomial norm equivalence also gives uniform doubling and separated packing. The preceding recursion therefore supplies the required phase-cell hierarchy. We use the spatial grid and its small-boundary property from [4]; the analytic tree estimates are proved below.

Let $\sigma_-(x), \sigma_+(x)$ enclose all shells meeting the selected witness in (2.4). Define

$$E_p = G \cap I_p \cap \{Q_x \in \Omega_p\} \cap \{\sigma_-(x) \le s(p) \le \sigma_+(x)\}. \tag{5.5}$$

Zero-selector scales between two nonzero intervals are retained in (5.5). Their kernels are zero, but deleting their tiles would destroy the interval property of the active scales in a convex tree. The local operator is

$$T_p f(x) = \mathbf{1}_{E_p}(x) \int k_{s(p),x}(x-y) e^{i(Q_x(y) - Q_x(x))} f(y)\, dy. \tag{5.6}$$

The common factor $e^{-iQ_x(x)}$ is harmless for spatial norms and is fixed for all tiles. The sum over all tiles is exactly the linearized witness on G.

We use the two densities in [4, (2.24)–(2.27)]. For precision, $E_2(\lambda,p)=I_p\cap G\cap\{d_p(Q_x,Q_p)<\lambda\}$; $\lambda p\lesssim\lambda p'$ means spatial inclusion and reverse inclusion of the corresponding phase balls. If $\mathfrak{P}(\mathfrak{C})$ is the set of tiles whose spatial cube is contained in the spatial cube of some member of $\mathfrak{C}$, set

$$\delta_1(\mathfrak{C})=\sup_{p'\in\mathfrak{C}}\sup_{\lambda\ge 2}\lambda^{-a}\sup_{p\in\mathfrak{P}(\mathfrak{C}):\,\lambda p'\lesssim\lambda p}\frac{|E_2(\lambda,p)|}{|I_p|},\tag{5.7}$$

$$\delta_2(\mathfrak{C})=\sup_{p\in\mathfrak{C}}\sup_{r\ge 4r_p}\frac{|F\cap B(c_p,r)|}{|B(c_p,r)|}.\tag{5.8}$$

Empty suprema are zero. Fixed changes of ball radii are absorbed into structural constants.

An antichain has no two comparable tiles. A tree $\mathfrak{T}(u)$ is convex in the tile order, satisfies $4p\lesssim u$ for every tile p, and has the support margin

$$B(c_p,8r_p)\subset I_u.\tag{5.9}$$

Its top polynomial is Q_u. A forest of density generation m has pairwise disjoint tile trees, top multiplicity at most 2^m, tree density at most $C2^{-m}$, and cross-top separation

$$d_p(Q_p,Q_u)>2^{Z(m+1)}\quad(p\in\mathfrak{T}(u'),\ u'\neq u,\ I_p\subset I_u).\tag{5.10}$$

Here Z is a fixed structural integer, chosen above the geometric threshold and the analytic threshold specified below. Increasing Z does not change the already chosen spatial grid, its base D, or the parameter a in the density definition. Effective output sets of distinct trees in one forest are disjoint: at a point belonging to two trees, the smaller spatial tile is inside the other top, while tree coherence makes both phases close in that tile metric, contradicting (5.10).

Proposition 5.1 (Geometric decomposition and count)**.** *The following conclusions hold for the preceding finite model.*

(i) Suppose $|G|>0$. There is a measurable ambient set $H\subset\mathbb{R}^n$, $|H|\le|G|/2$, such that the tiles whose spatial cube is not contained in H partition into $O((m+2)^2)$ forests and $O(Z(m+2)^3)$ antichains at every generation $m\ge 0$. The forests satisfy (5.10)*; the antichains satisfy $\delta_1\le C2^{-m}$. Every part has $\delta_2\le C\min\{1,|F|/|G|\}$. Put $G'=G\cap H$. Then $G'\subset G$ and $|G'|\le|G|/2$; the spatial discard condition is $I_p\subset H$, not $I_p\subset G'$.*

(ii) Given $\kappa>0$, a fixed $q_0>2$ can be chosen so that, for any antichain $\mathfrak{A}$, any subfamily $\mathfrak{B}$, and any polynomial Q,

$$\Big\|\sum_{p\in\mathfrak{B}}(1+d_p(Q_p,Q))^{-\kappa}\mathbf{1}_{E_p}\Big\|_{q_0}\le C\delta_1(\mathfrak{A})^{1/q_0}\Big|\bigcup_{p\in\mathfrak{B}}I_p\Big|^{1/q_0}.\tag{5.11}$$

The sets E_p of an antichain are pairwise disjoint.

(iii) For a nonempty tree, the input stopping partition $\mathscr{J}(\mathfrak{T})$ and the output stopping partition $\mathscr{L}(\mathfrak{T})$ described below satisfy

$$|F\cap J|\le C\delta_2(\mathfrak{T})|J|,\qquad|L\cap\textstyle\bigcup_{p\in\mathfrak{T}}E_p|\le C\delta_1(\mathfrak{T})|L|.\tag{5.12}$$

Empty trees act by zero raw and smoothed operators and may be omitted.

Proof. For (i), apply the geometric decomposition of [4, Proposition 2.4 and Section 4] to obtain the ambient exceptional set H. The counting and density estimates depend only on the grid, phase cells, and admissible scale interval in (5.5). Set $G'=G\cap H$. Every discarded tile has $I_p\subset H$ and hence vanishes on $G\setminus G'$. Thus the surviving operator agrees with the full finite operator on the required major subset of G.

For (ii), first recall the finite-reference consequence of the counting argument of [4, Lemmas 5.6–5.9]: if Q_0 belongs to the original finite phase range and $\mathfrak{C}$ is a sub-antichain satisfying $d_p(Q_p,Q_0) < 2^{M+1}$ for every $p \in \mathfrak{C}$, then

$$\sum_{p\in\mathfrak{C}} |E_p| \le C\delta_1(\mathfrak{A})2^{C_aM}\left|\bigcup_{p\in\mathfrak{C}} I_p\right|.$$

The ball version displayed here follows by summing the distance layers up to $M+1$; the resulting geometric sum only changes the structural constant.

To allow an arbitrary ambient polynomial Q without changing the grid, put

$$\mathfrak{B}_N = \{p \in \mathfrak{B} : 2^N \le 1 + d_p(Q_p,Q) < 2^{N+1}\}, \qquad N \ge 0.$$

If $\mathfrak{B}_N$ is empty, its contribution is zero. Otherwise, let $\mathscr{U}_N$ be the inclusion-maximal spatial cubes among $\{I_p : p \in \mathfrak{B}_N\}$. These cubes are pairwise disjoint and cover all spatial cubes in this finite collection. For each $L \in \mathscr{U}_N$ choose $p_L \in \mathfrak{B}_N$ with $I_{p_L} = L$, and let $\mathfrak{B}_{N,L} = \{p \in \mathfrak{B}_N : I_p \subset L\}$. The centers were selected from the original finite phase range, so Q_{p_L} is an admissible finite reference. For $I_p \subset L$, the ball defining d_p lies in $B(c_L, 5r_L)$. Polynomial norm equivalence on this fixed enlargement gives $d_p(R,S) \le C_{n,d} d_{p_L}(R,S)$. The triangle inequality therefore gives, for $p \in \mathfrak{B}_{N,L}$,

$$\begin{aligned} d_p(Q_p,Q_{p_L}) &\le d_p(Q_p,Q) + d_p(Q,Q_{p_L}) \\ &\le d_p(Q_p,Q) + C_{n,d} d_{p_L}(Q,Q_{p_L}) < (1+C_{n,d})2^{N+1}. \end{aligned}$$

Choose a fixed integer c_* with $2^{c_*} \ge 1 + C_{n,d}$. The finite-reference estimate with $M = N + c_*$ applies to this bucket. Since p_L belongs to the bucket, its spatial union is exactly L. Summing over the disjoint L yields

$$\sum_{p\in\mathfrak{B}_N} |E_p| \le C\delta_1(\mathfrak{A})2^{C_aN}\left|\bigcup_{p\in\mathfrak{B}} I_p\right|.$$

Every cube and phase cell used in this reduction belongs to the original grid. Antichain output sets are disjoint, so the q_0-th power of the left side of (5.11) is bounded by

$$C\delta_1(\mathfrak{A})\left|\bigcup_{p\in\mathfrak{B}} I_p\right| \sum_{N\ge 0} 2^{-\kappa q_0 N + C_a N}.$$

Choose $q_0 > \max\{2, (C_a+1)/\kappa\}$.

Part (iii) is the purely geometric local density argument of [4, Lemmas 6.8–6.9], with the effective output set in (5.5). □

The input partition $\mathscr{J}(\mathfrak{C})$ consists of the maximal grid cubes J for which either $s(J)$ is the bottom scale or no spatial cube of $\mathfrak{C}$ is contained in $B(c(J), 100D^{s(J)+1})$. The output partition consists of the maximal cubes lying in some I_p, $p \in \mathfrak{C}$, that contain no such spatial cube, with the bottom-scale convention. We write $P_{\mathscr{J}}$ and $P_{\mathscr{L}}$ for their averaging projections. In applications these partitions are restricted to the relevant top.

We will use one further geometric consequence, stated explicitly in Section 7.2: two separated trees admit a partition into a large-phase portion and a thin boundary portion, with uniform lower bounds on their active scales. These consequences are [4, Lemmas 6.13–6.24]. In the subsequent estimates the geometric parameter a is fixed and kernel regularity is denoted separately by L. For this fixed geometry, the actual small-boundary estimate and polynomial oscillation estimate supply positive decay exponents; their combination gives a fixed $\eta > 0$ in the row correlation estimate. Let Z_{geo} be a structural integer above which the same-grid geometric decomposition is valid. Choose integers $K \ge 1$ and Z so that

$$K\beta \ge 2, \qquad Z \ge \max\{Z_{\mathrm{geo}}, 4\}, \qquad \eta Z > 2K + 3. \tag{5.13}$$

Such choices exist by $\beta,\eta>0$. The geometric construction and its boundary budget remain valid for every larger Z, while a, D and the density exponent stay unchanged. With $\varepsilon_m=\frac{1}{8}2^{-K(m+1)}$, the two numerical estimates are

$$2^{m/2}\varepsilon_m^{\beta}\le(\tfrac{1}{8})^{\beta}2^{-3m/2},\qquad 2^m\varepsilon_m^{-2}2^{-\eta Z(m+1)}\le 64\,2^{-2m}.$$

Moreover $1+\log(1/\varepsilon_m)\le C_K(1+m)$. All these choices precede the finite cutoffs, angular atoms, witness lengths and density generations. No parameter is subsequently enlarged as a function of a density level or a smoothing radius.

6 Antichains and a collective high-frequency remainder

The central point is to estimate the remainder after grouping all phases at one spatial scale. Estimating one large tile at a time would introduce the multiplicity of enlarged spatial supports.

6.1 A single-shell estimate after frequency translation

Let A_h be a normalized unphased shell operator with an actual witness (2.4), restricted to that shell. Section 4 gives

$$\|A_h\Delta_m\|_{2\to2}\le C2^{-\beta_0 m}\quad(m\ge1),\qquad \|A_h\|_{2\to2}\le C,$$

with $\beta_0=1/(8\rho)$. The fixed radial factor $b(\rho_A)$ is included in A_h. The operator is not assumed translation invariant: it is pointwise dominated by the short variation of the translation-invariant partial-shell path. For the real, even mollifier $P_t=\phi_t*$ one consequently has

$$\|A_h[e^{i\ell\cdot y}(I-P_t)f]\|_2\le Ct^{\beta_0}(1+|\ell|)^{\beta_0}\|f\|_2,\qquad 0<t\le1/8. \tag{6.1}$$

Indeed, after modulation the multiplier of $I-P_t$ is $1-\widehat{\phi}(t(\xi-\ell))$. On the annulus $|\xi|\simeq2^m$ its modulus is at most $C\min\{1,t2^m\}+C\min\{1,t|\ell|\}$. The low-frequency piece is estimated as one piece; the high-frequency pieces are summed with $2^{-\beta_0 m}$. This gives (6.1).

Lemma 6.1 (Collective demodulated remainder)**.** *Let outputs be restricted to a fixed bounded cube I. Let Q_x,V_x be measurable polynomial fields of degree at most d such that $\|Q_x-V_x\|_{I^+,*}\le C_0$ on a fixed enlargement containing all kernel supports. Put*

$$\mathscr{R}_t f(x)=\mathbf{1}_I(x)\int k_x(y)e^{iQ_x(y)}\big[f(y)-e^{-iV_x(y)}P_t(e^{iV_x}f)(y)\big]\,dy,$$

where $k_x(y)=K_\Theta(x-y)b(\rho_A(x-y))h_x(\rho_A(x-y))$ is restricted to one fixed normalized shell. Then

$$\|\mathscr{R}_t\|_{2\to2}\le Ct^{\gamma} \tag{6.2}$$

for a fixed $\gamma>0$. The constant is independent of the number and size of the phase coefficients. The scale t is common to this operator.

Proof. Partition the finite-dimensional coefficient space modulo constants into cubes of sidelength $L\ge L_0$. Let E_ν be the outputs whose V_x belongs to the νth cube. Each coefficient cube has only a fixed number of neighbors within a fixed multiple of L.

On one cube, remove its center polynomial W_ν by an input modulation and an output factor of modulus one. We may then assume $\|Q_x\|_{I^+,*}+\|V_x\|_{I^+,*}\le CL$. We have the exact identity

$$\begin{aligned}f-e^{-iV_x}P_t(e^{iV_x}f)&=(I-P_t)f\\&\quad+\int\phi_t(v)[1-e^{i(V_x(y-v)-V_x(y))}]f(y-v)\,dv.\end{aligned}$$

The second term contributes an operator bounded pointwise by $CLtMf$. For the first term expand $e^{iQ_x(y)}$ on the fixed input support, using one fixed compact smooth extension:

$$e^{iQ_x(y)}\chi(y) = \sum_{\ell\in\mathbb{Z}^n} c_\ell(x)e^{i\lambda_\ell\cdot y}, \qquad |c_\ell(x)| \le C_M L^M(1+|\ell|)^{-M}.$$

Choose $M > n+\beta_0+2$ and apply (6.1). This proves

$$\|\mathbf{1}_{E_v}\mathscr{R}_t\|_{2\to 2} \le CL^M t^{\beta_0}. \tag{6.3}$$

The number of neighboring coefficient cubes is fixed, so all near-block contributions to $\mathscr{R}_t\mathscr{R}_t^*$ have norm at most $CL^{2M}t^{2\beta_0}$.

For far blocks set $r_x = Q_x - V_x$ and $d\mu_t(v) = d\delta_0(v) - \phi_t(v)\,dv$. Direct computation, using that ϕ is even, gives the adjoint kernel

$$\mathscr{R}_t^*(y,x) = e^{-iQ_x(y)}\int e^{-i[r_x(y-v)-r_x(y)]}\overline{k_x(y-v)}\,d\mu_t(v). \tag{6.4}$$

The coefficient norm of $r_x(\cdot - v) - r_x$ is $O(t)$ on the fixed support. For far coefficient blocks the nonconstant part of $Q_x - Q_{x'}$ has norm at least cL. The total phase in the correlation of (6.4), with shifts v, w, therefore still has norm at least cL. Translation from global normalized coordinates to the kernel center changes this norm by at most a fixed factor.

Proposition 4.9 gives the correlation majorant

$$C\iint \Psi_L(x-x'+v-w)\,d|\mu_t|(v)d|\mu_t|(w), \qquad \Psi_L(z) = \min\{1,(L|z|^d)^{-c}\}.$$

In dimension one use its stronger bound. On a fixed bounded region,

$$\int \Psi_L(z)\,dz \le CL^{-\tau}, \qquad \tau = \min\{n/(2d), c/2\} > 0.$$

To verify this split at $|z| = L^{-1/(2d)}$. The signed measure μ_t is common to all outputs and has total variation at most two. Tonelli therefore gives both row and column integral bounds $CL^{-\tau}$. Schur's test bounds the complete far-block operator by that quantity; no sum over the number of far coefficient cubes is taken.

Combining near and far blocks yields

$$\|\mathscr{R}_t\|_{2\to 2}^2 \le C(L^{2M}t^{2\beta_0} + L^{-\tau}).$$

Take $L = L_0 t^{-2\beta_0/(2M+\tau)}$. Then (6.2) holds with $\gamma = \beta_0\tau/(2M+\tau) > 0$. □

Localize a fixed spatial scale R_j into grid cubes. Their fixed Euclidean enlargements have bounded overlap. Scaling Lemma 6.1 gives

$$\left\| \sum_{p:\,s(p)=j} M_{-Q_p}(I - P_{tR_j})M_{Q_p}T_p^* g \right\|_2 \le Ct^\gamma \|g\|_2. \tag{6.5}$$

A common unit-modulus output factor in (5.6) is absorbed into g. At a given output the center field V_x is the phase of its unique active tile at that scale. Thus all hypotheses of the lemma hold. Spatial cube overlap, not phase-tile overlap, is used in this localization.

6.2 Mass-sensitive estimates for the principal correlation

Let $g_p = g\mathbf{1}_{E_p}$ and $M_p = \|g_p\|_1$. Put $F_p = M_{Q_p} T_p^* g_p$. Then

$$\|F_p\|_\infty \le C R_p^{-n} M_p. \tag{6.6}$$

If $r = R_q \le R = R_p$, choose $0 < \varepsilon \le 1$ and mollify F_p at the smaller scale εr. This is a choice of the auxiliary proof parameter; the fixed-scale remainder estimate above remains valid for every positive smoothing parameter. The size and derivative bounds are

$$\|P_{\varepsilon r} F_p\|_\infty + r\|\nabla P_{\varepsilon r} F_p\|_\infty \le C\varepsilon^{-1} R^{-n} M_p.$$

Apply Lemma 4.8 inside the qth tile operator. The comparison of local phase metrics, together with $Q_x \in \Omega_q$, gives

$$\left|\langle M_{-Q_p} P_{\varepsilon r} F_p, T_q^* g_q\rangle\right| \le C\varepsilon^{-1}(1 + d_q(Q_q, Q_p))^{-\kappa} R^{-n} M_p M_q. \tag{6.7}$$

The local norm comparison is uniform when the two supports meet: then I_q lies in a fixed enlargement of I_p, and translation between the centers of the small integration annulus and I_q is bounded in units of R_q. For bounded phase distance use the size estimate instead.

Proposition 6.2 (Antichain density estimate)**.** *There is $\vartheta > 0$ such that*

$$\|T_{\mathfrak{A}}\|_{2\to 2} \le C\delta_1(\mathfrak{A})^{\vartheta} \tag{6.8}$$

for every finite antichain. Constants are independent of the number of scales, phase cells, and selected jumps.

Proof. Write $\delta = \delta_1(\mathfrak{A})$ and choose $q_0 > 2$ as in (5.11), for the exponent in (6.7). Set $\sigma = 1/q_0$. For a fixed larger tile p, the raw adjoint is supported, up to a null set, in $B(c_p, 5R_p)$: the output cell lies in $B(c_p, 4R_p)$ and the raw kernel has displacement at most $R_p/4$. Mollification at $\varepsilon R_q \le R_p$ enlarges this to $B(c_p, 6R_p)$. If this support meets $B(c_q, 5R_q)$ and $R_q \le R_p$, the whole cell I_q lies in $B_p = B(c_p, 16R_p)$. Otherwise the Hilbert pairing is zero. Thus only the stated smaller tiles interact, and $|B_p| \simeq R_p^n$. Hölder and (5.11) give

$$R_p^{-n} \sum_q (1 + d_q(Q_q, Q_p))^{-\kappa} M_q \le C\delta^{\sigma} \left(\frac{1}{|B_p|}\int_{B_p} |g|^{q_0'}\right)^{1/q_0'}.$$

For $x \in E_p$ the right side is at most $C\delta^{\sigma} M_{q_0'} g(x)$. Since all E_p are disjoint and $q_0' < 2$, the sum of the absolute main correlations is at most

$$C\varepsilon^{-1}\delta^{\sigma}\|g\|_2^2. \tag{6.9}$$

For the errors, group by scales before applying any norm. Write $g_j = g\mathbf{1}_{\cup_{p\in\mathfrak{A}, s(p)=j} E_p}$ and $F_j = \sum_{s(p)=j} T_p^* g_p$. Disjointness gives $\sum_j \|g_j\|_2^2 \le \|g\|_2^2$. The corresponding forward operator is bounded pointwise by CMf, so $\|F_j\|_2 \le C\|g_j\|_2$. At scale gap $k \ge 0$, use (6.5) with $t = \varepsilon D^{-k}$. Then

$$\left|\sum_j \left\langle \sum_{s(p)=j} M_{-Q_p}(I - P_{\varepsilon R_{j-k}}) M_{Q_p} T_p^* g_p, F_{j-k}\right\rangle\right| \le C\varepsilon^{\gamma} D^{-\gamma k}\|g\|_2^2.$$

The sum over k is geometric. Expand $\|\sum_j F_j\|_2^2$ with all same-scale pairs in the diagonal and with $2\mathrm{Re}\langle F_j, F_{j-k}\rangle$ for $k > 0$. Its main terms are covered by (6.9); its errors are covered by the last display. Thus

$$\|T_{\mathfrak{A}}^* g\|_2^2 \le C(\varepsilon^{-1}\delta^{\sigma} + \varepsilon^{\gamma})\|g\|_2^2.$$

The density satisfies $0 \le \delta \le 1$, directly from its supremum definition. Fix $0 < \varepsilon_0 \le 1$. If $\delta > 0$, take $\varepsilon = \varepsilon_0 \delta^{\sigma/(1+\gamma)}$, which lies in the permitted range. If $\delta = 0$, let $\varepsilon \downarrow 0$ in the preceding quadratic bound instead. This proves (6.8) with $\vartheta = \sigma\gamma/[2(1+\gamma)]$. There is no enlarged-support multiplicity in this calculation. □

7 Trees, input smoothing, and local correlations

The definitions and densities are those of Section 5. In particular E_p includes G and the whole enclosing interval of witness scales. A tree is never replaced by its nonzero-selector subset. We first prove the local correlation estimate, retaining a separate kernel regularity constant; we then prove its single-tree norm inputs. For a tree with top phase Q_i and $x \in E_p$, the metric growth and coherence give

$$\|Q_x - Q_i\|_{B(c_p,20r_p),*} \le C. \tag{7.1}$$

These constants depend only on the fixed spatial and phase geometry.

Suppose $I_1 \subset I_2$; the case of disjoint tops is immediate by support. Put

$$\mathfrak{S} = \{p \in \mathfrak{T}_1 \cup \mathfrak{T}_2 : d_p(Q_1,Q_2) \ge A^{1/2}\}, \qquad \mathfrak{T}_2^+ = \mathfrak{T}_2 \cap \mathfrak{S}, \quad \mathfrak{T}_2^- = \mathfrak{T}_2 \setminus \mathfrak{S}. \tag{7.2}$$

The following are geometric facts. Their statements do not mention a kernel or its regularity constant.

Lemma 7.1 (Geometric separation and stopping regions)**.** *There are fixed positive constants c_*, κ_* and fixed finite constants C_*, N_* with the following properties.*

(a) $\mathfrak{T}_1 \subset \mathfrak{S}$, and every tile in $\mathfrak{T}_2^-$ has $I_p \cap I_1 = \varnothing$.

(b) Let $\mathscr{J}_+ = \{J \in \mathscr{J}(\mathfrak{S}) : J \subset I_1\}$. There is a partition of unity $\{\chi_J\}_{J\in\mathscr{J}_+}$ on I_1, supported in $B_J = B(c(J), 8r_J)$, with Lipschitz constants at most C_*/r_J. Put $B_J^0 = B(c(J), r_J/8)$. If the support ball $B(c(I_p), 5r_p)$ of a tile in $\mathfrak{T}_1$ or $\mathfrak{T}_2^+$ meets $2B_J$, then $s(p) \ge s(J) - C_*$. Moreover,

$$\|Q_2 - Q_1\|_{B_J,*} \ge c_* A^{1/2}. \tag{7.3}$$

Only at most N_* consecutive scales of $\mathfrak{T}_2^-$ can have their support balls meet B_J^0.

(c) Put $\mathscr{J}_0 = \{J \in \mathscr{J}(\mathfrak{T}_1) : J \subset I_1\}$. If $p \in \mathfrak{T}_2^-$ and $B(c(I_p), 5r_p)$ meets $J \in \mathscr{J}_0$, then

$$s(p) \le s(J) - c_* \log_D A + C_*. \tag{7.4}$$

At relative scale $k = s(J) - s(p)$, the union of these support balls inside J is a set $F_{J,k}$ satisfying

$$|F_{J,k}| \le C_* D^{-\kappa_* k} |J|. \tag{7.5}$$

At a fixed relative scale their overlap is bounded by C_*. The corresponding lower-scale exclusion in (b) holds for $\mathfrak{T}_1$ and $\mathscr{J}_0$ as well.

Proof. These are the geometric conclusions of [4, Lemmas 6.13–6.14, 6.17, 6.21, 6.23–6.24], with A denoting the cross-top separation parameter. We record why the use here does not alter their hypotheses. Membership in $\mathfrak{S}$ is determined by local phase distances; nestedness of spatial cubes, coherence, and cross-top separation give (a). Maximality of $\mathscr{J}(\mathfrak{S})$ supplies a nearby high-separation tile at the parent scale of each J; comparing the finitely enlarged ball metrics gives (7.3). The same maximality excludes smaller contributing tiles. A low-separation tile meeting B_J^0 cannot be smaller than J by the interior ball property, or more than a fixed number of scales larger without contradicting the high-separation tile at the parent scale. This gives the last assertion in (b).

For (c), a low-separation tile has spatial cube outside I_1. Comparing its phase metric with the metric of a nearby tile in $\mathfrak{T}_1$, provided by maximality of $\mathscr{J}(\mathfrak{T}_1)$, forces a scale gap proportional to $\log_D A$. Its support can enter J only through a boundary layer of relative width $O(D^{-k})$. The grid small-boundary property gives (7.5); the fixed-scale packing property gives bounded overlap. All comparisons use only the fixed grid, phase metrics, and the support radius $5r_p$. The parameter controlling kernel smoothness does not occur. □

7.1 The smoothed operators

First multiply the full linearized output by $e^{-iQ_x(x)}$. This is a common unit-modulus factor for all tiles and does not change any spatial norm. The operators below refer to that normalization.

In this subsection write $\rho(z) = |A_{\mathrm{cone}} z|$, where A_{cone} is the fixed matrix denoted by A in Theorem 2.2. The scalar A in the separated-tree estimates is a distinct phase-separation parameter. The Euclidean case is $A_{\mathrm{cone}} = I$. Choose disjoint radial scale intervals so that the associated raw kernels $k_{j,x}$ satisfy

$$\operatorname{supp} k_{j,x} \subset \{z : c_- D^j \le |z| \le D^j/4\}, \qquad |k_{j,x}(z)| \le C D^{-jn}. \tag{7.6}$$

Here $k_{j,x}$ includes the angular factor, the fixed amplitude $b(\rho(z))$, the sharp radial scale cutoff, and the actual equal-amplitude selector $h_x(\rho(z))$. The size estimate uses $\|b\|_\infty \le 1$ and $\|h_x\|_\infty \le 1$. The positive constant c_- may depend on D and the cone aperture, but is independent of the least singular value of A_{cone}.

Let $\phi \in C_c^\infty(B(0,1))$ be real, even, and nonnegative, with integral one. For $0 < \varepsilon \le \varepsilon_0 < 1/8$, put $P_{\varepsilon r} f = \phi_{\varepsilon r} * f$. For $p \in \mathfrak{T}_i$, let

$$\psi_{i,x}(z) = Q_x(z) - Q_i(z) - Q_x(x) + Q_i(x).$$

Define the phase-removed kernel

$$b_{i,p}^{\varepsilon}(x,y) = \mathbf{1}_{E_p}(x) \int k_{s(p),x}(x-z) e^{i\psi_{i,x}(z)} \phi_{\varepsilon r_p}(z-y)\, dz. \tag{7.7}$$

If $B_{i,p}^{\varepsilon}$ is the operator with this kernel, the actual smoothed tile is

$$U_{i,p}^{\varepsilon} = M_{-Q_i} B_{i,p}^{\varepsilon} M_{Q_i}, \qquad M_Q f = e^{iQ} f.$$

Thus it is the original tile conjugated by the top phase, smoothed on its input, and conjugated back. It is not obtained by smoothing only the angular function.

Lemma 7.2 (Uniform kernel bounds)**.** *The kernels* (7.7) *satisfy*

$$|b_{i,p}^{\varepsilon}(x,y)| \le C \mathbf{1}_{E_p}(x) r_p^{-n}, \tag{7.8}$$

$$|\nabla_y b_{i,p}^{\varepsilon}(x,y)| \le C \varepsilon^{-1} \mathbf{1}_{E_p}(x) r_p^{-n-1}. \tag{7.9}$$

Their supports in y lie in $B(c(I_p), 5r_p) \subset I_i$. The constants are independent of the top phase, all other phase coefficients, the number of jumps, and, in the conical case, τ.

Proof. The size bound follows from (7.6) and $\|\phi\|_1 = 1$. Differentiation in y falls only on $\phi_{\varepsilon r_p}$, whose derivative has L^1 norm $C/(\varepsilon r_p)$. This proves (7.9); no derivative of the rough angular factor or of h_x is taken. If the kernel is nonzero, $|x-y| \le r_p/4 + \varepsilon r_p < r_p/2$. Since $x \in I_p \subset B(c(I_p), 4r_p)$, the asserted support follows. Its inclusion in the top cube follows from (5.9). □

For $\mathfrak{C} \subset \mathfrak{T}_i$, write

$$U_{i,\mathfrak{C}}^{\varepsilon} = \sum_{p \in \mathfrak{C}} U_{i,p}^{\varepsilon}, \qquad H_{i,\mathfrak{C}} = e^{iQ_i} (U_{i,\mathfrak{C}}^{\varepsilon})^* g_i.$$

Because M_{Q_i} is an isometry and only introduces a unit-modulus factor on the input of the adjoint, Lemma 7.2 gives all estimates below for $H_{i,\mathfrak{C}}$ with $|g_i|$ on their right-hand sides.

7.2 A localized separated-tree theorem

The proof is stated for a general $L \geq 1$ in place of $C\varepsilon^{-1}$ in (7.9). Assume (7.8), the support conclusion of Lemma 7.2, and

$$|\nabla_y b_{i,p}(x,y)| \leq CL\mathbf{1}_{E_p}(x) r_p^{-n-1}. \tag{7.10}$$

All constants except L remain fixed. Denote the resulting tree operators by U_i and set $g_i^0 = g_i \mathbf{1}_{\cup_{p\in\mathfrak{T}_i} E_p}$. Define

$$\mathscr{S}_i g_i = |U_i^* g_i| + C_1 LM(|g_i^0|) + |g_i^0|, \tag{7.11}$$

where C_1 is a sufficiently large fixed structural constant.

Theorem 7.3 (Localized correlation with separate regularity)**.** *Under the fixed geometric conditions of Section 5, there is $\eta > 0$, independent of L, such that*

$$|\langle U_1^* g_1, U_2^* g_2\rangle| \leq CA^{-\eta} \prod_{i=1}^{2} \|\mathscr{S}_i g_i\|_{L^2(I_1\cap I_2)}. \tag{7.12}$$

The constant C and exponent η depend only on the fixed geometry, dimension, and degree. In particular, neither the dyadic base nor the phase-distance exponent changes with L.

7.3 Local control of phase-removed adjoint outputs

Lemma 7.4. *For $J \in \mathscr{J}_+$, $\mathfrak{C} = \mathfrak{T}_1$ or $\mathfrak{T}_2^+$, and $y, y' \in 2B_J$,*

$$|H_{i,\mathfrak{C}}(y) - H_{i,\mathfrak{C}}(y')| \leq CL \frac{|y-y'|}{r_J} \operatorname*{ess\,inf}_J M(|g_i^0|). \tag{7.13}$$

Moreover,

$$\sup_{B_J^0} |H_{2,\mathfrak{T}_2^-}| \leq C \operatorname*{ess\,inf}_J M(|g_2^0|). \tag{7.14}$$

The oscillation estimate for the full first tree also holds on $J \in \mathscr{J}_0$.

Proof. At a fixed scale, the output sets E_p are disjoint. For tiles whose support meets $2B_J$, their output sets lie in a ball about $c(J)$ of radius Cr_p. Consequently the sum of the gradient estimates (7.10) at that scale is at most

$$CLr_p^{-n-1} \int_{B(c(J),Cr_p)} |g_i^0| \leq CLr_p^{-1} \operatorname*{ess\,inf}_J M(|g_i^0|).$$

The lower-scale exclusion in Lemma 7.1(b) allows the geometric sum over $s(p) \geq s(J) - C_*$. It is at most $CLr_J^{-1} \operatorname{ess\,inf}_J M(|g_i^0|)$. Integrating the gradient on the segment from y to y' proves (7.13). The fixed enlargement can be increased once in the geometric construction to include these segments.

For (7.14), only N_* consecutive scales can contribute. At each such scale use (7.8) and fixed-scale disjointness instead of (7.10). Each scale is bounded by $C \operatorname{ess\,inf}_J M(|g_2^0|)$. The same lower-scale argument for $\mathscr{J}_0$ proves the final assertion. □

7.4 The contribution with large phase separation

Assume $I_1 \subset I_2$ and let $\Delta = Q_2 - Q_1$. Define

$$A_i(J) = \operatorname*{ess\,inf}_{B_J^0} |U_i^* g_i| + C_2 L \operatorname*{ess\,inf}_J M(|g_i^0|).$$

Lemma 7.4, including (7.14) for the second tree, gives

$$\sup_{2B_J} |H_{1,\mathfrak{T}_1}| + r_J \sup_{2B_J} |\nabla H_{1,\mathfrak{T}_1}| \leq CA_1(J),$$

$$\sup_{2B_J}|H_{2,\mathfrak{T}_2^+}| + r_J \sup_{2B_J}|\nabla H_{2,\mathfrak{T}_2^+}| \le CA_2(J).$$

Thus

$$a_J = \chi_J H_{1,\mathfrak{T}_1}\overline{H_{2,\mathfrak{T}_2^+}}$$

satisfies $\|a_J\|_\infty + r_J \operatorname{Lip}(a_J) \le CA_1(J)A_2(J)$. This function can be extended by zero off I_1 without changing the bound by more than a fixed constant. Indeed the first-tree kernel support lies in $B(c(I_p), 5r_p)$ while $B(c(I_p), 8r_p) \subset I_1$; the contributing first-tree scales are at least comparable to r_J. Hence the first-tree factor vanishes in a boundary neighborhood of I_1 of width comparable to r_J whenever this extension is relevant.

For completeness, on a normalized fixed ball a real polynomial S of bounded degree with nonconstant coefficient norm $\Lambda \ge 1$ satisfies

$$\left|\int e^{iS}a\right| \le C\Lambda^{-\xi}\big(\|a\|_\infty + \operatorname{Lip}(a)\big), \qquad \xi > 0, \tag{7.15}$$

for Lipschitz amplitudes supported in that ball. To see this, one derivative $\partial_j S$ has coefficient norm at least $c\Lambda$. The bounded-degree polynomial sublevel estimate bounds $\{|\partial_j S| \le \Lambda u\}$ by Cu^α for a fixed $\alpha > 0$. On its complement, divide the j-sections at the zeros of the relevant bounded-degree polynomials so that $\partial_j S$ is monotone and has fixed sign. One integration by parts yields $C(\Lambda u)^{-1}(\|a\|_\infty + \operatorname{Lip}(a))$. Optimizing u gives (7.15). The elementary sublevel estimate follows by one-variable Lagrange interpolation and induction on the number of variables.

Apply (7.15) after scaling B_J and use (7.3). It follows that

$$\left|\int e^{i\Delta}a_J\right| \le CA^{-\xi/2}|J|A_1(J)A_2(J).$$

For almost every $y \in B_J^0 \subset J$, the definition gives

$$A_i(J) \le C\mathscr{S}_i g_i(y).$$

One must use the inner ball here: a bound on the low part only on B_J^0 is not a pointwise bound on all of J. Since $|J| \le C|B_J^0|$ and the balls B_J^0 are pairwise disjoint, we have

$$\sum_J |J|A_1(J)A_2(J) \le C\sum_J \int_{B_J^0} \mathscr{S}_1 g_1\, \mathscr{S}_2 g_2 \le C\prod_i \|\mathscr{S}_i g_i\|_{L^2(I_1)}.$$

Thus the high-separation contribution satisfies

$$|\langle U_1^* g_1, (U_{2,\mathfrak{T}_2^+})^* g_2\rangle| \le CA^{-\xi/2}\prod_i \|\mathscr{S}_i g_i\|_{L^2(I_1)}. \tag{7.16}$$

No global norm has replaced the local norms in this step.

7.5 The small-boundary contribution

Let $H_2^- = e^{iQ_2}(U_{2,\mathfrak{T}_2^-})^* g_2$. For $J \in \mathscr{J}_0$, the final assertion of Lemma 7.4 gives

$$\sup_J |U_1^* g_1| \le \langle |U_1^* g_1|\rangle_J + CL \operatorname*{ess\,inf}_J M(|g_1^0|) \le C\langle \mathscr{S}_1 g_1\rangle_J.$$

Therefore

$$|\langle U_1^* g_1, (U_{2,\mathfrak{T}_2^-})^* g_2\rangle| \le C\|\mathscr{S}_1 g_1\|_{L^2(I_1)}\|P_{\mathscr{J}_0}|H_2^-|\|_{L^2(I_1)}. \tag{7.17}$$

Here $P_{\mathscr{J}_0}$ denotes averaging over the disjoint cubes of $\mathscr{J}_0$. This argument uses local oscillation of the first tree; it does not require an unstated projection estimate for a different tree operator.

We estimate the second factor explicitly. At a fixed relative scale $k = s(J) - s(p)$, the absolute sum of adjoint kernels from $\mathfrak{T}_2^-$ is bounded on J by

$$CM(|g_2^0|)\mathbf{1}_{F_{J,k}}.$$

This follows from the size bound and the disjointness of E_p at a fixed scale. By Lemma 7.1(c), only $k \geq c_* \log_D A - C_*$ occur. Jensen and Cauchy–Schwarz give

$$|J| \left(\frac{1}{|J|} \int_J M(|g_2^0|)\mathbf{1}_{F_{J,k}} \right)^2 \leq CD^{-\kappa_* k} \int_J M(|g_2^0|)^2.$$

Minkowski for the outer $\ell^2(J)$ norm and the geometric sum in k show that

$$\|P_{\mathscr{J}_0}|H_2^-|\|_{L^2(I_1)} \leq CA^{-\xi_1} \|M(|g_2^0|)\|_{L^2(I_1)}, \qquad \xi_1 > 0. \tag{7.18}$$

Combining (7.17)–(7.18) proves

$$|\langle U_1^* g_1, (U_{2,\mathfrak{T}_2^-})^* g_2 \rangle| \leq CA^{-\xi_1} \prod_i \|\mathscr{S} g_i\|_{L^2(I_1)}. \tag{7.19}$$

Equations (7.16) and (7.19) prove Theorem 7.3, with $\eta = \min(\xi/2, \xi_1)$. Disjoint tops give zero correlation, and nestedness handles the remaining order of the two tops.

8 Single-tree norm and density estimates

We establish the approximation and projection estimates for the same operators used in the local correlation theorem. Assume that the raw kernels at scale $R_j = D^j$ are disjoint annular pieces of $K_\Theta(z) b(\rho_A(z))$, with bounded mean-zero angular part and a fixed normalized radial amplitude $b \in \mathscr{V}^\rho$. The truncation may be Euclidean or conical elliptic. The selector is an actual equal-amplitude paired-jump witness, and restrictions to intervals of active scales are allowed. Constants below are uniform over finite cutoffs.

The radial function b is fixed, satisfies $\|b\|_\infty + V^\rho(b) \leq 1$, and is included in every raw shell. The path norm $\mathscr{N}$ is the paired-increment norm of Lemma 2.1. Lemma 4.5 gives

$$\|\mathscr{N}(\Phi_{R_j} * f : j \in \mathbb{Z})\|_2 \leq C\|f\|_2, \qquad R_j = D^j. \tag{8.1}$$

Here D is an integral power of two, so this is a subsequence of the smooth dyadic path. The pointwise martingale theorem and the proved square-function comparison supply this estimate with the jump-amplitude supremum inside the spatial norm.

8.1 Single-shell and frequency-ordered estimates

Write $H_{j,u}$ for a weighted partial raw shell, where u ranges over a fixed compact interval and the path vanishes at its upper endpoint. The fixed shell ratio may depend on D. Lemma 4.2 and its proof give, for $R_j|\xi| \simeq 2^m$,

$$\sup_u |\widehat{H}_{j,u}(\xi)| \leq C_\rho \min\{2^m, 2^{-m/(2\rho)}\}, \qquad \operatorname*{ess\,sup}_u |\partial_u \widehat{H}_{j,u}(\xi)| \leq C \min\{2^m, 1\}.$$

The shell still has zero mass because both its endpoints and the amplitude b are expressed in the same radial variable $|Az|$. The proof of (4.12) therefore gives

$$\|V^2(H_{j,u}\Delta_{m,j} f : u)\|_2 \leq C_\rho b_m \|f\|_2, \qquad b_m = \begin{cases} 2^m, & m \leq 0, \\ 2^{-m/(8\rho)}, & m > 0. \end{cases} \tag{8.2}$$

The constants are independent of the smallest singular value of A, the radial location of the variation, and all finite cutoffs.

We next prove a frequency-ordered tail estimate. Suppose that u_j has Fourier support in a fixed annulus at frequency $2^m/R_j$, initially for finitely many j. Then

$$\left\| \mathscr{N}\left(\sum_{\ell \ge j} u_\ell : j \in \mathbb{Z} \right) \right\|_2 \le C \left(\sum_j \|u_j\|_2^2 \right)^{1/2}. \tag{8.3}$$

To prove this, put $U = \sum_j u_j$ and compare the tail with $\Phi_{R_j 2^{-m}} * U$, taking $\widehat{\Phi} = 1$ near zero and $\widehat{\Phi} = 0$ outside a fixed ball. All annuli except a bounded number adjacent to the transition agree exactly. Since each annulus participates in only a bounded number of errors, Plancherel yields

$$\left\| \left(\sum_j \left| \sum_{\ell \ge j} u_\ell - \Phi_{R_j 2^{-m}} * U \right|^2 \right)^{1/2} \right\|_2 \le C \left(\sum_j \|u_j\|_2^2 \right)^{1/2}.$$

The elementary bound $\mathscr{N}(v_j : j) \le 2(\sum_j |v_j|^2)^{1/2}$ controls this error path. Because $R_j 2^{-m}$ is a subsequence of the positive dyadic scales, Lemma 4.5 controls the smooth path by $C\|U\|_2$. Frequency overlap gives $\|U\|_2^2 \le C\sum_j \|u_j\|_2^2$, proving (8.3).

Let $A_j^{b,h}$ denote the unphased weighted shell with the common output-dependent actual witness

$$h_x(r) = N(x)^{-1/2} \sum_{\nu=1}^{N(x)} \zeta_\nu(x) \mathbf{1}_{(a_\nu(x), b_\nu(x)]}(r), \qquad |\zeta_\nu(x)| \le 1.$$

The intervals are ordered and disjoint. For arbitrary $f_j \in L^2$,

$$\left\| \sum_j A_j^{b,h} \Delta_{m,j} f_j \right\|_2 \le C_\rho b_m \left(\sum_j \|f_j\|_2^2 \right)^{1/2}. \tag{8.4}$$

Here and below the sum may initially be finite. Concatenate the partial-shell paths $H_{j,u}\Delta_{m,j} f_j$ in radial order, with zero at the outer endpoint of all selected shells. Its complete-shell increments u_j have the frequency supports required by (8.3). The shell envelope gives

$$\left(\sum_j \|u_j\|_2^2 \right)^{1/2} \le C_\rho a_m \left(\sum_j \|f_j\|_2^2 \right)^{1/2}, \qquad a_m = \begin{cases} 2^m, & m \le 0, \\ 2^{-m/(2\rho)}, & m > 0. \end{cases}$$

Equation (8.3) controls the long part, and (8.2), summed in j, controls the short-square part. Since $a_m \le b_m$, the deterministic long–short inequality controls the full paired norm of this concatenated path by the right side of (8.4). For every x, the sum on its left is one normalized paired witness for that path, which proves the asserted estimate.

A restriction to an output-dependent consecutive interval of shell indices preserves the estimate. Intersect every interval of h_x with the corresponding radial interval and discard the empty intersections. If $N' \le N$ intervals remain, then $N^{-1/2} = \sqrt{N'/N}\,(N')^{-1/2}$; the resulting function is an actual paired witness multiplied by a scalar of modulus at most one. The case $N' = 0$ gives the zero operator. Tree convexity supplies exactly such restrictions. The weight b remains fixed in this argument; the product bh_x is not substituted for an unrestricted element of a variation space when applying (8.4).

8.2 Tree approximation in L^2

Let $j_+(x)$ be the largest active shell index at x in a tree; the operator is zero when no shell is active. Remove the top phase as in Section 7.1. Polynomial metric growth and tree coherence give, on normalized shell j, fixed-order derivative bounds $C_M 2^{j-j_+(x)}$, $j \le j_+(x)$, for the phase difference; see [4, (6.1)

and the proof of Lemma 6.2]. Multiplying by a fixed smooth cutoff on the normalized annulus gives the Fourier expansion

$$e^{i\psi_{i,x}(x-R_jv)}-1=\sum_{\ell\in\mathbb{Z}^n}c_{j,\ell}(x)e^{i\lambda_\ell\cdot v},\qquad |c_{j,\ell}(x)|\le C_M2^{j-j_+(x)}(1+|\ell|)^{-M}.$$

The frequencies λ_ℓ are constant multiples of ℓ. The constants do not depend on the radial amplitude.

For each ℓ, Cauchy–Schwarz in j uses $\sum_{j\le j_+(x)}2^{2(j-j_+(x))}\le C$. If $m>0$ and $|\ell|\le c2^m$ for a sufficiently small fixed c, modulation by λ_ℓ/R_j leaves the input frequency in an annulus comparable to $2^m/R_j$. Equation (8.2) then bounds the remaining square sum. For $|\ell|>c2^m$, use the uniform single-shell L^2 bound and sum $(1+|\ell|)^{-M}$ with $M>n+2$. For $m\le 0$ use this uniform bound for all Fourier modes. Combining these phase-error estimates with (8.4) proves

$$\left\|\sum_j B_j\Delta_{m,j}f_j\right\|_2\le C_\rho w_m\left(\sum_j\|f_j\|_2^2\right)^{1/2},\qquad w_m=\begin{cases}1, & m\le 0,\\ 2^{-\beta m}, & m>0,\end{cases}\qquad \beta=\frac{1}{16\rho}. \tag{8.5}$$

Here B_j is the phase-removed unsmoothed weighted tree shell. The use of $w_m=1$ at low frequencies records the loss caused by the phase error explicitly.

Apply a Littlewood–Paley resolution to $\sum_j B_j(I-P_{\varepsilon R_j})f$. At fixed relative frequency m, the input square sum is at most $C\min\{1,\varepsilon2^m\}\|f\|_2$, because the corresponding frequency annuli have bounded overlap and $|1-\widehat{\phi}(\varepsilon R_j\xi)|\le C\min\{1,\varepsilon2^m\}$ there. Equation (8.5) and summation in m give

$$\left\|\sum_j B_j(I-P_{\varepsilon R_j})\right\|_{2\to2}\le C_\rho\sum_m w_m\min\{1,\varepsilon2^m\}\le C_\rho\varepsilon^\beta. \tag{8.6}$$

The final series is geometric after division at $2^m\simeq\varepsilon^{-1}$, since $0<\beta<1$. This is the exact input smoothing in (7.7); the common output phase and top modulation are isometries.

The unphased part of one unsmoothed tree is controlled on L^2 by Proposition 4.6, because its active scales restrict one actual witness to a radial interval. The phase error is bounded pointwise by

$$C\sum_{j\le j_+(x)}2^{j-j_+(x)}Mf(x)\le CMf(x).$$

Thus the unsmoothed tree is uniformly bounded on L^2. Combining this with (8.6) gives

$$\|U_{\mathfrak{T}}\|_{2\to2}+\|U_{\mathfrak{T}}^{\varepsilon}\|_{2\to2}\le C_{n,d,c_0,\rho}. \tag{8.7}$$

8.3 The logarithmic projection estimate

Let $\mathscr{J}=\mathscr{J}(\mathfrak{T})$ and $\mathscr{L}=\mathscr{L}(\mathfrak{T})$ be the input and output stopping partitions of Section 5. Define $P_{\mathscr{J}}$ and $P_{\mathscr{L}}$ to average over their respective disjoint cubes. Then

$$|\langle U_{\mathfrak{T}}^{\varepsilon}f,g\rangle|\le C\big(1+\log(1/\varepsilon)\big)\|P_{\mathscr{J}}|f|\|_2\|P_{\mathscr{L}}|g|\|_2. \tag{8.8}$$

Here is the proof, including the dependence on ε.

Remove the top phase. Split the smoothed tree into its unphased part and its small phase error. The latter is bounded by $CMP_{\mathscr{J}}|f|(x')$ for any x' in the same output stopping cube as x. Indeed, its kernel is bounded by $C2^{j-j_+(x)}R_j^{-n}$ on $|x-y|\le CR_j$; every input stopping cube meeting that support has radius at most CR_j, by the definition of $\mathscr{J}$. Enlarge the ball to include those cubes, replace $|f|$ by $P_{\mathscr{J}}|f|$, and sum the geometric coefficients.

For the unphased part, write $f=P_{\mathscr{J}}f+(f-P_{\mathscr{J}}f)$. At fixed output x, freeze the actual selector h_x. Its smoothed shell kernel obeys

$$|k_{j,x}^{\varepsilon}(y)-k_{j,x}^{\varepsilon}(y')|\le CR_j^{-n}\min\{1,|y-y'|/(\varepsilon R_j)\}.$$

Using mean zero of $f - P_{\mathscr{J}} f$ on each input cube gives a positive boundary-average operator with relative-scale weights

$$\omega_\varepsilon(k) = \min\{1, C\varepsilon^{-1} D^{-k}\}, \qquad k \geq 0.$$

Indeed, with $r_J = D^{s(J)}$, a dominating operator at relative scale k is

$$S_k u(x) = \sum_{J \in \mathscr{J}} \frac{\mathbf{1}_{B(c(J), CD^k r_J)}(x)}{(D^k r_J)^n} \int_J u(y)\, dy.$$

Every input point belongs to at most one J. Hence, for $y \in J$,

$$S_k^* v(y) = (D^k r_J)^{-n} \int_{B(c(J), CD^k r_J)} v(x)\, dx \leq CMv(y),$$

with a constant independent of k. In this application $u = P_{\mathscr{J}}|f|$, since its integral on J equals that of $|f|$. Moving the output point inside its stopping cube changes the ball constant only: its distance to the original point is bounded by the smallest active radius. Thus $\|S_k\|_{2\to 2} \leq C$. Since

$$\sum_{k \geq 0} \omega_\varepsilon(k) \leq C(1 + \log(1/\varepsilon)),$$

the full boundary-average operator has this logarithmic L^2 bound.

For the averaged input $P_{\mathscr{J}} f$, use the path obtained by smoothing the input on each shell at scale εR_j. Its unphased L^2 jump norm is uniformly bounded: compare it to the original unphased path by (8.4), exactly as in (8.6). The non-tangential version, in which the output center is moved within the initial-scale cube, costs at most $C(1 + \log(1/\varepsilon))Mf$. To verify this, freeze the same selector and the same active-scale interval at both centers, use the last kernel difference bound at each scale, and sum $\omega_\varepsilon(k)$. At the second center the frozen expression is one paired witness for its full smoothed path, so it is bounded by that path norm. Convexity and the output partition guarantee that the distance between the centers is at most a fixed multiple of the smallest active scale. No continuity of h_x in x is used.

Consequently a nonnegative function b_f exists such that $|U_{\mathfrak{T}}^{\varepsilon} f(x)| \leq b_f(x')$ for all x, x' in the same output stopping cube, and

$$\|b_f\|_2 \leq C(1 + \log(1/\varepsilon)) \|P_{\mathscr{J}}|f|\|_2.$$

Average in x' on each cube and use Cauchy–Schwarz. This proves (8.8). The step replaces pointwise Hölder regularity by the displayed summable difference modulus; it does not replace the actual jump selector by an arbitrary bounded-V^2 function.

If $|E \cap L| \leq \delta |L|$ for every $L \in \mathscr{L}$, then

$$\|P_{\mathscr{L}}(|g|\mathbf{1}_E)\|_2 \leq \sqrt{\delta} \|g \mathbf{1}_E\|_2.$$

Thus (8.8) supplies the smoothed row diagonal bound at density $\delta \leq C2^{-m}$. Equations (8.6) and (8.8) verify both single-tree inputs used in the subsequent forest assembly.

9 Forests and the conical theorem

We combine the estimates while keeping the same kernel and normalization at every density generation.

9.1 A double-density estimate for one tree

Proposition 9.1. *Suppose the input and output supports satisfy $|F \cap J| \leq C\delta_2 |J|$ for $J \in \mathscr{J}(\mathfrak{T})$ and $|E \cap L| \leq C\delta_1 |L|$ for $L \in \mathscr{L}(\mathfrak{T})$, where E is the effective output set. Then*

$$\|\mathbf{1}_E U_{\mathfrak{T}} \mathbf{1}_F\|_{2\to 2} \leq C\sqrt{\delta_1 \delta_2} \left[1 + \log \frac{1}{\delta_1 \delta_2}\right] \tag{9.1}$$

when $0 < \delta_1, \delta_2 \leq 1$; the operator is zero if either density vanishes. Constants absorb fixed factors in the density conditions.

Proof. Cauchy–Schwarz on each stopping cube gives

$$\|P_{\mathscr{J}}|f|\|_2 \le C\sqrt{\delta_2}\|f\|_2, \qquad \|P_{\mathscr{L}}|g|\|_2 \le C\sqrt{\delta_1}\|g\|_2$$

for the restricted inputs. Combine (8.6) and (8.8). With $\delta_{12} = \sqrt{\delta_1\delta_2}$ this gives

$$|\langle U_{\mathfrak{T}} f, g\rangle| \le C\{\delta_{12}[1+\log(1/\varepsilon)] + \varepsilon^{\beta}\}\|f\|_2\|g\|_2.$$

Choose $\varepsilon = \varepsilon_0\delta_{12}^{1/\beta}$. This proves the assertion. □

9.2 Local correlation summed into rows

A row is a family of trees with pairwise disjoint tops. The support margin makes a row act diagonally on the corresponding orthogonal sum of input spaces. Let R_i^{ε} be its smoothed operator. For two different rows, Theorem 7.3 gives

$$\|R_i^{\varepsilon}(R_j^{\varepsilon})^*\|_{2\to 2} \le C\varepsilon^{-2}A^{-\eta}. \tag{9.2}$$

Here is the summation without a tree-count loss. Localize g to each tree's effective output set, and let $\mathscr{S}_{\mathfrak{T}}g = |U_{\mathfrak{T}}^*g| + C\varepsilon^{-1}M(|g^0|) + |g^0|$. The single-tree bound in (8.7) gives $\|\mathscr{S}_{\mathfrak{T}}g\|_2 \le C\varepsilon^{-1}\|g^0\|_2$. For a fixed tree in the first row, its intersections with tops of the second row are disjoint. Therefore

$$\sum_{\mathfrak{T},\mathfrak{T}'} \|\mathscr{S}_{\mathfrak{T}}g\|^2_{L^2(I_{\mathfrak{T}}\cap I_{\mathfrak{T}'})} \le \sum_{\mathfrak{T}} \|\mathscr{S}_{\mathfrak{T}}g\|_2^2 \le C\varepsilon^{-2}\|g\|_2^2.$$

The second row has the same bound. Cauchy–Schwarz over tree pairs proves (9.2). A global two-tree inequality would not justify this step; the intersection in the local theorem is used exactly.

Proposition 9.2 (Two forest estimates)**.** *Let $\mathfrak{F}_m$ be a forest of density generation m, and suppose $\delta_2(\mathfrak{F}_m) \le \zeta \le 1$. Then*

$$\|T_{\mathfrak{F}_m}\|_{2\to 2} \le C(1+m)2^{-m/2}, \tag{9.3}$$

$$\|T_{\mathfrak{F}_m}\mathbf{1}_F\|_{2\to 2} \le C\sqrt{\zeta}[1+m+\log(1/\zeta)]. \tag{9.4}$$

Proof. Fix $\varepsilon_0 = 1/8$ and use the common smoothing scale $\varepsilon_m = \varepsilon_0 2^{-K(m+1)}$ and separation $A_m = 2^{Z(m+1)}$. The constants satisfy (5.13). Because tops have multiplicity at most 2^m and effective outputs of distinct trees are disjoint,

$$\|T_{\mathfrak{F}_m} - T_{\mathfrak{F}_m}^{\varepsilon_m}\|_{2\to 2} \le C2^{m/2}\varepsilon_m^{\beta} \le C2^{-3m/2}.$$

The smoothed projection estimate and the output density give $\|R_i^{\varepsilon_m}\|_{2\to 2} \le C(1+m)2^{-m/2}$ for every row. There are at most 2^m rows. Using the disjoint effective output sets in the Gram matrix and (9.2),

$$\|T_{\mathfrak{F}_m}^{\varepsilon_m}\|_{2\to 2}^2 \le C(1+m)^2 2^{-m} + C2^m\varepsilon_m^{-2}A_m^{-\eta} \le C(1+m)^2 2^{-m}.$$

This proves (9.3).

For (9.4), use Proposition 9.1 with output density $C2^{-m}$ and input density ζ. Sum the squared norms over trees, using disjoint outputs and top overlap at most 2^m. The factors $2^{m/2}$ and $2^{-m/2}$ cancel. This proves the second estimate. These are estimates for the same unsmoothed operator; their proofs need not choose the same auxiliary smoothing scale. □

9.3 Strong bounds for $1 < p < 2$

Fix $1 < p < 2$. A geometric mean of the two bounds in Proposition 9.2, with powers $1 - 1/p$ and $1/p$, gives

$$\|T_{\mathfrak{F}_m}\mathbf{1}_F\|_{2\to 2} \le C_p(1+m)2^{-c_p m}\zeta^{1/p-1/2}, \qquad c_p = (p-1)/(2p) > 0. \tag{9.5}$$

To verify the logarithm absorption, the initial exponent of ζ is $1/(2p)$ and $1/(2p)-(1/p-1/2)=(p-1)/(2p)>0$. The positive difference of exponents absorbs the logarithmic factor.

For an antichain and $|f|\le \mathbf{1}_F$, size and input density give $\|T_{\mathfrak{A}}f\|_\infty \le C\zeta$. Since a point belongs to only one antichain output tile, $|T_{\mathfrak{A}}f|\le CMf$. Put $r=2-1/p>1$. Interpolation between these elementary L^r and L^∞ bounds gives

$$\|T_{\mathfrak{A}}f\|_2 \le C_p \zeta^{1/(2p)}|F|^{1/2}.$$

Take the geometric mean of this bound with Proposition 6.2, with powers $2-p$ and $p-1$. At density generation m,

$$\|T_{\mathfrak{A}_m}f\|_2 \le C_p 2^{-\vartheta(p-1)m}\zeta^{1/p-1/2}|F|^{1/2}. \tag{9.6}$$

For $|G|>0$, apply Proposition 5.1 with its ambient set H, put $G'=G\cap H$, and pair the surviving operator with $g\mathbf{1}_{G\setminus G'}$, $|g|\le 1$. The discarded tiles vanish on this set, so this is exactly the same restricted full operator. For $|G|=0$ the desired integral is zero. The number of families per level is polynomial in m, while (9.5) and (9.6) decay geometrically. With $\zeta \le C\min\{1,|F|/|G|\}$, this yields

$$\int_{G\setminus G'}|Tf| \le C_p|F|^{1/p}|G|^{1-1/p}, \qquad |G'|\le |G|/2. \tag{9.7}$$

Every constant is uniform over the fixed finite linearization. Apply this to finite subsets of level sets of $|Tf|$ to obtain restricted weak type (p,p). Restricted weak-type interpolation between two indices in $(1,2)$ proves the strong estimate throughout that interval. Apply the argument to the linearization maximizing the finite path norm; its sublinearity follows from Lemma 2.1.

9.4 Strong bounds for $2<p<\infty$

No adjoint identity for the maximally modulated operator is used. For each fixed $r>1$, remove a tree's top phase. Its unphased part is a restriction of one jump witness and is bounded by Proposition 4.6. Its phase error is bounded by $C\sum_{j\le j_+(x)} 2^{j-j_+(x)}Mf \le CMf$. Hence

$$\|T_{\mathfrak{T}}\|_{r\to r} \le C_r.$$

Rows have disjoint tops, and different rows have disjoint effective outputs, so a forest with at most 2^m rows satisfies

$$\|T_{\mathfrak{F}_m}\|_{r\to r} \le C_r 2^{m/r}.$$

For $p>2$ take $r=2p$ and interpolate with (9.3). The L^2 interpolation weight is $1/(p-1)$; consequently

$$\|T_{\mathfrak{F}_m}\|_{p\to p} \le C_p(1+m)^{1/(p-1)}2^{-m/[p(p-1)]}. \tag{9.8}$$

An antichain is pointwise bounded by CM, so interpolation with (6.8) gives

$$\|T_{\mathfrak{A}_m}\|_{p\to p} \le C_p 2^{-\vartheta m/(p-1)}. \tag{9.9}$$

For bounded compactly supported f, choose $F=\operatorname{supp} f$ in the geometric decomposition and discard its input-density improvement. Again use $G'=G\cap H$ for the output restriction and remove spatial tiles only when $I_p\subset H$; null G gives a zero integral. Summing (9.8)–(9.9) gives

$$\int_{G\setminus G'}|Tf| \le C_p\|f\|_p|G|^{1-1/p}.$$

The level-set argument proves weak type (p,p) for each $p>2$. Interpolation gives strong type for every $p>2$. Interpolation between an exponent below two and an exponent above two then yields the case $p=2$. This completes the proof of Theorem 2.2 for finite phase and parameter sets.

Remark 9.3 (Uniformity of constants). The dimension, degree, finite radial exponent, cone aperture, and normalized support annuli are fixed first. The positive shell and correlation exponents depend only on these data. In particular the single-tree exponent $\beta = 1/(16\rho)$ is independent of the geometric integer a and base D. Choose the fixed geometric a and $D = 2^{100a^2}$, then choose K and a sufficiently large structural Z by (5.13), using the positive correlation exponent for this same grid. The parameter a used in D and in the density is never changed in this selection. Splitting a fixed-ratio shell changes constants but does not change the stated positive exponents. The smoothness cost is recorded as $L = C\varepsilon^{-1}$ and never absorbed into a varying geometric parameter. The analytic estimates are uniform in the least singular value of A, in phase coefficients, in finite cutoffs, and in witness lengths. These uniformities permit the angular normalization in the next section.

10 Cancellative atomic decomposition on the sphere

Throughout the atomic argument assume $n \geq 2$. For $n = 1$, put $H_0^1(\mathbb{S}^0) = \{\Omega : \Omega(1) + \Omega(-1) = 0\}$ with its L^1 norm. Then $\|\Omega\|_\infty \leq \|\Omega\|_{H^1}$, and Theorem 2.2 with $A = I$ proves the main estimate, including every fixed radial profile $b \in \mathscr{V}^\rho$, $1 \leq \rho < \infty$. No strict angular enlargement is asserted in this two-point case. Put $m_n = \sigma(\mathbb{S}^{n-1})$, fix $r_* = \pi/16$, and let

$$A_\tau = \operatorname{diag}(1, \tau, \dots, \tau), \quad \Gamma = \{\theta : \theta_1 \geq 2^{-1/2}\}, \quad \tau_* = \tan r_*.$$

The angular Hardy norm is fixed below. The spatial input exponent remains $1 < p < \infty$. For comparison with the angular $L\log L$ class, write $\mathfrak{L}(\Omega) = \mathfrak{L}(\Omega)$ as in (1.2). A logarithm will be used only to prove the inclusion of mean-zero $L\log L$ data in the angular Hardy space; the Hardy-space estimate itself uses its atomic norm.

Definition 10.1 (The real angular Hardy space)**.** A cancellative spherical $(1,\infty)$-atom is a measurable function a supported in a geodesic ball $B \subset \mathbb{S}^{n-1}$ such that $\int a\, d\sigma = 0$ and $\|a\|_\infty \leq \sigma(B)^{-1}$. The whole sphere is allowed as a supporting ball. The exceptional constant atom is $a_* = m_n^{-1}\mathbf{1}_{\mathbb{S}^{n-1}}$. We use the standard real atomic definition on the sphere: $\Omega \in H^1(\mathbb{S}^{n-1})$ if

$$\Omega = c a_* + \sum_{\nu \geq 1} \lambda_\nu a_\nu, \qquad |c| + \sum_{\nu \geq 1} |\lambda_\nu| < \infty.$$

The norm $\|\Omega\|_{H^1}$ is the infimum of this coefficient sum. All these expansions converge absolutely in L^1, because every atom has L^1 norm at most one. This is the real atomic Hardy space of the angular variable; see [8, 9]. It is not a spatial $H^1(\mathbb{R}^n)$ assumption on the input. Write

$$H_0^1(\mathbb{S}^{n-1}) = \{\Omega \in H^1(\mathbb{S}^{n-1}) : \textstyle\int \Omega\, d\sigma = 0\}.$$

Integration of the absolutely convergent expansion shows that $c = \int \Omega\, d\sigma$. Thus every mean-zero element admits an expansion containing only cancellative atoms, and $\|\Omega\|_1 \leq \|\Omega\|_{H^1}$.

Definition 10.2. A small-cap atom is a measurable complex function a supported in a geodesic ball $B(e,r) \subset \mathbb{S}^{n-1}$, $0 < r \leq r_*$, such that

$$\int a\, d\sigma = 0, \qquad \|a\|_\infty \leq \sigma(B(e,r))^{-1}.$$

A global atom satisfies $\int a = 0$ and $\|a\|_\infty \leq m_n^{-1}$, without a support restriction.

Lemma 10.3 (A dyadic spherical filtration)**.** *There is a nested measurable partition of $\mathbb{S}^{n-1}$, beginning with the whole sphere, with the following properties. The number of children of each set is uniformly bounded; child and parent measures are comparable from below; the filtration differentiates L^1 functions; and each non-root set Q is contained in a spherical ball B_Q with $\sigma(B_Q) \leq C_n\sigma(Q)$. Sets with* $\operatorname{rad}(B_Q) > r_*$ *have measure bounded below by a positive dimensional constant.*

Proof. Partition the boundary of the cube $[-1,1]^n$ into its faces, assign shared boundaries by a fixed measurable rule, and project radially to the sphere. The radial maps on the faces are bi-Lipschitz with surface Jacobians bounded above and below by dimensional constants. Subdivide each face into its usual dyadic cubes and transfer the subdivisions to the sphere. Boundaries have surface measure zero and can retain the same fixed assignment convention at every generation. The assertions follow from Euclidean dyadic differentiation and the fixed bounds on these finitely many charts. Diameter and measure are comparable to the appropriate powers of the face-cube sidelength. Consequently only finitely many generations can have enclosing balls larger than r_*. □

Write $M_{\mathscr{D}}$ for the maximal operator of this filtration, applied to $|\Omega|$.

Lemma 10.4 (The single logarithmic estimate)**.** *For $\Omega \in L\log L(\mathbb{S}^{n-1})$,*

$$\|M_{\mathscr{D}}\Omega\|_1 \le C_n \mathfrak{L}(\Omega). \tag{10.1}$$

Proof. The zero function requires no argument. Put $A_0 = \|\Omega\|_1$ and $\lambda_0 = A_0/m_n$. For $\lambda \ge \lambda_0$, the part of $|\Omega|$ bounded by $\lambda/2$ has maximal function at most $\lambda/2$. The maximal disjoint stopping sets for the remaining part therefore give

$$\sigma\{M_{\mathscr{D}}\Omega > \lambda\} \le \frac{2}{\lambda}\int_{\{|\Omega|>\lambda/2\}} |\Omega|\, d\sigma.$$

Integrate in λ. Tonelli yields

$$\|M_{\mathscr{D}}\Omega\|_1 \le A_0 + 2\int |\Omega| \log^+(2|\Omega|/\lambda_0)\, d\sigma.$$

Since m_n depends only on n, this is bounded by the right-hand side of (10.1). □

Proposition 10.5 (Summable small-cap Hardy expansion)**.** *If $\Omega \in H^1_0(\mathbb{S}^{n-1})$, then*

$$\Omega = \sum_{\nu=1}^{\infty} \lambda_\nu a_\nu \quad \text{in } L^1(\mathbb{S}^{n-1}), \qquad \sum_\nu |\lambda_\nu| \le C_n \|\Omega\|_{H^1}, \tag{10.2}$$

where every a_ν is a small-cap atom. Conversely, every such absolutely summable series belongs to H^1_0, with Hardy norm at most its coefficient sum. In particular, restricting the atomic norm to small caps gives an equivalent norm on H^1_0.

Proof. The zero element has the zero expansion. Otherwise choose a mean-zero Hardy expansion with coefficient sum at most twice its positive Hardy norm; its exceptional constant term is zero. Atoms supported in caps of radius greater than r_* are uniformly bounded multiples of global atoms, by positivity and rotational invariance of the measure of a cap of the fixed radius r_*. It remains to decompose a global atom uniformly.

Choose a finite cover of the sphere by open geodesic caps B_i of a fixed radius at most r_*. The overlap graph is connected: otherwise the unions of two graph components would disconnect the connected sphere. Every nonempty overlap has positive surface measure. Disjointize the cover to measurable sets $P_i \subset B_i$ and put $\psi_i = \mathbf{1}_{B_i}/\sigma(B_i)$. For a global atom a, set $m_i = \int_{P_i} a$. The functions $\mathbf{1}_{P_i} a - m_i\psi_i$ are bounded, mean zero, supported in B_i, and normalize to finitely many small-cap atoms with uniformly bounded coefficient sum. Also $\sum m_i = 0$ and $\sum |m_i| \le \|a\|_1 \le 1$.

Fix a root of the overlap graph and, for each vertex i, one finite path from i to the root. Then

$$\sum_i m_i\psi_i = \sum_i m_i(\psi_i - \psi_{\text{root}}).$$

Telescope each difference along its fixed path. For an edge uv let $\psi_{uv} = \mathbf{1}_{B_u\cap B_v}/\sigma(B_u \cap B_v)$ and write $\psi_u - \psi_v = (\psi_u - \psi_{uv}) + (\psi_{uv} - \psi_v)$. Each difference is bounded, mean zero, and supported in one of the

small caps. All paths and all positive measures are fixed in advance, so normalization gives a uniform finite coefficient bound. No smooth partition, smooth bump, or spanning-tree existence is needed.

Apply this genuine finite decomposition to every atom of the original absolutely summable expansion. The resulting countable double series has absolutely summable coefficients; reindex it by the natural numbers. Its L^1 limit is the original datum, and its coefficient sum is bounded by a dimensional constant times the original coefficient sum. Absorb the initial factor two into the dimensional constant. Taking infima also gives the asserted comparison of atomic norms. The converse follows directly from the definition, absolute L^1 convergence, and termwise integration. □

Proposition 10.6 (The $L\log L$ inclusion)**.** *If $\int \Omega\, d\sigma = 0$ and $\mathfrak{L}(\Omega) < \infty$, then*

$$\Omega \in H^1_0(\mathbb{S}^{n-1}), \qquad \|\Omega\|_{H^1} \le C_n \mathfrak{L}(\Omega).$$

Proof. Write $d = n-1 \ge 1$, $M = \|\Omega\|_1$, and $E_A(f) = \int |f| \log^+(|f|/A)$. The case $M = 0$ is immediate. We give a finite-chart proof using ordinary Euclidean dyadic cubes; in particular, this proof does not require the preceding spherical filtration lemma.

Cover the sphere by the $2n$ closed coordinate patches $U_{i,\varepsilon} = \{\theta : |\theta| = 1, \varepsilon\theta_i \ge 1/n\}$, $\varepsilon \in \{-1,1\}$. Disjointize this finite cover to measurable sets $P_\alpha \subset U_\alpha$. For each patch choose the normalized indicator ψ_α of the geodesic cap of radius $\pi/16$ about its signed coordinate pole. This cap has positive surface measure and is contained in U_α. Set

$$m_\alpha = \int_{P_\alpha} \Omega, \qquad \Omega_\alpha = \mathbf{1}_{P_\alpha}\Omega - m_\alpha \psi_\alpha, \qquad R = \sum_\alpha m_\alpha \psi_\alpha.$$

Then $\sum |m_\alpha| \le M$, each Ω_α is supported in its patch and has mean zero, $\|\Omega_\alpha\|_1 \le 2M$, and $\Omega = R + \sum \Omega_\alpha$ in L^1. The remainder has mean zero and $\|R\|_\infty \le C_n M$, so it is a constant multiple, of size at most $C_n M$, of a global atom. The elementary inequality, for $x, y \ge 0$,

$$(x+y)\log^+((x+y)/M) \le 2\log 2\,(x+y) + 2x\log^+(x/M) + 2y\log^+(y/M)$$

and the fixed bounds for the normalized indicators give

$$E_M(\Omega_\alpha) \le C_n\{M + E_M(\Omega)\}.$$

It remains to treat one mean-zero function F on a coordinate patch, with $\|F\|_1 \le 2M$. By the antipodal isometry it suffices to use a positive patch. Its gnomonic parameterization is $T(z) = (1,z)/|(1,z)|$, with the designated coordinate inserted in the appropriate position. The inverse coordinate norm on the patch is at most n. Fix, dimensionally, a translated dyadic root box containing this inverse image in its interior, of volume at least one. Choose a fixed bounded source ball S containing the translated-back root box. The actual pullback measure satisfies

$$K_n^{-1}\, dz \le T^*(d\sigma) \le dz \quad \text{on } S.$$

Indeed the chart is Lipschitz with constant one, and its inverse is Lipschitz on this bounded domain; the already proved identification of surface measure with Euclidean-normalized Hausdorff measure gives both inequalities. The Radon–Nikodym derivative w therefore satisfies $K_n^{-1} \le w \le 1$ almost everywhere. The zero extension of $f(z) = w(z)F(Tz)$ has mean zero, L^1 mass $\|F\|_1$, and $E_M(f) \le E_M(F)$. Translating the source makes its support lie in the fixed ordinary dyadic root cube.

For the usual Euclidean dyadic grid perform the maximal-cube Calderón–Zygmund decomposition of f at $t_k = 2^{k+1}M$. Write its good part as g_k and its disjoint maximal bad cubes as $\mathscr{Q}_k$. These cubes lie inside the root cube: an ancestor of the root has average at most $\|f\|_1/|Q_0| \le 2M$, and any bad cube intersecting the support is nested with the root. The usual parent-cube estimate gives $|g_k| \le 2^d t_k$ almost everywhere. For $Q \in \mathscr{Q}_k$ put

$$h_{k,Q} = \mathbf{1}_Q(g_{k+1} - g_k).$$

Actual nesting of the stopping cubes gives $\int h_{k,Q} = 0$ and $|h_{k,Q}| \le 3 \cdot 2^d t_k$, and $g_{k+1} - g_k = \sum_{Q \in \mathcal{Q}_k} h_{k,Q}$. The good part preserves the integral. Moreover $\|f - g_k\|_1 \le 2 \int_{O_k} |f| \to 0$, since $|O_k| \le \|f\|_1 / t_k \to 0$.

The coefficient payment is a direct tail count, not an invocation of a spherical maximal theorem. For a bad cube at height $2a$, $a|Q| \le \int_Q |f| \mathbf{1}_{\{|f|>a\}}$. Summing the disjoint cubes and then $a = 2^k M$ yields

$$\sum_{k \ge 0} t_k \sum_{Q \in \mathcal{Q}_k} |Q| \le 2 \left(\|f\|_1 + (\log 2)^{-1} E_M(f) \right).$$

Consequently $f = g_0 + \sum_{k,Q} h_{k,Q}$ absolutely in L^1, with total amplitude-times-cube-volume bounded by $C_n(M + E_M(f))$; the initial term is bounded, mean zero, and supported in the fixed root box, with the same dimensional payment.

Transport signed densities back by $h \mapsto (h/w) \circ T^{-1}$, extended by zero. This is an L^1 isometry on the restricted source measure and preserves integrals, hence transports the countable series. The image of any translated cube of sidelength ℓ is contained in a spherical supporting ball of measure at most $C_n \ell^d$. For small ℓ this follows from the chart Lipschitz bound, the angle–chord comparison, and the proved cap measure bound; for large ℓ use the whole sphere. Since $w^{-1} \le K_n$, every transported $h_{k,Q}$ is a cancellative spherical atom times a coefficient at most $C_n t_k |Q|$. Thus $\|F\|_{H^1} \le C_n(M + E_M(F))$. Apply this to the finitely many Ω_α and add the global remainder. The result is $\|\Omega\|_{H^1} \le C_n(M + E_M(\Omega)) = C_n \mathfrak{L}(\Omega)$, with an actual absolutely summable atomic expansion. □

For completeness the inclusion holds without imposing mean zero. For $\Omega \in L \log L$, write $c = \int \Omega \, d\sigma$ and $\Omega = c a_* + (\Omega - c a_*)$. The first summand has Hardy norm $|c| \le \|\Omega\|_1$. The second has mean zero and satisfies $\mathfrak{L}(\Omega - c a_*) \le C_n \mathfrak{L}(\Omega)$ directly: with $M = \|\Omega\|_1$ and $M' = \|\Omega - c a_*\|_1 \le 2M$, the entropy subtraction inequality above bounds $E_M(\Omega - c a_*)$ by $C_n(M + E_M(\Omega))$. Changing the normalization from M to $M' > 0$ costs at most $M' \log^+(M/M') \le M$; the case $M' = 0$ is immediate. The gauge bound and the mean-zero inclusion therefore give $\|\Omega\|_{H^1} \le C_n \mathfrak{L}(\Omega)$.

Proposition 10.7 (Strictness of the angular extension)**.** *For $n \ge 2$ the inclusion $L \log L(\mathbb{S}^{n-1}) \cap \{\int \Omega = 0\} \subset H^1_0(\mathbb{S}^{n-1})$ is strict.*

Proof. Put $d = n - 1 \ge 1$ and fix a positive gnomonic chart $T : \mathbb{R}^d \to \mathbb{S}^d$. On each fixed bounded source ball its surface density w satisfies $0 < c_d \le w \le 1$. For $j \ge 0$ put $N_j = 4^j + j + 3$. Take the dyadic cube Q_j of sidelength 2^{-N_j} whose first-coordinate interval begins at 2^{-j} and whose other intervals begin at zero. These cubes are pairwise disjoint: their first-coordinate intervals are contained in $[2^{-j}, (1 + 1/8)2^{-j}]$. They lie in a fixed bounded source ball. Choose two distinct dyadic children Q_j^+, Q_j^- of Q_j and set

$$A_j = (2|Q_j^+|)^{-1}, \qquad u_j = A_j(\mathbf{1}_{Q_j^+} - \mathbf{1}_{Q_j^-}), \qquad g_j(Tx) = u_j(x)/w(x),$$

extending g_j by zero outside the chart images of these children. The exact change of measure gives $\int g_j = 0$ and $\|g_j\|_1 = 1$. Its support is contained in TQ_j, and the cap envelope bound $\sigma(B_j) \le C_d |Q_j|$, together with $w^{-1} \le c_d^{-1}$, shows that g_j is a multiple of an actual cancellative spherical atom with coefficient at most a fixed dimensional constant, since $A_j |Q_j| = 2^{d-1}$. Consequently

$$\Omega = \sum_{j \ge 0} 2^{-j} g_j$$

converges absolutely in L^1 and defines an element of H^1_0. The supports are disjoint, and $w \le 1$ implies $|g_j| \ge A_j$ wherever $g_j \ne 0$. Thus the entropy integral on the jth support alone is at least

$$2^{-j} \log^+(2^{-j} A_j) \ge 2^{-j} 4^j \log 2 = 2^j \log 2.$$

Indeed $\log A_j = (d(N_j + 1) - 1) \log 2$, so $\log(2^{-j} A_j) \ge 4^j \log 2$ for every $d \ge 1$. On this finite-measure sphere, $L \log L$ membership implies finiteness of the unnormalized entropy integral. The displayed lower bounds are unbounded, whereas each would then be bounded by that full integral. Hence $\Omega \notin L \log L$, proving strictness. □

11 Affine normalization and the exact sharp truncation

Lemma 11.1 (Spherical Jacobian)**.** *For an invertible real matrix L, the map $\theta \mapsto L\theta/|L\theta|$ has surface Jacobian $|\det L||L\theta|^{-n}$.*

Proof. For any bounded measurable a, make the substitution $w = Lz$ in

$$\int_{1<|Lz|<2} |\det L|\, a((Lz)')|Lz|^{-n}\, dz.$$

Integrating in polar coordinates first in w and then in z gives, respectively,

$$(\log 2)\int a(\omega)\, d\sigma(\omega), \qquad (\log 2)\int |\det L||L\theta|^{-n} a((L\theta)')\, d\sigma(\theta).$$

This identity for every a proves the formula. □

Proposition 11.2 (Uniform normalization of one cap atom)**.** *Let a be a small-cap atom supported in $B(e,r)$, $0 < r \le r_*$. Choose $U \in O(n)$ with $Ue_1 = e$, let $\tau = \tan r$, and set $L = UA_\tau$. Define*

$$\Theta_L(\theta) = |\det L||L\theta|^{-n} a\left(\frac{L\theta}{|L\theta|}\right). \tag{11.1}$$

Then

$$\operatorname{supp}\Theta_L \subset \Gamma, \qquad \|\Theta_L\|_\infty \le C_n, \qquad \int \Theta_L\, d\sigma = 0.$$

For $f_L(v) = f(Lv)$, the exact transformed trajectory is

$$T^{Q,b}_{a,t} f(Lu) = S^{Q\circ L, A_\tau, b}_{\Theta_L, t} f_L(u). \tag{11.2}$$

Proof. If $\Theta_L(\theta) \neq 0$, the cap condition gives $\theta_1 > 0$ and $\tau|\theta_\perp|/\theta_1 \le \tan r = \tau$. Hence $|\theta_\perp| \le \theta_1$, so $\theta_1 \ge 2^{-1/2}$ and $|A_\tau\theta| \ge 2^{-1/2}$. Since $|\det L| = \tau^{n-1}$ and $\sigma(B(e,r)) \asymp_n r^{n-1}$, (11.1) gives the uniform L^∞ bound. Mean zero follows from Lemma 11.1.

The change of variables $x = Lu$, $y = Lv$ gives

$$|\det L| K_a(L(u-v)) = K_{\Theta_L}(u-v), \qquad |L(u-v)| = |A_\tau(u-v)|.$$

It also sends $e^{iQ(y)}$ to $e^{iQ(Lv)}$ and the radial factor $b(|x-y|)$ to $b(|A_\tau(u-v)|)$, proving (11.2). Both the sharp cutoff and the radial profile are transformed exactly into functions of $|A_\tau(u-v)|$. No Euclidean cutoff or Euclidean radial profile is substituted for them. □

Corollary 11.3 (Uniform estimate for angular atoms)**.** *Let $d \ge 0$, $1 < p < \infty$, and $1 \le \rho < \infty$. For every small-cap atom, every finite $\mathscr{F} \subset \mathscr{P}_d(\mathbb{R}^n)$ and finite $D \subset (0,\infty)$, and every fixed $b \in \mathscr{V}^\rho$, the following holds for bounded compactly supported f:*

$$\left\| \max_{Q\in\mathscr{F}} \mathscr{N}_D(T^{Q,b}_{a,t} f : t \in D) \right\|_p \le C_{n,d,p,\rho} \|b\|_{\mathscr{V}^\rho} \|f\|_p.$$

Proof. Apply (2.8) to (11.2). Composition by L is an invertible map on $\mathscr{P}_d$, preserving total degree. The change of variables in the output norm contributes $|\det L|^{1/p}$, whereas $\|f_L\|_p = |\det L|^{-1/p}\|f\|_p$. These factors cancel exactly. Use the uniform angular bound in Proposition 11.2. □

12 Absolute annular integrals and local tail bounds

Lemma 12.1 (Local positive tail estimate)**.** *Let $\Omega \in L^1(\mathbb{S}^{n-1})$, $1 \le p < \infty$, $a > 0$, and $M > 0$. Set*

$$k_a^+(z) = |\Omega(z')||z|^{-n}\mathbf{1}_{\{|z|>a\}}.$$

Then

$$\|k_a^+ * |f|\|_{L^p(B(0,M))} \le C_{n,a,M,p}\|\Omega\|_1\|f\|_p. \tag{12.1}$$

Proof. The portion in any fixed bounded annulus is an L^1 convolution kernel. For the tail, write $k_j^+(z) = |\Omega(z')||z|^{-n}\mathbf{1}_{\{2^j<|z|\le 2^{j+1}\}}$. The row integral is at most $C\|\Omega\|_1$. For each fixed y, the intersection of a ray $y + r\theta$ with $B(0,M)$ has length at most $2M$. Consequently

$$\sup_y \int_{B(0,M)} k_j^+(x-y)\,dx \le 2M2^{-j}\|\Omega\|_1.$$

Thus the positive operator $f \mapsto \mathbf{1}_{B(0,M)}(k_j^+ * f)$ has $L^1 \to L^1$ norm at most $CM2^{-j}\|\Omega\|_1$ and $L^\infty \to L^\infty$ norm at most $C\|\Omega\|_1$. Interpolation gives an L^p norm at most $C(M2^{-j})^{1/p}\|\Omega\|_1$. This is summable for the large scales. Adding the bounded annuli proves (12.1). □

Corollary 12.2 (Common representatives)**.** *For $\Omega \in L^1$, a fixed bounded measurable b, and $f \in L^p$, $1 \le p < \infty$, there is a full-measure set X_f, independent of Q and t, on which every $T^{Q,b}_{\Omega,t}f$ is absolutely convergent. On X_f the path is jointly continuous in positive t and the finite-dimensional coefficient vector of Q. Its absolute tail tends to zero as $t \to \infty$, uniformly over all Q.*

Proof. Apply Lemma 12.1 on the countable collection of output balls $B(0,M)$, $M \in \mathbb{N}$, and lower cutoffs $a = 1/N$, $N \in \mathbb{N}$. At a point outside the resulting null set, all positive truncations are absolutely integrable after multiplying the positive majorant by $\|b\|_\infty$. For sequences of positive cutoffs and polynomial coefficients converging to a fixed cutoff and coefficients, use a fixed smaller positive cutoff as the dominating integrable kernel. Spheres have Lebesgue measure zero, so the cutoff indicators converge almost everywhere in the integration variable. Dominated convergence proves joint continuity. The uniform tail statement follows from $|e^{iQ(y)}| = 1$, boundedness of b, and the vanishing tail of an absolutely integrable function. All coordinates are set to zero on X_f^c. The construction depends on no individual phase and takes no uncountable union of null sets. □

13 Proof of the angular endpoint theorem

Proof of Theorem 1.1. Fix $b \in \mathscr{V}^\rho$, $1 \le \rho < \infty$. Start with bounded compactly supported f, and choose the atomic expansion from Proposition 10.5. Let $\Omega_N = \sum_{\nu\le N}\lambda_\nu a_\nu$. For finite parameter and phase sets, Lemma 2.1 and Corollary 11.3 give

$$\left\|\max_Q \mathscr{N}_D(T^{Q,b}_{\Omega_N,t}f)\right\|_p \le C\|b\|_{\mathscr{V}^\rho}\sum_{\nu\le N}|\lambda_\nu|\|f\|_p \le C\|b\|_{\mathscr{V}^\rho}\|\Omega\|_{H^1}\|f\|_p.$$

All constants are uniform over these finite sets.

Let $\operatorname{supp} f \subset B(0,R_f)$. For fixed x and $a > 0$,

$$\sup_Q \sup_{t\ge a}|T^{Q,b}_{\Omega_N,t}f(x) - T^{Q,b}_{\Omega,t}f(x)| \le \|b\|_\infty\|f\|_\infty\|\Omega_N - \Omega\|_1 \log^+\left(\frac{|x|+R_f}{a}\right) \longrightarrow 0. \tag{13.1}$$

This follows by polar coordinates and absolute values. Hence every finite strict jump witness for any fixed phase passes to the approximating trajectories. Equivalently, one can use continuity of $\mathscr{N}_D$ on the finite-dimensional coordinate space. Fatou gives the same bound for Ω, first for finite phase and

parameter sets. Take increasing finite sets $\mathscr{F}_m, D_m$ exhausting, respectively, polynomials with rational nonconstant coefficients and positive rational cutoffs. Joint continuity from Corollary 12.2 gives

$$\sup_{Q\in\mathscr{P}_d} \mathscr{N}_{(0,\infty)}(T^{Q,b}_{\Omega,t} f) = \sup_m \max_{Q\in\mathscr{F}_m} \mathscr{N}_{D_m}(T^{Q,b}_{\Omega,t} f).$$

Indeed, approximate the distinct endpoints of a finite witness by rational endpoints preserving their order, using a single rational approximation for each shared endpoint. Approximate its one phase by a single rational coefficient vector for the entire witness. This approximates the witness's value arbitrarily well. The reverse inequality is immediate. The right side is a countable supremum of measurable functions. Increasing finite exhaustions, monotone convergence, and $\mathbf{J} \le \mathscr{N}$ now give (1.6) for core data. In the equivalent argument with strict jumps, only the finitely many strict margins of one witness are used at a time.

For general $f \in L^p$, choose bounded compactly supported f_m with $\sum_m \|f_m - f\|_p < \infty$. For each positive rational a and integer $M > 0$, Minkowski's inequality and Lemma 12.1 give

$$\left\| \sum_m k_a^+ * |f_m - f| \right\|_{L^p(B(0,M))} < \infty.$$

Consequently, outside one null set, for every positive rational a,

$$k_a^+ * |f_m - f|(x) \longrightarrow 0.$$

Consequently

$$\sup_Q \sup_{t\ge a} |T^{Q,b}_{\Omega,t}(f_m - f)(x)| \le \|b\|_\infty k_a^+ * |f_m - f|(x) \longrightarrow 0.$$

Every finite strict witness therefore passes to the limit. Fatou and $\|f_m\|_p \to \|f\|_p$ prove (1.6). Corollary 12.2 identifies these trajectories with the original absolutely convergent positive truncations and supplies their common representative and zero anchor.

Finally, at a fixed output point x, define $Q_{P,x}(y) = P(x,y) - P(x,0)$. Then

$$T^{P,b}_{\Omega,t} f(x) = e^{iP(x,0)} T^{Q_{P,x},b}_{\Omega,t} f(x).$$

The scalar factor is independent of t. Conversely every polynomial $Q(y)$ occurs by setting $P(x,y) = Q(y)$. Thus the two phase suprema of the jump functional agree pointwise. For fixed x, the coefficient vector of $Q_{P,x}$ and the scalar $e^{iP(x,0)}$ depend continuously on the coefficients of P. The common representative's joint continuity therefore also holds for the complete two-variable phase class. This proves all assertions of the theorem. □

14 Maximal truncation, variation, and endpoint limits

The common anchored trajectories of Theorem 1.1 yield maximal and variational estimates, simultaneous principal values, and a separate angular improvement for odd kernels. The weak endpoint under a phase supremum is excluded by Proposition 14.6.

Corollary 14.1 (Annular maximal truncation)**.** *Define*

$$\mathscr{C}^{\mathrm{ann}}_{\Omega,b,d} f(x) = \sup_{Q\in\mathscr{P}_d} \sup_{0<a<R<\infty} \left| \int_{a<|x-y|\le R} e^{iQ(y)} b(|x-y|) K_\Omega(x-y) f(y)\, dy \right|.$$

Then

$$\|\mathscr{C}^{\mathrm{ann}}_{\Omega,b,d} f\|_p \le C_{n,d,p,\rho} \|b\|_{\mathscr{V}^\rho} \|\Omega\|_{H^1} \|f\|_p.$$

The same bound holds for the maximal positive-radius tail truncation.

Proof. Each annular integral is one increment of the tail trajectory, hence is bounded by its jump functional. The tail maximum is bounded by the same functional using the zero-at-infinity anchor. Take phase suprema before the spatial norm. □

Corollary 14.2 (Variation and simultaneous endpoint limits)**.** *For $r > 2$, let $\kappa_r = (\sum_{k\geq 1} k^{-r/2})^{1/r}$. Then*

$$\left\| \sup_{Q\in\mathscr{P}_d} V^r(T^{Q,b}_{\Omega,t} f : t > 0) \right\|_p \leq C_{n,d,p,\rho}\, \kappa_r \|b\|_{\mathscr{V}^\rho} \|\Omega\|_{H^1} \|f\|_p.$$

On one full-measure set the principal-value limit exists for every $Q \in \mathscr{P}_d$. This does not assert convergence uniform over the entire unbounded coefficient space.

Proof. For a fixed variation partition, rearrange the increment magnitudes as c_k^*. The argument of Lemma 2.1 gives $c_k^* \leq k^{-1/2}\mathbf{J}(F)$, whence $V^r(F) \leq \kappa_r \mathbf{J}(F)$. For almost every x, the phase supremum in Theorem 1.1 is finite. All trajectories have already been constructed on that same set. Failure of the Cauchy criterion at zero for any one trajectory would produce arbitrarily many disjoint jumps of a fixed positive size, contradicting finiteness of its jump functional. □

Remark 14.3 (Distinction between maximal, variational, and jump control)*.* A maximal truncation theorem alone does not imply the critical jump estimate. Nor does a fixed $r > 2$ variation theorem imply it. For example, the finite path obtained by repeating $0, 1, 0$ M times has maximum one and jump functional $\sqrt{2M}$. Replacing each peak by $M^{-1/r}$ gives $V^r = 2^{1/r}$ but jump functional $\sqrt{2}M^{1/2-1/r}$. Thus neither maximal control nor a fixed noncritical variation estimate implies critical-jump control for general scalar paths.

Corollary 14.4 (Odd angular kernels in L^1)**.** *Let $\Omega \in L^1(\mathbb{S}^{n-1})$ be odd, let $d \geq 0$, and let $1 < p < \infty$. Fix $1 \leq \rho < \infty$ and $b \in \mathscr{V}^\rho$. Then, for every $f \in L^p(\mathbb{R}^n)$,*

$$\left\| \sup_{\deg P\leq d} \mathbf{J}(T^{P,b}_{\Omega,t} f : t > 0) \right\|_p \leq C_{n,d,p,\rho} \|b\|_{\mathscr{V}^\rho} \|\Omega\|_1 \|f\|_p. \tag{14.1}$$

The maximal-truncation, $r > 2$ variation, and simultaneous principal-value conclusions of this section hold with $\|\Omega\|_{H^1}$ replaced by $\|\Omega\|_1$.

Proof. First let f be bounded and compactly supported, and restrict the phase and truncation parameters to finite sets. It is enough to treat input phases $Q(y)$ by the pointwise identity used in the proof of Theorem 1.1. For $\theta \in \mathbb{S}^{n-1}$ define

$$H^{Q,b}_{\theta,t} f(x) = \int_{|u|>t} e^{iQ(x-u\theta)} b(|u|) f(x-u\theta)\, \frac{du}{u}.$$

Polar coordinates and the oddness of Ω give the exact identity

$$T^{Q,b}_{\Omega,t} f(x) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} \Omega(\theta) H^{Q,b}_{\theta,t} f(x)\, d\sigma(\theta). \tag{14.2}$$

All integrals in this finite-parameter identity are absolutely convergent. Lemma 2.1 therefore yields

$$\max_Q \mathscr{N}_{D_0}(T^{Q,b}_{\Omega,t} f) \leq \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| \max_Q \mathscr{N}_{D_0}(H^{Q,b}_{\theta,t} f)\, d\sigma(\theta). \tag{14.3}$$

For fixed θ, write $x = z + s\theta$, $z \perp \theta$, and set $f_{z,\theta}(v) = f(z + v\theta)$. On this line,

$$H^{Q,b}_{\theta,t} f(z+s\theta) = \int_{|s-v|>t} e^{iQ(z+v\theta)} b(|s-v|) f_{z,\theta}(v)\, \frac{dv}{s-v}.$$

The function $v \mapsto Q(z+v\theta)$ is a real polynomial of degree at most d. The one-dimensional case of Theorem 1.1, applied to the Hilbert kernel, and the equivalence $\mathbf{J} \le \mathscr{N} \le 2\mathbf{J}$ give

$$\left\| \max_{Q} \mathscr{N}_{D_0}(H_{\theta,t}^{Q,b} f) \right\|_{L^p(\mathbb{R}^n)} \le C_{d,p,\rho} \|b\|_{\gamma^\rho} \|f\|_p.$$

Indeed, the bound holds on almost every line with a constant independent of z and θ, and integration in z gives the displayed estimate. Minkowski's inequality in (14.3) proves (14.1) for the finite model.

Finally, $\Omega \in L^1$ satisfies the positive-tail estimate in Lemma 12.1. The common representatives and approximation argument in Sections 12–13 therefore remove the finite parameter restrictions and extend the bound to every $f \in L^p$. The anchored path calculus gives the asserted maximal, variational, and convergence consequences. □

Remark 14.5 (Parity and the angular endpoint). For a general angular function with zero mean, Theorem 1.1 uses its angular H^1 norm. Corollary 14.4 replaces that assumption by L^1 in the odd subclass, with the same fixed radial amplitude. It neither supplies a phase-supremum weak $(1,1)$ estimate nor removes the angular assumption for a general even component. The inherited sharpness statement in Section 1.4 concerns the Orlicz scale, not a maximality claim for the Hardy class.

Proposition 14.6 (The phase supremum does not preserve the weak endpoint)**.** *For the Hilbert kernel in dimension one and any $d \ge 2$, there is no uniform weak $(1,1)$ estimate for* $\sup_{\deg P \le d} \mathbf{J}(T_t^P f : t > 0)$.

Proof. We give explicit bounded compactly supported counterexamples using only linear phases; the argument in fact applies to every $d \ge 1$. Fix an integer $m \ge 1$, put $N = 4m$, and define

$$s_j = \sum_{r=0}^{j-1} 8^{-r}, \qquad c_j = \frac{j + s_j/2}{N}, \qquad \delta = \frac{1}{100N8^N}, \qquad \xi_k = 2\pi N 8^k.$$

For $0 \le j < N$ let

$$I_j = [c_j - \delta, c_j + \delta], \qquad f_N(y) = \frac{1}{2\delta N} \sum_{j=0}^{N-1} \mathbf{1}_{I_j}(y).$$

Each summand has integral $1/N$, so $\|f_N\|_1 \le 1$. Moreover $0 \le c_j \le 1$ and $\delta \le 1$, hence f_N is bounded and supported in $[-2,2]$. Consider the disjoint output intervals

$$J_k = \left[\frac{k+3/4}{N}, \frac{k+5/4}{N} \right], \qquad m \le k < 3m, \qquad E_m = \bigcup_{k=m}^{3m-1} J_k.$$

Each has length $1/(2N)$, and therefore $|E_m| = 1/4$.

The centers encode the signs needed for the Hilbert denominator. Indeed, $0 \le s_j \le 8/7$, and direct summation of the geometric progression gives

$$\xi_k c_j \equiv \begin{cases} 0, & j \le k, \\ \pi + \dfrac{\pi}{8} s_{j-k-1}, & j > k \end{cases} \pmod{2\pi}.$$

The residual angle in the second case belongs to $[0, \pi/7]$. For $y \in I_j$ and $k < N$, $|\xi_k (y - c_j)| \le \xi_k \delta \le 1/4$. Using $\cos u \ge 1 - u^2/2$ and $\pi/7 + 1/4 < 1$, we obtain

$$\cos(\xi_k y) \ge \tfrac{1}{2} \quad (j \le k), \qquad \cos(\xi_k y) \le -\tfrac{1}{2} \quad (j > k).$$

On the other hand,

$$j \le N c_j \le j + \tfrac{4}{7}, \qquad N c_j \ge j + \tfrac{1}{2} \ \ (j > 0), \qquad N\delta \le \tfrac{1}{100}.$$

Consequently, for $x \in J_k$ and $y \in I_j$,

$$\frac{1}{16N} < |x-y| < 3, \qquad x-y > 0 \ \ (j \le k), \qquad x-y < 0 \ \ (j > k).$$

Thus every packet contributes a nonnegative real part to

$$R_k(x) := \int_{\mathbb{R}} \frac{e^{i\xi_k y}}{x-y} f_N(y)\, dy.$$

This is an absolutely convergent integral on J_k, since that interval is separated from every packet support. For each $0 \le r < m$ the left packet $j = k-r$ is present, and on that packet

$$0 < x-y \le \frac{r+2}{N}, \qquad \frac{\cos(\xi_k y)}{x-y} \ge \frac{N}{2(r+2)}.$$

Its real contribution is therefore at least $1/(2(r+2))$. Summing these m contributions and retaining the nonnegative contributions of all remaining packets yields

$$|R_k(x)| \ge \operatorname{Re} R_k(x) \ge H_m, \qquad H_m := \frac{1}{2} \sum_{r=0}^{m-1} \frac{1}{r+2} \longrightarrow \infty.$$

For the Hilbert kernel, take the original bivariate phase $P_k(x,y) = \xi_k y$, of degree at most one, and the constant radial amplitude $b = 1$. The support distances just proved give the exact sharp-tail identities

$$T^{P_k}_{1/(16N)} f_N(x) = R_k(x), \qquad T^{P_k}_3 f_N(x) = 0 \qquad (x \in J_k).$$

For every $0 < \lambda < H_m$ these two radii form a strict one-pair λ-jump. Taking the supremum in λ and then in the phase, with both suprema still pointwise before the spatial norm, gives

$$G_N(x) := \sup_{\deg P \le d} \mathbf{J}(T^P_t f_N : t > 0)(x) \ge H_m \qquad (x \in E_m).$$

In particular, testing the weak norm at the strict level $H_m/2$ gives

$$\|G_N\|_{L^{1,\infty}} \ge \frac{H_m}{2} |E_m| = \frac{H_m}{8} \longrightarrow \infty, \qquad \|f_N\|_1 \le 1.$$

This contradicts any uniform weak $(1,1)$ bound, already on bounded compactly supported inputs. No principal-value limit or external counterexample theorem is needed. Thus the fixed-phase weak theorem in Part II is complementary to Theorem 1.1; it is not a weak-type consequence of the phase-supremum estimate. □

Proposition 14.7 (Failure of the spatial strong Hardy endpoint)**.** *The angular Hardy conclusion of Theorem 1.1 does not imply a strong estimate from $H^1(\mathbb{R})$ to $L^1(\mathbb{R})$. Such an estimate fails even for the Hilbert kernel, the constant radial amplitude $b = 1$, and one fixed linear phase.*

Proof. Set

$$a(y) = \tfrac{1}{2}\big(\mathbf{1}_{(0,1)}(y) - \mathbf{1}_{(-1,0)}(y)\big).$$

This is a real $H^1(\mathbb{R})$ atom: it is supported in $[-1,1]$, has integral zero, and has L^∞ norm $1/2$. Use the phase $P(x,y) = \pi y$. For $x > 2$ the truncation at $t = 1$ is

$$T^{P,1}_1 a(x) = \int_{-1}^{1} \frac{e^{i\pi y} a(y)}{x-y}\, dy = \frac{2i}{\pi x} + E(x), \qquad |E(x)| \le \frac{1}{x(x-1)}.$$

Indeed $\int_{-1}^{1} e^{i\pi y} a(y)\, dy = 2i/\pi$ and $|(x-y)^{-1} - x^{-1}| \le |y|/[x(x-1)]$. For $x \ge 5$ this gives $|T^{P,1}_1 a(x)| \ge 1/(\pi x)$. The trajectory is zero when $t > x+1$, so one-pair tests and the strict jump definition give $\mathbf{J}(T^{P,1}_t a : t > 0) \ge |T^{P,1}_1 a|$. Its integral over $[5,\infty)$ is infinite. Thus even this fixed-phase critical-jump operator has no strong spatial $H^1 \to L^1$ bound. □

Part II
Fixed-phase endpoints, radial amplitudes, and sharpness

This part develops the fixed-phase weak endpoint and radial-amplitude theory independently of Theorem 1.1. The rational oscillatory and equal-shell correlation estimates used in Part I are proved here without invoking the phase-supremum conclusion. The independent strong estimates provide the local analytic input for the weak endpoint, the radial extension, and the completion of the corresponding trajectories.

Section 15 establishes the unphased strong input. Sections 16–17 develop the global phase flag, annular decay, and the independent fixed-phase strong theorem. Sections 18–20 prove the terminal and phase-adapted packet estimates and the concentration-stable correlations. The weak endpoint is obtained in Sections 21–24. Section 25 completes the trajectories and proves the direct bounded-variation transfer; Section 26 treats every finite radial variation exponent and the shellwise Hilbert refinement. Section 27 contains the sharpness results.

15 Abstract jump inequalities and the unphased strong endpoint

This section assembles the jump calculus and the one-dimensional estimates used throughout the paper, then combines them with a Riesz-transform factorization to prove the unphased strong critical-jump theorem.

Lemma 15.1 (Gauge normalization)**.** *For every $\Omega \in L\log L(\mathbb{S}^{n-1})$, the quantity $\mathfrak{L}$ in* (1.2) *is equivalent, with dimensional constants, to the homogeneous Luxemburg norm associated with $\Phi_0(t) = (1+t)\log(1+t) - t$. It also satisfies* (1.3)*.*

Proof. The assertion is immediate for $\Omega = 0$, so assume $\mathfrak{L}(\Omega) > 0$. Put $A = \|\Omega\|_1$ and $M = \mathfrak{L}(\Omega)$. The elementary bound

$$\Phi_0(t) \le C\big(t + t\log^+ t\big), \qquad t \ge 0,$$

and $M \ge A$ give, for a sufficiently large absolute C_0,

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} \Phi_0\left(\frac{|\Omega|}{C_0 M}\right) d\sigma &\le \frac{C}{C_0 M}\left(A + \int_{\mathbb{S}^{n-1}} |\Omega| \log^+\left(\frac{|\Omega|}{M}\right) d\sigma\right) \\ &\le \frac{C}{C_0 M}\left(A + \int_{\mathbb{S}^{n-1}} |\Omega| \log^+\left(\frac{|\Omega|}{A}\right) d\sigma\right) \le 1.\end{aligned}$$

This proves one direction of the Luxemburg comparison.

Conversely, suppose that $\int \Phi_0(|\Omega|/L)d\sigma \le 1$, and put $m_n = \sigma(\mathbb{S}^{n-1})$. Jensen's inequality yields

$$\Phi_0\left(\frac{A}{m_n L}\right) \le \frac{1}{m_n}\int \Phi_0(|\Omega|/L)d\sigma \le \frac{1}{m_n},$$

and hence $A \le C_n L$. Since

$$\log^+\left(\frac{|\Omega|}{A}\right) \le \log\left(1 + \frac{|\Omega|}{L}\right) + \log^+\left(\frac{L}{A}\right),$$

while $t\log(1+t) \le \Phi_0(t) + t$, we obtain

$$\begin{aligned}\int |\Omega| \log^+(|\Omega|/A) d\sigma &\le L\int \frac{|\Omega|}{L}\log\left(1 + \frac{|\Omega|}{L}\right) d\sigma + A\log^+(L/A) \\ &\le L\int \Phi_0(|\Omega|/L)d\sigma + A + L/e \le C_n L.\end{aligned}$$

Thus the two quantities are equivalent, with dimensional constants.

If $R\Omega(\theta) = \Omega(-\theta)$, then reflection is an isometry of the Luxemburg space. Convexity gives

$$\|\Omega_o\|_{L\log L,\mathrm{Lux}} + \|\Omega_e\|_{L\log L,\mathrm{Lux}} \le 2\|\Omega\|_{L\log L,\mathrm{Lux}}.$$

Homogeneity and the equivalence just proved give (1.3). □

15.1 Jump calculus and one-dimensional inputs

For a scalar path indexed by positive radii, write

$$N_\lambda^{\mathrm{dyad}}(F) := N_\lambda(F_{2^j} : j \in \mathbb{Z}), \qquad \mathbf{J}^{\mathrm{dyad}}(F) := \sup_{\lambda>0} \lambda N_\lambda^{\mathrm{dyad}}(F)^{1/2}. \tag{15.1}$$

For a finite ordered parameter set I, $\mathbf{J}_I(F)$ denotes the same pointwise functional with the path restricted to I.

Lemma 15.2 (Pointwise jump and variation calculus)**.** *Let E be a linearly ordered set. For scalar E-paths F, G and $\lambda > 0$,*

$$N_\lambda(F+G) \le N_{\lambda/2}(F) + N_{\lambda/2}(G), \tag{15.2}$$

$$\lambda N_\lambda(F)^{1/2} \le V^2(F), \tag{15.3}$$

$$V^2(F+G) \le V^2(F) + V^2(G). \tag{15.4}$$

Consequently,

$$\mathbf{J}(F+G) \le 2\mathbf{J}(F) + 2\mathbf{J}(G), \tag{15.5}$$

$$\mathbf{J}(F) \le V^2(F). \tag{15.6}$$

If $F_t \to 0$ at the upper endpoint of its ordered parameter interval (with value 0 there when that endpoint belongs to the parameter set), then

$$\sup_t |F_t| \le \mathbf{J}(F). \tag{15.7}$$

For every $r > 2$,

$$V^r(F) \le \kappa_r \mathbf{J}(F), \qquad \kappa_r = \left(\sum_{k=1}^{\infty} k^{-r/2}\right)^{1/r}. \tag{15.8}$$

Finally, if $F = (F_t)_{t>0}$ is a positive-radius path decomposed into dyadic parameter blocks, then

$$\lambda N_\lambda(F)^{1/2} \le C\left[\left(\sum_{j\in\mathbb{Z}} V^2(F_\varepsilon : 2^j \le \varepsilon \le 2^{j+1})^2\right)^{1/2} + \lambda N_{\lambda/3}^{\mathrm{dyad}}(F)^{1/2}\right], \tag{15.9}$$

and hence

$$\mathbf{J}(F) \le C\left[\left(\sum_{j\in\mathbb{Z}} V^2(F_\varepsilon : 2^j \le \varepsilon \le 2^{j+1})^2\right)^{1/2} + \mathbf{J}^{\mathrm{dyad}}(F)\right]. \tag{15.10}$$

Moreover $\lambda N_\lambda^{\mathrm{dyad}}(F)^{1/2} \le V^2(F_{2^j} : j \in \mathbb{Z})$.

Proof. For each strict λ-jump of $F+G$, one of the two corresponding increments has modulus $> \lambda/2$. Assign the pair to such a summand; each assigned subcollection remains ordered and paired, proving (15.2). Inserting the endpoints of a paired collection into a variation partition gives (15.3); Minkowski in the finite ℓ^2 sum gives (15.4). Taking the supremum in λ in (15.2) proves (15.5), and (15.6) is (15.3) followed by the same supremum.

For (15.7), first let t lie strictly below the upper endpoint and let $s > t$ tend to that endpoint. For every $0 < \lambda < |F_t - F_s|$ there is one strict λ-jump, so $\mathbf{J}(F) \geq |F_t - F_s|$; now let s tend to the endpoint. If the upper endpoint belongs to the parameter set, its value is zero by hypothesis, so the assertion at that endpoint is immediate.

To prove (15.8), fix a finite variation partition and let $a_1^* \geq a_2^* \geq \cdots$ be the decreasing rearrangement of the moduli of its increments. For every $0 < \lambda < a_k^*$, at least k partition increments are strict λ-jumps. Retain any k of them and restore their original temporal order. They are a subfamily of the consecutive partition intervals, so their interiors are disjoint and the right endpoint of each retained interval is no later than the left endpoint of the next. Hence they form an admissible paired family. Thus $\lambda\sqrt{k} \leq \mathbf{J}(F)$, and letting $\lambda \uparrow a_k^*$ gives $a_k^* \leq k^{-1/2}\mathbf{J}(F)$. Hence

$$\left(\sum_i a_i^r\right)^{1/r} \leq \mathbf{J}(F)\left(\sum_{k=1}^{\infty} k^{-r/2}\right)^{1/r}.$$

Taking the supremum over partitions proves the claim. The numerical bound in (1.10) follows from the integral test.

For (15.9), let $j(t)$ be the integer determined by $2^{j(t)} \leq t < 2^{j(t)+1}$. Fix an ordered paired family (s_i, t_i) of strict λ-jumps, and denote its finite cardinality by N. If $j(s_i) = j(t_i)$, assign the pair to that dyadic block. Otherwise put

$$\widehat{s_i} = 2^{j(s_i)+1}, \qquad \widehat{t_i} = 2^{j(t_i)}.$$

Then $s_i \leq \widehat{s_i} \leq \widehat{t_i} \leq t_i$, and

$$F_{t_i} - F_{s_i} = (F_{t_i} - F_{\widehat{t_i}}) + (F_{\widehat{t_i}} - F_{\widehat{s_i}}) + (F_{\widehat{s_i}} - F_{s_i}).$$

One of the three terms has modulus $> \lambda/3$. Choose exactly one such term, using any fixed tie-breaking rule. The selected middle terms form an ordered paired family on the dyadic sequence. The within-block terms and the selected initial and terminal terms assigned to any fixed dyadic block have pairwise disjoint interiors, because the original pairs are ordered. Insert all their endpoints, together with the two endpoints of that dyadic block, into a single increasing partition and delete repeated points. Every selected short segment is then one increment of this partition: two such segments may share an endpoint but cannot overlap in their interiors. Therefore the sum of their squared increments is bounded by the V^2 square of the path on that block. Consequently, if N_{sh} and N_{long} denote the numbers assigned to short and middle terms, respectively, then

$$N = N_{\mathrm{sh}} + N_{\mathrm{long}}, \qquad \frac{\lambda^2}{9} N_{\mathrm{sh}} \leq \sum_j V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2, \qquad N_{\mathrm{long}} \leq N_{\lambda/3}^{\mathrm{dyad}}(F).$$

It follows that

$$\lambda\sqrt{N} \leq 3\left(\sum_j V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2\right)^{1/2} + 3\frac{\lambda}{3} N_{\lambda/3}^{\mathrm{dyad}}(F)^{1/2}.$$

Taking the supremum over all finite strict λ-jump witnesses proves (15.9), including the case in which the jump count is infinite. Taking the supremum in λ then gives (15.10). The last assertion follows from (15.3) on the dyadic sequence. □

Normability at the critical jump exponent is essential for the continuum angular factorizations below. On a linearly ordered set E, define

$$N_{\lambda,E}^{\mathrm{ch}}(F) := \sup\Big(\{0\} \cup \big\{J \in \mathbb{N} : \exists\, t_0 < \cdots < t_J, \quad t_j \in E, \quad |F_{t_j} - F_{t_{j-1}}| > \lambda \ (1 \leq j \leq J)\big\}\Big) \in \mathbb{N}_0 \cup \{\infty\}.$$

As before, the index-set subscript is omitted when it is displayed or unambiguous. Then, for every ordered E,

$$N_{\lambda,E}^{\mathrm{ch}}(F) \leq 2N_{\lambda,E}(F), \qquad N_{\lambda,E}(F) \leq N_{\lambda/2,E}^{\mathrm{ch}}(F). \tag{15.11}$$

The first inequality follows by taking alternating edges of a chain. For the second, start from $r_0 = s_1$ and $r_1 = t_1$ in an admissible family of pairs. Inductively, after r_{i-1} has been selected from the preceding pair, at least one of s_i, t_i is more than $\lambda/2$ away from $F_{r_{i-1}}$; otherwise

$$|F_{t_i} - F_{s_i}| \le |F_{t_i} - F_{r_{i-1}}| + |F_{s_i} - F_{r_{i-1}}| \le \lambda.$$

Choose such an endpoint as r_i. The temporal ordering of the pairs, together with the strict increment, gives $r_{i-1} < r_i$. The resulting chain has one initial λ-increment and then strict $\lambda/2$-increments, hence has at least as many strict $\lambda/2$-increments as the original paired family has pairs. For a linearly ordered set E, put

$$\mathbf{J}_E^{\mathrm{ch}}(F) := \sup_{\lambda>0} \lambda N_{\lambda,E}^{\mathrm{ch}}(F)^{1/2}. \tag{15.12}$$

We omit E when it is the full parameter set. For a finite ordered set I and a spatially measurable I-path, also put

$$\mathscr{J}_{2,p}^{\mathrm{ch}}(F_t : t \in I) := \|\mathbf{J}_I^{\mathrm{ch}}(F)\|_p. \tag{15.13}$$

Thus (15.11) gives, pointwise and uniformly in every ordered set E,

$$\mathbf{J}_E^{\mathrm{ch}}(F) \le \sqrt{2}\,\mathbf{J}_E(F), \qquad \mathbf{J}_E(F) \le 2\mathbf{J}_E^{\mathrm{ch}}(F). \tag{15.14}$$

Lemma 15.3 (Pathwise critical-jump normability and integration)**.** *For every nonempty finite ordered set I, the scalar functional $\mathbf{J}_I^{\mathrm{ch}}$ is equivalent, with absolute constants uniform in I, to a subadditive seminorm on scalar I-paths modulo constant paths. Consequently, if μ is a complex measure on a measure space Z and $F^z = (F_t^z)_{t\in I}$ is coordinatewise $|\mu|$-integrable, then*

$$\mathbf{J}_I^{\mathrm{ch}}\left(\int_Z F^z\,d\mu(z)\right) \le C\int_Z \mathbf{J}_I^{\mathrm{ch}}(F^z)\,d|\mu|(z). \tag{15.15}$$

If the paths also depend on x, assume that, for every $t \in I$, the map $(z,x) \mapsto F_t^z(x)$ is jointly measurable and that, for almost every x,

$$\int_Z |F_t^z(x)|\,d|\mu|(z) < \infty, \qquad t \in I.$$

Then (15.15) *holds for almost every x. In particular, for $1 \le p < \infty$,*

$$\mathscr{J}_{2,p}^{\mathrm{ch}}\left(\int_Z F^z\,d\mu(z)\right) \le C\int_Z \mathscr{J}_{2,p}^{\mathrm{ch}}(F^z)\,d|\mu|(z), \tag{15.16}$$

whenever the right-hand side is finite.

Proof. Use the explicit paired-path norm $\mathscr{N}_I$ from Lemma 2.1. Together with (15.14), its equivalence to $\mathbf{J}_I$ gives

$$\mathbf{J}_I^{\mathrm{ch}}(F) \le \sqrt{2}\,\mathscr{N}_I(F), \qquad \mathscr{N}_I(F) \le 4\mathbf{J}_I^{\mathrm{ch}}(F).$$

Apply (2.3) and these two comparisons to obtain (15.15). On a finite parameter set the path functionals are continuous functions of their coordinates, so the stated joint measurability and sectionwise integrability hypotheses permit the same argument almost everywhere in the spatial variable. Minkowski's inequality in L^p, $1 \le p < \infty$, proves (15.16). □

We next record the pointwise critical-jump estimates for the one-dimensional model paths used in the factorization. If $u \in \mathbb{S}^{n-1}$, put

$$H_{\varepsilon,u}f(x) = \int_{|s|>\varepsilon} f(x-su)\,\frac{ds}{s}, \qquad A_{t,u}^{+}f(x) = \frac{1}{t}\int_0^t f(x-su)\,ds.$$

Lemma 15.4 (One-dimensional pointwise critical jumps)**.** *For every $1 < p < \infty$, every $f \in L^p(\mathbb{R}^n)$, and every $u \in \mathbb{S}^{n-1}$,*

$$\|\mathbf{J}(H_{\varepsilon,u}f : \varepsilon > 0)\|_p + \|\mathbf{J}(A_{t,u}^{+}f : t > 0)\|_p \le C_p\|f\|_p, \tag{15.17}$$

with a constant independent of u. The same estimate holds with the chain functional $\mathbf{J}^{\mathrm{ch}}$ in place of $\mathbf{J}$.

Proof. We first work on $\mathbb{R}$ and keep every jump-amplitude supremum inside the spatial norm. The Hilbert term is the proved scalar unphased specialization of Proposition 4.6, with $A = I$, $b \equiv 1$ and angular values $1, -1$ on $\mathbb{S}^0$. Counting measure on $\mathbb{S}^0$ gives the literal, unnormalized operator

$$H_\varepsilon f(x) = \int_{|s|>\varepsilon} f(x-s)\,\frac{ds}{s}.$$

That specialization uses the independent smooth-approximation and shell estimates, not the present averaging assertion or a jump-amplitude supremum outside L^p.

Here is the sharp-average argument. Write $A_t f(x) = t^{-1}\int_0^t f(x-s)\,ds$, let E_m be expectation on the dyadic intervals of length 2^m, and set $D_m f = E_{m-1}f - E_m f$. For $k, m \in \mathbb{Z}$ define

$$C_{k,m} f = (A_{2^{m+k}} - E_{m+k}) D_m f.$$

We first establish the uniform comparison-square estimate

$$\left\| \left(\sum_j |A_{2^j} f - E_j f|^2 \right)^{1/2} \right\|_p \le C_p \|f\|_p. \tag{15.18}$$

The dyadic almost-orthogonality estimate is

$$\|C_{k,m} f\|_2 \le C 2^{-|k|/2} \|D_m f\|_2. \tag{15.19}$$

For $k < 0$, $E_{m+k} D_m f = D_m f$. Translation of a function constant on intervals of length 2^{m-1} changes it only across cell boundaries, giving

$$\|D_m f(\cdot - h) - D_m f\|_2 \le C \min\{1, (|h|/2^m)^{1/2}\} \|D_m f\|_2.$$

Integration against the actual one-sided box kernel gives the required $2^{k/2}$ gain; only its finite half moment is used, not smoothness. For $k \ge 0$, the expectation vanishes. Subtract the box kernel at the left endpoint of each length-2^m cell, on which $D_m f$ has integral zero. The resulting boundary kernel has row and column integrals at most $C2^m/2^{m+k}$. Schur's inequality gives $C2^{-k}$, which is stronger than (15.19). All these convolutions are genuine almost-everywhere integrals.

Orthogonality of the D_m's now gives, uniformly over finite scale sets, the $L^2(\ell^2)$ bound $C2^{-|k|/2}\|f\|_2$. For every $1 < q < \infty$, the vector maximal theorem and the dyadic square-function theorem give the crude bound

$$\|\{C_{k,m} f\}_m\|_{L^q(\ell^2)} \le C_q \|f\|_q.$$

Indeed both the sharp average and the dyadic expectation are dominated by one-dimensional positive maximal operators on the inputs $D_m f$. Interpolate the actual finite-vector linear operators between this bound and L^2. For each target p the result is $C_p 2^{-\varepsilon_p |k|}\|f\|_p$, where $\varepsilon_p > 0$ is independent of the finite scale set. Increase the scale sets and then sum in k. For each j, the corresponding finite sums of $D_m f$ telescope, and $E_a f \to f$ as $a \to -\infty$, $E_b f \to 0$ as $b \to +\infty$ in L^p. Boundedness of $A_{2^j} - E_j$ identifies the coordinate limit as $A_{2^j} f - E_j f$. Countable-vector Minkowski proves (15.18) for the original arbitrary L^p input.

The elementary paired-path inequality $\mathscr{N}(v_j : j \in \mathbb{Z}) \le 2(\sum_j |v_j|^2)^{1/2}$ and the all-integer dyadic martingale jump estimate therefore give

$$\|\mathscr{N}(A_{2^j} f : j \in \mathbb{Z})\|_p \le C_p \|f\|_p. \tag{15.20}$$

This is a pointwise path-norm estimate, not a family of estimates for fixed jump amplitudes.

The continuum short-variation square function. For $g = D_m f$ and $R = 2^{m+k}$ anchor the path by $F_{k,m}(x,u) = A_{Ru} g(x) - A_{2R} g(x)$, $1 \le u \le 2$. It vanishes at $u = 2$. The preceding half-moment and boundary-cancellation arguments, uniformly for $1 \le u \le 2$, give

$$\|F_{k,m}\|_{L^2(dx\,du)} \le C 2^{-|k|/2} \|D_m f\|_2.$$

The primitive defining A_{Ru} is absolutely continuous, and its actual almost-everywhere derivative is

$$\partial_u A_{Ru} g(x) = \frac{g(x-Ru) - A_{Ru} g(x)}{u}.$$

Translations and the averaging contraction show that this derivative has joint $L^2(dx\,du)$ norm at most $C\|D_m f\|_2$. Fubini gives the scalar absolutely continuous paths and their Bochner-integrable derivatives on one conull spatial set. For an anchored absolutely continuous scalar path,

$$V^2(F)^2 \le 2 \sup|F| \int_1^2 |F'|, \qquad \sup|F|^2 \le 2 \int_1^2 |FF'|.$$

Hölder, first in u and then in x, consequently proves

$$\|V^2(A_{2^{m+k}u} D_m f : 1 \le u \le 2)\|_2 \le C 2^{-|k|/8} \|D_m f\|_2. \tag{15.21}$$

Squaring and summing in m uses the same dyadic orthogonality. For the crude all-q estimate, integrate the displayed derivative:

$$V^2(A_{Ru} g : 1 \le u \le 2) \le \int_1^2 |\partial_u A_{Ru} g|\, du \le C M|g|.$$

The vector maximal and dyadic square-function bounds are uniform in k. Apply genuine sublinear interpolation to the finite-scale square of variations, first using a countable dense set of parameters. Subadditivity follows from the variation triangle inequality and Minkowski in ℓ^2; all L^q inputs are in the domain by the integrability just proved. This yields a summable decay $C_p 2^{-\eta_p|k|}$, $\eta_p > 0$, with constants independent of all finite truncations. Continuity of the averaging paths identifies the dense-parameter variation with the full variation.

To reconstruct, retain finitely many scales and dense parameters, sum the D_m telescopes in L^p, and identify each averaged coordinate by boundedness of its box convolution. The summable offset bounds give an L^p majorant for the variation-square tails; finite-partition lower semicontinuity and Fatou then remove the cutoffs. Exhausting the countable dense parameters and using path continuity proves the literal all-integer continuum estimate

$$\left\| \left(\sum_{j \in \mathbb{Z}} V^2(A_t f : 2^j \le t \le 2^{j+1})^2 \right)^{1/2} \right\|_p \le C_p \|f\|_p. \tag{15.22}$$

The pointwise long–short inequality (15.10), combined with (15.20), now bounds $\|\mathcal{N}(A_t f : t > 0)\|_p$ by $C_p\|f\|_p$.

All radii and original representatives on each direction. For a fixed unit vector u, the map $(r, x_\perp) \mapsto x_\perp + ru$ preserves Lebesgue measure. Almost every complete fiber has input $g \in L^p(\mathbb{R})$. On each such fiber and at every point, local integrability defines all averages and all bounded Hilbert annuli, while

$$\int_{|s|>1} \frac{|g(r-s)|}{|s|}\, ds \le \|g\|_p \big\| |s|^{-1} \mathbf{1}_{\{|s|>1\}} \big\|_{p'} < \infty.$$

Both positive-parameter paths are locally absolutely continuous. Equality of ambient input representatives almost everywhere gives, by Fubini on complete rays, equality of these integrals for every positive radius on one common conull set. No uncountable intersection of exceptional sets is taken. Rational positive parameters detect all strict jump witnesses, making the pointwise functionals measurable. Integrating the real-line bounds over the fibers preserves the same constants, independently of n and u.

Finally $\mathbf{J} \le \mathcal{N}$ and (15.14) bound the paired and chain critical jumps by the same absolute multiple of the two path norms. Add the Hilbert and average constants and choose that constant before n, u, f. This proves the stated sum for both conventions, for every positive radius and every original L^p representative. □

Lemma 15.5 (Radial profiles). *Let $1 < p < \infty$, let $f \in L^p(\mathbb{R}^n)$, let $u \in \mathbb{S}^{n-1}$, and let φ be represented by a right-continuous function that is locally of bounded variation on $(0,\infty)$, $\varphi(t) \to 0$ as $t \to \infty$, and*

$$B(\varphi) = \int_{(0,\infty)} t\, d|D\varphi|(t) < \infty.$$

For $P^{\varphi}_{\varepsilon,u} f(x) = \int_0^\infty \varphi(s) f(x - \varepsilon s u)\, ds$, the pointwise path is represented by the compatible Stieltjes representative constructed in the proof, and

$$\|\mathbf{J}(P^{\varphi}_{\varepsilon,u} f : \varepsilon > 0)\|_p \le C_p B(\varphi) \|f\|_p. \tag{15.23}$$

Suppose, more generally, that $(\varphi_u)_{u \in \mathbb{S}^{n-1}}$ is a family satisfying the preceding hypotheses, with right-continuous representatives chosen so that $(u,s) \mapsto \varphi_u(s)$ is jointly measurable on $\mathbb{S}^{n-1} \times (0,\infty)$, and suppose that $\int_{\mathbb{S}^{n-1}} B(\varphi_u)\, d\sigma(u) < \infty$. Then, for every $\varepsilon > 0$, the map $u \mapsto P^{\varphi_u}_{\varepsilon,u} f$ is strongly measurable in L^p, its Bochner integral is well defined, and

$$\left\| \mathbf{J}\left(\int_{\mathbb{S}^{n-1}} P^{\varphi_u}_{\varepsilon,u} f\, d\sigma(u) : \varepsilon > 0 \right) \right\|_p \le C_p \left(\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) \right) \|f\|_p. \tag{15.24}$$

Both conclusions remain valid with the chain functional.

Proof. We apply the pointwise finite-pair norm $\mathscr{N}$ before taking the L^p norm. First take a Borel representative of f; the transfer back to an arbitrary L^p representative is justified below.

The right-continuous convention and the zero limit at infinity give

$$D\varphi((s,R]) = \varphi(R) - \varphi(s), \qquad \varphi(s) = -D\varphi((s,\infty)), \qquad s > 0.$$

Indeed, $|D\varphi|((s,\infty)) \le s^{-1} B(\varphi)$, so the tail limit is legitimate. Tonelli's theorem gives $\int_0^\infty |\varphi(s)|\, ds \le B(\varphi)$. Thus the literal profile integral is a Bochner integral in L^p, and Bochner–Fubini proves

$$P^{\varphi}_{\varepsilon,u} f = -\int_{(0,\infty)} t A^+_{\varepsilon t,u} f\, d\varphi(t) \qquad \text{in } L^p. \tag{15.25}$$

Let η_φ be the polar density of $D\varphi$ with respect to $|D\varphi|$. In logarithmic coordinates define

$$\mu_\varphi = (\log)_*\big(t\, d|D\varphi|(t)\big), \qquad h_\varphi(r) = \eta_\varphi(e^r).$$

Then $\mu_\varphi(\mathbb{R}) = B(\varphi)$ and $|h_\varphi| = 1$ μ_φ-almost everywhere. The compatible pointwise representative is

$$L^u_\varepsilon(x) = -\int_{\mathbb{R}} h_\varphi(r) A^+_{\varepsilon e^r,u} f(x)\, d\mu_\varphi(r).$$

On almost every line parallel to u, all positive-radius averages are continuous. The one-sided Hardy–Littlewood theorem and Fubini give

$$\mathscr{M}^+_u f(x) = \sup_{r \in \mathbb{Q}_+} |A^+_{r,u} f(x)|, \qquad \|\mathscr{M}^+_u f\|_p \le C_p \|f\|_p. \tag{15.26}$$

By continuity, this rational supremum bounds every positive-radius average on the same conull set. It dominates the Stieltjes integrand, so L^u_ε is defined for all $\varepsilon > 0$ there, and dominated convergence makes its entire path continuous.

For a finite ordered pair family, the sum of the absolute increments of an integral is at most the integral of the corresponding sums. Taking the supremum over the pair families proves the path-norm integral inequality. Since $\varepsilon \mapsto \varepsilon e^r$ preserves order, it follows pointwise that

$$\mathscr{N}(L^u_\varepsilon : \varepsilon > 0) \le B(\varphi) \mathscr{N}(A^+_{t,u} f : t > 0). \tag{15.27}$$

The full-radius path-norm bound proved in Lemma 15.4 now gives $\|\mathcal{N}(L^u)\|_p \le C_p B(\varphi)\|f\|_p$. The comparison with both critical-jump conventions proves the fixed-direction clause. A fixed countable dense radius set detects the path norm and all strict jump witnesses of these continuous paths, proving their measurability as well.

For the spherical clause we use the given jointly measurable profiles directly. For each fixed positive ε, joint measurability of $(u,s) \mapsto \varphi_u(s)$, translation of the Borel input, and parameter integration give a jointly measurable representative of

$$H_\varepsilon(u,x) = \int_0^\infty \varphi_u(s) f(x - \varepsilon s u)\, ds.$$

The translation action is strongly continuous in L^p; separability and simple-function approximation therefore give strong measurability of $u \mapsto H_\varepsilon(u,\cdot)$ in L^p. Moreover,

$$\|H_\varepsilon(u,\cdot)\|_p \le B(\varphi_u)\|f\|_p,$$

so the angular Bochner integral exists. Local spatial exhaustion by finite-measure sets and Fubini identify this Bochner integral with the literal scalar angular integral almost everywhere for every fixed ε.

Let $D = \mathbb{Q}_+$. On D the path norm is a countable supremum of measurable finite-pair tests. The fixed-direction result and the coordinatewise Stieltjes identities, intersected over D, give

$$\mathcal{N}(H_q(u,x) : q \in D) \le 3B(\varphi_u)\mathcal{N}(A^+_{q,u} f(x) : q \in D)$$

for almost every x for each direction u. The budget is measurable by rational partitions, and the average norm by countable pair tests. Both sides thus have jointly measurable versions, so Fubini reverses these almost-everywhere quantifiers. The path-norm integral inequality and weighted Minkowski, followed by the uniform directional estimate, give

$$\begin{aligned} &\left\| \mathcal{N}\left(\int H_q(u,\cdot)\, d\sigma(u) : q \in D \right) \right\|_p \\ &\quad \le 3 \int B(\varphi_u) \|\mathcal{N}(A^+_{q,u} f : q \in D)\|_p \, d\sigma(u) \le C_p \left(\int B(\varphi_u)\, d\sigma(u) \right) \|f\|_p. \end{aligned} \tag{15.28}$$

All suprema here remain inside the spatial norm.

To recover the continuum, the one-sided maximal theorem and Minkowski give

$$G(x) = \int B(\varphi_u) \mathscr{M}^+_u f(x)\, d\sigma(u), \qquad \|G\|_p \le C_p \left(\int B(\varphi_u)\, d\sigma(u) \right) \|f\|_p.$$

On one conull spatial set, G is finite and the individual Stieltjes paths are continuous for almost every direction. Their absolute values are bounded by $3B(\varphi_u)\mathscr{M}^+_u f(x)$. Dominated convergence extends the angular integral on D to a continuous path $L_\varepsilon(x)$ on all positive radii. At each fixed radius it agrees almost everywhere with both the literal angular integral and its Bochner L^p representative. The dense-set finite-witness principle transfers (15.28) to this entire path; comparison with $\mathbf{J}$ and $\mathbf{J}^{\mathrm{ch}}$ proves both claimed estimates with a constant depending only on p.

Finally, if the given f is merely an arbitrary measurable L^p representative, choose a Borel version and a measurable null cover of the set where they differ. Translation invariance and Fubini show that their directional averages agree on one conull set at every radius, hence so do their compatible Stieltjes paths. The same null-cover argument on the angular–radial product shows that the literal spherical profile integrals agree almost everywhere at each fixed radius. Thus the continuous representatives above represent the original input, not an additional regularity assumption on it. □

We repeatedly use the following elementary compactness principle.

Lemma 15.6 (Finite witnesses and dense parameter sets)**.** *Let I be a finite ordered set, and suppose that $F_t^{(m)} \to F_t$ for every $t \in I$. Then, for every $\lambda > 0$ and $r \geq 1$,*

$$N_\lambda(F_t : t \in I) \leq \liminf_{m\to\infty} N_\lambda(F_t^{(m)} : t \in I), \qquad N_\lambda^{\mathrm{ch}}(F_t : t \in I) \leq \liminf_{m\to\infty} N_\lambda^{\mathrm{ch}}(F_t^{(m)} : t \in I),$$

$$V^r(F_t : t \in I) \leq \liminf_{m\to\infty} V^r(F_t^{(m)} : t \in I),$$

and

$$\mathbf{J}_I(F) \leq \liminf_{m\to\infty} \mathbf{J}_I(F^{(m)}). \tag{15.29}$$

Moreover,

$$\mathbf{J}_I^{\mathrm{ch}}(F) \leq \liminf_{m\to\infty} \mathbf{J}_I^{\mathrm{ch}}(F^{(m)}). \tag{15.30}$$

If a path F is continuous on compact subintervals of an interval J and $D \subset J$ is dense, then its paired and chain strict jump counts, its paired and chain pointwise critical-jump functionals, and its r-variation on D agree with those on J.

Proof. Any finite family of strict λ-jumps of F has a positive minimum excess over λ; all of its witnessing increments therefore persist for sufficiently large m. This applies verbatim to both paired and chain witnesses and proves the two jump-count inequalities. Applying the same argument to a fixed finite partition and then taking the supremum proves the variation inequality. If $a < \mathbf{J}_I(F)$, choose a jump amplitude λ and a finite strict witness with $\lambda N_\lambda(F)^{1/2} > a$; that witness persists for all sufficiently large m, proving (15.29). The identical persistence argument for a finite chain witness proves (15.30). Finally, continuity permits the finitely many points in any paired witness, chain witness, or variation partition to be perturbed into D while preserving their strict order and changing every increment by an arbitrarily small amount. This proves all dense-set assertions. □

Lemma 15.7 (Homogeneous tail counting)**.** *Let $\Omega \in L\log L(\mathbb{S}^{n-1})$ and put $A = \|\Omega\|_1$. For every $\delta > 0$,*

$$\sum_{s\geq 0} \|\Omega \mathbf{1}_{\{|\Omega| > 2^{\delta s} A\}}\|_1 \leq C_\delta \mathfrak{L}(\Omega). \tag{15.31}$$

If $\int \Omega = 0$, put $m_n = \sigma(\mathbb{S}^{n-1})$ and $E_s = \{|\Omega| > 2^{\delta s} A\}$, $c_s = m_n^{-1} \int_{E_s} \Omega \, d\sigma$. Then

$$\Omega_s^\sharp = \Omega \mathbf{1}_{E_s} - c_s, \qquad \omega_s = \Omega \mathbf{1}_{E_s^c} + c_s \tag{15.32}$$

have mean zero and satisfy

$$\|\omega_s\|_\infty \leq C_n 2^{\delta s} A, \quad \|\omega_s\|_1 \leq 2A, \quad \|\Omega_s^\sharp\|_1 \leq 2\|\Omega \mathbf{1}_{E_s}\|_1. \tag{15.33}$$

Proof. If $A = 0$, all assertions are immediate. Hence assume $A > 0$. Tonelli gives

$$\sum_s \|\Omega \mathbf{1}_{E_s}\|_1 = \int |\Omega(\theta)| \#\{s : 2^{\delta s} A < |\Omega(\theta)|\} \, d\sigma(\theta),$$

and the counting function is bounded by $C_\delta(1 + \log^+(|\Omega|/A))$. This proves (15.31). The identities and bounds in (15.32)–(15.33) follow from $\int_{E_s^c} \Omega = -m_n c_s$ and $m_n |c_s| \leq \|\Omega \mathbf{1}_{E_s}\|_1 \leq A$. □

Lemma 15.8 (Compact $L\log L$ data in H^1)**.** *Suppose that F is supported in a fixed ball and $\int F = 0$. Define $\mathscr{E}(0) = 0$, while for $F \neq 0$ put*

$$\mathscr{E}(F) = \|F\|_1 + \int |F(x)| \log\left(e + \frac{|F(x)|}{\|F\|_1}\right) dx.$$

If $\mathscr{E}(F) < \infty$, then $\|F\|_{H^1(\mathbb{R}^n)} \leq C\mathscr{E}(F)$, with a constant depending only on the dimension and the supporting ball.

Proof. If $F = 0$ there is nothing to prove. Put $A = \|F\|_1$. By translation invariance we may suppose that $\operatorname{supp} F \subset B(0,R)$, where R is fixed. Let $\phi \in C_c^\infty(B(0,1))$ have integral one and let $\mathscr{M}_\phi F = \sup_{t>0} |F * \phi_t|$. On a fixed enlargement of the support, $\mathscr{M}_\phi F \le CMF$. The refined weak estimate

$$|\{MF > 2s\}| \le \frac{C}{s} \int_{\{|F|>s\}} |F|$$

follows by decomposing $F = F\mathbf{1}_{\{|F|\le s\}} + F\mathbf{1}_{\{|F|>s\}}$: the maximal function of the first summand is at most s, and the weak $(1,1)$ maximal theorem applies to the second. If B_R is a fixed enlargement of $B(0,R)$, layer cake, with threshold $s_0 = A$, gives

$$\begin{aligned}\int_{B_R} MF &= \int_0^\infty |B_R \cap \{MF > s\}|\, ds \\ &\le CA + C\int_A^\infty \frac{1}{s} \int_{\{|F|>s/2\}} |F(x)|\, dx\, ds \\ &\le CA + C\int |F(x)| \log^+\left(\frac{2|F(x)|}{A}\right) dx \le C\mathscr{E}(F),\end{aligned}$$

where Tonelli justifies the interchange. Outside that ball, cancellation gives

$$|F * \phi_t(x)| \le \int |F(y)|\, |\phi_t(x-y) - \phi_t(x)| dy \le C_{\phi,R}|x|^{-n-1}\|F\|_1.$$

Indeed, whenever the integrand is nonzero, $t \ge c_R|x|$, and the mean-value theorem gives

$$|\phi_t(x-y) - \phi_t(x)| \le |y|\|\nabla\phi_t\|_\infty \le C_\phi R t^{-n-1} \le C_{\phi,R}|x|^{-n-1}.$$

The last majorant is integrable off B_R. Thus $\|\mathscr{M}_\phi F\|_1 \le C\mathscr{E}(F)$. The radial maximal characterization of real H^1 is [24, Theorem 11 and the following Remark 2]; this proves the assertion. The Riesz-transform characterization $R_j : H^1 \to L^1$ is recorded in [24, Section II.2, p. 144]. □

Proposition 15.9 (Unphased strong critical-jump inequality)**.** *If $\int \Omega = 0$ and $\Omega \in L\log L(\mathbb{S}^{n-1})$, then, for every $1 < p < \infty$ and every $f \in L^p(\mathbb{R}^n)$,*

$$\mathscr{J}_{2,p}(T^0_{\Omega,\varepsilon} f : \varepsilon > 0) \le C_{n,p}\mathfrak{L}(\Omega)\|f\|_p. \tag{15.34}$$

Proof. Fix $1 < p < \infty$. We first take $f \in \mathscr{S}(\mathbb{R}^n)$, restrict the truncation parameter to a finite ordered set $I \subset (0,\infty)$, and use the chain convention. All constants below are independent of I.

We record the measurability convention for the angular integrations. For $M > \varepsilon$, put

$$H^M_{\varepsilon,u} f = \int_{\varepsilon<|s|<M} f(\cdot - su)\, \frac{ds}{s}.$$

Strong continuity of translations makes $u \mapsto H^M_{\varepsilon,u} f$ continuous as an L^p-valued map. The one-dimensional fixed-truncation theorem, followed by slicing along lines parallel to u, gives

$$\sup_{u\in\mathbb{S}^{n-1}} \|H_{\varepsilon,u} f\|_p \le C_p\|f\|_p,$$

and $H^M_{\varepsilon,u} f \to H_{\varepsilon,u} f$ in L^p for every u. Hence $u \mapsto H_{\varepsilon,u} f$ is strongly measurable and Bochner integrable against every L^1 angular density. Likewise, for every $a \in L^1(0,\infty)$, the map

$$u \longmapsto \int_0^\infty a(s) f(\cdot - \varepsilon su)\, ds$$

is strongly continuous in L^p, by dominated convergence. Thus every application of (15.16) below satisfies its strong measurability and Bochner-integrability hypotheses.

By density, both conclusions extend to every $h \in L^p(\mathbb{R}^n)$. Indeed, the fixed-truncation estimate for $H_{\varepsilon,u}$ is uniform in u, while

$$\left\| \int_0^\infty a(s) h(\cdot - \varepsilon s u)\, ds \right\|_p \le \|a\|_1 \|h\|_p$$

uniformly in u. Approximating h in L^p by Schwartz functions therefore makes each corresponding L^p-valued map a uniform L^p limit, in u, of strongly measurable maps. In particular these conclusions apply to the functions $g_j = -R_j f$ introduced below.

Decompose $\Omega = \Omega_o + \Omega_e$ into its odd and even parts. The gauge is stable under this decomposition: this is (1.3).

The odd component.

For odd Ω, polar coordinates give the exact rotation identity

$$T^0_{\Omega,\varepsilon} f = \frac{1}{2} \int_{\mathbb{S}^{n-1}} \Omega(u) H_{\varepsilon,u} f \, d\sigma(u). \tag{15.35}$$

Lemma 15.3 and Lemma 15.4 yield

$$\mathscr{J}^{\mathrm{ch}}_{2,p}(T^0_{\Omega,\varepsilon} f : \varepsilon \in I) \le C_p \|\Omega\|_1 \|f\|_p. \tag{15.36}$$

The even component and the Riesz-transform factorization.

Assume now that Ω is even; when $n = 1$ its mean-zero condition makes it identically zero. Let K be the homogeneous principal-value distribution

$$K(x) = \Omega(x/|x|)|x|^{-n}.$$

Its action on a Schwartz function is obtained by integrating over $\{\delta < |x| < R\}$ and passing to $\delta \downarrow 0$, $R \uparrow \infty$. More explicitly, angular cancellation gives the absolutely convergent formula

$$\langle K, \phi \rangle = \int_{\mathbb{S}^{n-1}} \Omega(u) \int_0^\infty \big[\phi(ru) - \phi(0)\mathbf{1}_{(0,1)}(r)\big] \frac{dr}{r}\, d\sigma(u), \qquad \phi \in \mathscr{S}(\mathbb{R}^n).$$

Near zero the bracket is $O(r)$, while at infinity Schwartz decay applies; thus this formula also proves that the principal-value limit exists and is tempered.

Choose a smooth radial function Φ with $\Phi = 0$ on $[0, 1/4]$ and $\Phi = 1$ on $[3/4, \infty)$, and put $\psi = 1 - \Phi$. We now fix the distributional normalization of every Riesz transform used below. Write

$$\mathfrak{r}_j(\xi) = -i \frac{\xi_j}{|\xi|}, \qquad \xi \neq 0,$$

so that $\sum_j \mathfrak{r}_j(\xi)^2 = -1$.

The following rough-multiplier estimate fixes all low-frequency choices:

$$\|\widehat{K}\|_{L^\infty(\mathbb{R}^n \setminus \{0\})} + \|\widehat{K\psi}\|_{L^\infty(\mathbb{R}^n)} \le C_n \mathfrak{L}(\Omega). \tag{15.37}$$

For completeness, if $e = \xi/|\xi|$, the polar-coordinate formula for the even kernel gives, up to a harmless Fourier-normalization constant,

$$\widehat{K}(\xi) = c_n \int_{\mathbb{S}^{n-1}} \Omega(u) \log|e \cdot u| \, d\sigma(u).$$

The term containing $\log|\xi|$ is annihilated by $\int \Omega = 0$, and the sine term vanishes because Ω is even. Also

$$\widehat{K\psi}(\xi) = \int_{\mathbb{S}^{n-1}} \Omega(u) a_\psi(\xi \cdot u)\, d\sigma(u), \qquad a_\psi(t) = \int_0^\infty \psi(r)(\cos(rt) - 1)\, \frac{dr}{r}.$$

One has $|a_\psi(t) + \log^+|t|| \le C_\psi$ and, uniformly for $R > 0$ and $|\tau| \le 1$,

$$|a_\psi(R\tau) + \log^+ R| \le C_\psi\big(1 + |\log|\tau||\big).$$

The functions $u \mapsto 1 + |\log|e \cdot u||$ have uniformly bounded exponential-Orlicz norm. The $L\log L$–exponential-Orlicz Hölder inequality first gives

$$\left| \int \Omega(u) \log|e \cdot u| \, d\sigma(u) \right| \le C_n \mathfrak{L}(\Omega).$$

Writing $\xi = Re$ and using $\int \Omega = 0$ also gives the exact cancellation

$$\widehat{K\psi}(Re) = \int_{\mathbb{S}^{n-1}} \Omega(u)[a_\psi(Re \cdot u) + \log^+ R] \, d\sigma(u),$$

so the preceding uniform estimate with $\tau = e \cdot u$ and the same Orlicz Hölder inequality prove (15.37).

To justify the formulas without a boundedness assumption on Ω, put $m_n = \sigma(\mathbb{S}^{n-1})$ and set

$$\Omega_M = \Omega \mathbf{1}_{\{|\Omega| \le M\}} - m_n^{-1} \int_{\mathbb{S}^{n-1}} \Omega \mathbf{1}_{\{|\Omega| \le M\}} \, d\sigma.$$

Then Ω_M is bounded, even, and mean zero, and $\mathfrak{L}(\Omega_M - \Omega) \to 0$ by absolute continuity of the $L\log L$ Luxemburg norm. The displayed distributional formulas hold for Ω_M; their uniform Orlicz bounds and convergence in $\mathscr{S}'$ allow $M \to \infty$, proving them for the original Ω.

We define, as tempered distributions,

$$\widehat{N_j} = \mathfrak{r}_j \widehat{K}, \qquad \widehat{R_j(K\psi)} = \mathfrak{r}_j \widehat{K\psi},$$

and

$$\widehat{R_j(K\Phi)} = \mathfrak{r}_j \big(\widehat{K} - \widehat{K\psi} \big).$$

Values assigned at $\xi = 0$ are irrelevant, since all three multipliers are locally integrable functions. In particular,

$$R_j(K\Phi) = N_j - R_j(K\psi)$$

holds exactly in $\mathscr{S}'$, with no term supported at the frequency origin. Because $\mathfrak{r}_j$ is homogeneous of degree zero, these operators commute with L^1-normalized dilations and with $\mathscr{A}_0 = x \cdot \nabla + n$. On separated supports the same normalization agrees with the absolutely convergent Riesz-kernel pairing, and on H^1 pieces it agrees with the usual $H^1 \to L^1$ Riesz transform. Both assertions follow by smoothing $\mathfrak{r}_j$ near $\xi = 0$ and passing to the limit.

We repeatedly use the following explicit fixed-annulus consequence of Lemma 15.8. If $\eta \in C_c^\infty((0,\infty))$ is supported in $c \le r \le C$, then

$$F_\eta(x) = \eta(|x|) K(x), \qquad \int F_\eta = 0, \qquad \|F_\eta\|_{H^1} \le C_{\eta,n} \mathfrak{L}(\Omega). \tag{15.38}$$

The cancellation follows in polar coordinates. Put $c_\eta = \int_0^\infty |\eta(r)| dr/r$. If $c_\eta \|\Omega\|_1 = 0$, then $F_\eta = 0$ and the claim is immediate. Otherwise

$$\|F_\eta\|_1 = c_\eta \|\Omega\|_1.$$

Since r stays in a fixed compact subinterval of $(0,\infty)$, polar coordinates also give

$$\begin{aligned} &\int |F_\eta| \log\left(e + \frac{|F_\eta|}{\|F_\eta\|_1} \right) \\ &\qquad \le C_{\eta,n} \int_{\mathbb{S}^{n-1}} |\Omega(u)| \log\left(e + C_{\eta,n} \frac{|\Omega(u)|}{\|\Omega\|_1} \right) d\sigma(u) \le C_{\eta,n} \mathfrak{L}(\Omega). \end{aligned}$$

The asserted H^1 bound now follows from Lemma 15.8.

We first establish

$$N_j(x) = \frac{\nu_j(x/|x|)}{|x|^n}, \qquad \nu_j \text{ odd}, \qquad \|\nu_j\|_1 \le C_n \mathfrak{L}(\Omega). \tag{15.39}$$

It suffices to estimate N_j on $\{3/4 \le |x| \le 3/2\}$. Choose a smooth radial partition $\chi_< + \chi_0 + \chi_> = 1$ with

$$\operatorname{supp}\chi_< \subset [0,2/3), \qquad \operatorname{supp}\chi_0 \subset (1/2,3), \qquad \operatorname{supp}\chi_> \subset (2,\infty),$$

so that the supports of the first and last pieces are separated from the compact x-annulus. If $\rho_j(z) = c_n z_j |z|^{-n-1}$ is the Riesz kernel, angular cancellation gives for the first piece

$$\int_0^{2/3} \chi_<(r) \int \Omega(u)[\rho_j(x-ru) - \rho_j(x)]\, d\sigma(u) \frac{dr}{r},$$

whose absolute value is at most $C\|\Omega\|_1$. The last piece is absolutely bounded by the same quantity, because $|x-y| \simeq |y|$ and the remaining radial integral is $\int_2^\infty r^{-n} dr/r$. Indeed, on the first piece the mean-value theorem gives $|\rho_j(x-ru) - \rho_j(x)| \le Cr$, so its radial integral is $C\int_0^{2/3} dr$. The middle piece is controlled by (15.38); hence its Riesz transform is in L^1 with the same bound. Integrating over the compact x-annulus, and then using homogeneity and parity of the Riesz multiplier, gives

$$\int_{3/4\le|x|\le 3/2} |N_j(x)|dx = \log 2 \int_{\mathbb{S}^{n-1}} |\nu_j(u)| d\sigma(u) \le C_n \mathfrak{L}(\Omega),$$

which proves (15.39). The annular calculation gives an actual locally integrable representative away from the origin. Two homogeneous extensions that agree there can differ only by a finite linear combination of derivatives of δ_0. Homogeneity of degree $-n$ permits only a multiple of δ_0, which is even, whereas the Fourier-defined N_j is odd. Thus there is no origin-supported ambiguity.

Define the commutator correction

$$E_j = R_j(K\psi) - \psi N_j. \tag{15.40}$$

By the Fourier normalization above and the identity $K\Phi = K - K\psi$ in $\mathscr{S}'$, one has the exact distributional identity

$$R_j(K\Phi) = \Phi N_j - E_j. \tag{15.41}$$

Let $\mathscr{A}_0 = x\cdot\nabla + n$, the generator of L^1-normalized dilations. We next prove

$$\|E_j\|_1 + \|(\mathscr{A}_0 - 1)E_j\|_1 \le C_n \mathfrak{L}(\Omega). \tag{15.42}$$

For $|x| < 1/8$, formula (15.41) reads $E_j = -R_j(K\Phi)$ and the support separation gives an absolutely convergent bound. More precisely, splitting at $|y| = 1$ gives

$$|R_j(K\Phi)(x)| \le C\|\Omega\|_1 \left(\int_{1/4}^1 \frac{dr}{r} + \int_1^\infty r^{-n} \frac{dr}{r} \right).$$

For $|x| > 2$, use $E_j = R_j(K\psi)$ and subtract $\rho_j(x)$ inside the angular integral. Here ψ is supported in $[0,3/4]$ and $\int K(y)\psi(|y|)dy = 0$ in the principal-value sense, so the mean-value theorem applied to

$$\int \Omega(u) \int_0^{3/4} \psi(r)[\rho_j(x-ru) - \rho_j(x)] \frac{dr}{r} d\sigma(u)$$

gives $C\|\Omega\|_1 |x|^{-n-1}$. On the intervening annulus, $E_j = \Phi N_j - R_j(K\Phi)$. The first term is integrable by (15.39). Choose $\zeta \in C_c^\infty((1/8,8))$ equal to one on $[1/4,4]$. The term $\zeta K\Phi$ is controlled by (15.38) and $R_j : H^1 \to L^1$. The remainder $(1-\zeta)K\Phi$ is separated from the intervening x-annulus and is absolutely integrable after applying the Riesz kernel. This proves $\|E_j\|_1 \le C\mathfrak{L}(\Omega)$.

Since $\mathscr{A}_0$ commutes with every R_j and annihilates the homogeneous distributions K, N_j,

$$\mathscr{A}_0 E_j = R_j(K(r\psi'(r))) - r\psi'(r) N_j.$$

The factor $r\psi'(r)$ is supported in the fixed annulus $1/4 \le r \le 3/4$. The first term is controlled by (15.38) and $R_j : H^1 \to L^1$, and the second by (15.39). Together with the bound for E_j, this proves (15.42).

We next choose the ray representatives in a jointly measurable way. In logarithmic polar coordinates put

$$G_j(s,u) = e^{(n-1)s}E_j(e^s u), \qquad D_j(s,u) = e^{(n-1)s}(\mathscr{A}_0 - 1)E_j(e^s u).$$

Both functions are locally integrable in (s,u), and $\partial_s G_j = D_j$ in the distributional sense, because, with $t = e^s$,

$$\partial_s[t^{n-1}E_j(tu)] = t^{n-1}(\mathscr{A}_0 - 1)E_j(tu).$$

Applying Fubini first on each compact s-interval gives this one-dimensional distributional identity for almost every u. Choose $\vartheta \in C_c^\infty(\mathbb{R})$ with $\int \vartheta = 1$, and, for every u for which the following integrals are defined, set

$$c_j(u) = \int_{\mathbb{R}} \vartheta(s)\left(G_j(s,u) - \int_0^s D_j(v,u)\,dv\right) ds,$$

$$\widetilde{G}_j(s,u) = c_j(u) + \int_0^s D_j(v,u)\,dv.$$

Let $\mathscr{N}_j^{(0)}$ be the exceptional null set of directions on which either these integrals are not defined or the one-dimensional distributional identity above fails, and set $\widetilde{G}_j = 0$ on $\mathscr{N}_j^{(0)}$. Fubini and the one-dimensional $W^{1,1}_{\mathrm{loc}}$ theorem show that $\widetilde{G}_j = G_j$ almost everywhere. This construction makes $(s,u) \mapsto \widetilde{G}_j(s,u)$ jointly measurable and makes $s \mapsto \widetilde{G}_j(s,u)$ locally absolutely continuous for almost every u.

Define

$$\varphi_{j,u}(t) = \widetilde{G}_j(\log t, u), \qquad t > 0.$$

The map $(u,t) \mapsto \varphi_{j,u}(t)$ is jointly measurable. For every u, the map $t \mapsto \varphi_{j,u}(t)$ is continuous (locally absolutely continuous for every nonexceptional direction and identically zero on the exceptional set), and hence is a right-continuous representative. Moreover,

$$\varphi'_{j,u}(t) = t^{n-2}(\mathscr{A}_0 - 1)E_j(tu)$$

for almost every (t,u). Polar coordinates give

$$\int_{\mathbb{S}^{n-1}} B(\varphi_{j,u})\,d\sigma(u) = \|(\mathscr{A}_0 - 1)E_j\|_1 \le C\mathfrak{L}(\Omega),$$

and also

$$\int_{\mathbb{S}^{n-1}} \int_0^\infty |\varphi_{j,u}(t)|\,dt\,d\sigma(u) = \|E_j\|_1.$$

For almost every u, the derivative of $\varphi_{j,u}$ is integrable on $(1,\infty)$, since

$$\int_1^\infty |\varphi'_{j,u}(t)|\,dt \le \int_1^\infty t^{n-1}|(\mathscr{A}_0 - 1)E_j(tu)|\,dt.$$

Integrating this inequality in u gives

$$\int_{\mathbb{S}^{n-1}} \int_1^\infty |\varphi'_{j,u}(t)|\,dt\,d\sigma(u) \le \|(\mathscr{A}_0 - 1)E_j\|_1 < \infty.$$

To ensure that every direction satisfies the hypotheses of Lemma 15.5, enlarge the exceptional set to

$$\mathscr{N}_j := \mathscr{N}_j^{(0)} \cup \{u : B(\varphi_{j,u}) = \infty\} \cup \left\{u : \int_0^\infty |\varphi_{j,u}(t)|\,dt = \infty\right\}$$
$$\cup \left\{u : \int_1^\infty |\varphi'_{j,u}(t)|\,dt = \infty\right\}.$$

For $u \notin \mathscr{N}_j^{(0)}$, local absolute continuity and the displayed derivative identity give

$$B(\varphi_{j,u}) = \int_0^\infty t^{n-1}|(\mathscr{A}_0 - 1)E_j(tu)|\,dt,$$

On $\mathscr{N}_j^{(0)}$ the profile is zero, so its B-value is zero; extend the displayed integral by zero there as well. Thus $u \mapsto B(\varphi_{j,u})$ is measurable by joint measurability of the integrand; the other two defining integrals are measurable for the same reason. Tonelli and the three displayed integral bounds now show that every set in the union is null. Redefine both $\widetilde{G}_j(s,u)$ and $\varphi_{j,u}(t)$ to be zero for $u \in \mathscr{N}_j$. This preserves joint measurability and changes the functions only on a product-null set. For $u \notin \mathscr{N}_j$, the integrable tail derivative gives a limit at infinity, and $L^1(0,\infty)$ integrability forces that limit to be zero; for $u \in \mathscr{N}_j$ the same conclusion is immediate from the zero definition. Thus every direction, not merely almost every direction, satisfies the profile hypotheses. The product-null modification does not change the polar-coordinate convolution identity, so all hypotheses of the parameterized form of Lemma 15.5 are satisfied.

For a distribution F write $F^{(\varepsilon)}(x) = \varepsilon^{-n}F(x/\varepsilon)$. Define

$$T_\varepsilon^0 f(x) = \int_{|y|<\varepsilon} K(y)\Phi(|y|/\varepsilon)f(x-y)dy.$$

Since $(K\Phi)^{(\varepsilon)}(x) = K(x)\Phi(|x|/\varepsilon)$,

$$T_{\Omega,\varepsilon}^0 f = (K\Phi)^{(\varepsilon)} * f - T_\varepsilon^0 f. \tag{15.43}$$

For the current Schwartz function f, put $g_j = -R_j f$, so that $f = \sum_j R_j g_j$. The multiplier normalization (15.37) makes the following calculation an L^2 identity. Since $\widehat{g_j} = -\mathfrak{r}_j \widehat{f}$ and $\mathfrak{r}_j(\varepsilon\xi) = \mathfrak{r}_j(\xi)$,

$$\begin{aligned}\sum_{j=1}^n \widehat{(R_j(K\Phi))^{(\varepsilon)} * g_j}(\xi) &= -\widehat{K\Phi}(\varepsilon\xi)\sum_{j=1}^n \mathfrak{r}_j(\xi)^2\widehat{f}(\xi)\\ &= \widehat{K\Phi}(\varepsilon\xi)\widehat{f}(\xi).\end{aligned}$$

Thus

$$(K\Phi)^{(\varepsilon)} * f = \sum_{j=1}^n (R_j(K\Phi))^{(\varepsilon)} * g_j$$

in L^2 and in $\mathscr{S}'$. Substituting (15.41) into this exact identity and then using (15.43) gives

$$T_{\Omega,\varepsilon}^0 f = \sum_{j=1}^n (\Phi N_j)^{(\varepsilon)} * g_j - \sum_{j=1}^n E_j^{(\varepsilon)} * g_j - T_\varepsilon^0 f. \tag{15.44}$$

Each of the three families is now controlled without a dyadic summation. First, let $q(t) = \Phi(t)t^{-1}\mathbf{1}_{(0,1)}(t)$, with the right-continuous value $q(1) = 0$. Since Φ vanishes near zero,

$$B(q) = \int_0^1 t\left|\frac{d}{dt}\frac{\Phi(t)}{t}\right|dt + |\Phi(1)| < \infty.$$

Moreover,

$$((\Phi N_j)^{(\varepsilon)} * g)(x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} \nu_j(u)\int_{\mathbb{R}} \Phi(|s|/\varepsilon)g(x-su)\frac{ds}{s}\,d\sigma(u),$$

while

$$\int_{\mathbb{R}} \Phi(|s|/\varepsilon)g(x-su)\frac{ds}{s} = H_{\varepsilon,u}g + P_{\varepsilon,u}^q g - P_{\varepsilon,-u}^q g.$$

Thus (15.39), Lemma 15.3, Lemma 15.4, and Lemma 15.5 control the first sum in (15.44) by $C\mathfrak{L}(\Omega)\sum_j \|g_j\|_p$, uniformly for $\varepsilon \in I$. For the second sum, polar coordinates give the explicit Bochner representation

$$(E_j^{(\varepsilon)} * g)(x) = \int_{\mathbb{S}^{n-1}}\int_0^\infty \varphi_{j,u}(t)g(x-\varepsilon tu)\,dt\,d\sigma(u).$$

The profiles $\varphi_{j,u}$ and Lemma 15.5 therefore control it by the same quantity, uniformly on I. Finally,

$$T_\varepsilon^0 f(x) = \int_{\mathbb{S}^{n-1}} \Omega(u)\int_0^1 q(t)f(x-\varepsilon tu)\,dt\,d\sigma(u),$$

which is the parameterized profile family $\varphi_u(t) = \Omega(u)q(t)$ and satisfies

$$\int_{\mathbb{S}^{n-1}} B(\varphi_u)d\sigma(u) = B(q)\|\Omega\|_1.$$

Thus Lemmas 15.3 and 15.5 give $C\|\Omega\|_1\|f\|_p$. The scalar L^p bounds for the Riesz transforms [37] prove the required estimate for the even component. Combining it with (15.36) by the subadditive jump seminorm in Lemma 15.3, and then using (1.3), proves (15.34) for Schwartz functions, uniformly over every finite ordered parameter set I.

Extension in the truncation parameter and in the input.

All estimates above are uniform over finite parameter sets, and the same factorization with ordinary L^p norms bounds every fixed coordinate. Fix a countable dense set $D \subset (0,\infty)$, and choose $f_m \in \mathscr{S}(\mathbb{R}^n)$ with $f_m \to f$ in L^p so rapidly that

$$\sum_m \|f_{m+1} - f_m\|_p < \infty.$$

For every $\varepsilon \in D$, the right-hand side of (15.44) is Cauchy in L^p; denote its limit by $T^0_{\Omega,\varepsilon}f$.

Let $I \subset D$ be finite. After passing to a subsequence, the coordinates for f_m converge almost everywhere simultaneously on I. Lemma 15.6, in the form (15.30), and Fatou's lemma give

$$\mathscr{J}^{\mathrm{ch}}_{2,p}(T^0_{\Omega,\varepsilon}f : \varepsilon \in I) \le C_{n,p}\mathfrak{L}(\Omega)\|f\|_p.$$

The subsequence may depend on I, which is harmless because the bound is uniform in I. If $I_1 \subset I_2 \subset \cdots$ are finite and exhaust D, then $\mathbf{J}^{\mathrm{ch}}_{I_k}$ increases pointwise to the chain functional on D: every strict jump witness is finite and is therefore contained in some I_k. Monotone convergence in the spatial variable gives the same estimate on all of D.

It remains to record compatibility of the chosen coordinates. If $\varepsilon < \eta$ lie in D, then for every m

$$T^0_{\Omega,\varepsilon}f_m - T^0_{\Omega,\eta}f_m = \int_{\varepsilon<|z|\le\eta} K_\Omega(z)f_m(\cdot - z)\,dz. \tag{15.45}$$

The finite-annulus kernel is in L^1, with norm $\|\Omega\|_1 \log(\eta/\varepsilon)$. Young's inequality passes this identity in L^p to the limiting coordinates. Since D is countable, one null set may be discarded so that the identity holds simultaneously for every pair in D.

Fix $\varepsilon_0 \in D$. For arbitrary $\varepsilon > 0$ define

$$T^0_{\Omega,\varepsilon}f = \begin{cases} T^0_{\Omega,\varepsilon_0}f + \displaystyle\int_{\varepsilon<|z|\le\varepsilon_0} K_\Omega(z)f(\cdot - z)dz, & \varepsilon < \varepsilon_0, \\ T^0_{\Omega,\varepsilon_0}f - \displaystyle\int_{\varepsilon_0<|z|\le\varepsilon} K_\Omega(z)f(\cdot - z)dz, & \varepsilon > \varepsilon_0, \\ T^0_{\Omega,\varepsilon_0}f, & \varepsilon = \varepsilon_0. \end{cases}$$

The preceding identity leaves all coordinates of D unchanged. On each $0 < a < b < \infty$ and every bounded measurable $U \subset \mathbb{R}^n$, Fubini and Hölder's inequality give

$$\begin{aligned} \int_U \int_a^b \int_{\mathbb{S}^{n-1}} |\Omega(u)||f(x - ru)|\,d\sigma(u)\frac{dr}{r}dx \\ \le |U|^{1/p'}\|f\|_p\|\Omega\|_1 \log(b/a) < \infty. \end{aligned}$$

Thus, for almost every x,

$$\int_a^b \int_{\mathbb{S}^{n-1}} |\Omega(u)||f(x - ru)|\,d\sigma(u)\,\frac{dr}{r} < \infty.$$

Taking a countable exhaustion by rational a, b and bounded balls U produces one full-measure set on which this holds on every compact subinterval of $(0,\infty)$. The resulting path is locally absolutely continuous, with almost-everywhere derivative

$$-\varepsilon^{-1}\int_{\mathbb{S}^{n-1}} \Omega(u)f(x - \varepsilon u)d\sigma(u).$$

Its strict jump counts on D equal those on all positive parameters by Lemma 15.6. Finally use (15.11).

The zero-at-infinity anchor.

For Schwartz data, the defining tail integral tends to zero pointwise as $\varepsilon \to \infty$ by absolute convergence and Schwartz decay. Equation (15.7) therefore gives

$$\sup_{\varepsilon>0} |T^0_{\Omega,\varepsilon} f| \le \mathbf{J}(T^0_{\Omega,\varepsilon} f : \varepsilon > 0).$$

The right-hand side belongs to L^p by the estimate already proved, so dominated convergence also gives the L^p anchor on the Schwartz core.

For arbitrary $f \in L^p$, use the rapidly convergent Schwartz sequence chosen above and put

$$M_m(x) = \sup_{\varepsilon>0} |T^0_{\Omega,\varepsilon}(f_{m+1} - f_m)(x)|.$$

Each difference is a Schwartz function and its trajectory is anchored at infinity; hence (15.7) and the pointwise jump estimate give

$$\sum_m \|M_m\|_p \le C_{n,p}\mathfrak{L}(\Omega) \sum_m \|f_{m+1} - f_m\|_p < \infty.$$

Thus $\sum_m M_m < \infty$ almost everywhere, and the core trajectories converge uniformly in $\varepsilon > 0$ outside one null set. Their limit agrees at every parameter in D with the coordinates constructed above, preserves the finite-annulus identities and local continuity, and is anchored at infinity: first approximate uniformly by one core trajectory and then send its truncation radius to infinity. The same estimate shows convergence in L^p uniformly over ε, so the anchor also holds in L^p. Finally use (15.14) to return to the paired convention. Corollary 12.2 constructs the absolutely convergent positive truncations for this $\Omega \in L^1$ and $f \in L^p$ on one common full-measure set. They have the same finite-annulus differences and zero limit at infinity as the path just constructed. Their difference is constant in the radius and has zero limit at infinity, so they agree simultaneously at every positive radius. This completes the proof. □

16 The polynomial phase flag and scale-covariant annular decay

This section develops the coefficient-uniform oscillatory input. Pure output and input phases do not affect operator norms or pointwise jump counts, because

$$P(x,y) = p(x) + q(y) + \widetilde{P}(x,y) \quad \Longrightarrow \quad T^P_{\Omega,\varepsilon} f = e^{ip} T^{\widetilde{P}}_{\Omega,\varepsilon}(e^{iq} f). \tag{16.1}$$

16.1 Algebraic flag

Let $\mathscr{P}_d$ be the space of real polynomials of degree at most d. Set

$$\Delta P(y,u,h) = P(y+u, y+h) - P(y+u, y), \qquad \mathfrak{D}P = \operatorname{Mix} \Delta P,$$

where Mix retains precisely the monomials having positive degree in both y and u.

Lemma 16.1 (Kernel of the transverse map)**.** *For every $P \in \mathscr{P}_d$, one has $\mathfrak{D}P = 0$ if and only if*

$$P(x,y) = p(x) + q(y) + Q(x-y) + x^T B y, \qquad B^T = -B. \tag{16.2}$$

Equivalently, for every $a \in \mathbb{R}^n$ there are polynomials p_a, q_a such that, for all $X, Y \in \mathbb{R}^n$,

$$P(a+X, a+Y) - P(X,Y) = p_a(X) + q_a(Y).$$

Proof. The phases in (16.2) are annihilated by $\mathfrak{D}$. Conversely, $\mathfrak{D}P = 0$ means that ΔP has no monomial of positive degree in both y and u. Equivalently,

$$\partial_{y_j}\partial_{u_k}\Delta P(y,u,h) = 0 \qquad (1 \le j,k \le n).$$

Differentiate this identity in h_i at $h = 0$ and put $x = y+u$. Since

$$\partial_{h_i}\Delta P(y,u,h)|_{h=0} = \partial_{y_i}P(y+u,y),$$

the chain rule shows that the mixed Hessian $M_{ki} = \partial_{x_k}\partial_{y_i}P$ satisfies $(\partial_{x_j} + \partial_{y_j})M_{ki} = 0$, so $M(x,y) = F(x-y)$. Commutation of x- and y-derivatives gives

$$\partial_{z_\ell}F_{ki} = \partial_{z_k}F_{\ell i}, \qquad \partial_{z_\ell}F_{ki} = \partial_{z_i}F_{k\ell}.$$

Put $B = (F - F^T)/2$. The two compatibility identities imply, for every i,k,ℓ,

$$\partial_{z_\ell}F_{ik} = \partial_{z_i}F_{\ell k} = \partial_{z_k}F_{\ell i} = \partial_{z_\ell}F_{ki}.$$

Thus every entry of B is constant; in particular, B is a constant skew-symmetric matrix. Put $S = F - B = (F + F^T)/2$. The symmetric potential is produced by the following polynomial Poincaré construction. For each k the polynomial one-form $\sum_i S_{ki}(z)\,dz_i$ is closed, and therefore

$$v_k(z) = \int_0^1 \sum_i S_{ki}(tz)z_i\,dt$$

satisfies $\partial_i v_k = S_{ki}$. Symmetry of S makes $\sum_k v_k(z)\,dz_k$ closed, so

$$Q(z) = -\int_0^1 \sum_k v_k(tz)z_k\,dt$$

is a polynomial with $-\partial_k\partial_i Q = S_{ki}$. Subtracting $Q(x-y) + x^T By$ leaves zero mixed Hessian, because

$$\partial_{x_k}\partial_{y_i}\big(Q(x-y) + x^T By\big) = -\partial_k\partial_i Q(x-y) + B_{ki} = S_{ki}(x-y) + B_{ki} = F_{ki}(x-y).$$

A polynomial with zero mixed Hessian is a sum of a pure x-polynomial and a pure y-polynomial, which proves the converse. The resulting decomposition plainly has the stated translation property for every a. Conversely, that property implies, after taking the mixed Hessian, that $M(x+a,y+a) = M(x,y)$ for every a; hence $M(x,y) = F(x-y)$, and the compatibility and polynomial Poincaré argument just given recovers (16.2), and therefore $\mathfrak{D}P = 0$. □

Order the mixed monomials in (y,u,h) first by decreasing y-degree, then by decreasing u-degree, decreasing h-degree, decreasing total degree, and finally by a fixed lexicographic order. Columns are listed in this order; throughout, the pivot of a nonzero row means its first nonzero column, normalized to have coefficient one, and rows are listed in the order of their pivots. The substitutions defining Δ are linear and contain no constant term, and Mix only deletes monomials. Consequently $\mathfrak{D}$ preserves total homogeneous degree and $\operatorname{im}\mathfrak{D}$ is the direct sum of its homogeneous components. In each such component take reduced row-echelon form with the preceding column and pivot convention. Merge the rows from the separate homogeneous components in the global order of their pivot monomials, and call them $F_1,\dots,F_M$. Here homogeneous means homogeneous in the total degree of (y,u,h), and the global monomial order restricts to each homogeneous component. To check the echelon property after merging, let $\pi_\nu \prec \pi_\mu$ be two pivot monomials. Every monomial in F_μ has the same total degree as π_μ. If one of them, say τ, also satisfied $\tau \prec \pi_\nu$, then $\tau \prec \pi_\mu$ and the reduced row-echelon construction in the component containing F_μ would force its coefficient to vanish. If the two rows have the same total degree, reduced row-echelon form also clears the π_ν column from F_μ; if their total degrees differ, F_μ

cannot contain π_ν at all. Thus no row later than F_ν contains a monomial preceding or equal to the pivot of F_ν. We record this global echelon property for later use: if π_ν is the pivot of F_ν, then

$$\mu > \nu \quad \Longrightarrow \quad \operatorname{supp} F_\mu \cap \{\tau : \tau \preceq \pi_\nu\} = \varnothing. \tag{16.3}$$

This includes rows of a different total degree, since a row and all of its monomials lie in the same homogeneous component. Homogeneous preimages may be chosen without an extra assumption: if F_μ has degree r and $\mathfrak{D}H = F_\mu$, write $H = \sum_k H^{[k]}$ as a sum of homogeneous parts. The graded property gives $\mathfrak{D}H^{[r]} = F_\mu$ and $\mathfrak{D}H^{[k]} = 0$ for $k \neq r$. We therefore take $H_\mu = H^{[r]}$, so that $\mathfrak{D}H_\mu = F_\mu$ and H_μ is homogeneous. Replacing $H_\mu(x,y)$ by $H_\mu(x,y) - H_\mu(y,y)$, which changes only a pure input phase, arrange

$$H_\mu(y,y) = 0. \tag{16.4}$$

Append to this list, in decreasing homogeneous degree and fixed monomial order, the convolution monomials $(x-y)^\alpha$ with $2 \le |\alpha| \le d$, and, when $d \ge 2$, a fixed basis $x_i y_j - x_j y_i$ of skew bilinear forms. Denote the full finite list again by $H_1, \dots, H_{N_d}$, and write $m_\mu = \deg H_\mu$. Modulo pure phases every $P \in \mathscr{P}_d$ has a unique first nonzero flag coordinate c_ν.

Theorem 16.2 (Global polynomial phase flag)**.** *For each $n \in \mathbb{N}$ and $d \in \mathbb{N}_0$ there is a finite ordered family of homogeneous real polynomials $H_1, \dots, H_{N_d}$ with the following properties.*

(i) *Every $P \in \mathscr{P}_d$ has a unique representation modulo pure input and output phases,*

$$P(x,y) = p(x) + q(y) + \sum_{\mu=1}^{N_d} c_\mu(P) H_\mu(x,y). \tag{16.5}$$

Thus every nonpure phase has a well-defined first nonzero flag coordinate.

(ii) *The coordinates lying in* $\ker \mathfrak{D}$ *occur after the transverse coordinates and, modulo pure phases, span exactly*

$$Q(x-y) + x^T B y, \qquad B^T = -B.$$

In particular, the same flag incorporates convolution and alternating-bilinear phases as its final nonterminal strata.

(iii) *If ν is the first nonzero coordinate of P and $m_\nu = \deg H_\nu$, then, for every dilation factor $\rho > 0$, the simultaneous dilation $(x,y) = (\rho X, \rho Y)$ changes that coordinate to $c_\nu \rho^{m_\nu}$. For every $a \in \mathbb{R}^n$,*

$$P(a+X, a+Y) = p_a(X) + q_a(Y) + c_\nu H_\nu(X,Y) + \sum_{\mu > \nu} c_{\mu,a} H_\mu(X,Y). \tag{16.6}$$

Thus translation preserves the first coordinate and creates only later coordinates.

(iv) *Suppose that H_ν is transverse and that $y^\alpha u^\beta h^\gamma$ is the pivot monomial of $\mathfrak{D}H_\nu$ in the reduced-row-echelon construction above. If P has first coordinate $c_\nu = \pm 1$, then the coefficient of $y^\alpha u^\beta$ in $\mathfrak{D}P$ is a polynomial $a_P(h)$ of degree $|\gamma|$. Its h^γ coefficient is exactly $c_\nu = \pm 1$, and its top homogeneous coefficient tensor has norm at least $c_{n,d} > 0$. This conclusion is uniform in every later flag coefficient.*

Proof. Lemma 16.1 gives spanning; we record explicitly why the resulting coordinates are unique modulo pure phases. The reduced row-echelon rows $F_1, \dots, F_M$ are linearly independent. Hence, if a linear combination of their preimages together with the appended kernel elements is pure, applying $\mathfrak{D}$ first forces every transverse coefficient to vanish. It remains to consider an identity

$$Q(x-y) + x^T B y = p(x) + q(y), \qquad B^T = -B,$$

where Q is a linear combination of the appended convolution monomials and therefore has no terms of degree at most one. Taking the mixed Hessian in (x,y) gives

$$-\nabla^2 Q(x-y)+B=0.$$

The first matrix is symmetric and the second is skew-symmetric, so both vanish. Thus $B=0$ and Q is affine; by the chosen degree range for Q, this means $Q=0$. This proves linear independence of the appended kernel basis in the quotient and, together with Lemma 16.1, proves the representation and the description of the kernel strata. Homogeneity gives the dilation statement. For a transverse row,

$$\mathfrak{D}(P(a+\cdot,a+\cdot))=\mathrm{Mix}[(\mathfrak{D}P)(y+a,u,h)].$$

Translation preserves the highest y-degree term and creates only lower y-degree terms, which are later in the chosen order. Reduced echelon form, specifically (16.3), prevents a later row from altering the pivot coefficient. A convolution monomial is unchanged by simultaneous translation, and a skew bilinear form changes only by pure input and output phases. This proves (iii).

For (iv), project $\mathfrak{D}P$ onto the scalar tensor component indexed by $y^\alpha u^\beta$. The pivot row contributes $\pm h^\gamma$. No later row contains the pivot monomial, by (16.3), and no later row contains a monomial with the same y- and u-component and higher h-order, because such a monomial would precede the pivot in the defining order. Thus the h^γ coefficient remains ±1, and later coefficients can change only terms of lower h-degree or other multiindices of the same h-degree. In particular, even in the latter case the fixed h^γ coordinate of the top homogeneous coefficient tensor remains ±1. The coordinate functional selecting h^γ has norm at most $C_{n,d}$ on each of the finitely many fixed coefficient spaces that occur. Consequently a_P has degree $|\gamma|$, and the norm of its top homogeneous coefficient tensor is at least $c_{n,d}>0$, uniformly in all later coefficients. □

For packet arguments it is convenient to use anchored representatives

$$\widehat{H}_\mu(X,Y)=H_\mu(X,Y)-H_\mu(0,Y). \tag{16.7}$$

They have the same flag coordinates modulo pure input phases and satisfy $\widehat{H}_\mu(0,Y)=0$.

16.2 Polynomial oscillation estimates

We give the elementary estimates needed for rough annular decay.

Lemma 16.3 (Sublevel and negative moments)**.** *Let $k\ge1$, let $p(t)=a_kt^k+\cdots+a_0$, $a_k\neq0$, on an interval I of length $L>0$, and let $\eta>0$. Then*

$$|\{t\in I:|p(t)|\le\eta\}|\le C_k(\eta/|a_k|)^{1/k}.$$

Consequently, if $0<\sigma<1/k$, then, uniformly in c and the lower coefficients,

$$\int_I|p(t)-c|^{-\sigma}dt\le C_{k,\sigma}|a_k|^{-\sigma}L^{1-k\sigma}. \tag{16.8}$$

If a is a polynomial of degree k on $\mathbb{R}^n$ whose top homogeneous coefficient norm is at least one, then, for every $R>0$ and every fixed $C_0<\infty$,

$$\int_{|h|\le C_0R}|a(h)|^{-\sigma}dh\le CR^{n-k\sigma},\qquad 0<\sigma<1/k. \tag{16.9}$$

Here the last constant may depend on C_0. If A_k is homogeneous of degree k and $\|A_k\|_{\mathrm{coeff}}=1$, then

$$\int_{\mathbb{S}^{n-1}}|A_k(\theta)|^{-\beta}d\sigma(\theta)\le C,\qquad 0<\beta<1/k. \tag{16.10}$$

Proof. Let $E = \{t \in I : |p(t)| \le \eta\}$ and suppose $|E| > 0$. Set $\delta = |E|/[2(k+1)]$. A greedy selection gives $k+1$ distinct points of E at mutual distance at least δ: after $j \le k$ points have been selected, the union of their open δ-neighborhoods has measure at most $2j\delta < |E|$ and therefore cannot cover E. After relabeling, we have $t_0 < \cdots < t_k$ and

$$|t_i - t_j| \ge \delta = c_k|E| \qquad (0 \le i, j \le k,\ i \ne j).$$

Lagrange interpolation at these points gives

$$|a_k| = \left| \sum_{j=0}^{k} \frac{p(t_j)}{\prod_{\ell \ne j}(t_j - t_\ell)} \right| \le C_k \eta |E|^{-k}.$$

This proves the sublevel estimate.

Apply it to $p - c$ and combine it with the trivial bound by L:

$$|\{t \in I : |p(t) - c| \le \eta\}| \le \min\left\{L, C_k(\eta/|a_k|)^{1/k}\right\}.$$

The layer-cake identity gives

$$\int_I |p(t) - c|^{-\sigma} dt = \sigma \int_0^\infty \eta^{-\sigma-1} |\{t \in I : |p(t) - c| < \eta\}|\, d\eta.$$

Split the last integral where the two bounds in the minimum agree, at a value $\eta_0 \asymp_k |a_k| L^k$. The part below η_0 is bounded by

$$C_k |a_k|^{-1/k} \int_0^{\eta_0} \eta^{1/k - \sigma - 1}\, d\eta,$$

and the part above it by $L \int_{\eta_0}^\infty \eta^{-\sigma-1}\, d\eta$. Both are at most $C_{k,\sigma} |a_k|^{-\sigma} L^{1-k\sigma}$ precisely when $0 < \sigma < 1/k$, proving (16.8).

For (16.9), write $a = A_k +$ lower-order terms. Finite-dimensional norm equivalence supplies $v \in \mathbb{S}^{n-1}$ such that

$$|A_k(v)| \ge c_{n,k} \|A_k\|_{\text{coeff}} \ge c_{n,k}.$$

Write $h = w + tv$ with $w \in v^\perp$. For fixed w, the polynomial $t \mapsto a(w + tv)$ has degree k and leading coefficient $A_k(v)$, independent of w. Its intersection with $|h| \le C_0 R$ is an interval of length at most $2C_0 R$. Hence (16.8), followed by integration over the w-projection of that ball, gives

$$\int_{|h| \le C_0 R} |a(h)|^{-\sigma} dh \le C R^{1-k\sigma} R^{n-1} = C R^{n-k\sigma}.$$

Finally apply this estimate with $a = A_k$ and $R = 2$ to the annulus $1 \le |x| \le 2$. Homogeneity and polar coordinates give

$$\int_{1 \le |x| \le 2} |A_k(x)|^{-\beta} dx = \left(\int_1^2 r^{n-1-k\beta}\, dr \right) \int_{\mathbb{S}^{n-1}} |A_k(\theta)|^{-\beta} d\sigma(\theta).$$

The radial factor is positive and finite, so (16.10) follows. □

Lemma 16.4 (Leading-coefficient van der Corput)**.** *Let $k \ge 1$, let I be an interval of length at most $L_0 < \infty$, and let*

$$\phi(t) = a_k t^k + \cdots + a_0, \qquad a_k \ne 0.$$

For every bounded-variation amplitude ψ supported in I,

$$\left| \int e^{i\phi(t)} \psi(t) dt \right| \le C_{k,L_0} \min(1, |a_k|^{-1/k}) \big(\|\psi\|_\infty + \mathrm{Var}(\psi) \big).$$

The constant is independent of all lower coefficients.

Proof. Extend ψ by zero off I. Its distributional derivative is a finite measure and

$$|D\psi|(\mathbb{R}) \le \mathrm{Var}_I(\psi) + 2\|\psi\|_\infty.$$

For $k = 1$, the trivial estimate applies when $|a_1| < 1$, while Stieltjes integration by parts applies when $|a_1| \ge 1$ and gives the asserted bound. Suppose $k \ge 2$. The case $|a_k| < 1$ follows from the trivial estimate $L_0\|\psi\|_\infty$, so assume $|a_k| \ge 1$ and let $\mu > 0$. Put $E = \{t \in I : |\phi'(t)| \le \mu\}$. Lemma 16.3, applied to ϕ', gives

$$|E| \le C_k(\mu/|a_k|)^{1/(k-1)}.$$

The boundary of E is contained in the roots of $\phi' = \mu$ and $\phi' = -\mu$ together with the endpoints of I. Consequently, after also splitting at the at most $k-2$ roots of ϕ'', $I \setminus E$ is the union of $O_k(1)$ intervals J on each of which $|\phi'| \ge \mu$ and ϕ' is monotone. On every such interval, with one-sided endpoint values (equivalently, on half-open intervals so that jumps are counted only once), Stieltjes integration by parts yields

$$\int_J e^{i\phi(t)}\psi(t)\,dt = \left[\frac{e^{i\phi(t)}\psi(t)}{i\phi'(t)}\right]_{\partial J} - \int_J \frac{e^{i\phi(t)}}{i\phi'(t)}\,d\psi(t) + \int_J \frac{e^{i\phi(t)}\psi(t)\phi''(t)}{i\phi'(t)^2}\,dt.$$

Since ϕ' is monotone on J,

$$\int_J \frac{|\phi''(t)|}{|\phi'(t)|^2}\,dt \le \frac{2}{\mu}.$$

Summing the endpoint, Stieltjes, and absolutely continuous terms over the $O_k(1)$ intervals, and estimating the integral over E trivially, gives

$$\left|\int e^{i\phi(t)}\psi(t)\,dt\right| \le C_{k,L_0}\left\{(\mu/|a_k|)^{1/(k-1)}\|\psi\|_\infty + \mu^{-1}\big(\|\psi\|_\infty + \mathrm{Var}_I(\psi)\big)\right\}.$$

Taking $\mu = |a_k|^{1/k}$ proves the result, uniformly in all lower coefficients. □

Lemma 16.5 (Compact polynomial operator)**.** *Let $R > 0$, let Q, U be cubes of diameter at most C_0R, and let $\Phi(y,u)$ be a real polynomial. Suppose the largest mixed y-degree is $m \ge 1$, and among those terms the largest u-degree is $\ell \ge 1$. If the Euclidean norm M of the coefficient tensor of bidegree (m,ℓ) is nonzero, then*

$$\left\|1_Q(y)\int_U e^{i\Phi(y,u)}g(u)du\right\|_2 \le CR^n \min\{1, (R^{m+\ell}M)^{-\varepsilon_{m,\ell}}\}\|g\|_2 \tag{16.11}$$

for every $g \in L^2(U)$, where $\varepsilon_{m,\ell} > 0$ and the constant may depend on C_0 but is independent of all pure terms and all lower mixed coefficients.

Proof. Remove pure terms by unitary multiplication. Let $\Phi_{m,\ell}(y,u)$ be the component homogeneous of degree m in y and degree ℓ in u. On this fixed finite-dimensional space, equivalence of the coefficient norm with the supremum norm on $\mathbb{S}^{n-1} \times \mathbb{S}^{n-1}$ gives unit vectors v, w such that

$$|\Phi_{m,\ell}(v,w)| \ge c_{n,m,\ell}M.$$

Slice $y = y_\perp + sv$, $u = u_\perp + tw$. Direct expansion shows that

$$[s^m t^\ell]\,\Phi(y_\perp + sv, u_\perp + tw) = \Phi_{m,\ell}(v,w). \tag{16.12}$$

In particular, this coefficient is independent of the transverse variables and has modulus comparable to M. We give the fiber calculation, including the T^*T square root that determines the exponent.

Fix $y_\perp, u_\perp$ and denote the resulting operator from the t-line to the s-line by $T_{y_\perp,u_\perp}$. Cross-sections of Q and U are intervals of length at most CR; their characteristic functions have variation at most two.

Write $p_m(t)$ for the coefficient of s^m in the sliced phase. It is a polynomial of degree ℓ whose leading coefficient has modulus at least cM. The kernel of $T^*_{y_\perp,u_\perp}T_{y_\perp,u_\perp}$ is

$$\mathbf{1}_{I_U}(t)\mathbf{1}_{I_U}(t')\int_{I_Q} e^{i[\Phi(s,t)-\Phi(s,t')]}\,ds.$$

After $s=R\tau$, Lemma 16.4, including its bounded-variation amplitude form, gives

$$|K(t,t')| \le CR\min\left\{1,\left(R^m|p_m(t)-p_m(t')|\right)^{-1/m}\right\}. \tag{16.13}$$

Choose

$$0<\delta<\min\{1/m,1/\ell\}.$$

Since $\min(1,X^{-1/m})\le X^{-\delta}$, the negative-moment estimate (16.8), applied to the polynomial in t' for fixed t, yields the required uniform bound: indeed, $t'\mapsto p_m(t)-p_m(t')$ has degree ℓ and leading coefficient $-\Phi_{m,\ell}(v,w)$, independent of $t,y_\perp,u_\perp$. Therefore

$$\begin{aligned}\sup_t\int|K(t,t')|\,dt' &\le CR^{1-m\delta}M^{-\delta}R^{1-\ell\delta}\\ &= CR^2(R^{m+\ell}M)^{-\delta}.\end{aligned}$$

The same estimate with t,t' interchanged follows by symmetry. The trivial kernel estimate gives CR^2 instead, so Schur's test and then the T^*T identity imply

$$\|T_{y_\perp,u_\perp}\|_{2\to2}\le CR\min\{1,(R^{m+\ell}M)^{-\delta/2}\}. \tag{16.14}$$

Reassemble the slices, without invoking a direct-integral identification. Minkowski's inequality in $u_\perp$ and (16.14) give, for fixed $y_\perp$,

$$\|Tg(y_\perp,\cdot)\|_{L^2_s}\le B_R\int_{\pi_{w^\perp}U}\|g(u_\perp,\cdot)\|_{L^2_t}\,du_\perp,\quad B_R=CR\min\{1,(R^{m+\ell}M)^{-\delta/2}\}.$$

Cauchy–Schwarz in $u_\perp$ makes the right side at most

$$CB_RR^{(n-1)/2}\left(\int_{\pi_{w^\perp}U}\|g(u_\perp,\cdot)\|^2_{L^2_t}\,du_\perp\right)^{1/2}=CB_RR^{(n-1)/2}\|g\|_2.$$

The $y_\perp$-projection of Q has measure at most CR^{n-1}. Squaring the preceding estimate, integrating in $y_\perp$, and taking a square root therefore give

$$\|Tg\|_2\le CR^{n-1}B_R\|g\|_2,$$

which is (16.11) with $\varepsilon_{m,\ell}=\delta/2$. Every bound used only the displayed leading coefficient, so all lower mixed coefficients remain unrestricted. □

For $R>0$ and $1/2\le u\le1$, put

$$A^{P,\Theta}_{R,u}f(x)=\int_{uR<|x-y|<R}e^{iP(x,y)}K_\Theta(x-y)f(y)dy. \tag{16.15}$$

Proposition 16.6 (Normalized rough annular decay)**.** *Let $P\in\mathscr{P}_d$. For every flag position μ there are $a_\mu>0$ and $C_{n,d}$ such that, if the first nonzero flag coordinate of P is ±1 at position μ, then, for $R\ge1$ and $\Theta\in L^\infty(\mathbb{S}^{n-1})$,*

$$\sup_{1/2\le u\le1}\|A^{P,\Theta}_{R,u}\|_{2\to2}\le C_{n,d}R^{-a_\mu}\|\Theta\|_\infty. \tag{16.16}$$

Every later coefficient is unrestricted.

Proof. We treat the three parts of the flag.

For a transverse pivot, partition the input into cubes of side comparable to R; the corresponding output enlargements have bounded overlap. On one input cube Q, write $A_Q = A_{R,u}\mathbf{1}_Q$. Its output is supported in $X_Q = Q + B(0,R)$, and the family $\{X_Q\}$ has bounded overlap. The kernel of $A_Q^* A_Q$ is

$$L_Q(y, y+h) = \mathbf{1}_Q(y)\mathbf{1}_Q(y+h)\int e^{i[P(x,y+h)-P(x,y)]}\overline{k_{R,u}(x-y)}k_{R,u}(x-y-h)dx,$$

where $k_{R,u} = K_\Theta \mathbf{1}_{\{uR<|\cdot|<R\}}$. Put $x = y+v$ and remove the pure y- and v-terms in the phase difference. Let ν be the current transverse flag position. Since its normalized coefficient is ± 1 and all earlier coordinates vanish,

$$\mathfrak{D}P(y,v,h) = \pm F_\nu(y,v,h) + \sum_{\nu<\mu\le M} c_\mu F_\mu(y,v,h). \tag{16.17}$$

Let $y^\alpha v^\beta h^\gamma$ be the pivot monomial of F_ν, and put

$$m = |\alpha|, \qquad \ell = |\beta|, \qquad k = |\gamma|.$$

Because Mix retains only terms of positive y- and v-degree, while $\Delta P(y,v,0) = 0$, one has $m,\ell,k \ge 1$. Thus every use below of the compact-operator and negative-moment lemmas lies in their stated positive-degree range. Write $\mathsf{T}_{m,\ell}(h)$ for the coefficient tensor of bidegree (m,ℓ) in (y,v) in the mixed part of $P(y+v,y+h) - P(y+v,y)$, and let $a(h)$ be its (α,β)-component. By Theorem 16.2(iv), a has degree k, the coefficient of h^γ is exactly ± 1, and no later flag coefficient can change that coefficient or introduce a term of higher h-degree in the same component. Consequently

$$\|\mathsf{T}_{m,\ell}(h)\|_{\mathrm{coeff}} \ge c_{n,d}|a(h)|, \tag{16.18}$$

and no mixed bidegree preceding (m,ℓ) occurs. Here is the precise link between the flag order and the hypothesis of Lemma 16.5. A mixed monomial with y-degree greater than m, or with y-degree m and v-degree greater than ℓ, precedes the pivot regardless of its h-degree. Such a monomial is absent from F_ν because its pivot is the first nonzero column, absent from every later transverse row by (16.3), and absent from the kernel strata because $\mathfrak{D}$ vanishes there. All earlier flag coordinates are zero by hypothesis. Thus (m,ℓ) is exactly the leading bidegree to which Lemma 16.5 applies, uniformly in all unrestricted later coefficients. Notice that this argument detects a fixed component of the full (m,ℓ) tensor; it never replaces a nonzero tensor by its diagonal contraction. Such a replacement would be false in general (the alternating-bilinear contraction vanishes identically), which is why the convolution and skew kernel strata are treated separately below. For fixed h, define

$$(E_h G)(y) = \mathbf{1}_Q(y)\mathbf{1}_Q(y+h)\int_{|v|\le R} e^{i[P(y+v,y+h)-P(y+v,y)]}G(v)\,dv.$$

To apply Lemma 16.5, extend $G\mathbf{1}_{\{|v|\le R\}}$ by zero to a cube of diameter $O(R)$; this leaves the operator unchanged. Put

$$g_h(v) = \overline{k_{R,u}(v)}k_{R,u}(v-h).$$

Let $E_h^{(0)}$ denote the operator in the preceding display with the factor $\mathbf{1}_Q(y+h)$ omitted. Since

$$E_h = M_{\mathbf{1}_{Q-h}}E_h^{(0)}, \qquad \|E_h\|_{2\to 2} \le \|E_h^{(0)}\|_{2\to 2},$$

the additional output cutoff is harmless. Moreover $L_Q(y,y+h) = E_h g_h(y)$. If $a(h) = 0$, Cauchy–Schwarz in the input and output cubes gives the trivial estimate $\|E_h\|_{2\to 2} \le CR^n$, which is the desired bound below with the minimum interpreted as 1. If $a(h) \ne 0$, then (16.18) makes the leading tensor nonzero, so Lemma 16.5 applies to $E_h^{(0)}$. Absorbing the fixed factor $c_{n,d}$ into the structural constant, the two cases give

$$\|E_h\|_{2\to 2} \le CR^n \min\{1, (R^{m+\ell}|a(h)|)^{-\varepsilon}\}.$$

The actual input g_h satisfies, uniformly in h and in $u \in [1/2,1]$,

$$\|g_h\|_2 \le \|k_{R,u}\|_\infty \|k_{R,u}\|_2 \le CR^{-3n/2}\|\Theta\|_\infty^2. \tag{16.19}$$

Consequently

$$\int |L_Q(y,y+h)|^2 dy \le CR^{-n}\|\Theta\|_\infty^4 \min\{1,(R^{m+\ell}|a(h)|)^{-2\varepsilon}\}.$$

Choose $0<\sigma<\min(2\varepsilon,1/k)$ and integrate in $|h|\lesssim R$. The top homogeneous coefficient tensor of a has norm at least $c_{n,d}$ by Theorem 16.2(iv); rescaling a by this fixed constant puts it in the normalization of (16.9). Thus, using $\min(1,X^{-2\varepsilon}) \le X^{-\sigma}$ and absorbing the resulting factor $c_{n,d}^{-\sigma}$, (16.9) gives

$$\begin{aligned} R^{-n}R^{-\sigma(m+\ell)} &\int_{|h|\lesssim R} |a(h)|^{-\sigma}\,dh \\ &\le CR^{-n}R^{-\sigma(m+\ell)}R^{n-k\sigma} = CR^{-\sigma(m+\ell+k)}. \end{aligned}$$

Consequently,

$$\iint |L_Q(y,y+h)|^2\,dy\,dh \le C\|\Theta\|_\infty^4 R^{-\sigma(m+\ell+k)}. \tag{16.20}$$

The change of variables $(y,h)\mapsto(y,y+h)$ has unit Jacobian, so the left side is exactly $\|A_Q^*A_Q\|_{\mathrm{HS}}^2$. Taking its square root first gives

$$\|A_Q^*A_Q\|_{2\to2} \le \|A_Q^*A_Q\|_{\mathrm{HS}} \le C\|\Theta\|_\infty^2 R^{-\sigma(m+\ell+k)/2},$$

and the $A_Q^*A_Q$ identity supplies the second square root:

$$\|A_Q\|_{2\to2} \le C\|\Theta\|_\infty R^{-\sigma(m+\ell+k)/4}.$$

Finally, bounded overlap of the X_Q and disjointness of the input cubes give, with $a=\sigma(m+\ell+k)/4$,

$$\begin{aligned} \Big\|\sum_Q A_Q(f\mathbf{1}_Q)\Big\|_2^2 &\le C\sum_Q \|A_Q(f\mathbf{1}_Q)\|_2^2 \\ &\le CR^{-2a}\|\Theta\|_\infty^2 \sum_Q \|f\mathbf{1}_Q\|_2^2 = CR^{-2a}\|\Theta\|_\infty^2\|f\|_2^2. \end{aligned}$$

This reassembles the cubes and proves (16.16) with a positive exponent independent of the later coefficients.

For a convolution pivot, after pure phases are removed write $P(x,x-z)=Q(z)-x^TBz$. Let Q_s be the first nonzero homogeneous convolution component. Its pivot monomial has coefficient ±1, and later convolution basis vectors do not change that coordinate; boundedness of the corresponding coordinate functional therefore gives $\|Q_s\|_{\mathrm{coeff}} \ge c_{n,d} > 0$. Here Q_s includes every later convolution coordinate of the same homogeneous degree s; those coordinates may change the other entries of its coefficient tensor but cannot remove the normalized pivot entry. Only components of degree strictly below s become lower coefficients in the one-dimensional van der Corput argument. Polar coordinates give

$$A_{R,u}f = \int_{\mathbb{S}^{n-1}} \Theta(\theta) \int_{uR}^{R} e^{i[Q(r\theta)-rx^TB\theta]} f(x-r\theta)\frac{dr}{r}\,d\sigma(\theta).$$

On lines parallel to θ this is a one-dimensional convolution multiplier; the skew modulation depends only on the transverse line since $\theta^TB\theta=0$. More explicitly, write $x=x_\perp+t\theta$, with $x_\perp\in\theta^\perp$, and use the Fourier convention $\widehat{g}(\xi)=\int e^{-it\xi}g(t)\,dt$ on that line. The fiber operator has multiplier

$$m_{\theta,x_\perp,u}(\xi) = \int_{uR}^{R} e^{i[Q(r\theta)-rx_\perp^TB\theta-r\xi]}\frac{dr}{r}. \tag{16.21}$$

Indeed, $(x_\perp + t\theta)^T B\theta = x_\perp^T B\theta$ because $\theta^T B\theta = 0$. After the change $r = R\rho$, both the Fourier frequency and the transverse skew modulation therefore enter only the coefficient of the linear term in ρ; neither can alter the leading coefficient. The amplitude

$$\psi_u(\rho) = \rho^{-1}\mathbf{1}_{[u,1]}(\rho)$$

satisfies $\|\psi_u\|_\infty + \mathrm{Var}(\psi_u) \le C$ uniformly for $1/2 \le u \le 1$. The leading coefficient of the resulting polynomial phase is $R^s Q_s(\theta)$. Lemma 16.4 and Plancherel therefore give the fiber norm

$$C\min\{1, (R^s|Q_s(\theta)|)^{-1/s}\}.$$

Choose $0 < \beta < 1/s$. The last display is bounded by $CR^{-s\beta}|Q_s(\theta)|^{-\beta}$. Minkowski in θ and (16.10) therefore give the required angular integral explicitly: writing $Q_s = M_s A_s$ with $\|A_s\|_{\mathrm{coeff}} = 1$ and $M_s \ge c_{n,d}$,

$$\int_{\mathbb{S}^{n-1}} |Q_s(\theta)|^{-\beta}\, d\sigma(\theta) = M_s^{-\beta} \int_{\mathbb{S}^{n-1}} |A_s(\theta)|^{-\beta}\, d\sigma(\theta) \le C_{n,s,\beta}.$$

Thus Minkowski gives $CR^{-s\beta}\|\Theta\|_\infty$. All lower homogeneous components of Q occur only as lower coefficients in van der Corput and are unrestricted.

For a skew pivot, all transverse and convolution coordinates vanish, because they precede the skew coordinates in the flag. Thus, modulo pure phases, the remaining phase is $x^T By$ with $B^T = -B$. The normalized pivot coefficient gives $\|B\| \ge c_{n,d} > 0$. Choose an orthogonal matrix O that puts B in real skew-normal form. The change of variables $x = O\widetilde{x}$, $y = O\widetilde{y}$ replaces Θ by $\Theta_O(\theta) = \Theta(O\theta)$; rotation invariance of surface measure and of the annulus gives $\|\Theta_O\|_2 = \|\Theta\|_2$ and $\|\Theta_O\|_\infty = \|\Theta\|_\infty$. We may therefore assume that one two-dimensional block is βJ with $|\beta| \gtrsim_{n,d} 1$. For the associated twisted convolution

$$Sg(u,v) = \int k(a,b) e^{i\beta(ub-va)} g(u-a, v-b)\, da\, db,$$

take the Fourier transform $\widehat{g}(u,\eta) = \int e^{-iv\eta} g(u,v)\, dv$ in v. A direct calculation gives

$$\widehat{Sg}(u,\eta) = \int_{\mathbb{R}} \widehat{k}^b\big(a, \eta + \beta a - \beta u\big)\widehat{g}(u-a, \eta + \beta a)\, da,$$

where $\widehat{k}^b(a,\xi) = \int e^{-ib\xi} k(a,b)\, db$. The quantity $q = \eta + \beta u$ is invariant under the passage from (u,η) to the input pair $(u-a, \eta+\beta a)$. Setting $s = u - a$, the operator on the fixed-q fiber has kernel

$$K_q(u,s) = \widehat{k}^b(u-s, q - \beta(u+s)).$$

The change of variables

$$(u,s) \longmapsto (a = u-s, \rho = q - \beta(u+s))$$

has Jacobian of absolute value $2|\beta|$. Therefore Plancherel in the b variable gives, uniformly in q,

$$\begin{aligned}\|K_q\|_{L^2_{u,s}}^2 &= \frac{1}{2|\beta|} \iint |\widehat{k}^b(a,\rho)|^2\, da\, d\rho \\ &\le C|\beta|^{-1}\|k\|_2^2.\end{aligned}$$

The operator norm of each fiber is bounded by its Hilbert–Schmidt norm. Since $(u,\eta) \mapsto (u,q)$ has unit Jacobian, direct integration in q and Plancherel yield

$$\|S\|_{2\to 2} \le C|\beta|^{-1/2}\|k\|_2.$$

For the remaining coordinates, write $B = \beta J \oplus B'$ and $z = (z_0, z') \in \mathbb{R}^2 \times \mathbb{R}^{n-2}$. For each fixed z', the factor coming from B' is unimodular and the translation $x' \mapsto x' - z'$ is an L^2 isometry. Minkowski

therefore reduces the estimate to the preceding two-dimensional bound for the sliced kernel $k_{z'}(z_0) = k_{R,u}(z_0, z')$. Cauchy–Schwarz gives

$$\int_{|z'|\le R} \|k_{z'}\|_2 \, dz' \le |\{|z'| \le R\}|^{1/2} \left(\int \|k_{z'}\|_2^2 \, dz' \right)^{1/2} \le CR^{(n-2)/2} \|k_{R,u}\|_2 \le CR^{-1} \|\Theta\|_2.$$

Therefore $\|A_{R,u}\|_{2\to 2} \le CR^{-1}\|\Theta\|_2$, hence $CR^{-1}\|\Theta\|_\infty$. This proves the proposition in every flag stratum. There are only finitely many strata. □

Lemma 16.7 (Short variation from partial-annulus decay)**.** *Under the hypotheses of Proposition 16.6, after reducing the exponent if necessary, every $f \in L^2(\mathbb{R}^n)$ satisfies*

$$\|V^2(A_{R,u}^{P,\Theta} f : 1/2 \le u \le 1)\|_2 \le CR^{-b_\mu} \|\Theta\|_\infty \|f\|_2 \tag{16.22}$$

for some $b_\mu > 0$.

Proof. Let $F(u) = A_{R,u} f$. Notice that $F(1) = 0$. For test functions,

$$F'(u,x) = -u^{-1} \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x, x-uR\theta)} f(x - uR\theta) d\sigma(\theta),$$

so $\sup_u \|F'(u)\|_2 \le C\|\Theta\|_1 \|f\|_2$. If $a = \sup_u \|F(u)\|_2$ and $b = \sup_u \|F'(u)\|_2$, the fundamental theorem of calculus and the anchor at 1 give pointwise

$$\sup_{1/2\le u\le 1} |F(u,x)|^2 \le 2 \int_{1/2}^1 |F(v,x)| \, |F'(v,x)| \, dv.$$

Integration in x and Cauchy–Schwarz therefore give $\|\sup_u |F(u)|\|_2 \le Ca^{1/2} b^{1/2}$. For every partition,

$$\sum |F(u_i) - F(u_{i-1})|^2 \le 2 \sup_u |F(u)| \int_{1/2}^1 |F'(u)| du.$$

To record the exponents, put

$$M(x) = \sup_{1/2\le u\le 1} |F(u,x)|, \qquad A(x) = \int_{1/2}^1 |F'(u,x)| \, du.$$

The preceding partition estimate says $V^2(F)^2 \le 2MA$. Minkowski gives $\|A\|_2 \le b/2$, while the anchored fundamental theorem of calculus gives $\|M\|_2 \le Ca^{1/2} b^{1/2}$. Consequently

$$\|V^2(F)\|_2^2 \le 2\|M\|_2 \|A\|_2 \le Ca^{1/2} b^{3/2},$$

and hence $\|V^2(F)\|_2 \le Ca^{1/4} b^{3/4}$. Insert (16.16); since $\|\Theta\|_1 \le C_n \|\Theta\|_\infty$, one may take $b_\mu = a_\mu/4$.

For completeness, the argument extends from test functions to all $f \in L^2$. Choose $f_j \in C_c^\infty(\mathbb{R}^n)$ converging to f in L^2 so rapidly that $\sum_j \|f_{j+1} - f_j\|_2 < \infty$, and let F_j be the corresponding paths. For the path associated with $f_j - f_\ell$,

$$V^2 \le V^1 \le \int_{1/2}^1 |F'_{j,\ell}(u)| \, du,$$

so Minkowski and the derivative bound give

$$\|V^2(F_j - F_\ell)\|_2 \le C\|\Theta\|_1 \|f_j - f_\ell\|_2.$$

In particular,

$$W(x) := \sum_j V^2(F_{j+1} - F_j)(x) \in L^2(\mathbb{R}^n).$$

All difference paths vanish at $u = 1$, and hence, off the single null set $\{W = \infty\}$,

$$\sup_{1/2\le u\le 1} |F_{j+1}(u,x) - F_j(u,x)| \le V^2(F_{j+1} - F_j)(x).$$

Thus $F_j(u,x)$ converges uniformly in the entire parameter interval on one common full-measure set. Denote its continuous limit by $\widetilde{F}(u,x)$ and set every coordinate equal to zero on the exceptional set.

This limit represents the intended operator at every parameter. Indeed, the absolute-value kernel on the fixed annulus has both row and column integrals bounded by $C\|\Theta\|_1$, so Schur's test gives, uniformly in u,

$$\|F_j(u) - A^{P,\Theta}_{R,u} f\|_2 \le C\|\Theta\|_1 \|f_j - f\|_2.$$

On the other hand, the preceding summable variation bound gives

$$\sup_u \|F_j(u) - \widetilde{F}(u)\|_2 \le \sum_{k\ge j} \|V^2(F_{k+1} - F_k)\|_2 \longrightarrow 0.$$

Consequently $\widetilde{F}(u)$ is an L^2 representative of $A^{P,\Theta}_{R,u} f$ for every u, with the representative chosen on one common spatial set.

For clarity, this common representative may also be taken locally absolutely continuous. With a fixed Borel representative of f, define in L^2_x

$$D_f(u,x) = -u^{-1} \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x,x-uR\theta)} f(x - uR\theta)\, d\sigma(\theta).$$

The map $u \mapsto D_f(u)$ is strongly measurable and $\sup_u \|D_f(u)\|_2 \le C\|\Theta\|_1 \|f\|_2$. Approximation by f_j in the fundamental theorem of calculus gives, in L^2_x,

$$A^{P,\Theta}_{R,v} f - A^{P,\Theta}_{R,u} f = \int_u^v D_f(w)\, dw.$$

Fubini therefore makes $-\int_u^1 D_f(w,x)\, dw$ an absolutely continuous pointwise representative for almost every x. It agrees with $\widetilde{F}$ at all rational u on one full-measure set and hence, by continuity, at every u.

Finally, uniform pointwise convergence implies for each finite partition π that its ℓ^2 increment sum for $\widetilde{F}$ is the limit of the corresponding sums for F_j. Taking the supremum over π, then using Fatou's lemma, yields

$$\|V^2(\widetilde{F})\|_2 \le \liminf_{j\to\infty} \|V^2(F_j)\|_2 \le CR^{-b_\mu} \|\Theta\|_\infty \|f\|_2,$$

which proves (16.22) for general f. □

Corollary 16.8 (Scale-covariant form)**.** *Let P be a real polynomial of degree at most d whose first nonzero flag coordinate is $c_\mu \neq 0$ of degree m_μ. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For a shell radius $R > 0$ set*

$$L_\mu = |c_\mu| R^{m_\mu}.$$

If $L_\mu \ge 1$, then

$$\sup_{1/2\le u\le 1} \|A^{P,\Theta}_{R,u}\|_{2\to 2} + \|V^2_u(A^{P,\Theta}_{R,u})\|_{2\to 2} \le C\|\Theta\|_\infty L_\mu^{-\beta_\mu} \tag{16.23}$$

for some $\beta_\mu > 0$. Here the variation-operator norm means

$$\|V^2_u(A^{P,\Theta}_{R,u})\|_{2\to 2} := \sup_{f\neq 0} \frac{\|V^2(A^{P,\Theta}_{R,u} f : 1/2 \le u \le 1)\|_2}{\|f\|_2}. \tag{16.24}$$

This is the norm of a homogeneous sublinear map; linearity of the variation map is not asserted. For the later packet rescaling only (the following variables are not hypotheses of (16.23)*), fix a packet side*

$q > 0$ and a truncation parameter $\varepsilon > 0$, and set $t = \log_2(R/q)$. Under the normalizing dilation $x = \rho X$, $y = \rho Y$, $\rho = |c_\mu|^{-1/m_\mu}$,

$$R \mapsto \widetilde{R} = R/\rho = L_\mu^{1/m_\mu}, \quad q \mapsto \widetilde{q} = q/\rho, \quad \varepsilon \mapsto \widetilde{\varepsilon} = \varepsilon/\rho, \quad \log_2(\widetilde{R}/\widetilde{q}) = t. \tag{16.25}$$

If $h_\rho(Y) = h(\rho Y)$, then

$$\|h_\rho\|_1 = \rho^{-n}\|h\|_1, \qquad \|h_\rho\|_2 = \rho^{-n/2}\|h\|_2, \qquad \|h_\rho\|_\infty = \|h\|_\infty. \tag{16.26}$$

Thus a normalized bound $\widetilde{R}^{-a_\mu}$ becomes $L_\mu^{-a_\mu/m_\mu}$. For any packet family h_Q supported on q-cubes, put $\widetilde{Q} = \rho^{-1}Q$ and $\widetilde{h}_Q(Y) = h_Q(\rho Y)$. Then

$$\|\widetilde{h}_Q\|_1 \le \Lambda|\widetilde{Q}| \quad \Longleftrightarrow \quad \|h_Q\|_1 \le \Lambda|Q|,$$

so the density condition is invariant under the same rescaling.

Proof. Homogeneity of K_Θ gives the exact conjugacy

$$A_{R,u}^{P,\Theta} h(\rho X) = A_{R/\rho,u}^{P_\rho,\Theta} h_\rho(X).$$

Here and below $P_\rho(X,Y) = P(\rho X, \rho Y)$ and $h_\rho(Y) = h(\rho Y)$. The displayed transformations and norm identities follow by change of variables. Apply Proposition 16.6 and Lemma 16.7 at radius $\widetilde{R}$ and take $\beta_\mu = \min(a_\mu, b_\mu)/m_\mu$. □

17 Far-field strong estimates and triangular induction

We convert the scale-covariant annular decay into an angular $L\log L$ estimate and close the finite induction over the phase flag. Proposition 15.9 provides the terminal unphased estimate, and Section 16 provides oscillatory decay on every nonterminal stratum.

Call a phase *normalized nonterminal* if its first nonzero flag coordinate is ± 1. Since the flag is finite, there is a number

$$a_0 = a_0(n,d) > 0 \tag{17.1}$$

not exceeding the exponent in Proposition 16.6 at any nonterminal flag position. If there is no such position, this section is vacuous and we set $a_0 = 1$.

17.1 Angular interpolation and short variation

Lemma 17.1 (q'-tracked partial-annulus estimates)**.** *Let P be a normalized nonterminal phase, let $R \ge 1$, and let $1 < q \le 2$. Suppose that $\Theta \in L^q(\mathbb{S}^{n-1})$. Uniformly in every later coefficient of P,*

$$\sup_{1/2\le u\le 1} \|A_{R,u}^{P,\Theta}\|_{2\to 2} \le C_{n,d} R^{-a_0/q'} \|\Theta\|_q, \qquad q' = \frac{q}{q-1}. \tag{17.2}$$

Moreover, for every $1 < p < \infty$ put

$$\gamma_p = \frac{a_0}{2\max(p,p')} > 0 \tag{17.3}$$

and let $f \in L^p(\mathbb{R}^n)$. Then

$$\sup_{1/2\le u\le 1} \|A_{R,u}^{P,\Theta} f\|_p \le C_{n,d,p} R^{-\gamma_p/q'} \|\Theta\|_q \|f\|_p, \tag{17.4}$$

$$\|V^2(A_{R,u}^{P,\Theta} f : 1/2 \le u \le 1)\|_p \le C_{n,d,p} R^{-\gamma_p/q'} \|\Theta\|_q \|f\|_p. \tag{17.5}$$

All prefactors are uniform for $1 < q \le 2$.

Proof. The trivial annular estimate is

$$\sup_u \|A_{R,u}^{P,\Theta}\|_{2\to 2} \le \|K_\Theta \mathbf{1}_{\{R/2<|\cdot|<R\}}\|_1 \le C\|\Theta\|_1. \tag{17.6}$$

For fixed P,R,u, the map $\Theta \mapsto A_{R,u}^{P,\Theta}$ is linear from the angular space into $\mathscr{B}(L^2)$. Complex interpolation of (17.6) with Proposition 16.6 gives (17.2). Indeed, if ϑ denotes the weight of the $L^\infty(\mathbb{S}^{n-1})$ endpoint, then

$$\frac{1}{q} = \frac{1-\vartheta}{1} + \frac{\vartheta}{\infty}, \qquad \text{so} \qquad \vartheta = \frac{1}{q'}.$$

Thus R^{-a_0} becomes $R^{-a_0/q'}$, with no q-dependent interpolation loss. This interpolation is first performed for each fixed u in the Banach target $\mathscr{B}(L^2)$; both endpoint bounds are uniform in u, so the supremum in u may then be taken. The complex interpolation constants are uniform in q.

For the short variation put $F(u) = A_{R,u}^{P,\Theta} f$. Initially take f and Θ bounded. The path is absolutely continuous, $F(1) = 0$, and polar coordinates give, for almost every $u \in [1/2, 1]$,

$$F'(u,x) = -\frac{1}{u}\int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x,x-uR\theta)} f(x-uR\theta)\, d\sigma(\theta). \tag{17.7}$$

Consequently, for every $1 \le s \le \infty$,

$$\sup_{1/2\le u\le 1} \|F'(u)\|_s \le C\|\Theta\|_1 \|f\|_s. \tag{17.8}$$

Set

$$a = \sup_u \|F(u)\|_2, \qquad b = \sup_u \|F'(u)\|_2.$$

Since $F(1) = 0$, the fundamental theorem of calculus yields pointwise

$$\sup_u |F(u,x)|^2 \le 2\int_{1/2}^1 |F(v,x)F'(v,x)|\, dv.$$

After integration in x,

$$\|\sup_u |F(u)|\|_2 \le Ca^{1/2}b^{1/2}. \tag{17.9}$$

For every partition $1/2 \le u_0 < \cdots < u_N \le 1$,

$$\sum_{i=1}^N |F(u_i) - F(u_{i-1})|^2 \le 2\sup_u |F(u)| \int_{1/2}^1 |F'(v)|\, dv.$$

Taking the supremum over partitions, integrating, and using (17.9) gives

$$\|V^2(F(u) : 1/2 \le u \le 1)\|_2 \le Ca^{1/4}b^{3/4}. \tag{17.10}$$

Equations (17.2), (17.8), and $\|\Theta\|_1 \le C_n\|\Theta\|_q$ therefore imply

$$\|V^2(F)\|_2 \le CR^{-a_0/(4q')}\|\Theta\|_q \|f\|_2. \tag{17.11}$$

The map

$$S_R f := V^2(A_{R,u}^{P,\Theta} f : 1/2 \le u \le 1)$$

is nonnegative, absolutely homogeneous, and subadditive. Moreover, by (17.8),

$$S_R f \le V^1(A_{R,u}^{P,\Theta} f : 1/2 \le u \le 1) \le \int_{1/2}^1 |F'(u)|\, du,$$

and hence S_R has nondecaying strong $(1,1)$ and (∞,∞) bounds with norm $C\|\Theta\|_1$. We may therefore apply the Marcinkiewicz interpolation theorem for homogeneous sublinear operators. For $1 < p < 2$,

interpolation between the strong $(1,1)$ bound and (17.11) puts weight $2/p'$ on the L^2 endpoint and gives decay $a_0/(2p'q')$. For $2<p<\infty$, interpolation between (17.11) and the strong (∞,∞) bound puts weight $2/p$ on the L^2 endpoint and gives decay $a_0/(2pq')$. These exponents are exactly γ_p/q', since

$$\frac{a_0}{4q'}\frac{2}{p'}=\frac{a_0}{2p'q'}\quad(1<p<2),\qquad \frac{a_0}{4q'}\frac{2}{p}=\frac{a_0}{2pq'}\quad(2<p<\infty).$$

This proves (17.5). The same spatial interpolation, now starting from (17.2), proves the stronger corresponding bound for the annular operator; weakening its exponent to γ_p/q' gives (17.4).

To remove the boundedness assumptions, choose bounded $f_m\to f$ in $L^p(\mathbb{R}^n)$ and bounded $\Theta_m\to\Theta$ in $L^q(\mathbb{S}^{n-1})$, and put $F_m(u)=A^{P,\Theta_m}_{R,u}f_m$. By (17.8), linearity, and $V^2\le V^1$,

$$\|V^2(F_m-F_\ell)\|_p\le C\|\Theta_m\|_1\|f_m-f_\ell\|_p+C\|\Theta_m-\Theta_\ell\|_1\|f_\ell\|_p.$$

The right-hand side tends to zero, since the sphere has finite measure. Thus the paths are Cauchy in the metric $\|V^2(F_m-F_\ell)\|_p$. We spell out the completion because the parameter set is uncountable. Every difference path is anchored at $u=1$. Choose a subsequence, still denoted by F_m, such that

$$\sum_m\|V^2(F_{m+1}-F_m:1/2\le u\le 1)\|_p<\infty.$$

After deleting one null set, the corresponding pointwise series of variations is finite. The anchor then gives

$$\sup_{1/2\le u\le 1}|F_{m+1}(u,x)-F_m(u,x)|\le V^2(F_{m+1}-F_m)(x),$$

so $F_m(u,x)$ converges uniformly in u on that common full-measure set. For a fixed rational u, Young's inequality on the fixed annulus identifies the limit with $A^{P,\Theta}_{R,u}f$ in L^p. Passing to a further diagonal subsequence and deleting one countable union of null sets makes these identities simultaneous for all rational $u\in[1/2,1]$. Polar coordinates and Fubini give the latter annular integrals a jointly measurable, locally absolutely continuous representative in u: its radial derivative is the right side of (17.7), whose $L^1_u(L^p_x)$ norm is bounded by $C\|\Theta\|_1\|f\|_p$. Continuity therefore identifies the uniform limit at every real u. On each finite partition the increment sum is continuous under this uniform convergence. Taking first the limit, then the supremum over the nested finite rational partitions, and using Fatou proves (17.5). The same common representative and coordinatewise convergence prove (17.4). □

17.2 The far path and the q' loss

For a normalized nonterminal phase define

$$\mathscr{F}^{P,\Theta}_\varepsilon f(x)=\int_{|x-y|>\max(\varepsilon,1)}e^{iP(x,y)}K_\Theta(x-y)f(y)\,dy \tag{17.12}$$

on bounded compactly supported data. Equivalently, put

$$B^{P,\Theta}_j=A^{P,\Theta}_{2^{j+1},1/2},\qquad j\ge 0. \tag{17.13}$$

If $2^j\le\varepsilon\le 2^{j+1}$, then

$$\mathscr{F}^{P,\Theta}_\varepsilon f=A^{P,\Theta}_{2^{j+1},\varepsilon/2^{j+1}}f+\sum_{k=j+1}^{\infty}B^{P,\Theta}_k f, \tag{17.14}$$

whereas for $0<\varepsilon\le 1$ the path is the constant $\sum_{k\ge 0}B^{P,\Theta}_k f$.

We use the elementary long–short inequality

$$V^2(F_\varepsilon:\varepsilon>0)\le CV^2(F_{2^j}:j\in\mathbb{Z})+C\left(\sum_{j\in\mathbb{Z}}V^2(F_\varepsilon:2^j\le\varepsilon\le 2^{j+1})^2\right)^{1/2}. \tag{17.15}$$

Indeed, insert the adjacent dyadic endpoints into an arbitrary variation partition. Each increment is the sum of two within-block increments and one increment between dyadic endpoints. The between-block increments form an ordered subpartition, and every dyadic block contributes at most two within-block increments. The inequality follows from $|z_1+z_2+z_3|^2 \le 3\sum |z_i|^2$.

Proposition 17.2 (q'-tracked far V^2)**.** *Let P be normalized and nonterminal, let $1<p<\infty$, and let $1<q\le 2$. Suppose that $\Theta \in L^q(\mathbb{S}^{n-1})$; no cancellation of Θ is required. Then, for every $f\in L^p(\mathbb{R}^n)$,*

$$\|\mathscr{F}_1^{P,\Theta} f\|_p + \|V^2(\mathscr{F}_\varepsilon^{P,\Theta} f : \varepsilon>0)\|_p \le C_{n,d,p} q' \|\Theta\|_q \|f\|_p. \tag{17.16}$$

The constant is uniform in q and in every unused coefficient of P.

Proof. Lemma 17.1 gives

$$\|B_j^{P,\Theta} f\|_p \le C_{n,d,p} 2^{-\gamma_p j/q'} \|\Theta\|_q \|f\|_p. \tag{17.17}$$

Thus the series in (17.14) converges in L^p and

$$\sum_{j\ge 0} \|B_j^{P,\Theta} f\|_p \le C_{n,d,p} q' \|\Theta\|_q \|f\|_p, \tag{17.18}$$

because, for $q'\ge 2$,

$$\sum_{j\ge 0} 2^{-\gamma_p j/q'} = \frac{1}{1-2^{-\gamma_p/q'}} \le C_{\gamma_p} q'.$$

This proves the asserted bound for $\mathscr{F}_1$. Choose measurable representatives of the countable family $B_j f$. By Minkowski and monotone convergence,

$$G(x) := \sum_{j\ge 0} |B_j^{P,\Theta} f(x)| \in L^p(\mathbb{R}^n), \qquad \|G\|_p \le \sum_{j\ge 0} \|B_j^{P,\Theta} f\|_p.$$

Consequently the series is absolutely convergent outside one null set, and all of its (countably many) dyadic tail identities hold there simultaneously. On every block we use the jointly measurable, locally absolutely continuous partial-annulus representative constructed in Lemma 17.1. Formula (17.14) then defines one common representative of the full far trajectory, with adjacent block definitions agreeing at their shared dyadic endpoint. All pointwise variation estimates below refer to this representative.

At the dyadic parameters,

$$\mathscr{F}_{2^j}^{P,\Theta} f = \sum_{k=j}^{\infty} B_k^{P,\Theta} f, \qquad j\ge 0,$$

and the sequence is constant for $j<0$. Hence, pointwise,

$$V^2(\mathscr{F}_{2^j}^{P,\Theta} f : j\in\mathbb{Z}) \le \sum_{j\ge 0} |B_j^{P,\Theta} f|. \tag{17.19}$$

Its L^p norm is bounded by (17.18).

On the block $[2^j, 2^{j+1}]$, the tail in (17.14) is constant and the nonconstant part is $A^{P,\Theta}_{2^{j+1},u}$, $1/2\le u\le 1$. Therefore (17.5) gives

$$\|V^2(\mathscr{F}_\varepsilon : 2^j\le\varepsilon\le 2^{j+1})\|_p \le C 2^{-\gamma_p j/q'} \|\Theta\|_q \|f\|_p.$$

Pointwise, $(\sum_j V_j^2)^{1/2} \le \sum_j V_j$; hence the L^p norm of the short term in (17.15) is again bounded by the geometric sum in (17.18). Combining the long and short parts proves (17.16). □

17.3 Finite-trajectory extrapolation to $L\log L$

Let

$$\Phi_0(t) = (1+t)\log(1+t) - t, \qquad \Psi_0(s) = e^s - s - 1.$$

These are complementary Young functions. On the finite measure space $\mathbb{S}^{n-1}$, the Luxemburg norm for Φ_0 is equivalent, with dimensional constants, to the homogeneous gauge $\mathfrak{L}$ in (1.2); the Luxemburg norm for Ψ_0 will be denoted by $\|\cdot\|_{\exp L}$. The Orlicz–Hölder inequality therefore gives

$$\int_{\mathbb{S}^{n-1}} |uv|\, d\sigma \le 2\|u\|_{L\log L}\|v\|_{\exp L}. \tag{17.20}$$

For the quantitative direction used below the assertion is trivial when $\Theta = 0$; otherwise let $A = \|\Theta\|_1$ and $B = \mathfrak{L}(\Theta)$. Since $B \ge A$ and $\Phi_0(t) \le Ct\log(e+t)$,

$$\int_{\mathbb{S}^{n-1}} \Phi_0\!\left(\frac{|\Theta|}{B}\right) d\sigma \le \frac{C}{B}\left[A + \int_{\mathbb{S}^{n-1}} |\Theta| \log^+\!\left(\frac{|\Theta|}{A}\right) d\sigma\right] \le C.$$

Convexity and $\Phi_0(0) = 0$ imply $\Phi_0(t/C_1) \le C_1^{-1}\Phi_0(t)$ for $C_1 \ge 1$. Choosing C_1 larger than the preceding absolute constant gives

$$\|\Theta\|_{L\log L} \le C_1\,\mathfrak{L}(\Theta). \tag{17.21}$$

Proposition 17.3 (Angular endpoint for the far path)**.** *Let P be normalized and nonterminal, let $1 < p < \infty$, and let $\Theta \in L\log L(\mathbb{S}^{n-1})$. No cancellation of Θ is required. Then, for every $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathscr{F}_1^{P,\Theta} f\|_p + \|V^2(\mathscr{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0)\|_p \le C_{n,d,p}\,\mathfrak{L}(\Theta)\|f\|_p. \tag{17.22}$$

Consequently,

$$\mathscr{J}_{2,p}(\mathscr{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0) \le C_{n,d,p}\,\mathfrak{L}(\Theta)\|f\|_p. \tag{17.23}$$

Proof. Fix f and a finite ordered set $D \subset \mathbb{Q}_+$ containing 1. On $\mathbb{C}^D$ define

$$\|z\|_{E_D} = |z_1| + V_D^2(z_\varepsilon : \varepsilon \in D). \tag{17.24}$$

This is a norm: vanishing variation makes the trajectory constant, and the value at 1 then detects that constant. Since D is finite, E_D is a finite-dimensional Banach space, hence $L^p(\mathbb{R}^n; E_D)$ is Banach and its norm is recovered by Hahn–Banach duality. For bounded angular data define, after complexification if necessary, the linear map

$$\mathscr{A}_{f,D}\Theta = (\mathscr{F}_\varepsilon^{P,\Theta} f)_{\varepsilon\in D} \quad \text{in } L^p(\mathbb{R}^n; E_D).$$

Proposition 17.2, uniformly in D, says

$$\|\mathscr{A}_{f,D}\|_{L^q(\mathbb{S}^{n-1})\to L^p(E_D)} \le C_{n,d,p}\,q'\|f\|_p, \qquad 1 < q \le 2. \tag{17.25}$$

Let $\Lambda \in [L^p(\mathbb{R}^n; E_D)]^*$ have norm one. For $s \ge 2$ take $q = s/(s-1)$ in (17.25). The scalar functional $\Theta \mapsto \Lambda(\mathscr{A}_{f,D}\Theta)$ is represented by a function $h_s \in L^s(\mathbb{S}^{n-1})$ satisfying

$$\|h_s\|_s \le Cs\|f\|_p. \tag{17.26}$$

The representatives h_s agree almost everywhere. Indeed, two of them represent the same functional on the common subspace of bounded angular functions: all of these extensions agree there with the original finite-trajectory formula. Since the sphere has finite measure, $h_s - h_t \in L^1$ for any $s,t \ge 2$. Its bounded complex sign belongs to every relevant L^q space, and testing with that sign gives $\int |h_s - h_t|\, d\sigma = 0$. Thus the representatives agree almost everywhere; denote their common representative by h.

Put $A = \|f\|_p$. If $A = 0$ there is nothing to prove. For every integer $k \ge 2$, $\|h\|_k \le CkA$. Hence, by Tonelli and the exponential series,

$$\int_{\mathbb{S}^{n-1}} \Psi_0\!\left(\frac{|h|}{LA}\right) d\sigma = \sum_{k=2}^{\infty} \frac{\|h\|_k^k}{(LA)^k k!} \le \sum_{k=2}^{\infty} \frac{(Ck/L)^k}{k!}. \tag{17.27}$$

Since $k^k/k! \le C_0 e^k \sqrt{k}$, the last series is at most one once $L = L(n,d,p)$ is sufficiently large. Thus

$$\|h\|_{\exp L} \le C_{n,d,p} \|f\|_p. \tag{17.28}$$

Equations (17.20) and (17.28) give

$$|\Lambda(\mathscr{A}_{f,D}\Theta)| \le C_{n,d,p} \|\Theta\|_{L\log L} \|f\|_p.$$

Taking the supremum over Λ proves

$$\|\mathscr{A}_{f,D}\Theta\|_{L^p(E_D)} \le C_{n,d,p}\mathfrak{L}(\Theta)\|f\|_p, \tag{17.29}$$

uniformly in the finite set D.

Bounded angular functions are dense in the Luxemburg $L\log L$ norm. Indeed, for $\Theta_N = \Theta \mathbf{1}_{\{|\Theta|\le N\}}$, the Δ_2 condition for Φ_0 and dominated convergence for the corresponding modular give

$$\|\Theta - \Theta_N\|_{L\log L} \longrightarrow 0.$$

Thus the finite-trajectory maps extend to all $\Theta \in L\log L$. To check consistency, if $D \subset D'$ and both contain 1, coordinate restriction satisfies

$$|z_1| + V_D^2(z) \le |z_1| + V_{D'}^2(z),$$

so it is a contraction from $E_{D'}$ to E_D. The two extensions agree on bounded angular functions and hence, by density, on all of $L\log L$. Thus the rational trajectory is independent of the exhausting sequence. Fix once and for all an enumeration of $\mathbb{Q}_+$ beginning with 1, and let D_N be its first N elements. Choose representatives of the countable family of rational coordinates simultaneously. For every N, (17.29) applies to these same coordinates, and

$$|\mathscr{F}_1 f| + V_{D_N}^2(\mathscr{F}_\varepsilon f) \uparrow |\mathscr{F}_1 f| + V_{\mathbb{Q}_+}^2(\mathscr{F}_\varepsilon f) \quad \text{pointwise.}$$

Monotone convergence (equivalently, Fatou followed by monotonicity) now passes the uniform finite-D_N estimate to all positive rational parameters. The finite-dimensional extension is completed first, and only then is $N \to \infty$ taken. For $0 < a < b < \infty$, finite-shell differences satisfy

$$\|(\mathscr{F}_a - \mathscr{F}_b)f\|_p \le \|\Theta\|_1 \log(b/a) \|f\|_p, \tag{17.30}$$

including when $[a,b]$ meets 1: the difference is zero if $b \le 1$, and if $a < 1 < b$ its kernel has L^1 norm $\|\Theta\|_1 \log b \le \|\Theta\|_1 \log(b/a)$.

The difference identity and estimate above are first obtained on rational parameters; approximation by bounded angular data and Young's inequality preserve that identity. Put

$$\mathscr{A}_x(r) = \int_{\mathbb{S}^{n-1}} e^{iP(x,x-r\theta)} \Theta(\theta) f(x - r\theta)\, d\sigma(\theta).$$

For every $0 < a < b < \infty$, polar coordinates give

$$\int_a^b |\mathscr{A}_x(r)| \frac{dr}{r} \le \int_{a<|z|<b} |K_\Theta(z)|\, |f(x-z)|\, dz.$$

The right-hand side belongs to L^p. Applying this observation to a countable family of annuli covering $(0,\infty)$, discard one null set on whose complement all these radial densities are locally integrable. Choose one representative of $\mathscr{F}_1 f$, and further remove the countable union of the null sets on which a rational finite-shell identity fails. On the resulting common full-measure set define

$$\mathscr{F}_\varepsilon f = \begin{cases} \mathscr{F}_1 f, & 0 < \varepsilon \le 1, \\ \mathscr{F}_1 f - \displaystyle\int_{1<|z|\le\varepsilon} e^{iP(x,x-z)} K_\Theta(z) f(x-z)\, dz, & \varepsilon > 1. \end{cases}$$

The rational finite-shell identities show that this leaves every rational coordinate unchanged. The resulting path is locally absolutely continuous and satisfies, for almost every ε,

$$\frac{d}{d\varepsilon}\mathscr{F}_\varepsilon f(x) = -\mathbf{1}_{(1,\infty)}(\varepsilon)\frac{\mathscr{A}_x(\varepsilon)}{\varepsilon}.$$

Lemma 15.6 therefore identifies the rational variation with the full-parameter variation. This proves (17.22). Finally (15.3) gives (17.23). □

Remark 17.4 (The far anchor). The construction above is anchored at infinity. For bounded Θ, take $q = 2$ in Lemma 17.1 and interpolate spatially as in its proof. More explicitly, for $\varepsilon \geq 1$ choose $j \geq 0$ with $2^j \leq \varepsilon \leq 2^{j+1}$. Then the current partial annulus is bounded in L^p by

$$C_{n,d,p}2^{-\gamma_p j/2}\|\Theta\|_2\|f\|_p,$$

and the remaining annular tail is bounded by

$$C_{n,d,p}\sum_{k\geq j+1} 2^{-\gamma_p k/2}\|\Theta\|_2\|f\|_p.$$

Both quantities tend to zero as $j \to \infty$, and hence as $\varepsilon \to \infty$. For general $\Theta \in L\log L$, choose bounded $\Theta_m \to \Theta$ in that norm. Proposition 17.3 and

$$\sup_{\varepsilon>0}|F_\varepsilon| \leq |F_1| + V^2(F_\varepsilon : \varepsilon > 0)$$

make the approximation uniform in L^p over ε. Hence $\mathscr{F}_\varepsilon^{P,\Theta} f \to 0$ in L^p. Finite V^2 supplies an almost-everywhere limit at infinity, and an L^p-convergent subsequence identifies that limit as zero.

17.4 Local triangular reduction

For a normalized phase put

$$\mathscr{L}_\varepsilon^{P,\Omega} f(x) = \begin{cases} \displaystyle\int_{\varepsilon<|x-y|<1} e^{iP(x,y)}K_\Omega(x-y)f(y)\,dy, & 0<\varepsilon<1,\\ 0, & \varepsilon \geq 1. \end{cases} \tag{17.31}$$

We record two elementary endpoint facts. First, if a path E is zero at one endpoint and $V^1(E) \leq W$, then

$$\lambda N_\lambda(E) \leq W, \qquad N_\lambda(E) > 0 \Longrightarrow \lambda \leq W, \qquad \lambda N_\lambda(E)^{1/2} \leq W. \tag{17.32}$$

Indeed, the sum of the increments in any ordered family of jump pairs is at most the total 1-variation. Second, if

$$L_\varepsilon = \begin{cases} F_\varepsilon - F_1, & 0<\varepsilon<1,\\ 0, & \varepsilon \geq 1, \end{cases} \tag{17.33}$$

and $F_\varepsilon \to F_1$ as $\varepsilon \uparrow 1$, then

$$N_\lambda(L) \leq N_\lambda(F). \tag{17.34}$$

To see this, retain every jump pair lying below 1; if a pair crosses 1, it is necessarily the last nonzero pair and may be replaced by the pair with upper endpoint 1.

Proposition 17.5 (Triangular local reduction in the strong range)**.** *Let P be a real polynomial of degree at most d whose first nonzero flag coordinate is $c_\nu = \pm 1$. Let $\Omega \in L^1(\mathbb{S}^{n-1})$ and let f be bounded and compactly supported. On each unit output cube, the local path* (17.31) *is, modulo pure input and output phases, the sum of*

(i) *a localized path of the form* (17.33) *whose residual phase either has all flag coordinates equal to zero, or has first nonzero flag coordinate strictly later than* ν*;*

(ii) *an error path* E *satisfying*

$$V^1(E)(x) \le C_{n,d} \int_{|x-y|<1} \frac{|\Omega((x-y)/|x-y|)|}{|x-y|^{n-1}} |f(y)|\,dy. \tag{17.35}$$

The majorant in (17.35) *has* L^p *norm at most* $C_{n,d}\|\Omega\|_1\|f\|_p$ *for every* $1 \le p \le \infty$.

Proof. Let Q_a be the output unit cube centered at a, and let Q_a^* be the fixed input enlargement relevant to Q_a. Write $x = a+X$, $y = a+Y$. Theorem 16.2(iii) gives

$$P(a+X,a+Y) = p_a(X) + q_a(Y) + c_\nu H_\nu(X,Y) + P_{a,>\nu}(X,Y),$$

where every flag coordinate of $P_{a,>\nu}$ is later than ν. Set

$$f_a(Y) = e^{iq_a(Y)} f(a+Y)\mathbf{1}_{Q_a^*-a}(Y).$$

The coordinate translation identifies the original path at $a+X$ with the translated path at X and preserves Lebesgue measure. Multiplication of the output path by $e^{ip_a(X)}$ leaves every pointwise jump unchanged, while the pure input phase is absorbed into f_a. In particular,

$$|f_a(Y)| = |f(a+Y)|\mathbf{1}_{Q_a^*-a}(Y), \qquad \|f_a\|_p = \|f\mathbf{1}_{Q_a^*}\|_p.$$

Thus the local path is the path with phase $c_\nu H_\nu + P_{a,>\nu}$ applied to f_a. If X lies in the reference output cube and $|X-Y|<1$, both variables belong to a fixed bounded set. Since $H_\nu(Y,Y)=0$,

$$|H_\nu(X,Y)| \le C_{n,d}|X-Y|. \tag{17.36}$$

Thus

$$|e^{ic_\nu H_\nu(X,Y)} - 1| \le C_{n,d}|X-Y|.$$

The total 1-variation of the difference of the two nested truncation paths is therefore bounded by (17.35). Its kernel has L^1 norm

$$C_{n,d}\int_0^1 \int_{\mathbb{S}^{n-1}} |\Omega(\theta)|\,d\sigma(\theta)\,dr = C_{n,d}\|\Omega\|_1,$$

which proves the asserted L^p bound. Removing this difference leaves the local truncation of $P_{a,>\nu}$ applied to f_a. To identify it precisely with (17.33), let

$$F_\varepsilon = T_{\Omega,\varepsilon}^{P_{a,>\nu}} f_a$$

at this bounded compactly supported stage. The finite-shell identity gives

$$\mathscr{L}_\varepsilon^{P_{a,>\nu},\Omega} f_a = F_\varepsilon - F_1 \quad (0<\varepsilon<1), \qquad \mathscr{L}_\varepsilon^{P_{a,>\nu},\Omega} f_a = 0 \quad (\varepsilon \ge 1).$$

Moreover $F_\varepsilon \to F_1$ as $\varepsilon \uparrow 1$ by the L^1 norm of the intervening shell kernel. Thus the remaining path is exactly the localized full path in (17.33), and (17.34) applies to it. □

17.5 Coefficient-uniform strong jumps for every phase

Proposition 17.6 (Strong polynomial critical jumps)**.** *Let* $1<p<\infty$*, let* P *be any real polynomial of degree at most* d*, and let* $\Omega \in L\log L(\mathbb{S}^{n-1})$ *have mean zero. Then for every* $f \in L^p(\mathbb{R}^n)$*,*

$$\mathscr{J}_{2,p}(T_{\Omega,\varepsilon}^P f : \varepsilon>0) \le C_{n,d,p}\mathfrak{L}(\Omega)\|f\|_p, \tag{17.37}$$

where the constant is independent of every coefficient of P*. Consequently, for every* $r>2$*,*

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon>0)\|_p \le C_{n,d,p,r}\mathfrak{L}(\Omega)\|f\|_p. \tag{17.38}$$

Proof. We first take bounded compactly supported data, for which every finite-shell identity is classical. At this stage the induction concerns only these classical finite-shell trajectories and uses no extension at the flag position currently being proved. We use backward induction on the finite phase flag. If every flag coordinate vanishes, then P is a sum of a pure input phase and a pure output phase. Equation (16.1) and Proposition 15.9 give (17.37).

Assume the estimate for every phase whose first coordinate is later than ν, and suppose that the first coordinate of P is $c_\nu \neq 0$, of homogeneous degree m_ν. Under $x = \rho X$, $y = \rho Y$, where

$$\rho = |c_\nu|^{-1/m_\nu}, \tag{17.39}$$

the first coordinate becomes ± 1. Work first with this normalized phase. Proposition 17.3 controls its far path.

For the local path partition the output into disjoint unit cubes Q_a. Only the restriction of f to a fixed enlargement Q_a^* contributes on Q_a. Proposition 17.5 gives a residual phase $P_{a,>\nu}$, the modulated translated input f_a used in its proof, and an error E_a. If all flag coordinates of $P_{a,>\nu}$ vanish, Proposition 15.9 together with the pure input/output factorization (16.1) supplies the base case. Otherwise its first nonzero coordinate is strictly later than ν, so the induction hypothesis applies. In either case, (17.34) and the supremum over the jump amplitude give

$$\|\mathbf{J}(\mathscr{L}_\varepsilon^{P_{a,>\nu},\Omega} f_a : \varepsilon > 0)\|_p \le C\mathfrak{L}(\Omega)\|f_a\|_p = C\mathfrak{L}(\Omega)\|f\mathbf{1}_{Q_a^*}\|_p \tag{17.40}$$

uniformly in a and in all translated residual coefficients. Equations (17.32), (15.6), and (17.35) give the same estimate for $\mathbf{J}(E_a)$, with $\|\Omega\|_1 \le C\mathfrak{L}(\Omega)$.

Apply the pointwise quasi-triangle inequality (15.5) on every output cube, sum the p-th powers over the disjoint Q_a, and use bounded overlap of the Q_a^*. More explicitly, the overlap multiplicity is bounded by a dimensional constant and hence

$$\sum_a \|f\mathbf{1}_{Q_a^*}\|_p^p = \int |f(x)|^p \sum_a \mathbf{1}_{Q_a^*}(x)\,dx \le C_n\|f\|_p^p.$$

Combining this identity with (17.40) and the corresponding error estimate gives

$$\mathscr{J}_{2,p}(\mathscr{L}_\varepsilon^{P,\Omega} f : \varepsilon > 0) \le C_{n,d,p}\mathfrak{L}(\Omega)\|f\|_p. \tag{17.41}$$

Adding the local and far paths and using (15.5) proves the normalized case.

It remains to undo the dilation. If $f_\rho(Y) = f(\rho Y)$, homogeneity of the kernel gives the exact conjugacy

$$T_{\Omega,\varepsilon}^P f(\rho X) = T_{\Omega,\varepsilon/\rho}^{P_\rho} f_\rho(X). \tag{17.42}$$

The map $\varepsilon \mapsto \varepsilon/\rho$ is an order-preserving bijection of $(0,\infty)$, so it leaves every jump number unchanged. Changing variables in the input and output norms gives the explicit identities

$$\|f_\rho\|_{L_X^p} = \rho^{-n/p}\|f\|_{L_x^p}, \qquad \|\mathbf{J}(T_{\Omega,\varepsilon/\rho}^{P_\rho} f_\rho)\|_{L_X^p} = \rho^{-n/p}\|\mathbf{J}(T_{\Omega,\varepsilon}^P f)\|_{L_x^p}.$$

Thus the input and output acquire the same factor $\rho^{-n/p}$. Thus the constant does not depend on c_ν. The finite backward induction proves (17.37) for every phase.

The variation estimate is now a pointwise consequence rather than a mixed-norm extrapolation: by (15.8), (1.10), and (17.37),

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_p \le \kappa_r\,\mathscr{J}_{2,p}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0) \le C_{n,d,p}\kappa_r\mathfrak{L}(\Omega)\|f\|_p.$$

This proves (17.38), with the stated r-dependence absorbed in its constant.

We next construct the anchored trajectory and pass to general data. For a normalized nonterminal phase, Proposition 17.3 defines the far path on L^p, including its value at 1; add the finite-shell local path (17.31). Their sum has the classical finite-shell increments, agrees with the far path for $\varepsilon \ge 1$,

and is anchored at infinity by Remark 17.4. For a nonnormalized nonterminal phase use the conjugacy (17.42). If every flag coordinate vanishes, use the anchored factorization of Proposition 15.9 and the pure modulations (16.1).

Choose bounded compactly supported $f_m \to f$ in L^p. Write

$$\widetilde{\Omega}_m = \Omega \mathbf{1}_{\{|\Omega|\le m\}}, \qquad c_m = \frac{1}{m_n}\int_{\mathbb{S}^{n-1}} \widetilde{\Omega}_m \, d\sigma, \qquad m_n = \sigma(\mathbb{S}^{n-1}),$$

and put $\Omega_m = \widetilde{\Omega}_m - c_m$. Since the defining Orlicz function satisfies the Δ_2 condition, $\widetilde{\Omega}_m \to \Omega$ in the Luxemburg $L\log L$ norm. Moreover, using $\int \Omega \, d\sigma = 0$ and $L\log L \hookrightarrow L^1$,

$$c_m = -\frac{1}{m_n}\int_{\mathbb{S}^{n-1}} \Omega \mathbf{1}_{\{|\Omega|>m\}} \, d\sigma \longrightarrow 0.$$

On the finite-measure sphere the Luxemburg norm of the constant function c_m is $C_n|c_m|$. Hence the bounded mean-zero functions Ω_m converge to Ω in $L\log L$. By Lemma 15.1, this convergence is equivalently

$$\mathfrak{L}(\Omega_m - \Omega) \longrightarrow 0. \tag{17.43}$$

The lattice property of the Luxemburg norm, the embedding $L\log L \hookrightarrow L^1$ on the sphere, and the triangle inequality also give

$$\|\widetilde{\Omega}_m\|_{L\log L} \le \|\Omega\|_{L\log L}, \qquad \|c_m \mathbf{1}\|_{L\log L} \le C_n|c_m| \le C_n\|\Omega\|_1 \le C_n\|\Omega\|_{L\log L}.$$

Lemma 15.1 therefore yields the uniform bound

$$\sup_{m\ge 1} \mathfrak{L}(\Omega_m) \le C_n \mathfrak{L}(\Omega). \tag{17.44}$$

For a normalized nonterminal phase, bilinearity gives

$$\mathscr{T}_\varepsilon^{P,\Omega_m} f_m - \mathscr{T}_\varepsilon^{P,\Omega} f = \mathscr{T}_\varepsilon^{P,\Omega_m}(f_m - f) + \mathscr{T}_\varepsilon^{P,\Omega_m - \Omega} f.$$

Proposition 17.3, (17.44), and (17.43) show that both terms tend to zero in the finite-trajectory $L^p(E_D)$ norm, uniformly over every fixed finite parameter set D. Thus the far coordinates for (Ω_m, f_m) converge in L^p to those for (Ω, f). The same bilinear decomposition and Young's inequality give convergence of every local finite-shell coordinate. The same conclusion follows after dilation, and in the pure case from the factorization estimates in Proposition 15.9. The finite-shell bound used here is, for $0 < a < b < \infty$,

$$\Big\| \int_{a<|z|<b} e^{iP(x,x-z)} K_\Theta(z) g(x-z)\, dz \Big\|_p \le \|\Theta\|_1 \log(b/a) \|g\|_p.$$

Enumerate $\mathbb{Q}_+ = \{r_1, r_2, \dots\}$ and put $D_N = \{r_1, \dots, r_N\}$. A single diagonal subsequence of the coordinatewise L^p convergence is almost everywhere simultaneous at every rational parameter. For each fixed N, apply (15.29) on the finite coordinate space indexed by D_N and then Fatou's lemma. This transfers (17.37) to the limit with a bound independent of N. Only after this finite-dimensional passage do we let $N \to \infty$. The pointwise functionals increase monotonically, and every strict rational jump witness is contained in one D_N; monotone convergence therefore yields the bound on $\mathbb{Q}_+$. The finite-shell integrability used in (17.30) gives local absolute continuity in the truncation parameter almost everywhere, so Lemma 15.6 recovers all positive parameters. The variation estimate then follows pointwise from (15.8). The construction above supplies the L^p and almost-everywhere anchor at infinity. □

18 Separated-packet estimates for the terminal phase

All packet families in this and the following packet sections are finite; we suppress this qualifier in individual statements. Rankings and randomizations are always performed at this finite level. In the endpoint arguments, cube, shell, and parameter cutoffs are imposed explicitly and removed only after the estimates have been established, by coordinatewise convergence on finite annuli and lower semicontinuity of strict jump counts. Thus no infinite randomization or infinite-rank argument is used.

For $n \geq 2$, we use the following two microlocal estimates of Seeger [36, Lemmas 2.1 and 2.2]. Let H_j be supported where $2^{j-2} \leq |x| \leq 2^{j+2}$, and assume that $r \mapsto H_j(r\theta)$ is C^N on $(0,\infty)$ for every θ, with the radial derivatives satisfying

$$M_N = \sup_{j,\theta,r} \sup_{0\leq \ell \leq N} r^{n+\ell} |\partial_r^\ell H_j(r\theta)| < \infty. \tag{18.1}$$

For an integer $s > 3$ and a fixed $0 < \kappa < 1$, Seeger's construction [36, equation (2.2)] is $H_j = \Gamma_j^s + (H_j - \Gamma_j^s)$. Put $L(Q) = m$ when $2^{m-1} < \ell(Q) \leq 2^m$. If the cubes have disjoint interiors, $F_m = \sum_{L(Q)=m} f_Q$, and $\operatorname{supp} f_Q \subset Q$, $\|f_Q\|_1 \leq \Lambda|Q|$, then, for $0 < \kappa < 1$,

$$\|\sum_j \Gamma_j^s * F_{j-s}\|_2^2 \leq C M_0^2 2^{-s(1-\kappa)} \Lambda \sum_Q \|f_Q\|_1. \tag{18.2}$$

If additionally $\operatorname{supp} b_Q \subset Q$, $\int b_Q = 0$, $\ell(Q) = 2^{j-s}$, $N \geq n+1$, and $0 \leq \varepsilon \leq 1$, then

$$\|(H_j - \Gamma_j^s) * b_Q\|_1 \leq C_N \left[M_0 2^{-s\varepsilon} + M_N 2^{s(n+(\varepsilon-\kappa)N)} \right] \|b_Q\|_1. \tag{18.3}$$

The constants may depend on the fixed value of κ and on Seeger's fixed auxiliary cutoffs, but not on j, s, the cubes, or the functions. These are precisely the support hypotheses in [36, Lemmas 2.1–2.2]: the first lemma requires no cancellation, whereas the second uses the displayed mean-zero condition. In dimension one, the corresponding one-shell estimates below follow directly from cancellation, radial derivative bounds, and the same positive third-moment estimate. They provide the one-dimensional input to the weak endpoint argument in Section 21.

We also need a geometric observation.

Lemma 18.1 (Thin radial layers)**.** *Fix $0 < c_- < c_+ < \infty$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, let $R > 0$, let $t \in \mathbb{N}_0$, put $q = R2^{-t}$, and let $\Lambda \geq 0$. Suppose $d_U \in L^1$ are supported on pairwise disjoint q-cubes and satisfy $\|d_U\|_1 \leq \Lambda|U|$. If $E \subset [c_-R, c_+R]$ is a union of M_E intervals of total length $|E|$, then*

$$\left\| K_\Theta \mathbf{1}_{\{|z|\in E\}} * \sum_U d_U \right\|_\infty \leq C_{n,c_-,c_+} \Lambda \|\Theta\|_\infty \left(\frac{|E|}{R} + M_E 2^{-t} \right), \tag{18.4}$$

$$\left\| K_\Theta \mathbf{1}_{\{|z|\in E\}} * \sum_U d_U \right\|_1 \leq C_{n,c_-,c_+} \|\Theta\|_1 \frac{|E|}{R} \sum_U \|d_U\|_1. \tag{18.5}$$

Here and in the proof the implicit constants depend only on n, c_-, c_+.

Proof. Write $E = \bigcup_{i=1}^{M_E} I_i$, allowing overlapping intervals and any choices of endpoints. For $w = \mathbf{1}_E$ one has

$$\|w\|_1 = |E|, \qquad \|w\|_\infty \leq 1, \qquad V^1(w) \leq 2M_E.$$

Indeed, along an increasing finite sequence the indicator of one interval changes at most twice, and $|\max_i a_i - \max_i b_i| \leq \sum_i |a_i - b_i|$ for zero–one values. Thus overlaps do not replace $|E|$ by the sum of the interval lengths. If $M_E = 0$ the kernel is zero; otherwise $\|w\|_\infty + V^1(w) \leq 3M_E$.

First take $R = 1$ and $\delta = 2^{-t}$. The radial upper-sum packing argument in Lemma 26.5, with variation exponent one, gives

$$\left\| K_\Theta w * \sum_U d_U \right\|_\infty \leq C\Lambda \|\Theta\|_\infty \big(\|w\|_1 + \delta(\|w\|_\infty + V^1(w)) \big).$$

For clarity, this argument requires only disjoint cube interiors and side length at most δ: group the cubes by the radial distance of their centers, bound their total volume in each enlarged radial cell, and apply the one-dimensional upper-sum estimate to w. Neither a common dyadic grid nor disjointness of the I_i is needed. Essential angular bounds are used only inside the polar integrals, so exceptional angular representatives do not produce an exceptional pointwise hypothesis. This proves the unit-scale L^∞ estimate. The actual kernel satisfies

$$\|K_\Theta \mathbf{1}_{\{|z|\in E\}}\|_1 = \|\Theta\|_1 \int_E \frac{dr}{r} \le c_-^{-1}\|\Theta\|_1 |E|,$$

which proves the L^1 estimate by Young's inequality.

For general $R>0$ use $x = Rx'$, $U' = R^{-1}U$, $d'_{U'}(y) = d_U(Ry)$ and $E' = R^{-1}E$. The density bound remains $\|d'_{U'}\|_1 \le \Lambda|U'|$, the cube sides become 2^{-t}, and $|E'| = |E|/R$. Homogeneity of K_Θ identifies the original convolution at Rx' with the unit-scale convolution. The R^n in its L^1 norm cancels the R^{-n} in the packet masses. This gives exactly (18.4)–(18.5). Finally, the same estimates apply to arbitrary countable cube families: Tonelli controls the sum of absolute packet contributions, finite-family exhaustion gives the bounds, and finite total mass gives the genuine L^1 convolution limit. The inequalities retain their extended-norm meaning when the total mass is infinite. □

Lemma 18.2 (Variational Rademacher–Menshov)**.** *For a finite sequence $g_1,\dots,g_M$ in L^2,*

$$\left\| V^2\left(\sum_{\ell\le m} g_\ell : 0\le m\le M\right)\right\|_2 \le C\log(2+M) \sup_{\varepsilon_\ell\in\{-1,0,1\}} \left\|\sum_\ell \varepsilon_\ell g_\ell\right\|_2. \tag{18.6}$$

Proof. The case $M=0$ is immediate. For $M\ge 1$, put $L = \lfloor \log_2 M\rfloor$, $P_m = \sum_{\ell\le m} g_\ell$, and $a_r(m) = 2^r\lfloor m/2^r\rfloor$ for $0\le m\le M$. Define $H_r(m) = P_{a_r(m)} - P_{a_{r+1}(m)}$. Since $a_0(m)=m$ and $a_{L+1}(m)=0$, the exact telescoping identity is $P_m = \sum_{r=0}^L H_r(m)$. At level r, let $\mathscr{D}_r$ partition $\{1,\dots,M\}$ into successive blocks of length 2^r, with a possible shorter last block, and put $G_I = \sum_{\ell\in I} g_\ell$. The path H_r is a monotone pullback of a finite state path alternating zero with some of these complete block sums. For any finite state path (z_q), the inequality $|z_b - z_a|^2 \le 2|z_b|^2 + 2|z_a|^2$, followed by counting each endpoint at most twice in an increasing variation partition, gives the convenient bound $V^2(z_q) \le \sqrt{8\sum_q |z_q|^2}$. Adding unused blocks, including the final incomplete block, only increases the bound. Subadditivity of variation therefore proves

$$V^2(P_m : 0\le m\le M) \le \sqrt{8}\sum_{r=0}^{L}\left(\sum_{I\in\mathscr{D}_r}|G_I|^2\right)^{1/2}. \tag{18.7}$$

For each level, the Hilbert-space parallelogram identity, applied inductively, supplies fixed signs $\varepsilon_I \in \{-1,1\}$ with

$$\sum_{I\in\mathscr{D}_r}\|G_I\|_2^2 \le \left\|\sum_{I\in\mathscr{D}_r}\varepsilon_I G_I\right\|_2^2.$$

These are signs of entire L^2 vectors, not spatially varying selectors. Every block-sign sum is an original-index sign sum. Thus each square function on the right of (18.7) has L^2 norm at most the supremum in (18.6). Minkowski and

$$L+1 \le \frac{2}{\log 2}\log(2+M)$$

prove (18.6), for example with $C = 2\sqrt{8}/\log 2$. In fact the proof uses only the smaller supremum over $\{-1,1\}$ signs, which is bounded by the displayed $\{-1,0,1\}$ supremum. □

We shall use the following elementary completion fact whenever a finite Fourier expansion is removed. Its anchored formulation avoids treating V^r as a norm modulo constants and also avoids any appeal to an uncountable-parameter Bochner space.

Lemma 18.3 (Anchored variation completion)**.** *Let $I=[a,b]$, let $1\le p,r<\infty$, and let $H_N=(H_{N,t})_{t\in I}$ be paths of measurable functions satisfying $H_{N,b}=0$. Assume that the variation functions of the paths and their differences are measurable; this holds, in particular, when their variation is detected by one fixed countable dense subset of I, as in all applications below. Suppose*

$$\|V^r(H_N-H_M:t\in I)\|_p\longrightarrow 0\qquad (N,M\to\infty). \tag{18.8}$$

Then there is an anchored path $H=(H_t)_{t\in I}$ such that

$$\|V^r(H_N-H:t\in I)\|_p\longrightarrow 0. \tag{18.9}$$

After passage to a subsequence, $H_{N,t}(x)\to H_t(x)$ uniformly in $t\in I$ for almost every x. Moreover,

$$\|V^r(H:t\in I)\|_p\le \liminf_{N\to\infty}\|V^r(H_N:t\in I)\|_p. \tag{18.10}$$

Consequently, if $P_N=G_N+E_N$, both (G_N) and (E_N) satisfy the corresponding anchored Cauchy hypotheses (possibly with different p,r), and $P_{N,t}\to P_t$ for every t almost everywhere, then the limits satisfy $P_t=G_t+E_t$ for every t on one common full-measure set whenever the convergence of P_N is uniform in t there.

Proof. Choose a subsequence N_k such that

$$\sum_k\|V^r(H_{N_{k+1}}-H_{N_k}:t\in I)\|_p<\infty.$$

Minkowski's inequality shows that, outside one null set,

$$\sum_k V^r(H_{N_{k+1}}-H_{N_k}:t\in I)<\infty.$$

Since every difference path vanishes at b,

$$\sup_{t\in I}|H_{N_{k+1},t}-H_{N_k,t}|\le V^r(H_{N_{k+1}}-H_{N_k}:t\in I).$$

Thus the subsequence converges uniformly in t almost everywhere to an anchored path H. We record explicitly the measurability needed below. On the same full-measure set put

$$S_k=\sum_{\ell\ge k}V^r(H_{N_{\ell+1}}-H_{N_\ell}:t\in I).$$

The functions S_k are measurable, finite almost everywhere, and decrease to zero. Applying the variation triangle inequality first to each finite tail and then passing to its uniform limit gives

$$V^r(H_{N_k}-H:t\in I)\le S_k.$$

Consequently, for every fixed N,

$$|V^r(H_N-H:t\in I)-V^r(H_N-H_{N_k}:t\in I)|\le S_k\longrightarrow 0.$$

Hence $V^r(H_N-H:t\in I)$ is the pointwise limit of measurable functions and is measurable. For every finite parameter partition, its increment sum passes to the pointwise limit. Taking the supremum over finite partitions and applying Fatou now gives

$$\|V^r(H_N-H:t\in I)\|_p\le\liminf_{k\to\infty}\|V^r(H_N-H_{N_k}:t\in I)\|_p.$$

Indeed, for every $\delta>0$ the Cauchy hypothesis supplies N_0 such that

$$\|V^r(H_N-H_M:t\in I)\|_p<\delta\quad\text{whenever } N,M\ge N_0.$$

For $N \geq N_0$, the preceding inequality with $M = N_k$ and then $k \to \infty$ yields $\|V^r(H_N - H : t \in I)\|_p \leq \delta$. This proves (18.9) for the full sequence. Pointwise, the variation triangle inequality gives

$$\big|V^r(H_N : t \in I) - V^r(H : t \in I)\big| \leq V^r(H_N - H : t \in I).$$

The reverse triangle inequality in L^p and (18.9) therefore give convergence of the two variation norms, which is stronger than (18.10). For the last assertion, choose one diagonal subsequence along which the required summability holds for both G_N and E_N; passing the finite identities to their common uniform pointwise limits gives the claimed identity. □

Because Seeger's microlocal kernels Γ_j^s have noncompact spatial support, localization and ranking are carried out with the original compactly supported kernels H_j.

Lemma 18.4 (Fixed-sign full-kernel estimate)**.** *Let $n \geq 1$, $N = 4n+1$, and let H_j be supported where $2^{j-2} \leq |x| \leq 2^{j+2}$, radially C^N as above, and satisfy* (18.1)*. Let $\mathscr{Q}$ be a finite family of cubes with disjoint interiors, and suppose*

$$\operatorname{supp} b_Q \subset Q, \qquad \int b_Q = 0, \qquad \|b_Q\|_1 \leq \Lambda |Q|.$$

For $s \in \mathbb{N}$, $s \geq 5$, pair H_j with the cubes of side 2^{j-s}. Then, for every fixed choice $|c_Q| \leq 1$,

$$\left\| \sum_j H_j * \sum_{\ell(Q) = 2^{j-s}} c_Q b_Q \right\|_2^2 \leq C_n M_N^2 \, 2^{-s/24} \Lambda \sum_Q \|b_Q\|_1. \tag{18.11}$$

Proof. Put $\Sigma = \sum_Q \|b_Q\|_1$. The cases $M_N = 0$ or $\Sigma = 0$ give the zero operator. Otherwise normalize $M_N = 1$ by homogeneity. The density assumption then implies $\Lambda > 0$ whenever $\Sigma > 0$. Write

$$F = \sum_Q c_Q H_{j(Q)} * b_Q, \qquad R_Q = 2^{j(Q)}, \quad m_Q = \|b_Q\|_1, \quad u_Q = |H_{j(Q)}| * |b_Q|.$$

Here $\ell(Q) = R_Q 2^{-s} \leq R_Q$. The zeroth radial bound and the annular support imply, with constants depending only on n,

$$|F| \leq \sum_Q u_Q, \qquad \|u_Q\|_1 \leq C_n m_Q, \qquad 0 \leq u_Q \leq C_n R_Q^{-n} m_Q \mathbf{1}_{E_Q},$$

where E_Q is the axis-parallel closed box about the center of Q with coordinate half-length $20R_Q$. These statements hold for arbitrary integrable packets; no pointwise bound on b_Q is used.

We prove the required third moment directly. For each fixed Q, all cubes U satisfying $R_U \leq R_Q$ and $E_U \cap E_Q \neq \varnothing$ are contained in a ball of radius $C_n' R_Q$ about the center of Q. Their interiors are disjoint and their boundaries have measure zero. Consequently

$$\sum_{\substack{U : R_U \leq R_Q \\ E_U \cap E_Q \neq \varnothing}} m_U \leq \Lambda \sum_{\substack{U : R_U \leq R_Q \\ E_U \cap E_Q \neq \varnothing}} |U| \leq C_n \Lambda R_Q^n. \tag{18.12}$$

Expand $(\sum_Q u_Q)^3$ and order the three radii. Each ordered triple accounts for at most six permutations, including ties. If $R_U \leq R_V \leq R_W$, the triple integral is zero unless both $E_U \cap E_V$ and $E_V \cap E_W$ are nonempty; otherwise the preceding pointwise bounds give

$$\int u_U u_V u_W \leq C_n m_U m_V m_W R_V^{-n} R_W^{-n}.$$

First sum over U and apply (18.12) at V; then sum over V and apply the same estimate at W. This gives

$$\sum_{\substack{U,V,W \\ R_U \leq R_V \leq R_W}} \int u_U u_V u_W \leq C_n \Lambda^2 \sum_W m_W. \tag{18.13}$$

All sums are finite and all functions are nonnegative, so the expansion and integrations require no conditional interchange. It follows that

$$\|F\|_3^3 \le \int \Big(\sum_Q u_Q\Big)^3 \le C_n \Lambda^2 \Sigma. \tag{18.14}$$

This proof works in every positive dimension and at all the relevant scales, without an interpolation theorem involving BMO.

Suppose first that $n = 1$. Put $a_j = 2^{j-2}$ and normalize the kernel by $\Phi_j(x) = a_j H_j(a_j x)$. The positive and negative radial restrictions, together with the inner support gap, show that Φ_j is globally Lipschitz with constant at most 2, is supported in $[-16, 16]$, and has absolute value at most 1. Thus

$$\int_{\mathbb{R}} |\Phi_j(x-h) - \Phi_j(x)|\, dx \le C|h|.$$

Indeed, for $|h| \le 1$ use the Lipschitz bound on the union of the two supports, whose length is bounded; for $|h| > 1$ use twice the L^1 norm and $1 < |h|$. Undoing the scaling gives

$$\|H_j(\cdot - y) - H_j(\cdot - x_Q)\|_1 \le C 2^{-j}|y - x_Q|.$$

Cancellation, with x_Q the center of Q, now yields

$$\begin{aligned}\|H_j * (c_Q b_Q)\|_1 &\le \int_Q |b_Q(y)| \|H_j(\cdot - y) - H_j(\cdot - x_Q)\|_1 \, dy \\ &\le C 2^{-j} \ell(Q) \|b_Q\|_1.\end{aligned}$$

Since $\ell(Q) = 2^{j-s}$, summing yields

$$\|F\|_1 \le C 2^{-s} \Sigma. \tag{18.15}$$

For $u > 0$, split at $|F| = u$ and use the valid, slightly loose bound

$$\int |F|^2 \le 2u \|F\|_1 + u^{-1} \|F\|_3^3.$$

With $u = \Lambda 2^{s/2}$ this gives

$$\|F\|_2^2 \le C 2^{-s/2} \Lambda \Sigma \le C 2^{-s/24} \Lambda \Sigma.$$

For $n \ge 2$, use Seeger's construction with $\kappa = 1/2$ and write $F = A + R$. Equation (18.2) gives

$$\|A\|_2^2 \le C_n 2^{-s/2} \Lambda \Sigma. \tag{18.16}$$

Summing (18.3) with $\varepsilon = 1/4$ gives

$$\|R\|_1 \le C_n 2^{-s/4} \Sigma, \tag{18.17}$$

because $N = 4n + 1$ and $n - N/4 = -1/4$. Fixed coefficients of absolute value at most one preserve cancellation and the density bound.

For any $u > 0$, the elementary inequality $|F|^2 \le |A|^2 + 2|F||R|$ holds. Use it on $\{|F| \le u\}$, and use $|F|^2 \le u^{-1}|F|^3$ on the complement. Thus

$$\int |F|^2 \le \|A\|_2^2 + 2u \|R\|_1 + u^{-1} \|F\|_3^3.$$

Choosing $u = \Lambda 2^{s/8}$ and using the three proved bounds yields

$$\|F\|_2^2 \le C_n \Lambda \Sigma \big(2^{-s/2} + 2^{-s/8}\big) \le C_n 2^{-s/8} \Lambda \Sigma \le C_n 2^{-s/24} \Lambda \Sigma.$$

The inequality itself establishes square integrability; it was not assumed beforehand. Finally undoing the normalization restores M_N^2 and proves the original estimate in every dimension. □

Proposition 18.5 (Selector-free terminal packet trajectory)**.** *There are $t_0 \in \mathbb{N}$ and $\delta_T > 0$, depending only on n, such that the following holds. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, let $R > 0$, $t \in \mathbb{N}$, $q = R2^{-t}$, $t \geq t_0$, and let $\{U\}$ be pairwise disjoint q-cubes. Let $\Lambda \geq 0$ and suppose*

$$\operatorname{supp} d_U \subset U, \quad \int d_U = 0, \quad \|d_U\|_1 \leq \Lambda|U|, \quad M = \sum_U \|d_U\|_1.$$

For fixed complex coefficients $|a_U| \leq 1$, define

$$F_\varepsilon(x) = \int_{\varepsilon<|x-y|\leq 2R} K_\Theta(x-y) \sum_U a_U d_U(y)\,dy, \qquad R \leq \varepsilon \leq 2R.$$

Then

$$\|V^2(F_\varepsilon : R \leq \varepsilon \leq 2R)\|_2^2 \leq C\|\Theta\|_\infty^2 2^{-\delta_T t}\Lambda M. \tag{18.18}$$

The same estimate holds after multiplication of the entire output path by $e^{i\xi\cdot x}$, for any fixed $\xi \in \mathbb{R}^n$. It also holds after replacing each $d_U(y)$ by $e^{i\eta_U\cdot y}d_U(y)$, with arbitrary $\eta_U \in \mathbb{R}^n$, provided $\int e^{i\eta_U\cdot y}d_U(y)\,dy = 0$ for every U. In the phase-adapted application below, the input modulation is frozen packet by packet and the resulting error is estimated separately.

Proof. We use a single argument in every dimension $n \geq 1$; its full-kernel input is Lemma 18.4, including the one-dimensional annular-kernel case proved there. All signs introduced below are fixed before evaluation in x.

First replace Θ by a measurable representative bounded pointwise by $\|\Theta\|_\infty$ on the sphere. Polar coordinates, followed by translation of the exceptional null set, show that this replacement preserves each literal annular integral for every x and every radius. Put $\theta = \|\Theta\|_\infty$. Normalize $R = 1$ by $x = RX$, $y = RY$, $\widetilde{U} = R^{-1}U$ and $\widetilde{d}_U(Y) = d_U(RY)$. Then

$$F_\varepsilon(RX) = \widetilde{F}_{\varepsilon/R}(X), \qquad \|\widetilde{d}_U\|_1 = R^{-n}\|d_U\|_1, \qquad \|\widetilde{d}_U\|_1 \leq \Lambda|\widetilde{U}|.$$

Thus the input mass is multiplied by R^{-n}, whereas the squared output norm is multiplied by R^n. These factors cancel exactly. We initially work with a finite packet family at unit scale.

Put

$$N = 4n+1, \qquad k = \left\lfloor \frac{t}{384N} \right\rfloor, \qquad L = 2^k,$$
$$w = L^{-1}, \qquad h = (8L^2)^{-1}, \qquad a_i = 1 + \frac{i}{L} \quad (0 \leq i \leq L).$$

Choose $t_0 = t_0(n) > 3$ large enough that $\sqrt{n}\,2^{-t} \leq 1/4$ whenever $t \geq t_0$. In particular $h \leq 1/8$ and $4h \leq w$. For fixed complex numbers β_i with $|\beta_i| \leq 1$, let

$$A(r) = \sum_{i=0}^{L-1} \beta_i \mathbf{1}_{(a_i, a_{i+1}]}(r).$$

Choose once and for all a smooth nondecreasing function χ on $\mathbb{R}$, equal to zero on $(-\infty, 0]$ and one on $[1,\infty)$, and define the explicit endpoint regularization

$$A_h(r) = \sum_{i=0}^{L-1} \beta_i \left[\chi\left(\frac{r-a_i+h}{2h}\right) - \chi\left(\frac{r-a_{i+1}+h}{2h}\right)\right].$$

The bracketed terms are nonnegative and telescope in their unsigned sum, so $|A_h| \leq 1$. The endpoint transition regions are disjoint because $4h \leq w$. Consequently, for every $0 \leq \ell \leq N$,

$$\|A_h^{(\ell)}\|_\infty \leq C_n h^{-\ell}, \qquad \operatorname{supp} A_h \subset [1-h, 2+h],$$

with no factor depending on L. The constants here depend on the fixed transition function, not on any freely varying cutoff. Leibniz' rule for $r^{-n}A_h(r)$ gives normalized radial C^N size at most $C_n\theta h^{-N}$ for

$H(z) = K_\Theta(z)A_h(|z|)$. Embed this one-shell kernel in its dyadically rescaled annular family and apply Lemma 18.4 at the single active scale. It yields

$$\left\| \sum_U a_U H * d_U \right\|_2^2 \le C_n \theta^2 h^{-2N} 2^{-t/24} \Lambda M \le C_n \theta^2 w \Lambda M.$$

Indeed $h^{-2N} = 8^{2N} 2^{4Nk}$, $4Nk \le t/96$, and $2^{-t/32} \le 2^{-k} = w$. This estimate is uniform in the fixed β_i and the original fixed packet coefficients a_U.

Here is the sharp-profile error estimate needed to remove the smoothing. Write $E_{\mathrm{bdry}} = \bigcup_{i=0}^{L} [a_i - h, a_i + h]$. Then

$$\operatorname{supp}(A_h - A) \subset E_{\mathrm{bdry}}, \qquad |E_{\mathrm{bdry}}| \le 2(L+1)h, \tag{18.19}$$

$$|A_h - A| \le 2\mathbf{1}_{E_{\mathrm{bdry}}}, \qquad E_{\mathrm{bdry}} \subset [3/4, 9/4]. \tag{18.20}$$

If $G_{\mathrm{bdry}} = \sum_U a_U [K_\Theta(A_h - A)(|\cdot|)] * d_U$, then

$$|G_{\mathrm{bdry}}(x)| \le 2 \left(|K_\Theta| \mathbf{1}_{\{|z| \in E_{\mathrm{bdry}}\}} * \sum_U |d_U| \right)(x). \tag{18.21}$$

The endpoint-layer volume bound gives the right-hand kernel L^1 norm at most $C_n\theta(L+1)h$. The broad fixed-annulus cube-density estimate gives its convolution with $\sum_U |d_U|$ an L^∞ bound at most $C_n\theta\Lambda$. This latter bound uses only the original packet L^1 density, not a pointwise bound on the packets. Therefore

$$\|G_{\mathrm{bdry}}\|_2^2 \le C_n \theta^2 (L+1) h \Lambda M \le C_n \theta^2 w \Lambda M. \tag{18.22}$$

Combining the smooth estimate with this error bound proves the genuine sharp-cell fixed-sign estimate

$$\left\| \sum_{i=0}^{L-1} \beta_i G_i \right\|_2^2 \le C_n \theta^2 w \Lambda M, \qquad G_i = K_\Theta \mathbf{1}_{\{a_i < |z| \le a_{i+1}\}} * \sum_U a_U d_U.$$

In particular it holds for every spatially constant choice $\beta_i \in \{-1, 1\}$.

The grid values F_{a_i} are, up to a constant and a sign, the prefixes of the actual cell outputs G_i. Lemma 18.2 consequently gives

$$\|V^2(F_{a_i} : 0 \le i \le L)\|_2^2 \le C_n (1 + \log_2 L)^2 \theta^2 w \Lambda M. \tag{18.23}$$

This is the inside pointwise variation estimate; it does not permit signs depending on x.

For the continuum part set

$$W_i(x) = \int_{a_i < |x-y| \le a_{i+1}} |K_\Theta(x-y)| \sum_U |d_U(y)| \, dy.$$

The local V^1 of F on $[a_i, a_{i+1}]$ is at most W_i. The thin-annulus cube-density bound gives $\|W_i\|_\infty \le C_n\theta\Lambda(w + 2\sqrt{n}\,2^{-t})$. The radial cells are disjoint, so Tonelli and the fixed-annulus kernel mass give $\sum_i \|W_i\|_1 \le C_n\theta M$. Hence

$$\int \sum_i W_i(x)^2 \, dx \le C_n \theta^2 \Lambda M (w + 2\sqrt{n}\,2^{-t}) \le C_n \theta^2 w \Lambda M.$$

An exact deterministic grid–cell decomposition, using anchored local errors supported in their respective parameter cells, gives

$$V^2(F_\varepsilon : 1 \le \varepsilon \le 2) \le V^2(F_{a_i} : 0 \le i \le L) + 40\sqrt{2} \left(\sum_i W_i^2 \right)^{1/2}. \tag{18.24}$$

At a cell boundary the next anchored error is zero, so the active errors remain disjoint. Each partition increment meets at most two active errors; Cauchy–Schwarz proves the displayed square-sum bound. The harmless constant 40 includes zero extension at the endpoints. Thus the original continuum trajectory, not merely a finite grid, satisfies

$$\int V^2(F_\varepsilon : 1 \le \varepsilon \le 2)^2 \le C_n \theta^2 (1+k)^2 2^{-k} \Lambda M \le C_n \theta^2 2^{-t/(768N)} \Lambda M. \tag{18.25}$$

For the last inequality, $(1+k)^2 2^{-k/2}$ is bounded and $k > t/(384N) - 1$. We may therefore choose, in every positive dimension,

$$\delta_T = \frac{1}{768(4n+1)} > 0.$$

For completeness, the elementary bounded-output estimate used in the later smooth-profile argument can also be recorded here. Whenever $S = A + R$ and $\|S\|_\infty \le H$, the pointwise triangle inequality and $(|S| - |A|)^2 \ge 0$ imply

$$\|S\|_2^2 \le \|A\|_2^2 + 2H\|R\|_1. \tag{18.26}$$

This estimate requires no exceptional-set interpolation and is consistent with all fixed-sign bounds above.

We now remove the finite-family restriction. Disjoint positive cube interiors form a countable family. If $M < \infty$, exhaust it by finite subfamilies. The clipped annular kernel is bounded, so the output paths converge for every x, uniformly over the radius interval, with error bounded by its kernel L^∞ norm times the omitted L^1 packet mass. Each finite path is continuous in its positive radius parameter: dominated convergence applies because translated spheres have measure zero. Each radius coordinate is measurable in x; a fixed countable dense restriction therefore proves measurability of its continuum variation. Finite-partition lower semicontinuity and Fatou pass the uniform squared-energy bound to the literal integral of the full packet sum. If $M = \infty$ and $\theta\Lambda > 0$, the asserted extended-value inequality is immediate. If $\theta = 0$ or $\Lambda = 0$, the angular kernel or every packet, respectively, is zero almost everywhere, and all the literal paths vanish. Returning to the original angular representative and radius proves (18.18) in its full stated scope.

Finally, multiplication of the whole trajectory by $e^{i\xi\cdot x}$ preserves every increment norm and hence its variation. Replacing $d_U(y)$ by $e^{i\eta_U\cdot y} d_U(y)$ preserves support, integrability, L^1 mass and density. The explicitly assumed mean zero after this replacement is exactly the remaining packet hypothesis. Applying the proved estimate to these packets establishes both modulation clauses. □

19 An anchored phase–packet induction

After removal of the parent-dependent input modulation, the phase-adapted decomposition recovers a mean-zero packet structure. We formulate the corresponding estimate in this anchored normalization.

Fix a real polynomial $P \in \mathscr{P}_d$ and $\Theta \in L^\infty(\mathbb{S}^{n-1})$. Let $t \in \mathbb{N}$, and let $R > 0$ and $q = R2^{-t}$ be dyadic. Let $\mathscr{Q}$ be a family of pairwise disjoint dyadic cubes, each contained in a dyadic R-cube, and for $Q \in \mathscr{Q}$ let c_Q denote its center. Suppose d_U are mean-zero descendants, supported on pairwise disjoint q-cubes $U \subset Q$, with

$$\|d_U\|_\infty \le \Lambda, \qquad M = \sum_U \|d_U\|_1. \tag{19.1}$$

Define the phase-adapted shell trajectory

$$\mathscr{P}_\varepsilon(x) = \sum_{Q\in\mathscr{Q}} \sum_{U\subset Q} \int_{\varepsilon<|x-y|\le 2R} e^{i[P(x,y)-P(c_Q,y)]} K_\Theta(x-y) d_U(y)\,dy, \quad R \le \varepsilon \le 2R. \tag{19.2}$$

We fix the Fourier localization once, rather than separately for each block. After the normalization by an interacting R-block used below, Y and every parent center ξ_Q lie in one fixed cube, while X lies in one of only $O_n(1)$ unit cubes whose difference from that cube meets $\{1 < |Z| \le 2\}$. Choose structural boxes $\mathfrak{X}, \mathfrak{Y}$ containing all these cubes with a fixed margin, and smooth cutoffs equal to one on the boxes.

The anchored flag contains only finitely many polynomials (depending on n,d); throughout the proof we use these same boxes and cutoffs and take the maximum of the resulting Fourier constants over that finite list. Fix a single period cell large enough for these boxes, their cutoff supports, and the finitely many block localizations below. Its structural period T and frequency $\omega_F = 2\pi/T$ are kept unchanged in every low expansion.

Lemma 19.1 (Fourier expansion of a low pivot)**.** *Let H be one of the fixed anchored flag polynomials, and let X,Y,ξ range in the structural boxes just fixed. Let $\chi,\widetilde{\chi}$ be the corresponding fixed smooth cutoffs. Choose one structural period $T>0$ and a translated fundamental cell containing the original boxes and cutoff supports in its interior, and put $\omega_F = 2\pi/T$. The same T is used for the entire finite flag. Then, for every $L\in\mathbb{R}$, on the original X,Y boxes,*

$$\chi(X)\widetilde{\chi}(Y)e^{iL[H(X,Y)-H(\xi,Y)]} = \sum_{u,v\in\mathbb{Z}^n} a_{u,v}(L,\xi)e^{i\omega_F u\cdot X}e^{i\omega_F v\cdot Y}.$$

The series is absolutely convergent. There exist nonnegative numbers $A_{u,v}(L)$ independent of ξ such that $|a_{u,v}(L,\xi)|\le A_{u,v}(L)$ and

$$\sum_{u,v} A_{u,v}(L)(1+|v|) \le C(1+|L|)^{C_F}, \qquad C_F = 2n+2. \tag{19.3}$$

The constants are uniform in L and ξ apart from the displayed power and, with the preceding convention, depend only on n,d. Outside the chosen cell the series represents the periodic extension, not the compactly supported expression on the left.

Proof. Take compact closures of all the original bounded boxes. Their union and the fixed cutoff supports fit strictly inside one translated cube of side T in each of the $2n$ joint coordinates. This choice depends only on the fixed geometry, not on L, ξ, or the flag member. The cutoff expression vanishes near the boundary, so its periodization is a smooth function on $(\mathbb{R}/T\mathbb{Z})^{2n}$. Define its actual Fourier coefficients by integrating over that cell $\mathscr{C}$:

$$a_{u,v}(L,\xi) = T^{-2n}\int_{\mathscr{C}} \chi(X)\widetilde{\chi}(Y)e^{iL[H(X,Y)-H(\xi,Y)]}e^{-i\omega_F(u\cdot X+v\cdot Y)}\,dX\,dY.$$

Apply $(1-\Delta_{X,Y})^{n+1}$ under this integral. There are no boundary terms. Every derivative of order at most $2n+2$ is bounded by $C(1+|L|)^{2n+2}$ uniformly in the compact pivot set, since H is a fixed polynomial. Taking the maximum over the finite flag gives

$$|a_{u,v}(L,\xi)| \le \frac{C(1+|L|)^{2n+2}}{(1+|(u,v)|)^{2n+2}} =: A_{u,v}(L).$$

The fixed frequency scale ω_F changes only C. Since $1+|v|\le 1+|(u,v)|$ and $\sum_{k\in\mathbb{Z}^{2n}}(1+|k|)^{-(2n+1)}<\infty$, this is the claimed weighted common envelope. Absolute uniform convergence and Fourier reconstruction of the smooth periodic function identify the sum at every point of the cell, hence at every original relevant point, including the boundaries of the original boxes. Translation of the cell is already incorporated into the coefficients; the displayed characters are in the original coordinates. □

Proposition 19.2 (Anchored phase–packet principle)**.** *There are $c_P,c_E>0$ and t_0, depending only on n,d, such that every trajectory* (19.2) *under the finite-packet setup above, with $P\in\mathscr{P}_d$, $\Theta\in L^\infty(\mathbb{S}^{n-1})$, and $t\ge t_0$, admits a decomposition $\mathscr{P} = G+E$ satisfying*

$$\|V^2(G_\varepsilon : R\le\varepsilon\le 2R)\|_2 \le C\|\Theta\|_\infty 2^{-c_P t}(\Lambda M)^{1/2}, \tag{19.4}$$

$$\|V^1(E_\varepsilon : R\le\varepsilon\le 2R)\|_1 \le C\|\Theta\|_1 2^{-c_E t}M. \tag{19.5}$$

The constants are independent of R, q, every coefficient of P, the locations of the cubes, and the parent labels Q.

Proposition 19.2 is included as a self-contained bounded-density model; it is not used in the weak-endpoint induction. Its purpose is to isolate the phase–packet mechanism from the additional concentration difficulty. The actual Calderón–Zygmund bad functions satisfy only cancellation and an average mass bound. They are handled instead by the concentration-stable estimates of Section 20.

Proof. Let $\mathscr{D}_R$ be the dyadic partition into cubes of side R, and let $\mathscr{J}_R \subset \mathscr{D}_R$ be the finite family of cubes containing at least one parent $Q \in \mathscr{Q}$. Define the finite set of interacting block pairs by

$$\mathscr{A}_R = \{(I,J) \in \mathscr{D}_R \times \mathscr{J}_R : \{x-y : x \in I,\ y \in J\} \cap \{z : R < |z| \le 2R\} \neq \varnothing\}. \tag{19.6}$$

For $(I,J) \in \mathscr{A}_R$, set

$$\mathscr{P}_\varepsilon^{I,J}(x) = \mathbf{1}_I(x) \sum_{\substack{Q \in \mathscr{Q},\ Q \subset J \\ U \subset Q}} \int_{\varepsilon < |x-y| \le 2R} e^{i[P(x,y)-P(c_Q,y)]} K_\Theta(x-y) d_U(y)\, dy. \tag{19.7}$$

The annular support gives, almost everywhere, the exact block decomposition $\mathscr{P}_\varepsilon = \sum_{(I,J)\in\mathscr{A}_R} \mathscr{P}_\varepsilon^{I,J}$. All block sums in this proof are henceforth restricted to $\mathscr{A}_R$.

For a fixed $(I,J) \in \mathscr{A}_R$, let a be the center of J and scale $x = a + RX$, $y = a + RY$. Put

$$M_{I,J} = \sum_{\substack{Q \in \mathscr{Q},\ Q \subset J \\ U \subset Q}} \|d_U\|_1.$$

Because I and J have side R and their difference meets a fixed annulus of radii R and $2R$, elementary lattice geometry gives the two-sided adjacency estimate

$$\sup_{I \in \mathscr{D}_R} \#\{J \in \mathscr{J}_R : (I,J) \in \mathscr{A}_R\} + \sup_{J \in \mathscr{J}_R} \#\{I \in \mathscr{D}_R : (I,J) \in \mathscr{A}_R\} \le C_n. \tag{19.8}$$

In particular, each packet mass is counted for only $O_n(1)$ output blocks, and hence

$$\sum_{(I,J)\in\mathscr{A}_R} M_{I,J} \le C_n M. \tag{19.9}$$

More explicitly, if every path $H_{I,J}$ is supported in I, the variation triangle inequality, Cauchy–Schwarz in the at most C_n adjacent cubes J, and disjointness of the output cubes give

$$\begin{aligned} \left\| V^2\left(\sum_{(I,J)\in\mathscr{A}_R} H_{I,J} \right) \right\|_2^2 &= \sum_I \left\| V^2\left(\sum_{J:(I,J)\in\mathscr{A}_R} H_{I,J} \right) \right\|_{L^2(I)}^2 \\ &\le C_n \sum_{(I,J)\in\mathscr{A}_R} \|V^2(H_{I,J})\|_2^2. \end{aligned} \tag{19.10}$$

Similarly,

$$\left\| V^1\left(\sum_{(I,J)\in\mathscr{A}_R} H_{I,J} \right) \right\|_1 \le \sum_{(I,J)\in\mathscr{A}_R} \|V^1(H_{I,J})\|_1. \tag{19.11}$$

For $Q \subset J$ and $U \subset Q$, set

$$\widetilde{U} = R^{-1}(U-a), \qquad \widetilde{d}_U(Y) = d_U(a+RY), \qquad \widetilde{I} = R^{-1}(I-a).$$

Then $\widetilde{U}$ has side 2^{-t} and

$$\|\widetilde{d}_U\|_\infty = \|d_U\|_\infty \le \Lambda, \quad \|\widetilde{d}_U\|_1 = R^{-n}\|d_U\|_1, \quad \|\widetilde{d}_U\|_1 \le \Lambda|\widetilde{U}|, \quad \sum_{\substack{Q \in \mathscr{Q},\ Q \subset J \\ U \subset Q}} \|\widetilde{d}_U\|_1 = R^{-n} M_{I,J}. \tag{19.12}$$

Homogeneity, $K_\Theta(RZ)R^n = K_\Theta(Z)$, cancels the Jacobian, so the scaled block trajectory has the unit-annulus kernel. Since the output cutoff is independent of ε, for every $r \geq 1$ one has $V^r(\mathbf{1}_I H_\varepsilon) = \mathbf{1}_I V^r(H_\varepsilon)$ pointwise. Thus restricting a unit-scale output to $\widetilde{I}$ cannot increase its variation norm. If $\widetilde{G}$ and $\widetilde{E}$ are paths in the scaled variables, then

$$\|\mathbf{1}_I V^2(G)\|_{L^2_x} = R^{n/2}\|\mathbf{1}_{\widetilde{I}} V^2(\widetilde{G})\|_{L^2_X}, \qquad \|\mathbf{1}_I V^1(E)\|_{L^1_x} = R^n \|\mathbf{1}_{\widetilde{I}} V^1(\widetilde{E})\|_{L^1_X}. \tag{19.13}$$

Thus unit-scale factors $(\Lambda R^{-n} M_{I,J})^{1/2}$ and $R^{-n}M_{I,J}$ become $(\Lambda M_{I,J})^{1/2}$ and $M_{I,J}$ in physical variables. Put

$$\Phi(X,Y) = P(a+RX, a+RY) - P(a, a+RY).$$

Then $\Phi(0,Y) = 0$. If $Q \subset J$ and $\xi_Q = (c_Q - a)/R$, the phase in (19.2) is exactly

$$\Phi(X,Y) - \Phi(\xi_Q, Y). \tag{19.14}$$

Write the anchored flag decomposition, retaining its pure output part,

$$\Phi(X,Y) = p(X) + \sum_{\mu=1}^{N_d} C_\mu \widehat{H}_\mu(X,Y). \tag{19.15}$$

All C_μ are dimensionless; they may depend arbitrarily on a and R. The list and its length depend only on n, d. The exact phase difference is

$$\Phi(X,Y) - \Phi(\xi_Q,Y) = p(X) - p(\xi_Q) + \sum_{\mu=1}^{N_d} C_\mu [\widehat{H}_\mu(X,Y) - \widehat{H}_\mu(\xi_Q,Y)]. \tag{19.16}$$

Hence $e^{ip(X)}$ is one common output modulation, whereas $s_Q := e^{-ip(\xi_Q)}$ is a parent-dependent scalar of modulus one. We work with the demodulated block, retain s_Q in every packet coefficient below, and at the end multiply both constructed component paths by $e^{ip(X)}$. This last operation preserves pointwise V^2, pointwise V^1, and the identity $\mathscr{P} = G + E$.

If $N_d = 0$, the common output modulation is harmless and the scalars s_Q are admissible fixed packet coefficients in Proposition 18.5; that proposition gives the conclusion directly. We therefore assume $N_d \geq 1$ below.

Let $\beta_* = \min_\mu \beta_\mu > 0$, where the β_μ are from Corollary 16.8. Let C_F be from Lemma 19.1. Choose $B > 1 + 8C_F/\beta_*$, then choose $\varepsilon_* > 0$ so small that

$$C_F N_d \varepsilon_* < \min(\delta_T/8, 1/8), \qquad \kappa_\mu = \varepsilon_* B^{\mu - N_d}. \tag{19.17}$$

Thus the κ_μ increase with μ and

$$C_F \sum_{r<\mu} \kappa_r \leq \frac{1}{4}\beta_* \kappa_\mu. \tag{19.18}$$

Starting at the first flag coordinate, call a coordinate low if $|C_\mu| \leq 2^{\kappa_\mu t}$. Expand every low factor

$$e^{iC_\mu[\widehat{H}_\mu(X,Y) - \widehat{H}_\mu(\xi_Q,Y)]}$$

by Lemma 19.1. To record the product estimate explicitly, let S be any set of low positions already expanded and index a term by tuples $\mathbf{u} = (u_r)_{r\in S}$ and $\mathbf{v} = (v_r)_{r\in S}$. Its parent-dependent coefficient and combined frequencies are

$$a_Q(\mathbf{u},\mathbf{v}) = s_Q \prod_{r\in S} a^{(r)}_{u_r,v_r}(C_r, \xi_Q), \quad U(\mathbf{u}) = \sum_{r\in S} u_r, \quad V(\mathbf{v}) = \sum_{r\in S} v_r.$$

Their physical frequencies are $\omega_F U(\mathbf{u})$ and $\omega_F V(\mathbf{v})$. The common period preserves integer frequency addition; moreover $1 + \omega_F|V| \leq \max\{1, \omega_F\}(1+|V|)$, so the original integer-frequency weight absorbs

the structural scale. With $A(\mathbf{u},\mathbf{v}) = \prod_{r\in S} A^{(r)}_{u_r,v_r}(C_r)$, if $A(\mathbf{u},\mathbf{v}) = 0$, then $|a_Q(\mathbf{u},\mathbf{v})| \le A(\mathbf{u},\mathbf{v})$ forces $a_Q(\mathbf{u},\mathbf{v}) = 0$ for every Q, and the corresponding term is omitted. Thus every division by $A(\mathbf{u},\mathbf{v})$ below is made only when $A(\mathbf{u},\mathbf{v}) > 0$. Moreover, the inequality $1+|\sum_{r\in S} v_r| \le \prod_{r\in S}(1+|v_r|)$ and Lemma 19.1 give

$$\begin{aligned}\sum_{\mathbf{u},\mathbf{v}} A(\mathbf{u},\mathbf{v})(1+|V(\mathbf{v})|) &\le \prod_{r\in S}\sum_{u_r,v_r} A^{(r)}_{u_r,v_r}(C_r)(1+|v_r|)\\ &\le C\prod_{r\in S}(1+|C_r|)^{C_F} \le C2^{C_F t\sum_{r\in S}\kappa_r}. \end{aligned} \tag{19.19}$$

The same estimate without the weight controls the ordinary Fourier loss. Keeping the tuple indexing and applying the triangle inequality yields the same bound while preserving uniformity in every ξ_Q. In particular, the product of the expansions preceding position μ has coefficient ℓ^1 norm at most

$$C2^{C_F t\sum_{r<\mu}\kappa_r}. \tag{19.20}$$

We now make the limiting convention needed for the nonlinear variation norms explicit. No infinite Fourier series is inserted directly into a V^2 or V^1 norm. For $N\in\mathbb{N}$, whenever the factor at a low position r is expanded, first retain only

$$\mathfrak{F}_N = \{(u_r,v_r)\in\mathbb{Z}^n\times\mathbb{Z}^n : |u_r|+|v_r|\le N\}.$$

For a set S of expanded positions, put

$$\omega_N(S) = \sum_{\mathbf{u},\mathbf{v}:\ \max_{r\in S}(|u_r|+|v_r|)>N} A(\mathbf{u},\mathbf{v})(1+|V(\mathbf{v})|), \qquad \omega_N(\varnothing) = 0. \tag{19.21}$$

The weighted ℓ^1 estimate (19.19) implies $\omega_N(S)\to 0$. All component estimates below are first applied to these finite mode sums; their passage to the full Fourier expansions is carried out at the end of the proof.

Suppose first that μ is the first high coordinate. After fixing a Fourier mode from the preceding low factors, the remaining common operator phase is $\sum_{r\ge\mu} C_r\widehat{H}_r(X,Y)$; its first flag coordinate is C_μ. The factor obtained by evaluating the remaining phase at $X=\xi_Q$ is a unimodular input multiplier on the descendants of Q. After division by $A(\mathbf{u},\mathbf{v})$, the parent-dependent coefficients have modulus at most one. Absorb them and the preceding unimodular factor into input multipliers $\vartheta_{Q,\mathbf{u},\mathbf{v}}$, including the common input modulation $e^{i\omega_F V(\mathbf{v})\cdot Y}$. Disjointness of the descendants and (19.12) give, in scaled variables,

$$\left\|\sum_{\substack{Q\in\mathscr{Q},\ Q\subset J\\ U\subset Q}} \vartheta_{Q,\mathbf{u},\mathbf{v}}(Y)\widetilde{d}_U(Y)\right\|_2^2 \le \sum_{\substack{Q\in\mathscr{Q},\ Q\subset J\\ U\subset Q}} \|\widetilde{d}_U\|_\infty\|\widetilde{d}_U\|_1 \le \Lambda R^{-n}M_{I,J}. \tag{19.22}$$

Thus Corollary 16.8 applies to one common operator phase. The fixed output modulation $e^{i\omega_F U(\mathbf{u})\cdot X}$ does not change variation. Summing the majorants produces exactly the loss in (19.20), and (19.13) restores physical scale. The scaled truncation interval is $1\le\varepsilon/R\le 2$, so it is the partial annulus of radius 2 in Corollary 16.8. Its exact scale parameter is therefore

$$L_\mu = 2^{m_\mu}|C_\mu| \asymp_{n,d} |C_\mu|,$$

not merely $|C_\mu|$. Consequently we obtain

$$C\|\Theta\|_\infty(2^{m_\mu}|C_\mu|)^{-\beta_\mu}(\Lambda M_{I,J})^{1/2} \le C\|\Theta\|_\infty 2^{-\beta_*\kappa_\mu t}(\Lambda M_{I,J})^{1/2}$$

for the entire short trajectory. Multiplication by the loss (19.20) and (19.18) leave decay at least $2^{-\beta_*\kappa_\mu t/2}$. Equations (19.10) and (19.9) prove (19.4) in every high branch, with $E=0$.

If every coordinate is low, expand all factors. For a combined tuple and a descendant U, let ζ_U be the center of $\widetilde{U}$. Write

$$e^{i\omega_F V(\mathbf{v})\cdot Y}\widetilde{d}_U(Y) = e^{i\omega_F V(\mathbf{v})\cdot\zeta_U}\widetilde{d}_U(Y) + \widetilde{r}_{U,\mathbf{v}}(Y), \qquad \|\widetilde{r}_{U,\mathbf{v}}\|_1 \le C\omega_F|V(\mathbf{v})|2^{-t}\|\widetilde{d}_U\|_1. \tag{19.23}$$

The first term is still mean zero. For each fixed mode, its coefficients are fixed before evaluation in x. More precisely, divide the parent-dependent coefficient by $A(\mathbf{u},\mathbf{v})$; the resulting packet coefficients have modulus at most one, so Proposition 18.5 applies to the scaled unit-annulus packets. Equation (19.19) makes the summation uniform in every parent label ξ_Q. The total Fourier loss is at most $2^{C_F N_d \varepsilon_* t}$. Proposition 18.5 gives $2^{-\delta_T t/2}$ at the (unsquared) L^2 level, so (19.17) leaves at least $2^{-3\delta_T t/8}$ after the Fourier summation. The remainders in (19.23) are controlled by total radial variation and Young's inequality. Using (19.19) and (19.12), their total L^1 variation in scaled variables is

$$C\|\Theta\|_1 2^{-t} 2^{C_F N_d \varepsilon_* t} R^{-n} M_{I,J}.$$

After (19.13) and (19.17), this is at most $C\|\Theta\|_1 2^{-3t/4} M_{I,J}$. The good term similarly returns with the factor $(\Lambda M_{I,J})^{1/2}$. Apply (19.10) to the squared L^2 bounds and (19.11) to the L^1 errors, followed by (19.9). This proves (19.4)–(19.5).

The exponents can now be fixed uniformly rather than left implicit. Take

$$c_P = \min\left\{ \min_{1\le\mu\le N_d} \frac{\beta_*\kappa_\mu}{2}, \frac{3\delta_T}{8} \right\}, \qquad c_E = \frac{3}{4}. \tag{19.24}$$

The first term in c_P covers every high branch, the second covers the all-low branch, and c_E is permitted because $1 - C_F N_d \varepsilon_* > 7/8$. All quantities in (19.24), as well as the single system of Fourier cutoffs fixed above, depend only on n,d. In the already settled $N_d = 0$ terminal case one takes $c_P = 3\delta_T/8$ and $c_E = 3/4$; its actual good decay $\delta_T/2$ is stronger.

It remains to remove the finite-mode truncation. Let $\mathscr{P}_N$ be the trajectory in which every low Fourier expansion is restricted to $\mathfrak{F}_N$, and let $\mathscr{P}_N = G_N + E_N$ be the decomposition just constructed, summed over $(I,J) \in \mathscr{A}_R$. These are genuine finite sums. If a block has first high position μ and $S = \{r : r < \mu\}$, then for $N' > N$ the same component estimate used above gives

$$\|G^{I,J}_{N'} - G^{I,J}_N\|_{L^2(V^2)} \le C\|\Theta\|_\infty (2^{m_\mu}|C_\mu|)^{-\beta_\mu} (\Lambda M_{I,J})^{1/2} \omega_N(S). \tag{19.25}$$

Here and below the superscript I,J includes the fixed output restriction $\mathbf{1}_I$. If every position is low, the terminal-packet estimate and the remainder estimate in (19.23) give instead

$$\|G^{I,J}_{N'} - G^{I,J}_N\|_{L^2(V^2)} \le C\|\Theta\|_\infty 2^{-\delta_T t/2} (\Lambda M_{I,J})^{1/2} \omega_N(S), \tag{19.26}$$

$$\|E^{I,J}_{N'} - E^{I,J}_N\|_{L^1(V^1)} \le C\|\Theta\|_1 2^{-t} M_{I,J} \omega_N(S). \tag{19.27}$$

The harmless Fourier losses already displayed in (19.20) and (19.19) may be retained on the right sides; for the fixed block they do not depend on N, while $\omega_N(S) \to 0$. There are only finitely many interacting blocks. Therefore (19.10), (19.11), and (19.25)–(19.27) show that (G_N) is Cauchy in $L^2(V^2)$ and (E_N) is Cauchy in $L^1(V^1)$. Every component path is anchored at the upper shell endpoint: $G_N(2R) = E_N(2R) = 0$. Hence

$$\sup_{R\le\varepsilon\le 2R} |H_\varepsilon| \le V^a(H_\varepsilon : R \le \varepsilon \le 2R), \qquad a \in \{1,2\},$$

for every component or difference path H in the corresponding variation space. Lemma 18.3 therefore constructs anchored paths G,E and gives convergence of the full sequences in the displayed variation norms, with no uncountable-parameter Bochner-space assumption.

On the other hand, the Weierstrass majorant in (19.19) shows that the Fourier products defining $\mathscr{P}^{I,J}_N$ converge absolutely and locally uniformly in (X,Y,ξ_Q) to the original phase factor. Since the packets are finite, bounded, and supported in the fixed block, a constant multiple of the integrable majorant

$$\mathbf{1}_I(x)\left(K_{|\Theta|}\mathbf{1}_{\{R<|z|\le 2R\}} * \sum_{\substack{Q\in\mathscr{Q},\ Q\subset J\\ U\subset Q}} |d_U| \right)(x)$$

is independent of N and of ε, where $K_{|\Theta|}(z) = |\Theta(z/|z|)|/|z|^n$. Indeed, disjointness and (19.1) give $\sum_U |d_U| \le \Lambda$ almost everywhere, so the displayed function is bounded by $C\Lambda\|\Theta\|_1 \mathbf{1}_I$ and belongs to both L^1 and L^2. The same majorant gives convergence uniformly in ε for almost every x (and in L^1+L^2 on the fixed output block). The last assertion of Lemma 18.3 therefore identifies the anchored trajectory limit with the original one: $\mathscr{P} = G+E$. Its Fatou conclusion gives

$$\|V^2(G)\|_2 \le \liminf_{N\to\infty} \|V^2(G_N)\|_2, \qquad \|V^1(E)\|_1 \le \liminf_{N\to\infty} \|V^1(E_N)\|_1.$$

Thus the uniform finite-mode bounds pass to G, E with no change in any decay exponent, completing the proof. □

The proof is a simultaneous phase–packet induction. At every later phase, regardless of the sizes of its coefficients, one either reaches a high coordinate and obtains scale-covariant oscillatory decay, or expands a low coordinate and proceeds. The finite hierarchy (19.18) ensures that Fourier losses accumulated before a later high coordinate are strictly smaller than its oscillatory gain. Translation-generated coefficients enter as unrestricted later coordinates in (19.15), and the flag hierarchy supplies their control.

20 Concentration-stable degree-block packet estimates

The original phase-adapted Calderón–Zygmund blocks do not satisfy the pointwise packet-density hypothesis of Proposition 19.2. This section replaces that hypothesis by estimates using only cancellation and an average L^1 mass bound. Throughout, coefficient norms are fixed Euclidean norms on the relevant bounded-degree polynomial spaces.

20.1 Polynomial incidence and rational finite type

Lemma 20.1 (Multivariable sublevels and discrete smearing)**.** *Let $I \subset \mathbb{R}^N$ be a fixed compact box with nonempty interior, and let $\mathbf{p}$ be a nonzero vector of polynomials of degree at most D on I. Put $A = \|\mathbf{p}\|_{\mathrm{coeff}}$. There are $c, C > 0$, depending only on I, N, D and the vector length, such that*

$$|\{x \in I : |\mathbf{p}(x)| \le \tau A\}| \le C\tau^c, \tag{20.1}$$

$$\int_I \min\left\{1, \left(\frac{L|\mathbf{p}(x)|}{A}\right)^{-\sigma}\right\} dx \le CL^{-\delta_\sigma}, \qquad \delta_\sigma := \frac{c\sigma}{c+\sigma} > 0, \tag{20.2}$$

for $0 < \tau \le 1$, $L \ge 1$, and $\sigma > 0$. At a zero of $\mathbf{p}$, the integrand in (20.2) *is defined to be* 1.

Let $0 < q \le 1$, let $\mathscr{Q}$ be a family of pairwise disjoint q-cubes in I, choose $y_Q \in Q$ for every $Q \in \mathscr{Q}$, and attach weights $0 \le w_Q \le \Lambda q^N$. If $M = \sum_{Q\in\mathscr{Q}} w_Q$, then every polynomial p on $\mathbb{R}^N \times \mathbb{R}^N$ of degree at most D and coefficient norm one satisfies, for $0 < \tau \le 1$,

$$\sum_{Q,Q'\in\mathscr{Q}} w_Q w_{Q'} \mathbf{1}_{\{|p(y_Q, y_{Q'})| \le \tau\}} \le C\Lambda M(\tau+q)^c. \tag{20.3}$$

There is also the following chart version. Let $U_\nu \Subset U_\nu^* \subset \mathbb{R}^N$ be fixed boxes and let $\chi_\nu : U_\nu^* \to \mathscr{M}_\nu^*$ be a fixed finite family of rational charts. Assume, uniformly in ν, that

$$c|z-z'| \le |\chi_\nu(z) - \chi_\nu(z')| \le C|z-z'|, \qquad c \le J_\nu(z) \le C, \tag{20.4}$$

where J_ν is the density of the induced measure under χ_ν. Thus, for a coordinate q-cube $Q^\flat \subset U_\nu$ and its image $Q = \chi_\nu(Q^\flat)$,

$$cq^N \le |Q|_{\mathscr{M}_\nu} = \int_{Q^\flat} J_\nu(z)\, dz \le Cq^N. \tag{20.5}$$

The chart pair version permits selected points in such chart cubes and weights $w_Q \le \Lambda|Q|_{\mathscr{M}_\nu}$; by (20.5) *this is, up to a fixed constant, the same cap as $w_Q \le \Lambda q^N$. It also permits pairwise interior-disjoint physical q-cubes in $\mathbb{R}^m$, with selected points on the chart images and upper weight cap $0 \le w_Q \le C\Lambda q^N$; the ambient dimension m need not equal N. Assign each selected point to one chart. A fixed proximity coloring with at most 4^m colors in each chart makes the selected coordinate centers cq-separated. They therefore admit disjoint coordinate smearing cubes of side $c'q$ in U_ν^*, with c' also chosen below the fixed chart collar. Only the upper weight cap is used. It follows, in particular, from $w_Q \le \Lambda|Q\cap\chi_\nu(U_\nu)|_{\mathscr{M}_\nu}$, since* (20.4) *gives $|Q\cap\chi_\nu(U_\nu)|_{\mathscr{M}_\nu} \le Cq^N$. No lower volume bound for the inverse image of an arbitrary physical cube is asserted.*

In a given application let $\mathscr{V}_\nu$ be a specified fixed finite-dimensional space of polynomial vectors on $\mathscr{M}_\nu$, and let $\mathscr{V}_{\nu,\nu'}^{(2)}$ be the specified fixed finite-dimensional pair-polynomial space on $\mathscr{M}_\nu \times \mathscr{M}_{\nu'}$. Choose a fixed common denominator $\mathscr{D}_\nu$ which clears every pullback of an element of $\mathscr{V}_\nu$ and is bounded above and away from zero on U_ν^*, and assume explicitly that

$$T_\nu : \mathscr{V}_\nu \longrightarrow \mathscr{P}_{\le D_\nu}(U_\nu), \qquad T_\nu \mathbf{p} = \mathscr{D}_\nu\, \mathbf{p}\circ\chi_\nu, \quad \textit{is injective}. \tag{20.6}$$

Then (20.1)–(20.2) *hold after pullback to these charts. The pair estimate* (20.3) *also holds provided the analogous cleared pullback on $\mathscr{V}_{\nu,\nu'}^{(2)}$ is injective. The constants may depend on the finitely many least singular values in* (20.6)*, in addition to the fixed chart data. No injectivity assertion is made for all ambient polynomials restricted to a lower-dimensional algebraic chart.*

Proof. We first prove (20.1). A scalar polynomial of degree at most D on a fixed interval satisfies a uniform sublevel estimate. Indeed, when a sublevel set E has positive measure, one may choose $D+1$ of its points with consecutive separations bounded below by a constant multiple of $|E|$; Lagrange interpolation then bounds the coefficient norm by $C\tau|E|^{-D}$. The case $D=0$ is immediate, so assume henceforth that $D \ge 1$. For several variables, write $p(x',t) = \sum_{j=0}^{D} a_j(x')t^j$ and choose a_j with coefficient norm bounded below. If the $(N-1)$-variable exponent already obtained is c_{N-1}, split the base according to $|a_j(x')| \le s$. The induction hypothesis bounds the first part by $Cs^{c_{N-1}}$, while on the complement the one-dimensional estimate bounds every fibre by $C(\tau/s)^{1/D}$. Taking $s = \tau^{1/(1+Dc_{N-1})}$ gives a positive exponent depending only on N and D, and proves (20.1). For a polynomial vector, one component has coefficient norm at least a fixed multiple of A, and normalization of that component completes the proof. Choose c in the statement no larger than this exponent and no larger than one half of the corresponding scalar sublevel exponent in dimension $2N$; for the fixed finite chart family, take the minimum of the finitely many resulting exponents.

For (20.2), set $\theta = \sigma/(c+\sigma)$ and

$$E_L = \{x \in I : |\mathbf{p}(x)| \le AL^{-\theta}\}.$$

Equation (20.1) gives $|E_L| \le CL^{-c\theta} = CL^{-\delta_\sigma}$. On $I \setminus E_L$ the integrand is at most $L^{-\sigma(1-\theta)} = L^{-\delta_\sigma}$. This proves (20.2), with a constant uniform in σ.

For (20.3), write a_Q for the lower corner of Q and color it by $(\lfloor a_{Q,k}/q \rfloor \bmod 4)_{k=1}^N$. Distinct cubes of the same color have selected points at sup-norm distance at least q, even when the points lie on closed faces. Indeed, distance less than q would force the lower-corner floor indices to differ by less than 3 in every coordinate. Congruence modulo 4 then makes all these indices equal, forcing the cube interiors to intersect. Spread each weight uniformly on a coordinate cube of side $q/2$ centered at its selected point. Within one color these smearing cubes are disjoint. The resulting density on a fixed enlargement of I is bounded by $C\Lambda$ and has integral at most M. A normalized bounded-degree polynomial has bounded gradient on a fixed enlargement of I^2, so every bad selected pair smears into the $C(\tau+q)$-sublevel. For any ordered pair of colors, if F, G are the two smeared densities and E is this sublevel, then

$$\iint_E F(x)G(y)\,dx\,dy \le \min\{M^2, C\Lambda^2|E|\} \le C\Lambda M|E|^{1/2}.$$

If $\tau + q$ is bounded below by a fixed constant, the desired estimate follows directly from $M \le C_I\Lambda$. Otherwise apply the scalar sublevel estimate just proved, now in $2N$ variables, to E and sum the finite color pairs. The choice of c made above proves (20.3).

For the chart assertion, fix one of the specified rational charts and the application-specific space $\mathscr{V}_\nu$. The denominator and its reciprocal are uniformly bounded on the enlarged box. The required injectivity is precisely hypothesis (20.6); it is not inferred merely from vanishing on a possibly lower-dimensional chart image. The least singular value of this fixed map and finite-dimensional norm equivalence give

$$c_\nu\|\mathbf{p}\|_{\mathrm{coeff}} \le \|T_\nu\mathbf{p}\|_{\mathrm{coeff}} \le C_\nu\|\mathbf{p}\|_{\mathrm{coeff}}, \qquad |T_\nu\mathbf{p}(\zeta)| \simeq |\mathbf{p}(\chi_\nu(\zeta))|.$$

The change-of-variables formula and (20.4) give, for every measurable $E \subset \mathscr{M}_\nu$ contained in the chart,

$$c|\chi_\nu^{-1}(E)| \le |E|_{\mathscr{M}_\nu} = \int_{\chi_\nu^{-1}(E)} J_\nu(z)\,dz \le C|\chi_\nu^{-1}(E)|.$$

Consequently (20.1) and (20.2) transfer to the chart. For (20.3), use the product chart and the explicitly assumed injectivity of its cleared pullback on $\mathscr{V}^{(2)}_{\nu,\nu'}$. In coordinate variables a chart cube has mass cap $C\Lambda q^N$ by (20.5), for any measurable choice of its open, closed or half-open faces. For a physical cube Q, the set $E_Q = \chi_\nu^{-1}(Q \cap \chi_\nu(U_\nu))$ may instead be a sliver or even empty. Its coordinate diameter is at most Cq by the inverse Lipschitz bound, so it lies in a coordinate cube of side Cq. The upper Jacobian bound gives only the estimate needed here:

$$|Q \cap \chi_\nu(U_\nu)|_{\mathscr{M}_\nu} = \int_{E_Q} J_\nu(z)\,dz \le Cq^N.$$

This converts an induced-volume upper weight cap into $C\Lambda q^N$ without any lower bound for $|E_Q|$.

For physical cubes, use the same lower-corner coloring modulo 4, now in the m ambient coordinates, separately on each assigned chart. The forward Lipschitz bound in (20.4) converts the resulting physical separation into coordinate separation at least cq. Choose disjoint coordinate smearing cubes of side $c'q$ centered at these coordinate points. Since $q \le 1$, a fixed decrease of c' puts them inside U_ν^*; no extension of a rational chart across a denominator zero is involved. Spreading w_Q uniformly on its coordinate smearing cube produces density at most

$$C\frac{\Lambda q^N}{q^N} = C\Lambda.$$

The normalized cleared polynomial has bounded coordinate gradient on the fixed enlarged product box. The bounded denominator and the coefficient coercivity above show that a bad selected pair, and then the product of its smearing cubes, lie in the $C(\tau+q)$-sublevel. The preceding coordinate proof of (20.3) therefore applies with the fixed chart, Jacobian and injectivity constants, independently of whether $m = N$. Taking the minimum over the finite chart family proves all asserted chart versions. □

We first fix the rational quotient and its norm. This is needed because all later rational phases have parameters which may approach degenerate values, and a qualitative assertion that a phase is not separated does not by itself give a uniform oscillatory constant.

Fix compact intervals I, J and slightly larger intervals I^*, J^*. For fixed $D_0 \in \mathbb{N}$ and $0 < \eta < 1 < M$, call

$$\Psi(u,v) = \frac{N(u,v)}{Q(u,v)} \tag{20.7}$$

an admissible rational presentation if N, Q are real polynomials of degree at most D_0,

$$\|Q\|_{\mathrm{coeff}} \le M, \qquad \eta \le |Q(u,v)| \le M \quad ((u,v) \in I^* \times J^*). \tag{20.8}$$

The denominator in (20.7) is part of the presentation. In every application below it is the displayed common denominator produced by the coordinate change.

For an admissible presentation define the polynomial

$$\mathfrak{C}_Q\Psi(u,u',v) = \{Q(u,v)Q(u',v)\}^2\partial_v\{\Psi(u,v)-\Psi(u',v)\} \tag{20.9}$$

and set

$$\|[\Psi]\|_{\mathrm{rat},Q} := \|\mathfrak{C}_Q\Psi\|_{\mathrm{coeff},(u,u',v)}. \tag{20.10}$$

This definition is invariant under addition of $F(u)+G(v)$. Conversely, if (20.9) vanishes, then $\partial_v\Psi(u,v)$ is independent of u, and integration on the rectangle gives $\Psi(u,v)=F(u)+G(v)$. Thus (20.10) is a genuine norm on the quotient by separated phases. If the same rational function is written using two admissible common denominators of bounded degree, the resulting norms are uniformly equivalent. Indeed, if the denominators are Q_1,Q_2, then pointwise on the fixed enlarged box

$$\mathfrak{C}_{Q_2}\Psi(u,u',v) = \left\{\frac{Q_2(u,v)Q_2(u',v)}{Q_1(u,v)Q_1(u',v)}\right\}^2\mathfrak{C}_{Q_1}\Psi(u,u',v). \tag{20.11}$$

The factor in braces and its reciprocal are bounded by constants depending only on η,M. All difference numerators belong to one fixed bounded-degree polynomial space, on which coefficient and supremum norms are uniformly equivalent. Consequently

$$\|\mathfrak{C}_{Q_2}\Psi\|_{\mathrm{coeff}} \gtrsim \|\mathfrak{C}_{Q_2}\Psi\|_\infty \gtrsim \|\mathfrak{C}_{Q_1}\Psi\|_\infty \gtrsim \|\mathfrak{C}_{Q_1}\Psi\|_{\mathrm{coeff}}.$$

Interchanging Q_1,Q_2 gives the reverse inequality and proves the claimed uniform equivalence.

We will also use the mixed numerator

$$\mathfrak{M}_Q\Psi(u,v) = Q(u,v)^3\partial_u\partial_v\Psi(u,v). \tag{20.12}$$

It is a polynomial of uniformly bounded degree and satisfies

$$\|[\Psi]\|_{\mathrm{rat},Q} \geq c\|\mathfrak{M}_Q\Psi\|_{\mathrm{coeff}}. \tag{20.13}$$

To verify this, differentiate (20.9) in u' and then put $u'=u$. Since the phase difference and its v derivative vanish on the diagonal, derivatives falling on the prefactor make no contribution, and one obtains exactly

$$\partial_{u'}\mathfrak{C}_Q\Psi(u,u',v)\big|_{u'=u} = -Q(u,v)^4\partial_u\partial_v\Psi(u,v) = -Q(u,v)\mathfrak{M}_Q\Psi(u,v).$$

Differentiation and diagonal restriction are bounded maps on coefficient spaces. Moreover $\|QH\|_{\mathrm{coeff}} \geq c\|H\|_{\mathrm{coeff}}$ for every polynomial H of the relevant bounded degree: this follows from $\|QH\|_{L^\infty(I^*\times J^*)} \geq \eta\|H\|_{L^\infty(I^*\times J^*)}$ and finite-dimensional norm equivalence. This proves (20.13).

Lemma 20.2 (Rational van der Corput with a BV amplitude)**.** *Let $\psi = N/q$ on a fixed compact interval, where N,q have degree at most D_0, $\|q\|_{\mathrm{coeff}} \leq M$, and $\eta \leq |q| \leq M$ on a fixed enlargement. Put*

$$G = \|q^2\psi'\|_{\mathrm{coeff}} = \|N'q - Nq'\|_{\mathrm{coeff}}.$$

If a is a zero-extended bounded-variation function supported in the interval, then

$$\left|\int e^{i\psi(v)}a(v)\,dv\right| \leq C(\|a\|_\infty + \mathrm{Var}(a))\min\{1,G^{-c}\}. \tag{20.14}$$

When $G=0$, the minimum on the right is understood to be 1*. The same assertion holds after a bounded partition into intervals, with the sum of the sectional $L^\infty+\mathrm{BV}$ norms on the right. The constants and the positive exponent depend only on D_0,η,M and the fixed interval and its enlargement.*

Proof. Only $G > 1$ requires proof; the case $G \le 1$ is the trivial L^1 bound. Write $p = q^2\psi'$. By Lemma 20.1, applied to p/G, the set

$$E = \{v : |p(v)| \le G^{1/2}\}$$

has measure at most CG^{-c_0}. Its contribution is bounded by $C\|a\|_\infty G^{-c_0}$. On the complement, $|\psi'| \ge cG^{1/2}$. The numerator of ψ'' has bounded degree. Apart from the two endpoints of the ambient interval, the boundary of E is contained in the union of the zero sets of the two bounded-degree polynomials $p - G^{1/2}$ and $p + G^{1/2}$; hence E has only $O_{D_0}(1)$ components. Partition its complement at those boundary points and at the zeros of the numerator of ψ''; the number of intervals is therefore bounded in terms of D_0. On each remaining interval ψ' is monotone and bounded away from zero. The first derivative van der Corput estimate, proved for a BV amplitude by one Stieltjes integration by parts, gives $C(\|a\|_\infty + \mathrm{Var}(a))G^{-1/2}$. Summing the boundedly many pieces and decreasing the exponent proves (20.14). If the numerator of ψ'' vanishes identically, ψ' is already monotone and the same argument applies without this partition. □

Lemma 20.3 (Coefficient-norm polynomial van der Corput)**.** *Let I be a fixed compact interval, let $D \ge 1$, let p be a real polynomial of degree at most D, and put*

$$\Lambda_p = \|p - p(0)\|_{\mathrm{coeff}}.$$

If a is supported in I and is of bounded variation, then

$$\left| \int_I e^{ip(t)} a(t)\, dt \right| \le C_D \{\|a\|_\infty + \mathrm{Var}(a)\} \min\{1, \Lambda_p^{-c_D}\}. \tag{20.15}$$

The displayed norm uses the canonical representative with zero constant coefficient. Equivalently, Λ_p may be any fixed coefficient norm on the space of real polynomials of degree at most D, modulo constants. Changing p by an additive constant leaves this quotient class and its norm unchanged. When $\Lambda_p = 0$, the minimum in (20.15) *is understood to be* 1.

Proof. Apply Lemma 20.2 with denominator identically one. Differentiation is injective from the finite-dimensional quotient of real polynomials of degree at most D by the constants into the space of polynomials of degree at most $D-1$, and therefore $\|p'\|_{\mathrm{coeff}} \ge c_D \Lambda_p$ by finite-dimensional norm equivalence. Decreasing the exponent in (20.14) proves (20.15). □

Lemma 20.4 (Uniform rational oscillatory operator)**.** *Let I, J be fixed compact intervals, let $A : I \times J \to \mathbb{C}$ be measurable, and define $S : L^2(J) \to L^2(I)$ initially on bounded functions by*

$$Sf(u) = \int_J e^{i\Psi(u,v)} A(u,v) f(v)\, dv.$$

Assume that (20.7)– (20.8) *hold with fixed data. Suppose $B \ge 1$ and*

$$\sup_{u \in I} \int_J |A(u,v)|\, dv \le B, \qquad \sup_{v \in J} \int_I |A(u,v)|\, du \le B.$$

Assume also that, for every $u \in I$, there is a partition of J into at most M_0 intervals on each of which

$$\|A(u,\cdot)\|_\infty + \mathrm{Var}(A(u,\cdot)) \le B.$$

The endpoints may depend on u. Put $L = \|[\Psi]\|_{\mathrm{rat},Q}$. Then

$$\|S\|_{2\to 2} \le C B^C \min\{1, L^{-c}\}. \tag{20.16}$$

When $L = 0$, the minimum is again understood to be 1. *The constants depend only on the fixed intervals, D_0, η, the admissibility bound M, and M_0.*

Proof. The trivial part follows from Schur's test, so assume $L \geq 1$. The kernel of SS^* is

$$K(u,u') = \int_J e^{i[\Psi(u,v)-\Psi(u',v)]} A(u,v)\overline{A(u',v)}\,dv.$$

For fixed (u,u'), an admissible common denominator of the phase difference is $q_{u,u'}(v) = Q(u,v)Q(u',v)$; its degree, coefficient norm, and upper and lower bounds are controlled by the fixed admissibility data. Its derivative numerator is precisely (20.9). If

$$G(u,u') = \|\mathfrak{C}_Q \Psi(u,u',\cdot)\|_{\mathrm{coeff},v},$$

then refine the two partitions of J belonging to u and u'. There are at most $2M_0$ resulting intervals, and

$$\|A(u,\cdot)\overline{A(u',\cdot)}\|_{L^\infty+\mathrm{BV}} \leq CB^2$$

on each of them. Lemma 20.2, summed over this partition, gives

$$|K(u,u')| \leq CB^2 \min\{1, G(u,u')^{-c_0}\}. \tag{20.17}$$

The entries of the coefficient vector defining G are polynomials in (u,u') of uniformly bounded degree, and their joint coefficient norm is exactly L. Consequently (20.2), applied on $I \times I$, yields

$$\iint_{I\times I} |K(u,u')|^2\,du\,du' \leq CB^C L^{-c_1}.$$

There is no deletion of the diagonal or of any algebraically degenerate branch: all such points are included in this negative-moment integral. Finally,

$$\|S\|_{2\to 2}^2 = \|SS^*\|_{2\to 2} \leq \|K\|_{L^2(I\times I)}.$$

Taking the two square roots and decreasing c_1 proves (20.16). □

20.2 Rough two-ray correlation and a flat high block

Lemma 20.5 (Universal rough polynomial correlation)**.** *Let $n \geq 1$, let $D \geq 1$, let $K = [r_-, r_+] \Subset (0,\infty)$, and let a,b be radial* BV *profiles supported in K with $L^\infty + \mathrm{BV}$ norm at most $B \geq 1$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$ and put*

$$k_a(z) = |z|^{-n}\Theta(z/|z|)a(|z|), \qquad k_b(z) = |z|^{-n}\Theta(z/|z|)b(|z|).$$

Let S be a real polynomial of degree at most D, and let Λ be the coefficient norm of its nonconstant part. Then, for every $h \in \mathbb{R}^n$, if $n \geq 2$,

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h)\,dz \right| \leq CB^C \|\Theta\|_\infty^2 \min\{1, (\Lambda|h|^D)^{-c}\}. \tag{20.18}$$

For $n \geq 2$ the trivial bound is understood when $h = 0$ or $\Lambda = 0$. In dimension one, for every h, the precise unit-scale analogue is

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h)\,dz \right| \leq CB^C \|\Theta\|_\infty^2 \min\{1, \Lambda^{-c}\}.$$

Lemma 20.6 (Quantitative equal-shell pullback)**.** *Let $D \geq 1$ and let $\phi(t,r) = \sum_{m=0}^D \phi_m(t,r)$, where ϕ_m is homogeneous of degree m, and put*

$$\Gamma = \|\phi - \phi_0\|_{\mathrm{coeff}}.$$

Let $0 < \alpha \leq 1$, $0 < H \leq C_0$, $|U_0| \leq C_0$, let $\Delta \in \mathbb{R}$, and let $q(X,Y) = \Delta + X - Y$ satisfy $c_0 \leq |q| \leq C_0$ on a fixed enlargement of a fixed rectangle. Define

$$\Psi(X,Y) = \phi\left(\frac{U_0 + HX}{q(X,Y)}, \frac{\alpha}{q(X,Y)}\right) = \sum_{m=0}^D \frac{\phi_m(U_0+HX,\alpha)}{q(X,Y)^m}. \tag{20.19}$$

Use the canonical common denominator $Q = q^D$. Then

$$\|[\Psi]\|_{\mathrm{rat},Q} \geq c(\alpha H)^D \Gamma. \tag{20.20}$$

The constant depends only on D, c_0, C_0 and the fixed rectangles.

Proof of Lemma 20.6. Put $A(X) = U_0 + HX$ and use (X,q) rather than (X,Y) as polynomial coordinates. Since X,Y range in fixed rectangles and $c_0 \leq |q| \leq C_0$ on their enlargement, Δ is uniformly bounded. Thus $(X,Y) \mapsto (X,q)$ and its inverse have uniformly bounded coefficient matrices on every bounded-degree polynomial space. For $m \geq 1$, direct differentiation gives

$$q^{D+2}\partial_X\partial_Y\left\{\phi_m(A(X),\alpha)q^{-m}\right\} = q^{D-m}\left\{mHq\,\partial_t\phi_m(A(X),\alpha) - m(m+1)\phi_m(A(X),\alpha)\right\}. \tag{20.21}$$

This identity also identifies all possible cancellation between adjacent homogeneous degrees. After summing (20.21), the coefficient of q^{D-m} is

$$-m(m+1)\phi_m(A,\alpha) + (m+1)H\partial_t\phi_{m+1}(A,\alpha), \qquad \phi_{D+1} = 0. \tag{20.22}$$

Starting with $m = D$ and descending, the triangular system (20.22) recovers every $\phi_m(A,\alpha)$, $1 \leq m \leq D$. More explicitly, on the finite direct sum $\bigoplus_{m=1}^{D} \mathscr{P}_{\leq m}(X)$ the diagonal maps are the nonzero scalars $-m(m+1)$ and the only off-diagonal maps are $(m+1)H\partial_t$ from level $m+1$ to level m. Descending substitution and $0 < H \leq C_0$ therefore give

$$\sum_{m=1}^{D} \|\phi_m(A,\alpha)\|_{\mathrm{coeff},X} \leq C_{D,C_0}\|q^{D+2}\partial_X\partial_Y\Psi\|_{\mathrm{coeff},(X,q)}.$$

The uniformly invertible affine coefficient change above makes the last norm equivalent to its (X,Y) coefficient norm. Hence

$$\|q^{D+2}\partial_X\partial_Y\Psi\|_{\mathrm{coeff},(X,Y)} \geq c\left(\sum_{m=1}^{D}\|\phi_m(U_0+HX,\alpha)\|_{\mathrm{coeff},X}^2\right)^{1/2}. \tag{20.23}$$

In particular, no choice of a largest homogeneous term is being made: all homogeneous degrees are recovered simultaneously.

The affine map $t = U_0 + HX$ and evaluation at $r = \alpha$ are quantitatively injective on homogeneous two-variable polynomials. Indeed, recover $\phi_m(t,\alpha)$ from its composition with $U_0 + HX$ and then recover the coefficient of $t^j r^{m-j}$ by division by α^{m-j}. Since $|U_0| \leq C_0$, $0 < \alpha \leq 1$, and $0 < H \leq C_0$, finite-dimensional norm equivalence gives

$$\|\phi_m(U_0+HX,\alpha)\|_{\mathrm{coeff},X} \geq c(\alpha H)^D\|\phi_m\|_{\mathrm{coeff}}. \tag{20.24}$$

Indeed, the inverse of the affine substitution $t = U_0 + HX$ has norm $O_{m,C_0}(H^{-m})$, while evaluation at $r = \alpha$ multiplies the coefficient of $t^j r^{m-j}$ by α^{m-j}. The left side is thus at least $cH^m\alpha^m\|\phi_m\|_{\mathrm{coeff}}$. If $\alpha H \leq 1$, this is at least the right side of (20.24); if $\alpha H > 1$, the bounded upper ranges of α and H allow the remaining factor to be absorbed into c. Combining (20.23) and (20.24) gives the desired lower bound for the reduced mixed numerator $q^{D+2}\Psi_{XY}$.

For the presentation with denominator $Q = q^D$, the mixed numerator $Q^3\Psi_{XY}$ differs from the reduced numerator by multiplication by q^{2D-2}. This factor is bounded above and away from zero on the enlarged rectangle, so their coefficient norms are uniformly equivalent. Equation (20.13) now proves (20.20). □

Proof of Lemma 20.5. Assume $n \geq 2$. Since the radial profiles are supported in a fixed compact subset of $(0,\infty)$, kernel overlap implies $0 < |h| \leq C$. The cases $h = 0$ and $\Lambda = 0$ are covered by the stated trivial bound. Put $H = |h|$ and choose $\mathscr{O} \in O(n)$ such that $h = H\mathscr{O}e_1$. Under the change of variables $z = \mathscr{O}\widetilde{z}$ replace

$$S(z) \quad \text{by} \quad \widetilde{S}(\widetilde{z}) = S(\mathscr{O}\widetilde{z}), \qquad \Theta(\xi) \quad \text{by} \quad \widetilde{\Theta}(\xi) = \Theta(\mathscr{O}\xi).$$

The radial profiles are unchanged, Lebesgue measure is preserved, and $\|\widetilde{\Theta}\|_\infty = \|\Theta\|_\infty$. On the finite-dimensional space of nonconstant polynomials of degree at most D, composition with $\mathscr{O}$ is invertible with inverse composition by $\mathscr{O}^T$. Compactness of $O(n)$ therefore gives, uniformly in $\mathscr{O}$,

$$c_{n,D}\Lambda \le \|\widetilde{S} - \widetilde{S}(0)\|_{\mathrm{coeff}} \le C_{n,D}\Lambda.$$

Thus the desired estimate is invariant, up to its allowed structural constant, under this rotation. We may rename the transformed quantities, assume $h = He_1$, and write

$$z = (t, \rho\omega), \quad u = t/\rho, \quad v = (t-H)/\rho, \quad \omega \in \mathbb{S}^{n-2}.$$

Then exactly

$$\rho = \frac{H}{u-v}, \qquad z = \frac{H(u,\omega)}{u-v}, \qquad z-h = \frac{H(v,\omega)}{u-v}. \tag{20.25}$$

The two rough factors are respectively functions of (u,ω) and (v,ω), and the Jacobian times the two homogeneous kernels is

$$H^{-n}|u-v|^{n-1}(1+u^2)^{-n/2}(1+v^2)^{-n/2}\,du\,dv\,d\omega. \tag{20.26}$$

Thus the rough angular factors are input and output L^2 functions and are never differentiated.

Choose once and for all a nonnegative $\beta \in C_c^\infty((c,C))$ such that, on the relevant positive range,

$$\sum_{k\in\mathbb{Z}} \beta(2^k\rho) = 1.$$

After combining the boundedly many largest pieces, write $\alpha = 2^{-k} \le 1$ and $\beta_\alpha(\rho) = \beta(\rho/\alpha)$. This is a genuine partition of unity on the overlap region, its supports have uniformly bounded overlap, and

$$\sup_\alpha \left\{ \|\beta_\alpha\|_\infty + \mathrm{Var}_\rho(\beta_\alpha) \right\} \le C. \tag{20.27}$$

Thus the transverse radius is decomposed into bands $\rho \simeq \alpha$, $0 < \alpha \le 1$. Put $U = \alpha u, V = \alpha v$. In these variables (20.26) becomes

$$H^{-n}\alpha^{n-1}|U-V|^{n-1}(\alpha^2+U^2)^{-n/2}(\alpha^2+V^2)^{-n/2}\,dU\,dV\,d\omega. \tag{20.28}$$

Annular support gives $|U-V| \simeq H$ in a fixed bounded set and also $\alpha^2+U^2 \simeq \alpha^2+V^2 \simeq 1$. Thus, after the factor α^{n-1} is removed, the kernel in (U,V) has size $O(H^{-1})$ and is supported in a strip of width $O(H)$.

Choose a fixed $\varepsilon_0 > 0$, smaller than one tenth of the lower constant in $|U-V| \simeq H$, and partition the U- and V-axes into half-open intervals of length $\varepsilon_0 H$. Each input interval interacts with only boundedly many output intervals and conversely. On an interacting pair write

$$U = U_0 + HX, \qquad V = V_0 + HY, \qquad \Delta = \frac{U_0 - V_0}{H};$$

then X, Y range over fixed intervals of length ε_0 and

$$q(X,Y) = \Delta + X - Y$$

is bounded above and away from zero on a fixed enlargement of their full product. This last assertion concerns the whole product rectangle, not only the support of the amplitude, and is the reason for the small fixed choice of ε_0.

The phase on this block is exactly

$$\Psi_{\omega,U_0,V_0,H,\alpha}(X,Y) = S\left(\frac{(U_0+HX)e_1 + \alpha\omega}{q(X,Y)}\right). \tag{20.29}$$

The factor $dV = H\,dY$ cancels the H^{-1} in the normalized kernel. The two angular factors are, up to the fixed sign of q,

$$\overline{\Theta\left(\frac{Ue_1+\alpha\omega}{\sqrt{U^2+\alpha^2}}\right)}, \qquad \Theta\left(\frac{Ve_1+\alpha\omega}{\sqrt{V^2+\alpha^2}}\right);$$

they are respectively functions of the output and input variables and are never differentiated. The radial factors are

$$\overline{a\left(\frac{\sqrt{U^2+\alpha^2}}{|q|}\right)}, \qquad b\left(\frac{\sqrt{V^2+\alpha^2}}{|q|}\right).$$

Here is the uniformity in this assertion. The sign of q is fixed on the enlarged block. Put $s_U = (U^2+\alpha^2)^{1/2}$ and $s_V = (V^2+\alpha^2)^{1/2}$. Up to nonzero factors of fixed sign, the four sectional derivatives of $s_U/|q|$ and $s_V/|q|$ are therefore controlled by those of s_U/q and s_V/q. Directly,

$$\partial_X \frac{s_U}{q} = \frac{HUq - s_U^2}{s_U q^2}, \qquad \partial_Y \frac{s_U}{q} = \frac{s_U}{q^2},$$
$$\partial_X \frac{s_V}{q} = -\frac{s_V}{q^2}, \qquad \partial_Y \frac{s_V}{q} = \frac{HVq + s_V^2}{s_V q^2}.$$

Thus a derivative can vanish only through

$$HUq - s_U^2 = 0 \quad \text{or} \quad HVq + s_V^2 = 0;$$

the other two derivatives never vanish. Since $U = U_0 + HX$, $V = V_0 + HY$, and $q = \Delta + X - Y$, the two displayed critical-point equations are polynomial equations of degree at most two. Their zero sets therefore split every section into $O(1)$ monotonicity intervals (with an identically vanishing derivative requiring no split). On an interval I on which a profile argument γ is monotone, the one-dimensional BV pullback inequality gives

$$\mathrm{Var}_I(a\circ\gamma) \le \mathrm{Var}_{\gamma(I)}(a) \le \mathrm{Var}(a),$$

and the same conclusion is immediate if γ is constant. Summing over the uniformly bounded number of intervals gives

$$\mathrm{Var}(a\circ\gamma) + \mathrm{Var}(b\circ\widetilde{\gamma}) \le C_D\{\mathrm{Var}(a) + \mathrm{Var}(b)\}.$$

To extend the remaining geometric factors to the whole rectangle, choose a fixed $\eta \in C_c^\infty((0,\infty))$ equal to one on the possible values of s_U and s_V on the radial support. These values lie in a fixed compact subinterval of $(0,\infty)$, because the profiles have fixed annular support and $|q|$ is bounded above and away from zero. Replace s_U^{-n} and s_V^{-n} in the geometric amplitude by $\eta(s_U)s_U^{-n}$ and $\eta(s_V)s_V^{-n}$. The integrand is unchanged, and the extended factors and their first sectional derivatives are uniformly bounded on the enlarged rectangle, including as $\alpha \downarrow 0$. Moreover, on the block $\rho/\alpha = H/(U-V) = 1/q(X,Y)$, so a smooth dyadic cutoff in ρ/α is the fixed function $\beta(1/q)$ of q. Since q is bounded above and away from zero on the enlarged rectangle, this cutoff and all of its sectional derivatives are uniformly bounded, consistently with (20.27). Thus the amplitude in each normalized block satisfies Lemma 20.4 with bound CB^C.

For completeness, the block synthesis has no hidden dependence on H. If T_{ij} is the operator between an interacting V-interval and U-interval and

$$(\mathscr{U}_i g)(X) = H^{1/2} g(U_i + HX), \qquad (\mathscr{V}_j f)(Y) = H^{1/2} f(V_j + HY),$$

then $\mathscr{U}_i T_{ij} \mathscr{V}_j^{-1}$ is precisely the normalized operator above: the factor $dV = H\,dY$ cancels the kernel factor H^{-1}. If d_{in} and d_{out} are the two degrees of the block interaction graph, Cauchy–Schwarz gives

$$\|T\|_{2\to 2} \le (d_{\mathrm{in}} d_{\mathrm{out}})^{1/2} \sup_{i,j} \|T_{ij}\|_{2\to 2}.$$

Both graph degrees are absolute constants. The same calculation with Schur's test proves the trivial band bound $CB^C\alpha^{n-1}$.

For fixed ω, put

$$\phi_\omega(t,r) = S(te_1 + r\omega), \qquad \Gamma(\omega) = \|\phi_\omega - \phi_\omega(0,0)\|_{\mathrm{coeff}}.$$

Formula (20.29) is (20.19). Lemma 20.6 gives, for its canonical denominator q^D,

$$\|[\Psi_{\omega,U_0,V_0,H,\alpha}]\|_{\mathrm{rat},q^D} \geq c\Gamma(\omega)(\alpha H)^D.$$

Lemma 20.4 gives the fixed-ω block gain, and the preceding almost-orthogonal synthesis preserves it. We now spell out the angular synthesis. Let $I_0 \subset \mathbb{R}$ be a fixed compact interval containing every possible U- and V-coordinate on the band, and let β denote the fixed zero-extended cutoff defining this band in $\rho/\alpha = H/(U-V)$. After the factor α^{n-1} in (20.28) is removed, put

$$A_{\alpha,H}(U,V) = \beta\left(\frac{H}{U-V}\right) H^{-n}(U-V)^{n-1}(\alpha^2+U^2)^{-n/2}(\alpha^2+V^2)^{-n/2} \times \overline{a\left(\frac{H\sqrt{U^2+\alpha^2}}{U-V}\right)}\, b\left(\frac{H\sqrt{V^2+\alpha^2}}{U-V}\right),$$

extended by zero off the band support. Here $U-V>0$ by (20.25). Define

$$(T_{\alpha,\omega}f)(U) = \int_{I_0} e^{iS(H(Ue_1+\alpha\omega)/(U-V))} A_{\alpha,H}(U,V) f(V)\, dV,$$
$$f_{\alpha,\omega}(V) = \Theta\left(\frac{Ve_1+\alpha\omega}{\sqrt{V^2+\alpha^2}}\right)\mathbf{1}_{I_0}(V),$$
$$g_{\alpha,\omega}(U) = \Theta\left(\frac{Ue_1+\alpha\omega}{\sqrt{U^2+\alpha^2}}\right)\mathbf{1}_{I_0}(U).$$

With the convention $\langle F,G\rangle = \int F\overline{G}$, the contribution $\mathscr{I}_\alpha$ of this transverse band is exactly

$$\mathscr{I}_\alpha = \alpha^{n-1}\int_{\mathbb{S}^{n-2}} \langle T_{\alpha,\omega}f_{\alpha,\omega}, g_{\alpha,\omega}\rangle_{L^2(I_0)}\, d\omega. \tag{20.30}$$

The functions in this formula are jointly measurable, and $\omega \mapsto f_{\alpha,\omega}, g_{\alpha,\omega}$ are strongly measurable $L^2(I_0)$-valued maps. Moreover

$$\|f_{\alpha,\omega}\|_2 + \|g_{\alpha,\omega}\|_2 \leq C\|\Theta\|_\infty.$$

The block argument above, Lemma 20.6, and Lemma 20.4 give, uniformly in ω,α,H,

$$\|T_{\alpha,\omega}\|_{2\to 2} \leq CB^C \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\}. \tag{20.31}$$

The trivial Schur bound supplies an integrable majorant in (20.30); hence ordinary Fubini and Cauchy–Schwarz yield

$$|\mathscr{I}_\alpha| \leq CB^C\|\Theta\|_\infty^2\alpha^{n-1}\int_{\mathbb{S}^{n-2}} \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\}\, d\omega. \tag{20.32}$$

No measurable-field or abstract direct-integral assertion is being used.

It remains to estimate the scalar angular integral. When $n=2$, $\mathbb{S}^0 = \{-1,1\}$, and the two restrictions are related by $r \mapsto -r$; either restriction has coefficient norm comparable to Λ, so the required bound follows by summing the two points. Assume henceforth in this paragraph that $n \geq 3$. Split $\mathbb{S}^{n-2}$ by a fixed measurable partition subordinate to a finite rational atlas $\omega = \omega_\nu(\zeta)$; the chart Jacobians are bounded above and away from zero on fixed enlargements. In the ν-th chart let $\mathbf{g}_\nu(\zeta)$ be the vector of all nonconstant (t,r) coefficients of $S(te_1 + r\omega_\nu(\zeta))$, after multiplication by one fixed common chart denominator. The denominator and its reciprocal are bounded on the enlargement, so pointwise

$$|\mathbf{g}_\nu(\zeta)| \simeq \Gamma(\omega_\nu(\zeta)). \tag{20.33}$$

For each nonempty chart, the restriction map

$$S - S(0) \longmapsto \mathbf{g}_\nu$$

is injective: if its nonconstant (t,r) coefficients vanish throughout the chart, then $S-S(0)$ vanishes on the corresponding open cone and therefore identically. Thus, for the application-specific polynomial space consisting of these restrictions, this verifies exactly the injectivity hypothesis (20.6); no injectivity of all ambient polynomials restricted to the sphere is being used. With $A_\nu = \|\mathbf{g}_\nu\|_{\mathrm{coeff},\zeta}$, finite-dimensional norm equivalence and the finite atlas give

$$A_\nu \geq c\Lambda \qquad \text{for every } \nu. \tag{20.34}$$

If $(\alpha H)^D A_\nu \geq 1$, apply (20.2) to $\mathbf{g}_\nu/A_\nu$, with $L = (\alpha H)^D A_\nu$ and $\sigma = c_0$; if $(\alpha H)^D A_\nu < 1$, use the trivial bound. In view of (20.33)– (20.34), summing the finitely many charts gives

$$\int_{\mathbb{S}^{n-2}} \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\}\, d\omega \leq C \min\{1, [\Lambda(\alpha H)^D]^{-c_1}\}. \tag{20.35}$$

All constants here are uniform as $\alpha, H \downarrow 0$: the normalized rectangles, denominator bounds, block-graph degrees, and numbers of sectional monotonicity intervals are fixed. Combining (20.32) and (20.35) proves the band bound

$$CB^C \|\Theta\|_\infty^2 \alpha^{n-1} \min\{1, [\Lambda(\alpha H)^D]^{-c}\}.$$

The original correlation is the sum of these band contributions because the preceding cutoffs form a partition of unity. Taking absolute values only after this identity and using their bounded overlap, write $A = \Lambda H^D$. The dyadic band sum is bounded by

$$\sum_{\alpha \in 2^{-\mathbb{N}_0}} \alpha^{n-1} \min\{1, (A\alpha^D)^{-c}\}.$$

If $A \leq 1$, this is $O(1)$. If $A > 1$, choose $\alpha_0 \simeq A^{-1/D}$. The bands below the threshold satisfy

$$\sum_{\alpha \leq \alpha_0} \alpha^{n-1} \lesssim A^{-(n-1)/D}.$$

After decreasing the oscillatory exponent so that $Dc < n-1$, the remaining bands satisfy

$$\sum_{\alpha > \alpha_0} A^{-c} \alpha^{n-1-Dc} \lesssim A^{-c}.$$

Thus the complete sum is $O(\min\{1, A^{-c'}\})$, where $c' = \min\{c, (n-1)/D\} > 0$, which is the right side of (20.18).

When $n = 1$, split the support into the finitely many signed intervals on which the two homogeneous kernel factors and radial profiles are of bounded variation. Their product has $L^\infty + \mathrm{BV}$ norm at most $CB^C \|\Theta\|_\infty^2$. Lemma 20.3 gives the asserted unit-scale bound uniformly in h. □

For a homogeneous polynomial $H(x,y)$, let $\mathscr{D}H$ denote the vector of all nonconstant-in-z coefficients of

$$H(y+z, y') - H(y+z, y). \tag{20.36}$$

Lemma 20.7 (Same-shell degree-block injectivity)**.** *For every integer $m \geq 2$, on homogeneous degree-m polynomials modulo pure phases, the map $[H] \mapsto \mathscr{D}H$ is injective. Hence $\|\mathscr{D}H\|_{\mathrm{coeff}} \geq c_{n,m} \|[H]\|_{\mathrm{coeff}}$.*

Proof. If (20.36) is independent of z, then $\partial_x H(x,y') - \partial_x H(x,y) = 0$. Thus $\partial_x H$ is independent of its second variable, and integration makes H pure. Quantitative injectivity is finite-dimensional norm equivalence. □

Proposition 20.8 (Flat-measure high block)**.** *Fix a translate B_0 of a prescribed bounded unit-scale cube, and assume that every cube below is contained in B_0; the constants are uniform in the location of B_0. Let $t \geq 0$, let $\mathscr{Q}$ be a finite family of pairwise disjoint q-cubes with $q = 2^{-t}$, and, for each $Q \in \mathscr{Q}$, let ν_Q be a complex finite measure supported on Q. Assume $\Lambda \geq 0$ and*

$$|\nu_Q|(Q) \leq \Lambda |Q|, \qquad M = \sum_Q |\nu_Q|(Q). \tag{20.37}$$

Suppose P is a real polynomial of degree at most d, its highest nonpure homogeneous block has coefficient norm $L \geq 1$, and $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For every $|e_Q| \leq 1$ and every radial BV *profile a, supported in a fixed compact subinterval of $(0,\infty)$ and of $L^\infty + \mathrm{BV}$ norm at most B, where $B \geq 1$, write*

$$A_a^{P,\Theta}\nu(x) = \int e^{iP(x,y)} k_a(x-y)\, d\nu(y).$$

Then

$$\left\| A_a^{P,\Theta} \sum_Q e_Q \nu_Q \right\|_2^2 \leq C B^C \|\Theta\|_\infty^2 \Lambda M (L^{-c} + 2^{-ct}). \tag{20.38}$$

Proof. Put $w_Q = |\nu_Q|(Q)$ and $F = A_a^{P,\Theta} \sum_Q e_Q \nu_Q$. The mass cap and disjointness give

$$M = \sum_Q w_Q \leq \Lambda \sum_Q |Q| \leq C\Lambda, \qquad M^2 \leq C\Lambda M.$$

We keep the original measures in their physical coordinates. If b is the center of B_0, set $P_0(u,v) = P(b+u, b+v)$ and $Q_0 = Q - b$. All Q_0 lie in one fixed reference box. Simultaneous translation preserves the highest nonpure homogeneous block and its norm L: higher homogeneous blocks are pure and remain pure under translation. It also transports the actual Gram integrals by translation of the output variable. We make this recentering before extracting any scalar polynomial. The nonvacuous hypotheses already imply $d \geq 2$.

The given radial budget controls the original pointwise values of a; indeed, comparison with a zero value beyond its compact support also bounds each such value by its original variation. The profile is measurable. Thus $k_a \in L^2$, and the rows

$$f_y(x) = e^{iP(x,y)} k_a(x-y) \quad \text{satisfy} \quad \|f_y\|_2 = \|k_a\|_2 \leq CB\|\Theta\|_\infty \quad \text{for every } y.$$

The row family is jointly measurable modulo $|\nu_Q| \otimes dx$ null sets, even for singular ν_Q. Here an essentially bounded angular representative can be used without changing these L^2 rows or their product-measure classes. Writing $d\nu_Q = h_Q\, d|\nu_Q|$ with $|h_Q| \leq 1$ almost everywhere uses only a polar factor relative to total variation, not a density relative to spatial volume. The finite measures and uniform row bound make $\int h_Q(y) f_y\, d|\nu_Q|(y)$ genuine Bochner integrals in L^2. Their scalar representatives agree almost everywhere with the original complex-measure integrals; complex linearity gives

$$F = \sum_Q e_Q \int h_Q(y) f_y\, d|\nu_Q|(y) \quad \text{in } L^2.$$

In particular the operator on the literal combined measure is genuinely in L^2, not merely a formal expression whose extended norm is bounded.

For $u, v \in \mathbb{R}^n$, write

$$\Psi_{u,v}(z) = P_0(u+z,v) - P_0(u+z,u), \qquad \Gamma(u,v) = \|\Psi_{u,v} - \Psi_{u,v}(0)\|_{\mathrm{coeff},z}.$$

The retained polynomial $R(u,v,z) = \Psi_{u,v}(z) - \Psi_{u,v}(0)$ depends only on the mixed part of P_0. Its map from the bounded-degree mixed polynomial space is injective: vanishing of R makes $\partial_x P_0(x,v) - \partial_x P_0(x,u) = 0$, so P_0 is pure. This is the same injectivity underlying Lemma 20.7, now on the full mixed

space. Finite-dimensional norm equivalence, followed by evaluation on a fixed unit box in (u,v,z), therefore provides a single $z_* \in [-1,1]^n$ such that the scalar polynomial $S(u,v) = R(u,v,z_*)$ has coefficient norm at least a fixed multiple of the full mixed coefficient norm of P_0, and hence at least cL. Evaluation at this fixed z_* also gives $|S(u,v)| \le C\Gamma(u,v)$ for every pair (u,v). Consequently

$$\begin{gathered} N := \|S\|_{\mathrm{coeff},(u,v)} \ge cL > 0, \qquad p := S/N, \qquad \|p\|_{\mathrm{coeff}} = 1, \qquad \deg p \le d, \\ L|p(u,v)| \le K\Gamma(u,v) \qquad (u,v \in \mathbb{R}^n), \end{gathered} \tag{20.39}$$

where $K > 0$ depends only on the fixed degree and dimension. By the fixed norm-equivalence convention, we use the standard Euclidean coefficient norms here; equivalence with the maximum coefficient norm only changes fixed constants. In particular no bound on lower or pure coefficients is being imposed, and no previously frozen scalar polynomial is translated.

Hilbert-space integration of the actual rows, followed by the triangle inequality and $|e_Q h_Q| \le 1$, gives

$$\|F\|_2^2 \le \sum_{Q,Q'} \int_Q \int_{Q'} |\mathscr{K}(y,y')|\, d|\nu_{Q'}|(y')\, d|\nu_Q|(y), \tag{20.40}$$

$$\mathscr{K}(y,y') = \int e^{i\Psi_{u,v}(z)} \overline{k_a(z)} k_a(z-(v-u))\, dz, \quad u = y-b, \quad v = y'-b. \tag{20.41}$$

The latter is exactly $\int \overline{f_y(x)} f_{y'}(x)\, dx$ after $x = y+z$. The preceding Hilbert integrals justify the Gram expansion and Fubini before absolute values are taken. In particular y and y' remain independent variables even when $Q = Q'$; they are not replaced by a single selected point of that cell.

We next cover bad pairs by sets of cell indices fixed before either measure variable is integrated. Write a_Q for the lower corner of Q_0. On the fixed reference box, every degree-at-most-d scalar polynomial f of coefficient norm one satisfies

$$|f(u,v) - f(a_Q, a_{Q'})| \le Hq \qquad (u \in \overline{Q_0},\ v \in \overline{Q_0'}),$$

with one structural $H \ge 0$. For $0 < \tau \le 1$ define

$$\mathscr{E}_\tau(f) = \{(Q,Q') : |f(a_Q, a_{Q'})| \le \tau + Hq\}.$$

Thus any point pair in these cells with $|f(u,v)| \le \tau$ has its index pair in $\mathscr{E}_\tau(f)$, including two independent points in the same closed cell. Apply (20.3) only at the fixed corners, with threshold $\tau + Hq$. The Lipschitz enlargement gives

$$\sum_{(Q,Q') \in \mathscr{E}_\tau(f)} w_Q w_{Q'} \le C\Lambda M(\tau+q)^\alpha$$

for a structural $\alpha > 0$. If the enlarged threshold exceeds one, the same bound follows from $M^2 \le C\Lambda M$, since then $\tau + q$ is bounded below. Corners on closed faces are admissible; only disjoint interiors are used in this estimate.

Let $\mathscr{B} = \mathscr{E}_\tau(p)$ and fix one coordinate, say the first. The normalized linear polynomial $\ell(u,v) = (u_1 - v_1)/\sqrt{2}$ has coefficient norm one, and $|\ell(u,v)| \le |u-v|$. The second cover $\mathscr{N} = \mathscr{E}_\tau(\ell)$ therefore contains the index pair of every point pair with $|u-v| \le \tau$, and satisfies

$$\sum_{(Q,Q') \in \mathscr{N}} w_Q w_{Q'} \le C\Lambda M(\tau+q)^\alpha. \tag{20.42}$$

The same bound holds for $\mathscr{B}$. This positive-power estimate, rather than a sharp $(\tau+q)^n$ estimate, is all that is needed.

Put $s(h) = \min\{1, |h|\}$. Lemma 20.5, including its stronger one-dimensional conclusion, implies in every $n \ge 1$

$$|\mathscr{K}(y,y')| \le A\big[\max\{1, s(v-u)^d \Gamma(u,v)\}\big]^{-c_0}, \qquad A = CB^C \|\Theta\|_\infty^2,$$

for some structural $c_0 > 0$. In particular $|\mathscr{K}| \le A$. Outside $\mathscr{B} \cup \mathscr{N}$, the covering property holds for every pair of points in the relevant cells and gives $|p(u,v)| > \tau$ and $|u-v| > \tau$. By (20.39) and $\tau \le 1$,

$$s(v-u)^d \Gamma(u,v) \ge \frac{L\tau^{d+1}}{K}.$$

Consequently, with $\delta = [\max\{1, L\tau^{d+1}/K\}]^{-c_0}$, we have the fixed-index majorant

$$|\mathscr{K}(y,y')| \le A\{\delta + \mathbf{1}_{\mathscr{B}}(Q,Q') + \mathbf{1}_{\mathscr{N}}(Q,Q')\} \qquad (y \in Q,\ y' \in Q').$$

Integrating this bound in (20.40) and using the two cover estimates yields

$$\|F\|_2^2 \le A\{\delta M^2 + C\Lambda M(\tau+q)^\alpha\} \le CA\Lambda M\{\delta + (\tau+q)^\alpha\}.$$

Finally choose

$$\sigma = \frac{1}{2(d+1)}, \qquad \tau = L^{-\sigma}, \qquad c = \min\{\sigma\alpha, c_0/2\} > 0.$$

Then $0 < \tau \le 1$, $L\tau^{d+1} = L^{1/2}$, and

$$\delta + (\tau+q)^\alpha \le C\{L^{-c_0/2} + L^{-\sigma\alpha} + q^\alpha\} \le C(L^{-c} + q^c).$$

Since $q = 2^{-t}$ for every real $t \ge 0$, this proves (20.38). No division by a packet mass, M, or Λ has occurred, so zero measures, an empty family, and $\Lambda = 0$ are included without normalization or a separate limiting argument. Only the original radial budget has been used. □

20.3 A one-shell phase-adapted packet theorem

Lemma 20.9 (Fourier expansion of a low homogeneous block)**.** *Let $\mathbf{H} = (H_1, \dots, H_N)$ be a fixed basis of one homogeneous degree block modulo pure phases. Fix smooth cutoffs $\chi, \widetilde{\chi}$ supported on fixed bounded cubes and equal to one on the relevant smaller cubes. Choose one structural period $T > 0$ and a translated fundamental cell containing the original input and output boxes and the cutoff supports in its interior. Put $\omega_F = 2\pi/T$; the same T is used for all the finitely many degree blocks and the finite flag of Lemma 19.1. Then, for every $\mathbf{C} \in \mathbb{R}^N$ and every ξ in the specified fixed cube, on the original X, Y boxes,*

$$\chi(X)\widetilde{\chi}(Y) e^{i\mathbf{C}\cdot[\mathbf{H}(X,Y) - \mathbf{H}(\xi,Y)]} = \sum_{u,v \in \mathbb{Z}^n} a_{u,v}(\mathbf{C},\xi) e^{i\omega_F u\cdot X} e^{i\omega_F v\cdot Y}.$$

The series is absolutely convergent, and there are nonnegative numbers $A_{u,v}(\mathbf{C})$, independent of ξ, such that $|a_{u,v}(\mathbf{C},\xi)| \le A_{u,v}(\mathbf{C})$ and

$$\sum_{u,v} A_{u,v}(\mathbf{C})(1+|v|) \le C(1+\|\mathbf{C}\|)^{C_F}, \qquad C_F = 2n+2. \tag{20.43}$$

Outside the chosen cell the series represents the periodic extension, not the compactly supported expression on the left.

Proof. Choose the period cell to contain the compact closures of all original relevant boxes and the fixed cutoff supports, without changing any of these boxes or cutoffs. Since there are only finitely many fixed polynomial bases, the same cell and one maximum constant work for all of them. Periodize the cutoff expression and take its actual torus Fourier coefficients, as in Lemma 19.1. The map

$$\mathbf{C} \longmapsto \sum_{r=1}^{N} C_r[H_r(X,Y) - H_r(\xi,Y)]$$

is linear into the fixed finite-dimensional polynomial space in (X,Y,ξ). Its derivatives on the fixed boxes are bounded by $C\|\mathbf{C}\|$, uniformly in ξ. Applying $(1-\Delta_{X,Y})^{n+1}$ in the coefficient integral therefore yields

$$|a_{u,v}(\mathbf{C},\xi)| \le \frac{C(1+\|\mathbf{C}\|)^{2n+2}}{(1+|(u,v)|)^{2n+2}} =: A_{u,v}(\mathbf{C}).$$

This nonnegative envelope is chosen before ξ. Multiplication by $1+|v|$ leaves the summable lattice weight of exponent $2n+1$. Smooth torus reconstruction and this common majorant prove the stated identity and absolute convergence on the entire original boxes. In particular, no supremum in ξ is interchanged with an infinite sum, and no compact amplitude is asserted to be globally periodic. □

Proposition 20.10 (Phase-adapted one-shell packet principle)**.** *There are $c_P, c_E > 0$ and t_0, depending only on n,d, such that the following holds. Let P be a real polynomial of degree at most d, let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and let $R>0$, $t\in\mathbb{N}$, $q=R2^{-t}$, $t\ge t_0$, and let $\mathcal{Q}$ be a finite family of pairwise disjoint dyadic q-cubes (with R and q dyadic scales), with center c_Q for each $Q\in\mathcal{Q}$. Suppose*

$$\operatorname{supp} d_Q \subset Q, \qquad \int d_Q = 0, \qquad \|d_Q\|_1 \le \Lambda|Q|, \qquad M = \sum_{Q\in\mathcal{Q}} \|d_Q\|_1. \tag{20.44}$$

For $R\le\varepsilon\le 2R$ put

$$\mathscr{P}_\varepsilon(x) = \sum_{Q\in\mathcal{Q}} \int_{\varepsilon<|x-y|\le 2R} e^{i[P(x,y)-P(c_Q,y)]} K_\Theta(x-y) d_Q(y)\,dy.$$

Then $\mathscr{P} = G+E$ with

$$\|V^2(G_\varepsilon : R\le\varepsilon\le 2R)\|_2^2 \le C\|\Theta\|_\infty^2 2^{-c_P t}\Lambda M, \tag{20.45}$$

$$\|V^1(E_\varepsilon : R\le\varepsilon\le 2R)\|_1 \le C\|\Theta\|_1 2^{-c_E t} M. \tag{20.46}$$

No pointwise bound on d_Q is assumed. Both estimates remain uniform after replacing every d_Q by $e_Q d_Q$, where $|e_Q|\le 1$.

Proof. The optional factors e_Q may be absorbed into the packets, so suppress them. Partition input into dyadic R-cubes J and output into dyadic R-cubes I; only $O_n(1)$ relative positions interact. For one such pair put

$$M_{I,J} = \sum_{Q\subset J} \|d_Q\|_1.$$

Every input block has boundedly many interacting output blocks, and hence

$$\sum_{I,J} M_{I,J} \le C_n M. \tag{20.47}$$

The same annular support gives

$$\sup_I \#\{J : I\sim J\} + \sup_J \#\{I : I\sim J\} \le C_n,$$

where $I\sim J$ denotes an interacting block pair. Let a be the center of J, set $x=a+RX$, $y=a+RY$, and define

$$\Phi(X,Y) = P(a+RX,a+RY) - P(a,a+RY), \qquad \xi_Q = \frac{c_Q-a}{R}. \tag{20.48}$$

Then $\Phi(0,Y)=0$, and the packet phase is exactly $\Phi(X,Y)-\Phi(\xi_Q,Y)$. The rescaled packets $\widetilde{d}_Q(Y) = d_Q(a+RY)$ are supported on disjoint dyadic 2^{-t}-cubes and satisfy

$$\|\widetilde{d}_Q\|_1 = R^{-n}\|d_Q\|_1, \qquad \|\widetilde{d}_Q\|_1 \le \Lambda|\widetilde{Q}|, \qquad \widetilde{M}_{I,J} := \sum_{Q\subset J} \|\widetilde{d}_Q\|_1 = R^{-n} M_{I,J}. \tag{20.49}$$

All normalized packet cubes lie in the fixed normalized input block $B_0 = R^{-1}(J-a)$, as required in Proposition 20.8. We prove the unit-scale estimate and suppress tildes on the packets, but retain $\widetilde{M}_{I,J}$, until the last paragraph. Insert fixed smooth cutoffs which equal one on the normalized interacting input and output blocks; these are the cutoffs in Lemma 20.9.

Write the pure part of Φ as $p(X)+q(Y)$. Its mixed representative vanishes at $X=0$, so $\Phi(0,Y)=0$ forces q to be constant. In the anchored difference $p(X)-p(\xi_Q)$ is a common output modulation times a packet scalar and may be removed. Modulo these factors, write the common block phase in descending nonpure homogeneous degrees as

$$\Phi(X,Y) = H_1(X,Y)+\cdots+H_N(X,Y), \qquad \deg H_1 > \cdots > \deg H_N \geq 2,$$

so the anchored phase is $\Phi(X,Y)-\Phi(\xi_Q,Y)$. Put $C_r = \|H_r\|_{\mathrm{coeff}}$.

If $N=0$, only the removed output modulation and packet scalars remain; Proposition 18.5 gives the required unit-block bound with its positive terminal decay exponent and $E=0$. The rescaling and bounded-overlap summation in the last paragraph then give (20.45). Hence assume $N \geq 1$ below.

Let c_{flat}, C_B be respectively the decay exponent and the BV power in Proposition 20.8, let δ_T be from Proposition 18.5, and choose $C_{\mathrm{Four}} \geq C_F$ large enough to cover all Fourier losses in Lemma 20.9. Decrease c_{flat} if necessary so that $0 < c_{\mathrm{flat}} \leq 1$. Set

$$\theta = \frac{c_{\mathrm{flat}}}{16(C_B+1)}.$$

Choose $B_* > 1$ so large that

$$\frac{2C_{\mathrm{Four}}}{B_*-1} \leq \min\{\theta/2, c_{\mathrm{flat}}/8\}, \tag{20.50}$$

and then choose $0 < \kappa_* \leq 1$ so small that, with

$$\kappa_r = \kappa_* B_*^{r-N}, \qquad S_r = \sum_{i<r} \kappa_i, \qquad \gamma_r = \theta\kappa_r, \tag{20.51}$$

one has

$$2C_{\mathrm{Four}} \sum_{r=1}^{N} \kappa_r \leq \min\{\delta_T/2, 1/2\}. \tag{20.52}$$

Then $S_r \leq \kappa_r/(B_*-1)$ and

$$\gamma_r - 2C_{\mathrm{Four}} S_r \geq \gamma_r/2, \tag{20.53}$$

$$c_{\mathrm{flat}}\kappa_r - C_B\gamma_r - 2C_{\mathrm{Four}} S_r \geq c_{\mathrm{flat}}\kappa_r/2, \tag{20.54}$$

$$c_{\mathrm{flat}} - C_B\gamma_r - 2C_{\mathrm{Four}} S_r \geq c_{\mathrm{flat}}/2. \tag{20.55}$$

Call block r low if $C_r \leq 2^{\kappa_r t}$ and expand it by Lemma 20.9. The product of the preceding low expansions has weighted coefficient ℓ^1 norm at most

$$C 2^{C_{\mathrm{Four}} S_r t}. \tag{20.56}$$

For the nonlinear variation norms we use only finite Fourier sums and pass to the limit afterwards. Write $a^{(i)}_{u_i,v_i}$ and $A^{(i)}_{u_i,v_i}$ for the coefficient and its common majorant in the expansion of a low block i. For $K \in \mathbb{N}$ retain only

$$\mathfrak{F}_K = \{(u_i,v_i) \in \mathbb{Z}^n \times \mathbb{Z}^n : |u_i|+|v_i| \leq K\}$$

in each low expansion. If S is a set of expanded low blocks, define

$$\omega_K(S) = \sum_{(u_i,v_i)_{i\in S}:\ \max_{i\in S}(|u_i|+|v_i|)>K} \left(\prod_{i\in S} A^{(i)}_{u_i,v_i}\right)\left(1+\left|\sum_{i\in S} v_i\right|\right), \qquad \omega_K(\varnothing) = 0. \tag{20.57}$$

Since

$$1+\left|\sum_{i\in S} v_i\right| \le \prod_{i\in S}(1+|v_i|),$$

Lemma 20.9 and dominated convergence for the resulting absolutely summable product series give

$$\omega_K(S) \longrightarrow 0 \qquad (K\to\infty), \tag{20.58}$$

uniformly in all packet centers ξ_Q. Every estimate below is first applied with this finite-mode convention. Its constants are independent of K. For a fixed tuple write $u=\sum_i u_i$ and $v=\sum_i v_i$; the common output and input characters have frequencies $\omega_F u$ and $\omega_F v$. Since $1+\omega_F|v| \le \max\{1,\omega_F\}(1+|v|)$, this structural scale changes no Fourier-loss exponent. Suppose r is the first high block. After a preceding Fourier tuple is fixed, denote its parent-dependent Fourier coefficient by $a_Q(\mathbf{u},\mathbf{v})$ and the product of the corresponding common majorants in Lemma 20.9 by $A(\mathbf{u},\mathbf{v})$. Then $|a_Q/A| \le 1$, and summation of A gives exactly (20.56). Absorb this normalized scalar together with the factor

$$e^{-i\sum_{i\ge r} H_i(\xi_Q,Y)} e^{i\omega_F v\cdot Y}$$

into the complex measure on packet Q. This multiplier need not be constant; total variation and (20.44) are unchanged. The remaining common operator phase has H_r as its highest nonpure block, and all lower blocks are unrestricted.

For this fixed Fourier tuple denote the resulting measures by μ_Q, and write $\Phi_r=\sum_{i\ge r} H_i$ for the common phase. Partition $[1,2]$ into $K_r \simeq 2^{\gamma_r t}$ radial cells I_ℓ and set

$$\mathscr{G}_\ell(X) = \sum_Q \int_{\{|X-Y|\in I_\ell\}} e^{i\Phi_r(X,Y)} K_\Theta(X-Y)\, d\mu_Q(Y).$$

Inserting all cell endpoints into an arbitrary variation partition gives the pointwise deterministic bound

$$\begin{aligned} V^2(\mathscr{H}_\varepsilon : 1\le\varepsilon\le 2) \le CV^2\left(\sum_{\ell\le j}\mathscr{G}_\ell : 0\le j\le K_r\right) \\ + C\left(\sum_\ell V^2(\mathscr{H}_\varepsilon:\varepsilon\in I_\ell)^2\right)^{1/2}, \end{aligned} \tag{20.59}$$

where $\mathscr{H}$ is this component trajectory. Reversing the order of the grid partial sums, if necessary, does not change their variation. Inside a cell use $V^2\le V^1$. Lemma 18.1, whose proof uses only the average mass cap, gives

$$\sup_\ell \|W_\ell\|_\infty \le C\Lambda\|\Theta\|_\infty(2^{-\gamma_r t}+2^{-t}), \qquad \sum_\ell \|W_\ell\|_1 \le C\|\Theta\|_1 \widetilde{M}_{I,J}.$$

Thus the squared L^2 norm of the within-cell square function is at most $C\|\Theta\|_\infty^2\Lambda\widetilde{M}_{I,J}2^{-\gamma_r t}$.

For the grid partial sums, fix $(\varepsilon_\ell)_\ell \in \{-1,0,1\}^{K_r}$ before evaluation in X. The corresponding sharp radial profile $a_\varepsilon=\sum_\ell \varepsilon_\ell \mathbf{1}_{I_\ell}$ satisfies

$$\|a_\varepsilon\|_\infty + \mathrm{Var}(a_\varepsilon) \le CK_r.$$

Proposition 20.8 and Lemma 18.2 therefore give, uniformly over these fixed signs, using $\log^2(2+K_r)\lesssim (1+\gamma_r t)^2 \lesssim t^2$,

$$Ct^2\|\Theta\|_\infty^2 2^{C_B\gamma_r t}\Lambda\widetilde{M}_{I,J}\{2^{-c_{\mathrm{flat}}\kappa_r t}+2^{-c_{\mathrm{flat}}t}\}. \tag{20.60}$$

In particular, no X-dependent selector is inserted into the fixed-sign estimate; all such dependence was already confined to the positive within-cell term in (20.59). Denote by $G^{(r),I,J}$ the full component obtained after summing all Fourier tuples preceding the first high block r. The variation triangle inequality and the square of the common-majorant sum (20.56) give the complete estimate

$$\|V^2(G_\varepsilon^{(r),I,J})\|_2^2 \le C\|\Theta\|_\infty^2\Lambda\widetilde{M}_{I,J}2^{2C_{\mathrm{Four}}S_r t}\Big[2^{-\gamma_r t}+t^2 2^{C_B\gamma_r t}\{2^{-c_{\mathrm{flat}}\kappa_r t}+2^{-c_{\mathrm{flat}}t}\}\Big]. \tag{20.61}$$

There are no suppressed exponential factors in this line: the first term is the within-cell square function and the other two are respectively the large-coefficient and mesh terms in (20.60). The three hierarchy inequalities say, term by term,

$$\begin{aligned} 2C_{\mathrm{Four}}S_r-\gamma_r &\le -\gamma_r/2,\\ 2C_{\mathrm{Four}}S_r+C_B\gamma_r-c_{\mathrm{flat}}\kappa_r &\le -c_{\mathrm{flat}}\kappa_r/2,\\ 2C_{\mathrm{Four}}S_r+C_B\gamma_r-c_{\mathrm{flat}} &\le -c_{\mathrm{flat}}/2. \end{aligned}$$

Since $t^2 2^{-at}\le C_a 2^{-at/2}$ for $t\ge t_0$, the right side of (20.61) is bounded by

$$C\|\Theta\|_\infty^2\Lambda\widetilde{M}_{I,J}2^{-c_r t},\qquad c_r=\frac{1}{4}\min\{\gamma_r,c_{\mathrm{flat}}\kappa_r,c_{\mathrm{flat}}\}>0.$$

This proves every high branch with $E=0$ without changing any exponent used later.

If every block is low, expand them all. Under the normalized variables in (20.48), the center of the normalized packet cube is

$$\zeta_Q:=\xi_Q=\frac{c_Q-a}{R}. \tag{20.62}$$

For a combined input frequency v, write

$$e^{i\omega_F v\cdot Y}d_Q(Y)=e^{i\omega_F v\cdot\zeta_Q}d_Q(Y)+r_{Q,v}(Y),\qquad r_{Q,v}(Y)=(e^{i\omega_F v\cdot Y}-e^{i\omega_F v\cdot\zeta_Q})d_Q(Y),$$

and, since $|Y-\zeta_Q|\le C2^{-t}$ on the normalized cube,

$$\|r_{Q,v}\|_1\le C\omega_F|v|2^{-t}\|d_Q\|_1.$$

The frozen term remains mean zero, so Proposition 18.5 applies with constant packet coefficients of modulus at most one: divide the parent-dependent Fourier coefficient by its product common majorant and absorb the normalized scalar into the packet. Summation of the remaining majorants is governed by (20.43), and the output Fourier factor is the fixed modulation $e^{i\omega_F u\cdot X}$. The factor ω_F in the remainder is structural and is absorbed into C. Equation (20.52) leaves at least $2^{-\delta_T t/2}$ after the squared Fourier loss. Young's inequality and the weighted Fourier sum give total L^1 variation at most $C\|\Theta\|_1 2^{-t/2}\widetilde{M}_{I,J}$ for the remainders.

We now remove the finite-mode cutoff. For a normalized interacting block pair (I,J), let

$$\mathscr{P}^{I,J,K}=G^{I,J,K}+E^{I,J,K}$$

be the finite-mode trajectory and decomposition constructed above, with the output restricted to the normalized cube corresponding to I. If r is the first high block, put $\mathscr{I}_r=\{i:i<r\}$ and

$$\mathfrak{B}_{r,t}=2^{-\gamma_r t}+t^2 2^{C_B\gamma_r t}\{2^{-c_{\mathrm{flat}}\kappa_r t}+2^{-c_{\mathrm{flat}}t}\}.$$

For $K'>K$, applying the same within-cell and grid estimates only to the Fourier tuples omitted at level K, and then using the variation triangle inequality, gives

$$\|G^{I,J,K'}-G^{I,J,K}\|_{L^2(V^2)}\le C\|\Theta\|_\infty(\Lambda\widetilde{M}_{I,J})^{1/2}\mathfrak{B}_{r,t}^{1/2}\,\omega_K(\mathscr{I}_r),\qquad E^{I,J,K'}-E^{I,J,K}=0. \tag{20.63}$$

If every block is low, put $S=\{1,\dots,N\}$. The terminal-packet estimate for the frozen terms and Young's inequality for the remainders give

$$\|G^{I,J,K'}-G^{I,J,K}\|_{L^2(V^2)}\le C\|\Theta\|_\infty 2^{-\delta_T t/2}(\Lambda\widetilde{M}_{I,J})^{1/2}\omega_K(S), \tag{20.64}$$

$$\|E^{I,J,K'}-E^{I,J,K}\|_{L^1(V^1)}\le C\|\Theta\|_1 2^{-t}\widetilde{M}_{I,J}\omega_K(S). \tag{20.65}$$

Indeed, in the first line the unweighted coefficient tail is bounded by (20.57); in the second line the factor $1+|\sum_i v_i|$ absorbs the freezing error. These are the same estimates used above, with the full coefficient sum replaced by its tail, so no decay exponent is changed.

There are only finitely many interacting block pairs. Hence (20.58) and the bounded-overlap inequalities show that (G^K) is Cauchy in $L^2(V^2)$ and (E^K) is Cauchy in $L^1(V^1)$. All finite-mode component paths vanish at the upper endpoint $\varepsilon = 2R$. Consequently the same is true of every difference path, and

$$\sup_{R\le\varepsilon\le 2R} |H_\varepsilon| \le V^a(H_\varepsilon : R\le\varepsilon\le 2R), \qquad a\in\{1,2\},$$

so convergence in the anchored variation spaces also controls the supremum of the trajectories. Lemma 18.3 constructs anchored limits G,E for the full finite-mode sequences and gives the corresponding $L^2(V^2)$ and $L^1(V^1)$ convergence.

It remains to identify this limit. The common majorants in Lemma 20.9 imply that the truncated Fourier products converge absolutely and uniformly on the fixed normalized input, output, and packet-center boxes. Since the kernel is restricted to $1<|X-Y|\le 2$ and there are only finitely many packets, for each block pair

$$\sup_{1\le\varepsilon\le 2} |\mathscr{P}_\varepsilon^{I,J,K}(X) - \mathscr{P}_\varepsilon^{I,J}(X)| \le o_K(1)C\|\Theta\|_\infty \sum_{Q\subset J}\|d_Q\|_1, \qquad o_K(1)\longrightarrow 0,$$

on its normalized output cube. The right side belongs to both L^1 and L^2 there, and the convergence is uniform in ε almost everywhere. The last assertion of Lemma 18.3 identifies $\mathscr{P}^{I,J} = G^{I,J} + E^{I,J}$ (after rescaling, at the physical scale). The Fatou conclusion of that lemma passes the uniform finite-mode bounds to these limits without changing an exponent.

Let $G_\varepsilon^{I,J} = \mathbf{1}_I G_{I,J,\varepsilon}$ and $E_\varepsilon^{I,J} = \mathbf{1}_I E_{I,J,\varepsilon}$ denote the resulting physical paths. Define

$$G_\varepsilon = \sum_I \sum_{J:J\sim I} G_\varepsilon^{I,J}, \qquad E_\varepsilon = \sum_I \sum_{J:J\sim I} E_\varepsilon^{I,J}.$$

The block family is finite, and the limiting blockwise identities are exact on I. Therefore $\mathscr{P}_\varepsilon = G_\varepsilon + E_\varepsilon$ for the original trajectory.

Take $c_P > 0$ to be the minimum of $\delta_T/2$ and the finitely many high-branch exponents c_r, over all possible $N \le d-1$, and take $c_E = 1/2$. At unit scale $\widetilde{M}_{I,J} = R^{-n}M_{I,J}$; on returning to x, an L^2-square and an L^1 norm both acquire the Jacobian R^n. Thus

$$\|V^2(G_\varepsilon^{I,J})\|_2^2 \le C\|\Theta\|_\infty^2 2^{-c_P t}\Lambda M_{I,J},$$
$$\|V^1(E_\varepsilon^{I,J})\|_1 \le C\|\Theta\|_1 2^{-c_E t} M_{I,J}.$$

Since the output cubes I are disjoint and each has only $O_n(1)$ input neighbors, the variation triangle inequality and Cauchy–Schwarz give

$$\|V^2(G_\varepsilon)\|_2^2 \le C_n \sum_{I,J}\|V^2(G_\varepsilon^{I,J})\|_2^2, \qquad \|V^1(E_\varepsilon)\|_1 \le \sum_{I,J}\|V^1(E_\varepsilon^{I,J})\|_1.$$

Using (20.47) proves (20.45)–(20.46). □

Independent parent anchors. For descendant packets, the parent anchor and the packet center play different roles. Let z_Q be the center of the normalized packet cube and let a_Q be its anchor in a fixed structural box. The phase is $\Phi(X,Y) - \Phi(a_Q,Y)$, and the input is frozen at z_Q. The estimates below are uniform in both points.

Put $F = 2n+2$, let $\delta > 0$ be the squared terminal-packet decay, and let $\beta > 0$ be the flat high-block decay. All these numbers are chosen before the phase coefficients or packets. There are positive rates $\kappa_m, \gamma_m \le 1$, depending only on the structural data, such that, with $A_m = 2F\sum_{k>m}\kappa_k$,

$$2F\sum_{k=0}^{d}\kappa_k \le \tfrac{1}{2}\min\{\delta,1\}, \qquad A_m \le \gamma_m/2,$$
$$A_m + 2\gamma_m - \beta\kappa_m \le -\beta\kappa_m/2, \qquad A_m + 2\gamma_m - \beta \le -\beta/2.$$

For example take $\kappa_m = ar^m$ and $\gamma_m = \theta\kappa_m$: first choose $\theta > 0$ sufficiently small in terms of β, then $r > 0$ sufficiently small in terms of θ, F, d, and finally $a > 0$ sufficiently small in terms of δ, F, d. The finite geometric tail $\sum_{k>m} \kappa_k \le (d+1)r\kappa_m$ proves all four conditions. These choices include the degrees whose homogeneous blocks vanish. Increase t_0 also so that $\sqrt{n}\,2^{-t} \le 1/4$ for $t \ge t_0$, as required by the high-block cube geometry.

Write $D = \|\operatorname{Mix}\Phi\|_{\mathrm{coeff}}$. If every mixed homogeneous block has norm at most $2^{\kappa_m t}$, then $D \le \sum_{m=0}^{d} 2^{\kappa_m t}$. In this branch define

$$G_u(X) = \sum_Q e_Q e^{i[\Phi(X,z_Q)-\Phi(a_Q,z_Q)]} \int_{u<|X-Y|\le 2} K_\Theta(X-Y) d_Q(Y)\, dY, \qquad E_u = \mathscr{P}_u - G_u.$$

These are literal integral paths, defined on a common conull set of X simultaneously for every $1 \le u \le 2$. Changing the anchor in the frozen phase from z_Q to a_Q multiplies packet Q by the scalar $e^{i[\Phi(z_Q,z_Q)-\Phi(a_Q,z_Q)]}$, whose modulus is one. Consequently the same terminal-packet estimate and a single Fourier expansion of the full mixed polynomial give

$$\|V^2(G)\|_2^2 \le C\|\Theta\|_\infty^2 (1+D)^{2F} 2^{-\delta t} \Lambda \widetilde{M}.$$

Pure input and output terms disappear from the freezing difference:

$$\begin{aligned}&[\Phi(X,Y)-\Phi(a_Q,Y)] - [\Phi(X,z_Q)-\Phi(a_Q,z_Q)]\\ &\quad = [\operatorname{Mix}\Phi(X,Y)-\operatorname{Mix}\Phi(X,z_Q)] - [\operatorname{Mix}\Phi(a_Q,Y)-\operatorname{Mix}\Phi(a_Q,z_Q)].\end{aligned}$$

The coefficient Lipschitz bound on the fixed boxes therefore bounds its absolute value by $CD|Y-z_Q|$. The total 1-variation of the annular indicator is at most one. Tonelli and $|Y-z_Q| \le C2^{-t}$ yield

$$\|V^1(E)\|_1 \le C\|\Theta\|_1 D 2^{-t}\widetilde{M}.$$

The first rate condition bounds both displays by their respective budgets times $C2^{-\frac{1}{2}\min\{\delta,1\}t}$.

Otherwise select the largest mixed degree m whose block exceeds its threshold, and write $\Phi = \Phi_0 + S$, where S is the sum of the higher mixed blocks. The mixed degree of Φ_0 is at most m, its top block has norm $L > 2^{\kappa_m t}$, and $\|S\|_{\mathrm{coeff}} \le \sum_{k>m} 2^{\kappa_k t}$. All pure terms and all lower mixed coefficients remain in Φ_0; no bound is placed on them. Expand $e^{i[S(X,Y)-S(a_Q,Y)]}$ once on the fixed period cell. The common coefficient sum is at most $C(1+\|S\|_{\mathrm{coeff}})^F \le C2^{F\sum_{k>m}\kappa_k t}$. For each Fourier mode absorb the input character and $e^{-i\Phi_0(a_Q,Y)}$ into d_Q, and divide its packet coefficient by its common majorant. This preserves support and the average L^1 cap, but need not preserve mean zero; the flat high-block bound requires no such cancellation.

Take a grid with $N = 2^{\lfloor \gamma_m t \rfloor}$ cells on $[1,2]$. By homogeneity in the radial profile, the $B = 1$ case of Proposition 20.8, applied to a/B, gives quadratic dependence B^2 on its $L^\infty + \mathrm{BV}$ budget; the zero-budget case is immediate. The signed-grid profiles have budget $O(N)$. The signed-grid estimate, Rademacher–Menshov, and the positive within-cell bound therefore give the squared energy at most

$$C\|\Theta\|_\infty^2 \Lambda \widetilde{M} 2^{A_m t}\left[(1+\log_2 N)^2 N^2 (L^{-\beta} + 2^{-\beta t}) + N^{-1} + C_n 2^{-t}\right].$$

The three remaining rate inequalities and absorption of the polynomial factor in t give $C2^{-\eta_m t}$ for some $\eta_m > 0$. Absolute coefficient summability justifies interchange with the literal annular integrals on a common conull set of X, simultaneously for all radii; the same envelope bounds finite variation sums and their tails. Thus the estimate holds for the actual path, not a surrogate finite-mode operator. Set $G = \mathscr{P}$ and $E = 0$ in this branch. Taking the minimum of the low exponent and the finitely many η_m gives one positive rate before the polynomial is chosen.

To globalize, group whole packet cubes by the R-grid address of their centers; do not clip a packet by an output block, which would destroy its mean zero. Each group's normalized packets and anchors lie in a fixed box, and its trajectories have support in a fixed enlargement. These enlargements have

bounded overlap. The squared L^2 and L^1 Jacobian factors R^n cancel $\widetilde{M} = R^{-n}M$ exactly. This proves the original estimates with independent bounded parent anchors. For the bounded-density setup of Proposition 19.2, $\|d_U\|_1 \le \Lambda|U|$ and a parent center is within $\sqrt{n}R$ of its descendant center. The same construction therefore applies there; taking a square root divides the good decay exponent by two and leaves a coefficient-uniform constant. The argument here uses the terminal and flat high-block estimates, not Proposition 19.2, so this alternative derivation is not a circular input to the concentration-stable estimate.

20.4 Exact oriented algebra and the collective high-shell estimate

Let

$$\mathscr{V}_m = \{\text{homogeneous degree-}m\text{ polynomials in } (x,y)\}/\{p(x)+q(y)\}.$$

Each class has a unique representative consisting only of monomials which have positive degree in both x and y. We use this representative whenever an element of $\mathscr{V}_m$ is evaluated. Let $\mathscr{W}_m$ be the space of homogeneous degree-m polynomials in (W,X). For such a representative put

$$\mathscr{E}H(W,X) = \mathrm{Mix}_{W,X}\{H(W+X,0) - H(W+X,W)\}. \tag{20.66}$$

Lemma 20.11 (Oriented graph injectivity)**.** *For every integer $m \ge 2$, the map $\mathscr{E} : \mathscr{V}_m \to \mathscr{W}_m$ is injective and*

$$\|\mathscr{E}H\|_{\mathrm{coeff}} \ge c_{n,m}\|[H]\|_{\mathrm{coeff}}. \tag{20.67}$$

Moreover

$$\mathscr{E}H = \sum_{\substack{A+B=m\\A,B\ge 1}} E_{A,B}(W,X), \tag{20.68}$$

where $E_{A,B}$ is bihomogeneous of bidegree (A,B).

Proof. If the mixed part in (20.66) vanishes, the expression in braces is $A(W)+B(X)$. At $W=0$ it vanishes, and hence B is constant. Substituting $x = W+X$ and $y = W$ shows that $H(x,0)-H(x,y)$ is a function of y alone. Thus H is pure. Quantitative injectivity gives (20.67) by norm equivalence on the fixed finite-dimensional spaces. Homogeneity and the identities at $W=0$ and $X=0$ give (20.68). □

We next record the exact projective calculation used below. Its purpose is to identify ordinary coefficients of one common numerator; no formal asymptotic expansion in the scale ratio is used.

Lemma 20.12 (Exact projective numerator tags)**.** *Fix $D_0 \ge 1$, a multi-index γ with $q = |\gamma|$, and a polynomial $Q(s) = \sum_{p=0}^{P} c_p s^p$, where $P \in \mathbb{N}_0$ and $P+q \le D_0 - 1$. Let $\delta > 0$, $0 < \alpha \le 1$, and let $\omega \in \mathbb{R}^k$, where k is the length of γ. On the set $D := V - \delta u \ne 0$, put*

$$t = \frac{\delta u}{D}, \qquad r_\perp = \frac{\delta\alpha\omega}{D},$$

and put

$$F_{q,Q}(u,V,\omega) = r_\perp^\gamma Q(t).$$

Then

$$D^{D_0+2}\partial_u\partial_V F_{q,Q} = \alpha^q\omega^\gamma \sum_{j=0}^{D_0-q} \delta^{q+1+j}\tau_{q,j}(Q)u^j V^{D_0-q-j}, \tag{20.69}$$

where the functionals $\tau_{q,j}$ depend only on D_0, q, j. If $q = 0$ and $P \ge 1$, the map

$$(c_1,\dots,c_P) \longmapsto (\tau_{0,0}(Q),\dots,\tau_{0,P-1}(Q))$$

is invertible. If $q \ge 1$, the map

$$(c_0,\dots,c_P) \longmapsto (\tau_{q,0}(Q),\dots,\tau_{q,P}(Q))$$

is invertible. The inverse norms are bounded in terms of D_0 only. When $q = P = 0$, the first map is the empty map and the assertion is vacuous. In particular, every detecting tag in the case $q = 0$, $P \geq 1$ has order $\kappa = q+1+j \leq P$, and in the case $q \geq 1$ it has order $\kappa = q+1+j \leq q+P+1$. After multiplication by D^{D_0+2}, the displayed equality is a polynomial identity. Although the derivation below uses the dense open set on which $DV \neq 0$, the cleared identity therefore extends across both $D = 0$ and $V = 0$.

Proof. Set $z = \delta u/V$ and

$$H_{q,Q}(z) = (1-z)^{-q} Q\left(\frac{z}{1-z}\right).$$

Apart from the factor $\alpha^q \omega^\gamma$, one has

$$F_{q,Q} = \delta^q V^{-q} H_{q,Q}(z).$$

Consequently

$$\partial_u \partial_V F_{q,Q} = -\alpha^q \omega^\gamma \delta^{q+1} V^{-q-2} \big[(q+1)H'_{q,Q}(z) + zH''_{q,Q}(z)\big]. \tag{20.70}$$

Multiplication by $D^{D_0+2} = V^{D_0+2}(1-z)^{D_0+2}$ gives a polynomial. The coefficient of z^j becomes the ordinary coefficient of $u^j V^{D_0-q-j}$, with the exact factor δ^{q+1+j}. This proves (20.69).

Write $H_{q,Q}(z) = \sum_{s\geq 0} h_s z^s$. The coefficient of z^{s-1} in the bracket in (20.70) is $s(q+s)h_s$. Multiplication by $(1-z)^{D_0+2}$ is triangular with diagonal one. It therefore suffices to inspect $(h_1, \dots, h_P)$ when $q = 0$, and $(h_1, \dots, h_{P+1})$ when $q \geq 1$. For $q = 0$, $Q(z/(1-z))$ is triangular with diagonal one on $(c_1, \dots, c_P)$. For $q \geq 1$, if the latter $P+1$ coefficients vanished, then, after $s = z/(1-z)$,

$$(1+s)^q Q(s) = c_0 + O(s^{P+2}).$$

The first P equations make Q equal to c_0 times the degree-P Taylor polynomial of $(1+s)^{-q}$, while the coefficient of s^{P+1} is a nonzero multiple of $c_0\binom{q+P}{P+1}$. Hence $c_0 = 0$, and then all $c_p = 0$. Finite-dimensional norm equivalence gives the quantitative assertion. □

We next define the unequal-shell phase appearing in the correlation. Throughout the remainder of this subsection assume that P has a highest nonpure homogeneous block P_m; necessarily $m \geq 2$. Write, modulo pure phases,

$$P(a+X, a+Y) = p_a(X) + q_a(Y) + \sum_{\ell=2}^{m} H_{\ell,a}(X,Y), \tag{20.71}$$

where every $H_{\ell,a}$ is homogeneous of degree ℓ, and where $H_{m,a} = P_m$. For $W, X \in \mathbb{R}^n$, set

$$\Phi_{a,S}(W,X) = P(a+S(W+X), a) - P(a+S(W+X), a+SW). \tag{20.72}$$

Modulo a function independent of X,

$$\Phi_{a,S}(W,X) \equiv \sum_{\ell=2}^{m} S^\ell \mathscr{E} H_{\ell,a}(W,X). \tag{20.73}$$

Indeed, the pure output phase cancels, the pure input phase is independent of X, and a mixed representative has no pure-X term in the difference defining $\mathscr{E}$.

Fix once and for all annular constants $0 < c_* < C_* < \infty$ containing all values of $|W|$ allowed by the two radial supports, and then choose C_0 sufficiently large, depending only on those fixed supports and annular constants. Thus $R \leq S/C_0$ makes $\delta = R/(S\rho)$ uniformly small throughout the overlap region.

For completeness we now give a concrete rational atlas and frame. For $1 \leq j \leq n$ and $s \in \{-1, 1\}$ put $c_{j,s} = se_j$ and

$$U_{j,s} = \{\theta \in \mathbb{S}^{n-1} : c_{j,s} \cdot \theta > (2\sqrt{n})^{-1}\}, \qquad U^*_{j,s} = \{\theta \in \mathbb{S}^{n-1} : c_{j,s} \cdot \theta > (4\sqrt{n})^{-1}\}.$$

The smaller sets cover the sphere and $U_{j,s} \Subset U_{j,s}^*$. Choose a fixed signed permutation $C_{j,s} \in O(n)$ with $C_{j,s}e_1 = c_{j,s}$. In stereographic coordinates $\vartheta \in \mathbb{R}^{n-1}$ from the pole $-c_{j,s}$, write

$$\theta_{j,s}(\vartheta) = C_{j,s}\left(\frac{1-|\vartheta|^2}{1+|\vartheta|^2}, \frac{2\vartheta}{1+|\vartheta|^2}\right).$$

On $U_{j,s}^*$ the coordinate ϑ ranges over a fixed bounded set; the chart Jacobian and its inverse are therefore uniformly bounded. Let $a_{j,s} = -c_{j,s}$, choose a fixed signed permutation $B_{j,s} \in O(n)$ satisfying $B_{j,s}e_1 = a_{j,s}$, and define the Householder frame

$$O_{j,s}(\vartheta) = \left(I - 2\frac{(a_{j,s} - \theta_{j,s}(\vartheta))(a_{j,s} - \theta_{j,s}(\vartheta))^T}{|a_{j,s} - \theta_{j,s}(\vartheta)|^2}\right) B_{j,s}. \tag{20.74}$$

This matrix is orthogonal and $O_{j,s}(\vartheta)e_1 = \theta_{j,s}(\vartheta)$. Moreover,

$$|a_{j,s} - \theta|^2 = 2(1 + c_{j,s}\cdot\theta) \geq 2\big(1 + (4\sqrt{n})^{-1}\big) \quad \text{on } U_{j,s}^*.$$

Thus every matrix entry is rational in ϑ; after taking the fixed products needed as common denominators, those denominators and their reciprocals are uniformly bounded on the enlarged chart domains. This also remains true for the finitely many fixed powers that occur below.

Order these $2n$ charts as $U_1, \dots, U_{2n}$, relabeling the corresponding maps as θ_χ and O_χ, and make the smaller cover disjoint by putting

$$E_1 = U_1, \qquad E_\chi = U_\chi \setminus \bigcup_{\nu<\chi} U_\nu.$$

These are measurable, disjoint, and cover the sphere. Write $\chi(W)$ for the unique active index satisfying $W/|W| \in E_{\chi(W)}$. Finally, the annular coordinate map $(\rho, \vartheta) \mapsto W = \rho\theta_\chi(\vartheta)$ is uniformly bi-Lipschitz for $c_* \leq \rho \leq C_*$ on each enlarged chart. We use the following precise convention. A *cleared chart-polynomial vector* on this annulus is a vector F_χ for which

$$\widetilde{F}_\chi(\rho, \vartheta) := F_\chi(\rho\theta_\chi(\vartheta))$$

is an ordinary polynomial vector of degree at most a fixed structural bound, after the fixed common denominator has been incorporated into F_χ. Its chart coefficient norm is

$$\|F_\chi\|_{\mathrm{ch},\chi} := \|\widetilde{F}_\chi\|_{\mathrm{coeff},(\rho,\vartheta)}. \tag{20.75}$$

On the fixed enlarged coordinate box, finite-dimensional norm equivalence gives

$$\|\widetilde{F}_\chi\|_\infty + \|\nabla_{\rho,\vartheta}\widetilde{F}_\chi\|_\infty \leq C\|F_\chi\|_{\mathrm{ch},\chi}. \tag{20.76}$$

Any two fixed clearing conventions give equivalent norms: after comparing supremum norms their bounded, nowhere-vanishing denominator ratio is absorbed, and coefficient/supremum norm equivalence is used on the fixed polynomial spaces. In particular, all later polynomial sublevel arguments are made in (ρ, ϑ) coordinates with (20.75); they do not regard $\rho = |W|$ as an ambient polynomial in W.

In frame coordinates define

$$G_{\ell,a,\chi}(\vartheta; T) = \rho^{-\ell}\mathscr{E}H_{\ell,a}\big(\rho O_\chi e_1, \rho O_\chi T\big). \tag{20.77}$$

This is independent of ρ, is a polynomial in T with no constant term, and has degree at most $\ell - 1$.

Proposition 20.13 (Oriented unequal-shell coefficient)**.** *Assume $n \geq 2$. Let P_m be the highest nonpure homogeneous block of a real polynomial P of degree at most d, put*

$$L_R = R^m\|P_m\|_{\mathrm{coeff}},$$

and suppose $0 < R \le S/C_0$. *For* $y = a + SW$, *assume* $c_* \le |W| \le C_*$, *where* $a, W \in \mathbb{R}^n$; *put* $\rho = |W|$. *On every chart* χ *there is a bounded-degree cleared chart-polynomial vector* $\mathbf{p}_{a,R,S,\chi}(W)$, *independent of the polar band parameter* α, *such that*

$$\|\mathbf{p}_{a,R,S,\chi}\|_{\mathrm{coeff}} \ge 1. \tag{20.78}$$

On the disjoint active-chart partition define

$$\mathbf{p}_{a,R,S}(W) := \mathbf{p}_{a,R,S,\chi(W)}(W). \tag{20.79}$$

Thus $\mathbf{p}$ *is piecewise chart-polynomial, with a unique value at every* W. *Every later incidence argument is applied after splitting the atoms into the finitely many active sets* E_χ. *Here* $\|\mathbf{p}\|_{\mathrm{coeff}}$ *means the chart norm* (20.75), *whereas* $|\mathbf{p}(W)|$ *denotes the Euclidean norm in its fixed finite-dimensional value space. The coefficient lower bound in* (20.78) *is not asserted to be a pointwise lower bound; small values of* $|\mathbf{p}(W)|$ *are handled by the incidence estimate.*

More precisely, put $\delta = R/(S\rho)$. *For* $0 < \alpha \le 1$, $\omega \in \mathbb{S}^{n-2}$, *and on each member of the fixed finite family of* (u,V)*-rectangles arising from the radial supports, use variables*

$$D = V - \delta u, \qquad T = T_{\delta,\alpha,\omega}(u,V) = \frac{(\delta u, \delta\alpha\omega)}{D}. \tag{20.80}$$

Every such rectangle has a fixed enlargement on which $D = V - \delta u$ *is uniformly bounded above and away from zero. The actual unequal-shell phase is*

$$\begin{aligned}\Psi_{a,R,S,W,\alpha,\omega}(u,V) &= \Phi_{a,S}\big(W, \rho O_\chi T_{\delta,\alpha,\omega}(u,V)\big)\\ &= P\left(a + S\rho O_\chi \frac{(V,\delta\alpha\omega)}{D}, a\right)\\ &\quad - P\left(a + S\rho O_\chi \frac{(V,\delta\alpha\omega)}{D}, a + S\rho O_\chi e_1\right).\end{aligned} \tag{20.81}$$

Since $\deg P \le d$, *the phase in* (20.81) *has the canonical common denominator*

$$Q(u,V) = D^d = (V - \delta u)^d.$$

Define its literal mixed numerator by

$$\mathscr{N}_\Psi(u,V) := D^{d+2}\partial_u\partial_V\Psi(u,V). \tag{20.82}$$

This is a polynomial of bounded degree. For the canonical presentation,

$$\mathfrak{M}_{D^d}\Psi = D^{3d}\partial_u\partial_V\Psi = D^{2d-2}\mathscr{N}_\Psi.$$

Multiplication by D^{2d-2} *is uniformly bounded below in coefficient norm on these fixed polynomial spaces: this follows from the equivalence of coefficient and supremum norms and the lower bound for* $|D|$ *on the enlarged rectangle. Equation* (20.13) *therefore gives*

$$\|[\Psi]\|_{\mathrm{rat},D^d} \ge c\|\mathscr{N}_\Psi\|_{\mathrm{coeff(u,V)}}. \tag{20.83}$$

For every α *there is a bounded-degree vector* $\mathbf{q}_{\alpha,W}(\omega)$, *made from selected ordinary coefficients of* $\mathscr{N}_\Psi$, *such that*

$$\|[\Psi_{a,R,S,W,\alpha,\omega}]\|_{\mathrm{rat},D^d} \ge c|\mathbf{q}_{\alpha,W}(\omega)|, \tag{20.84}$$

$$\|\mathbf{q}_{\alpha,W}\|_{\mathrm{coeff},\omega} \ge c\alpha^d L_R|\mathbf{p}_{a,R,S,\chi}(W)|. \tag{20.85}$$

The constants are independent of a *and of all coefficients of total degree less than* m.

Proof. Use the fixed signed-cap atlas with smaller and enlarged thresholds $1/(2\sqrt{n})$ and $1/(4\sqrt{n})$. It has $2n$ charts, and taking the first admissible chart in a fixed order gives the measurable disjoint active partition. The rational frame in stereographic coordinates is exactly the displayed Householder reflection composed with a fixed signed permutation: both send the distinguished pole to the same unit direction, and substitution gives equality of every matrix entry. Thus the chart computation uses the literal phase in (20.81), not an unrelated rotation. If an enlarged radial rectangle satisfies $|u| \le U$ and $\eta \le |V| \le V_*$, choose the structural separation C_0 so that $\delta U \le \eta/2$ for $\rho \ge c_*$. Then

$$\eta/2 \le |V - \delta u| \le V_* + \eta/2.$$

This works for both signs of V and simultaneously for the fixed rectangle family and its enlargements.

Pure terms in the first variable cancel in the correlation; pure terms in the second variable are independent of (u,V). They therefore contribute no mixed numerator even if their degree exceeds m. The highest nonpure block is nonzero, so $2 \le m \le d$ follows from the original degree hypotheses. All remaining lower coefficients and all pure higher coefficients stay arbitrary. By (20.73) and (20.77), modulo $F(u)+G(V)$ the phase in (20.81) is

$$\sum_{\ell=2}^{m} S^{\ell}\rho^{\ell} G_{\ell,a,\chi}\left(\vartheta; \frac{(\delta u, \delta\alpha\omega)}{D}\right). \tag{20.86}$$

Expand

$$G_{\ell,a,\chi}(\vartheta;t,r_\perp) = \sum_{q,|\gamma|=q} r_\perp^{\gamma} Q_{\ell,a,\chi,\gamma}(\vartheta;t).$$

Here $\deg Q_{\ell,a,\chi,\gamma} + q \le \ell - 1$, and the constant coefficient of $Q_{\ell,a,\chi,0}$ is zero because $G_{\ell,a,\chi}(\vartheta;0) = 0$. Lemma 20.12, used with $D_0 = d$, shows that the coefficient of $\omega^{\gamma} u^{j} V^{d-q-j}$ in $\mathscr{N}_\Psi$ is exactly

$$C_{q,\gamma,j,\alpha}(\rho,\vartheta) = \alpha^q \sum_{\ell=2}^{m} R^{\kappa} S^{\ell-\kappa} \rho^{\ell-\kappa} \tau_{q,j}(Q_{\ell,a,\chi,\gamma}), \qquad \kappa = q+1+j. \tag{20.87}$$

Thus different tag orders are different ordinary numerator monomials; there is no appeal to a formal expansion in δ.

We make the finite-dimensional coefficient step explicit. Fix the mixed monomial basis of $\mathscr{V}_m$, and use ordinary monomial coefficients in the chart variable ϑ. Choose the canonical common clearing factor $\Delta_\chi(\vartheta) = (1+|\vartheta|^2)^d$ in the rational Householder coordinates (with the fixed signed permutation understood). It clears every frame denominator in degrees at most d, and both it and its reciprocal are bounded on U_χ^*. For $T = (t, r_\perp) \in \mathbb{R} \times \mathbb{R}^{n-1}$ define the cleared restriction map

$$(\mathsf{R}_\chi H)(\vartheta;T) = \Delta_\chi(\vartheta)\mathscr{E}H\big(O_\chi(\vartheta)e_1, O_\chi(\vartheta)T\big). \tag{20.88}$$

By homogeneity this is precisely $\Delta_\chi G_{m,a,\chi}$ when $H = P_m$; in particular it is independent of a and ρ. Write

$$\mathsf{R}_\chi H = \sum_{\gamma} r_\perp^{\gamma} \widetilde{Q}_{\chi,\gamma}(\vartheta;t), \qquad q = |\gamma|.$$

The $q = 0$ polynomial has zero constant coefficient, and $\deg_t \widetilde{Q}_{\chi,\gamma} + q \le m-1$.

Let

$$\mathfrak{I}_m = \{(0,j) : 0 \le j \le m-2\} \cup \{(\gamma,j) : 1 \le |\gamma| \le m-1,\ 0 \le j \le m-1-|\gamma|\}, \tag{20.89}$$

where 0 denotes the zero multi-index, and put $\kappa(\gamma,j) = |\gamma| + 1 + j$. Let D_χ be a fixed degree bound for the cleared chart polynomials and set

$$\mathscr{Z}_{m,\chi} = \bigoplus_{(\gamma,j)\in\mathfrak{I}_m} \mathscr{P}_{\le D_\chi}(\vartheta),$$

with the Euclidean norm of all ordinary monomial coefficients. All tag functionals below are those of Lemma 20.12 with $D_0 = d$, extended coefficientwise in ϑ. Define

$$\mathsf{T}_\chi : \mathscr{V}_m \longrightarrow \mathscr{Z}_{m,\chi}, \qquad \mathsf{T}_\chi H = \big(\tau_{|\gamma|,j}(\widetilde{Q}_{\chi,\gamma}(\vartheta;\cdot))\big)_{(\gamma,j)\in\mathfrak{I}_m}. \tag{20.90}$$

Multiplication by Δ_χ commutes with the tag functionals because it is independent of t and $r_\perp$. Lemma 20.12 therefore recovers, separately for each γ, all coefficients of $\widetilde{Q}_{\chi,\gamma}$: for $q = 0$ it recovers the nonconstant t-coefficients, and for $q \geq 1$ it recovers all of them. Thus vanishing of $\mathsf{T}_\chi H$ implies $\mathsf{R}_\chi H = 0$.

The restriction map R_χ is injective. Indeed, its vanishing implies $\mathscr{E}H(\theta, X) = 0$ for every X and every direction θ in the open chart. In the bidegree decomposition (20.68), distinct X-degrees separate the terms $E_{A,B}$; each is homogeneous in W, so vanishing on an open patch of the unit sphere extends to the corresponding open cone. Hence $\mathscr{E}H = 0$, and Lemma 20.11 gives $H = 0$ in $\mathscr{V}_m$. Consequently T_χ is injective. Its uniform injectivity verifies the application-specific hypothesis (20.6) for the oriented tag space; no blanket injectivity statement for ambient polynomials on the sphere is used. Its uniform constant is the genuine least singular value

$$\sigma_\chi = \min_{\|H\|_{\mathrm{coeff}}=1} \|\mathsf{T}_\chi H\|_{\mathrm{coeff}} > 0, \qquad \sigma_* := \min_\chi \sigma_\chi > 0. \tag{20.91}$$

The minimum is taken on the unit sphere of the fixed finite-dimensional space $\mathscr{V}_m$; the maps depend only on n, m, d and the fixed atlas, not on a, R, S, α or any coefficient of a lower homogeneous block.

Now retain exactly the coordinates in (20.89) and multiply (20.87) by ρ^κ. This is a uniformly invertible diagonal change on $c_* \leq \rho \leq C_*$. For $\alpha = 1$, after the fixed denominator is cleared, the selected polynomial vector is

$$\widehat{C}_{q,\gamma,j}(\rho,\vartheta) = \Delta_\chi(\vartheta) \sum_{\ell=2}^{m} R^\kappa S^{\ell-\kappa} \rho^\ell \tau_{q,j}(Q_{\ell,a,\chi,\gamma}). \tag{20.92}$$

Let π_m be ordinary coefficient projection onto radial degree ρ^m. Since the chart denominator and the tag selection are independent of ρ, both commute with π_m, and therefore

$$\pi_m \widehat{\mathbf{C}}_{a,R,S,\chi} = D_{R,S}\, \mathsf{T}_\chi P_m, \qquad (D_{R,S} v)_{\gamma,j} = R^{\kappa(\gamma,j)} S^{m-\kappa(\gamma,j)} v_{\gamma,j}. \tag{20.93}$$

This identity also shows directly why no translated lower-degree term can enter a protected coordinate: the block of total degree ℓ carries exactly the radial monomial ρ^ℓ, while $H_{m,a} = P_m$. For every retained coordinate, $1 \leq \kappa(\gamma, j) \leq m$; hence, since $R \leq S$,

$$\|D_{R,S} v\|_{\mathrm{coeff}} \geq R^m \|v\|_{\mathrm{coeff}}. \tag{20.94}$$

Combining (20.91)– (20.94) with the bounded coefficient projection gives

$$\|\widehat{\mathbf{C}}_{a,R,S,\chi}\|_{\mathrm{coeff}} \geq c_0 R^m \|P_m\|_{\mathrm{coeff}} = c_0 L_R, \tag{20.95}$$

uniformly in all lower blocks. For orientation, the retained orders are $\{1,2\}$ when $m = 2$ and $\{1,2,3\}$ when $m = 3$, giving respectively the scale factors RS, R^2 and $RS^2, R^2 S, R^3$. Define

$$\widetilde{\mathbf{p}}_{a,R,S,\chi}(\rho,\vartheta) = \frac{\widehat{\mathbf{C}}_{a,R,S,\chi}}{c_0 L_R}.$$

This is an ordinary polynomial vector in (ρ, ϑ) because the fixed factor Δ_χ has already cleared all frame denominators. Counting degrees in the quadratic frame numerators and in the radial powers gives the uniform total-degree bound $4d$. The norm here is the Euclidean norm of all ordinary coefficients, as in (20.75); coefficient projection is contractive and the scale diagonal has least singular value at least R^m.

Choose c_0 as the minimum of the positive structural constants over the finite atlas and the finitely many $2 \le m \le d$, before P, a, R, S, α. Define

$$\mathbf{p}_{a,R,S,\chi}(\rho\theta_\chi(\vartheta)) := \widetilde{\mathbf{p}}_{a,R,S,\chi}(\rho,\vartheta).$$

This proves (20.78).

For fixed W, group the selected ordinary coefficients of $\mathscr{N}_\Psi$ according to (q,j), keeping the degree-q polynomial in ω intact, and call this vector $\mathbf{q}_{\alpha,W}(\omega)$. Thus its entries are the actual $u^j V^{d-q-j}$ coefficient polynomials, without inserting clearing factors into the definition of $\mathbf{q}$. Their Euclidean value norm is bounded by a structural multiple of $\|\mathscr{N}_\Psi\|_{\mathrm{coeff(u,V)}}$. The exact identity $\mathfrak{M}_{D^d}\Psi = D^{2d-2}\mathscr{N}_\Psi$ and the denominator bounds therefore give (20.84) for the genuine Euclidean quotient norm.

To compare coefficients with the protected vector, now multiply its (q,γ,j) coordinate by $\rho^\kappa \Delta_\chi(\vartheta)$. This is a uniformly bounded and uniformly invertible diagonal map on the fixed radial annulus and enlarged chart. By (20.87), the resulting coordinate is $\alpha^q \widehat{C}_{q,\gamma,j}(\rho,\vartheta)$. Distinct γ are distinct ordinary angular monomials, so the Euclidean coefficient norm retains their full square sum; there is no cancellation between different q's. Since $0<\alpha\le 1$ and $0 \le q \le d$, this diagonal scaling has least singular value at least α^d. Consequently

$$\|\mathbf{q}_{\alpha,W}\|_{\mathrm{coeff},\omega} \ge c\alpha^d |\widehat{\mathbf{C}}_{a,R,S,\chi}(W)| \ge c\alpha^d L_R |\mathbf{p}_{a,R,S,\chi}(W)|.$$

This proves (20.85). The normalization of $\mathbf{p}$ was made before α, so one chart vector works on every polar band and every angular vector, in particular on the whole stated sphere. Select its value on the disjoint active chart to obtain $\mathbf{p}_{a,R,S}(W)$. At no step is its coefficient lower bound converted into a pointwise lower bound on $|\mathbf{p}(W)|$. □

Lemma 20.14 (Homogeneous angular restriction)**.** *Let $k \ge 1$ and*

$$\mathbf{Q} = \bigoplus_{q=0}^{d} \mathbf{Q}^{[q]},$$

where the summands occupy disjoint value coordinates and every entry of $\mathbf{Q}^{[q]}$ is a homogeneous degree-q polynomial on $\mathbb{R}^k$. If $k \ge 2$, fix a finite rational atlas $\omega = \omega_\nu(\zeta)$ on $\mathbb{S}^{k-1}$ and fixed denominators Δ_ν clearing all pullbacks through degree d. Then, on every nonempty chart,

$$\|\bigoplus_{q=0}^{d} \Delta_\nu(\zeta)\mathbf{Q}^{[q]}(\omega_\nu(\zeta))\|_{\mathrm{coeff},\zeta} \ge c_{k,d,\nu}\|\mathbf{Q}\|_{\mathrm{coeff}}. \tag{20.96}$$

For $k=1$, evaluation on $\mathbb{S}^0 = \{-1,1\}$ has the same property.

Proof. The displayed pullback is a linear map between fixed finite-dimensional coefficient spaces. If it vanishes on a nonempty chart, the disjoint value coordinates show separately that every homogeneous block $\mathbf{Q}^{[q]}$ vanishes on an open patch of the sphere. Homogeneity extends this vanishing to the corresponding open cone in $\mathbb{R}^k$, so every block is the zero polynomial. Thus the pullback is injective, and its least singular value gives (20.96). When $k=1$, an entry is $c\omega^q$, whose value at either point of $\mathbb{S}^0$ has modulus $|c|$. Notice that keeping distinct degrees in disjoint value coordinates is essential; no quotient by the sphere relation $|\omega|^2=1$ is being taken. □

Corollary 20.15 (Rough unequal-shell alternative)**.** *Let P be a real polynomial of degree at most d with highest nonpure homogeneous block P_m, let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and let $0 < R \le S/C_0$. Put $L_R = R^m \|P_m\|_{\mathrm{coeff}}$, and let b_R, b_S be supported in a fixed compact subinterval of $(0,\infty)$, with $L^\infty + \mathrm{BV}$ norm at most B, where $B \ge 1$. Put*

$$k_R(z) = |z|^{-n}\Theta(z/|z|)b_R(|z|/R), \qquad k_S(z) = |z|^{-n}\Theta(z/|z|)b_S(|z|/S).$$

Fix $a, W \in \mathbb{R}^n$, put $y = a + SW$, and assume $c_* \le |W| \le C_*$. Use the active-chart vector $\mathbf{p}(W) = \mathbf{p}_{a,R,S,\chi(W)}(W)$ from (20.79)*. Then, for $n \ge 2$,*

$$\left| \int e^{i[P(y+z,a)-P(y+z,y)]} \overline{k_R(z)} k_S(z+y-a)\, dz \right| \le CB^C \|\Theta\|_\infty^2 S^{-n} \min\{1, (L_R|\mathbf{p}(W)|)^{-c}\}. \tag{20.97}$$

If $\mathbf{p}(W) = 0$, the last factor is understood to be 1. For $n = 1$ the same conclusion holds, with S^{-1}, for a bounded-degree polynomial vector $\mathbf{p}(W)$ satisfying $\|\mathbf{p}\|_{\mathrm{coeff}} \ge 1$.

Proof. If the two radial supports have no overlap after the translation by SW, the integral is zero in every dimension. Assume that the overlap is nonempty, and first let $n \ge 2$. Then $|z| \simeq R$, $|z+SW| \simeq S$, and the contributing set has measure $O(R^n)$; hence

$$\int |k_R(z) k_S(z+SW)|\, dz \le CB^2 \|\Theta\|_\infty^2 S^{-n}. \tag{20.98}$$

This proves (20.97) when $\mathbf{p}(W) = 0$. Henceforth assume that $\mathbf{p}(W) \ne 0$. Write $W = \rho\theta$, choose the active chart so that $\theta = O_\chi(\vartheta) e_1$, and let $\omega \in \mathbb{S}^{n-2} \subset \mathbb{R}^{n-1}$ denote a transverse frame coordinate. Put

$$\omega_\chi = O_\chi(\vartheta)(0, \omega) \in \theta^\perp$$

and decompose $z = t\theta + r\omega_\chi$, where $r > 0$. Put

$$\xi = \frac{t}{r}, \qquad \eta = \frac{t + S\rho}{r}.$$

Thus the two rays are $z = r(\xi\theta + \omega_\chi)$ and $z + SW = r(\eta\theta + \omega_\chi)$. Choose a zero-extended bounded-variation partition $\{\beta_k\}_{k \ge 0}$ in the dimensionless transverse radius such that, on the relevant range,

$$\sum_{k \ge 0} \beta_k(r/R) = 1, \qquad \operatorname{supp} \beta_k \subset [c2^{-k}, C2^{-k}], \qquad \sup_k \big(\|\beta_k\|_\infty + \operatorname{Var}(\beta_k) \big) \le C,$$

and the supports have uniformly bounded overlap. Absorb the boundedly many largest transverse pieces into $k = 0$, put $\alpha = 2^{-k}$, and on the k-th band write $r \simeq \alpha R$. Set

$$\delta = \frac{R}{S\rho}, \qquad u = \alpha\xi, \qquad V = \alpha\delta\eta, \qquad D = V - \delta u. \tag{20.99}$$

Then, exactly,

$$t = \frac{Ru}{D}, \qquad r = \frac{\alpha R}{D}, \qquad z = \frac{R(u\theta + \alpha\omega_\chi)}{D}, \qquad z + SW = \frac{S\rho(V\theta + \alpha\delta\omega_\chi)}{D}. \tag{20.100}$$

Here $D = \alpha R / r > 0$. Moreover $z/S = \rho O_\chi T_{\delta,\alpha,\omega}(u, V)$, so the phase in these variables is exactly (20.81).

Indeed, cylindrical measure is $r^{n-2}\, dt\, dr\, d\omega$, and

$$\left| \frac{\partial(t, r)}{\partial(u, V)} \right| = \frac{\alpha R^2}{D^3}.$$

Since

$$|z| = \frac{R\sqrt{u^2 + \alpha^2}}{D}, \qquad |z + SW| = \frac{S\rho\sqrt{V^2 + \alpha^2\delta^2}}{D},$$

the cylindrical Jacobian and the two homogeneous kernels give the exact factor

$$(S\rho)^{-n} \alpha^{n-1} D^{n-1} (\alpha^2 + u^2)^{-n/2} (V^2 + \alpha^2\delta^2)^{-n/2}\, du\, dV\, d\omega. \tag{20.101}$$

The radial profiles become

$$\overline{b_R\left(\frac{\sqrt{u^2 + \alpha^2}}{D} \right)}, \qquad b_S\left(\frac{\rho\sqrt{V^2 + \alpha^2\delta^2}}{D} \right), \tag{20.102}$$

while the two rough factors are, respectively,

$$\overline{\Theta\left(\frac{u\theta+\alpha\omega_\chi}{\sqrt{u^2+\alpha^2}}\right)}, \qquad \Theta\left(\frac{V\theta+\alpha\delta\omega_\chi}{\sqrt{V^2+\alpha^2\delta^2}}\right). \tag{20.103}$$

Thus neither rough angular factor is differentiated.

On the support of one transverse band and of the two radial profiles, u,V lie in fixed compact sets and D is bounded above and away from zero. The profile support also gives $\sqrt{u^2+\alpha^2}\gtrsim 1$ and $\sqrt{V^2+\alpha^2\delta^2}\gtrsim 1$; hence the two possibly singular weights in (20.101) and all of their sectional derivatives are uniformly bounded. After a bounded number of sign and monotonicity pieces, the remaining amplitude in (20.101)– (20.102), including the transverse band cutoff, has uniform row and column L^1 bounds and sectional $L^\infty+\mathrm{BV}$ norm at most CB^C. This follows from the one-dimensional BV chain rule on each monotonicity piece. Explicitly, the two profile arguments are

$$\gamma_1(u,V)=\frac{\sqrt{u^2+\alpha^2}}{V-\delta u}, \qquad \gamma_2(u,V)=\frac{\rho\sqrt{V^2+\alpha^2\delta^2}}{V-\delta u}.$$

Writing $s_1=(u^2+\alpha^2)^{1/2}$, $s_2=(V^2+\alpha^2\delta^2)^{1/2}$, and $D=V-\delta u$, direct differentiation gives

$$\partial_u\gamma_1=\frac{uV+\delta\alpha^2}{s_1D^2}, \qquad \partial_V\gamma_1=-\frac{s_1}{D^2}, \qquad \partial_u\gamma_2=\frac{\rho\delta s_2}{D^2}, \qquad \partial_V\gamma_2=-\frac{\rho\delta(uV+\delta\alpha^2)}{s_2D^2}.$$

Thus the only nontrivial sectional critical-point equation is the linear equation $uV+\delta\alpha^2=0$ in the varying variable. Each section therefore has a uniformly bounded number of monotonicity pieces; on each piece $\mathrm{Var}(b_R\circ\gamma_1)\le\mathrm{Var}(b_R)$ and $\mathrm{Var}(b_S\circ\gamma_2)\le\mathrm{Var}(b_S)$. More explicitly, on an interval I on which γ is monotone,

$$\mathrm{Var}_I(b\circ\gamma)\le\mathrm{Var}_{\gamma(I)}(b)\le\mathrm{Var}(b),$$

and the conclusion is trivial if γ is constant. The numerator $uV+\delta\alpha^2$ is affine in either varying coordinate, so each section requires at most one additional cut, independently of α,δ,ρ. Products are controlled by

$$\mathrm{Var}(fg)\le\|f\|_\infty\mathrm{Var}(g)+\|g\|_\infty\mathrm{Var}(f).$$

After extracting the explicit factor α^{n-1} in (20.101), the displayed lower bounds for D and the two square roots give uniformly bounded sectional $W^{1,\infty}$ norms for all remaining algebraic factors. Hence the sectional $L^\infty+\mathrm{BV}$ bounds and the Schur row and column bounds remain uniform as $\alpha,\delta\downarrow 0$.

Fix a compact rectangle containing all the (u,V)-supports above, and partition it by a rectangular grid of sufficiently small fixed mesh. For each α,δ,ρ, retain only the grid rectangles meeting the corresponding support. Since on that support

$$D, \qquad \sqrt{u^2+\alpha^2}, \qquad \sqrt{V^2+\alpha^2\delta^2}$$

are bounded below uniformly, and these functions are uniformly Lipschitz in (u,V) for $0<\alpha\le 1$ and $0<\delta\le\delta_0$, the mesh may be chosen so that the same quantities remain bounded below on a fixed enlargement of every retained rectangle. A fixed tensor-product partition subordinate to the grid has uniformly bounded sectional $L^\infty+\mathrm{BV}$ norms. Refining each section at the at most one zero of $uV+\delta\alpha^2$ gives the monotonicity partitions required by Lemma 20.4. The number and interaction degree of all rectangles are uniform in α,δ,ρ, so their L^2 bounds synthesize by Cauchy–Schwarz with only a fixed loss.

Let $\mathscr{R}^{\mathrm{rect}}_{\alpha,\delta,\rho}$ denote the uniformly finite family of retained rectangles, including the tensor-product partition just constructed. For a rectangle $\mathfrak{b}\in\mathscr{R}^{\mathrm{rect}}_{\alpha,\delta,\rho}$ and fixed ω, let $T^{\mathfrak{b}}_{\alpha,W,\omega}$ be the corresponding operator from the V-interval to the u-interval, after the explicit factor $(S\rho)^{-n}\alpha^{n-1}$ in (20.101) has been removed. Put, with zero extension to the fixed ambient intervals,

$$f_{\alpha,W,\omega}(V)=\Theta\left(\frac{V\theta+\alpha\delta\omega_\chi}{\sqrt{V^2+\alpha^2\delta^2}}\right),$$
$$g_{\alpha,W,\omega}(u)=\Theta\left(\frac{u\theta+\alpha\omega_\chi}{\sqrt{u^2+\alpha^2}}\right).$$

If the tensor cutoff of $\mathfrak{b}$ is $\eta_{\mathfrak{b}}(u)\widetilde{\eta}_{\mathfrak{b}}(V)$, set $f^{\mathfrak{b}}_{\alpha,W,\omega}=\widetilde{\eta}_{\mathfrak{b}}f_{\alpha,W,\omega}$ and $g^{\mathfrak{b}}_{\alpha,W,\omega}=\eta_{\mathfrak{b}}g_{\alpha,W,\omega}$; their L^2 norms on the fixed intervals are at most $C\|\Theta\|_\infty$. Lemma 20.4, (20.84), and the finite interaction degree give

$$\sup_{\mathfrak{b}\in\mathscr{R}^{\mathrm{rect}}_{\alpha,\delta,\rho}}\|T^{\mathfrak{b}}_{\alpha,W,\omega}\|_{2\to2}\le CB^C\min\{1,|\mathbf{q}_{\alpha,W}(\omega)|^{-c_0}\}. \tag{20.104}$$

Here and below the value at a zero of $\mathbf{q}_{\alpha,W}$ is defined to be one. If $\mathscr{I}_\alpha(W)$ is the contribution of the current transverse band, the changes of variables above and the tensor partition give the exact synthesis identity

$$\mathscr{I}_\alpha(W)=(S\rho)^{-n}\alpha^{n-1}\int_{\mathbb{S}^{n-2}}\sum_{\mathfrak{b}\in\mathscr{R}^{\mathrm{rect}}_{\alpha,\delta,\rho}}\Big\langle T^{\mathfrak{b}}_{\alpha,W,\omega}f^{\mathfrak{b}}_{\alpha,W,\omega},g^{\mathfrak{b}}_{\alpha,W,\omega}\Big\rangle\,d\omega. \tag{20.105}$$

All functions here are jointly measurable. The trivial Schur bound is a uniform integrable majorant, so ordinary Fubini justifies (20.105). Cauchy–Schwarz, (20.104), the uniformly finite rectangle sum, and $c_*\le\rho\le C_*$ now yield the explicit first half of the angular synthesis:

$$|\mathscr{I}_\alpha(W)|\le CB^C\|\Theta\|_\infty^2S^{-n}\alpha^{n-1}\int_{\mathbb{S}^{n-2}}\min\{1,|\mathbf{q}_{\alpha,W}(\omega)|^{-c_0}\}\,d\omega. \tag{20.106}$$

Write

$$\mathbf{q}_{\alpha,W}=\bigoplus_{q=0}^{d}\mathbf{q}^{[q]}_{\alpha,W},$$

where different homogeneous degrees q occupy disjoint value coordinates, and every entry of $\mathbf{q}^{[q]}_{\alpha,W}$ is homogeneous of degree q in ω. Set

$$A_\alpha=\|\mathbf{q}_{\alpha,W}\|_{\mathrm{coeff},\omega}.$$

Lemma 20.14, applied before any degrees are combined, shows that on every nonempty rational chart the cleared chart-polynomial vector has coefficient norm at least cA_α. Let $A_*\ge1$ be fixed so large that the chart lower bound cA_α is at least one whenever $A_\alpha\ge A_*$. If $A_\alpha<A_*$, the integral in (20.106) is bounded by the surface measure. If $A_\alpha\ge A_*$, normalize the cleared vector on each chart by its own coefficient norm and apply (20.2) with $\sigma=c_0$. The chart Jacobians and clearing denominators are bounded above and away from zero, so summing the finite atlas gives

$$\begin{aligned}\int_{\mathbb{S}^{n-2}}\min\{1,|\mathbf{q}_{\alpha,W}(\omega)|^{-c_0}\}\,d\omega&\le C\min\{1,A_\alpha^{-c_1}\}\\&\le C\min\{1,(\alpha^dL_R|\mathbf{p}(W)|)^{-c_1}\},\end{aligned} \tag{20.107}$$

where the last inequality is (20.85). When $n=2$, the same calculation is the finite sum over $\mathbb{S}^0$: the $k=1$ clause of Lemma 20.14 gives the required lower bound at either point. Combining (20.106) and (20.107) gives the band bound

$$CB^C\|\Theta\|_\infty^2S^{-n}\alpha^{n-1}\min\{1,(L_R\alpha^d|\mathbf{p}(W)|)^{-c}\}. \tag{20.108}$$

Here ρ is confined to a fixed annulus and has been absorbed into the constant. If $A=L_R|\mathbf{p}(W)|\le1$, summability of α^{n-1} gives the trivial alternative. If $A>1$, split the dyadic sum at $\alpha\simeq A^{-1/d}$; the two resulting geometric series are $O(A^{-c'})$ after decreasing the exponent. This proves (20.97).

For $n=1$, work on one of the finitely many signed overlap intervals and write $z=Rt$. Let $\mathbf{C}(W)$ be the vector of all nonconstant-in-t coefficients of the complete translated correlation phase. Its degree-m part is

$$\sum_{\substack{A+B=m\\A,B\ge1}}c_{A,B}S^AR^BW^At^B. \tag{20.109}$$

For a fixed t^B coefficient, every degree $\ell<m$ contribution has W-degree at most $\ell-B<m-B$, and cannot alter the coefficient of W^{m-B}. Since $S^AR^B\ge R^m$, Lemma 20.11 gives

$$\|\mathbf{C}\|_{\mathrm{coeff},W}\ge cR^m\|P_m\|_{\mathrm{coeff}}=cL_R.$$

Put $\mathbf{p} = \mathbf{C}/(cL_R)$. The product of the two kernels and dz is $S^{-1}b(t)dt$, on finitely many intervals, with $\|b\|_{L^\infty+\mathrm{BV}} \le CB^C\|\Theta\|_\infty^2$. Lemma 20.3 gives the asserted estimate when $\mathbf{p}(W) \neq 0$; when $\mathbf{p}(W) = 0$, the same conclusion is the trivial S^{-1} amplitude bound. □

Lemma 20.16 (One-sided variable-mesh incidence)**.** *Let $0 < \varepsilon \le 1$, let $q_R = \varepsilon R$, let the scale-R packet cubes form a finite disjoint family $\mathcal{Q}_R$ of dyadic cubes of common side length $q_R = \varepsilon R$ in one aligned grid, where $R > 0$. For each $Q \in \mathcal{Q}_R$ choose a point $y_Q \in Q$, and attach a weight $0 \le w_Q \le \Lambda q_R^n$, where $\Lambda \ge 0$. Fix $a \in \mathbb{R}^n$ and a large shell scale $S \ge C_0 R$. If $\mathbf{p}$ is as in Proposition 20.13 for $n \ge 2$, or as constructed in Corollary 20.15 for $n = 1$, then, for the fixed annular-overlap constants $0 < c_* < C_* < \infty$ and every $0 < \tau \le 1$,*

$$\sum_{\substack{Q:\ c_*S \le |y_Q - a| \le C_*S \\ |\mathbf{p}((y_Q-a)/S)| \le \tau}} w_Q \le C\Lambda S^n\{\tau^c + (\varepsilon R/S)^c\}. \tag{20.110}$$

Proof. Throughout this proof retain only cubes with $c_*S \le |y_Q - a| \le C_*S$; every sum and smearing density below uses this annular subfamily. For $n \ge 2$, first split the sum into the finitely many disjoint active sets E_χ from (20.79); on the χ-th piece use $\mathbf{p}_{a,R,S,\chi}$. (For $n = 1$ no chart split is needed.) There is no second denominator clearing at this stage. Indeed, for $n \ge 2$, the normalized vector constructed after (20.95) satisfies

$$\widetilde{\mathbf{p}}_{a,R,S,\chi}(\rho, \vartheta) = \mathbf{p}_{a,R,S,\chi}(\rho\,\theta_\chi(\vartheta))$$

as an ordinary coordinate-polynomial vector after the fixed denominator has already been cleared. For $n = 1$ the vector constructed in the proof of Corollary 20.15 is already an ordinary polynomial vector in W. In either case its vector coefficient norm is at least one. Hence one fixed scalar component on the current chart, called p_0, satisfies $A_0 = \|p_0\|_{\mathrm{coeff}} \ge c > 0$, where c depends only on the fixed vector length. Put $\widetilde{p}_0 = p_0/A_0$. This last polynomial has coefficient norm one. At a bad point it satisfies $|\widetilde{p}_0| \le \tau/A_0 \le C\tau$, regardless of how large the translated lower blocks make A_0. The cubes have the common side $q_R = \varepsilon R$, not merely membership in one dyadic grid. Color their lower-corner indices modulo 4 as in Lemma 20.1; after division by S, put disjoint balls of radius $c\varepsilon R/S$ around the selected points. Work in the fixed enlarged radial annulus $c_*/2 < |W| < 2C_*$. If this radius exceeds a fixed fraction of either the radial collar or, when $n \ge 2$, the angular chart collar $\mathrm{dist}(U_\chi, \partial U_\chi^*)$, the mesh term on the right of (20.110) is already bounded below by a fixed constant. Disjointness and the weight cap give total weight at most $C\Lambda S^n$ in the fixed annulus, so the assertion is then trivial. Otherwise the balls stay in the enlarged radial annulus and, for $n \ge 2$, in the enlarged chart U_χ^*. Uniform bi-Lipschitz equivalence between physical W-balls and chart-coordinate balls (ordinary W-intervals when $n = 1$) changes their radii and volumes by fixed factors only. The normalized, already-cleared polynomial $\widetilde{p}_0$ has uniformly bounded gradient on the enlarged coordinate domain by (20.76), so a ball centered at a bad point lies in the $C(\tau + \varepsilon R/S)$-sublevel. To record the density calculation, put

$$h = \frac{\varepsilon R}{S}, \qquad W_Q = \frac{y_Q - a}{S}.$$

For $n \ge 2$ write $z_Q = (\rho_Q, \vartheta_Q)$ for the active chart coordinate of W_Q; for $n = 1$ put $z_Q = W_Q$. For one color $\mathfrak{c}$ the coordinate balls $B_Q = B(z_Q, ch)$ are disjoint, after a fixed decrease of c, and

$$F_{\mathfrak{c}}(z) = \sum_{Q \in \mathfrak{c}} \frac{w_Q}{|B_Q|} \mathbf{1}_{B_Q}(z)$$

satisfies

$$\|F_{\mathfrak{c}}\|_\infty \le C\frac{\Lambda(\varepsilon R)^n}{h^n} = C\Lambda S^n. \tag{20.111}$$

Every ball centered at a bad point is contained in the $C_1(\tau + h)$-sublevel of $\widetilde{p}_0$. If $C_1(\tau + h) \ge 1$, then $\tau^c + h^c \ge (\tau + h)^c \ge C_1^{-c}$. The disjoint packet cubes whose centers lie in the fixed S-annulus have total volume $O(S^n)$, and hence

$$\sum_{Q\ \mathrm{bad}} w_Q \le \sum_Q w_Q \le C\Lambda S^n \le C\Lambda S^n(\tau^c + h^c).$$

This is the required trivial branch for the current color and chart, without applying a sublevel estimate outside its stated threshold range. For the remaining color–chart pieces we may therefore assume $C_1(\tau+h)<1$. Lemma 20.1, now legitimately applied in the coordinate box, gives

$$\sum_{\substack{Q\in\mathfrak{c}\\ Q\text{ bad}}} w_Q \le \int_{\{|\widetilde{p}_0|\le C_1(\tau+h)\}} F_{\mathfrak{c}}(z)\,dz \le C\Lambda S^n(\tau+h)^{c_0}.$$

Sum the boundedly many colors and active charts, and decrease the exponent so that $(\tau+h)^{c_0}\le C(\tau^c+h^c)$. This proves (20.110). The large atom was fixed before smearing, so $h=\varepsilon R/S$ and no mesh term of size $\varepsilon S/S$ occurs. □

Proposition 20.17 (Collective fixed-sign high-shell estimate)**.** *Let P be a real polynomial of degree at most d, let P_m be its highest nonpure homogeneous block, let $\Theta\in L^\infty(\mathbb{S}^{n-1})$, fix $\kappa>0$, let $s\in\mathbb{N}$, let $\varepsilon=2^{-s}$, and suppose that $\mathscr{R}$ is a finite set of positive dyadic radii such that every $R\in\mathscr{R}$ satisfies*

$$L_R = R^m\|P_m\|_{\mathrm{coeff}} \ge \varepsilon^{-\kappa}. \tag{20.112}$$

At radius $R\in\mathscr{R}$, let $\mathscr{Q}_R$ be a finite family of disjoint dyadic cubes of side εR in one aligned grid; assume the cubes over all radii are globally disjoint and each complex measure ν_Q is supported on Q and satisfies $|\nu_Q|(Q)\le\Lambda|Q|$, where $\Lambda\ge 0$. Let

$$T_R\nu(x)=\int e^{iP(x,y)}K_\Theta(x-y)b_R(|x-y|/R)\,d\nu(y),$$

where $B\ge 1$ and all normalized radial profiles are supported in one fixed compact subinterval of $(0,\infty)$ with $L^\infty+\mathrm{BV}$ norm at most B. Then, uniformly for arbitrary complex coefficients $|e_Q|\le 1$,

$$\Big\|\sum_{R\in\mathscr{R}} T_R\sum_{Q\in\mathscr{Q}_R} e_Q\nu_Q\Big\|_2^2 \le CB^C\|\Theta\|_\infty^2\Lambda M 2^{-c_\kappa s},\qquad M=\sum_{R\in\mathscr{R}}\sum_{Q\in\mathscr{Q}_R}|\nu_Q|(Q). \tag{20.113}$$

Here $c_\kappa>0$, and the implicit constant may depend on κ (in addition to the fixed structural data).

Proof. We prove the stronger bound with B^2 in place of B^C. All constants below are chosen from the fixed dimension, degree bound, annular support and κ, before the polynomial, its highest mixed degree, the scales, the profiles, and the measures. Let $\beta_{\mathrm{eq}}>0$ be the global same-shell row exponent constructed below and let $\gamma_\kappa>0$ be the exponent of the complete separated-shell row. Set

$$c_0=\beta_{\mathrm{eq}}\min\{1,\kappa\}>0,\qquad c_\kappa=\min\{c_0,\gamma_\kappa\}>0. \tag{20.114}$$

The empty scale family is immediate. Otherwise the high threshold forces a nonzero highest mixed block, of degree $2\le m\le d$. All row constants will be uniform over these finitely many degrees. Put $\eta_Q=|\nu_Q|$, $w_Q=\eta_Q(Q)$, and $M_R=\sum_{Q\in\mathscr{Q}_R}w_Q$. The support assumption identifies w_Q with the full variation mass, without changing the physical cap $w_Q\le\Lambda(\varepsilon R)^n$.

Write $k_R(z)=K_\Theta(z)b_R(|z|/R)$ and $f_{R,y}(x)=e^{iP(x,y)}k_R(x-y)$. The polar decomposition and complex linearity realize the actual shell output in the Hilbert space L^2 as

$$F_R:=T_R\sum_{Q\in\mathscr{Q}_R}e_Q\nu_Q=\sum_{Q\in\mathscr{Q}_R}e_Q\int_Q h_Q(y)f_{R,y}\,d\eta_Q(y),\qquad d\nu_Q=h_Q\,d\eta_Q. \tag{20.115}$$

These are genuine Bochner integrals, with $|h_Q|\le 1$ almost everywhere. Indeed the original radial budget gives measurable profiles and the uniform row bound

$$\|f_{R,y}\|_2^2=\|k_R\|_2^2\le CB^2\|\Theta\|_\infty^2R^{-n}. \tag{20.116}$$

The row family is jointly measurable modulo $\eta_Q\otimes dx$ null sets, including when η_Q is singular. An essentially bounded angular representative gives the same L^2 rows and product-measure classes. The

scalar representatives of the Hilbert integrals agree almost everywhere with the literal complex-measure operators, on one common full-measure output set for the finite family. No spatial density or probability normalization is introduced.

Define the actual Gram kernel

$$\mathscr{K}_{R,S}(y,a) = \int \overline{f_{R,y}(x)} f_{S,a}(x)\,dx.$$

Hilbert integration and $|e_Q h_Q| \le 1$ give

$$\|F_R\|_2^2 \le \sum_{Q,A\in\mathscr{Q}_R} \int_Q \int_A |\mathscr{K}_{R,R}(y,a)|\,d\eta_A(a)\,d\eta_Q(y). \tag{20.117}$$

Both variables remain independent even in the same cube. We establish the needed same-shell bound globally, rather than applying a fixed-box statement to the whole input family.

Fix $a \in \mathbb{R}^n$ and arbitrary selected points $y_Q \in \overline{Q}$ at one radius R. Normalize by $v = (y-a)/R$ and use the full phase $P_{a,R}(X,Y) = P(a+RX, a+RY)$. Its highest mixed block has norm L_R; translation preserves that block and dilation multiplies it by R^m, while all lower and pure coefficients remain unrestricted. The Gram change of variables contributes exactly R^{-n}. The normalized cubes have side ε, disjoint interiors, and weights $\widetilde{w}_Q = w_Q/R^n \le \Lambda\varepsilon^n$. If $[r_*, r^*] \subset (0,\infty)$ contains the normalized profile support, then $\mathscr{K}_{R,R}(y_Q,a) = 0$ for $|y_Q - a| > 2r^*R$. Every remaining normalized cube lies in the fixed box $[-2r^* - 1, 2r^* + 1]^n$. This box is local to the fixed row center; there is no common physical box assumption on all packets.

For completeness, the scalar polynomial used in this unary row is extracted after this recentering. Put $D = d+1$. By Hermitian symmetry, $|\mathscr{K}_{R,R}(y_Q,a)| = |\mathscr{K}_{R,R}(a,y_Q)|$; in the latter integral set $x = a + RZ$. Thus the nonconstant normalized phase has coefficient norm

$$\Gamma_a(v) = \big\|P_{a,R}(Z,v) - P_{a,R}(Z,0) - P_{a,R}(0,v) + P_{a,R}(0,0)\big\|_{\mathrm{coeff},Z}.$$

Evaluation of the full mixed polynomial on a fixed unit box, followed by finite-dimensional norm equivalence, supplies Z_* such that normalizing the scalar polynomial $\mathrm{mix}(P_{a,R})(Z_*,v)$ gives a polynomial $p_a(v)$ with

$$\deg p_a \le D, \qquad \|p_a\|_{\mathrm{coeff}} = 1, \qquad L_R|p_a(v)| \le K\Gamma_a(v) \quad \text{for every } v.$$

The constants do not depend on a, R or the lower coefficients. The weighted cube sublevel estimate, applied in the preceding local box to p_a and to the coordinate polynomial v_1, gives, for $0 < \tau \le 1$, mass at most $C\Lambda(\tau+\varepsilon)^\alpha$ for each of the sets $|p_a(v_Q)| \le \tau$ and $|v_{Q,1}| \le \tau$. The total interacting normalized mass is at most $C\Lambda$. Fixed-dimensional coefficient norm equivalence accounts for the scalar normalization, and the Lipschitz enlargement of the sublevel includes arbitrary selected boundary points.

Outside these two bad sets, the clipped separation $\min\{1, |v_Q|\}$ is at least τ and the actual coefficient norm is at least $L_R\tau/K$. Lemma 20.5, with its one-dimensional branch, therefore bounds the normalized Gram norm by

$$CB^2\|\Theta\|_\infty^2[\max\{1, L_R\tau^{D+1}/K\}]^{-c_{\mathrm{eq}}}.$$

Here the quadratic radial factor follows by normalizing the two radial profiles by B before applying the correlation estimate. The crude bound has the same prefactor. Choosing $\sigma = 1/[2(D+1)]$, $\tau = L_R^{-\sigma}$ and $\beta_{\mathrm{eq}} = \min\{\sigma\alpha, c_{\mathrm{eq}}/2\} > 0$ yields

$$\sum_{Q\in\mathscr{Q}_R} w_Q|\mathscr{K}_{R,R}(y_Q,a)| \le C_{\mathrm{eq}}B^2\|\Theta\|_\infty^2\Lambda(L_R^{-\beta_{\mathrm{eq}}} + \varepsilon^{\beta_{\mathrm{eq}}}).$$

The factor R^{-n} is exactly absorbed into $\widetilde{w}_Q = w_Q/R^n$; no output-block counting is needed.

An all-selections estimate transfers to the original measures as follows. For a finite positive measure η supported in a nonempty cube and an integrable real function g, some supported point satisfies $\int g\,d\eta \le \eta(Q)g(y)$; the zero-measure case is immediate. Apply this separately to the finitely many

functions $g_Q(y) = |\mathscr{K}_{R,R}(y,a)|$, then use the row bound for the resulting selection. These functions are integrable by the uniform L^2 row bounds. Thus the same inequality holds with $\sum_Q \int_Q |\mathscr{K}_{R,R}(y,a)|\, d\eta_Q(y)$ on the left. Integrating in a and summing A, justified by the actual Gram integrability and Fubini, proves

$$\|F_R\|_2^2 \le C_{\mathrm{eq}} B^2 \|\Theta\|_\infty^2 \Lambda M_R (L_R^{-\beta_{\mathrm{eq}}} + \varepsilon^{\beta_{\mathrm{eq}}}) \le 2 C_{\mathrm{eq}} B^2 \|\Theta\|_\infty^2 \Lambda M_R 2^{-c_0 s}.$$

This is a first-moment selection argument for a scalar row, not a replacement of a double measure integral by a single point in each cube.

Write $R = 2^j$ and let $\mathscr{J}$ be the finite exponent set. The genuine Hilbert vectors satisfy

$$\Big\| \sum_{j\in\mathscr{J}} F_{2^j} \Big\|_2^2 = \sum_{j\in\mathscr{J}} \|F_{2^j}\|_2^2 + 2\,\mathrm{Re} \sum_{\substack{j,J\in\mathscr{J} \\ j<J}} \langle F_{2^j}, F_{2^J} \rangle. \tag{20.118}$$

Choose a fixed integer $N \ge 0$ so that $j + N < J$ implies the scale separation required by Corollary 20.15. The band $|j - J| \le N$ has at most $2N+1$ neighbors per row and column. Cauchy–Schwarz, or $2uv \le u^2 + v^2$, bounds its absolute Gram sum by $(2N+1)\sum_j \|F_{2^j}\|_2^2$.

For the separated rows, fix the large-shell input point a at radius S before making any small-shell selection. After $x = y + z$ the exact correlation is

$$\mathscr{K}_{R,S}(y,a) = \int e^{i[P(y+z,a) - P(y+z,y)]} \overline{k_R(z)} k_S(z + y - a)\, dz.$$

Annular overlap restricts $(y-a)/S$ to a fixed annulus. Use the actual oriented coefficient vector of Corollary 20.15 and put

$$\tau_R = L_R^{-1/2}. \tag{20.119}$$

Lemma 20.16 bounds the bad small-shell mass by $C\Lambda S^n \{\tau_R^{\alpha_{\mathrm{inc}}} + (\varepsilon R/S)^{\alpha_{\mathrm{inc}}}\}$. The actual correlation has crude size $CB^2\|\Theta\|_\infty^2 S^{-n}$ and, off this bad set, gains $(L_R \tau_R)^{-c_{\mathrm{uneq}}}$. Packing bounds the total interacting small-shell mass by $C\Lambda S^n$. Consequently, for every selected family,

$$\sum_{Q\in\mathscr{Q}_R} w_Q |\mathscr{K}_{R,S}(y_Q, a)| \le CB^2 \|\Theta\|_\infty^2 \Lambda \{L_R^{-\beta_{\mathrm{far}}} + (\varepsilon R/S)^{\beta_{\mathrm{far}}}\},$$

where one may first take $\beta_{\mathrm{far}} \le \min\{\alpha_{\mathrm{inc}}/2, c_{\mathrm{uneq}}/2\}$ and then decrease it uniformly over $2 \le m \le d$. In dimension one the direct two-point-sphere correlation and unary incidence estimate give the same conclusion. No rough angular factor is differentiated.

The row sum is geometric, not a scale-count estimate. If R_0 is the least retained radius, then $L_R = L_{R_0}(R/R_0)^m$ and

$$\sum_{R\in\mathscr{R}} L_R^{-\beta_{\mathrm{far}}} \le C\varepsilon^{\kappa\beta_{\mathrm{far}}}, \qquad \sum_{R \le S/C_{\mathrm{sep}}} (\varepsilon R/S)^{\beta_{\mathrm{far}}} \le C\varepsilon^{\beta_{\mathrm{far}}}.$$

Taking positive lower bounds over the finitely many mixed degrees therefore gives a uniform $\gamma_\kappa > 0$ and a bound $C_{\mathrm{far}} B^2 \|\Theta\|_\infty^2 \Lambda 2^{-\gamma_\kappa s}$ for the complete small-shell row, at every fixed a. The first-moment selection argument transfers this whole finite scale-by-packet row to the original η_Q. Integrating against the large-shell measures, using Fubini for the actual Gram norm, gives

$$\sum_{\substack{j\in\mathscr{J} \\ j+N<J}} \sum_{Q\in\mathscr{Q}_{2^j}} \sum_{A\in\mathscr{Q}_{2^J}} \int_Q \int_A |\mathscr{K}_{2^j,2^J}(y,a)|\, d\eta_A(a)\, d\eta_Q(y)$$
$$\le C_{\mathrm{far}} B^2 \|\Theta\|_\infty^2 \Lambda M_{2^J} 2^{-\gamma_\kappa s}.$$

Thus the large-shell mass occurs once, with no additional scale or packet factor and with arbitrary singular input measures.

Finally set $A = B^2\|\Theta\|_\infty^2\Lambda$, $D_0 = 2C_{\mathrm{eq}}A2^{-c_\kappa s}$ and $E_0 = C_{\mathrm{far}}A2^{-c_\kappa s}$. The diagonal bounds are $\|F_{2^j}\|_2^2 \le D_0M_{2^j}$; the separated absolute inner-product row is at most $E_0M_{2^J}$ by Hilbert integration and the preceding triple-Gram bound. Combining the band and the two conjugate far orientations in (20.118) gives

$$\Big\|\sum_{R\in\mathscr{R}}F_R\Big\|_2^2 \le \{(2N+1)D_0+2E_0\}\sum_R M_R \le CB^2\|\Theta\|_\infty^2\Lambda M2^{-c_\kappa s}.$$

Since $B\ge 1$, enlarging the harmless exponent proves (20.113). The Hilbert representatives equal the literal grouped operator in the statement. All arguments retain arbitrary packetwise coefficients of modulus at most one; zero masses and $\Lambda = 0$ require no division or limiting argument. Only per-scale disjoint interiors were needed, so the stated global disjointness and grid hypotheses are more than sufficient. □

21 The unphased weak endpoint

We prove the unphased weak endpoint simultaneously in every dimension $n \ge 1$. The terminal one-shell estimate already has a direct one-dimensional branch in Proposition 18.5, and the long-variation fixed-sign estimate has the direct one-dimensional branch of Lemma 18.4. Thus the selector-free rank argument below applies in dimension one as well. The good part of the Calderón–Zygmund decomposition is controlled by the $p = 2$ case of Proposition 15.9:

$$\|\mathbf{J}(T^0_{\Omega,\varepsilon}h : \varepsilon > 0)\|_2 \le C_n\mathfrak{L}(\Omega)\|h\|_2, \qquad \int\Omega = 0. \tag{21.1}$$

Proposition 21.1 (Unphased weak critical-jump inequality on core data)**.** *For mean-zero* $\Omega \in L\log L(\mathbb{S}^{n-1})$ *and every bounded compactly supported* f*,*

$$\|\mathbf{J}(T^0_{\Omega,\varepsilon}f : \varepsilon > 0)\|_{L^{1,\infty}} \le C_n\mathfrak{L}(\Omega)\|f\|_1. \tag{21.2}$$

Proof. We prove the stronger assertion for every $f \in L^1$, with the paired-path seminorm $\mathscr{N}$ of (2.1) in place of $\mathbf{J}$. Write

$$A = \|\Omega\|_{L^1(\mathbb{S}^{n-1})}, \qquad G = \mathfrak{L}(\Omega), \qquad F = \|f\|_1.$$

If $G = 0$, the angular function vanishes almost everywhere, and polar coordinates make every literal truncation zero. We may therefore assume $G > 0$. Constants below depend only on n; the angular gauge is not normalized away.

Dyadic decomposition and finite gap paths. Fix a height $\alpha > 0$. Select the maximal dyadic cubes Q with $|Q|^{-1}\int_Q|f| > \alpha$, and put

$$g = f\mathbf{1}_{(\cup Q)^c} + \sum_Q f_Q\mathbf{1}_Q, \qquad b_Q = (f - f_Q)\mathbf{1}_Q.$$

Dyadic differentiation, maximality, and disjointness give

$$\sum_Q|Q| \le F/\alpha, \quad \|g\|_2^2 \le 2^n\alpha F, \quad \int b_Q = 0, \quad \|b_Q\|_1 \le 2^{n+1}\alpha|Q|, \quad \Sigma := \sum_Q\|b_Q\|_1 \le 2F. \tag{21.3}$$

Only g, not f or the individual packets, needs to belong to L^2. Let $N = 4n+1$ and fix the exact parameters

$$\delta = \frac{1}{6144N}, \qquad \Theta_s = A2^{\delta s}, \qquad h_s = \frac{1}{8}2^{-s/(192N)}. \tag{21.4}$$

Choose a dimensional integer $s_0 \ge 4$ above the terminal-packet threshold. For each retained gap $s \ge s_0$, let $\mathscr{Q}_s$ be any finite subfamily of the selected bad cubes. Write $B^s_m = \sum_{Q\in\mathscr{Q}_s:\ell(Q)=2^m} b_Q$. For an angular function Ξ, its actual finite gap path is

$$P^\Xi_s(\varepsilon) = \sum_j K_\Xi\mathbf{1}_{\{\max(\varepsilon,2^j)<|z|\le 2^{j+1}\}} * B^s_{j-s}, \qquad \varepsilon > 0.$$

The sum in j is finite, since every retained cube occurs at exactly one shell $j = m+s$. All estimates will be uniform in these finite families and in the finite set $\mathscr{H}$ of retained gaps.

Raw angular tails and the actual short and smoothing terms. Split Ω into its raw tail $\Omega_s^{\mathrm{tail}} = \Omega \mathbf{1}_{\{|\Omega|>\Theta_s\}}$ and the complementary bounded function. Choose a measurable representative Ψ_s of the latter with $|\Psi_s| \le \Theta_s$ everywhere. Neither summand is corrected to have mean zero. The angular almost-everywhere identity pulls back under polar coordinates to the original kernels and hence, on one conull set, to every positive radius of every finite gap path. Put

$$U_{s,\varepsilon} = P_s^{\Psi_s}(\varepsilon), \qquad \mathrm{Tail}_{s,\varepsilon} = P_s^{\Omega_s^{\mathrm{tail}}}(\varepsilon). \tag{21.5}$$

For any increasing finite list of radii, the annular regions of its successive increments are disjoint. Absolute convolution therefore provides a measurable envelope $\mathscr{T}_s$ such that

$$V^1(\mathrm{Tail}_{s,\varepsilon} : \varepsilon > 0) \le \mathscr{T}_s \quad \text{a.e.}, \qquad \|\mathscr{T}_s\|_1 \le C\|\Omega_s^{\mathrm{tail}}\|_1 \Sigma. \tag{21.6}$$

Indeed its terms are $|K_{\Omega_s^{\mathrm{tail}}}|\mathbf{1}_{\{2^j<|z|\le 2^{j+1}\}} * |b_Q|$; the kernel integral is $\log 2\, \|\Omega_s^{\mathrm{tail}}\|_1$. There is no shell-counting loss. The homogeneous entropy count gives

$$\sum_{s\ge 0} \|\Omega_s^{\mathrm{tail}}\|_1 \le C_\delta G, \qquad \Big\| \sum_{s\in\mathscr{H}} \mathscr{T}_s \Big\|_1 \le CGF.$$

To see the count directly, a value $|\Omega|/A$ can exceed $2^{\delta s}$ for at most $C_\delta \log(2+|\Omega|/A)$ nonnegative integers s; integrate this inequality against $|\Omega|$. The case $A = 0$ was already excluded.

Let

$$\mathscr{S}_s = \left(\sum_j V^2(U_{s,\varepsilon} : 2^j \le \varepsilon \le 2^{j+1})^2 \right)^{1/2}.$$

On each dyadic parameter interval the other shells are constant, so its variation is exactly the corresponding terminal-shell variation. Proposition 18.5, applied to the actual mean-zero spatial packets and the bounded angular function, yields

$$\|\mathscr{S}_s\|_2^2 \le C\Theta_s^2 2^{-s/(768N)} \alpha F. \tag{21.7}$$

Angular mean zero is not required for this packet estimate.

Let ϑ be the fixed smooth increasing transition, equal to zero on $(-\infty, 0]$ and one on $[1,\infty)$, and set

$$a_s(r) = \vartheta\left(\frac{r-1+h_s}{2h_s}\right) - \vartheta\left(\frac{r-2+h_s}{2h_s}\right), \qquad H_j^s(z) = K_{\Psi_s}(z) a_s(2^{-j}|z|).$$

These are the actual one-cell smoothed kernels. Their radial support is in a fixed enlargement of the shell and their full-kernel budget satisfies

$$M_N(H^s) \le C_n \Theta_s h_s^{-N}. \tag{21.8}$$

This follows by differentiating the scaled transitions and the radial factor r^{-n}, with $r \simeq 2^j$ on the support. The endpoint collars have total relative length $O(h_s)$; consequently

$$\|K_{\Psi_s}\mathbf{1}_{\{2^j<|z|\le 2^{j+1}\}} - H_j^s\|_1 \le C_n \Theta_s h_s.$$

Define the literal smoothed suffix and its sharp-suffix error by

$$L_s(k) = \sum_{j\ge k} H_j^s * B_{j-s}^s, \qquad \mathscr{L}_s = V^2(L_s(k) : k \in \mathbb{Z}), \qquad \mathscr{E}_s = V^1(U_{s,2^k} - L_s(k) : k \in \mathbb{Z}).$$

A suffix of shell errors has consecutive differences equal to individual shell errors. Thus Young's inequality and the one-shell assignment of each packet give

$$\|\mathscr{E}_s\|_1 \le C_n \Theta_s h_s F. \tag{21.9}$$

No variation estimate for continuously truncated error kernels is used. The deterministic dyadic long–short decomposition for V^2, obtained by splitting each increment at its first and last dyadic endpoints, gives pointwise

$$V^2(U_{s,\varepsilon} : \varepsilon > 0) \le 3(\mathscr{S}_s + \mathscr{L}_s + \mathscr{E}_s). \tag{21.10}$$

Here all three quantities have measurable representatives: for the short term use continuity of finite annular integrals on compact positive-radius intervals and rational partitions; the other two are variations on a countable parameter set.

Finite fixed-rank control, with direct binary summation. We record the adjacent-grid construction needed also in subsequent proofs. The global nested systems

$$\mathscr{D}^a = \left\{ 2^{-k}\big([0,1)^n + m + (-1)^k a\big) : k \in \mathbb{Z},\ m \in \mathbb{Z}^n \right\}, \qquad a \in \{0, 1/3, 2/3\}^n, \tag{21.11}$$

are nested because $3a$ is integral. At any fixed scale their three one-dimensional boundary classes are separated by one third of a side. Choosing a side strictly larger than three times a closed input interval therefore puts it in one of these cells. First enlarge the input box slightly; taking coordinate products then places the original closed box in the interior of an output cube of comparable size. This proves the closed-enlargement and essential-support construction used below, with a fixed finite number of grid and generation-offset classes.

For the present finite packet family one can use the following exact 2^n-color construction. Choose $M \in \mathbb{Z}$ with $j + 5 \le M$ for every occurring shell. Translate the ordinary dyadic grid by vectors whose coordinates are either 0 or $2^M/3$. At every scale 2^{j+5} the two boundary lattices are separated by at least $2^{j+5}/3$: a power of two is not divisible by three. The support of $H_j^s * b_Q$ lies in the box $|x_i - c_{Q,i}| \le 5\,2^j$, whose side $10\,2^j$ is strictly less than $2^{j+5}/3$. The coordinatewise covering argument assigns it to a cube $R(Q)$ of side 2^{j+5} in one of these fixed translated grids. The strict slack also allows a slightly enlarged box, so, if needed, the original support is inside the interior. In particular,

$$\operatorname{ess\,supp}(H_j^s * b_Q) \subset R(Q). \tag{21.12}$$

All grids, assignments, and subsequent ranks are fixed before evaluating x or choosing signs. The covering estimate, valid uniformly in M, is

$$\sum_{Q \in \mathscr{Q}_s} |R(Q)| \le 2^{(s+6)n} \sum_{Q \in \mathscr{Q}_s} |Q|.$$

Group identical output cubes within each color c and write $F_{R,c} = \sum_{Q : R(Q) = R} H^s_{m(Q)+s} * b_Q$. Let $\mathscr{R}_{s,c}$ be the resulting finite set of distinct cubes. Set $D_s = 2^{(n+1)s}$ and

$$X_{s,c} = \left\{ x : \sum_{R \in \mathscr{R}_{s,c}} \mathbf{1}_R(x) > D_s \right\}, \qquad X_s = \bigcup_c X_{s,c}.$$

Counting distinct supports only decreases the covering sum, and hence

$$|X_s| \le 2^{6n} 2^{-s} F/\alpha. \tag{21.13}$$

Discard $R \subset X_{s,c}$. Every retained cube has at most D_s retained ancestors, since a point in $R \setminus X_{s,c}$ belongs to all of them. For the retained family define the reversed ancestor rank

$$\mathrm{rk}_c(R) = D_s + 1 - \#\{R' \in \mathscr{R}^{\mathrm{ret}}_{s,c} : R \subseteq R'\}. \tag{21.14}$$

Ranks lie in $\{1, \dots, D_s\}$ and strictly increase under strict inclusion. This is just the reverse of the ancestor-count rank; reversing a finite sequence interchanges prefixes and suffixes without changing variation.

Put $g_{r,c} = \sum_{\mathrm{rk}_c(R)=r} F_{R,c}$. For $x \notin X_{s,c}$, active cubes form a chain and discarded cubes contribute zero. Consequently, for a suitable rank threshold $r = r(k,x,c)$,

$$\sum_{j\ge k} \sum_{\ell(R)=2^{j+5}} F_{R,c}(x) = \sum_{q\ge r} g_{q,c}(x). \tag{21.15}$$

Missing ranks merely repeat suffix values. The same active-chain proof works for the global grids (21.11), with 5 replaced by a fixed generation offset in each class.

For clarity, the Rademacher–Menshov step here uses only deterministic signed L^2 bounds. If every fixed signed sum of a finite sequence has L^2 norm at most B, averaging the signs bounds the sum of squared L^2 norms of its disjoint block sums by B^2. Decompose each interval of a variation partition into binary blocks, at most two at each level. Cauchy–Schwarz over the at most $2(\lfloor \log_2(D_s+1)\rfloor + 1)$ blocks per interval, followed by the preceding sign average at each level, gives the prefix (and suffix) bound

$$\|V^2(\text{rank suffixes})\|_2 \le \sqrt{8}\,(\lfloor \log_2(D_s+1)\rfloor + 1)B.$$

No spatial selector, BMO estimate, or additional rank-energy hypothesis enters this argument. A fixed rank sign gives a fixed coefficient of modulus at most one on each original packet, with zero coefficient on discarded packets and packets of other colors. Thus the genuine full-kernel estimate of Lemma 18.4, with (21.8), gives

$$B^2 \le C_n(\Theta_s h_s^{-N})^2\, 2^{-s/24}\alpha F.$$

Restricting to $X_{s,c}^c$ can only decrease this norm. Summing over the 2^n colors, and using $\lfloor \log_2(D_s+1)\rfloor + 1 \le (n+1)s+2$, proves

$$\|\mathscr{L}_s\|_{L^2(X_s^c)} \le C_n(s+2)\Theta_s h_s^{-N} 2^{-s/48}\sqrt{\alpha F}. \tag{21.16}$$

The full-kernel and terminal-packet estimates used here include $n = 1$; there is no dimension restriction beyond $n \ge 1$.

Measurable majorants and uniform finite-gap distributions. The exact scalar rates in the three estimates are

$$\begin{aligned} \Theta_s 2^{-s/(1536N)} &= A 2^{-s/(2048N)}, \\ \Theta_s h_s &= (A/8) 2^{-31s/(6144N)}, \\ (s+2)\Theta_s h_s^{-N} 2^{-s/48} &= 8^N A (s+2) 2^{-(1/64-\delta)s}. \end{aligned}$$

Each series converges, and $A \le CG$. For a finite gap set $\mathscr{H}$, take $X = \bigcup_{s\in\mathscr{H}} X_s$ and set

$$\mathscr{W} = \sum_{s\in\mathscr{H}} (\mathscr{T}_s + 3\mathscr{E}_s), \qquad \mathscr{B} = 3\sum_{s\in\mathscr{H}} (\mathscr{S}_s + \mathscr{L}_s), \qquad \mathscr{G} = \mathbf{1}_{X^c}\mathscr{B}.$$

These are measurable majorants, and restricted-measure Minkowski gives

$$\|\mathscr{W}\|_1 \le C_n GF, \qquad \|\mathscr{B}\|_{L^2(X^c)} = \|\mathscr{G}\|_2 \le C_n G\sqrt{\alpha F}, \qquad |X| \le C_n F/\alpha. \tag{21.17}$$

In particular the original normalization $G = 1$ gives the same global L^1/L^2 majorant form. The raw split and (21.10) imply, almost everywhere,

$$V^2\left(\sum_{s\in\mathscr{H}} P_s^{\Omega}(\varepsilon) : \varepsilon > 0\right) \le \mathscr{W} + \mathscr{B}.$$

Chebyshev at $v/2$ for each majorant yields, for every $v > 0$,

$$\left|\left\{V^2\left(\sum_{s\in\mathscr{H}} P_s^{\Omega}\right) > v\right\}\right|^* \le C_n\left\{\frac{F}{\alpha} + \frac{GF}{v/2} + \frac{G^2\alpha F}{(v/2)^2}\right\}. \tag{21.18}$$

The star permits outer measure: no measurability assumption about an uncountable supremum is needed to obtain or pass this estimate.

Remove gaps first, then packets, on common sets. Let E_0 be the union of the selected closed bad cubes enlarged concentrically by the side factor 2^{s_0+3}. Then $|E_0| \le 2^{n(s_0+3)}F/\alpha$. If $x \notin E_0$, a packet supported in a cube of side 2^m cannot meet any shell with $j-m < s_0$. This follows coordinatewise from $|x_i - y_i| \le |x-y|$ and is an exact vanishing of every truncated-shell integrand, for all positive radii. The exception is fixed independently of all finite cutoffs.

We use the original L^1 input, rather than adding an L^2 hypothesis, to justify every-radius convergence. If $q \in L^1$ and $a, R > 0$, polar coordinates, Tonelli, and the length at most $2R$ of a line inside $B(0,R)$ give

$$\int_{B(0,R)} \int_{|z|>a} |K_\Omega(z)|\,|q(x-z)|\,dz\,dx \le \frac{2R}{a} A\|q\|_1.$$

Here $dr/r \le dr/a$ on the radial tail. Transfer from measurable representatives preserves the original integrands almost everywhere. Taking the countable cutoffs $a = 1/(k+1)$ and integer radii R produces one conull set on which the absolute tail is finite for *every* positive cutoff. Apply this to f, g, and $f-g$. Since disjoint packet supports give $\sum_Q |b_Q| = |f-g|$, the same set controls every finite packet subfamily.

Fix such a finite subfamily S and take $\mathscr{Q}_s = S$ for all gaps. Outside E_0, the consecutive cells with gaps $s_0, \dots, s_0+L-1$ exhaust the entire sharp tail of each packet. Absolute integrability and dominated convergence therefore give, simultaneously for every $\varepsilon > 0$,

$$\sum_{s=s_0}^{s_0+L-1} P_s^\Omega(\varepsilon, x) \longrightarrow T^0_{\Omega,\varepsilon}\Big(\sum_{Q\in S} b_Q\Big)(x).$$

Every strict superlevel of V^2 has a finite partition witness. Convergence on this single common set preserves that witness eventually; outer-measure lower semicontinuity passes (21.18). Adding $|E_0|$ gives the same bound, with a new dimensional constant, for the full literal trajectory of $b_S = \sum_{Q\in S} b_Q$. The moving rank exceptions have already been paid for in the uniform distribution bound; no common intersection of their complements is required.

For the good part, (21.1), (2.2), and (21.3) give

$$\big|\{x : \mathscr{N}(T^0_{\Omega,\varepsilon} g : \varepsilon > 0) > u/2\}\big| \le C_n G^2 \alpha F/u^2. \tag{21.19}$$

The paired-path triangle inequality and $\mathscr{N} \le 2\mathbf{J} \le 2V^2$ reduce the superlevel at u for $T^0 g + T^0 b_S$ to the good superlevel at $u/2$ and the bad V^2 superlevel at $u/4$. Choose $\alpha = u/G$. The three terms in the bad distribution and the good bound are then all at most $C_n GF/u$, uniformly in S.

Finally exhaust the countable family of bad cubes by finite sets S_L. The identity $\sum_Q |b_Q| = |f-g|$ and the common absolute-tail bound show by dominated convergence that

$$T^0_{\Omega,\varepsilon} g(x) + T^0_{\Omega,\varepsilon} b_{S_L}(x) \longrightarrow T^0_{\Omega,\varepsilon} f(x) \quad \text{for all } \varepsilon > 0 \text{ on one conull set.}$$

The finite-pair witness definition of $\mathscr{N}$ passes the same uniform outer-distribution bound to the limit. Thus

$$\big|\{x : \mathscr{N}(T^0_{\Omega,\varepsilon} f : \varepsilon > 0) > u\}\big|^* \le C_n GF/u.$$

On the common conull set these literal tails are continuous on each compact positive-radius interval, so rational finite witnesses also show measurability of the resulting path functionals. Since $\mathbf{J} \le \mathscr{N}$, this proves (21.2), in particular for the stated bounded compactly supported data. The amplitude supremum has remained inside the measured spatial superlevel throughout; no union over jump amplitudes, density argument, or completed-trajectory assumption is used. □

22 Phase-adapted Calderón–Zygmund decomposition

We modify the Calderón–Zygmund decomposition so that each bad block cancels against the input modulation generated by its parent cube. This is exactly the cancellation condition required by the packet estimates in the next section.

Fix a polynomial P and take the maximal dyadic family $\mathscr{Q}$ constructed in the proof of Proposition 21.1, at height α for f. For each $Q \in \mathscr{Q}$, let c_Q be its center and put

$$a_Q = |Q|^{-1} \int_Q e^{iP(c_Q,y)} f(y)dy, \quad g = f\mathbf{1}_{(\cup Q)^c} + \sum_Q a_Q e^{-iP(c_Q,\cdot)}\mathbf{1}_Q, \quad b_Q = (f - a_Q e^{-iP(c_Q,\cdot)})\mathbf{1}_Q. \tag{22.1}$$

Then

$$\|g\|_\infty \le C\alpha, \quad \|g\|_2^2 \le C\alpha\|f\|_1, \quad \sum_Q \|b_Q\|_1 \le C\|f\|_1, \quad \|b_Q\|_1 \le C\alpha|Q|. \tag{22.2}$$

Indeed, maximality gives $|Q|^{-1}\int_Q |f| \le 2^n\alpha$, and hence $|a_Q| \le 2^n\alpha$ and $\|b_Q\|_1 \le 2\int_Q |f|$. The cubes are disjoint. Off their union $|f| \le \alpha$ almost everywhere. Moreover,

$$|a_Q||Q| = \left| \int_Q e^{iP(c_Q,y)} f(y)\,dy \right| \le \int_Q |f(y)|\,dy$$

and, on Q, $|g|^2 = |a_Q|^2 \le 2^n\alpha|a_Q|$. Consequently

$$\|g\|_2^2 \le \alpha \int_{(\cup Q)^c} |f| + 2^n\alpha \sum_Q |a_Q||Q| \le C\alpha\|f\|_1.$$

These observations give all four estimates in (22.2). Most importantly, phase adaptation gives the exact cancellation

$$h_Q(y) := e^{iP(c_Q,y)} b_Q(y), \qquad \int_Q h_Q(y)\,dy = 0, \qquad |h_Q(y)| = |b_Q(y)|. \tag{22.3}$$

Indeed,

$$\int_Q h_Q(y)\,dy = \int_Q e^{iP(c_Q,y)} f(y)\,dy - a_Q|Q| = 0, \qquad \|h_Q\|_1 = \|b_Q\|_1.$$

No pointwise bound for b_Q or h_Q is asserted. The packet estimates below are deliberately formulated with only the density $\|h_Q\|_1/|Q|$ and are applied directly to the original blocks; thus no charge-neutralized stopping-tree decomposition is used. In the one-shell estimate below we take $d_Q = h_Q$ and use its cancellation. In the collective high-shell estimate we instead take $d\nu_Q = b_Q(y)\,dy$; that proposition requires the same mass-density bound but no cancellation.

23 The far-field weak endpoint

This section proves the weak critical-jump estimate for the normalized far-field trajectory. Assume that the highest nonpure homogeneous block P_m has been normalized by

$$\|P_m\|_{\mathrm{coeff}} = 1. \tag{23.1}$$

Define the far family by

$$\mathscr{F}_\varepsilon^P f(x) = \int_{\substack{|x-y|>\varepsilon \\ |x-y|>1}} e^{iP(x,y)} K_\Omega(x-y) f(y)\,dy,$$

extended constantly for $0 < \varepsilon \le 1$. Equivalently,

$$\mathscr{F}_\varepsilon^P f = T^P_{\Omega,\max\{\varepsilon,1\}} f. \tag{23.2}$$

Thus the far trajectory is a restriction of the full trajectory, with repeated values for $0 < \varepsilon \le 1$, and hence

$$N_\lambda(\mathscr{F}_\varepsilon^P f : \varepsilon > 0) \le N_\lambda(T^P_{\Omega,\varepsilon} f : \varepsilon > 0), \qquad \mathbf{J}(\mathscr{F}_\varepsilon^P f : \varepsilon > 0) \le \mathbf{J}(T^P_{\Omega,\varepsilon} f : \varepsilon > 0). \tag{23.3}$$

The strong $p = 2$ theorem gives

$$\|\mathbf{J}(T^P_{\Omega,\varepsilon}h : \varepsilon > 0)\|_2 \le C_{n,d}\mathfrak{L}(\Omega)\|h\|_2. \tag{23.4}$$

Proposition 23.1 (Far weak endpoint on core data)**.** *Let P be a real polynomial of degree at most d with highest nonpure homogeneous block P_m, and assume $\|P_m\|_{\mathrm{coeff}} = 1$. Let $\Omega \in L\log L(\mathbb{S}^{n-1})$ have mean zero. Then, for every bounded compactly supported f,*

$$\|\mathbf{J}(\mathscr{F}^P_\varepsilon f : \varepsilon > 0)\|_{L^{1,\infty}} \le C_{n,d}\mathfrak{L}(\Omega)\|f\|_1. \tag{23.5}$$

Proof. We first take bounded compactly supported f and impose finite cube, shell, and rational-parameter cutoffs. Every estimate below is uniform in those cutoffs.

If $\mathfrak{L}(\Omega) = 0$, the assertion is immediate. By angular homogeneity normalize $\mathfrak{L}(\Omega) = 1$, fix a Calderón–Zygmund height $\alpha > 0$, and use (22.1). By (23.3), (23.4), (22.2), and Chebyshev,

$$\big|\{x : \mathbf{J}(\mathscr{F}^P_\varepsilon g : \varepsilon > 0) > \alpha\}\big| \le \alpha^{-2}\|\mathbf{J}(\mathscr{F}^P_\varepsilon g)\|_2^2 \le C\alpha^{-2}\|g\|_2^2 \le C\alpha^{-1}\|f\|_1. \tag{23.6}$$

Thus the good part is controlled outside a set of the required measure. Enlarge the bad cubes by a fixed factor chosen as follows. Let $s_0 \ge \max\{5, t_0\}$, where t_0 is the threshold in Proposition 20.10, and use dilation factor $C_n 2^{s_0}$. This produces another exceptional set E_0 of measure $C\alpha^{-1}\|f\|_1$. Outside E_0, a cube meeting a shell $R = 2^j \ge 1$ has

$$\ell(Q) = R2^{-s}, \qquad s \ge s_0. \tag{23.7}$$

Put $b = \sum_Q b_Q$ and $\Sigma = \sum_Q \|b_Q\|_1$. For every fixed s, the cubes in (23.7), as j varies, form a subfamily of the original disjoint Calderón–Zygmund cubes and

$$\sum_j \sum_{\ell(Q)=2^{j-s}} \|h_Q\|_1 \le \Sigma, \qquad h_Q = e^{iP(c_Q,\cdot)}b_Q. \tag{23.8}$$

Split $\Omega = \omega_s + \Omega^\sharp_s$ by (15.32). With the finite shell cutoff understood, define the bounded-angular and angular-tail trajectories

$$U^P_{s,\varepsilon}(x) = \sum_{j\ge 0}\int_{\max\{\varepsilon,2^j\}<|x-y|\le 2^{j+1}} e^{iP(x,y)}K_{\omega_s}(x-y)\sum_{\ell(Q)=2^{j-s}} b_Q(y)\,dy, \tag{23.9}$$

$$\mathrm{Tail}^P_{s,\varepsilon}(x) = \sum_{j\ge 0}\int_{\max\{\varepsilon,2^j\}<|x-y|\le 2^{j+1}} e^{iP(x,y)}K_{\Omega^\sharp_s}(x-y)\sum_{\ell(Q)=2^{j-s}} b_Q(y)\,dy. \tag{23.10}$$

Since $|e^{iP(x,y)}| = 1$, the same Fubini–Young calculation as in (21.6), together with (23.8), gives

$$\|V^1(\mathrm{Tail}^P_{s,\varepsilon} : \varepsilon > 0)\|_1 \le C\|\Omega^\sharp_s\|_1\Sigma. \tag{23.11}$$

Here again the reindexing $(j,m) \mapsto (j, s = j-m)$ assigns each bad cube to at most one j for fixed s, so no shell-counting factor occurs. Summing (23.11) in s and using Lemma 15.7 proves that the angular tails have total V^1L^1 norm at most $C\mathfrak{L}(\Omega)\Sigma$. For the bounded pieces,

$$\|\omega_s\|_\infty \le C2^{\delta_{as}}\|\Omega\|_1, \qquad \|\omega_s\|_1 \le 2\|\Omega\|_1. \tag{23.12}$$

Short variation.

We first control the short variation inside every dyadic annulus. Apply Proposition 20.10 directly to the h_Q at each (j,s). The required phase matching is the exact identity

$$e^{iP(x,y)}b_Q(y) = e^{i[P(x,y)-P(c_Q,y)]}h_Q(y). \tag{23.13}$$

Thus the packet datum in that proposition is $d_Q = h_Q$; by (22.3) and (22.2) it is supported in Q, has integral zero, satisfies $\|d_Q\|_1 \le C\alpha|Q|$, and the packet cubes are globally disjoint. We may therefore take $\Lambda = C\alpha$ and $t = s$. Square-summing in j and using (23.8) gives

$$\left\| \left(\sum_j V^2(G_{j,s})^2 \right)^{1/2} \right\|_2^2 \le C\|\omega_s\|_\infty^2 2^{-c_P s}\alpha\Sigma, \tag{23.14}$$

and the packet errors satisfy

$$\sum_j \|V^1(E_{j,s})\|_1 \le C\|\omega_s\|_1 2^{-c_E s}\Sigma. \tag{23.15}$$

Long variation: low shells.

It remains to control the dyadic long variation. Fix a small $\kappa > 0$ and split the shells by the single global top block:

$$\text{low: } 1 \le R^m \le 2^{\kappa s}, \qquad \text{high: } R^m > 2^{\kappa s}. \tag{23.16}$$

Since $R = 2^j \ge 1$, the low condition implies $0 \le j \le \kappa s/m$; hence there are at most $1 + \kappa s/m \le C_d(1+s)$ low shells. If $F_{j,s}$ is their full annular output, then

$$V^2\left(\sum_{j\ge k,\ j \text{ low}} F_{j,s} : k \right) \le 2 \sum_{j \text{ low}} |F_{j,s}|.$$

The endpoint difference of a one-shell trajectory is dominated by its full variation. More explicitly write $F_{j,s} = F^G_{j,s} + F^E_{j,s}$ using the endpoint differences of the decomposition in Proposition 20.10. Thus, with $R = 2^j$,

$$F_{j,s} = \mathscr{P}_R - \mathscr{P}_{2R} = (G_R - G_{2R}) + (E_R - E_{2R}),$$

and consequently

$$|F^G_{j,s}| \le V^2(G_{j,s}), \qquad |F^E_{j,s}| \le V^1(E_{j,s}).$$

Cauchy–Schwarz in the $O(1+s)$ low shells and (20.45)–(20.46) therefore give

$$\left\| 2 \sum_{j \text{ low}} |F^G_{j,s}| \right\|_2^2 \le C(1+s) \sum_{j \text{ low}} \|V^2(G_{j,s})\|_2^2 \le C\|\omega_s\|_\infty^2 2^{-cs}\alpha\Sigma, \qquad c = c_P/2, \tag{23.17}$$

where $(1+s)2^{-c_P s} \lesssim 2^{-c_P s/2}$ has been absorbed, together with a summable V^1L^1 error. This is the only place where translated lower blocks are Fourier-expanded across low shells.

Long variation: high shells.

On the high shells no phase block is expanded. Return to the original form $e^{iP(x,y)}b_Q(y)$; every packet and shell now has the same global phase P, and (23.16) is precisely the threshold (20.112). Their short variation has already been included in the square sum (23.14); the collective estimate is needed only for the dyadic long variation.

For the high dyadic long variation use the output localization and fixed rank from the proof of Proposition 21.1. For every Q at a high shell 2^j, assign its natural annular output to a cube $R(Q)$ in one of the finitely many nested adjacent systems (21.11), with $\ell(R(Q)) \simeq 2^j$. As in the closed-enlargement construction preceding (21.12), choose $R(Q)$ so that the essential support of this sharp-annular output is contained in its interior. Let $\mathfrak{C}$ be the resulting finite set of adjacent-system and generation- offset classes; it depends only on the dimension. Make one fixed class assignment for each pair (j,Q) and write $\mathscr{Q}^{\text{high}}_{s,c}$ for the packets assigned to c. Within a fixed class, all $R(Q)$ lie in one nested system and $\ell(R(Q)) = 2^{j+\iota_c}$ for one fixed integer ι_c. Define the current bounded-angular annular operator by

$$T_j^{P,\omega_s} g(x) = \int_{2^j < |x-y| \le 2^{j+1}} e^{iP(x,y)} K_{\omega_s}(x-y) g(y)\, dy,$$

and, for the classwise output cubes, put

$$F_{R,c} = T_j^{P,\omega_s} \sum_{\substack{Q\in\mathscr{Q}_{s,c}^{\mathrm{high}}:\ R(Q)=R \\ \ell(Q)=2^{j-s}}} b_Q, \qquad \ell(R) = 2^{j+\iota_c}.$$

Let $\mathscr{R}_{s,c}$ be the finite family of nonzero classwise output cubes. Then $\operatorname{ess\,supp} F_{R,c} \subset R$ and

$$\sum_{R\in\mathscr{R}_{s,c}} |R| \le C2^{ns} \sum_{Q\in\mathscr{Q}_{s,c}^{\mathrm{high}}} |Q|.$$

Let $D_s = \lceil 2^{(n+1)s} \rceil$ and define

$$X_{s,c} = \left\{ \sum_{R\in\mathscr{R}_{s,c}} \mathbf{1}_R > D_s \right\}, \qquad X_s = \bigcup_{c\in\mathfrak{C}} X_{s,c}.$$

Since $\mathfrak{C}$ is finite, the packing estimate gives

$$|X_s| \le C2^{-s}\alpha^{-1}\|f\|_1. \tag{23.18}$$

For each class discard the cubes $R \subset X_{s,c}$ and rank the retained finite family by strict inclusion, as in (21.14). Put

$$\mathscr{Q}_{s,c}^{\mathrm{high,ret}} = \{Q \in \mathscr{Q}_{s,c}^{\mathrm{high}} : R(Q) \not\subset X_{s,c}\}.$$

At every $x \notin X_s$ outside the fixed null set of adjacent-grid boundaries, the high-shell tail in each class is a suffix of its rank sequence by the active-chain argument proving (21.15); gaps caused by low shells only repeat a suffix value. A fixed rank sign induces a fixed coefficient in $\{-1,0,1\}$ on each original packet, with coefficient zero on discarded cubes and on packets outside the current class. Thus no rank is ever invoked for a discarded or differently classified packet.

We verify explicitly the hypotheses of Proposition 20.17 class by class. For fixed s and c, the retained set of high radii is finite because the shell cutoff is finite. Across all of those radii, the packet cubes remain a subfamily of the original globally disjoint Calderón–Zygmund cubes. Take $d\nu_Q = b_Q(y)\,dy$; unlike the short estimate, this application uses no zero-mean hypothesis. Then

$$\operatorname{supp}\nu_Q \subset Q, \qquad |\nu_Q|(Q) = \|b_Q\|_1 = \|h_Q\|_1 \le C\alpha|Q|.$$

The normalized profile is the sharp function $\mathbf{1}_{(1,2]}$, whose $L^\infty + \mathrm{BV}$ norm is bounded by an absolute constant. Moreover, normalization of the top block and the high-shell condition give $R^m\|P_m\|_{\mathrm{coeff}} = R^m > 2^{\kappa s}$, exactly (20.112). The Rademacher–Menshov signs are fixed packet by packet before the output point x is evaluated. If

$$\Sigma_{s,c}^{\mathrm{high}} = \sum_{Q\in\mathscr{Q}_{s,c}^{\mathrm{high,ret}}} \|b_Q\|_1,$$

then Lemma 18.2 and Proposition 20.17 give

$$\|\mathbf{1}_{X_{s,c}^c} V^2(\text{high dyadic tails in class } c)\|_2^2 \le Cs^2\|\omega_s\|_\infty^2 2^{-c_\kappa s}\alpha\Sigma_{s,c}^{\mathrm{high}}.$$

The V^2 triangle inequality and Cauchy–Schwarz over the finite class set, together with $\sum_c \Sigma_{s,c}^{\mathrm{high}} \le \Sigma$, yield the full high-shell estimate

$$\|\mathbf{1}_{X_s^c} V^2(\text{high dyadic tails})\|_2^2 \le Cs^2\|\omega_s\|_\infty^2 2^{-c_\kappa s}\alpha\Sigma. \tag{23.19}$$

This is where the estimate must allow packetwise, rather than merely shellwise, signs.

Choose $0 < \delta_a < \frac{1}{4}\min(c_P, c_E, c_\kappa)$. Equations (23.14)–(23.19), polynomial factors in s, and (23.12) imply the following precise majorant bounds. Put

$$\mathscr{S}_s = \left(\sum_j V^2(G_{j,s})^2\right)^{1/2},$$
$$\mathscr{L}_s = 2\sum_{j\ \mathrm{low}} |F^G_{j,s}|,$$
$$\mathscr{H}_s = \mathbf{1}_{X_s^c} V^2(\text{high dyadic tails}).$$

For later use retain explicitly the error produced by the one-shell decompositions and put

$$\mathscr{E}^*_s = \sum_j V^1(E_{j,s}). \tag{23.20}$$

At the present finite-cutoff stage this is a finite pointwise sum. In particular, no compatibility of the choices of $G_{j,s}$ and $E_{j,s}$ from one shell to another is being assumed. Let $\mathscr{W}$ be the total 1-variation majorant of all angular tail paths plus $C\sum_s \mathscr{E}^*_s$; the constant also covers the low-shell endpoint errors $2\sum_{j\,\mathrm{low}} |F^E_{j,s}|$. Set

$$\mathscr{G} = \sum_s (\mathscr{S}_s + \mathscr{L}_s + \mathscr{H}_s), \qquad X = \bigcup_s X_s.$$

Minkowski's inequality, applied to the square roots of the fixed-s estimates, now gives

$$\|\mathscr{G}\|_2^2 \le C\alpha\Sigma, \qquad \|\mathscr{W}\|_1 \le C\Sigma, \qquad |X| \le C\alpha^{-1}\|f\|_1. \tag{23.21}$$

The geometric decay makes all three sums finite. More precisely, at the finite-cutoff stage these are finite sums, while summability of their L^2, respectively L^1, norms makes the partial sums of $\mathscr{G}$, respectively $\mathscr{W}$, converge in those spaces with the displayed bounds; monotone subsequences may be used for the nonnegative majorants. We now spell out the passage from the shellwise decompositions to the full trajectory. Denote the moving bounded-angular one-shell path on $[2^j, 2^{j+1}]$ by $\mathscr{P}_{j,s} = G_{j,s} + E_{j,s}$, and denote its endpoint difference by $F_{j,s} = F^G_{j,s} + F^E_{j,s}$, as above. The outer-shell terms in (23.9) are constant while $\varepsilon \in [2^j, 2^{j+1}]$. Hence the variation triangle inequality, $V^2(E_{j,s}) \le V^1(E_{j,s})$, and ℓ^2-Minkowski give the pointwise estimate

$$\begin{aligned}
\left(\sum_j V^2(U^P_{s,\varepsilon} : 2^j \le \varepsilon \le 2^{j+1})^2\right)^{1/2} &= \left(\sum_j V^2(\mathscr{P}_{j,s})^2\right)^{1/2} \\
&\le \left(\sum_j \big(V^2(G_{j,s}) + V^1(E_{j,s})\big)^2\right)^{1/2} \\
&\le \mathscr{S}_s + \left(\sum_j V^1(E_{j,s})^2\right)^{1/2} \le \mathscr{S}_s + \mathscr{E}^*_s.
\end{aligned} \tag{23.22}$$

For the dyadic endpoint path, split the full suffix sum into its high- and low-shell parts. The exact endpoint identity is

$$U^P_{s,2^k} = \sum_{j \ge k} F_{j,s}.$$

Here and below $F_{j,s} = 0$ for $j < 0$, so the identity also records the constant part of the far trajectory for $2^k \le 1$. On X_s^c, the high part is bounded by $\mathscr{H}_s$. The elementary suffix bound used in (23.17), together with $|F_{j,s}^E| \le V^1(E_{j,s})$, gives

$$\begin{aligned}\mathbf{1}_{X_s^c} V^2\left(\sum_{j\ge k} F_{j,s} : k\right) &\le \mathscr{H}_s + V^2\left(\sum_{\substack{j\ge k\\ j \text{ low}}} F_{j,s} : k\right)\\ &\le \mathscr{H}_s + 2\sum_{j \text{ low}} |F_{j,s}|\\ &\le \mathscr{H}_s + \mathscr{L}_s + 2\mathscr{E}_s^*. \end{aligned} \tag{23.23}$$

Thus the deterministic full-variation long–short inequality (17.15), applied to the *original* bounded-angular path rather than to separately chosen global G/E paths, yields on X_s^c

$$V^2(U_{s,\varepsilon}^P : \varepsilon > 0) \le C(\mathscr{S}_s + \mathscr{L}_s + \mathscr{H}_s + \mathscr{E}_s^*). \tag{23.24}$$

Summing (23.24) over s with (15.4), and adding the angular-tail V^1 paths, gives on $E_0^c \cap X^c$ a 2-variation majorant bounded by $C(\mathscr{G} + \mathscr{W})$. This construction uses only the displayed shellwise decompositions and therefore makes no unstated cross-shell selection. Since the majorants do not depend on the jump amplitude, equations (15.5) and (15.6) give directly

$$\mathbf{1}_{E_0^c \cap X^c} \mathbf{J}(\mathscr{F}_\varepsilon^P b : \varepsilon > 0) \le C(\mathscr{G} + \mathscr{W}). \tag{23.25}$$

Chebyshev applied to (23.25), together with (23.6), the measures of E_0 and X, (23.21), and the pointwise jump quasi-triangle inequality, gives

$$\left|\left\{x : \mathbf{J}(\mathscr{F}_\varepsilon^P f : \varepsilon > 0) > C_0\alpha\right\}\right| \le C\alpha^{-1}\|f\|_1.$$

Here C_0 is fixed and independent of every cutoff. Once the cutoffs are removed, setting $\alpha = \tau/C_0$ and taking the supremum over $\tau > 0$ gives precisely the $L^{1,\infty}$ quasi-norm of the pointwise critical-jump functional. The proof in dimension one is identical after using the one-dimensional parts of Corollary 20.15 and Proposition 20.17; mean zero on $\mathbb{S}^0$ makes the angular function a Hilbert-kernel multiple, so no angular-chart issue arises.

Removal of cutoffs.

Finally remove the cutoffs as follows. With the shell interval and the finite rational parameter set I fixed, exhaust the phase-adapted bad-cube family. Since $\sum_Q \|b_Q\|_1 < \infty$, the corresponding inputs converge in L^1; Young's inequality for each finite-annulus coordinate, followed by a subsequence, gives pointwise convergence simultaneously on the finite parameter set; denote the approximating and limiting finite paths by $F^{(\nu)}$ and F. The finite functional is Borel in its coordinates. If $\mathbf{J}_I(F) > \tau$, one amplitude and one finite strict witness exceed τ by a positive amount, and coordinatewise convergence preserves that witness for all sufficiently large ν. Hence the level-set form of (15.29),

$$\mathbf{1}_{\{\mathbf{J}_I(F) > \tau\}} \le \liminf_{\nu\to\infty} \mathbf{1}_{\{\mathbf{J}_I(F^{(\nu)}) > \tau\}},$$

followed by Fatou preserves the distribution estimate for every $\tau > 0$.

Next exhaust the dyadic shell interval, still with I fixed. For a fixed output point x and truncation $q > 0$, every shell with $2^{j+1} \le q$ lies below the truncation, while for all sufficiently large j the annulus centered at x misses the compact support of the input. Thus each coordinate is eventually a finite shell sum, and the shell-cutoff limit is pointwise on the finite set I. Apply the same strict-witness lower-semicontinuity and level-set Fatou argument once more. Finally enumerate $\mathbb{Q}_+ = \{q_1, q_2, \ldots\}$, put $I_N = \{q_1, \ldots, q_N\}$, and use

$$\{\mathbf{J}(\mathscr{F}_q^P f : q \in \mathbb{Q}_+) > \tau\} = \bigcup_{N\ge 1} \{\mathbf{J}_{I_N}(\mathscr{F}_q^P f : q \in I_N) > \tau\}.$$

The equality follows because every strict witness is finite; continuity from below therefore preserves the uniform distribution bound. For bounded compactly supported data, Fubini on finite annuli gives local continuity in the truncation parameter. Every full-parameter witness is contained in a compact interval, and Lemma 15.6 therefore identifies the rational and full-parameter functionals. The ranks, signs, and exceptional sets are auxiliary devices used only to prove estimates uniform in the cutoffs; none of them is passed to the limit. Only the original finite-cutoff trajectories are passed to the limit. In particular, no convergence of the ranks, signs, or exceptional sets is asserted or needed. This proves (23.5) for core data. Angular homogeneity restores $\mathfrak{L}(\Omega)$. □

24 Local weak estimates and homogeneous-degree induction

We close the weak endpoint with a coarser triangular order than in the strong argument: each step removes the entire highest nonpure homogeneous class rather than a single flag coordinate.

Proposition 24.1 (Local homogeneous-degree reduction)**.** *Let P be a real polynomial of degree at most d whose highest nonpure homogeneous block P_m satisfies $\|P_m\|_{\mathrm{coeff}} = 1$. Let $\Omega \in L^1(\mathbb{S}^{n-1})$ and let f be bounded and compactly supported. On each unit output cube, the local path* (17.31) *is, modulo pure input and output phases, the sum of*

(i) *a localized path of the form* (17.33) *whose phase has nonpure total degree at most $m-1$;*

(ii) *an error E satisfying*

$$V^1(E)(x) \le C_{n,d} \int_{|x-y|<1} \frac{|\Omega((x-y)/|x-y|)|}{|x-y|^{n-1}} |f(y)|\, dy. \tag{24.1}$$

The majorant has L^1 norm at most $C_{n,d}\|\Omega\|_1\|f\|_1$.

Proof. Let Q_a be the output unit cube centered at a, and let Q_a^* be the fixed input enlargement relevant to Q_a. Write $x = a+X, y = a+Y$. Simultaneous translation preserves the degree-m class and creates only lower degrees. After separating pure terms, write

$$P(a+X,a+Y) = p_a(X) + q_a(Y) + H_m(X,Y) + R_{a,<m}(X,Y),$$

where H_m is the mixed representative of the normalized top class and $R_{a,<m}$ has nonpure degree at most $m-1$. Replace H_m by the anchored representative

$$\widehat{H}_m(X,Y) = H_m(X,Y) - H_m(Y,Y),$$

and absorb the pure input polynomial $H_m(Y,Y)$ into q_a. Set

$$f_a(Y) = e^{iq_a(Y)} f(a+Y)\mathbf{1}_{Q_a^*-a}(Y).$$

The coordinate translation identifies the path at $a+X$ with the path at X and preserves Lebesgue measure. Multiplication of the output path by $e^{ip_a(X)}$ leaves pointwise jump counts unchanged, whereas the pure input factor is now part of f_a. In particular,

$$|f_a(Y)| = |f(a+Y)|\mathbf{1}_{Q_a^*-a}(Y), \qquad \|f_a\|_1 = \|f\mathbf{1}_{Q_a^*}\|_1.$$

The anchored representative satisfies $\widehat{H}_m(Y,Y) = 0$. On the fixed bounded set relevant to $|X-Y|<1$,

$$|\widehat{H}_m(X,Y)| \le |X-Y| \int_0^1 |\nabla_X H_m(Y+t(X-Y),Y)|\, dt \le C_{n,d}|X-Y|.$$

The last constant is uniform because the coefficient space $\mathscr{V}_m$ is finite dimensional, its normalized unit sphere is compact, and X,Y range over one fixed bounded set. Suppressing the harmless pure output factor, define, for $0<\varepsilon<1$,

$$E_{a,\varepsilon}(X)=\int_{\varepsilon<|X-Y|<1}\bigl(e^{i\widehat{H}_m(X,Y)}-1\bigr)e^{iR_{a,<m}(X,Y)}K_\Omega(X-Y)f_a(Y)\,dY,$$

and put $E_{a,\varepsilon}=0$ for $\varepsilon\ge 1$. For an arbitrary finite increasing truncation partition, the radial layers occurring in the successive increments are disjoint. Using $|e^{iu}-1|\le|u|$ and the preceding bound therefore gives

$$V^1(E_a)(X)\le C_{n,d}\int_{|X-Y|<1}\frac{|\Omega((X-Y)/|X-Y|)|}{|X-Y|^{n-1}}|f_a(Y)|\,dY.$$

After translating back and discarding the input restriction this is (24.1). Since the output cubes are disjoint, the piecewise error satisfies the same pointwise majorant, and polar coordinates give

$$\int_{|z|<1}\frac{|\Omega(z/|z|)|}{|z|^{n-1}}\,dz=\|\Omega\|_1.$$

Tonelli thus proves the asserted global L^1 estimate. What remains after subtracting E_a is exactly the localized path with phase $R_{a,<m}$ applied to f_a, whose nonpure degree is at most $m-1$. □

Proposition 24.2 (Weak polynomial critical jumps on core data)**.** *For every real polynomial P of degree at most d, every mean-zero $\Omega\in L\log L(\mathbb{S}^{n-1})$, and every bounded compactly supported f,*

$$\|\mathbf{J}(T^P_{\Omega,\varepsilon}f:\varepsilon>0)\|_{L^{1,\infty}}\le C_{n,d}\mathfrak{L}(\Omega)\|f\|_1,$$

with a constant independent of the coefficients of P.

Proof. All weak estimates invoked in this induction are the core-data versions proved above. More precisely, the base case uses the bounded compactly supported conclusion of Proposition 21.1, and the far part uses the bounded compactly supported conclusion of Proposition 23.1. The modulated translated inputs f_a arising in Proposition 24.1 remain bounded and compactly supported, and satisfy $\|f_a\|_1=\|f\mathbf{1}_{Q_a^*}\|_1$. Thus no operator value for a general L^1 input, and no result from Section 25, is used in the induction.

Induct on the largest total degree m carrying a nonpure homogeneous class. If there is no such class, the phase is pure and Proposition 21.1 applies after input and output conjugation. Assume the assertion for nonpure degree at most $m-1$ and let

$$C_m=\|P_m\|_{\mathrm{coeff}}>0,\qquad \rho=C_m^{-1/m}.$$

Under $x=\rho X, y=\rho Y$, the entire degree-m class, rather than one scalar coordinate, has norm one. Proposition 23.1 controls the normalized far path.

For the local path partition the output into disjoint unit cubes Q_a, and let Q_a^* denote the fixed input enlargement relevant to Q_a. Proposition 24.1 gives on Q_a a residual localized path L_a with nonpure degree at most $m-1$, applied to the corresponding modulated translated input f_a, and an error E_a. The piecewise error $E=E_a$ on Q_a satisfies, by (24.1),

$$\|V^1(E)\|_1\le C_{n,d}\|\Omega\|_1\|f\|_1\le C_{n,d}\mathfrak{L}(\Omega)\|f\|_1.$$

For the residual phase, subtracting the fixed truncation at one does not change increments below 1, and adjoining zero for $\varepsilon\ge 1$ creates no additional jump by (17.34). Taking the supremum in the jump amplitude in that pointwise domination, the induction hypothesis gives, for every $\alpha>0$,

$$|\{x\in Q_a:\mathbf{J}(L_a:\varepsilon>0)>\alpha\}|\le C_{n,d}\alpha^{-1}\mathfrak{L}(\Omega)\|f_a\|_1=C_{n,d}\alpha^{-1}\mathfrak{L}(\Omega)\|f\mathbf{1}_{Q_a^*}\|_1.$$

Let $L_{\mathrm{res}} = L_a$ on Q_a. Since the output cubes are disjoint and the input enlargements have bounded overlap,

$$\begin{aligned}|\{x : \mathbf{J}(L_{\mathrm{res}} : \varepsilon > 0) > \alpha\}| &= \sum_a |\{x \in Q_a : \mathbf{J}(L_a : \varepsilon > 0) > \alpha\}| \\ &\le C_{n,d}\alpha^{-1}\mathfrak{L}(\Omega)\sum_a \|f\mathbf{1}_{Q_a^*}\|_1 \\ &\le C_{n,d}\alpha^{-1}\mathfrak{L}(\Omega)\|f\|_1.\end{aligned}$$

For the error, (15.6) and $V^2 \le V^1$ give $\mathbf{J}(E) \le V^1(E)$; hence its L^1 majorant yields the required weak estimate. Thus the local residual and error paths have the desired pointwise critical-jump bound.

Combine local and far paths with (15.5). More explicitly, the two weak distribution estimates and the L^1 error majorant give a fixed C_0 such that, with $P_\rho(X,Y) = P(\rho X, \rho Y)$ and $f_\rho(X) = f(\rho X)$,

$$\left|\left\{X : \mathbf{J}(T^{P_\rho}_{\Omega,\varepsilon} f_\rho : \varepsilon > 0)(X) > C_0\alpha\right\}\right| \le C\alpha^{-1}\mathfrak{L}(\Omega)\|f_\rho\|_1.$$

Putting $\alpha = \tau/C_0$ gives the normalized weak quasi-norm estimate. The degree-m block of P_ρ has coefficient norm $\rho^m C_m = 1$, and the exact conjugacy is

$$T^P_{\Omega,\varepsilon} f(\rho X) = T^{P_\rho}_{\Omega,\varepsilon/\rho} f_\rho(X), \qquad f_\rho(X) = f(\rho X),$$

where the reparameterization $\varepsilon \mapsto \varepsilon/\rho$ preserves every jump count. The two scaling identities are, with H denoting any measurable scalar functional on $\mathbb{R}^n$ (in the application, the pointwise jump functional),

$$\|f_\rho\|_{L^1_X} = \rho^{-n}\|f\|_{L^1_x}, \qquad \|H(\rho\,\cdot)\|_{L^{1,\infty}_X} = \rho^{-n}\|H\|_{L^{1,\infty}_x}.$$

Thus the factors ρ^{-n} on the normalized input and output sides cancel exactly, and the bound is independent of C_m. The finite degree induction proves the proposition for every polynomial phase. □

25 Completion, consequences, and the direct radial BV transfer

25.1 Completion of the critical-jump trajectory

For core data, the weak endpoint propositions already remove the auxiliary cube, shell, and rational-parameter cutoffs. We pass to general L^1 data and construct the complete trajectory over all positive truncation parameters.

For bounded compactly supported f, put, for $N \ge 2$,

$$J_N(x) = \int_{N^{-1}<|z|<N} |K_\Omega(z)|\,|f(x-z)|\,dz.$$

Fubini gives

$$\int_{\mathbb{R}^n} J_N(x)\,dx = \|f\|_1 \|K_\Omega \mathbf{1}_{\{N^{-1}<|\cdot|<N\}}\|_1 < \infty.$$

After intersecting the countably many sets $\{J_N < \infty\}$, we obtain one full-measure set on which every finite-annulus integral is finite. Every compact interval $[a,b] \Subset (0,\infty)$ is contained in $[N^{-1},N]$ for some N; polar coordinates then show that, on this single set, the truncation path is locally absolutely continuous, hence continuous, simultaneously on all compact parameter intervals. Therefore the paired jump counts, and hence $\mathbf{J}$, computed on positive rationals agree there with those computed on all positive parameters by Lemma 15.6.

For such core data the path is zero for sufficiently large truncation at each fixed x. Equation (15.7) therefore gives pointwise

$$\sup_{\varepsilon>0} |T^P_{\Omega,\varepsilon} h| \le \mathbf{J}(T^P_{\Omega,\varepsilon} h : \varepsilon > 0).$$

The weak pointwise critical-jump estimate implies the maximal estimate

$$\| \sup_{q\in\mathbb{Q}_+} |T^P_{\Omega,q}h|\|_{L^{1,\infty}} \le C_{n,d}\mathfrak{L}(\Omega)\|h\|_1. \tag{25.1}$$

Start with any bounded compactly supported approximating sequence and pass to a subsequence, still denoted by f_k, such that

$$f_k \to f \quad \text{in } L^1, \qquad \|f_{k+1}-f_k\|_1 \le 2^{-3k}.$$

In particular $\|f_k\|_1 \to \|f\|_1$. By (25.1),

$$\left|\left\{ \sup_{q\in\mathbb{Q}_+} |T^P_{\Omega,q}(f_{k+1}-f_k)| > 2^{-k} \right\}\right| \le C\mathfrak{L}(\Omega)2^k\|f_{k+1}-f_k\|_1 \le C\mathfrak{L}(\Omega)2^{-2k}.$$

The right side is summable in k. Borel–Cantelli gives almost-everywhere uniform convergence on $\mathbb{Q}_+$. More precisely, let

$$E_{\mathrm{BC}} = \limsup_{k\to\infty}\left\{ x : \sup_{q\in\mathbb{Q}_+} |T^P_{\Omega,q}(f_{k+1}-f_k)(x)| > 2^{-k} \right\}.$$

This is a measurable null set, since the supremum is countable. For every rational $q>0$, define on all of $\mathbb{R}^n$

$$T^P_{\Omega,q}f(x) = \begin{cases} \lim_{k\to\infty} T^P_{\Omega,q}f_k(x), & x\notin E_{\mathrm{BC}}, \\ 0, & x\in E_{\mathrm{BC}}. \end{cases} \tag{25.2}$$

Thus all rational coordinates are defined on the same set, every one is measurable, and every countable supremum formed from them is measurable. Outside E_{BC} there is $k_0(x)$ such that, for every $k\ge k_0(x)$,

$$\sup_{q\in\mathbb{Q}_+} |T^P_{\Omega,q}(f_{k+1}-f_k)(x)| \le 2^{-k}.$$

Consequently the series

$$\mathscr{R}_m(q,x) := \sum_{k\ge m} T^P_{\Omega,q}(f_{k+1}-f_k)(x)$$

converges uniformly in $q\in\mathbb{Q}_+$ outside that exceptional set, and, for every $m\ge k_0(x)$,

$$\sup_{q\in\mathbb{Q}_+} |\mathscr{R}_m(q,x)| \le \sum_{k\ge m} 2^{-k}. \tag{25.3}$$

By (25.2), outside E_{BC}, $T^P_{\Omega,q}f = T^P_{\Omega,q}f_m + \mathscr{R}_m(q,\cdot)$; thus no operator value on $f-f_m$ is being invoked before the completion is constructed. For the core function f_m, the trajectory is zero at x for all sufficiently large q. Combining this fact with (25.3) gives the stronger uniform rational anchor

$$\lim_{R\to\infty} \sup_{\substack{q\in\mathbb{Q}_+ \\ q\ge R}} |T^P_{\Omega,q}f(x)| = 0 \tag{25.4}$$

for almost every x: given m, take R beyond the support-dependent vanishing threshold for f_m, and then let $m\to\infty$. Equivalently, for every $m\ge k_0(x)$,

$$\limsup_{R\to\infty} \sup_{\substack{q\in\mathbb{Q}_+ \\ q\ge R}} |T^P_{\Omega,q}f(x)| \le \sum_{k\ge m} 2^{-k};$$

letting $m\to\infty$ proves (25.4) with the quantifiers in the asserted order.

Select a further subsequence solely to obtain the pointwise finite-annulus identities below, and do not relabel it. This selection does not alter any of the rational limits already obtained, because the

convergence on $\mathbb{Q}_+$ was uniform along the preceding sequence. For every rational pair $0 < a < b < \infty$, the absolute-value Young bound for the finite-annulus kernel,

$$\|K_\Omega \mathbf{1}_{\{a<|\cdot|\le b\}}\|_1 = \|\Omega\|_1 \log(b/a),$$

gives L^1 convergence of

$$T^P_{\Omega,a} f_k - T^P_{\Omega,b} f_k \quad \text{to} \quad \int_{a<|z|\le b} e^{iP(x,x-z)} K_\Omega(z) f(x-z)\, dz.$$

A diagonal choice over the countable set of rational pairs gives pointwise convergence of all these identities on one full-measure set. Separately, Fubini applied to the countable annuli $\{N^{-1} < |z| < N\}$, $N \ge 2$, gives a common full-measure set on which every finite-annulus integral is finite and depends locally absolutely continuously on its radial endpoints. Intersecting these sets with the complement of E_{BC} preserves full measure; denote the resulting measurable set by $\mathscr{X}$.

On $\mathscr{X}$ define the value at arbitrary ε from the base point 1 by

$$T^P_{\Omega,\varepsilon} f = T^P_{\Omega,1} f + \begin{cases} \displaystyle\int_{\varepsilon<|z|\le 1} e^{iP(x,x-z)} K_\Omega(z) f(x-z) dz, & \varepsilon < 1, \\ \displaystyle -\int_{1<|z|\le\varepsilon} e^{iP(x,x-z)} K_\Omega(z) f(x-z) dz, & \varepsilon > 1. \end{cases}$$

On $\mathscr{X}^c$ define $T^P_{\Omega,\varepsilon} f = 0$ simultaneously for every $\varepsilon > 0$. For each fixed ε, this is a measurable representative: the finite-annulus integral is measurable by the parameter integral theorem and $T^P_{\Omega,1} f$ is measurable by (25.2). The modification on $\mathscr{X}^c$ merely replaces all rational representatives on one additional common null set. The resulting path is locally absolutely continuous, hence continuous, in ε and agrees with the already constructed rational values on $\mathscr{X}$ by the rational finite-annulus identities.

Fix an enumeration $\mathbb{Q}_+ = \{q_1, q_2, \dots\}$ and set $D_N = \{q_1, \dots, q_N\}$. For every N, (15.29) gives on $\mathscr{X}$, hence almost everywhere,

$$\mathbf{J}_{D_N}(T^P_{\Omega,\varepsilon} f : \varepsilon \in D_N) \le \liminf_{k\to\infty} \mathbf{J}_{D_N}(T^P_{\Omega,\varepsilon} f_k : \varepsilon \in D_N).$$

The finite functional is Borel in its finitely many coordinates. The lower-semicontinuity statement is exactly the strict-witness statement: if the left side is $> \tau$, one may choose a jump amplitude and a finite strict witness with a positive excess over τ; coordinatewise convergence preserves that same witness for all sufficiently large k. Consequently, for every $\tau > 0$,

$$\mathbf{1}_{\{\mathbf{J}_{D_N}(T^P_{\Omega,\varepsilon} f:\varepsilon\in D_N)>\tau\}} \le \liminf_{k\to\infty} \mathbf{1}_{\{\mathbf{J}_{D_N}(T^P_{\Omega,\varepsilon} f_k:\varepsilon\in D_N)>\tau\}}.$$

Fatou's lemma and $\|f_k\|_1 \to \|f\|_1$ therefore transfer the core distribution estimate with the same right-hand side $C\tau^{-1}\mathfrak{L}(\Omega)\|f\|_1$, uniformly in N. Because the sets D_N are nested and every strict rational jump witness is finite,

$$\{\mathbf{J}(T^P_{\Omega,q} f : q \in \mathbb{Q}_+) > \tau\} = \bigcup_{N=1}^{\infty} \{\mathbf{J}_{D_N}(T^P_{\Omega,q} f : q \in D_N) > \tau\}. \tag{25.5}$$

Continuity from below therefore transfers the same distribution estimate to the measurable rational pointwise functional. Every finite full-parameter witness lies in a compact interval; local continuity and Lemma 15.6 then identify it almost everywhere with the full positive-parameter functional. Finally, for every $R > 0$, continuity and density of $\mathbb{Q}_+$ imply

$$\sup_{\varepsilon\ge R} |T^P_{\Omega,\varepsilon} f(x)| \le \sup_{\substack{q\in\mathbb{Q}_+ \\ q\ge R}} |T^P_{\Omega,q} f(x)|.$$

Letting $R \to \infty$ in (25.4) proves the zero-at-infinity anchor for all real radii. In particular the construction is independent of the chosen core approximation: two resulting paths have the same finite-annulus increments, so on the intersection of their full-measure defining sets their difference is independent of ε; their common zero anchor forces that difference to vanish. This also proves uniqueness among all paths with the finite-annulus identities and the zero anchor. This completes the unweighted weak critical-jump construction and, together with the strong theorem already proved, establishes Theorem 1.2.

25.2 Abstract consequences

The pointwise inequality (15.8) applied to the completed trajectory gives, for every $r > 2$,

$$V^r(T^P_{\Omega,\varepsilon} f : \varepsilon > 0) \le \kappa_r \mathbf{J}(T^P_{\Omega,\varepsilon} f : \varepsilon > 0) \qquad \text{almost everywhere.}$$

Taking the strong or weak spatial norm and using Theorem 1.2 proves (1.11) and

$$\|V^r(T^P_{\Omega,\varepsilon} f : \varepsilon > 0)\|_{L^{1,\infty}} \le C_{n,d}\, \kappa_r \mathfrak{L}(\Omega) \|f\|_1. \tag{25.6}$$

No interchange of a spatial norm with the jump-amplitude supremum is used here. The zero anchor and (15.7) give the maximal estimate, while finite r-variation gives the limit at zero. This proves Corollary 1.3.

25.3 Radial BV transfer

Write

$$F_\varepsilon = T^P_{\Omega,\varepsilon} f, \qquad G_\varepsilon = T^{P,b}_{\Omega,\varepsilon} f.$$

Fix throughout the right-continuous representative of b specified after (1.5). A right-continuous BV function determines a finite complex Borel measure by

$$db((s,t]) = b(t) - b(s), \qquad |db|(0,\infty) \le \mathrm{Var}_{(0,\infty)}(b).$$

For test data the path F is locally absolutely continuous. The Lebesgue–Stieltjes product formula on $0 < \varepsilon < R < \infty$ reads

$$b(R)F_R - b(\varepsilon)F_\varepsilon = \int_{(\varepsilon,R]} F_s\, db(s) + \int_{(\varepsilon,R]} b(s-)\, dF_s.$$

Since $dF \ll ds$ has no atoms, the last integrand may equivalently be written $b(s)$. Moreover $G_\varepsilon - G_R = -\int_\varepsilon^R b(s)\, dF_s$. Rearranging gives

$$G_\varepsilon - G_R = b(\varepsilon)F_\varepsilon - b(R)F_R + \int_{(\varepsilon,R]} F_s\, db(s). \tag{25.7}$$

Altering b on a countable set of radii does not change the original Lebesgue shell integral, whereas the Stieltjes formula depends on the chosen representative. We next justify the passage to general data on one common full-measure set.

Use the representatives of F_ε constructed in the strong completion and in the preceding weak endpoint completion. In both cases the base coordinate F_1 is the almost-everywhere limit of core-data coordinates, and every other coordinate is fixed by its signed finite-annulus increment from 1. Choose the corresponding core approximation f_k as follows. In the strong L^p case, pass to a subsequence such that

$$\sum_k \|f_{k+1} - f_k\|_p < \infty.$$

Put

$$M_k = \sup_{q \in \mathbb{Q}_+} |T^P_{\Omega,q}(f_{k+1} - f_k)|.$$

The strong maximal estimate and Minkowski's inequality give

$$\sum_k \|M_k\|_p \le C\mathfrak{L}(\Omega) \sum_k \|f_{k+1} - f_k\|_p < \infty.$$

Applying this bound to finite partial sums and then using monotone convergence shows that $\sum_k M_k \in L^p$, and hence is finite almost everywhere. In the weak L^1 case, choose instead $\|f_{k+1} - f_k\|_1 \le 2^{-3k}$. By (25.1),

$$\sum_k \left| \left\{ \sup_{q \in \mathbb{Q}_+} |T^P_{\Omega,q}(f_{k+1} - f_k)| > 2^{-k} \right\} \right| < \infty,$$

so Borel–Cantelli gives the same uniform Cauchy conclusion outside one null set. Thus in either case $F_q^{(k)}$ converges uniformly in $q \in \mathbb{Q}_+$ on a common full-measure set. Since every core-data difference is locally continuous in the truncation parameter, the same sequence is uniformly Cauchy on each compact parameter interval: for a continuous difference, the supremum on such an interval is its supremum on the rational parameters there.

The locally uniform limit of the continuous core trajectories is locally continuous. Fix at this point Borel representatives of f and Ω. Also choose a Borel function F_1^{B} which agrees almost everywhere with the already completed base coordinate F_1. All finite-annulus integrals below are formed with these fixed Borel representatives. For every rational pair $0 < a < c < \infty$, Young's inequality uses

$$\|K_\Omega \mathbf{1}_{\{a<|\cdot|\le c\}}\|_1 \le \|\Omega\|_1 \log(c/a) \tag{25.8}$$

to pass the core-data finite-annulus identity to the limit in L^p in the strong case and in L^1 in the weak case. Pass to one further diagonal subsequence. The locally uniform convergence is unaffected, and outside one null set the identity holds simultaneously for every rational pair.

For $j \ge 1$, put

$$J_j(x) = \int_{2^{-j}<|z|<2^j} |K_\Omega(z)|\,|f(x-z)|\,dz.$$

The parameter integral theorem makes every J_j Borel measurable, and Young's inequality shows that it belongs to L^p in the strong case and to L^1 in the weak case. Form one exceptional set from the following null sets: the failure set for the locally uniform core limit; the set on which $F_1^{\mathrm{B}} \neq F_1$; the simultaneous failure set for the rational finite-annulus identities just obtained; the sets where some J_j is infinite; and the failure sets for the zero anchor and for finiteness of the pointwise critical-jump functional. This is a countable union of Lebesgue null sets. Contain it in a Borel null set $\mathscr{N}_{\mathrm{BV}}$, and put

$$\mathscr{X}_{\mathrm{BV}} = \mathbb{R}^n \setminus \mathscr{N}_{\mathrm{BV}}.$$

Replace F_1^{B} by zero on $\mathscr{N}_{\mathrm{BV}}$, and henceforth denote this Borel base coordinate simply by F_1. Thus, at every point of $\mathscr{X}_{\mathrm{BV}}$, it is the previously completed base coordinate, all rational finite-annulus identities hold for the fixed Borel versions of f and Ω, the core trajectories converge locally uniformly, and the zero anchor and finite critical-jump bound hold.

The function

$$\Phi(x,r,\theta) = e^{iP(x,x-r\theta)}\Omega(\theta) f(x-r\theta), \qquad (x,r,\theta) \in \mathbb{R}^n \times (0,\infty) \times \mathbb{S}^{n-1},$$

is jointly Borel measurable. By the parameter integral theorem, both $(x,r) \mapsto \int_{\mathbb{S}^{n-1}} |\Phi(x,r,\theta)|\,d\sigma(\theta)$ and the integrals of the real and imaginary parts of Φ, wherever absolutely convergent, are Borel measurable. Define on the entire product space

$$\mathscr{A}(x,r) = \begin{cases} \displaystyle\int_{\mathbb{S}^{n-1}} \Phi(x,r,\theta)\,d\sigma(\theta), & x \in \mathscr{X}_{\mathrm{BV}} \text{ and } \displaystyle\int_{\mathbb{S}^{n-1}} |\Phi(x,r,\theta)|\,d\sigma(\theta) < \infty, \\ 0, & \text{otherwise.} \end{cases} \tag{25.9}$$

Thus $(x,r) \mapsto \mathscr{A}(x,r)$ is a jointly Borel version, not merely a separately defined family of radial sections. Polar coordinates and the finiteness of the J_j give, for $x \in \mathscr{X}_{\mathrm{BV}}$ and on every compact $0 < a < c < \infty$,

$$\int_a^c |\mathscr{A}(x,r)|\,\frac{dr}{r} \le \int_{a<|z|<c} |K_\Omega(z)|\,|f(x-z)|\,dz < \infty.$$

Thus the sphere integral is absolutely convergent for dr/r-almost every r, which is exactly the representative needed below; no assertion is made about every individual spherical restriction of an L^p function. For rational $a < c$, the finite-annulus identity is

$$F_a(x) - F_c(x) = \int_a^c \mathscr{A}(x,r)\,\frac{dr}{r}.$$

The left-hand side is continuous in the endpoints, while the indefinite integral on the right is locally absolutely continuous. Density of the rational endpoints therefore extends the identity to all real $0 < a < c < \infty$. Thus, on this same full-measure set, F is locally absolutely continuous and

$$\int_a^c |\mathscr{A}(x,r)| \, \frac{dr}{r} < \infty, \qquad F_a(x) - F_c(x) = \int_a^c \mathscr{A}(x,r) \, \frac{dr}{r}. \tag{25.10}$$

Using the Borel base coordinate F_1 fixed above, define on the entire product space

$$F(x,s) = F_1(x) + \mathbf{1}_{\{s<1\}} \int_s^1 \mathscr{A}(x,r) \, \frac{dr}{r} - \mathbf{1}_{\{s>1\}} \int_1^s \mathscr{A}(x,r) \, \frac{dr}{r}. \tag{25.11}$$

The parameter integral theorem makes $(x,s) \mapsto F(x,s)$ jointly Borel measurable. On $\mathscr{X}_{\mathrm{BV}}$ it agrees with every previously constructed coordinate by (25.10); on its complement it is identically zero. We henceforth use this common representative. Finally, the zero anchor and (15.7) give

$$F_R \longrightarrow 0 \quad (R \to \infty), \qquad M_F := \sup_{s>0} |F_s| \le \mathbf{J}(F) < \infty.$$

On the entire product space define

$$G(x,\varepsilon) = b(\varepsilon) F(x,\varepsilon) + \int_{(\varepsilon,\infty)} F(x,s) \, db(s). \tag{25.12}$$

The Stieltjes integral is absolutely convergent because $|db|(0,\infty) < \infty$ and $M_F < \infty$ on $\mathscr{X}_{\mathrm{BV}}$, and both F and G are zero on its complement. Moreover, the integrand $(x,\varepsilon,s) \mapsto \mathbf{1}_{\{s>\varepsilon\}} F(x,s)$ is jointly Borel. Applying the parameter integral theorem to the total-variation measure $|db|$, and then to the real and imaginary parts of the complex measure db, proves that $(x,\varepsilon) \mapsto G(x,\varepsilon)$ is jointly Borel measurable. For $0 < \varepsilon < R < \infty$, integration by parts for the locally absolutely continuous path F and the fixed right-continuous representative of b uses the product formula

$$b(R) F_R - b(\varepsilon) F_\varepsilon = \int_{(\varepsilon,R]} F_s \, db(s) + \int_\varepsilon^R b(s) \, dF_s. \tag{25.13}$$

There is no ambiguity between $b(s)$ and $b(s-)$ in the second integral, because dF has no atoms. By (25.10),

$$dF_s = -\mathscr{A}(x,s) \, \frac{ds}{s}.$$

This equality is an equality of locally finite absolutely continuous measures and uses the preceding $L^1_{\mathrm{loc}}(ds/s)$ representative of $\mathscr{A}$; in particular it is only an almost-everywhere density assertion. Consequently, (25.13) gives

$$\begin{aligned} G_\varepsilon - G_R &= b(\varepsilon) F_\varepsilon - b(R) F_R + \int_{(\varepsilon,R]} F_s \, db(s) \\ &= -\int_{(\varepsilon,R]} b(s) \, dF_s \\ &= \int_{\varepsilon<|z|\le R} e^{iP(x,x-z)} K_\Omega(z) b(|z|) f(x-z) \, dz. \end{aligned} \tag{25.14}$$

Indeed, $-\int_\varepsilon^R b(s) \, dF_s = \int_\varepsilon^R b(s) \mathscr{A}(x,s) \, ds/s$, and polar coordinates give exactly the last line, including its sign and half-open endpoint convention. The last integral is absolutely meaningful on the same set; its kernel has L^1 norm at most $\|b\|_\infty \|\Omega\|_1 \log(R/\varepsilon)$. Hence (25.12) has exactly the required finite-annulus increments for general data. For clarity, it also agrees with the original operator on core data, rather than merely defining an abstract extension. Indeed, if f is bounded and compactly supported, then

$$\widetilde{G}_\varepsilon(x) = \int_{|z|>\varepsilon} e^{iP(x,x-z)} K_\Omega(z) b(|z|) f(x-z) \, dz$$

is absolutely convergent for each $\varepsilon > 0$, has the same finite-annulus increments as G by (25.14), and satisfies $\widetilde{G}_R(x) = 0$ for all sufficiently large R. Since $G_R(x) \to 0$, letting $R \to \infty$ in the common increment identity gives $G_\varepsilon(x) = \widetilde{G}_\varepsilon(x)$ on the common full-measure set. For general data, the finite-annulus increments and zero anchor uniquely determine the extension. Right continuity fixes the Stieltjes representative, while altering b at one radius does not change the original Lebesgue shell integral. Finally,

$$|G_\varepsilon| \le \|b\|_\infty |F_\varepsilon| + M_F |db|((\varepsilon,\infty)) \longrightarrow 0 \qquad (\varepsilon \to \infty), \tag{25.15}$$

so the zero anchor uniquely determines the extension.

Put $B = \|b\|_\infty$, $V = \mathrm{Var}(b)$, and $M_F = \sup_{s>0} |F_s|$. The zero anchor gives $M_F \le \mathbf{J}(F)$ by (15.7). For every $r \ge 1$, the same anchor also gives

$$M_F \le V^r(F). \tag{25.16}$$

Indeed, for fixed $s > 0$ and $R > s$, the two-point partition gives $|F_s - F_R| \le V^r(F)$; letting $R \to \infty$ and then taking the supremum in s proves (25.16). For a variation partition $t_0 < \cdots < t_N$, use the exact product increment

$$b(t_i)F_{t_i} - b(t_{i-1})F_{t_{i-1}} = b(t_i)(F_{t_i} - F_{t_{i-1}}) + (b(t_i) - b(t_{i-1}))F_{t_{i-1}}.$$

Minkowski's inequality and $(\sum_i |\Delta_i b|^r)^{1/r} \le \sum_i |\Delta_i b| \le V$ now give the first line below. Together with (25.12) this yields, pointwise,

$$\begin{aligned} V^r(bF) &\le BV^r(F) + VM_F, \\ V^r\left(\int_{(\varepsilon,\infty)} F_s\, db(s) : \varepsilon > 0\right) &\le VM_F, \\ V^r(G) &\le (B+2V)V^r(F). \end{aligned} \tag{25.17}$$

Indeed, an increment of the Stieltjes-tail path over $(t_{i-1}, t_i]$ is the integral of F over that half-open interval. These intervals are pairwise disjoint, so its ℓ^r variation is bounded by the sum of their $|db|$-masses times M_F, hence by VM_F.

We record separately the critical-jump transfer. If $M_F = 0$, then $F_s = 0$ for every $s > 0$, and (25.12) gives $G_\varepsilon = 0$; all assertions are then immediate. Hence assume $M_F > 0$. Suppose first that $B > 0$. For an ordered family of λ-jumps of bF, write

$$b(t)F_t - b(s)F_s = b(t)(F_t - F_s) + (b(t) - b(s))F_s.$$

Assign the pair to the first term if its modulus is $> \lambda/2$, and otherwise to the second. The first subfamily contributes at most $N_{\lambda/(2B)}(F)$. The parameter intervals in the second subfamily are disjoint, and hence

$$\frac{\lambda}{2M_F} N_{\text{second}} < \sum_{\text{second}} |b(t) - b(s)| \le V.$$

This proves

$$N_\lambda(bF) \le N_{\lambda/(2B)}(F) + \frac{2M_F V}{\lambda}. \tag{25.18}$$

Use $\sqrt{u+v} \le \sqrt{u} + \sqrt{v}$. On the set where the left side is nonzero, $\lambda < 2BM_F$. With $\mu = \lambda/(2B)$, the contribution of the second term after taking square roots is bounded by

$$\sqrt{2\lambda M_F V} \le 2M_F \sqrt{BV} \le (B+V)M_F.$$

It follows that

$$\lambda N_\lambda(bF)^{1/2} \le 2B\mu N_\mu(F)^{1/2} + C(B+V)M_F. \tag{25.19}$$

For $B = 0$ the path is zero. Define the Stieltjes-tail path by

$$H_\varepsilon = \int_{(\varepsilon,\infty)} F_s\, db(s).$$

It satisfies $V^1(H) \le VM_F$. The disjoint jump intervals give $\lambda N_\lambda(H) < V^1(H) \le VM_F$, whereas a nonzero jump also forces $\lambda < 2\sup_\varepsilon |H_\varepsilon| \le 2VM_F$. Hence

$$\lambda N_\lambda(H)^{1/2} \le CVM_F. \tag{25.20}$$

Taking the supremum in λ in (25.19)–(25.20) gives

$$\mathbf{J}(bF) \le 2B\mathbf{J}(F) + C(B+V)M_F, \qquad \mathbf{J}(H) \le CVM_F.$$

Since $G = bF + H$, equations (15.5) and (15.7) imply the pointwise transfer

$$\mathbf{J}(G) \le C(B+V)\mathbf{J}(F). \tag{25.21}$$

The strong and weak pointwise critical-jump estimates for G now follow directly from Theorem 1.2. Formula (25.17) gives the asserted strong and weak r-variation bounds. This proves the estimates in Corollary 1.5.

The almost-everywhere zero anchor was proved in (25.15). For $f \in L^p$, the strong zero anchor for F, the strong maximal estimate, and continuity from above for the finite measure $|db|$ give the explicit bound

$$\|G_\varepsilon\|_p \le B\|F_\varepsilon\|_p + |db|((\varepsilon,\infty))\|M_F\|_p \longrightarrow 0.$$

Thus both terms also tend to zero in L^p. Since a scalar path of finite r-variation has one-sided limits at both endpoints, G_ε has an almost-everywhere limit as $\varepsilon \downarrow 0$. The zero anchor gives $\sup_{\varepsilon>0}|G_\varepsilon| \le \mathbf{J}(G)$ by (15.7), and hence the maximal truncation assertion. This completes the proof of Corollary 1.5.

26 Finite radial variation beyond bounded variation

For $1 \le \rho < \infty$ recall the radial norm

$$\|b\|_{\mathscr{V}^\rho} = \|b\|_\infty + V^\rho(b;(0,\infty)).$$

The purpose of this section is twofold. First, we record a direct pathwise Young–Stieltjes transfer when $\rho < 2$. Second, we prove Theorem 1.4 for every fixed finite ρ by showing that each oscillatory, packet, endpoint, and completion component of the unweighted proof is stable under a radial V^ρ amplitude. The latter argument is genuinely operator-specific at and above $\rho = 2$.

26.1 A pathwise Young–Stieltjes transfer below the radial exponent two

For radial variation below two, the full operator-specific proof is not needed. The following abstract statement isolates a direct consequence of the pointwise critical-jump theorem. It is also useful for identifying the precise obstruction at the exponent two.

Throughout this subsection, for a right-continuous integrator b we use the Young increment notation

$$db((s,t]) = b(t) - b(s).$$

This is only a half-open interval-increment convention; when $\rho > 1$, it does not assert that db is a countably additive Stieltjes measure. Such a measure is available in the BV case, whereas the argument below uses Young sums and the Young–Loeve estimate. Young sums on a finite annulus use the corresponding half-open interval $(a,c]$: a tagged sum over $a = t_0 < \cdots < t_N = c$ is

$$\sum_{j=1}^{N} F(\xi_j)\{b(t_j) - b(t_{j-1})\}, \qquad \xi_j \in [t_{j-1}, t_j].$$

Thus a jump of the right-continuous integrator at the right endpoint is included and one at the left endpoint is not. For $F = F_1 + iF_2$ and $b = b_1 + ib_2$, the complex Young integral is defined by

$$\int F\,db := \int F_1\,db_1 - \int F_2\,db_2 + i\left(\int F_1\,db_2 + \int F_2\,db_1\right).$$

Thus it is the complex-bilinear extension of the four real integrals; the real Young–Loeve estimate therefore gives the same bound up to an absolute constant. Since the path F below is locally absolutely continuous and hence continuous, there is no common-jump correction in integration by parts. The integral against dF is the ordinary Lebesgue–Stieltjes integral; since local absolute continuity makes dF atomless, its value is unchanged if $b(s)$ is replaced by $b(s-)$.

Proposition 26.1 (Young–Stieltjes critical-jump transfer)**.** *Let $F = (F_\varepsilon)_{\varepsilon>0}$ be a locally absolutely continuous scalar path such that*

$$F_\varepsilon \longrightarrow 0 \quad (\varepsilon \to \infty), \qquad \mathbf{J}(F) < \infty.$$

Fix $1 \le \rho < 2$, and let $b : (0,\infty) \to \mathbb{C}$ be bounded and of finite ρ-variation. Choose its right-continuous representative. Then the improper Young integral

$$H_\varepsilon := \int_{(\varepsilon,\infty)} F_s\,db(s) \tag{26.1}$$

exists, and the anchored path

$$G_\varepsilon := b(\varepsilon)F_\varepsilon + H_\varepsilon \tag{26.2}$$

satisfies

$$\mathbf{J}(G) \le C_\rho\big(\|b\|_\infty + V^\rho(b)\big)\mathbf{J}(F). \tag{26.3}$$

Moreover $G_\varepsilon \to 0$ as $\varepsilon \to \infty$, and for every $0 < a < c < \infty$,

$$G_a - G_c = -\int_a^c b(s)\,dF_s. \tag{26.4}$$

Proof. Use the norm-nonincreasing right-continuous representative constructed after (1.5). To keep every constant quantitative, fix once and for all

$$u = u_\rho := \begin{cases} 3, & \rho = 1, \\ \frac{1}{2}(2+\rho'), & 1 < \rho < 2, \end{cases} \qquad \rho' = \frac{\rho}{\rho - 1}.$$

Then

$$2 < u < \rho', \qquad \frac{1}{u} + \frac{1}{\rho} > 1, \tag{26.5}$$

where $\rho' = \infty$ when $\rho = 1$. Put

$$J = \mathbf{J}(F), \quad R = V^u(F), \quad Q = V^\rho(b), \quad B = \|b\|_\infty, \quad M = \sup_{s>0}|F_s|.$$

Equations (15.8) and (15.7) give

$$R \le \kappa_u J, \qquad M \le J. \tag{26.6}$$

We construct the compact Young integral directly. On $[a,c]$, put $R_0 = V^u(F;[a,c])$, $Q_0 = V^\rho(b;[a,c])$, and $\theta = 1/u + 1/\rho > 1$. If either variation is zero, the construction is immediate. For a partition $a = t_0 < \cdots < t_m = c$, set

$$w_i = R_0^{-u} V^u(F;[t_{i-1},t_i])^u + Q_0^{-\rho} V^\rho(b;[t_{i-1},t_i])^\rho.$$

Concatenation gives $\sum_i w_i \le 2$. When $m \ge 2$, some interior index i has $w_i + w_{i+1} \le 4/(m-1)$. Removing t_i changes the left Young sum by

$$(F_{t_i} - F_{t_{i-1}})(b(t_{i+1}) - b(t_i)),$$

whose modulus is at most $R_0 Q_0 (4/(m-1))^\theta$. Apply the same argument to each reduced partition. Since $\sum_{j\ge1} j^{-\theta} < \infty$, every left sum differs from $F_a(b(c)-b(a))$ by at most $4^\theta \sum_{j\ge1} j^{-\theta} R_0 Q_0$. This partition-removal argument also allows jumps of b. Finite integration by parts expresses that left sum as $b(c)F_c - b(a)F_a$ minus the integral of F' times the right-endpoint step approximation to b. Right continuity and boundedness of b, local absolute continuity of F, and dominated convergence show that, as the mesh tends to zero, these sums converge to

$$b(c)F_c - b(a)F_a - \int_a^c b(s) F_s' \, ds.$$

Finally let ω be the largest oscillation of F on a partition cell and set $\zeta = 1 - u(1-1/\rho) > 0$. The difference between any tagged sum and its left sum is bounded by $C\omega^\zeta R_0^{1-\zeta} Q_0$ by finite Hölder and concatenation of the local variation powers. Uniform continuity of F makes this correction tend to zero for every choice of tags. Thus the actual all-tag Young integral exists with the stipulated half-open convention and the integration-by-parts identity. Passing the partition-removal estimate to its limit gives the classical Young–Loeve bound [39]

$$\left| \int_{(a,c]} (F_s - F_a) \, db(s) \right| \le C_{u,\rho} V^u(F;[a,c]) V^\rho(b;[a,c]). \tag{26.7}$$

Consequently,

$$\left| \int_{(a,c]} F_s \, db(s) \right| \le \sup_{a\le s\le c} |F_s| \, |b(c)-b(a)| + C_{u,\rho} V^u(F;[a,c]) V^\rho(b;[a,c]). \tag{26.8}$$

Finite ρ-variation gives a limit of b at infinity, while finite u-variation and the zero anchor imply that both $\sup_{s\ge R}|F_s|$ and $V^u(F;[R,\infty))$ tend to zero as $R\to\infty$. For completeness, the latter assertion follows directly from finite variation. If it failed, there would be an $\eta > 0$ and, recursively, disjoint finite intervals $[R_k, S_k]$, tending to infinity, with

$$V^u(F;[R_k,S_k]) > \eta.$$

Choose on each such interval a finite partition whose u-power sum is larger than η^u. Concatenating the first N partitions gives $V^u(F)^u \ge N\eta^u$, a contradiction. Thus the two terms on the right of (26.8), with $a, c \ge R$, tend to zero uniformly as $R\to\infty$. Hence that estimate makes the compact Young integrals Cauchy and proves (26.1). It also yields

$$\sup_{\varepsilon>0} |H_\varepsilon| \le C_{u,\rho} Q(M+R) \le C_\rho Q J. \tag{26.9}$$

With the half-open convention fixed above, Young integration by parts (valid because F is continuous) gives for $0 < \varepsilon < R_0 < \infty$,

$$b(R_0)F_{R_0} - b(\varepsilon)F_\varepsilon = \int_{(\varepsilon,R_0]} F_s \, db(s) + \int_\varepsilon^{R_0} b(s) \, dF_s.$$

As $R_0\to\infty$, the first integral on the right converges to H_ε by its Cauchy construction, while $|b(R_0)F_{R_0}| \le B|F_{R_0}| \to 0$. The identity therefore also proves the existence of the improper integral against dF and gives

$$G_\varepsilon = b(\varepsilon)F_\varepsilon + H_\varepsilon = -\int_\varepsilon^\infty b(s) \, dF_s. \tag{26.10}$$

Separately, as $\varepsilon\to\infty$,

$$|b(\varepsilon)F_\varepsilon| \le B \sup_{s\ge\varepsilon} |F_s| \longrightarrow 0, \qquad |H_\varepsilon| \le Q \sup_{s\ge\varepsilon} |F_s| + C_{u,\rho} V^u(F;[\varepsilon,\infty)) Q \longrightarrow 0.$$

Thus $G_\varepsilon \to 0$ at infinity. For two finite endpoints, the tail definition and finite-interval integration by parts give explicitly

$$\begin{aligned} G_a - G_c &= b(a)F_a - b(c)F_c + \int_{(a,c]} F_s\, db(s) \\ &= -\int_a^c b(s)\, dF_s. \end{aligned}$$

This proves (26.4), including its sign and half-open endpoint convention.

It remains to estimate jumps. Consider an ordered family of strict λ-jumps of the product path bF. On each jump interval (s,t),

$$b(t)F_t - b(s)F_s = b(t)(F_t - F_s) + (b(t) - b(s))F_s.$$

Assign the interval to the first term when its modulus exceeds $\lambda/2$, and otherwise to the second. For $B > 0$ this gives

$$N_\lambda(bF) \le N_{\lambda/(2B)}(F) + \left(\frac{2MQ}{\lambda}\right)^\rho. \tag{26.11}$$

A nonzero jump also forces $\lambda \le 2BM$. Since $\rho < 2$,

$$\lambda \left(\frac{2MQ}{\lambda}\right)^{\rho/2} \le C_\rho M B^{1-\rho/2} Q^{\rho/2} \le C_\rho M(B+Q).$$

Moreover,

$$\lambda N_{\lambda/(2B)}(F)^{1/2} = 2B\frac{\lambda}{2B} N_{\lambda/(2B)}(F)^{1/2} \le 2BJ.$$

Using $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$ in (26.11), and then $M \le J$, therefore gives

$$\mathbf{J}(bF) \le C_\rho (B+Q)J. \tag{26.12}$$

The case $B = 0$ is immediate.

Now let (s_i,t_i) be ordered λ-jump intervals for H, and put

$$A_i = V^u(F;[s_i,t_i]), \qquad C_i = V^\rho(b;[s_i,t_i]).$$

By (26.8),

$$|H_{s_i} - H_{t_i}| \le M|b(t_i) - b(s_i)| + C_{u,\rho} A_i C_i.$$

If $Q = 0$, the fixed right-continuous representative of b is constant, the Young tail H vanishes, and there is nothing to prove. Hence assume $Q > 0$ below. Assign each interval to a family $\mathscr{I}_1$ or $\mathscr{I}_2$ according as the first or the second term on the right exceeds a fixed positive multiple of λ, breaking ties arbitrarily. If $N_\nu = \#\mathscr{I}_\nu$, then

$$N_1 \le C\left(\frac{MQ}{\lambda}\right)^\rho.$$

Whenever a jump exists, (26.9) gives $\lambda \le C_\rho QJ$. Since $\rho < 2$ and $M \le J$, the precise interpolation is

$$\lambda N_1^{1/2} \le C(MQ)^{\rho/2} \lambda^{1-\rho/2} \le C_\rho (MQ)^{\rho/2} (QJ)^{1-\rho/2} \le C_\rho QJ.$$

For the second family define $0 < \theta < 1$ by

$$\frac{1}{\theta} = \frac{1}{u} + \frac{1}{\rho}.$$

Disjointness of the jump intervals and Hölder's inequality give

$$N_2(c\lambda)^\theta \le \sum_{i\in\mathscr{I}_2} (A_i C_i)^\theta \le \left(\sum_{i\in\mathscr{I}_2} A_i^u\right)^{\theta/u} \left(\sum_{i\in\mathscr{I}_2} C_i^\rho\right)^{\theta/\rho} \le R^\theta Q^\theta, \qquad N_2 \le C_{u,\rho}\left(\frac{RQ}{\lambda}\right)^\theta.$$

Using again $\lambda \le C_\rho QJ$ and $R \le \kappa_u J$ gives

$$\lambda N_2^{1/2} \le C_{u,\rho}(RQ)^{\theta/2}\lambda^{1-\theta/2} \le C_\rho (RQ)^{\theta/2}(QJ)^{1-\theta/2} \le C_\rho QJ.$$

For the original witness $N = N_1 + N_2$, and hence $\lambda\sqrt{N} \le \lambda\sqrt{N_1} + \lambda\sqrt{N_2} \le C_\rho QJ$. Taking the supremum over all finite strict witnesses and then over $\lambda > 0$ gives

$$\mathbf{J}(H) \le C_\rho QJ. \tag{26.13}$$

Finally apply the pointwise quasi-triangle inequality (15.5) to $G = bF + H$ and combine (26.12) with (26.13). □

26.2 The operator-specific theorem for every finite radial exponent

We now prove Theorem 1.4. The argument retains the notation of the preceding sections so that every dependency can be checked directly. By homogeneity in b, it is enough to impose

$$\|b\|_{\mathscr{V}^\rho} \le 1. \tag{26.14}$$

The factor $\|b\|_{\mathscr{V}^\rho}$ is restored only at the end. Notice also the exact dilation invariance

$$\|b(R\cdot)\|_\infty = \|b\|_\infty, \qquad V^\rho(b(R\cdot)) = V^\rho(b), \qquad R > 0. \tag{26.15}$$

Quantitative convention. Fix n, d, and $1 \le \rho < \infty$ for the remainder of this section. Every polynomial space, rational chart, compact interval, cutoff, and finite-dimensional norm below belongs to one of the finitely many structural families already fixed in Sections 16–20. We therefore take the minimum of the finitely many positive sublevel exponents in Lemma 20.1 and denote it by $c_{\mathrm{sub}} = c_{\mathrm{sub}}(n,d) > 0$. Constants denoted by C_ρ may increase from line to line and depend only on n, d, ρ and the fixed structural data. Polynomial powers of an amplitude norm are structural and finite; whenever such a power enters a later parameter choice, it is fixed once and for all at that point. Positive exponents are always named before they are used. Replacing an exponent by a smaller positive number is allowed without further comment. This convention makes the parameter choices below finite and noncircular: first the oscillatory exponents are fixed, then the packet hierarchy, and only afterwards the angular splitting parameters.

26.3 Finite ρ-variation calculus

We first collect the elementary facts used everywhere below. Throughout this subsection a compactly supported radial profile is first replaced by its fixed right-continuous representative and then extended by zero to the real line. Variations without an indicated interval refer to this zero extension. The endpoint convention for consecutive cells is specified in each displayed partition, and no endpoint value is counted twice. Thus every ordinary supremum and every endpoint jump refers to one fixed representative.

Lemma 26.2 (Disjoint intervals, pullbacks, products, and step profiles)**.** *Let $1 \le \rho < \infty$.*

(i) *If a is a scalar function with $V^\rho(a) < \infty$ and $I_1, \dots, I_M$ are pairwise disjoint intervals arranged from left to right, then*

$$\sum_{m=1}^{M} V^\rho(a; I_m)^\rho \le V^\rho(a)^\rho. \tag{26.16}$$

(ii) *If $a : I \to \mathbb{C}$ has finite ρ-variation and $\gamma : J \to I$ is monotone, then*

$$V^\rho(a \circ \gamma; J) \le V^\rho(a; I). \tag{26.17}$$

(iii) *For bounded f, g with $V^\rho(f) + V^\rho(g) < \infty$,*

$$V^\rho(fg) \le \|f\|_\infty V^\rho(g) + \|g\|_\infty V^\rho(f). \tag{26.18}$$

(iv) *Let $x_0 < \cdots < x_K$ and $I_k = [x_{k-1}, x_k)$ for $1 \le k \le K$. Extend σ by zero off $[x_0, x_K)$, and suppose that σ is constant on each I_k and $|\sigma| \le 1$. Then*

$$V^\rho(\sigma) \le 2K^{1/\rho}. \tag{26.19}$$

Consequently, if additionally $0 < x_0 < \cdots < x_K < \infty$ and (26.14) holds, then, with $b\sigma$ extended by zero off $[x_0, x_K)$,

$$\|b\sigma\|_\infty + V^\rho(b\sigma) \le C_\rho(1 + K^{1/\rho}). \tag{26.20}$$

Proof. For (i), choose on every I_m a partition which is within an arbitrary ε of its variation and concatenate these partitions in increasing order. The within-I_m increments are a subcollection of the increments in the resulting global partition. Letting $\varepsilon \downarrow 0$ proves (26.16). Assertion (ii) follows by pulling an ordered partition of J through γ and deleting repetitions; if γ is decreasing, reverse the resulting list, which does not change the moduli of its successive increments. For (iii), on every increment

$$f(t)g(t) - f(s)g(s) = f(t)(g(t) - g(s)) + g(s)(f(t) - f(s)),$$

and Minkowski in the finite ℓ^ρ norm gives (26.18). Finally, fix an arbitrary variation partition for the zero-extended step profile. Delete zero increments and all but one point from each constant cell that the partition visits. The remaining ordered list has at most $K-1$ increments between two cells, each of modulus at most two, and at most two increments between the exterior zero value and an interior cell, each of modulus at most one. Thus every such partition satisfies

$$\sum_j |\sigma(t_j) - \sigma(t_{j-1})|^\rho \le 2 + (K-1)2^\rho \le K2^\rho.$$

Taking the supremum proves (26.19). More explicitly, the zero extension creates at most two endpoint increments, and hence

$$V^\rho(b\sigma; \mathbb{R}) \le \|b\|_{L^\infty(\operatorname{supp}\sigma)} V^\rho(\sigma; \mathbb{R}) + \|\sigma\|_\infty \{V^\rho(b; \operatorname{supp}\sigma) + 2\|b\|_\infty\}.$$

This proves (26.20) without extending b itself outside $(0, \infty)$. □

Lemma 26.3 (Translation modulus and mollification)**.** *Let $I \Subset \mathbb{R}$ be fixed and let a be bounded, supported in the interior of I, and extended by zero to $\mathbb{R}$. All variation seminorms in this lemma are taken for that zero extension. Then for $|h| \le 1$,*

$$\|a(\cdot + h) - a\|_{L^1(\mathbb{R})} \le C_{I,\rho} |h|^{1/\rho} \big(\|a\|_\infty + V^\rho(a; I)\big). \tag{26.21}$$

If $0 < h \le 1$, ϕ is a standard compactly supported mollifier, $\phi_h(t) := h^{-1}\phi(t/h)$, and $a_h := a * \phi_h$, then, for every integer $N \ge 0$,

$$\|a - a_h\|_1 \le C_{I,\rho} h^{1/\rho} \big(\|a\|_\infty + V^\rho(a; I)\big), \tag{26.22}$$

$$V^\rho(a_h) \le V^\rho(a), \tag{26.23}$$

$$\|\partial^N a_h\|_\infty \le C_N h^{-N} \|a\|_\infty. \tag{26.24}$$

Proof. Write $I = [u, v]$; the same argument applies to any interval with these endpoints, since a vanishes near them. Monotone clipping to $[u, v]$ leaves the zero-extended function unchanged. Pulling finite partitions through this clipping map and deleting repeated points gives exactly

$$V^\rho(a; \mathbb{R}) = V^\rho(a; [u, v]).$$

Thus the global and restricted-interval readings of the displayed variation agree. The translation bound at $h = 0$ is immediate. Put $V = V^\rho(a; \mathbb{R})$, and first suppose $V < \infty$. The cumulative power variation

$$H(t) = V^\rho(a; (-\infty, t])^\rho$$

is nondecreasing, lies in $[0,V^\rho]$, and satisfies $|a(y)-a(x)|^\rho \le H(y)-H(x)$ for $x\le y$. Indeed append x,y to a finite partition ending at or before x, then take suprema. This also proves Borel measurability of the literal path: a is continuous at every continuity point of H, and a monotone function has only countably many discontinuities. For $h\ge 0$, integration of the monotone increment gives

$$\int_{\mathbb{R}} |a(x+h)-a(x)|^\rho\,dx \le \int_{\mathbb{R}} (H(x+h)-H(x))\,dx \le hV^\rho.$$

The last inequality follows first on finite intervals by cancellation of the overlapping integrals and then by monotone convergence. Translation invariance gives the same estimate with $|h|$ for negative h. If $|h|\le 1$, the difference is supported in $[u-1,v+1]$, so Hölder (including its endpoint $\rho=1$) yields the stronger estimate

$$\|a(\cdot+h)-a\|_1 \le (v-u+2)^{1-1/\rho}|h|^{1/\rho}V.$$

This proves (26.21).

A standard mollifier is a nonnegative smooth kernel of integral one, supported, after fixing its scale, in $[-1,1]$. Then $\phi_h(t)\,dt$ is a probability measure supported in $[-h,h]$. Tonelli and the translation estimate give

$$\|a-a_h\|_1 \le \int \phi_h(t)\|a-a(\cdot-t)\|_1\,dt \le (v-u+2)^{1-1/\rho}h^{1/\rho}V,$$

which implies (26.22). For every increasing finite partition (t_i), convexity and positivity give

$$\sum_i |a_h(t_i)-a_h(t_{i-1})|^\rho \le \int \phi_h(y)\sum_i |a(t_i-y)-a(t_{i-1}-y)|^\rho\,dy \le V^\rho.$$

Taking the partition supremum proves (26.23) with constant exactly one. If $V=\infty$, the estimates containing V are immediate; the fixed right-continuous representative stipulated in this subsection is still measurable.

For the remaining estimate only boundedness and measurability of a are needed. Such an a is locally integrable, so convolution with the actual smooth compactly supported kernel is smooth and

$$\partial^N a_h = (\partial^N\phi_h)*a, \qquad \partial^N\phi_h(t) = h^{-N-1}(\partial^N\phi)(t/h).$$

Consequently $\|\partial^N\phi_h\|_1 = h^{-N}\|\partial^N\phi\|_1$, and Young's inequality proves (26.24) with $C_N=\|\partial^N\phi\|_1$, also for $N=0$. □

The next elementary upper-sum estimate is the missing ingredient in the terminal packet argument. Small L^1 error alone is not enough there, because packet inputs are controlled by average mass rather than pointwise density.

Lemma 26.4 (Upper Riemann sums for finite variation)**.** *Let $I\Subset\mathbb{R}$ be a fixed bounded interval, and let $a:\mathbb{R}\to\mathbb{C}$ be bounded, supported in I, and satisfy $V^\rho(a;\mathbb{R})<\infty$. Partition I into consecutive intervals J_m whose lengths are comparable to $\delta\in(0,1]$. Then*

$$\delta\sum_m \sup_{J_m}|a| \le C_I\Big[\|a\|_{L^1(\mathbb{R})}+\delta^{1/\rho}\big(\|a\|_\infty+V^\rho(a;\mathbb{R})\big)\Big]. \tag{26.25}$$

If J_m^* is obtained from J_m by an enlargement about its centre by a fixed factor, then the same estimate holds with $\sup_{J_m^*}|a|$ on the left; the constant may additionally depend on that fixed factor.

Proof. Write the partition endpoints as $x_0<\cdots<x_M$, with $c\delta\le x_{m+1}-x_m\le C\delta$, and put $V=V^\rho(a;\mathbb{R})$. The cumulative power-variation control used above is a bounded nondecreasing function G satisfying $0\le G\le V^\rho$ and $|a(y)-a(x)|^\rho\le G(y)-G(x)$ for $x\le y$. For $\eta\ge 0$, consider the padded cells $J_m^\eta=[x_m-\eta\delta,x_{m+1}+\eta\delta]$ and set $A_m=G(x_{m+1}+\eta\delta)-G(x_m-\eta\delta)$. For $t\in[x_m,x_{m+1}]$, monotonicity gives

$$A_m \le G(t+(C+\eta)\delta)-G(t-(C+\eta)\delta).$$

The integrated-increment bound for a bounded monotone function therefore implies, after integration over the base cells,

$$c\delta\sum_m A_m \le 2(C+\eta)\delta V^\rho.$$

For $s \in J_m^\eta$ and $t \in [x_m, x_{m+1}]$, the control gives $|a(s)-a(t)| \le A_m^{1/\rho}$. Taking the literal pointwise supremum in s and averaging in t yields

$$\delta \sup_{J_m^\eta} |a| \le c^{-1} \int_{x_m}^{x_{m+1}} |a(t)|\,dt + \delta A_m^{1/\rho}.$$

Since $M \le |I|/(c\delta)$, finite Hölder, also valid at $\rho = 1$, gives

$$\delta\sum_m A_m^{1/\rho} \le (|I|/c)^{1-1/\rho}\big(2(C+\eta)/c\big)^{1/\rho}\delta^{1/\rho}V.$$

The case of no cells is immediate. Summing proves the desired bound, even without its additional nonnegative supremum-norm term. The factors are bounded uniformly for $\rho \ge 1$ using $z^e \le \max(1,z)$ for $z \ge 0$ and $0 \le e \le 1$. For an enlargement factor $L \ge 1$, the actual centre enlargement of each cell is contained in J_m^η with $\eta = (L-1)C/2$. The enlarged assertion follows by monotonicity of the pointwise supremum. This argument includes all endpoint values and uses neither a coloring nor an extra overlap hypothesis. □

Lemma 26.5 (Radial discretization for packet inputs)**.** *Fix $0 < c_- < c_+ < \infty$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, $\Lambda \ge 0$, and let w be a bounded measurable function supported in $[c_-, c_+]$ with $V^\rho(w) < \infty$. Let $0 < \delta \le 1$, let $\{U\}$ be pairwise disjoint dyadic δ-cubes in $\mathbb{R}^n$, and suppose functions $d_U \in L^1(\mathbb{R}^n)$ satisfy*

$$\operatorname{supp} d_U \subset U, \qquad \|d_U\|_1 \le \Lambda|U|.$$

Then

$$\left\| K_\Theta(\cdot)w(|\cdot|) * \sum_U d_U \right\|_\infty \le C\Lambda\|\Theta\|_\infty \Big[\|w\|_1 + \delta^{1/\rho}(\|w\|_\infty + V^\rho(w))\Big], \tag{26.26}$$

$$\left\| K_\Theta(\cdot)w(|\cdot|) * \sum_U d_U \right\|_1 \le C\|\Theta\|_1\|w\|_1 \sum_U \|d_U\|_1. \tag{26.27}$$

Here the L^1 norm of w is one-dimensional; on the fixed annulus it is equivalent to the natural dr/r norm.

Proof. Only (26.26) is new. Fix x. Since $|z|$ stays in a fixed annulus,

$$\left| K_\Theta w * \sum_U d_U(x) \right| \le C\Lambda\|\Theta\|_\infty \delta^n \sum_U \sup_{y\in U} |w(|x-y|)|.$$

Group the cubes according to intervals of length $C\delta$ containing the distance from x to their centers. If their centers lie in one such layer, the union of the corresponding disjoint cubes is contained in a fixed enlargement of a radial layer of volume $O(\delta)$. Since every cube has volume δ^n, the layer contains at most $C\delta^{1-n}$ cubes. Hence the last display is at most

$$C\Lambda\|\Theta\|_\infty \delta \sum_m \sup_{J_m^*} |w|,$$

where the J_m^* are fixed enlargements of consecutive radial intervals of length comparable to δ. Lemma 26.4 proves (26.26). Equation (26.27) is Young's inequality and polar coordinates. □

26.4 Oscillatory integrals with a finite-variation amplitude

Lemma 26.6 (First derivative estimate with a V^ρ amplitude)**.** *Let I be a fixed bounded interval, let $\phi \in C^1(I;\mathbb{R})$, and let $a : I \to \mathbb{C}$ be bounded with $V^\rho(a;I) < \infty$. Suppose ϕ' is monotone and*

$$|\phi'(t)| \geq L \geq 1 \qquad (t \in I).$$

Then

$$\left| \int_I e^{i\phi(t)} a(t)\, dt \right| \leq C_{I,\rho} L^{-1/\rho} \big(\|a\|_\infty + V^\rho(a;I) \big). \tag{26.28}$$

The same constant is valid when I is replaced by any subinterval of the fixed ambient interval.

Proof. Partition I into consecutive intervals $I_1, \dots, I_M$ of length at most L^{-1} and, except possibly for the last interval, at least $(2L)^{-1}$. Thus $M \leq C_I L$ because $L \geq 1$. Choose $t_m \in I_m$ in increasing order. The error after replacing a by $a(t_m)$ on I_m is bounded by

$$\sum_m |I_m| V^\rho(a;I_m) \leq C L^{-1} M^{1-1/\rho} \left(\sum_m V^\rho(a;I_m)^\rho \right)^{1/\rho} \leq C L^{-1/\rho} V^\rho(a;I),$$

using Lemma 26.2(i). Put $w_m = \int_{I_m} e^{i\phi(t)} dt$. The ordinary first derivative van der Corput estimate implies

$$\left| \sum_{m=r}^{s} w_m \right| = \left| \int_{\cup_{m=r}^{s} I_m} e^{i\phi(t)} dt \right| \leq C L^{-1}$$

for every consecutive block. Put $a_m = a(t_m)$ and $W_m = \sum_{j=1}^{m} w_j$. Discrete Abel summation gives

$$\sum_{m=1}^{M} a_m w_m = a_M W_M + \sum_{m=1}^{M-1} (a_m - a_{m+1}) W_m, \qquad |W_m| \leq C L^{-1}.$$

Consequently,

$$\left| \sum_m a(t_m) w_m \right| \leq C L^{-1} \left(\|a\|_\infty + \sum_{m=1}^{M-1} |a(t_{m+1}) - a(t_m)| \right).$$

Since the samples are ordered,

$$\left(\sum_{m=1}^{M-1} |a(t_{m+1}) - a(t_m)|^\rho \right)^{1/\rho} \leq V^\rho(a;I).$$

Hölder therefore bounds the last sum by $M^{1-1/\rho} V^\rho(a;I)$, proving (26.28). □

Lemma 26.7 (Rational van der Corput with a V^ρ amplitude)**.** *Let $\psi = N/Q$ on a fixed compact interval, where N, Q are real polynomials of degree at most D, no bound is imposed on the coefficients of N, and*

$$\|Q\|_{\mathrm{coeff}} \leq M, \qquad 0 < \eta \leq |Q| \leq M$$

on a fixed enlargement of the interval. Put

$$G := \|Q^2 \psi'\|_{\mathrm{coeff}} = \|N'Q - NQ'\|_{\mathrm{coeff}}.$$

Let $c_0 = c_0(D) > 0$ be a valid normalized one-dimensional polynomial sublevel exponent for the derived degree bound of $N'Q - NQ'$ (at most $2D-1$ when $D \geq 1$). Then, with

$$c_{\mathrm{vdc},\rho,D} := \frac{c_0}{1 + c_0 \rho} > 0, \tag{26.29}$$

every bounded amplitude a supported in the interval with $V^\rho(a) < \infty$ satisfies

$$\left| \int e^{i\psi(v)} a(v)\, dv \right| \le C_{\rho,D,\eta,M}(\|a\|_\infty + V^\rho(a)) \min\{1, G^{-c_{\mathrm{vdc},\rho,D}}\}. \tag{26.30}$$

When $G = 0$, the minimum on the right is understood to be 1. *The same assertion holds after a uniformly bounded sectional partition. In particular, the constants are independent of the coefficients of the numerator N.*

Proof. Put $p = Q^2\psi' = N'Q - NQ'$ and choose

$$\vartheta = \frac{c_0\rho}{1 + c_0\rho} \in (0,1).$$

The trivial L^1 bound proves the assertion for $G \le 1$. If $1 < G < (M^2)^{1/\vartheta}$, the same bound is still dominated by the right side of (26.30) after increasing its structural constant. We may therefore assume $M^{-2}G^\vartheta \ge 1$. The normalized polynomial p/G has coefficient norm one. Hence the fixed-degree sublevel estimate gives

$$E = \{|p| \le G^\vartheta\}, \qquad |E| \le CG^{-c_0(1-\vartheta)}.$$

The contribution of E is at most $C\|a\|_\infty G^{-c_0(1-\vartheta)}$. On E^c the denominator bound implies

$$|\psi'| = |p|/|Q|^2 \ge M^{-2}G^\vartheta.$$

The boundary of E is contained in the zero sets of the two bounded-degree polynomials $p \pm G^\vartheta$, and the numerator of ψ'' also has uniformly bounded degree. If that numerator vanishes identically, then ψ' is constant and no additional cut is needed; otherwise it has only $O_D(1)$ zeros. Splitting at all these points produces $O_D(1)$ intervals on each of which ψ' is monotone and bounded away from zero. Lemma 26.6 on every such interval yields

$$C_{\rho,D,\eta,M}(\|a\|_\infty + V^\rho(a))G^{-\vartheta/\rho}.$$

The chosen value of ϑ gives the exact equality

$$c_0(1-\vartheta) = \vartheta/\rho = \frac{c_0}{1 + c_0\rho} = c_{\mathrm{vdc},\rho,D}.$$

Summing the boundedly many intervals proves (26.30). Indeed, if these intervals are J_k, then

$$\sum_k V^\rho(a; J_k) \le C_D^{1-1/\rho} \left(\sum_k V^\rho(a; J_k)^\rho \right)^{1/\rho} \le C_{D,\rho} V^\rho(a),$$

and the finitely many supremum terms cost at most $C_D\|a\|_\infty$. Equivalently, if the restrictions are zero extended, their endpoint jumps cost at most two additional supremum terms per interval. A uniformly bounded initial partition is handled by the same refinement. □

Corollary 26.8 (Coefficient-norm polynomial van der Corput)**.** *For every fixed degree D, every real polynomial p of degree at most D, and every bounded measurable amplitude a supported on a fixed compact interval with $V^\rho(a) < \infty$,*

$$\left| \int e^{ip(t)} a(t)\, dt \right| \le C_{D,\rho}(\|a\|_\infty + V^\rho(a)) \min\{1, \|p - p(0)\|_{\mathrm{coeff}}^{-c_{D,\rho}}\}. \tag{26.31}$$

When p is constant, the minimum on the right is understood to be 1.

Proof. Apply Lemma 26.7 with denominator identically one. That lemma imposes no bound on the numerator coefficients, so the resulting estimate is uniform over all coefficients of p. Finally use finite-dimensional norm equivalence between the coefficient norm of p' and that of p modulo constants. □

Lemma 26.9 (Uniform rational oscillatory operator with sectional V^ρ)**.** *In Lemma 20.4, retain $B \geq 1$ and replace the sectional hypothesis $\|A(u,\cdot)\|_\infty + \mathrm{Var}(A(u,\cdot)) \leq B$ by*

$$\|A(u,\cdot)\|_\infty + V^\rho(A(u,\cdot)) \leq B$$

on at most M_0 intervals, and retain the same row and column L^1 bounds. Let $c_{\mathrm{vdc},\rho} > 0$ be the minimum of the finitely many exponents in (26.29) *that occur in the structural rational families, and put*

$$\delta_{\mathrm{rat},\rho} := \frac{2c_{\mathrm{sub}}c_{\mathrm{vdc},\rho}}{c_{\mathrm{sub}} + 2c_{\mathrm{vdc},\rho}}, \qquad c_{\mathrm{rat},\rho} := \frac{\delta_{\mathrm{rat},\rho}}{4} > 0. \tag{26.32}$$

Then

$$\|S\|_{2\to 2} \leq C_\rho B \min\{1, L^{-c_{\mathrm{rat},\rho}}\}, \tag{26.33}$$

where $L = \|[\Psi]\|_{\mathrm{rat},Q}$. The constants are independent of all phase coefficients.

Proof. The Schur bound handles $L \leq 1$, so assume $L > 1$. The kernel of SS^* is

$$K(u,u') = \int e^{i[\Psi(u,v)-\Psi(u',v)]} A(u,v)\overline{A(u',v)}\, dv.$$

Refine the two sectional partitions belonging to u and u'. On each refined interval Lemma 26.2(iii) gives

$$\|A(u,\cdot)\overline{A(u',\cdot)}\|_\infty + V^\rho(A(u,\cdot)\overline{A(u',\cdot)}) \leq CB^2.$$

With $G(u,u') = \|\mathfrak{C}_Q\Psi(u,u',\cdot)\|_{\mathrm{coeff},v}$, Lemma 26.7 therefore yields

$$|K(u,u')| \leq C_\rho B^2 \min\{1, G(u,u')^{-c_{\mathrm{vdc},\rho}}\}. \tag{26.34}$$

Let $\mathbf{p}(u,u')$ be the vector of coefficients in v whose Euclidean length is $G(u,u')$. Regrouping coefficients is an isomorphism between two fixed finite-dimensional coefficient spaces, so

$$A_0 := \|\mathbf{p}\|_{\mathrm{coeff},(u,u')} \asymp \|\mathfrak{C}_Q\Psi\|_{\mathrm{coeff},(u,u',v)} = L.$$

Apply (20.2) on $I \times I$ to the square of (26.34), with $\sigma = 2c_{\mathrm{vdc},\rho}$ and scale parameter $L \geq 1$. Since $A_0 \asymp L$, its normalized factor $L|\mathbf{p}(u,u')|/A_0$ is comparable to $G(u,u')$, so its integrand is comparable, with structural constants, to $\min\{1, G(u,u')^{-2c_{\mathrm{vdc},\rho}}\}$. The definition (26.32) gives

$$\iint |K(u,u')|^2\, du\, du' \leq C_\rho B^4 L^{-\delta_{\mathrm{rat},\rho}}.$$

Consequently

$$\|S\|_{2\to 2}^2 = \|SS^*\|_{2\to 2} \leq \|K\|_{L^2(I\times I)} \leq C_\rho B^2 L^{-\delta_{\mathrm{rat},\rho}/2}.$$

Thus the negative-moment estimate has first been square-rooted to pass from $\iint |K|^2$ to $\|K\|_2$, and the SS^* identity supplies the second square root. Taking that final square root proves (26.33). Algebraically degenerate points and the diagonal are already included in the negative-moment integral, so no coefficient-dependent deletion is made. □

Lemma 26.10 (Sectional pullback calculus)**.** *Let I, J be fixed compact intervals and let*

$$A(u,v) = w(u,v)\prod_{\ell=1}^{m} a_\ell(\gamma_\ell(u,v)).$$

Assume that w and every γ_ℓ are jointly Borel measurable on $I \times J$; in particular, after choosing the fixed regulated representatives of the a_ℓ, the function A is jointly measurable. Assume that every a_ℓ is extended by zero from one fixed compact radial interval and

$$\|a_\ell\|_\infty + V^\rho(a_\ell;\mathbb{R}) \leq B, \qquad B \geq 1.$$

Suppose that, in either variable with the other fixed, each γ_ℓ has at most J_0 monotonicity intervals, uniformly in the fixed variable. Suppose also that w is a specified bounded piecewise-Lipschitz Borel representative on a fixed partition into at most J_0 rectangles: its pointwise absolute value is uniformly bounded, and, for every fixed row or column, including boundary rows and columns, its restriction to each open section interior has a uniformly bounded pointwise Lipschitz constant. Boundary values are prescribed bounded one-sided or half-open values, not unspecified values of a Sobolev equivalence class. Fixed half-open cutoffs with a uniformly bounded number of sectional jumps may be included in w; singleton boundary pieces are allowed in the sectional partitions below. Then, for each fixed $u \in I$, the interval J has a partition into at most C_{m,J_0} pieces $J_\alpha(u)$ on each of which

$$\|A(u,\cdot)\|_{L^\infty(J_\alpha(u))} + V^\rho(A(u,\cdot);J_\alpha(u)) \le C_{m,J_0,\rho} B^m. \tag{26.35}$$

Separately, for each fixed $v \in J$, the interval I has a partition into at most C_{m,J_0} pieces $I_\beta(v)$ on each of which

$$\|A(\cdot,v)\|_{L^\infty(I_\beta(v))} + V^\rho(A(\cdot,v);I_\beta(v)) \le C_{m,J_0,\rho} B^m.$$

Moreover

$$\sup_u \int_J |A(u,v)|\,dv + \sup_v \int_I |A(u,v)|\,du \le C_{m,J_0} B^m. \tag{26.36}$$

Proof. First note that the stated profile budget controls the pointwise values of a_ℓ, not merely their essential supremum. Choose a point outside the compact radial support, where its zero extension is exactly zero. The two-point variation inequality then gives $|a_\ell(r)| \le V^\rho(a_\ell;\mathbb{R}) \le B$ for every r. Finite variation also makes the profile Borel measurable; hence the displayed product is jointly measurable on $I \times J$. No replacement of a profile at a jump or endpoint is needed.

For a fixed row take all endpoints from the structural partition, the finitely many cutoff jumps, and the monotonicity partitions of $v \mapsto \gamma_\ell(u,v)$. Partition J into the intervening open intervals and the singleton endpoints, deleting empty pieces. This is a genuine disjoint interval partition with cardinality bounded solely in terms of m, J_0. On every nonsingleton piece monotone pullback gives

$$V^\rho(a_\ell \circ \gamma_\ell) \le V^\rho(a_\ell;\mathbb{R}) \le B.$$

The zero extension is essential here: crossings of the radial support are already among the increments counted by its global variation. For the specified representative of w, its pointwise Lipschitz bound on that open section gives

$$V^\rho(w) \le V^1(w) \le \mathrm{Lip}(w)\,|J| \le C.$$

Repeated application of Lemma 26.2(iii), together with the pointwise bounds on the factors, proves the row estimate in (26.35). On a singleton the variation is zero, and the prescribed bounded boundary value gives the same size bound. Thus no endpoint value is inferred from an almost-everywhere Sobolev bound. The column argument is independent and identical. Finally $|A(u,v)| \le CB^m$ at every point, and the fixed interval lengths give the two separate Schur bounds (26.36), with no dependence on ρ. No two-variable or mixed variation norm is used. □

26.5 Stability of the correlation and high-shell estimates

Proposition 26.11 (Rough correlation with finite radial variation)**.** *Fix $1 \le \rho < \infty$ and put $d_* = \max(d,1)$. Uniformly for every polynomial degree $1 \le D \le d_*$, there are positive exponents $c_{\mathrm{eq},\rho}$ and $c_{\mathrm{uneq},\rho}$ and a finite power $C_{\mathrm{corr},\rho}$, depending only on n, d, ρ, such that the following two statements hold.*

First, the conclusion of Lemma 20.5 remains valid for phases of degree at most D and profiles satisfying $\|a\|_\infty + V^\rho(a) \le B$ and $\|b\|_\infty + V^\rho(b) \le B$, where the profiles are zero extended and $B \ge 1$, with

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h)\,dz \right| \le C_\rho B^{C_{\mathrm{corr},\rho}} \|\Theta\|_\infty^2 \min\{1, (\Lambda|h|^D)^{-c_{\mathrm{eq},\rho}}\} \tag{26.37}$$

when $n \ge 2$, and with $\min\{1, \Lambda^{-c_{\mathrm{eq},\rho}}\}$ in dimension one.

Second, Corollary 20.15 remains valid for V^ρ profiles, with

$$\left|\int e^{i[P(y+z,a)-P(y+z,y)]}\overline{k_R(z)}k_S(z+y-a)\,dz\right| \le C_\rho B^{C_{\mathrm{corr},\rho}}\|\Theta\|_\infty^2 S^{-n}\min\{1,(L_R|\mathbf{p}(W)|)^{-c_{\mathrm{uneq},\rho}}\} \quad (26.38)$$

for $n \ge 2$, with the stated S^{-1} analogue in dimension one.

Proof. We follow the coordinate calculations in the two cited proofs and record all places where radial regularity enters.

For the equal-shell estimate, on each normalized (X,Y) block the two profile arguments are, with the notation of the proof of Lemma 20.5,

$$\gamma_1(X,Y) = \frac{\sqrt{U^2+\alpha^2}}{|q(X,Y)|}, \qquad \gamma_2(X,Y) = \frac{\sqrt{V^2+\alpha^2}}{|q(X,Y)|},$$

where $U = U_0 + HX$, $V = V_0 + HY$, and $q = \Delta + X - Y$ has fixed sign and is bounded away from zero on a fixed enlargement. More precisely, if $K = [r_-, r_+] \Subset (0,\infty)$ is the common fixed support from Lemma 20.5, then the zero-extended amplitude is supported where both pullbacks lie in K. Choose $\chi \in C_c^\infty((0,\infty))$ equal to one on K and supported in a slightly larger fixed interval $K^+ \Subset (0,\infty)$. Because the profiles are zero outside K, inserting $\chi(\gamma_1)\chi(\gamma_2)$ does not change the amplitude; it does, however, give a smooth support factor on whose transition region the same lower bounds remain valid. The fixed block cutoffs and this enlarged support condition give, uniformly on a fixed enlargement,

$$|U|+|V| \le C_K, \qquad c_K \le |q| \le C_K, \qquad c_K \le s_U, s_V \le C_K, \quad s_U = (U^2+\alpha^2)^{1/2},\ s_V = (V^2+\alpha^2)^{1/2}.$$

Thus every denominator occurring in the normalized Jacobian and in the fixed rational cutoffs is separated from zero on the actual support. Equations displayed immediately before (20.31) give

$$\partial_X \frac{s_U}{q} = \frac{HUq - s_U^2}{s_U q^2}, \qquad \partial_Y \frac{s_U}{q} = \frac{s_U}{q^2},$$

and the analogous two formulas for s_V/q. Thus the only possible critical-point equations are polynomials of degree at most two. Each row and column therefore splits into at most $C_{n,d}$ intervals on which both arguments are monotone. On every such interval Lemma 26.2(ii) gives

$$V^\rho(a\circ\gamma) \le V^\rho(a), \qquad V^\rho(b\circ\widetilde{\gamma}) \le V^\rho(b).$$

Consequently the quotient rule, the preceding lower bounds, and the fixed size of the normalized rectangles give a uniform sectional $W^{1,\infty}$ bound for every smooth algebraic Jacobian and fixed cutoff (including cutoffs of the form $\beta(1/q)$). These smooth factors are differentiated only inside their fixed support; a sharp half-open cutoff contributes instead only a bounded number of sectional jumps. Lemma 26.10 therefore gives both the required sectional $L^\infty + V^\rho$ bound and the separate row and column L^1 bounds, with size $C_\rho B^{C_{\mathrm{corr},\rho}}$. The two rough angular factors remain the input and output L^2 functions in (20.30); they are not part of this amplitude and are never differentiated. Thus (20.31) is replaced by

$$\|T_{\alpha,\omega}\|_{2\to2} \le C_\rho B^{C_{\mathrm{corr},\rho}}\min\{1,[\Gamma(\omega)(\alpha H)^D]^{-c_{\mathrm{rat},\rho}}\}.$$

When $n=2$, the transverse sphere is $\mathbb{S}^0$ and the two restrictions have coefficient norm comparable to the full coefficient norm, exactly as in the proof of Lemma 20.5. When $n \ge 3$, the chart coefficient lower bound (20.34) and (20.2), with $\sigma = c_{\mathrm{rat},\rho}$, give an angular exponent

$$c_{\mathrm{ang},\rho} := \frac{c_{\mathrm{sub}}c_{\mathrm{rat},\rho}}{c_{\mathrm{sub}}+c_{\mathrm{rat},\rho}} > 0. \quad (26.39)$$

The $n=2$ estimate is stronger, so after decreasing its exponent the same $c_{\mathrm{ang},\rho}$ is valid in every dimension $n \ge 2$. Hence one transverse band is bounded by

$$C_\rho B^{C_{\mathrm{corr},\rho}}\|\Theta\|_\infty^2 \alpha^{n-1}\min\{1,[\Lambda(\alpha H)^D]^{-c_{\mathrm{ang},\rho}}\}.$$

For $n \geq 2$ choose

$$c_{\mathrm{eq},\rho}^{(n\geq 2)} := \frac{1}{2}\min\left\{c_{\mathrm{ang},\rho}, \frac{n-1}{d_*}\right\} > 0. \tag{26.40}$$

The two geometric sums in the proof of Lemma 20.5 then give (26.37). In dimension one, Corollary 26.8 applies on the finitely many signed overlap intervals and gives an exponent $c_{\mathrm{eq},\rho}^{(n=1)} > 0$. Define, for the fixed ambient dimension,

$$c_{\mathrm{eq},\rho} := \begin{cases} c_{\mathrm{eq},\rho}^{(n=1)}, & n = 1, \\ c_{\mathrm{eq},\rho}^{(n\geq 2)}, & n \geq 2. \end{cases}$$

This number is strictly positive.

For unequal shells, the two zero-extended profile arguments are exactly (20.102). In the formulas cited in this paragraph, write the geometric parameter $|W|$ as r_W (so the earlier ρ, $S\rho$, and $\delta = R/(S\rho)$ are read as r_W, Sr_W, and $R/(Sr_W)$); this keeps it distinct from the fixed radial-variation exponent ρ of the present section. On the support of a transverse band and both profiles, the calculation following (20.101) gives

$$|u| + |V| \leq C, \qquad 0 < c \leq D = V - \delta u \leq C, \qquad 0 < \delta \leq C,$$
$$(u^2 + \alpha^2)^{1/2} \geq c, \qquad (V^2 + \alpha^2\delta^2)^{1/2} \geq c.$$

As in the equal-shell case, insert fixed smooth cutoffs of the two pullback variables which equal one on the radial profile support and are supported in a slightly larger compact subinterval of $(0,\infty)$. Zero extension makes this insertion exact, and all the displayed lower bounds persist on the cutoff transition regions. All constants here are independent of the shell radii and of α, δ. It follows directly from the product and quotient rules that the factor

$$D^{n-1}(u^2 + \alpha^2)^{-n/2}(V^2 + \alpha^2\delta^2)^{-n/2}$$

and every smooth rational/chart cutoff have uniformly bounded sectional $W^{1,\infty}$ norm on each fixed normalized rectangle. A half-open rectangle or band boundary produces only one of a bounded number of sectional jumps. The two pullbacks have only the nontrivial sectional critical equation $uV + \delta\alpha^2 = 0$, so each section requires at most one additional cut. Take the common refinement for the two arguments, the fixed rational rectangles, and the transverse cutoff. Lemma 26.10 then supplies both sectional V^ρ and the independent Schur row/column bounds. The zero-extended transverse cutoffs satisfy $V^\rho(\beta_k) \leq V^1(\beta_k) \leq C$. Thus the normalized rational operator contributes the same exponent $c_{\mathrm{rat},\rho}$. For $n = 2$ the transverse sphere again consists of two points; for $n \geq 3$ use the finite rational atlas and (20.2). The oriented coefficient lower bound and this angular argument give, in either case after decreasing the exponent, a band estimate

$$C_\rho B^{C_{\mathrm{corr},\rho}} \|\Theta\|_\infty^2 S^{-n}\alpha^{n-1}\min\{1, (L_R\alpha^d|\mathbf{p}(W)|)^{-c_{\mathrm{ang},\rho}}\}.$$

For $n \geq 2$ set

$$c_{\mathrm{uneq},\rho}^{(n\geq 2)} := \frac{1}{2}\min\left\{c_{\mathrm{ang},\rho}, \frac{n-1}{d_*}\right\} > 0. \tag{26.41}$$

Splitting the dyadic α-sum at $\alpha \simeq (L_R|\mathbf{p}(W)|)^{-1/d_*}$ proves (26.38). The one-dimensional proof uses Corollary 26.8 and gives an exponent $c_{\mathrm{uneq},\rho}^{(n=1)} > 0$. Define, for the fixed ambient dimension,

$$c_{\mathrm{uneq},\rho} := \begin{cases} c_{\mathrm{uneq},\rho}^{(n=1)}, & n = 1, \\ c_{\mathrm{uneq},\rho}^{(n\geq 2)}, & n \geq 2. \end{cases}$$

This number is strictly positive and completes the proof. No rough angular factor has been differentiated. □

Corollary 26.12 (Quantitative flat high block and collective high shells)**.** *There are $c_{\mathrm{flat},\rho} > 0$ and a finite power $C_{B,\rho}$ such that Proposition 20.8, with radial profiles controlled in $L^\infty + V^\rho$ by $B \geq 1$, has the quantitative form*

$$\left\| A_a^{P,\Theta} \sum_Q e_Q \nu_Q \right\|_2^2 \leq C_\rho B^{C_{B,\rho}} \|\Theta\|_\infty^2 \Lambda M (L^{-c_{\mathrm{flat},\rho}} + 2^{-c_{\mathrm{flat},\rho} t}). \tag{26.42}$$

For every fixed $\kappa > 0$ there is $c_{\kappa,\rho} > 0$ such that Proposition 20.17 has the form

$$\left\| \sum_{R \in \mathscr{R}} T_R \sum_{Q \in \mathscr{Q}_R} e_Q \nu_Q \right\|_2^2 \leq C_{\rho,\kappa} B^{C_{B,\rho}} \|\Theta\|_\infty^2 \Lambda M 2^{-c_{\kappa,\rho} s}. \tag{26.43}$$

Both estimates are uniform under packetwise coefficients $|e_Q| \leq 1$.

Proof. Fix a finite $\rho \geq 1$. We use the same actual complex-measure Hilbert integrals as in the proofs of Propositions 20.8 and 20.17. A compactly supported profile of finite V^ρ is measurable, and comparison with a zero value beyond its support bounds each original pointwise value by its variation. Thus the stated $L^\infty + V^\rho$ budget supplies all row measurability and L^2 bounds needed there, without a new regularity assumption or a change of profile. The finite-ρ correlation estimates, normalized at radial budget one, have prefactor $C_\rho B^2 \|\Theta\|_\infty^2$ by bilinearity in the two radial profiles.

For the flat estimate put $D = d+1$, and let $\alpha_{\mathrm{pair}} > 0$ be the exponent of the fixed-index pair-cell covers in the proof of Proposition 20.8, with degree bound D. Let $c^*_{\mathrm{eq},\rho} > 0$ be an admissible clipped-separation correlation exponent in that degree bound, decreased if necessary from the finite-ρ equal-correlation exponent. Recenter the original phase before scalar extraction. Both the bad coefficient cover and the coordinate-difference cover have weighted mass $C\Lambda M(\tau+q)^{\alpha_{\mathrm{pair}}}$, where $q = 2^{-t}$. Outside their union, the actual oscillatory scale is at least $L\tau^{D+1}/K$. The integrated Gram bound is therefore

$$\|A_a^{P,\Theta} \sum_Q e_Q \nu_Q\|_2^2 \leq C_\rho B^2 \|\Theta\|_\infty^2 \Lambda M \left\{ [\max\{1, L\tau^{D+1}/K\}]^{-c^*_{\mathrm{eq},\rho}} + (\tau+q)^{\alpha_{\mathrm{pair}}} \right\}.$$

In every dimension $n \geq 1$ we may take

$$\sigma = \frac{1}{2(D+1)}, \qquad \tau = L^{-\sigma}, \qquad c_{\mathrm{flat},\rho} = \min\{\sigma\alpha_{\mathrm{pair}}, c^*_{\mathrm{eq},\rho}/2\} > 0. \tag{26.44}$$

Indeed $L\tau^{D+1} = L^{1/2}$ and the braces are bounded by $C_\rho(L^{-c_{\mathrm{flat},\rho}} + q^{c_{\mathrm{flat},\rho}})$. This proves (26.42), including every real $t \geq 0$. The physical cube caps and masses are unchanged, and the operator acts on the literal sum of the original complex measures.

For the collective estimate use the global same-shell unary row, not a global fixed-box hypothesis. Let $\alpha_{\mathrm{row}} > 0$ be the weighted cube sublevel exponent on its local interaction box. The two scalar bad sets in that proof and the same clipped correlation give the uniform exponent

$$\beta_{\mathrm{eq},\rho} = \min\{\sigma\alpha_{\mathrm{row}}, c^*_{\mathrm{eq},\rho}/2\} > 0.$$

Simultaneous translation preserves the highest mixed block of the full original phase, and dilation gives its norm $L_R = R^m \|P_m\|_{\mathrm{coeff}}$. The exact Gram factor R^{-n} absorbs into the normalized weights. Supported first-moment selection and Hilbert integration then give the actual same-shell energy

$$\|F_R\|_2^2 \leq C_\rho B^2 \|\Theta\|_\infty^2 \Lambda M_R (L_R^{-\beta_{\mathrm{eq},\rho}} + 2^{-\beta_{\mathrm{eq},\rho} s}).$$

The separated row uses the original choice $\tau_R = L_R^{-1/2}$. The finite-ρ unequal correlation and original one-sided incidence give a positive row exponent, uniformly over the finitely many possible highest mixed degrees. The two geometric sums of $L_R^{-\beta}$ and $(2^{-s}R/S)^\beta$ give a complete far-row exponent $\gamma_{\kappa,\rho} > 0$. This estimate is transferred to the arbitrary singular small-shell measures, then integrated

against the large-shell total variations; its right side is $C_{\rho,\kappa}B^2\|\Theta\|_\infty^2\Lambda M_S 2^{-\gamma_{\kappa,\rho}s}$. All constants precede P,m and the packet data, and the radial power is still two, independent of κ.

Choose

$$c_{\kappa,\rho}=\min\{\beta_{\mathrm{eq},\rho}\min\{1,\kappa\},\gamma_{\kappa,\rho}\}>0. \tag{26.45}$$

The high threshold gives the diagonal decay with this exponent. The fixed-width neighboring-scale Cauchy–Schwarz bound and the oriented far Gram rows then give (26.43), exactly as in the final Hilbert assembly of Proposition 20.17, without a scale-count loss. The same proof yields B^2 in both conclusions, so one may take $C_{B,\rho}=2$ (or any larger fixed finite power). Complex linearity identifies the literal grouped shell operators, and every estimate uses only $|e_Q|\le 1$. No probability-space Jensen step, density assumption, regularization of the measures, or supplied Gram bound is needed. □

26.6 Terminal packets with a finite-variation radial profile

Lemma 26.13 (Smooth fixed-sign packet kernel)**.** *Put $N_0=4n+1$ and $\eta=(16N_0)^{-1}$. In the unit-scale packet setting of Proposition 18.5, let c_t be supported in one fixed annular enlargement and satisfy*

$$|c_t|\le 1,\qquad \|\partial_r^\ell c_t\|_\infty\le C_\ell 2^{\eta\ell t}\quad(0\le\ell\le N_0).$$

For arbitrary fixed packet coefficients of modulus at most one, let S_t be the convolution output with kernel $K_\Theta(z)c_t(|z|)$. Then

$$\|S_t\|_2^2\le C_n\|\Theta\|_\infty^2 2^{-3t/32}\Lambda M. \tag{26.46}$$

The assertion is unchanged by the input and output modulations allowed in Proposition 18.5.

Proof. This is the smooth part of the proof of Proposition 18.5, but we record precisely which hypotheses are used. The fixed annular enlargement is $[1/4,4]$, as in the full-kernel convention, and the implicit constants may depend on the prescribed structural constants C_ℓ. Only radial C^{N_0} regularity is used. Initially take a finite packet family and a pointwise bounded angular representative as in Proposition 18.5. Put $H_t(z)=K_\Theta(z)c_t(|z|)$. Its normalized radial source budget is at most

$$M_{N_0}\le C\|\Theta\|_\infty 2^{\eta N_0 t}=C\|\Theta\|_\infty 2^{t/16}.$$

In dimension one, the cancellation argument in the proof of Lemma 18.4 gives squared energy at most $CM_{N_0}^2 2^{-t/2}\Lambda M$. Consequently

$$\|S_t\|_2^2\le C\|\Theta\|_\infty^2 2^{-3t/8}\Lambda M,$$

which is stronger than (26.46). We may therefore assume $n\ge 2$. The one-shell kernel has normalized radial C^{N_0} size at most $C\|\Theta\|_\infty 2^{\eta N_0 t}$. Seeger's decomposition with $\kappa=1/2$ and $\varepsilon=1/4$ therefore gives a microlocal part with squared L^2 norm

$$C\|\Theta\|_\infty^2 2^{2\eta N_0 t}2^{-t/2}\Lambda M=C\|\Theta\|_\infty^2 2^{-3t/8}\Lambda M$$

and a remainder with L^1 norm

$$C\|\Theta\|_\infty 2^{-(1/4-\eta N_0)t}M=C\|\Theta\|_\infty 2^{-3t/16}M.$$

The source budget enters quadratically in the microlocal energy above and linearly in the remainder bound. Independently, the support and the bound $|c_t|\le 1$ give, by full-annulus packing, the simultaneous L^∞ bound $C\Lambda\|\Theta\|_\infty$. The bounded-output estimate (18.26) gives squared L^2 decay $2^{-3t/16}$, which is stronger than the required $2^{-3t/32}$; this proves (26.46) for finite families. For an arbitrary disjoint cube family, its positive interiors imply countability. If $M<\infty$, the bounded compactly supported kernel makes the finite convolution sums converge pointwise, and Fatou gives the same energy bound. Infinite total mass is immediate when the right-hand coefficient is positive; zero angular norm or zero density gives zero output separately. Angular representative equality follows for every convolution point by polar coordinates and the map $y\mapsto x-y$. No property of a step function, or of the set on which an unsmoothing error is supported, enters these three estimates. The proof uses absolute packet coefficients and packet cancellation, so the modulation statements are preserved. □

Proposition 26.14 (Radially weighted selector-free terminal packet)**.** *Fix $1 \le \rho < \infty$ and put*

$$N_0 = 4n+1, \qquad \eta = \frac{1}{16N_0}, \qquad \gamma = \frac{\eta}{2}, \qquad c_{\mathrm{sm}} = \frac{3}{32}. \tag{26.47}$$

There is $t_{0,\rho} \in \mathbb{N}$ such that Proposition 18.5 remains valid for $t \ge t_{0,\rho}$ after inserting a radial factor $a(|x-y|/R)$ satisfying $\|a\|_\infty + V^\rho(a;[1,2]) \le 1$. More precisely, one may take

$$\delta_{T,\rho} := \frac{1}{4}\min\left\{\gamma, c_{\mathrm{sm}}, \frac{\eta}{\rho}\right\} > 0 \tag{26.48}$$

so that, with the same packet hypotheses and $R \le \varepsilon \le 2R$,

$$\|V^2(F_\varepsilon : R \le \varepsilon \le 2R)\|_2^2 \le C_{n,\rho}\|\Theta\|_\infty^2 2^{-\delta_{T,\rho}t}\Lambda M. \tag{26.49}$$

The modulation invariances in Proposition 18.5 remain valid.

Proof. We first retain the full scope of the local radial hypothesis. For every $r \in [1,2]$, comparison with a point where $|a|$ is arbitrarily close to its essential supremum gives

$$|a(r)| \le \|a\|_{L^\infty([1,2])} + V^\rho(a;[1,2]) \le 1.$$

This includes both endpoints and any exceptional point values. Extend a constantly outside $[1,2]$ by the monotone clamp $r \mapsto \min(2,\max(1,r))$, and divide that extension by two. The resulting profile b has global pointwise $\|b\|_\infty + V^\rho(b) \le 1$ and equals $a/2$ on $[1,2]$. It suffices to prove the estimate for b and restore the factor four in squared energy. This normalization imposes no extra hypothesis on a and changes none of its values in the original integral. As in Proposition 18.5, a pointwise bounded angular representative preserves every literal annular integral for every spatial point and every radius. We initially work with that representative, finitely many packets, and $R = 1$.

Put $k = \lceil \rho \rceil$, $J = N_0 k$, and choose the binary grid

$$L_t = \left\lfloor \frac{t}{40J} \right\rfloor, \qquad K_t = 2^{L_t}, \qquad h = (8K_t^2)^{-k}.$$

Let $I_\ell = (1+(\ell-1)/K_t, 1+\ell/K_t]$ and let G_ℓ be the full shell output over I_ℓ with radial profile b. Inserting all cell endpoints into an arbitrary variation partition gives the deterministic decomposition

$$V^2(F_\varepsilon : 1 \le \varepsilon \le 2) \le CV^2\left(\sum_{\ell \le m} G_\ell : 0 \le m \le K_t\right) + C\left(\sum_\ell V^2(F_\varepsilon : \varepsilon \in \bar{I}_\ell)^2\right)^{1/2}.$$

The within-cell argument uses $V^2 \le V^1$, the positive thin-layer majorant, and only $|b| \le 1$. The layer thickness is $K_t^{-1} + C_n 2^{-t}$; since $2^{-t} \le K_t^{-1}$, it gives

$$\left\| \left(\sum_\ell V^2(F_\varepsilon : \varepsilon \in \bar{I}_\ell)^2\right)^{1/2} \right\|_2^2 \le C_n\|\Theta\|_\infty^2 \Lambda M K_t^{-1}. \tag{26.50}$$

For the grid partial sums fix signs $\varepsilon_\ell \in \{-1,0,1\}$ and set

$$c(r) = b(r)\sum_{\ell=1}^{K_t} \varepsilon_\ell \mathbf{1}_{I_\ell}(r).$$

The literal half-open step factor has supremum at most one and ρ-variation at most $2K_t^{1/\rho}$. The product rule gives

$$\|c\|_\infty \le 1, \qquad V^\rho(c) \le 1 + 2K_t^{1/\rho}, \qquad \operatorname{supp} c \subset [1,2]. \tag{26.51}$$

Mollify this zero-extended profile by a nonnegative smooth unit-mass kernel at scale h. Lemma 26.3 yields

$$\|c-c_h\|_1 \le C_\rho h^{1/\rho}(1+K_t^{1/\rho}) \le C_\rho K_t^{-1}. \tag{26.52}$$

Indeed $k \ge \rho \ge 1$ implies $h^{1/\rho} \le (8K_t^2)^{-1}$ and $K_t^{1/\rho} \le K_t$. Both profiles are bounded by one and supported in a fixed annular enlargement, since $h \le 1/8$. Their difference need not be supported near the grid endpoints: the radial profile may vary throughout every cell. For its actual convolution output D, full-annulus packing and the polar-coordinate kernel L^1 bound give, respectively,

$$\|D\|_\infty \le C_n \Lambda \|\Theta\|_\infty, \tag{26.53}$$

$$\|D\|_1 \le C_n \|\Theta\|_\infty M \|c-c_h\|_1 \le C_{n,\rho}\|\Theta\|_\infty M K_t^{-1}. \tag{26.54}$$

The packing estimate uses only the pointwise kernel bound, the fixed annular support, disjoint packet interiors, and $\|d_U\|_1 \le \Lambda|U|$; no small L^∞ error is asserted. Consequently

$$\|D\|_2^2 \le C_{n,\rho}\|\Theta\|_\infty^2 \Lambda M K_t^{-1}. \tag{26.55}$$

For the smoothed signed sum S, positivity of the mollifier gives $|c_h| \le 1$ and $\|\partial_r^\ell c_h\|_\infty \le C_\ell h^{-\ell}$ with no additional factor K_t. Thus the actual one-shell kernel $K_\Theta(z)c_h(|z|)$ has radial source budget at most $C_n\|\Theta\|_\infty h^{-N_0}$. The proof of Lemma 18.4 explicitly establishes the stronger decay $2^{-t/8}$ in every positive dimension before weakening it to its stated exponent. Applying that proved bound at this single scale gives

$$\|S\|_2^2 \le C_n\|\Theta\|_\infty^2 \Lambda M h^{-2N_0} 2^{-t/8} \le C_{n,\rho}\|\Theta\|_\infty^2 \Lambda M K_t^{-1}. \tag{26.56}$$

For the last inequality, $L_t \le t/(40J)$ implies

$$h^{-2N_0}2^{-t/8} = 8^{2J}2^{4JL_t - t/8} \le 8^{2J}2^{-t/40} \le 8^{2J}K_t^{-1}.$$

This argument does not require the different smoothing scale in Lemma 26.13.

The fixed-sign bound obtained from (26.55) and (26.56), followed by Lemma 18.2, costs at most $C(1+L_t)^2$ in squared energy. Absorb just one sixteenth of the binary-depth exponent:

$$(1+L_t)^2 2^{-L_t} \le C 2^{-15L_t/16} \le 2C 2^{-3t/(128N_0k)}.$$

Together with (26.50) and the deterministic decomposition, this proves the desired continuum estimate with decay $3/(128N_0k)$. The displayed exponent in the statement is exactly

$$\delta_{T,\rho} = \min\left\{\frac{1}{128N_0}, \frac{1}{64N_0\rho}\right\} \le \frac{3}{128N_0\lceil\rho\rceil}.$$

For $1 \le \rho \le 2$ use $\lceil\rho\rceil \le 2$; for $\rho \ge 2$ use $2\lceil\rho\rceil \le 3\rho$. Hence no reduction of the original displayed decay is needed. Choose $t_{0,\rho}$ large enough for the fixed packet-diameter and full-kernel thresholds.

The physical change of variables $x = Rx'$, $y = Ry'$ proves the result for every $R > 0$: use packets $d_U(R\cdot)$ on the dilated cubes, whose total mass is $R^{-n}M$, and cancel this factor against the output Jacobian R^n. The normalized radial profile is unchanged. Disjoint positive cube interiors make any original cube family countable. If $M < \infty$, finite packet sums converge in L^1; the bounded clipped annular kernels give path convergence uniformly in $R \le \varepsilon \le 2R$ at every x. Lower semicontinuity on each finite variation partition and Fatou's lemma extend the estimate to the full family. If $M = \infty$ and $\Lambda\|\Theta\|_\infty > 0$, the extended-valued bound is immediate. The zero-density and zero-angular-norm cases instead have identically zero output and are treated before this multiplication, so no ambiguous zero-times-infinity argument is used. Restoring $a = 2b$ on $[1,2]$ only multiplies the energy constant by four. Finally packetwise input modulations preserve all absolute-value hypotheses and use precisely the stipulated cancellation of the modulated packets; common output modulation preserves every variation increment. This proves all the claimed invariances and (26.49). □

26.7 The unphased strong theorem for every finite radial exponent

We next prove the terminal strong input in a form with the exact angular conjugate-exponent loss needed for the $L\log L$ endpoint.

For $j \in \mathbb{Z}$ define the annular measure

$$d\nu_j(x) = \frac{\Omega(x/|x|)}{|x|^n} b(|x|)\mathbf{1}_{\{2^j<|x|\le 2^{j+1}\}}\,dx. \tag{26.57}$$

The mean-zero condition on Ω implies $\nu_j(\mathbb{R}^n) = 0$.

Lemma 26.15 (Shell Fourier envelope)**.** *Assume* (26.14) *and* $\int_{\mathbb{S}^{n-1}} \Omega\, d\sigma = 0$*. There exists* $\beta_\rho > 0$ *such that* $\widehat{\nu}_j(0) = 0$ *and, for* $t = 2^j|\xi| > 0$*,*

$$|\widehat{\nu}_j(\xi)| \le Ct\|\Omega\|_1, \qquad 0 < t \le 1, \tag{26.58}$$

$$|\widehat{\nu}_j(\xi)| \le C_\rho t^{-\beta_\rho}\|\Omega\|_2, \qquad t \ge 1. \tag{26.59}$$

One may take

$$\beta_\rho = \min\left\{\frac{1}{4}, \frac{1}{2\rho}\right\}.$$

Consequently, if $1 < s \le 2$*,* $s' = s/(s-1)$*, and* Ω *has mean zero, then, for* $t > 0$*,*

$$|\widehat{\nu}_j(\xi)| \le C_\rho \|\Omega\|_s\, a_{s,\rho}(\log_2 t), \tag{26.60}$$

where $a_{s,\rho} : \mathbb{R} \to (0,\infty)$ *is defined by*

$$a_{s,\rho}(r) = \begin{cases} C2^r, & r \le 0, \\ C_\rho 2^{-2\beta_\rho r/s'}, & r > 0. \end{cases} \tag{26.61}$$

For every $\delta > 0$*,*

$$\sum_{r\in\mathbb{Z}} a_{s,\rho}(r)^\delta \le C_{\rho,\delta} s'. \tag{26.62}$$

Proof. Use the angular Fourier convention $\widehat{f}(\xi) = \int f(z)e^{-iz\cdot\xi}\,dz$. In the convention with exponential $e^{-2\pi iz\cdot\eta}$ the same transform is evaluated at $\eta = \xi/(2\pi)$; this conversion does not change the shell or its radial representative. Polar coordinates and $r = 2^j u$ give, for $\xi \ne 0$, $e = \xi/|\xi|$ and $t = 2^j|\xi|$,

$$\widehat{\nu}_j(\xi) = \int_{\mathbb{S}^{n-1}} \Omega(\theta) H_{j,t}(e\cdot\theta)\, d\sigma(\theta), \qquad H_{j,t}(\tau) = \int_{(1,2]} b(2^j u) e^{-itu\tau}\frac{du}{u}.$$

These are integrals of the literal shell density on $(2^j, 2^{j+1}]$; the radial normalization ensures their integrability. At zero frequency the radial integral is independent of θ, giving $\widehat{\nu}_j(0) = 0$. Subtract this value before using $|e^{-iz} - 1| \le |z|$. Since $\|b\|_\infty \le 1$, the result is $|\widehat{\nu}_j(\xi)| \le t\|\Omega\|_1$, without evaluating a logarithm at zero.

Lemma 26.2 bounds the variation of $u \mapsto b(2^j u)/u$ on the fixed interval uniformly in j. The finite-variation first-derivative estimate of Lemma 26.6 therefore gives

$$|H_{j,t}(\tau)| \le C_\rho, \qquad |H_{j,t}(\tau)| \le C_\rho (t|\tau|)^{-1/\rho} \quad (\tau \ne 0).$$

Here and below the constants are fixed before j, b, s, Ω, ξ. For $t \ge 1$ split the sphere into the strip $F_t = \{\theta : |e\cdot\theta| < t^{-1/2}\}$ and its complement. The spherical strip bound is $\sigma(F_t) \le C_n t^{-1/2}$; in dimension one the narrow strip is empty, so the same estimate is valid. This uses no assertion about a density of the spherical projection in that dimension. For $1 < s \le 2$ put $a = 1 - 1/s = 1/s'$. Hölder on the strip and the radial bound off it yield directly

$$|\widehat{\nu}_j(\xi)| \le C_{n,\rho}\|\Omega\|_s\big(t^{-a/2} + t^{-1/(2\rho)}\big) \le C_{n,\rho}\|\Omega\|_s t^{-2\beta_\rho/s'}.$$

The last inequality follows from $0 < a \le 1/2$, $\beta_\rho \le 1/4$ and $\beta_\rho \le 1/(2\rho)$. The finite-measure factors are uniform in s, since $z^a \le \max(1,z)$ for $0 \le a \le 1$. At $s = 2$ this is exactly (26.59); together with the low estimate and $L^s \hookrightarrow L^1$ on the sphere it proves (26.60) with the stated asymmetric envelope.

For completeness, the nonpositive integer part of the envelope sum is bounded by $C^\delta/(1-2^{-\delta})$. The positive part is bounded by

$$C_\rho^\delta \sum_{r=1}^{\infty} e^{-(2\beta_\rho \delta \log 2) r/s'} = \frac{C_\rho^\delta}{e^{(2\beta_\rho \delta \log 2)/s'} - 1} \le \frac{C_\rho^\delta s'}{2\beta_\rho \delta \log 2}.$$

This proves both convergence and (26.62), with one linear factor s' and no hidden s-dependence. □

We shall also use the following standard vector-valued shell estimate. A proof is included because it is uniform down to angular L^1 and therefore is exactly what allows spatial interpolation below.

Lemma 26.16 (Uniform vector-valued annular estimate)**.** *For $1 < p < \infty$ and arbitrary functions f_j,*

$$\left\| \left(\sum_j |\nu_j * f_j|^2 \right)^{1/2} \right\|_p \le C_p \|b\|_\infty \|\Omega\|_1 \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_p . \tag{26.63}$$

Proof. Let $B = \|b\|_\infty$, interpreted as an essential radial norm, and let $M_\theta^+ g(x) = \sup_{a>0} a^{-1} \int_0^a |g(x - r\theta)|\, dr$. For each shell, polar coordinates and $r^{-1} \le 2^{-j}$ on $(2^j, 2^{j+1}]$ give, almost everywhere,

$$|\nu_j * f_j(x)| \le 2B \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| M_\theta^+ f_j(x)\, d\sigma(\theta).$$

Only the absolute inputs $|f_j|$ enter this domination. The essential radial inequality holds inside the radial integral; changing b on a radial null set leaves the kernel unchanged almost everywhere. The same applies to the original angular and input representatives.

For a finite shell set, Minkowski in ℓ^2 and then in L^p gives

$$\big\| \{\nu_j * f_j\}_j \big\|_{L^p(\ell^2)} \le 2B \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| \, \big\| \{M_\theta^+ f_j\}_j \big\|_{L^p(\ell^2)}\, d\sigma(\theta).$$

The one-dimensional vector maximal theorem, lifted along the orthogonal fibers $x = x_\perp + r\theta$, bounds the inner norm by $C_p \|\{f_j\}_j\|_{L^p(\ell^2)}$. The coordinate change preserves Lebesgue measure, so C_p is independent of both direction and ambient dimension. Angular integration contributes exactly $\|\Omega\|_1$; no normalized-sphere factor, variation budget, or angular supremum norm is introduced. In particular this is a bound for the vector of maximal functions, not for a scalar maximal function of the vector norm.

Increasing finite shell sets proves the countable inequality by monotone convergence. Equivalently, countable integral Minkowski follows here by Tonelli and finite-dimensional Cauchy–Schwarz followed by exhaustion. If the input square has finite L^p norm, every coordinate is in L^p; the L^1 shell kernel then gives genuine almost-everywhere convolution integrability. These integrability and representative identities hold on one conull set for all integer coordinates. If the input norm is infinite and $B\|\Omega\|_1 > 0$, the extended-norm assertion is immediate. If either factor is zero, the literal shell kernels vanish almost everywhere and so do all their convolutions. This also covers the zero-times-infinite case. □

Use the explicit cutoffs $\psi, \widetilde{\psi}$ from (4.4). Let Δ_ℓ and $\widetilde{\Delta}_\ell$ have multipliers $\psi(2^{-\ell}\xi)$ and $\widetilde{\psi}(2^{-\ell}\xi)$. Thus $\widetilde{\Delta}_\ell \Delta_\ell = \Delta_\ell$ and $\sum_\ell \widetilde{\Delta}_\ell \Delta_\ell f = f$ in L^p for $1 < p < \infty$. The square-function bounds below therefore apply to $\widetilde{\Delta}_\ell \Delta_\ell f = \Delta_\ell f$; no identity $\sum_\ell \psi(2^{-\ell}\xi)^2 = 1$ is assumed. For a relative frequency $r \in \mathbb{Z}$ put

$$g_{j,r} := \nu_j * \widetilde{\Delta}_{r-j} \Delta_{r-j} f.$$

On the Fourier support of $g_{j,r}$ one has $2^j|\xi| \simeq 2^r$.

Lemma 26.17 (Frequency-separated tail representation)**.** *For every $1 < p < \infty$ and every $f \in L^p(\mathbb{R}^n)$, there is a fixed integer $M \geq 5$ such that, after splitting the shell indices j into their M residue classes, the tail*

$$\sum_{\substack{j \geq k \\ j \equiv c \pmod M}} g_{j,r}$$

is, up to repetitions as k varies, exactly a subsequence of the smooth low-pass family $\{\phi_\ell * h_{r,c}\}_{\ell \in \mathbb{Z}}$ of Lemma 4.5, applied to

$$h_{r,c} := \sum_{j \equiv c \pmod M} g_{j,r}.$$

Here $\widehat{\phi} = 1$ on $|\xi| \leq 2$ and $\widehat{\phi} = 0$ on $|\xi| > 4$, and $\widehat{\phi_\ell}(\xi) = \widehat{\phi}(2^\ell \xi)$. Consequently, Lemma 4.5 gives

$$\left\| \mathbf{J} \left(\sum_{\substack{j \geq k \\ j \equiv c \pmod M}} g_{j,r} : k \in \mathbb{Z} \right) \right\|_p \leq C_p \|h_{r,c}\|_p. \tag{26.64}$$

Proof. We use the angular Fourier convention $e^{-ix\cdot\xi}$. The annular cutoff ψ and low-pass kernel ϕ have the stated multipliers; kernels defined using the cyclic-frequency convention are dilated spatially by 2π. Their plateaux are therefore $2, 4$ and their annular radii are $1/2, 2$. The dual-cutoff identity gives $g_{j,r} = \nu_j * \widetilde{\Delta}_{r-j} \Delta_{r-j} f = \nu_j * \Delta_{r-j} f$.

We first construct the sums in L^p. The annular vector estimate and the Littlewood–Paley square estimate imply

$$\left\| \left(\sum_j |g_{j,r}|^2 \right)^{1/2} \right\|_p < \infty \quad \text{for every fixed } r.$$

Choose a fixed wider smooth annular reproducer equal to one on the support of each Δ_{r-j}. It reproduces $g_{j,r}$ as an L^p identity, by convolution associativity, first on a dense class and then by the bounded L^1 kernel operators. The dual Littlewood–Paley estimate therefore bounds every finite sum over a set F by $C_p \|(\sum_{j \in F} |g_{j,r}|^2)^{1/2}\|_p$. The latter tends to zero for sets escaping finite windows, by dominated convergence. Consequently the series over j converges unconditionally in L^p, as do its residue subseries and their tails.

Take $M = 5$. The Fourier support of $g_{j,r}$ lies in $2^{r-j-1} \leq |\xi| \leq 2^{r-j+1}$. For $k \in \mathbb{Z}$ let $j_0(k)$ be the least integer at least k congruent to c modulo 5. The multiplier of $\phi_{j_0(k)-r}$ equals one on every retained support with $j \geq j_0(k)$. The preceding retained support starts at $2^{r-j_0(k)+4}$, strictly beyond the low-pass support endpoint $2^{r-j_0(k)+2}$; hence the multiplier vanishes on all earlier retained supports. Apply the bounded convolution operator to the unconditional L^p sum to obtain exactly

$$F_{r,c,k} := \sum_{\substack{j \geq k \\ j \equiv c \ (5)}} g_{j,r} = \phi_{j_0(k)-r} * h_{r,c} \quad \text{in } L^p.$$

The map $k \mapsto j_0(k) - r$ is nondecreasing. Repetitions cannot create jumps, so Lemma 4.5 gives (26.64) for representatives on a common conull set. Its paired-path estimate dominates $\mathbf{J}$; the comparison (15.14) also supplies the chain version.

We record precisely the pointwise meaning of these series. All shell convolutions and all displayed low-pass convolutions are absolutely integrable almost everywhere. Countability of r, c, k, j puts these integrability assertions, their representative identities, and every finite-window identity on one conull set. Unconditional L^p convergence gives $F_{r,c,k} \to 0$ as $k \to +\infty$ and $F_{r,c,k} \to h_{r,c}$ as $k \to -\infty$ in L^p. The finite pointwise jump functional supplied by the preceding bound ensures pointwise limits at both ends; subsequences of the L^p limits identify them as 0 and $h_{r,c}$ almost everywhere. On the same conull set,

$$\sum_{\substack{k \leq j < l \\ j \equiv c \ (5)}} g_{j,r} = F_{r,c,k} - F_{r,c,l}.$$

Letting $l \to +\infty$ proves convergence of every ordered tail; letting $k = -N$, $l = N$ proves convergence of the centered ordered totals. These conclusions hold simultaneously for all r, c, k. They do not assert unconditional or absolute convergence pointwise in space. □

Lemma 26.18 (s'-tracked dyadic terminal jump)**.** *Let $1 < s \le 2$, assume $\Omega \in L^s(\mathbb{S}^{n-1})$ has mean zero, and fix $1 < p < \infty$. Then, for every $f \in L^p(\mathbb{R}^n)$, under* (26.14)*,*

$$\Big\| \mathbf{J}(T^{0,b}_{\Omega,2^k} f : k \in \mathbb{Z}) \Big\|_p \le C_{n,p,\rho} s' \|\Omega\|_s \|f\|_p. \tag{26.65}$$

Proof. First impose finite shell and frequency cutoffs. For one residue class, Littlewood–Paley theory, Lemma 26.16, and bounded frequency overlap give, for every $1 < p_0 < \infty$,

$$\|h_{r,c}\|_{p_0} \le C_{p_0} \|\Omega\|_1 \|f\|_{p_0} \le C_{p_0} \|\Omega\|_s \|f\|_{p_0}. \tag{26.66}$$

On L^2, the separated Fourier supports and Lemma 26.15 give the following bound. Indeed, on $\operatorname{supp} \widehat{\Delta_{r-j} f}$ one has $|\log_2(2^j|\xi|) - r| \le 1$. The two exponential slopes in (26.61) are uniformly bounded for $1 < s \le 2$; including the possible crossing of $r = 0$, this gives $a_{s,\rho}(\log_2(2^j|\xi|)) \le C_\rho a_{s,\rho}(r)$ with a constant independent of s. Hence

$$\|h_{r,c}\|_2 \le C_\rho a_{s,\rho}(r) \|\Omega\|_s \|f\|_2. \tag{26.67}$$

If $p < 2$, interpolate (26.67) with (26.66) at one fixed $p_0 < p$; if $p > 2$, use one fixed $p_0 > p$. For $p = 2$ use (26.67). Thus for some $\theta_p > 0$ independent of s, r,

$$\|h_{r,c}\|_p \le C_{p,\rho} a_{s,\rho}(r)^{\theta_p} \|\Omega\|_s \|f\|_p. \tag{26.68}$$

Lemma 26.17 gives the same estimate for the paired jump of the corresponding residue-class tail. By $\mathbf{J}^{\mathrm{ch}} \le \sqrt{2}\mathbf{J}$ from (15.14), and then the uniform equivalence in Lemma 15.3, the same bound holds in the norm $\|\cdot\|_{\mathfrak{J}_D}$ for every finite ordered parameter set D. Summing the finitely many residue classes changes only the constant.

We now remove the cutoffs rather than merely invoking lower semicontinuity. Fix a finite ordered set $D \subset \mathbb{Z}$ of dyadic truncation indices and use the norm $\|\cdot\|_{\mathfrak{J}_D}$ supplied by Lemma 15.3. The preceding paired-to-chain conversion is what permits this normed summation. If $R_2 > R_1$, its triangle inequality and (26.68) give, uniformly in the finite shell cutoff,

$$\Big\| \sum_{R_1 < |r| \le R_2} \Big[\sum_{j \ge k} g_{j,r} \Big]_{k \in D} \Big\|_{L^p(\mathfrak{J}_D)} \le C_{p,\rho} \|\Omega\|_s \|f\|_p \sum_{R_1 < |r| \le R_2} a_{s,\rho}(r)^{\theta_p}.$$

By (26.62) the right side tends to zero as $R_1 \to \infty$, so the frequency sums are Cauchy in $L^p(\mathfrak{J}_D)$. For a fixed shell j, the Littlewood–Paley identity $\sum_r \widetilde{\Delta}_{r-j} \Delta_{r-j} f = f$ holds in L^p; convolution with the finite-annulus measure ν_j identifies every finite-shell coordinate of the limit.

Next remove the shell cutoff with the order of the two tails made explicit. Let $\mathscr{J}_J \uparrow \mathbb{Z}$ be the growing finite shell sets and put $E_J = \mathbb{Z} \setminus \mathscr{J}_J$. Given $\varepsilon > 0$, first choose R so large that

$$C_{p,\rho} \|\Omega\|_s \|f\|_p \sum_{|r| > R} a_{s,\rho}(r)^{\theta_p} < \frac{\varepsilon}{2};$$

this estimate is uniform in the shell cutoff. For the finitely many $|r| \le R$ and residue classes c, put

$$h^{E_J}_{r,c} = \sum_{\substack{j \in E_J \\ j \equiv c \pmod M}} \nu_j * \widetilde{\Delta}_{r-j} \Delta_{r-j} f.$$

The separated-tail estimate applies to an arbitrary subset $E \subset \{j : j \equiv c \pmod M\}$, not only to the full residue class: the low-pass multiplier used in its proof still gives, term by term,

$$\phi_{j_0-r} * h^E_{r,c} = \sum_{\substack{j\in E,\ j\ge j_0\\ j\equiv c \pmod M}} g_{j,r}.$$

Thus Lemma 4.5, with the same constant, controls the tail for every such E. Interpret this sum first over finite subsets of E_J. The following uniform estimate makes those partial sums Cauchy in L^p and hence defines $h^{E_J}_{r,c}$; the finite-parameter tail estimate then passes to that limit through the norm $\mathfrak{J}_D$. The proof of (26.66), restricted to E_J, gives

$$\|h^{E_J}_{r,c}\|_p \le C_p\|\Omega\|_1 \left\| \left(\sum_{j\in E_J} |\widetilde{\Delta}_{r-j}\Delta_{r-j}f|^2 \right)^{1/2} \right\|_p \longrightarrow 0.$$

Indeed, for fixed r the map $j \mapsto r-j$ is a bijection of $\mathbb{Z}$, and

$$\left(\sum_{j\in E_J} |\widetilde{\Delta}_{r-j}\Delta_{r-j}f|^2 \right)^{1/2} \longrightarrow 0 \quad \text{pointwise}, \qquad \left(\sum_{j\in E_J} |\widetilde{\Delta}_{r-j}\Delta_{r-j}f|^2 \right)^{1/2} \le \left(\sum_{\ell\in\mathbb{Z}} |\widetilde{\Delta}_{\ell}\Delta_{\ell}f|^2 \right)^{1/2}.$$

The dominating full Littlewood–Paley square function belongs to L^p; dominated convergence proves the asserted norm limit. This also shows that the limit is independent of the particular exhaustion $\mathscr{J}_J$. Choose J so that the sum of these finitely many norms is less than $\varepsilon/2$. Lemma 26.17, the triangle inequality in $L^p(\mathfrak{J}_D)$, and the preceding frequency-tail choice then show that the full omitted contribution has $L^p(\mathfrak{J}_D)$ norm below ε. This proves joint Cauchy convergence, independently of the order in which the two cutoffs are removed.

To retain the constant coordinate, fix $k_0 \in D$. For each residue c, the exact low-pass representation in Lemma 26.17 writes F_{r,c,k_0} as a convolution of $h_{r,c}$ with a kernel of fixed L^1 norm. Thus (26.68) and the summability of $a_{s,\rho}(r)^{\theta_p}$ show that $\sum_{r,c} F_{r,c,k_0}$ converges absolutely in L^p. The omitted-shell estimate proved above applies to this representation as well, so the finite shell sums converge to the same coordinate. The Littlewood–Paley identity identifies it with the original tail on bounded compactly supported inputs. For general L^p inputs, the local absolute-tail estimate of Section 12 and finite-annulus exhaustion give the same identification. Choose this L^p limit as the k_0-coordinate of the quotient representative. Finite-dimensional quotient convergence controls every difference between a k-coordinate and the k_0-coordinate; the same coordinatewise Littlewood–Paley identity therefore identifies all coordinates with $T^{0,b}_{\Omega,2^k}f$. Convergence in $L^p(\mathfrak{J}_D)$ gives, after a subsequence, convergence in the finite-dimensional quotient almost everywhere. The equivalence in Lemma 15.3 first returns the resulting bound to $\mathbf{J}^{\mathrm{ch}}_D$; the inequality $\mathbf{J}_D \le 2\mathbf{J}^{\mathrm{ch}}_D$ in (15.14), or equivalently finite strict-witness lower semicontinuity, then gives the paired estimate

$$\|\mathbf{J}(T^{0,b}_{\Omega,2^k}f : k \in D)\|_p \le C_{p,\rho} \sum_r a_{s,\rho}(r)^{\theta_p}\|\Omega\|_s\|f\|_p \le C_{p,\rho}s'\|\Omega\|_s\|f\|_p.$$

Finally exhaust D. Lemma 4.5 estimates the paired norm and hence the paired jump count. The paired-to-chain comparison is used only to enter the explicit finite-path quotient norm for summation, and the chain-to-paired comparison is used on exit. Both losses are absolute and independent of D. □

Lemma 26.19 (s'-tracked short terminal variation)**.** *Under the hypotheses of Lemma 26.18, and for every $f \in L^p(\mathbb{R}^n)$, let $S_2(T^{0,b}_\Omega f)$ denote the square sum of the within-dyadic-shell 2-variations. Then*

$$\|S_2(T^{0,b}_\Omega f)\|_p \le C_{n,p,\rho}s'\|\Omega\|_s\|f\|_p. \tag{26.69}$$

Proof. We first estimate the literal partial-shell paths

$$A_{j,u}f(x) = \int_{u2^j<|z|\le 2^{j+1}} b(|z|)K_\Omega(z)f(x-z)\,dz, \qquad 1 \le u \le 2.$$

These vanish at $u = 2$. Their kernels are genuinely in L^1 under angular L^1 and the normalized radial bound; no angular supremum bound is used. Put

$$S_r f = \left(\sum_{j\in\mathbb{Z}} V^2(A_{j,u}\widetilde{\Delta}_{r-j}\Delta_{r-j} f : 1 \le u \le 2)^2 \right)^{1/2}.$$

The dual cutoff is a reproducer, so the input is exactly $\widetilde{\Delta}_{r-j}\Delta_{r-j} f = \Delta_{r-j} f$.

For almost every x, full-shell absolute integrability makes the path absolutely continuous, with actual derivative

$$\partial_u A_{j,u} g(x) = -\frac{b(2^j u)}{u} \int_{\mathbb{S}^{n-1}} \Omega(\theta) g(x - 2^j u\theta)\, d\sigma(\theta)$$

for almost every u. Integrating its absolute value bounds V^2 by $C\|b\|_\infty \int |\Omega(\theta)| M_\theta g(x)\, d\sigma(\theta)$. The countable vector maximal estimate used in Lemma 26.16 and the Littlewood–Paley square bound give, for every $1 < p_0 < \infty$,

$$\|S_r f\|_{p_0} \le C_{n,p_0} \|\Omega\|_1 \|f\|_{p_0} \le C_{n,p_0} \|\Omega\|_s \|f\|_{p_0}. \tag{26.70}$$

The constants are fixed before r, s, Ω, b, f.

To obtain decay, apply the finite-variation radial estimate on the actual interval $(u, 2]$, uniformly for $1 \le u \le 2$. The zero-extension of $v \mapsto b(2^j v)/v$ on this interval has uniformly bounded ρ-variation, including the two endpoint increments. Mean-zero angular cancellation gives the low-frequency factor $2^j|\xi|$; the direct strip–Hölder argument of Lemma 26.15 gives the high-frequency factor $(2^j|\xi|)^{-2\beta_\rho/s'}$. On the localized frequency annulus this proves

$$\sup_{1\le u\le 2} |m_{j,u}(\xi)| \le C_{n,\rho} \|\Omega\|_s a_{s,\rho}(r). \tag{26.71}$$

The angular/cyclic Fourier conversion is $\xi = 2\pi\eta$; its fixed scale shift is absorbed into $C_{n,\rho}$, uniformly in s.

For a localized input g, Plancherel and the preceding uniform multiplier estimate bound the joint path norm in $L^2(dx\,du)$ by $C_{n,\rho}\|\Omega\|_s a_{s,\rho}(r)\|g\|_2$. The actual derivative has joint L^2 norm at most $C_n\|b\|_\infty\|\Omega\|_1\|g\|_2$, by translation invariance and Minkowski. Fubini and the fundamental theorem of calculus justify these as the path and its derivative, not merely formal multiplier derivatives. Since $A_{j,2} g = 0$, the anchored Sobolev–variation inequality gives

$$\|V^2(A_{j,u} g : 1 \le u \le 2)\|_2 \le C_{n,\rho} a_{s,\rho}(r)^{1/4} \|\Omega\|_s \|g\|_2.$$

Squaring, summing over every integer j, and applying the localized L^2 square estimate proves

$$\|S_r f\|_2 \le C_{n,\rho} a_{s,\rho}(r)^{1/4} \|\Omega\|_s \|f\|_2. \tag{26.72}$$

Thus the L^2 assertion concerns the genuine continuum V^2 square inside the spatial norm, not a supremum of separate operator norms.

The square of dense-parameter variations is a measurable sublinear operator on every L^{p_0}: use the variation triangle inequality, countable-vector Minkowski, and the actual convolution domain above. Interpolate this operator between (26.70) and (26.72). Choose the second spatial exponent using only the target p. For some $\vartheta_p > 0$ this gives

$$\|S_r f\|_p \le C_{n,p,\rho} a_{s,\rho}(r)^{\vartheta_p} \|\Omega\|_s \|f\|_p.$$

One may normalize by $\|\Omega\|_s$ when it is positive; when it is zero the original kernels vanish almost everywhere. Consequently the angular factor stays to exactly the first power, and neither the interpolation exponent nor its constant depends on s. Lemma 26.15 now gives

$$\sum_{r\in\mathbb{Z}} \|S_r f\|_p \le C_{n,p,\rho} s' \|\Omega\|_s \|f\|_p.$$

Here the reconstruction is at the level of the actual paths. Finite Littlewood–Paley sums converge to f in L^p; convolution with each fixed L^1 partial-shell kernel preserves that limit. The preceding summable bounds give absolute summability of the path coordinates almost everywhere for each j and each dense endpoint, since a coordinate is bounded by its variation anchored at $u = 2$. Thus the coordinate sums agree with $A_{j,u}f$, simultaneously for all integer j and dense endpoints on one conull set. The variation triangle inequality on finite partitions, countable Minkowski, and exhaustion yield

$$\left(\sum_j V^2(A_{j,u}f : 1 \le u \le 2)^2\right)^{1/2} \le \sum_r S_r f \quad \text{almost everywhere.}$$

Full-shell integrability makes each original path continuous, so the dense-endpoint variations here equal the continuum variations. This proves the required bound for the partial-shell square.

Finally identify that square with the original global zero-phase truncation. There is one conull set on which every positive-cutoff tail is absolutely integrable. To see this, prove integrability for countably many cutoffs using the positive radial tail and angular L^1 integration; domination by a smaller rational cutoff then gives all positive cutoffs. On that same set, for every integer j and every $u \in [1,2]$,

$$T^{0,b}_{\Omega,2^j u}f(x) = A_{j,u}f(x) + T^{0,b}_{\Omega,2^{j+1}}f(x).$$

Subtracting at any two real endpoints cancels the identical outer tail. The increasing parametrization $u \mapsto 2^j u$ is onto the whole dyadic shell; hence it preserves its V^2 exactly. Taking the all-integer square therefore identifies, almost everywhere, the partial-shell square with $S_2(T^{0,b}_{\Omega}f)$, not just with a majorant or a chosen finite-window trajectory. The displayed sum bound proves (26.69) with one factor s' and $C_{n,p,\rho}$ chosen before s. □

Proposition 26.20 (Terminal s'-tracked critical jump)**.** *For $1 < s \le 2$, mean-zero $\Omega \in L^s(\mathbb{S}^{n-1})$, and $1 < p < \infty$, the following holds for every $f \in L^p(\mathbb{R}^n)$:*

$$\|\mathbf{J}(T^{0,b}_{\Omega,\varepsilon}f : \varepsilon > 0)\|_p \le C_{n,p,\rho}\, s' \|\Omega\|_s \|f\|_p \tag{26.73}$$

under normalization (26.14).

Proof. The deterministic long–short inequality (15.10) gives pointwise

$$\mathbf{J}(T^{0,b}_{\Omega,\varepsilon}f) \le C\left[S_2(T^{0,b}_{\Omega}f) + \mathbf{J}(T^{0,b}_{\Omega,2^k}f : k \in \mathbb{Z})\right].$$

Apply Lemmas 26.18 and 26.19. □

Proposition 26.21 (Unphased terminal $L\log L$ theorem)**.** *For mean-zero $\Omega \in L\log L(\mathbb{S}^{n-1})$, $1 < p < \infty$, and every $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathbf{J}(T^{0,b}_{\Omega,\varepsilon}f : \varepsilon > 0)\|_p \le C_{n,p,\rho}\, \mathfrak{L}(\Omega) \|f\|_p \tag{26.74}$$

under (26.14).

Proof. We use exactly the finite-trajectory Orlicz extrapolation used in the proof of Proposition 17.3, but record the cancellation issue. Let

$$\Pi\Theta = \Theta - \sigma(\mathbb{S}^{n-1})^{-1}\int_{\mathbb{S}^{n-1}} \Theta\, d\sigma.$$

For a finite ordered truncation set D, let E_D be the scalar finite-path quotient endowed with a norm uniformly equivalent to the pointwise critical jump functional. Define the linear map

$$\mathscr{A}_{f,D}\Theta = \left[T^{0,b}_{\Pi\Theta,\varepsilon}f\right]_{\varepsilon\in D} \in L^p(\mathbb{R}^n; E_D).$$

Since Π is bounded on every L^s on the finite-measure sphere, Proposition 26.20 gives uniformly in D

$$\|\mathscr{A}_{f,D}\|_{L^s(\mathbb{S}^{n-1})\to L^p(E_D)} \le C_{p,\rho}\, s' \|f\|_p, \qquad 1 < s \le 2.$$

Pair with a norm-one functional in $[L^p(E_D)]^*$ and take $s = k/(k-1)$ for integers $k \geq 2$. For each k the resulting scalar functional on the sphere is represented by $h_k \in L^k$ satisfying $\|h_k\|_k \leq C_{p,\rho} k \|f\|_p$. These representatives agree almost everywhere: two of them represent the same functional on bounded angular functions, their difference lies in L^1 on the finite-measure sphere, and testing against its bounded complex sign makes that difference zero. Denote the common representative by h. If $\|f\|_p > 0$, Tonelli and the exponential series give, for a sufficiently large structural L,

$$\int_{\mathbb{S}^{n-1}} \Psi_0\left(\frac{|h|}{L\|f\|_p}\right) d\sigma \leq \sum_{k=2}^{\infty} \frac{(C_{p,\rho}k/L)^k}{k!} \leq 1.$$

The case $f = 0$ is immediate. Hence

$$\|h\|_{\exp L} \leq C_{p,\rho}\|f\|_p.$$

Orlicz–Hölder therefore yields

$$\|\mathscr{A}_{f,D}\Omega\|_{L^p(E_D)} \leq C_{p,\rho}\,\mathfrak{L}(\Omega)\|f\|_p$$

for mean-zero Ω, uniformly in D. Increase finite rational sets and use finite-annulus local absolute continuity to recover all positive parameters. This proves (26.74). □

26.8 Strong polynomial phases

Proposition 26.22 (Radially weighted annular decay)**.** *Let P be a real polynomial on $\mathbb{R}^n \times \mathbb{R}^n$ of total degree at most d, and let $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For $R > 0$ and $1/2 \leq u \leq 1$ put*

$$A_{R,u}^{P,\Theta,b} f(x) = \int_{uR<|x-y|<R} e^{iP(x,y)} b(|x-y|) K_\Theta(x-y) f(y)\, dy.$$

Under (26.14)*, for every nonterminal flag position μ there are $a_{\mu,\rho} > 0$ and $C_{n,d,\rho}$ such that, if the first nonzero flag coordinate of P is ± 1 at position μ, then*

$$\sup_{1/2\leq u\leq 1} \|A_{R,u}^{P,\Theta,b}\|_{2\to 2} \leq C_{n,d,\rho} R^{-a_{\mu,\rho}} \|\Theta\|_\infty, \qquad R \geq 1. \tag{26.75}$$

Every later phase coefficient remains unrestricted.

Proof. We revisit the three branches in Proposition 16.6.

For a transverse pivot, the kernel entering $A_Q^* A_Q$ is obtained from the original $g_h(v)$ by multiplication by $\overline{b(|v|)}b(|v-h|)$. The compact polynomial operator is applied to this function only through its L^2 norm. Since $\|b\|_\infty \leq 1$, the bound (16.19), the polynomial negative moment in h, and both Hilbert–Schmidt square roots are unchanged. Thus the original transverse exponent remains valid.

For a skew pivot, the twisted-convolution argument uses only the L^2 size of the annular kernel. Inserting b decreases that size by at most $\|b\|_\infty$; the original R^{-1} decay is unchanged.

For a convolution pivot, write the first nonzero homogeneous convolution term as Q_s, with $\|Q_s\|_{\mathrm{coeff}} \geq c_{n,d}$. After the change $r = Rv$, the one-dimensional fibre amplitude is

$$v \longmapsto b(Rv)v^{-1}\mathbf{1}_{[u,1]}(v).$$

Its zero extension has uniformly bounded $L^\infty + V^\rho$ norm by Lemma 26.2. Let $c_{\mathrm{poly},\rho} > 0$ be the exponent in Corollary 26.8 for degrees at most d. Plancherel on each line therefore gives the fibre bound

$$C_\rho \min\{1, (R^s|Q_s(\theta)|)^{-c_{\mathrm{poly},\rho}}\}.$$

Choose

$$\beta_{s,\rho} := \frac{1}{2}\min\left\{c_{\mathrm{poly},\rho}, \frac{1}{s}\right\} > 0. \tag{26.76}$$

The last display is bounded by $C_\rho R^{-s\beta_{s,\rho}}|Q_s(\theta)|^{-\beta_{s,\rho}}$, and (16.10) makes the angular integral finite uniformly in all lower coefficients. Thus the convolution branch has exponent $s\beta_{s,\rho}>0$.

Take $a_{\mu,\rho}$ to be the corresponding branch exponent. There are only finitely many flag positions, so all constants are structural and coefficient-uniform. □

For test functions, polar coordinates show that $u\mapsto A_{R,u}^{P,\Theta,b}f$ is an absolutely continuous L^2-valued path even when b has jumps. For almost every u,

$$\partial_u A_{R,u}^{P,\Theta,b}f(x)=-u^{-1}b(uR)\int_{\mathbb{S}^{n-1}}\Theta(\theta)e^{iP(x,x-uR\theta)}f(x-uR\theta)\,d\sigma(\theta).$$

Consequently

$$\sup_{1/2\le u\le 1}\|\partial_u A_{R,u}^{P,\Theta,b}f\|_q\le C\|b\|_\infty\|\Theta\|_1\|f\|_q,\qquad 1\le q\le\infty.$$

The anchored Sobolev–variation argument of Lemma 16.7 therefore gives

$$\|V^2(A_{R,u}^{P,\Theta,b}f:1/2\le u\le 1)\|_2\le C_{n,d,\rho}R^{-a_{\mu,\rho}/4}\|\Theta\|_\infty\|f\|_2. \tag{26.77}$$

Put

$$a_{0,\rho}:=\min\Big(\Big\{\frac{a_{\mu,\rho}}{4}:\mu\text{ nonterminal}\Big\}\cup\{1\}\Big)>0. \tag{26.78}$$

The added value 1 is used only when there is no nonterminal position.

Lemma 26.23 (Weighted q'-tracked partial-annulus estimates)**.** *Let P be normalized and nonterminal, $R\ge 1$, $1<q\le 2$, and $q'=q/(q-1)$. Let $\Theta\in L^q(\mathbb{S}^{n-1})$. Under* (26.14)*, for every $1<p<\infty$ set*

$$\gamma_{p,\rho}:=\frac{a_{0,\rho}}{2\max(p,p')}>0.$$

Then, for every $f\in L^p(\mathbb{R}^n)$,

$$\sup_{1/2\le u\le 1}\|A_{R,u}^{P,\Theta,b}f\|_p\le C_{n,d,p,\rho}R^{-\gamma_{p,\rho}/q'}\|\Theta\|_q\|f\|_p, \tag{26.79}$$

$$\|V^2(A_{R,u}^{P,\Theta,b}f:1/2\le u\le 1)\|_p\le C_{n,d,p,\rho}R^{-\gamma_{p,\rho}/q'}\|\Theta\|_q\|f\|_p. \tag{26.80}$$

All constants are uniform in the later phase coefficients and in the location of the variation of b.

Proof. For fixed P,R,u,b, the map $\Theta\mapsto A_{R,u}^{P,\Theta,b}$ is linear into $\mathscr{B}(L^2)$. Its angular L^1 bound follows from the annular L^1 kernel and $\|b\|_\infty\le 1$; its angular L^∞ bound is Proposition 26.22. Complex interpolation, whose L^∞ weight is $1/q'$, gives, after weakening the finite minimum of the nonterminal exponents to $a_{0,\rho}$,

$$\sup_u\|A_{R,u}^{P,\Theta,b}\|_{2\to 2}\le C_{n,d,\rho}R^{-a_{0,\rho}/q'}\|\Theta\|_q.$$

The preceding a.e. derivative estimate and the anchor $A_{R,1}^{P,\Theta,b}=0$ give, exactly as in (17.10),

$$\|V^2(A_{R,u}^{P,\Theta,b}f:u)\|_2\le C_{n,d,\rho}R^{-a_{0,\rho}/(4q')}\|\Theta\|_q\|f\|_2.$$

On the other hand $V^2\le V^1$ and the same derivative formula give nondecaying strong $(1,1)$ and (∞,∞) bounds with norm $C\|\Theta\|_1$. Marcinkiewicz interpolation puts weight $2/p'$ on the L^2 endpoint when $p<2$ and weight $2/p$ when $p>2$. This proves (26.80) with the displayed, slightly weakened, $\gamma_{p,\rho}$. The same spatial interpolation starting from the fixed-u L^2 estimate proves the stronger fixed-u bound and hence (26.79). Approximation in f and Θ is justified by the nondecaying V^1 estimate. □

Apply this lemma to the dyadic far path. The annular pieces and their short variations have L^p norm $C2^{-\gamma_{p,\rho}j/q'}\|\Theta\|_q\|f\|_p$; hence their geometric sum is at most $Cq'\|\Theta\|_q\|f\|_p$. The deterministic dyadic long–short decomposition therefore proves the weighted analogue of Proposition 17.2. On every finite truncation set, pair the resulting linear angular map with a norm-one dual functional. Taking $q = k/(k-1)$ gives the compatible bounds $\|h\|_k \le Ck\|f\|_p$; the common-representative and exponential-series argument used in Proposition 26.21 yields the weighted analogue of Proposition 17.3.

Finally, the local triangular reduction of Proposition 17.5 acquires the factor $b(|x-y|)$ in its error kernel; its absolute value is bounded by one, and the residual later-phase path carries the same b. Backward induction over the finite flag, with the unphased terminal estimate Proposition 26.21, therefore proves

$$\|\mathbf{J}(T^{P,b}_{\Omega,\varepsilon}f)\|_p \le C_{n,d,p,\rho}\,\mathfrak{L}(\Omega)\|f\|_p \tag{26.81}$$

under (26.14). The normalizing dilation sends b to $b(R\cdot)$, and (26.15) preserves its radial norm; hence the estimate is uniform in every coefficient of P.

26.9 Phase-adapted packets with finite radial variation

Proposition 26.24 (Radially weighted phase-adapted one-shell packet)**.** *For every fixed $1 \le \rho < \infty$, Proposition 20.10 remains valid with the radial factor $b(|x-y|)$ under* (26.14)*. Namely, there are $c_{P,\rho}, c_{E,\rho} > 0$ and $t_{0,\rho}$ such that, whenever the packet separation parameter satisfies $t \ge t_{0,\rho}$,*

$$\|V^2(G_\varepsilon : R \le \varepsilon \le 2R)\|_2^2 \le C_{n,d,\rho}\|\Theta\|_\infty^2 2^{-c_{P,\rho}t}\Lambda M, \tag{26.82}$$

$$\|V^1(E_\varepsilon : R \le \varepsilon \le 2R)\|_1 \le C_{n,d,\rho}\|\Theta\|_1 2^{-c_{E,\rho}t}M. \tag{26.83}$$

The estimates are uniform under packetwise scalar coefficients of modulus at most one.

Proof. We repeat the finite-mode low/high phase-block induction in Proposition 20.10, but choose all parameters after the radial exponents have been fixed. Let $c_{\mathrm{flat},\rho}$ and $C_{B,\rho}$ be the constants in Corollary 26.12, let $\delta_{T,\rho}$ be given by (26.48), and let C_{Four} dominate all powers in Lemma 20.9. Decrease $c_{\mathrm{flat},\rho}$ if necessary so that it is at most one. Here and below $C_{B,\rho}$ has first been enlarged once to dominate every radial-amplitude power in the equal-shell, unequal-shell, flat-block, and collective-shell estimates; it is not enlarged after the next parameter is chosen. Put

$$\theta_\rho := \frac{c_{\mathrm{flat},\rho}}{16(C_{B,\rho}+1)}. \tag{26.84}$$

Choose $B_{*,\rho} > 1$ so large that

$$\frac{2C_{\mathrm{Four}}}{B_{*,\rho}-1} \le \min\{\theta_\rho/2, c_{\mathrm{flat},\rho}/8\}. \tag{26.85}$$

Put $N_d = \max(d-1,0)$ and choose $0 < \kappa_{*,\rho} \le 1$ so small that, for every possible number $0 \le N \le N_d$ of nonpure homogeneous blocks,

$$2C_{\mathrm{Four}}\sum_{r=1}^{N}\kappa_{*,\rho}B_{*,\rho}^{r-N} \le \min\{\delta_{T,\rho}/2, 1/2\}. \tag{26.86}$$

Set

$$\kappa_r = \kappa_{*,\rho}B_{*,\rho}^{r-N}, \qquad S_r = \sum_{i<r}\kappa_i, \qquad \gamma_r = \theta_\rho\kappa_r.$$

Then $S_r \le \kappa_r/(B_{*,\rho}-1)$ and

$$\gamma_r - 2C_{\mathrm{Four}}S_r \ge \gamma_r/2, \tag{26.87}$$

$$c_{\mathrm{flat},\rho}\kappa_r - C_{B,\rho}\gamma_r - 2C_{\mathrm{Four}}S_r \ge c_{\mathrm{flat},\rho}\kappa_r/2, \tag{26.88}$$

$$c_{\mathrm{flat},\rho} - C_{B,\rho}\gamma_r - 2C_{\mathrm{Four}}S_r \ge c_{\mathrm{flat},\rho}/2. \tag{26.89}$$

These are the complete compatibility conditions; the choice is noncircular because all quantities on their right were fixed before $B_{*,\rho}$ and $\kappa_{*,\rho}$.

Suppose block r is the first high block. The preceding low blocks have squared Fourier loss at most $2^{2C_{\mathrm{Four}}S_r t}$. Partition the normalized shell into $K_r \simeq 2^{\gamma_r t}$ cells. The within-cell square function uses only $\|b\|_\infty \le 1$ and contributes $C2^{-\gamma_r t}\|\Theta\|_\infty^2 \Lambda M$. A fixed Rademacher–Menshov choice of cell signs produces the radial profile

$$b(Ru)\sum_{\ell=1}^{K_r} \varepsilon_\ell \mathbf{1}_{I_\ell}(u).$$

By (26.20), its $L^\infty + V^\rho$ norm is at most $C_\rho(1+K_r^{1/\rho})$. Applying (26.42) and Lemma 18.2 therefore gives, before the preceding Fourier loss is included,

$$C_\rho t^2 \|\Theta\|_\infty^2 2^{(C_{B,\rho}/\rho)\gamma_r t}\Lambda M\{2^{-c_{\mathrm{flat},\rho}\kappa_r t} + 2^{-c_{\mathrm{flat},\rho} t}\}. \tag{26.90}$$

Since $\rho \ge 1$, replace $C_{B,\rho}/\rho$ by $C_{B,\rho}$. Equations (26.87)– (26.89), after multiplication by the squared Fourier loss, leave the three exponents $\gamma_r/2$, $c_{\mathrm{flat},\rho}\kappa_r/2$, and $c_{\mathrm{flat},\rho}/2$. Thus, after absorbing the polynomial factor t^2, one may take

$$c_{r,\rho} := \frac{1}{4}\min\{\gamma_r, c_{\mathrm{flat},\rho}\kappa_r, c_{\mathrm{flat},\rho}\} > 0. \tag{26.91}$$

Every high branch therefore has $E = 0$ and geometric $L^2(V^2)$ decay.

If all blocks are low, expand them all with the same period T. The combined characters are $e^{i\omega_F u\cdot X}$ and $e^{i\omega_F v\cdot Y}$. Freezing the latter at each packet centre leaves a mean-zero packet; its remainder has the structural factor $\omega_F|v|2^{-t}$, absorbed by the unchanged weighted Fourier sum. Thus Proposition 26.14 applies. The squared Fourier loss is at most $2^{2C_{\mathrm{Four}}\sum_r \kappa_r t}$, and (26.86) gives, at the level of the full squared estimate,

$$2^{-\delta_{T,\rho} t}\left(2^{C_{\mathrm{Four}}\sum_r \kappa_r t}\right)^2 \le 2^{-\delta_{T,\rho} t/2}.$$

The input-modulation remainder is estimated in total truncation 1-variation exactly as in Proposition 20.10; multiplication by b costs only $\|b\|_\infty \le 1$. The weighted Fourier sum and (26.86) leave the bound $C\|\Theta\|_1 2^{-t/2}M$.

All Fourier expansions are first truncated. If $K' > K$ and block r is the first high block, define

$$\mathfrak{B}_{r,t,\rho} := 2^{-\gamma_r t} + t^2 2^{(C_{B,\rho}/\rho)\gamma_r t}\{2^{-c_{\mathrm{flat},\rho}\kappa_r t} + 2^{-c_{\mathrm{flat},\rho} t}\}.$$

Applying the within-cell and signed-grid estimates only to Fourier tuples omitted at level K gives, on every normalized interacting block pair,

$$\|G^{K'} - G^K\|_{L^2(V^2)} \le C_\rho \|\Theta\|_\infty (\Lambda\widetilde{M})^{1/2}\mathfrak{B}_{r,t,\rho}^{1/2}\,\omega_K(\{i : i < r\}), \qquad E^{K'} - E^K = 0. \tag{26.92}$$

If all blocks are low, the radial terminal-packet estimate and the freezing remainder give the following sharper tail estimates. Here one does *not* replace the omitted Fourier coefficients by the global majorant $2^{C_{\mathrm{Four}}\sum_r \kappa_r t}$: the exact common tail ω_K is retained. Consequently the terminal estimate contributes its raw square-root factor $2^{-\delta_{T,\rho} t/2}$ before the omitted tuples are summed:

$$\|G^{K'} - G^K\|_{L^2(V^2)} \le C_\rho \|\Theta\|_\infty 2^{-\delta_{T,\rho} t/2}(\Lambda\widetilde{M})^{1/2}\omega_K(\{1,\dots,N\}), \tag{26.93}$$

$$\|E^{K'} - E^K\|_{L^1(V^1)} \le C\|\Theta\|_1 2^{-t}\widetilde{M}\omega_K(\{1,\dots,N\}). \tag{26.94}$$

Here $\widetilde{M}$ is the normalized block mass. Since $\omega_K(S) \to 0$ by (20.58), the finite-mode decompositions are Cauchy in the anchored $L^2(V^2)$ and $L^1(V^1)$ spaces. The absolute common Fourier majorants give uniform convergence of the phase factors on the compact normalized blocks; dominated convergence therefore identifies the variation-space limit with the original trajectory. Fatou on finite parameter partitions preserves all displayed bounds.

For the branch $N = 0$ there is no high block and we assign its squared decay exponent to be $\delta_{T,\rho}/2$. Thus the final minimum is never over an empty set: take $c_{P,\rho}$ to be the minimum of $\delta_{T,\rho}/2$ and all high-branch exponents $c_{r,\rho}$ arising for the finitely many cases $1 \le N \le N_d$, and take $c_{E,\rho} = 1/2$. Increase $t_{0,\rho}$ so that it dominates the terminal-packet threshold and absorbs the finitely many polynomial factors used above. Rescaling and the bounded-overlap block summation are exactly those in the final paragraph of Proposition 20.10; the L^2-square and L^1 Jacobians cancel the normalized mass factor. This proves (26.82)–(26.83). □

Rescaling the radial profile. The direct realization following Proposition 20.10 works with the radial factor as well. On a physical shell of radius $R > 0$, use the actual local profile

$$b_R(u) = b(Ru), \qquad 1 \le u \le 2,$$

not the unscaled profile $b(u)$. Under $x = a + RX$, $y = a + RY$, the homogeneity of the angular kernel gives the exact identity

$$K_\Theta(x-y)b(|x-y|)\,dy = K_\Theta(X-Y)b_R(|X-Y|)\,dY.$$

The essential supremum of b_R on $[1,2]$ is no greater than that of b on $(0,\infty)$: multiplication by R and its inverse preserve Lebesgue null sets. Every increasing finite sequence in $[1,2]$ maps to an increasing finite sequence in $[R,2R]$, so its radial ρ-variation is also bounded by the global variation. Hence

$$\|b_R\|_{L^\infty([1,2])} + V^\rho(b_R;[1,2]) \le \|b\|_{\Upsilon^\rho}.$$

This is an essential-norm statement and does not replace the norm by a pointwise supremum. The radially weighted terminal and flat estimates can thus be applied to b_R with the same constants on every shell. Choose their positive decay exponents first, then the finite hierarchy of phase thresholds as in the direct realization. The low branch uses the same frozen integral, now including b_R, and its physical remainder; the high branch uses the full literal weighted trajectory and zero remainder. Fourier summation commutes with these integrals by the common absolutely summable envelope. Finally the whole-cube grouping and the cancelling mass Jacobians are unchanged. This proves (26.82)–(26.83) for the original physical radial factor with no additional normalization hypothesis.

26.10 The weak endpoint

We first treat the unphased long path, where one additional modification is essential.

Lemma 26.25 (Whole-profile smoothing in the unphased weak long path)**.** *Fix a shell $2^j < |z| \le 2^{j+1}$ and put*

$$a_j(u) = b(2^j u)\mathbf{1}_{(1,2]}(u), \qquad 1/2 < u < 3.$$

Under (26.14)*, $\|a_j\|_\infty + V^\rho(a_j) \le C$ uniformly in j. If $a_{j,h} = a_j * \phi_h$, then*

$$\|a_j - a_{j,h}\|_1 \le C_\rho h^{1/\rho}, \tag{26.95}$$

$$\|\partial^N a_{j,h}\|_\infty \le C_N h^{-N}. \tag{26.96}$$

Consequently, in the long-variation part of the proof of Proposition 21.1 one may replace the smoothing of the sharp shell indicator by smoothing the entire profile a_j. The sharp–smooth difference has total V^1L^1 norm $O_\rho(h^{1/\rho}\Sigma)$, while the smoothed full-kernel estimate has exactly the derivative hypothesis used in the proof of Lemma 18.4.

Proof. Extend a_j by zero outside $(1,2]$ and use a nonnegative unit-mass C^∞ mollifier supported in $[-1,1]$. The normalization (26.14) gives $|b(r)| \le 1$ at every positive radius. Thus $\|a_j\|_\infty \le 1$ and $V^\rho(a_j) \le 3$, the two added endpoint jumps accounting for the extra variation. In particular positive convolution gives the sharp bound $|a_{j,h}| \le 1$; no factor-two renormalization is used.

The translation estimate of Lemma 26.3, integrated against the positive mollifier, gives

$$\|a_j - a_{j,h}\|_1 \le 4\,3^{1-1/\rho} h^{1/\rho}.$$

Differentiating the convolution gives $\|\partial^N a_{j,h}\|_\infty \le \|\partial^N \phi\|_1 h^{-N}$. For the small smoothing range $0 < h \le 1/4$ used in the long path, $\operatorname{supp} a_{j,h} \subset [3/4, 9/4]$. This is a fixed enlarged annulus: the difference need not be supported near the two shell endpoints, nor near finitely many jump boundaries of b.

Let E_j be the literal sharp-minus-smooth kernel with angular datum ω_s. Polar coordinates, using $u^{-1} \le 4/3$ on this enlarged annulus, give

$$\|E_j\|_1 \le \frac{16}{3}\, 3^{1-1/\rho} \|\omega_s\|_1 h^{1/\rho}.$$

For a finite family, each increment of a suffix path is the sum over one interval of indices; those intervals are disjoint in a variation partition. Hence

$$V^1\left(\sum_{j\ge k} E_j * B_{j-s} : k\right) \le \sum_j |E_j * B_{j-s}|.$$

Young's inequality and the assignment of each bad cube to at most one shell, with its fixed coefficient of modulus at most one, prove

$$\big\| V^1(\text{sharp long path} - \text{smooth long path}) \big\|_1 \le C_\rho \|\omega_s\|_1 h^{1/\rho} \sum_j \|B_{j-s}\|_1 \le C_\rho \|\omega_s\|_1 h^{1/\rho} \Sigma.$$

Finite-family exhaustion gives the same estimate when $\Sigma < \infty$; the summable L^1 majorant justifies the actual suffix limits. There is no loss depending on the number or overlap of shells.

At physical scale the smooth kernel is exactly $H_j^s(z) = K_{\omega_s}(z) a_{j,h}(2^{-j}|z|)$. For $r > 0$, a unit direction θ, and $0 \le \ell \le N$, apply Leibniz' rule to $\omega_s(\theta) r^{-n} a_{j,h}(2^{-j} r)$. Every nonzero term has $r \simeq 2^j$ and is bounded after multiplication by $r^{n+\ell}$ by $C_{n,N} |\omega_s(\theta)| h^{-N}$. Therefore

$$\sup_{r>0,\, 0\le \ell \le N} r^{n+\ell} |\partial_r^\ell H_j^s(r\theta)| \le C_{n,N} |\omega_s(\theta)| h^{-N}, \qquad \|H_j^s\|_1 \le C_n \|\omega_s\|_1.$$

These are the required weighted ray-derivative and mass bounds for the very same whole-profile family. They impose no angular derivative assumption, and give the full-kernel constant $M_N \le C_{n,N} \|\omega_s\|_\infty h^{-N}$ used in Lemma 18.4. □

Lemma 26.26 (One-dimensional fixed-sign full-kernel estimate)**.** *Let H_j be odd C^2 kernels on $\mathbb{R}$, supported where $2^{j-2} \le |x| \le 2^{j+2}$, and assume*

$$\|H_j\|_1 \le M, \qquad \|H_j'\|_\infty \le M 2^{-2j}, \qquad \|H_j''\|_\infty \le M 2^{-3j}.$$

Let $\mathcal{Q}$ be a finite family of pairwise disjoint intervals and suppose

$$\operatorname{supp} b_Q \subset Q, \qquad \int b_Q = 0, \qquad \|b_Q\|_1 \le \Lambda |Q|.$$

For $s \ge 5$, pair H_j with the intervals of length 2^{j-s}. Then for arbitrary fixed coefficients $|\varepsilon_Q| \le 1$,

$$\left\| \sum_j H_j * \sum_{|Q| = 2^{j-s}} \varepsilon_Q b_Q \right\|_2^2 \le C M^2 2^{-s} \Lambda \sum_Q \|b_Q\|_1. \tag{26.97}$$

Proof. We prove the stronger squared-energy bound with $600\, 2^{-2s}$. Put $\delta = 2^{-s}$, and for each packet set $R_Q = 2^{j(Q)}$, where $|Q| = \delta R_Q$; let c_Q be its center. Since H_j' is continuous, its essential supremum bound

is a bound at every point. Mean-zero cancellation and the fundamental theorem of calculus consequently give

$$H_{j(Q)} * b_Q(x) = \int_Q [H_{j(Q)}(x-y) - H_{j(Q)}(x-c_Q)] b_Q(y)\, dy, \qquad |H_{j(Q)}(x-y) - H_{j(Q)}(x-c_Q)| \le M\delta / R_Q.$$

Here $|y - c_Q| \le \delta R_Q$ is sufficient. If $|x-y| > 5R_Q$, then $|x - c_Q| > 4R_Q$, because $\delta \le 1$; both kernel values vanish by the outer support assumption. Define the positive envelope

$$k_Q(z) = \frac{M\delta}{R_Q} \mathbf{1}_{\{|z| \le 5R_Q\}}, \qquad v_Q = k_Q * |b_Q|.$$

Thus $|\varepsilon_Q H_{j(Q)} * b_Q| \le v_Q$ almost everywhere and $\|k_Q\|_1 = 10M\delta$.

We spell out the multiscale packing estimate, so that no interaction budget is being assumed. Order packets by R_Q. If $R_{Q'} \le R_Q$, the positive correlation $k_Q * \widetilde{k}_{Q'}$ is supported in $[-10R_Q, 10R_Q]$ and is bounded by $(M\delta/R_Q)(10M\delta)$. For fixed y, only intervals Q' meeting $[y - 10R_Q, y + 10R_Q]$ can contribute to its convolution with $|b_{Q'}|$. Such intervals have length at most R_Q, and hence are contained in $[y - 15R_Q, y + 15R_Q]$. Their disjoint interiors and the density hypothesis therefore give, simultaneously across all these scales,

$$\sum_{R_{Q'} \le R_Q} \int (k_Q * \widetilde{k}_{Q'})(y-z) |b_{Q'}(z)|\, dz \le (10M\delta)(30M\delta\Lambda).$$

Tonelli now expands the square of $\sum_Q v_Q$ into nonnegative pairings. Split the ordered pairs according to which radius is larger and charge the diagonal at most twice. Integrating the last bound against $|b_Q(y)|$ proves

$$\int \left(\sum_Q v_Q \right)^2 \le 2(10M\delta)(30M\delta\Lambda) \sum_Q \|b_Q\|_1 = 600M^2\delta^2\Lambda \sum_Q \|b_Q\|_1.$$

The family is finite, so linearity of the genuine L^1 convolutions regroups the packet sum as the displayed sum over j in the statement. Pointwise domination by $\sum_Q v_Q$, followed by $\delta^2 \le \delta = 2^{-s}$, proves (26.97). Oddness, the inner support bound, and the second-derivative bound are compatible with this argument but are not needed for this stronger estimate.

The same reasoning also justifies the countable grouping used when exhausting a decomposition. If $\sum_Q \|b_Q\|_1 < \infty$, then $\sum_Q \|H_{j(Q)} * (\varepsilon_Q b_Q)\|_1 \le M \sum_Q \|b_Q\|_1$. Tonelli gives absolute summability almost everywhere and permits both same-scale packet summation inside convolution and subsequent summation over scales, on one common conull set. Fatou transfers the stronger energy bound to that actual output. When the total mass is infinite and $M\Lambda > 0$, the extended energy bound has infinite right-hand side, with no assertion of uncontrolled convergence. If $M = 0$, continuity and the L^1 bound force every H_j to be identically zero; if $\Lambda = 0$, every packet vanishes almost everywhere. Thus the zero-parameter and empty-family cases are included as well. □

Proposition 26.27 (Unphased weak endpoint with radial V^ρ)**.** *For every mean-zero* $\Omega \in L\log L(\mathbb{S}^{n-1})$ *and every bounded compactly supported* f*, under* (26.14)*,*

$$\|\mathbf{J}(T^{0,b}_{\Omega,\varepsilon} f)\|_{L^{1,\infty}} \le C_{n,\rho}\, \mathfrak{L}(\Omega) \|f\|_1. \tag{26.98}$$

Proof. We prove the estimate for every $f \in L^1(\mathbb{R}^n)$ and every $n \ge 1$. Initially use the pointwise radial budget

$$B(b) := \sup_{r>0} |b(r)| + V^\rho(b; (0,\infty)) \le 1.$$

This is exactly the normalization in (26.14). Write $F = \|f\|_1$ and $G = \mathfrak{L}(\Omega)$. If $G = 0$, polar coordinates give the zero truncation at every positive radius and the conclusion is immediate. Suppose $G > 0$, fix $\lambda > 0$, and perform the dyadic Calderón–Zygmund decomposition at height $\alpha = \lambda / G$:

$$f = g + \sum_Q b_Q, \qquad \sum_Q \|b_Q\|_1 \le 2F, \quad \sum_Q |Q| \le F/\alpha, \quad \|g\|_2^2 \le 2^n \alpha F.$$

Here the sum is the actual L^1-convergent sum of the mean-zero bad functions on the disjoint maximal dyadic cubes; in particular $\|b_Q\|_1 \le 2^{n+1}\alpha|Q|$. Let $\mathscr{N}$ be the pointwise finite-pair path norm in (2.1). Proposition 26.21 at $p=2$, together with the pointwise comparison $\mathscr{N} \le 2\mathbf{J}$ in (2.2), gives

$$|\{\mathscr{N}(T^{0,b}_{\Omega,\varepsilon}g) > u\}| \le C_{n,\rho}G^2 2^n \alpha F/u^2 \qquad (u>0).$$

Only this strong L^2 estimate is used for the good function.

The original bad functions and the raw angular split. For now retain a finite set of bad cubes. Put $B_k = \sum_{\ell(Q)=2^k} b_Q$, with this finite restriction understood. The gap-s path pairs B_{j-s} with the literal shell $2^j < |z| \le 2^{j+1}$, retaining the factor $b(|z|)$ and the positive cutoff ε. Fix a structural starting gap $s_0 \ge 5$ large enough for Proposition 26.14. Set

$$N = 4n+1, \quad k_\rho = \lceil \rho \rceil, \quad \delta_a = \frac{1}{6144Nk_\rho}, \quad h_s = \frac{1}{8}2^{-s/(192N)}, \quad \Theta_s = \|\Omega\|_1 2^{\delta_a s}. \tag{26.99}$$

Split Ω into its raw bounded part $\omega_s = \Omega \mathbf{1}_{\{|\Omega|\le\Theta_s\}}$ and raw tail $\Omega^\sharp_s = \Omega\mathbf{1}_{\{|\Omega|>\Theta_s\}}$. Neither part is required to have mean zero. A bounded measurable angular representative of ω_s may be used in the kernel estimates: polar coordinates and absolute convolution identify the resulting paths on one common conull set for all the gaps and positive radii. No radial representative is changed. The raw angular-tail summation and the absolute-convolution V^1 envelope give, uniformly in every finite set $\mathscr{I}$ of retained gaps,

$$\left\| V^1\left(\sum_{s\in\mathscr{I}} \mathrm{Tail}^b_{s,\varepsilon} : \varepsilon > 0 \right) \right\|_1 \le C_{n,\delta_a} GF.$$

This estimate uses only $|b| \le 1$ and the original angular $L\log L$ gauge, not cancellation of either split piece.

The three majorants for a bounded angular piece. Let $U^b_{s,\varepsilon}$ be its actual gap-s continuum path, and put

$$\mathscr{S}_s = \left(\sum_j V^2(U^b_{s,\varepsilon} : 2^j \le \varepsilon \le 2^{j+1})^2 \right)^{1/2}.$$

The unique shell assignment of each bad cube, followed by the full terminal estimate of Proposition 26.14, gives

$$\|\mathscr{S}_s\|_2^2 \le C_{n,\rho}\Theta_s^2 2^{-\delta_{T,\rho}s}\alpha F, \qquad \delta_{T,\rho} = \min\left\{ \frac{1}{128N}, \frac{1}{64N\rho} \right\}. \tag{26.100}$$

This is the original terminal decay. For scalar summation only we use $\delta_{T,\rho} \ge 1/(768Nk_\rho)$, so

$$\|\mathscr{S}_s\|_2 \le C_{n,\rho}\|\Omega\|_1 2^{-s/(2048Nk_\rho)}(\alpha F)^{1/2}.$$

Smooth the entire profile $a_j(u) = b(2^j u)\mathbf{1}_{(1,2]}(u)$ at width h_s, as in Lemma 26.25, and write

$$H^{s,b}_j(z) = K_{\omega_s}(z)(a_j * \phi_{h_s})(2^{-j}|z|), \qquad \widetilde{U}^b_s(m) = \sum_{j\ge m} H^{s,b}_j * B_{j-s}.$$

Define the literal discrete long and error majorants by

$$\mathscr{L}_s = V^2(\widetilde{U}^b_s(m) : m\in\mathbb{Z}), \qquad \mathscr{E}_s = V^1(U^b_{s,2^m} - \widetilde{U}^b_s(m) : m\in\mathbb{Z}).$$

The deterministic dyadic long–short inequality yields pointwise

$$V^2(U^b_{s,\varepsilon} : \varepsilon > 0) \le 3(\mathscr{S}_s + \mathscr{L}_s + \mathscr{E}_s).$$

These three majorants have measurable representatives. The sharp–smooth error is a difference of *discrete suffix paths*. Its V^1 norm is at most twice the sum of the absolute shell errors, hence Young's inequality and the unique shell assignment give

$$\|\mathscr{E}_s\|_1 \le C_{n,\rho}\Theta_s h_s^{1/\rho} F \le C_{n,\rho}\|\Omega\|_1 2^{-31s/(6144Nk_\rho)}F. \tag{26.101}$$

In particular this is not a radius-truncated thin-boundary estimate. The last exponent follows from $1/(192N\rho)-\delta_a \ge 31/(6144Nk_\rho) > 0$.

The smoothed full-kernel source has normalized radial derivative budget $M_N \le C_{n,N}h_s^{-N}\Theta_s$, uniformly in j. Apply Lemma 18.4 to the actual original bad-cube packets in each translated dyadic grid, with fixed coefficients chosen before the output point. Truncate the ranks at $2^{(n+1)s}$; the rank-overlap exception X_s satisfies

$$|X_s| \le 2^{6n-s}F/\alpha.$$

The finite translated-grid rank decomposition and the binary fixed-sign Rademacher–Menshov inequality give the explicit norm factor $D_{n,s} = 2^n((n+1)s+2)\sqrt{8}$. Consequently

$$\begin{aligned}\|\mathbf{1}_{X_s^c}\mathscr{L}_s\|_2 &\le D_{n,s}\big(C_n(h_s^{-N}\Theta_s)^2 2^{-s/24}\alpha F\big)^{1/2}\\ &\le C_{n,\rho}(s+2)\|\Omega\|_1 2^{-(1/64-\delta_a)s}(\alpha F)^{1/2}.\end{aligned} \tag{26.102}$$

Indeed $h_s^{-N} = 8^N 2^{s/192}$, so the square-root source decay $2^{-s/48}$ leaves $2^{-s/64}$ before the angular growth is included. The exponent $1/64-\delta_a$ is positive. This argument applies in every positive dimension and does not require Lemma 26.26, a sharper one-dimensional estimate, or an output-dependent choice of signs.

Finite-gap distribution and the two exhaustions. For a finite gap set $\mathscr{I} \subset \{s : s \ge s_0\}$ let $X = \bigcup_{s\in\mathscr{I}} X_s$. Sum the actual L^2 majorants by Minkowski on X^c and sum the error and raw-tail envelopes in L^1. The four scalar series for exceptional volume, short variation, long variation, and smoothing error are finite by the displayed exponents. Thus there are nonnegative measurable majorants $\mathscr{A},\mathscr{B}$ such that, almost everywhere,

$$V^2\left(\sum_{s\in\mathscr{I}} U_{s,\varepsilon}^{\mathrm{full}} : \varepsilon > 0\right) \le \mathscr{A}+\mathscr{B}, \quad \|\mathscr{A}\|_1 \le C_1 GF, \quad \|\mathbf{1}_{X^c}\mathscr{B}\|_2 \le \sqrt{C_2}G(\alpha F)^{1/2}, \quad |X| \le C_0 F/\alpha.$$

Here U_s^{full} includes both raw angular pieces. Chebyshev therefore gives, for $u>0$, constants independent of both the finite packet set and the finite gap set,

$$\left|\left\{V^2\left(\sum_{s\in\mathscr{I}} U_{s,\varepsilon}^{\mathrm{full}}\right) > u\right\}\right| \le C_0F/\alpha + C_1GF/(u/2) + C_2G^2\alpha F/(u/2)^2.$$

This sums strong majorants, not weak-L^1 quasi-norms or a sequence of per-gap level sets.

There is one enlarged original Calderón–Zygmund set E_0, independent of the finite cutoffs, with $|E_0| \le 2^{n(s_0+3)}F/\alpha$. Outside E_0 every prohibited near-gap cell vanishes at every radius. Fix the finite packet set first and exhaust the gaps $s \ge s_0$. Literal dyadic-cell exhaustion and absolute cutoff integrability give convergence to its original truncation on one common conull set, simultaneously for every positive radius. Every strict superlevel witness for V^2 uses finitely many radii, so it persists under this convergence. The resulting lower semicontinuity of distribution sets, with the payment for E_0 absorbed into C_0, gives

$$|\{V^2(T_{\Omega,\varepsilon}^{0,b}b_{\mathscr{Q}}) > u\}| \le C_0F/\alpha + C_1GF/(u/2) + C_2G^2\alpha F/(u/2)^2$$

for every finite original bad-cube sum $b_{\mathscr{Q}}$.

Now $\mathscr{N}(H+K) \le \mathscr{N}(H)+2V^2(K)$, so the good estimate at level $\lambda/2$ and this bad estimate at level $\lambda/4$, with $\alpha = \lambda/G$, imply

$$|\{\mathscr{N}(T_{\Omega,\varepsilon}^{0,b}g + T_{\Omega,\varepsilon}^{0,b}b_{\mathscr{Q}}) > \lambda\}| \le C_{n,\rho}GF/\lambda.$$

Finally exhaust the original bad cubes. Their actual L^1 series converges to $f-g$; common all-radius absolute cutoff integrability gives convergence of these original truncations to $T^{0,b}_{\Omega,\varepsilon}f$ on a single conull set for every $\varepsilon>0$. Strict finite-pair witnesses for $\mathscr{N}$ pass the last distribution estimate to the full path by the same lower-semicontinuity argument, which is valid for outer measure and assumes no level-set measurability. No L^2 approximation of f, finite parameter restriction, or uncountable intersection of exceptional sets is used. This establishes the pointwise-inside bound, since $\mathbf{J}\le\mathscr{N}$.

The same argument also gives a bound under the weaker essential-supremum budget. Put $B_{\mathrm{ess}}(b)=\|b\|_{L^\infty(0,\infty)}+V^\rho(b;(0,\infty))$. For each $r>0$, comparison with an essential-norm anchor gives

$$|b(r)|\le\|b\|_{L^\infty(0,\infty)}+V^\rho(b;(0,\infty)),\qquad B(b)\le 2B_{\mathrm{ess}}(b).$$

Indeed take an anchor outside the essential exceptional set with value at most the essential supremum plus any prescribed positive error, apply the two-point variation bound, and let that error tend to zero. If $B_{\mathrm{ess}}(b)=0$, this also proves $b(r)=0$ at every positive radius. Otherwise exact scalar normalization by the finite pointwise budget and homogeneity of the literal truncations restore a factor at most $2B_{\mathrm{ess}}(b)$. In particular, since $B_{\mathrm{ess}}(b)\le B(b)$, the asserted estimate follows under (26.14), throughout $1\le\rho<\infty$. □

Proposition 26.28 (Polynomial weak endpoint with radial V^ρ)**.** *Under* (26.14)*, for every polynomial P of degree at most d, every mean-zero $\Omega\in L\log L(\mathbb{S}^{n-1})$, and every bounded compactly supported f,*

$$\|\mathbf{J}(T^{P,b}_{\Omega,\varepsilon}f)\|_{L^{1,\infty}}\le C_{n,d,\rho}\,\mathfrak{L}(\Omega)\|f\|_1. \tag{26.103}$$

Proof. If $\mathfrak{L}(\Omega)=0$, then $\Omega=0$ and the assertion is immediate. By angular homogeneity assume throughout the induction that $\mathfrak{L}(\Omega)=1$. At each inductive stage impose finite cube and shell cutoffs and fix a finite rational truncation set I; every bound below is uniform in these choices. We induct on the largest total degree m carrying a nonpure homogeneous class. If no such class exists, write $P(x,y)=p(x)+q(y)$ and use Proposition 26.27 after input and output modulation.

Suppose the result is known through nonpure degree $m-1$. Let P_m be the highest nonpure homogeneous block and put

$$C_m=\|P_m\|_{\mathrm{coeff}},\qquad r_0=C_m^{-1/m}.$$

The exact conjugacy

$$T^{P,b}_{\Omega,r_0\varepsilon}f(r_0X)=T^{\widetilde{P},\widetilde{b}}_{\Omega,\varepsilon}f_{r_0}(X),\qquad \widetilde{b}(u)=b(r_0u), \tag{26.104}$$

and (26.15) reduce the proof to $\|P_m\|_{\mathrm{coeff}}=1$ without changing the normalized radial bound.

We now make the far-field parameter choice explicit. Put

$$d_*=\max(d,1),\qquad \kappa_0=\frac{1}{100d_*},\qquad c_{H,\rho}=c_{\kappa_0,\rho}, \tag{26.105}$$

where $c_{\kappa_0,\rho}>0$ is given by (26.45), and choose

$$0<\delta_a<\frac{1}{8}\min\{c_{P,\rho},c_{E,\rho},c_{H,\rho}\}. \tag{26.106}$$

Perform the phase-adapted Calderón–Zygmund decomposition and write

$$b_{\mathrm{CZ}}=\sum_Q b_Q,\qquad \Sigma=\sum_Q\|b_Q\|_1\le C\|f\|_1.$$

The good part is controlled by (26.81): since the Calderón–Zygmund decomposition gives $\|g\|_2^2\le C\alpha\|f\|_1$, the finite-path set

$$E^P_{\mathrm{good}}=\{x:\mathbf{J}_I(T^{P,b}_{\Omega,\varepsilon}g)>\alpha\}$$

has measure at most $C_{n,d,\rho}\alpha^{-1}\|f\|_1$. Use the mean-zero angular split (15.32) with the value δ_a in (26.106). The angular tails have the same summable V^1L^1 bound as in the unweighted proof.

For the bounded angular part, Proposition 26.24 gives the short square function with squared L^2 decay $2^{-[c_{P,\rho}-2\delta_a]s}$ and the packet error with L^1 decay $2^{-c_{E,\rho}s}$. Split the dyadic shells by

$$1 \le R^m \le 2^{\kappa_0 s} \quad \text{and} \quad R^m > 2^{\kappa_0 s}.$$

There are $O(1+s)$ low shells. Cauchy–Schwarz and the endpoint differences supplied by Proposition 26.24 give a low-shell squared L^2 bound

$$C(1+s)2^{-[c_{P,\rho}-2\delta_a]s}\alpha\Sigma,$$

plus a summable V^1L^1 error.

On the high shells use the same output localization, fixed adjacent-grid classes, exceptional sets, and rank suffixes as in Proposition 23.1. The normalized sharp-annulus profiles are products of $b(R\cdot)$ with fixed step cutoffs and hence have uniformly bounded $L^\infty + V^\rho$ norm. Corollary 26.12, followed by Lemma 18.2, gives

$$Cs^2 2^{-[c_{H,\rho}-2\delta_a]s}\alpha\Sigma$$

for the high dyadic tails. By (26.106), every displayed exponent is strictly positive and all polynomial factors in s are summable.

Here is the precise comparison with the majorants in Proposition 23.1. Write $G^b_{j,s} + E^b_{j,s}$ for the weighted one-shell decomposition supplied by Proposition 26.24, and $F^{G,b}_{j,s} + F^{E,b}_{j,s}$ for its endpoint difference. Define

$$\mathscr{S}^b_s = \left(\sum_j V^2(G^b_{j,s})^2\right)^{1/2}, \qquad \mathscr{L}^b_s = 2\sum_{j\ \mathrm{low}} |F^{G,b}_{j,s}|,$$

$$\mathscr{H}^b_s = \mathbf{1}_{X_s^c} V^2(\text{weighted high dyadic tails}), \qquad \mathscr{E}^{*,b}_s = \sum_j V^1(E^b_{j,s}).$$

Let $\mathscr{W}$ be the total V^1 majorant of the weighted angular-tail paths plus $C\sum_s \mathscr{E}^{*,b}_s$, including the low-shell endpoint errors, and set

$$\mathscr{G} = \sum_s (\mathscr{S}^b_s + \mathscr{L}^b_s + \mathscr{H}^b_s), \qquad X = \bigcup_s X_s.$$

These are exactly the four objects in (23.20)–(23.21), with the corresponding weighted direct-packet and collective-high estimates; multiplication by $|b| \le 1$ leaves the angular-tail comparison unchanged. Hence the resulting nonnegative majorants satisfy

$$\|\mathscr{G}\|_2^2 \le C_{n,d,\rho}\alpha\Sigma, \qquad \|\mathscr{W}\|_1 \le C_{n,d,\rho}\Sigma, \qquad |X| \le C\alpha^{-1}\|f\|_1,$$

and, off the fixed complement,

$$\mathbf{J}(\mathscr{F}^{P,b} b_{\mathrm{CZ}}) \le C_{n,d,\rho}(\mathscr{G} + \mathscr{W}).$$

The signs are fixed packetwise before the output point is evaluated, and none of these objects depends on the jump threshold.

For the local path apply Proposition 24.1. Its error kernel is simply multiplied by $|b(|x-y|)| \le 1$, while the residual phase has nonpure degree at most $m-1$ and carries the same radial function. On each unit output cube the induction hypothesis applies to the translated, purely modulated input; the fixed input enlargements have bounded overlap. The piecewise error satisfies

$$\|V^1(E)\|_1 \le C_{n,d}\|\Omega\|_1\|f\|_1.$$

Combining the far path, local residual, and error with the pointwise jump quasi-triangle inequality now gives a literal threshold argument. Namely, choose a fixed $A_{n,d,\rho}$ larger than all quasi-triangle constants. Outside E^P_{good}, the enlarged-CZ set, and X, a strict $A_{n,d,\rho}\alpha$ witness for the full path forces either $\mathscr{G} > c_{n,d,\rho}\alpha$, $\mathscr{W} > c_{n,d,\rho}\alpha$, a strict $c_{n,d,\rho}\alpha$ witness for the lower-degree residual, or $V^1(E) > c_{n,d,\rho}\alpha$.

Chebyshev, the two displayed majorant bounds, the induction hypothesis, and the $V^1(E)$ estimate therefore give

$$|\{\mathbf{J}_I(T^{P,b}_{\Omega,\varepsilon}f : \varepsilon \in I) > A_{n,d,\rho}\alpha\}| \le C_{n,d,\rho}\alpha^{-1}\|f\|_1.$$

Putting $\alpha = \tau/A_{n,d,\rho}$ proves the required distribution estimate for every strict threshold $\tau > 0$, uniformly at the finite-cutoff stage and for the normalized top block.

All estimates above were first proved with finite cube, shell, and rational-parameter cutoffs. To remove them, fix $\tau > 0$ and a finite rational parameter set I. Finite-annulus coordinate convergence gives, after a subsequence, $F_N \to F$ pointwise on I. A strict witness for $\mathbf{J}_I(F) > \tau$ has positive excess and survives for all sufficiently large N, so

$$\mathbf{1}_{\{\mathbf{J}_I(F)>\tau\}} \le \liminf_N \mathbf{1}_{\{\mathbf{J}_I(F_N)>\tau\}}.$$

Fatou preserves the finite-parameter distribution estimate without changing its threshold. Increasing finite sets exhaust $\mathbb{Q}_+$, and local absolute continuity recovers all positive parameters, exactly as in the final part of Proposition 23.1. Finally the input L^1 norm and weak output quasi-norm acquire the same factor under the dilation r_0, so the constant is independent of C_m. This closes the finite degree induction under the normalization $\mathfrak{L}(\Omega) = 1$; angular homogeneity restores the factor $\mathfrak{L}(\Omega)$. □

Remark 26.29 (Quantitative closure of the finite-radial proof). For clarity, the order of every small-parameter choice in this section is as follows. Once n, d, ρ are fixed, the polynomial sublevel constants are structural. Equation (26.29) first fixes $c_{\mathrm{vdc},\rho} > 0$, then (26.32) fixes $c_{\mathrm{rat},\rho} > 0$. These determine $c_{\mathrm{eq},\rho}$ and $c_{\mathrm{uneq},\rho}$ in (26.40) and (26.41), hence $c_{\mathrm{flat},\rho}$ and $c_{\kappa,\rho}$ in (26.44) and (26.45). Independently, (26.48) fixes $\delta_{T,\rho} > 0$. Only after these exponents are known are $B_{*,\rho}$ and $\kappa_{*,\rho}$ selected in (26.85)–(26.86); the three inequalities (26.87)–(26.89) then fix $c_{P,\rho}, c_{E,\rho} > 0$. Finally the angular splitting parameters are chosen in (26.99) and (26.106). Thus no decay exponent is defined using a parameter that is chosen later, and every geometric series in the strong and weak arguments has a strictly positive exponent for each fixed finite ρ. The resulting constants may deteriorate as $\rho \to \infty$, as they must in view of Proposition 27.3.

26.11 Completion of the proof

We give the completion because it also explains why point jumps of b do not create jumps in the truncation trajectory. Assume first (26.14). For bounded compactly supported f set

$$\mathscr{A}_x(r) := \int_{\mathbb{S}^{n-1}} e^{iP(x,x-r\theta)}\Omega(\theta)f(x-r\theta)\,d\sigma(\theta).$$

For every $0 < a < c < \infty$,

$$\|b(|\cdot|)K_\Omega(\cdot)\mathbf{1}_{\{a<|\cdot|\le c\}}\|_1 \le \|b\|_\infty\|\Omega\|_1\log(c/a). \tag{26.107}$$

Fubini, first on the countable annuli $\{N^{-1} < |z| < N\}$, gives one full-measure set on which

$$\int_a^c |b(r)\mathscr{A}_x(r)|\,\frac{dr}{r} < \infty, \qquad T^{P,b}_{\Omega,a}f(x) - T^{P,b}_{\Omega,c}f(x) = \int_a^c b(r)\mathscr{A}_x(r)\,\frac{dr}{r} \tag{26.108}$$

for every $0 < a < c < \infty$. Thus the core path is locally absolutely continuous. In particular, a jump of the regulated function b at one radius changes neither the Lebesgue integral nor the truncation path: no derivative of b occurs in (26.108). For core data the path is also zero for all sufficiently large truncations at each fixed x.

We next construct all coordinates for general data. In the strong case choose bounded compactly supported $f_m \to f$ in L^p so that

$$\sum_m \|f_{m+1} - f_m\|_p < \infty.$$

For $g_m = f_{m+1} - f_m$, the zero anchor for core data and (15.7), followed by (26.81), give

$$\left\| \sup_{q\in\mathbb{Q}_+} |T^{P,b}_{\Omega,q} g_m| \right\|_p \le \|\mathbf{J}(T^{P,b}_{\Omega,\varepsilon} g_m)\|_p \le C_{n,d,p,\rho}\, \mathfrak{L}(\Omega)\|g_m\|_p. \tag{26.109}$$

Hence the sum of the rational suprema is finite almost everywhere, and $T^{P,b}_{\Omega,q} f_m$ is uniformly Cauchy in $q \in \mathbb{Q}_+$ outside one null set.

In the weak case choose instead bounded compactly supported $f_m \to f$ in L^1 with $\|g_m\|_1 \le 2^{-3m}$. The core weak estimate and the same anchor imply

$$\sum_m \left| \left\{ x : \sup_{q\in\mathbb{Q}_+} |T^{P,b}_{\Omega,q} g_m(x)| > 2^{-m} \right\} \right| \le C_{n,d,\rho}\, \mathfrak{L}(\Omega) \sum_m 2^m \|g_m\|_1 < \infty. \tag{26.110}$$

Borel–Cantelli again gives almost-everywhere uniform Cauchy convergence on $\mathbb{Q}_+$. In either case we define every rational coordinate by this uniform limit. Moreover, if

$$\mathscr{R}_m(q,x) := \sum_{k\ge m} T^{P,b}_{\Omega,q} g_k(x),$$

then $\sup_{q\in\mathbb{Q}_+} |\mathscr{R}_m(q,x)| \to 0$ almost everywhere. For fixed m the core path of f_m vanishes for all sufficiently large q; consequently

$$\lim_{R\to\infty} \sup_{\substack{q\in\mathbb{Q}_+ \\ q\ge R}} |T^{P,b}_{\Omega,q} f(x)| = 0 \tag{26.111}$$

almost everywhere in both completions.

It remains to identify these coordinates with the desired finite-shell increments. By (26.107), for every rational $0 < a < c < \infty$ the core finite-annulus expressions converge to the corresponding expression for f in L^p in the strong case and in L^1 in the weak case. A diagonal subsequence over the rational pairs therefore gives, without changing the preceding uniform rational limit, the identity

$$T^{P,b}_{\Omega,a} f(x) - T^{P,b}_{\Omega,c} f(x) = \int_a^c b(r)\mathscr{A}_x(r)\,\frac{dr}{r}$$

on one full-measure set. Fubini on the same countable annuli shows that the right-hand density is locally integrable for general f as well (Young's inequality applies in the strong case). Define the real coordinates from the base point 1 by these finite-annulus integrals. The resulting path is locally absolutely continuous, agrees with the rational coordinates, and (26.111), by continuity and density of $\mathbb{Q}_+$, yields its zero-at-infinity anchor for all real radii.

For every finite rational parameter set, coordinatewise convergence and (15.29) transfer the strong estimate by Fatou. For the weak estimate, fix its threshold $\tau > 0$ before taking the limit: if the limiting finite path has $\mathbf{J} > \tau$, one of its finite strict witnesses has positive excess and therefore persists along the coordinatewise convergent subsequence. The corresponding level-set indicator is bounded by the liminf of the approximating indicators, so Fatou transfers the distribution estimate at the same τ. Increasing through finite sets exhausting $\mathbb{Q}_+$ and then using local continuity and Lemma 15.6 recovers the full positive-parameter functional. This also settles measurability: all coordinates are measurable, and the completed jump functional is a countable supremum over finite rational witnesses. Two locally continuous paths with the same finite-annulus increments differ by a constant in the parameter, and the zero anchor makes the completion unique.

We have proved (1.13) and (1.14) under (26.14). If $\|b\|_{\mathscr{V}^\rho} = 0$, then $b = 0$ almost everywhere and the theorem is immediate. Otherwise apply the normalized result to $b/\|b\|_{\mathscr{V}^\rho}$; linearity of every coordinate and homogeneity of $\mathbf{J}$ restore exactly one factor $\|b\|_{\mathscr{V}^\rho}$. Finally the pointwise weak-2 jump-to-r-variation inequality from Lemma 15.2 gives every $r > 2$ consequence, while the zero anchor gives the maximal assertion. This completes the proof of Theorem 1.4.

Remark 26.30 (Dependence on the radial exponent). The proof uses the positive exponents $1/\rho$ in both the translation modulus and the first-derivative oscillatory estimate. These exponents tend to zero as $\rho \to \infty$, and no constant uniform in ρ is asserted. Proposition 27.3 shows that a merely bounded radial endpoint is false even for the one-dimensional Hilbert kernel.

26.12 A shellwise quadratic radial-variation refinement for the Hilbert transform

The abstract transfer in Proposition 26.1 stops at $\rho = 2$. In the one-dimensional Hilbert model, Fourier orthogonality allows a sharper shellwise statement.

For bounded $b : (0,\infty) \to \mathbb{C}$, put

$$H^b_\varepsilon f(x) = \int_{|y|>\varepsilon} b(|y|) f(x-y)\,\frac{dy}{y},$$

where, under the finiteness condition below, the integral is absolutely convergent at every x for every L^2 input and agrees with the compatible strong-L^2 limit of finite upper cutoffs, as proved below, and define

$$q_j = V^2(b;(2^j,2^{j+1})), \qquad R_2(b) = \left(\sum_{j\in\mathbb{Z}} q_j^2\right)^{1/2}. \tag{26.112}$$

Concatenating disjoint shell partitions gives $R_2(b) \le V^2(b)$ whenever the latter is finite.

Proposition 26.31 (Shellwise radial V^2 Hilbert estimate)**.** *For every bounded b with $R_2(b) < \infty$ and every $f \in L^2(\mathbb{R})$,*

$$\|\mathbf{J}(H^b_\varepsilon f : \varepsilon > 0)\|_2 \le C\big(\|b\|_\infty + R_2(b)\big)\|f\|_2. \tag{26.113}$$

In particular, the right-hand side may be replaced by $C(\|b\|_\infty + V^2(b))\|f\|_2$.

Proof. Write $B = \|b\|_\infty$ for the essential supremum and $R = R_2(b)$. We use the paired-path norm $\mathscr{N}$ from Lemma 2.1; in particular, $\mathbf{J} \le \mathscr{N}$, and $\mathscr{N}$ is subadditive with constant one. All path norms below are taken pointwise before the spatial norm.

An open-shell decomposition. Put $I_j = (2^j,2^{j+1})$, $r_j = 3\cdot 2^j/2$, and $c_j = b(r_j)$. For any $s,t \in I_j$, $|b(s)-b(t)| \le q_j$. Since I_j has positive measure, it contains a point t with $|b(t)| \le B$; hence

$$|c_j| \le B + q_j \le B + R. \tag{26.114}$$

This pays for possible isolated spikes at the chosen midpoint using the original open-shell variation, rather than assuming a pointwise bound by the essential supremum. Define, on the whole real radial line,

$$e_j(r) = \mathbf{1}_{I_j}(r)\big(b(r)-c_j\big), \qquad e(r) = \sum_{j\in\mathbb{Z}} e_j(r), \qquad a(r) = \sum_{j\in\mathbb{Z}} c_j \mathbf{1}_{(2^j,2^{j+1}]}(r),$$

where each summand is zero outside its indicated positive shell. Each sum has at most one nonzero term at every radius. In particular, e is zero at every dyadic radius and on $(-\infty,0]$, while $b = a+e$ at every positive nondyadic radius. No endpoint limits of b are used.

We record explicitly the global variation bound for the residual. Zero extension of the anchored open-interval error gives

$$V^2(e_j;\mathbb{R}) \le 40 q_j.$$

Indeed, on any finite variation partition having interior shell points, let l,r be the first and last such points and clamp the radial variable to $[l,r] \subset I_j$. The clamped path $b-c_j$ has supremum at most q_j and variation at most q_j. Multiplication by $\mathbf{1}_{[l,r]}$ reproduces e_j at every point of the partition. The product-variation inequality and $V^2(\mathbf{1}_{[l,r]};\mathbb{R}) \le 39$ give the displayed bound. A partition with no interior shell point contributes zero. For disjoint-support paths the quadratic-variation estimate is

$$V^2\left(\sum_j e_j;\mathbb{R}\right)^2 \le 2\sum_j V^2(e_j;\mathbb{R})^2.$$

To see that this also holds for the countable sum, fix a finite partition: only finitely many supports meet its nonzero values, and the finite-family estimate applies. That estimate follows by splitting each increment into its supported components and using $|z-w|^2 \le 2|z|^2+2|w|^2$ when its two endpoints lie in different supports. Taking the supremum over partitions therefore gives, with $K=\sqrt{3200}$,

$$V^2(e;\mathbb{R}) \le KR, \qquad \sup_r |e(r)| \le KR, \qquad \sup_{r>0} |e(r)| + V^2(e;(0,\infty)) \le 2KR. \tag{26.115}$$

The second bound follows by comparing each value with $e(0)=0$. Finite variation makes e measurable; a is a measurable step function, bounded pointwise by $B+R$.

Arbitrary bounded dyadic step coefficients. We next prove, independently of any variation of the coefficient sequence, that for

$$a_d(r) = \sum_{j\in\mathbb{Z}} d_j \mathbf{1}_{(2^j,2^{j+1}]}(r), \qquad |d_j| \le M,$$

one has

$$\|\mathscr{N}(H_\varepsilon^{a_d} f : \varepsilon > 0)\|_2 \le CM\|f\|_2. \tag{26.116}$$

First take $M=1$ and a finite shell set $S \subset \mathbb{Z}$. Write

$$A_{j,u}f(x) = \int_{2^j u<|y|\le 2^{j+1}} f(x-y)\,\frac{dy}{y}, \quad 1 \le u \le 2, \qquad \nu_j(y) = \frac{\mathbf{1}_{\{2^j<|y|\le 2^{j+1}\}}}{y}. \tag{26.117}$$

Thus $A_{j,1}f = \nu_j * f$ and $A_{j,2}f = 0$. Choose the smooth dyadic frequency pieces P_k at frequency $|\xi| \simeq 2^{-k}$, with their exact reproducing dual pieces, and set

$$U_j^{(m)} = d_j A_{j,1} P_{j-m} f, \qquad S_m f = \left(\sum_j V^2(A_{j,u}P_{j-m}f : 1 \le u \le 2)^2\right)^{1/2}.$$

The unweighted localized short-square estimates already proved in Lemma 26.19, specifically (26.72) and the summation following it, give

$$\sum_{m\in\mathbb{Z}} \|S_m f\|_2 \le C\|f\|_2.$$

Here we use only the constant radial amplitude 1 and the Hilbert angular kernel. Scalar multiplication by d_j preserves exact frequency reproduction. The frequency-ordered tail estimate (8.3) therefore applies for each offset m:

$$\left\| \mathscr{N}\left(\sum_{\substack{j\in S\\ j\ge J}} U_j^{(m)} : J \in \mathbb{Z} \right) \right\|_2 \le C \left\| \left(\sum_{j\in S} |U_j^{(m)}|^2 \right)^{1/2} \right\|_2 \le C\|S_m f\|_2.$$

The last pointwise square domination uses $|d_j| \le 1$ and the zero anchor $A_{j,2}=0$. The actual shell reconstruction $A_{j,1}f = \sum_m A_{j,1}P_{j-m}f$ holds simultaneously for all integer j on a common full-measure set: the summed short squares give absolute coordinate convergence there, and the Littlewood–Paley identity identifies the limit in L^2. Applying the triangle inequality for entire trajectories and then Minkowski gives

$$\left\| \mathscr{N}\left(\sum_{\substack{j\in S\\ j\ge J}} d_j \nu_j * f : J \in \mathbb{Z} \right) \right\|_2 \le C\|f\|_2. \tag{26.118}$$

There is no factor depending on $|S|$ or the number of frequency offsets: finite sums use the constant-one triangle inequality for $\mathscr{N}$, and coordinate lower semicontinuity removes the offset cutoff.

For the short part, the same unweighted theorem gives

$$\left\| \left(\sum_{j\in S} V^2(d_j A_{j,u} f : 1 \le u \le 2)^2 \right)^{1/2} \right\|_2 \le \left\| \left(\sum_{j} V^2(A_{j,u} f : 1 \le u \le 2)^2 \right)^{1/2} \right\|_2 \le C\|f\|_2. \quad (26.119)$$

Concatenate the partial shells, using zero at the outer endpoint of each shell. The pointwise long–short inequality (15.10), together with (2.2), bounds the full finite-shell path

$$F_S(\varepsilon, x) = \sum_{j\in S} d_j \int_{\max\{\varepsilon, 2^j\} < |y| \le 2^{j+1}} f(x-y)\,\frac{dy}{y}$$

by a fixed multiple of the short square in (26.119) plus the dyadic path norm in (26.118). This is a measurable majorant with L^2 norm at most $C\|f\|_2$, uniformly in S.

For every original L^2 representative, every $x \in \mathbb{R}$, and every $\varepsilon > 0$, Cauchy–Schwarz gives

$$\int_{|y|>\varepsilon} \frac{|f(x-y)|}{|y|}\,dy \le \left(\frac{2}{\varepsilon}\right)^{1/2} \|f\|_2 < \infty. \quad (26.120)$$

Consequently the finite-shell paths with $S = [-N, N] \cap \mathbb{Z}$ converge to the literal integral $H_\varepsilon^{a_d} f(x)$ at *every* x and *every* positive radius. For each finite paired witness, its value converges; hence the limiting full path norm is bounded by the liminf of the finite-shell majorants. Fatou proves (26.116) for $M = 1$. Scaling gives the result for $M > 0$, and $M = 0$ is immediate. In particular, arbitrary jumps between the coefficients are allowed; no global-variation theorem has been applied to a_d.

Identification with the sharp truncations. Apply (26.116) to $d_j = c_j$ with $M = B + R$. For the residual, Theorem 1.4, in its one-dimensional zero-phase Hilbert specialization at $\rho = p = 2$, and (2.2) give

$$\|\mathscr{N}(H_\varepsilon^e f : \varepsilon > 0)\|_2 \le C\big(\sup_{r>0} |e(r)| + V^2(e; (0,\infty))\big)\|f\|_2 \le CR\|f\|_2.$$

The specialization agrees with the literal integral: by (26.120) its tails are absolutely convergent, its finite-annulus increments are the stated integrals, and its zero-at-infinity anchor identifies it with the completed path.

The set $\{y : |y| = 2^j \text{ for some } j \in \mathbb{Z}\}$ is countable and null. Since $b = a + e$ away from these radii and a, e are measurable and bounded, the original integrand is absolutely integrable and

$$H_\varepsilon^b f(x) = H_\varepsilon^a f(x) + H_\varepsilon^e f(x) \qquad (x \in \mathbb{R},\ \varepsilon > 0). \quad (26.121)$$

This is an identity of actual integrals, not only of selected L^2 representatives. For each of a, e, the integral is measurable in x and continuous in the positive cutoff at every x, by dominated convergence using the absolute tail at half the cutoff. Thus its full path norm equals its countable-dense-radius version and is measurable. The same facts hold for b by (26.121); all positive radii and the jump-amplitude supremum remain inside the spatial norm. Finally, subadditivity of $\mathscr{N}$, Minkowski, and $\mathbf{J} \le \mathscr{N}$ yield

$$\|\mathbf{J}(H_\varepsilon^b f : \varepsilon > 0)\|_2 \le \|\mathscr{N}(H_\varepsilon^a f : \varepsilon > 0)\|_2 + \|\mathscr{N}(H_\varepsilon^e f : \varepsilon > 0)\|_2 \le C(B+R)\|f\|_2,$$

as required. The argument also bounds $\|\mathscr{N}(H_\varepsilon^b f)\|_2$. Equation (26.120), applied to $a + e$, gives $H_T^b f(x) \to 0$ as $T \to \infty$ at every x. The zero anchor implies $|H_T^b f(x)| \le \mathscr{N}(H_\varepsilon^b f(x) : \varepsilon > 0)$, so dominated convergence gives $H_T^b f \to 0$ strongly in L^2. Hence, for each fixed $\varepsilon > 0$ and $T \ge \varepsilon$,

$$\int_{\varepsilon<|y|\le T} b(|y|) f(x-y)\,\frac{dy}{y} = H_\varepsilon^b f(x) - H_T^b f(x) \longrightarrow H_\varepsilon^b f \quad \text{strongly in } L^2.$$

Thus the literal coordinates also retain the compatible strong-L^2 interpretation. □

Remark 26.32. The condition $R_2(b) < \infty$ is strictly weaker than global finite $V^2(b)$. A bounded function may be constant on each half-open dyadic shell, so that $R_2(b) = 0$, while its jumps from shell to shell have infinite global 2-variation. Thus even this basic model identifies a radial class strictly larger than the global V^2 class of Theorem 1.4.

27 Obstructions and sharpness

This section records four complementary negative results. They distinguish the pointwise critical-jump functional from the weaker mixed quasi-seminorm, show why the soft radial transfer cannot reach exponent two, prove that the finite-radial theorem has no $\rho = \infty$ endpoint, and show that the output critical jump cannot be strengthened to true 2-variation.

27.1 The mixed-norm exchange is false

Proposition 27.1 (Mixed-norm counterexamples for finite jump paths)**.** *For every $1 \le p < \infty$ and every integer $M \ge 2$ there exist a finite measure space X_M and a two-point scalar path F such that*

$$\sup_{\lambda>0} \|\lambda N_\lambda(F)^{1/2}\|_{L^p(X_M)} \le C_p, \tag{27.1}$$

whereas

$$\left\| \sup_{\lambda>0} \lambda N_\lambda(F)^{1/2} \right\|_{L^p(X_M)} \simeq (\log M)^{1/p}. \tag{27.2}$$

In addition, for every $M \ge 2$ there is a scalar path G on one common finite ordered parameter set such that

$$\sup_{\lambda>0} \|\lambda N_\lambda(G)^{1/2}\|_{L^{1,\infty}(X_M)} \le \sqrt{2}, \tag{27.3}$$

but

$$\left\| \sup_{\lambda>0} \lambda N_\lambda(G)^{1/2} \right\|_{L^{1,\infty}(X_M)} = \sqrt{2}M. \tag{27.4}$$

Thus neither the strong nor the weak spatial norm can in general be interchanged with the supremum in the jump amplitude.

Proof. For the strong example, take $X_M = \{1,\dots,M\}$ with counting measure and set $a_k = k^{-1/p}$. On the ordered parameter set $\{0,1\}$ define $F_0(k) = 0$ and $F_1(k) = a_k$. Then $N_\lambda(F(k)) = 1$ for $0 < \lambda < a_k$ and vanishes for $\lambda \ge a_k$. Thus the pointwise supremum equals a_k, and its pth power sums to $\sum_{k\le M} k^{-1} \simeq \log M$. For fixed λ,

$$\|\lambda N_\lambda(F)^{1/2}\|_p = \lambda \#\{k : a_k > \lambda\}^{1/p} \le C_p,$$

which proves (27.1)– (27.2).

For the weak endpoint use the same counting space and put

$$I_M = \{0,1,\dots,2\cdot 4^M\}, \qquad a_k = 2^{-k}, \qquad L_k = 4^k.$$

For $j \in I_M$ define

$$G_j(k) = a_k \mathbf{1}_{\{j<2L_k\}} \mathbf{1}_{\{j \text{ odd}\}}.$$

Thus, at the point k, the path alternates $0, a_k, 0, a_k, \dots, 0$ through the first $2L_k + 1$ coordinates and is zero thereafter. For $0 < \lambda < a_k$, its $2L_k$ consecutive changes are admissible paired strict jumps (successive pairs may share an endpoint). Conversely, every strict jump interval must contain one of these changes, and ordered jump intervals cannot use the same change twice. Hence, exactly,

$$N_\lambda(G(k)) = \begin{cases} 2L_k, & 0 < \lambda < a_k, \\ 0, & \lambda \ge a_k. \end{cases}$$

It follows that

$$\mathbf{J}(G)(k) = a_k \sqrt{2L_k} = \sqrt{2}$$

for every $1 \le k \le M$, proving (27.4).

Fix $\lambda > 0$. If the path is nonzero at no spatial point there is nothing to prove. Otherwise let K be the largest integer in $\{1,\dots,M\}$ for which $\lambda < 2^{-K}$. The nonzero values of $\lambda N_\lambda(G)^{1/2}$ have decreasing rearrangement

$$A, \frac{A}{2}, \dots, \frac{A}{2^{K-1}}, \qquad A = \sqrt{2}\,\lambda 2^K < \sqrt{2}.$$

Consequently

$$\|\lambda N_\lambda(G)^{1/2}\|_{\ell^{1,\infty}} = \sup_{1\le m\le K} m\frac{A}{2^{m-1}} \le A < \sqrt{2},$$

uniformly in λ and M. This proves (27.3). □

The first construction shows that an outer supremum in the jump amplitude cannot be upgraded to the pointwise functional by an abstract exchange of strong spatial norms. The second construction proves the same strict failure in $L^{1,\infty}$ with a genuine jump trajectory on a common finite parameter set. In particular, the pointwise-inside weak endpoint in Theorems 1.2 and 1.4 is genuinely stronger than the conventional mixed weak quasi-seminorm.

27.2 Failure of an abstract endpoint radial-variation transfer

Proposition 27.2 (Abstract endpoint transfer obstruction)**.** *For every $N \ge 2$ there exist a finite scalar path F, a scalar coefficient path β, and the weighted-tail path*

$$G_i = \sum_{k=i}^{2N-1} \beta_k (F_k - F_{k+1}), \qquad G_{2N} = 0, \tag{27.5}$$

such that

$$\mathbf{J}(F) \le C, \qquad \|\beta\|_\infty + V^2(\beta) \le C, \tag{27.6}$$

but

$$\mathbf{J}(G) \ge c\sqrt{\log N}. \tag{27.7}$$

Hence no universal scalar-path estimate of the form

$$\mathbf{J}(G) \le C(\|\beta\|_\infty + V^2(\beta))\mathbf{J}(F)$$

can hold for the operation (27.5)*. Moreover, with absolute changes in the constants, this discrete example embeds into the same weighted-tail structural class that appears in Proposition 26.1, now at the formal endpoint $\rho = 2$: there are a locally absolutely continuous path $\widetilde{F} : (0,\infty) \to \mathbb{C}$ with $\widetilde{F}_s \to 0$ as $s \to \infty$ and a bounded right-continuous $\widetilde{\beta}$ of finite 2-variation such that*

$$\widetilde{G}_s = -\int_s^\infty \widetilde{\beta}(u)\, d\widetilde{F}_u$$

has the analogues of (27.6) *and* (27.7)*. Thus local absolute continuity of the input path does not restore an abstract endpoint transfer at $\rho = 2$.*

Proof. Fix $N \ge 2$ and put

$$H_N = \sum_{j=1}^N \frac{1}{j}, \qquad a_j = j^{-1/2}, \qquad \delta_j = \frac{a_j}{\sqrt{H_N}} \quad (1 \le j \le N).$$

Define the positive finite path

$$F_{2j} = 0 \quad (0 \le j \le N), \qquad F_{2j-1} = a_j \quad (1 \le j \le N). \tag{27.8}$$

Take $\beta_0 = 0$ and prescribe

$$\beta_{2j-1} - \beta_{2j-2} = \delta_j, \qquad \beta_{2j} - \beta_{2j-1} = -\delta_j \quad (1 \le j \le N). \tag{27.9}$$

Thus $\beta_{2j}=0$ and $\beta_{2j-1}=\delta_j$; in particular the coefficient returns to zero after each positive pulse. We have

$$\sum_{j=1}^{N}\delta_j^2=1,\qquad 0<\delta_j\le 1.$$

We construct the continuous paths at the same time. Let $r(v)=\min\{1,\max\{0,v\}\}$ and set, for $s\in\mathbb{R}$,

$$T_j(s)=r(s-(2j-1))-r(s-2j),\qquad \widetilde{F}(s)=\sum_{j=1}^{N}a_jT_j(s),$$

$$\widetilde{\beta}(s)=\sum_{j=1}^{N}\delta_j\mathbf{1}_{[2j,2j+1)}(s).$$

Here T_j is the unit positive triangular tent, rising on $(2j-1,2j)$ and falling on $(2j,2j+1)$, with $T_j(2j)=1$ and $T_j(s)=0$ outside $(2j-1,2j+1)$. The sets where the tents are nonzero are pairwise disjoint, as are the coefficient intervals. The finite sum $\widetilde{F}$ is globally Lipschitz, hence locally absolutely continuous, and vanishes for $s\le 1$ and for $s\ge 2N+1$. The half-open convention makes $\widetilde{\beta}$ right-continuous, including at the last endpoint, and $\|\widetilde{\beta}\|_\infty\le 1$. At every grid point,

$$\widetilde{F}(i+1)=F_i,\qquad \widetilde{\beta}(i+1)=\beta_i\qquad (0\le i\le 2N).$$

We first bound the genuine paired jump functional of the input. Fix any $m\ge 1$ and any ordered family $s_1<t_1\le s_2<t_2\le\cdots\le s_m<t_m$ in $\mathbb{R}$. Let U be the set of tent indices that are nonzero at at least one of these $2m$ endpoints. Since at most one tent is nonzero at any point, $|U|\le 2m$. For a nondecreasing function with values in $[0,1]$, the sum of its increments over these ordered intervals is at most one: the intervening gaps have nonnegative increments, so the sum is bounded by the increment from s_1 to t_m. Applying this to the two ramps in T_j, and then using the triangle inequality, gives

$$\sum_{\ell=1}^{m}|a_jT_j(t_\ell)-a_jT_j(s_\ell)|\le 2a_j.$$

Only indices in U can contribute to the endpoint differences. Moreover, any K distinct heights have sum at most $\sum_{j=1}^{K}j^{-1/2}\le\int_0^K x^{-1/2}\,dx=2\sqrt{K}$, with the empty sum interpreted as zero. Consequently

$$\sum_{\ell=1}^{m}|\widetilde{F}(t_\ell)-\widetilde{F}(s_\ell)|\le 2\sum_{j\in U}a_j\le 4\sqrt{|U|}\le 8\sqrt{m}.$$

If all m pairs are strict λ-jumps, their left-hand side exceeds $m\lambda$, so $\lambda\sqrt{m}\le 8$. Taking the supremum over all finite paired witnesses and all $\lambda>0$ proves $\mathbf{J}(\widetilde{F})\le 8$. This argument allows gaps and shared endpoints; it does not merely estimate the consecutive changes of the path. Restriction to positive parameters and then to the grid therefore gives $\mathbf{J}(F)\le 8$ as well.

For the coefficient, write $p_j(s)=\delta_j\mathbf{1}_{[2j,2j+1)}(s)$ and take an arbitrary variation partition $x_0<\cdots<x_L$ in $\mathbb{R}$. For each fixed j, deleting consecutive repeated values of $p_j(x_\ell)$ leaves a subsequence of $(0,\delta_j,0)$ and does not change its squared increment sum. The elementary bound $|z-w|^2\le 4(|z|^2+|w|^2)$, with each retained state counted at most twice, thus gives the convenient non-optimal estimate

$$\sum_{\ell=1}^{L}|p_j(x_\ell)-p_j(x_{\ell-1})|^2\le 8\delta_j^2.$$

At any pair of endpoints at most two pulses have a nonzero increment. Cauchy–Schwarz on those at most two terms yields

$$\begin{aligned}\sum_{\ell=1}^{L}|\widetilde{\beta}(x_\ell)-\widetilde{\beta}(x_{\ell-1})|^2&\le 2\sum_{j=1}^{N}\sum_{\ell=1}^{L}|p_j(x_\ell)-p_j(x_{\ell-1})|^2\\&\le 16\sum_{j=1}^{N}\delta_j^2=16.\end{aligned}$$

Taking the supremum over all partitions proves the true variation bound $V^2(\widetilde{\beta}) \le 4$, not just a bound on the sum of squares of its consecutive jumps. Hence

$$\|\widetilde{\beta}\|_\infty + V^2(\widetilde{\beta}) \le 5, \qquad \|\beta\|_\infty + V^2(\beta) \le 5,$$

where the second estimate follows by sampling. These estimates remain valid on $(0,\infty)$.

Define G by (27.5). Its even-edge terms vanish, and every odd-edge term at $k = 2j-1$ equals $\delta_j a_j$. Thus

$$G_0 = \sum_{j=1}^{N} \delta_j a_j = \sqrt{H_N}, \qquad G_{2N} = 0.$$

For every $0 < \lambda < \sqrt{H_N}$ these two endpoints form one strict jump. Taking the supremum in λ, and using $H_N \ge \log(N+1) \ge \log N$, gives

$$\mathbf{J}(G) \ge \sqrt{H_N} \ge \sqrt{\log N}.$$

Finally define the continuous output by the actual derivative integral

$$\widetilde{G}(s) = -\int_{(s,\infty)} \widetilde{\beta}(u)\widetilde{F}'(u)\,du = -\int_s^\infty \widetilde{\beta}(u)\,d\widetilde{F}_u.$$

The second equality uses the absolute continuity of $\widetilde{F}$. On each falling open interval $(2j, 2j+1)$ its derivative is $-a_j$; off the falling intervals the coefficient is zero, apart from the finitely many irrelevant endpoints. Hence, almost everywhere,

$$\widetilde{\beta}(u)\widetilde{F}'(u) = -\sum_{j=1}^{N} \delta_j a_j \mathbf{1}_{[2j,2j+1)}(u).$$

In particular this product is globally integrable, so the displayed tail is well defined and is zero for $s \ge 2N+1$. On every half-open unit interval $[k+1, k+2)$ the coefficient equals β_k, while absolute continuity gives $\int_{k+1}^{k+2} \widetilde{F}'(u)\,du = F_{k+1} - F_k$. Splitting the tail into these unit intervals therefore proves, for every $0 \le i \le 2N$, the exact identity

$$\begin{aligned} \widetilde{G}(i+1) &= -\int_{i+1}^{\infty} \widetilde{\beta}(s)\,d\widetilde{F}_s \\ &= \sum_{k=i}^{2N-1} \beta_k (F_k - F_{k+1}) = G_i. \end{aligned} \tag{27.10}$$

In particular $\widetilde{G}(1) = \sqrt{H_N}$ and $\widetilde{G}(2N+1) = 0$, so the same single-pair argument yields $\mathbf{J}(\widetilde{G}) \ge \sqrt{\log N}$ on $(0,\infty)$. Together with the input bounds 8 and 5, these are all the asserted discrete and continuous estimates, with the zero far anchor. A universal transfer constant C in either class would imply $\sqrt{\log N} \le 40C$ for every $N \ge 2$, which is impossible. □

This obstruction is the sequence-space failure $\ell^{2,\infty} \cdot \ell^2 \not\subset \ell^1$. Proposition 26.31 and the general argument of Section 26 escape it by using Fourier and packet orthogonality absent from an arbitrary scalar path.

27.3 Failure at the bounded-radial endpoint

For the limiting seminorm put $V^\infty(b) = \sup_{s<t} |b(t) - b(s)|$.

Proposition 27.3 (No uniform merely bounded radial theory)**.** *There is no uniform L^2 bound for the one-dimensional radial Hilbert operator in terms of $\|b\|_\infty + V^\infty(b)$ alone. Consequently, Theorem 1.4 cannot be extended to $\rho = \infty$.*

Proof. For $N \geq 1$ put

$$R_N = (N+1)\pi, \qquad b_N(r) = \mathbf{1}_{[\pi,R_N]}(r)\sin r, \qquad f_N(x) = \mathbf{1}_{[-2R_N,2R_N]}(x)e^{ix}.$$

The amplitude b_N is continuous, since its sine factor vanishes at both endpoints of its support. Moreover, $\|b_N\|_\infty \leq 1$ and $V^\infty(b_N) \leq 2$, while $b_N(|y|)/y$ is integrable. Set

$$T_N f(x) = \int_{\mathbb{R}} \frac{b_N(|y|)}{y} f(x-y)\,dy, \qquad A_N = \int_\pi^{R_N} \frac{\sin^2 r}{r}\,dr.$$

For each $k = 0,\dots,N-1$,

$$\int_{(k+1)\pi}^{(k+2)\pi} \frac{\sin^2 r}{r}\,dr \geq \frac{1}{(k+2)\pi}\int_{(k+1)\pi}^{(k+2)\pi} \sin^2 r\,dr = \frac{1}{2(k+2)}.$$

Consequently $A_N \geq \sum_{k=0}^{N-1}[2(k+2)]^{-1} \to \infty$. If $|x| \leq R_N$ and $b_N(|y|) \neq 0$, then $|x-y| \leq 2R_N$. Splitting the absolutely convergent integral into positive and negative y therefore gives the exact identity

$$T_N f_N(x) = e^{ix}\int_\pi^{R_N} \frac{\sin r}{r}(e^{-ir} - e^{ir})\,dr = -2ie^{ix}A_N, \qquad |x| \leq R_N.$$

Thus

$$\|f_N\|_2 = \sqrt{4R_N}, \qquad \|T_N f_N\|_2 \geq 2A_N\sqrt{2R_N}, \qquad \frac{\|T_N f_N\|_2}{\|f_N\|_2} \geq \sqrt{2}A_N \longrightarrow \infty.$$

This contradicts any uniform bound with radial factor $\|b_N\|_\infty + V^\infty(b_N) \leq 3$.

Finally, take $n = 1$, $P = 0$, and $\Omega(1) = 1$, $\Omega(-1) = -1$, so that $K_\Omega(y) = 1/y$. Write $T_\varepsilon^{b_N} = T_{\Omega,\varepsilon}^{0,b_N}$. Since $1 < \pi$ and b_N vanishes beyond R_N, the original sharp truncations satisfy

$$T_1^{b_N} f_N = T_N f_N, \qquad T_{R_N+1}^{b_N} f_N = 0.$$

At every x, these two parameters witness a strict λ-jump whenever $0 < \lambda < |T_N f_N(x)|$. Taking the supremum over such λ yields

$$|T_N f_N(x)| \leq \mathbf{J}(T_\varepsilon^{b_N} f_N(x) : \varepsilon > 0).$$

Hence the same diverging L^2 norm ratio rules out the $\rho = \infty$ critical-jump extension of Theorem 1.4, already at zero phase with this fixed angular datum. □

27.4 Failure of quadratic output variation for sharp Hilbert truncations

Let

$$H_\varepsilon f(x) = \int_{|y|>\varepsilon} f(x-y)\,\frac{dy}{y}, \qquad \widehat{P_s f}(\xi) = e^{-s|\xi|}\widehat{f}(\xi).$$

For $u,t > 0$ define

$$\Phi(u) = \int_u^\infty \frac{\sin v}{v}\,dv, \qquad K(t) = \frac{1}{\pi}\operatorname{sech}^2(\log t) = \frac{4t^2}{\pi(1+t^2)^2}. \tag{27.11}$$

Lemma 27.4 (Mellin identity)**.** *For every $s > 0$,*

$$e^{-s} = \int_0^\infty K(t)\Phi(s/t)\,\frac{dt}{t}, \qquad \int_0^\infty K(t)\,\frac{dt}{t} = \frac{2}{\pi}. \tag{27.12}$$

Proof. Put $\rho(t) = K(t)/t = 4t/[\pi(1+t^2)^2]$. All sine-tail integrals in this proof are improper limits, not totalized Lebesgue integrals. Integration by parts gives, for $u > 0$,

$$\Phi(u) = \frac{\cos u}{u} - \int_u^\infty \frac{\cos v}{v^2}\,dv, \qquad |\Phi(u)| \leq \frac{2}{u}, \qquad \Phi'(u) = -\frac{\sin u}{u}.$$

The remainder integral is absolutely convergent; the displayed formula also proves the existence of the original improper sine tail. The primitive $-2/[\pi(1+t^2)]$ gives

$$\int_a^\infty \rho(t)\,dt = \frac{2}{\pi(1+a^2)} \quad (a \geq 0).$$

In particular the asserted mass is $2/\pi$. Moreover $M := \int_0^\infty t\rho(t)\,dt < \infty$, since $t^2/(1+t^2)^2 \leq (1+t^2)^{-1}$.

Let $I(s) = \int_0^\infty \rho(t)\Phi(s/t)\,dt$. The bound $|\rho(t)\Phi(s/t)| \leq (2/s)t\rho(t)$ proves absolute convergence and $|I(s)| \leq 2M/s$, hence $I(s) \to 0$ as $s \to \infty$. For s in a neighborhood of any $s_0 > 0$, say $s > s_0/2$,

$$|\partial_s\{\rho(t)\Phi(s/t)\}| = \frac{\rho(t)}{s}|\sin(s/t)| \leq \frac{2}{s_0}\rho(t).$$

Differentiation under the integral is therefore justified by an integrable local majorant. The substitution $t = 1/y$ preserves $\rho(t)\,dt$, and gives

$$I'(s) = -\frac{1}{s}\int_0^\infty \rho(y)\sin(sy)\,dy.$$

For completeness, the required Cauchy cosine integral follows from Fourier inversion, without a contour-integral assertion. In cyclic frequency normalization, elementary half-line exponential integration gives

$$\mathscr{F}^{-1}(e^{-a|\xi|})(x) = \frac{1}{a-2\pi ix} + \frac{1}{a+2\pi ix} = \frac{2a}{a^2+(2\pi x)^2} \qquad (a > 0).$$

Both sides are integrable. Fourier inversion at $a = 2\pi$, followed by taking real parts and using evenness, yields

$$\int_0^\infty \frac{\cos(sy)}{1+y^2}\,dy = \frac{\pi}{2}e^{-s} \qquad (s > 0).$$

Now integrate the derivative of $\sin(sy)/(1+y^2)$ on $(0,\infty)$. Both derivative terms are absolutely integrable, and the primitive vanishes at both endpoints, so

$$0 = s\int_0^\infty \frac{\cos(sy)}{1+y^2}\,dy - 2\int_0^\infty \frac{y\sin(sy)}{(1+y^2)^2}\,dy.$$

Thus $\int_0^\infty \rho(y)\sin(sy)\,dy = se^{-s}$ and $I'(s) = -e^{-s}$. The difference $I(s) - e^{-s}$ is constant on $(0,\infty)$, and its limit at infinity is zero. Therefore $I(s) = e^{-s}$, as required. □

The following pointwise representation will be justified by an absolutely convergent physical-space Fubini calculation in the proof below. In angular frequency normalization, for Schwartz functions with Fourier support compactly contained in $(0,\infty)$, it reads

$$P_s f(x) = \frac{i}{2}\int_0^\infty K(t)H_{s/t}f(x)\,\frac{dt}{t} \qquad (s > 0,\ x \in \mathbb{R}). \tag{27.13}$$

In cyclic frequency normalization the left-hand side is $P_{2\pi s}f(x)$; this fixed positive reparametrization does not alter the variation comparison. The integral is absolutely convergent for every x, not merely a representative of an L^2 Bochner integral.

Proposition 27.5 (Poisson variation dominated by Hilbert variation on the core)**.** *If $f \in \mathscr{S}(\mathbb{R})$ and $\operatorname{supp}\widehat{f} \Subset (0,\infty)$, then*

$$V^2(P_s f : s > 0) \leq \frac{1}{\pi}V^2(H_\varepsilon f : \varepsilon > 0) \tag{27.14}$$

pointwise.

Proof. We establish an everywhere physical-space identity, so no choice of L^2 representatives or countable exceptional set is needed. Write $\rho(t) = K(t)/t$ and define the conjugate-Poisson convolution

$$Q_a f(x) = \frac{i}{\pi} \int_{\mathbb{R}} \frac{y}{a^2+y^2} f(x-y)\, dy \qquad (a>0).$$

For every $f \in L^1(\mathbb{R})$ and every $x \in \mathbb{R}$, Tonelli gives

$$\begin{aligned} &\int_0^\infty \rho(t) \int_{|y|>a/t} \frac{|f(x-y)|}{|y|}\, dy\, dt \\ &\quad = \frac{2}{\pi} \int_{\mathbb{R}} \frac{|y|}{a^2+y^2} |f(x-y)|\, dy \le \frac{\|f\|_1}{\pi a} < \infty. \end{aligned}$$

Indeed, the inner t-integral is the exact tail mass $\int_{a/|y|}^\infty \rho(t)\, dt = 2y^2/[\pi(a^2+y^2)]$ for $y \neq 0$; the zero point contributes nothing. Ordinary absolutely convergent Fubini therefore proves, for every a, x,

$$Q_a f(x) = \frac{i}{2} \int_0^\infty \rho(t) H_{a/t} f(x)\, dt.$$

Elementary inverse Fourier transformation of the signed exponential multiplier gives, in cyclic normalization,

$$\mathscr{F}^{-1}\left(e^{-2\pi a\xi} \mathbf{1}_{\{\xi>0\}} - e^{2\pi a\xi} \mathbf{1}_{\{\xi\le 0\}} \right)(y) = \frac{iy}{\pi(a^2+y^2)}.$$

The multiplier is integrable. Fubini for its inverse Fourier integral and the L^1 function f proves the convolution–multiplier identity at every x. Since $\widehat{f}$ vanishes on nonpositive frequencies, this signed multiplier times $\widehat{f}$ is $e^{-2\pi a|\xi|}\widehat{f}(\xi)$. Consequently $Q_a f = P_{2\pi a} f$ in cyclic normalization, equivalently $Q_a f = P_a f$ in angular normalization. A fixed positive rescaling of the Poisson parameter does not increase its variation (and in fact preserves it).

Fix x and a finite increasing partition $0 < a_0 < \cdots < a_J$. The everywhere identity above and the triangle inequality for the Bochner integral with values in ℓ^2_J give

$$\begin{aligned} \left(\sum_{j=1}^J |Q_{a_j} f(x) - Q_{a_{j-1}} f(x)|^2 \right)^{1/2} &\le \frac{1}{2} \int_0^\infty \rho(t) \left(\sum_{j=1}^J |H_{a_j/t} f(x) - H_{a_{j-1}/t} f(x)|^2 \right)^{1/2} dt \\ &\le \frac{1}{2} \left(\int_0^\infty \rho(t)\, dt \right) V^2(H_\varepsilon f(x) : \varepsilon > 0) \\ &= \frac{1}{\pi} V^2(H_\varepsilon f(x) : \varepsilon > 0). \end{aligned}$$

Taking the supremum over all finite partitions proves the asserted inequality at this arbitrary x. The normalization is that of the kernel $1/y$, not $1/(\pi y)$. □

We record the finite-dimensional form of the restriction principle of de Leeuw [19] used below, including its elementary proof.

Lemma 27.6 (Finite-dimensional de Leeuw restriction)**.** *Let E be a finite-dimensional Banach space and let $M : \mathbb{R} \to E$ be continuous. If*

$$\left\| \mathscr{F}^{-1}(M\widehat{f}) \right\|_{L^2(\mathbb{R};E)} \le C \|f\|_{L^2(\mathbb{R})}$$

for Schwartz f, then for every trigonometric polynomial g,

$$\left\| \sum_{n\in\mathbb{Z}} M(n)\widehat{g}(n) e^{inx} \right\|_{L^2(\mathbb{T};E)} \le C \|g\|_{L^2(\mathbb{T})}. \tag{27.15}$$

Proof. We give the restriction argument with an auxiliary Haar phase, so that periodic averaging is exact rather than asymptotic. Write

$$\mathscr{F}f(\xi)=\int_{\mathbb{R}} e^{-2\pi ix\xi}f(x)\,dx, \qquad N(\xi)=M(2\pi\xi).$$

The angular-frequency inverse has its factor $(2\pi)^{-1}$, and the change of variables $\xi=2\pi\zeta$ gives the exact identity

$$\mathscr{F}^{-1}(M\widehat{f})=\mathscr{F}^{-1}(N\mathscr{F}f).$$

Thus the assumed Euclidean bound has the same constant C in the cyclic-frequency convention. Let ν be Haar probability measure on $\mathbb{T}=\mathbb{R}/(2\pi\mathbb{Z})$, and write

$$g(x)=\sum_{n\in\Lambda} c_n e^{inx}, \qquad G(x)=\sum_{n\in\Lambda} c_n M(n)e^{inx}, \qquad \lambda_n=\frac{n}{2\pi},$$

where $\Lambda\subset\mathbb{Z}$ is finite. Orthogonality of the characters gives $c_n=\widehat{g}(n)$ on Λ and zero Fourier coefficients outside Λ.

Choose a linear isomorphism $L:E\to\mathbb{C}^m$, with the Euclidean norm on $\mathbb{C}^m$. Coordinatewise scalar Plancherel, and commutation of fixed continuous linear maps with the Fourier integral, give

$$\|\mathscr{F}^{-1}H\|_{L^2(\mathbb{R};E)}\le K_E\|H\|_{L^2(\mathbb{R};E)}, \qquad K_E=\|L^{-1}\|\,\|L\|, \quad H\in L^1(\mathbb{R};E)\cap L^2(\mathbb{R};E).$$

No Hilbert-space norm is imposed on E; this comparison will be used only for an error tending to zero.

Fix $\varepsilon>0$. Continuity at the finitely many frequencies λ_n gives $r>0$ such that

$$\|N(\lambda_n+\zeta)-N(\lambda_n)\|_E\le\varepsilon \quad (n\in\Lambda,\ |\zeta|<r).$$

Choose a nonzero compactly supported scalar Schwartz function ϕ that vanishes for $|\zeta|\ge r$, and put $\eta=\mathscr{F}^{-1}\phi$. For each $n\in\Lambda$, set

$$h_n(\zeta)=\phi(\zeta)[N(\lambda_n+\zeta)-N(\lambda_n)], \qquad u_n=\mathscr{F}^{-1}h_n.$$

These frequency errors are continuous and compactly supported, hence belong to $L^1\cap L^2$, and

$$\|u_n\|_{L^2(\mathbb{R};E)}\le K_E\varepsilon\|\phi\|_2.$$

For $a\in\mathbb{T}$, use the Schwartz test function

$$f_a(x)=\eta(x)g(a+x), \qquad \mathscr{F}f_a(\xi)=\sum_{n\in\Lambda} c_n e^{ina}\phi(\xi-\lambda_n).$$

In particular every tested Fourier transform is compactly supported; no growth assumption on M is needed. Frequency translation and linearity of the actual integrals give

$$\mathscr{F}^{-1}(N\mathscr{F}f_a)(x)=\eta(x)G(a+x)+R(a,x), \qquad R(a,x)=\sum_{n\in\Lambda} c_n e^{in(a+x)}u_n(x).$$

The functions in this identity are jointly continuous. Since $\nu(\mathbb{T})=1$ and each character is unimodular, the finite-sum triangle inequality yields

$$\|R\|_{L^2(\nu\otimes dx;E)}\le K_E\varepsilon\|\phi\|_2\sum_{n\in\Lambda}|c_n|.$$

For every continuous E-valued periodic function A, Tonelli and Haar translation invariance give the exact factorization

$$\|\eta(x)A(a+x)\|_{L^2(\nu\otimes dx;E)}=\|\eta\|_2\|A\|_{L^2(\nu;E)}.$$

Apply the assumed Euclidean inequality to each f_a and integrate its square with respect to $\nu(a)$. The preceding identity, applied to g and G, and the error bound imply

$$\|\eta\|_2\|G\|_{L^2(\nu;E)} \le C\|\eta\|_2\|g\|_{L^2(\nu)} + K_E\varepsilon\|\phi\|_2 \sum_{n\in\Lambda}|c_n|.$$

Scalar Plancherel gives $\|\eta\|_2 = \|\phi\|_2 > 0$. Cancel this common factor and let $\varepsilon \downarrow 0$. The result is $\|G\|_{L^2(\nu;E)} \le C\|g\|_{L^2(\nu)}$. Passing from ν to ordinary torus volume multiplies both norms by $(2\pi)^{1/2}$, so the constant remains exactly C. This proves (27.15). □

Theorem 27.7 (Failure of true V^2 for sharp Hilbert truncations)**.** *There is no constant C such that*

$$\|V^2(H_\varepsilon f : \varepsilon > 0)\|_2 \le C\|f\|_2 \tag{27.16}$$

for every $f \in L^2(\mathbb{R})$.

Proof. Assume (27.16), with constant $C_H \ge 0$. We first give an elementary finite Gaussian obstruction. Let Z_0 be a standard real Gaussian and $Q_0 = Z_0^2$. On the product of two independent copies of the preceding probability space set

$$Z_{n+1} = \frac{Z_n' + Z_n''}{\sqrt{2}}, \qquad Q_{n+1} = \max\left\{Z_{n+1}^2, \frac{Q_n' + Q_n''}{2}\right\}.$$

Each Z_n is again standard Gaussian. Fix $a \ge 0$, and recursively put

$$U_0 = \mathbf{1}_{\{|Z_0|\le a\}}, \qquad U_{n+1} = \mathbf{1}_{\{|Z_{n+1}|\le a\}}\frac{U_n' + U_n''}{2}.$$

Induction gives $0 \le U_n \le 1$, $U_n = 0$ when $|Z_n| > a$, and $Q_n \ge a^2(1-U_n)$. If $|Z_{n+1}| \le a$ and $U_n' > 0$, then $|Z_n''| \le (\sqrt{2}+1)a \le 3a$. The symmetric assertion holds with the children exchanged. Consequently, writing $q_a = \mathbb{P}\{|Z_0| \le 3a\} < 1$, independence gives

$$\mathbb{E}U_{n+1} \le q_a\,\mathbb{E}U_n, \qquad \mathbb{E}Q_n \ge a^2(1-q_a^n).$$

The strict inequality $q_a < 1$ follows directly from the strictly positive Gaussian density on the complement of the bounded interval. All these random variables are integrable (use the recursion and Gaussian second moments). Taking first a and then n proves

$$\sup_n \mathbb{E}Q_n = \infty, \qquad \mathbb{E}Z_n^2 = 1. \tag{27.17}$$

Here is the literal path statistic represented by Q_n. For $d = 2^n$ independent standard Gaussians $X_0,\dots,X_{d-1}$, let $S_l = \sum_{j<l} X_j$ $(0 \le l \le d)$. The largest sum of squared block increments over all dyadic prunings of $[0,d]$ is exactly dQ_n. Indeed, a pruning either keeps the root interval or prunes both children, which is precisely the displayed recursion. In particular, this is a lower bound for the square of genuine strong quadratic variation.

For each d let $\mathscr{W}_d$ be the finite set of all ordered families $([a_j,b_j])_{j=0}^{d-1}$ of intervals with endpoints in $\{0,\dots,d\}$, with $a_j \le b_j$ for $j < d$ and $b_j \le a_{j+1}$ for $j+1 < d$; degenerate intervals are allowed. The ordinary finite-dimensional Banach space

$$E_d = \prod_{w\in\mathscr{W}_d} \ell_d^2(\mathbb{C})$$

is given its maximum norm. Define the complex linear map

$$L_d z = \big((z_{b_j} - z_{a_j})_j\big)_{w\in\mathscr{W}_d}.$$

It kills constant paths and satisfies

$$\|L_d z\| \le V^2(z_l : 0 \le l \le d). \tag{27.18}$$

Padding each dyadic pruning by degenerate terminal intervals gives

$$\mathbb{E}\|L_d(S_l)_l\|^2 \ge d\,\mathbb{E}Q_n, \qquad \mathbb{E}|S_d|^2 = d. \tag{27.19}$$

We transfer the assumed Hilbert estimate to these finite path maps. The physical-space Mellin identity and integral Minkowski give, for every integrable f, the all-frequency conjugate-Poisson inequality

$$V^2(Q_s f : s > 0) \le \pi^{-1} V^2(H_\varepsilon f : \varepsilon > 0), \qquad Q_s f = \frac{i}{\pi}\int_{\mathbb{R}} \frac{y}{s^2+y^2} f(\cdot - y)\,dy. \tag{27.20}$$

In the cyclic Fourier convention its multiplier is $\sigma(\xi)e^{-2\pi s|\xi|}$ almost everywhere, where $\sigma(\xi) = 1$ for $\xi > 0$ and -1 otherwise. Fix positive strictly decreasing times $t_0, \dots, t_d$. On Schwartz inputs the assumed bound therefore, using conjugate-Poisson parameters $t_l/(2\pi)$, yields

$$\|\mathscr{F}^{-1}(M\mathscr{F}f)\|_{L^2(\mathbb{R};E_d)} \le \frac{C_H}{\pi}\|f\|_2, \qquad M(\xi) = \sigma(\xi)L_d\big((e^{-t_l|\xi|})_l\big), \tag{27.21}$$

where $\mathscr{F}$ denotes the cyclic Fourier transform. This symbol is continuous, including at zero: subtract the constant vector inside L_d and use $L_d\mathbf{1} = 0$ and $|\sigma| = 1$. We use the unit circle $\mathbb{R}/\mathbb{Z}$, its characters $e^{2\pi ikx}$, and normalized Haar measure. The cyclic-frequency form of Lemma 27.6, obtained by rescaling its 2π-periodic formulation, applies with the same constant. On this circle replace every Fourier coefficient c_k by $\sigma(k)c_k$. Parseval leaves the scalar L^2 norm unchanged, while $\sigma(k)^2 = 1$ removes the sign from the output. Thus, for every trigonometric polynomial g,

$$\big\|L_d((P_{t_l}g)_l)\big\|_{L^2(\mathbb{T};E_d)} \le \frac{C_H}{\pi}\|g\|_{L^2(\mathbb{T})}, \qquad \widehat{P_t g}(k) = e^{-t|k|}\widehat{g}(k), \tag{27.22}$$

uniformly in the finite positive time set and in d. No spectral-density passage is needed here.

Let p be any zero-mean trigonometric polynomial. On the product of d independent circles use phase variables $a = (a_j)_{j<d}$ and, for integers $M \ge 1$, set

$$N_j = (M+1)^{2j+1}, \qquad t_l = (M+1)^{-2l}, \qquad g_{M,a}(x) = \sum_{j<d} p(a_j + N_j x).$$

Apply (27.22), square, and integrate in a. For every x, translation $a_j \mapsto a_j + N_j x$ preserves product Haar measure. Thus the scalar right side is exactly $(C_H/\pi)^2\mathbb{E}|\sum_j p(a_j)|^2$, and the left side equals the squared product-Haar norm of

$$\sum_{j<d}\sum_{k\neq 0} \widehat{p}(k)e^{2\pi i k a_j} L_d\big((e^{-t_l|kN_j|})_l\big).$$

As $M \to \infty$, its multiplier tends to $1_{j<l}$ for every fixed j,k,l with $k \neq 0$. The sums are finite, so Fatou's lemma gives

$$\left\| L_d\left(\left(\sum_{j<l} p(a_j)\right)_l\right)\right\|_2 \le \frac{C_H}{\pi}\left\|\sum_{j<d} p(a_j)\right\|_2.$$

Realize one standard Gaussian as a measurable function of one Haar circle. It belongs to L^2 and has mean zero. Finite Fourier sums of this function, with the zero coefficient omitted, approximate it in complex L^2. For $e_j = (\mathbf{1}_{j<l})_{l=0}^d$, coordinate pullback is an L^2 isometry, so the triangle inequality gives, for every scalar $h \in L^2(\mathbb{T})$,

$$\left\| L_d\left(\left(\sum_{j<l} h(a_j)\right)_l\right)\right\|_2 \le \left(\sum_{j<d}\|L_d e_j\|\right)\|h\|_2, \qquad \left\|\sum_{j<d} h(a_j)\right\|_2 \le d\|h\|_2. \tag{27.23}$$

Both maps are thus bounded linear maps for each fixed d; their continuity bounds need not be uniform in d. Hence the preceding inequality passes to the actual Gaussian function. Product measure preservation now gives, for independent standard Gaussian leaves,

$$\mathbb{E}\|L_d(S_l)_l\|^2 \le (C_H/\pi)^2 \mathbb{E}|S_d|^2.$$

Together with (27.19) this implies $\mathbb{E}Q_n \le (C_H/\pi)^2$ for every n, contradicting (27.17). Thus no asserted Hilbert constant exists. □

Remark 27.8 (Sharpness of the output fluctuation exponent). The failure already occurs for $n = 1$, $P = 0$, $b \equiv 1$, and the odd kernel on $\mathbb{S}^0$. Thus the pointwise critical jump $\mathbf{J}(F)$—the paired/chain weak-ℓ^2 increment functional, up to the absolute comparison constants in (15.14)—cannot be replaced in the main theorems by the true strong variation $V^2(F)$.

28 Remarks on earlier proofs

Remark 28.1 (Independence from prior proofs). Our proofs do not use the coefficient-detection and annular-decay arguments in [2, 17, 25, 30, 31, 32] or the endpoint fluctuation arguments in [20, 6]. The observations in this remark concern the cited proofs as written, not the truth of their theorem statements. The discussion below refers to the preprint versions specified in the references.

In the preprint arXiv:2303.00594v1 corresponding to [17], the relevant coefficient-detection step follows the one-ray reduction of Lu–Zhang [32, pp. 210–213], which [17, Lemma 3.2] explicitly adapts. For the top bidegree block

$$P_{m,\ell}(x,y) = \sum_{\substack{|\alpha|=m \\ |\beta|=\ell}} a_{\alpha,\beta} x^\alpha y^\beta,$$

that reduction detects the coefficient tensor only through its diagonal restriction

$$A_{m,\ell}(\theta) = P_{m,\ell}(\theta,\theta) = \sum_{\substack{|\alpha|=m \\ |\beta|=\ell}} a_{\alpha,\beta} \theta^{\alpha+\beta}.$$

In the genuinely mixed setting $n \ge 2$ and $m, \ell \ge 1$, the map $P_{m,\ell} \mapsto A_{m,\ell}$ is in general not injective. For example, the alternating bilinear phase $P(x,y) = x_1y_2 - x_2y_1$ has a nonzero top bidegree block, but $A_{1,1} \equiv 0$. Thus the nonzero top-bidegree hypothesis in Lemma 3.2 does not supply the nonvanishing needed to apply Corollary 3.4 in the passage to (4.7). The related annular-decay reduction is invoked by Lu–Wu [30, p. 54, (3.1)] and [31, p. 308, Lemma 4], both referring to the argument of Lu–Zhang. The global phase flag used here avoids this diagonal compression and retains the full mixed coefficient data.

In the preprint version of [20], the squared-multiplier geometric tail preceding (2.16) carries a factor comparable to m that is not retained in the displayed estimate; after the Plancherel square root, the operator-norm estimate (2.16) is therefore missing a factor comparable to $\sqrt{m}$. Independently, Lemma 2.11 is false as stated for $0 < \theta < 1$: as $t \downarrow 0$, the ratio of its left-hand side to its right-hand side is comparable to $t^{\theta-1}$. The lemma is used in precisely this range to reduce a loss of order $m^{2-\theta}$ to the m-weighted angular summability available at the $L\log L$ endpoint. Consequently, that proof does not justify the endpoint summation in its displayed form.

In version 3 of [6, §2.2.2], the displayed fixed-shift families $\mathscr{D}_u = \{2^j([0,1)^n + m + u) : j \in \mathbb{Z},\ m \in \mathbb{Z}^n\}$ are not nested, although the subsequent stopping argument uses laminarity. For $n = 1$ and $u = 1/3$, the intervals $[1/3, 4/3)$ and $[2/3, 8/3)$ belong to the stated family and overlap without containment. A genuinely nested grid or another stopping argument is required. The distinct output-selector issue in the short-jump argument of version 2 is no longer an objection to the replacement argument in version 3. Neither observation is a counterexample to the endpoint theorem itself.

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